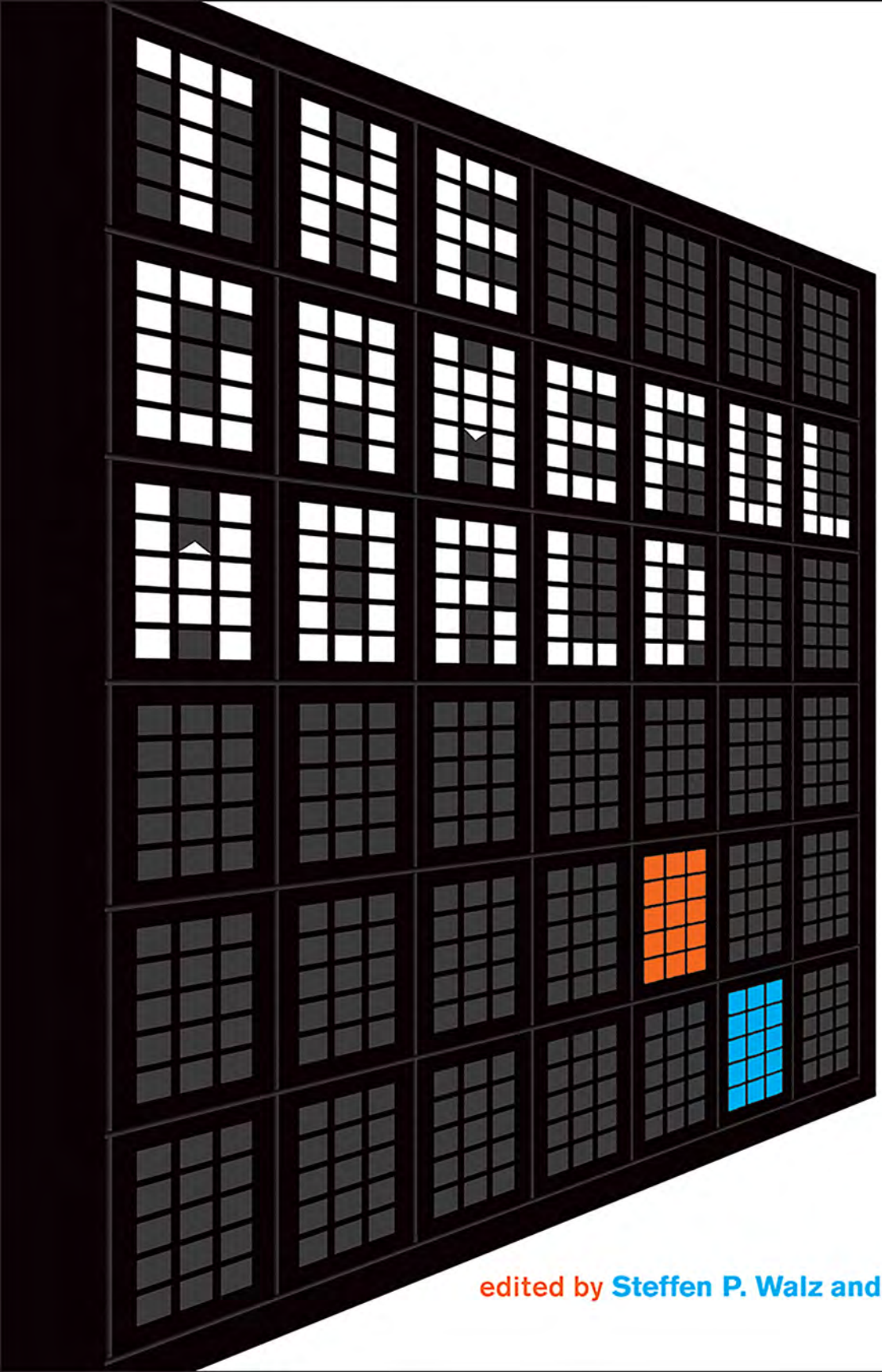


Approaches, Issues, Applications

edited by Steffen P. Walz and Sebastian Deterding

THE GAMEFUL WORLD

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AN INTRODUCTION TO THE GAMEFUL WORLD

Steffen P. Walz and Sebastian Deterding

Prelude

A gigantic conference table made of a single, thickly cut slice of wood (or so it seemed). Around it sat a client in a suit who represented a loyalty card program about to enter the market and the designers and developers of a web agency that had grown from four to more than one hundred employees in less than two years—all wearing T-shirts. Among them was one of the editors of this volume, then an aspiring interaction designer.

It was 1999, in the midst of the dot-com bubble. NASDAQ had not yet peaked. Many of us were speculating (intellectually and financially) what the Internet could become and how we would be interacting with people, products, and environments through networked computers in the near future. We dreamed that most of us would soon be carrying some form of handheld, networked device similar to a *Star Trek* Tricorder or IBM's proto-smartphone, the Simon Personal Communicator, which was released in 1991 (Sager 2012). And something else captivated the imagination of most of us designers around that table: in 1997, Ultima Online had been launched—the first massively multiplayer online role-playing game (MMORPG) to reach a subscriber base of 100,000 players within a mere six months (Electronic Arts 2006). Practically the entire agency played Ultima Online—during office hours. We hopped on and off game sessions between meetings and used the game's text chat to talk about game challenges and work projects alike. For most of us, Ultima Online had become part of our work culture. And to some extent, what looked and felt like a game had in fact turned into labor: the game extended into the office space, provided a crucial communication channel for job tasks, helped to bring aboard new employees, and kept us even busier than we were already. The directors of the agency had taken notice: the agency officially subscribed to several Ultima Online accounts, and certainly not just to keep the staff entertained.

At that table back in 1999, it had just been decided that the agency would design scenarios of how to engage online customers in the client's planned service using certain elements. Game elements. In the months prior, our agency had built a reputation for what was then called "advergaming"—little Flash-based web games that carried advertising messages. And we believed that in the suit sitting across the table was the ideal client with the ideal service to use gaming on a much grander scale: Loyalty Partner's Payback, which later not only survived the dot-com

crash but also went on to become Germany's largest loyalty card scheme, ultimately to be acquired by American Express in 2010 (Loyalty Partner 2010).

Everyone around the tree slice was excited about the new project: "So if customers collect discounts in the form of 'Payback bonus points,' and you market the service as 'life brings points,' clearly, you need a leaderboard," called out said coeditor. Added an agency colleague, "And we need badges, like we had them in church scouting. If Payback knows exactly what people shop, when and where, and in which sequence, couldn't we reward shopping patterns with badges, while collecting and mining that data? A 'toddler' badge if people consistently shop baby products. ... People will be super motivated!" And we went on to create and show a prototype of these ideas.

Yet in the end, nothing of that transpired. A couple of weeks after the initial launch of the Payback card in 2000, Loyalty Partner and its Payback card "won" the first German Big Brother Award (BBA), a negative price issued by the privacy and digital rights organization FoeBuD to raise public awareness about the most egregious violations of privacy rights. FoeBuD's rationale: Payback, albeit looking like a discount card, served the sole purpose of obtaining and commercially using personalized data related to the purchasing behaviors of consumers without properly informing the consumers about this fact (Big Brother Awards 2000). Soon after, and likely in response to the negative press stirred by the award, Payback became more transparent about its data collection; at the same time, federal privacy regulations prevented those leaderboards and badges we had conceived while chatting in Ultima Online, steering our avatars.

Payback worked with the agency for a while, but said coeditor left the agency soon after the BBA disaster to pursue an academic career, shocked into awareness of what could happen if game elements as simple as points were applied for non-entertainment purposes, even if conceived in a playful spirit.

Space Invaders: The Rise of Gamification

A few years ago, speaking about a *gameful world* would likely have conjured images of MMORPGs such as Ultima Online or World of Warcraft, not systems like Payback. The popular imagination of the time was (and still is) filled with utopian hopes and dystopian fears of an *Exodus to the Virtual World* (Castronova 2008): at best, we would escape the drudgery of twentieth-century schools and business training via serious games. At worst, people would regress from reality into the *Otherland* (Williams 1996) of games. Today, the direction is reversed: not people escaping into the virtual world of games, but games escaping into everyday life. On one side are utopian visions of re-engineering a supposedly "broken reality" (McGonigal 2011) into happiness engines: game design will allow us to maximize our individual potential, organize our "cognitive surplus" (Clay Shirky), and energize society to solve humankind's collective challenges. On the other side are dystopian reveries of Frankensteinian daemons and Skinnerian dictatorships: as algorithms increasingly rule the world (Slavin, this volume), we will wake up one day to find that our computers have become the game masters and we the pawns in one big Dream Park that isn't

just a game anymore (Suarez, this volume). Neuropsychological knowledge about the claviature of our pleasures will enable our future (robot) overlords to control our every step and action with a seamless, fine-tuned, pervasive mesh of incentive systems (Pesce, this volume).

Close observers could get a glimpse of the impending ludic reality invasion in the early 2000s, when live action role-playing, location-based, augmented reality, persistent alternate reality, and similar *pervasive games* began to extend the magic circle of play spatially, temporally, and socially (Montola 2005). But most of them remained too avant-garde and experimental to gain acceptance beyond the niches of academic laboratories, art exhibits, and game design festivals. It was only in the late 2000s that the reach of games and game design into everyday life appeared in the public consciousness, with "gamification" emerging around mid-2010 to overtake "serious games" in global web search interest in mid-2011.[1] The arguable blueprint for the gamification movement has been foursquare, a social, mobile, location-based service launched in 2009: foursquare allows users to "check in" at venues, see check-ins of members of their social graph, and browse venue suggestions based on check-in data. To motivate check-ins, foursquare implemented a series of game-like design elements:

- *Points* Every time a user checks in, she earns a variable amount of points depending on factors such as novelty (first check-in to a location of this type) or distance (check-in far away from last check-in).
- *Badges* Certain types or combinations of check-ins unlock "badges," virtual marks of achievement, like the "Gym Rat" badge earned for checking into gyms ten times in the course of thirty days.
- *Leaderboards* Points are compared with members of one's social graph on a weekly point leaderboard, aiming to spur competition between users.
- *Mayorships* Those users who have checked in the most often at a location in the past sixty days are acknowledged as its "mayor," which again can spur competition between users.
- *Rewards* Business owners of a location may offer location-specific rewards (e.g., a free drink when a user checks in at a certain time).

Today, five years after foursquare's launch, a whole cottage industry of gamification consultants, agencies, and software providers has emerged, spanning from "white label" platform providers such as Bunchball, Badgeville, or Lithium to more specialized platforms such as (now defunct) Greengoose, a sensor and software package for self-tracking everyday activities. Business consultancies publish rosy predictions, all arrows predictably pointing up and to the right: according to one forecast, the gamification market will grow from US$100 million in 2011 to US$2.6 billion by 2016 (Meloni & Gruener 2012). The tone is nothing if not evangelical: business books promise a *Gamification Revolution* (Zichermann & Linder 2013), explaining *How to Revolutionize Customer and Employee Engagement with Big Data and Gamification* (Paharia 2013).

Implementations abound across all domains of life, the overwhelming majority using some variation of the points-badges-leaderboards model of foursquare (Hamari, Koivisto & Sarsa 2014). In marketing, one finds digital loyalty programs and sweepstakes built around "customer engagement": checking into a store; sharing or liking posts and product pages of brands on social media platforms. Examples are platforms like Lockerz, Getglue, or the Buffalo Wild

Wings in-store mobile phone challenge, which prompts users to, for example, check in at the store or take a picture of a food and share it online to earn points redeemable for drinks and food. Another common form is a new breed of advertising games like the Heineken StarPlayer: while watching a soccer match on television, players of the application can bet on how certain game situations (such as a free kick) will be resolved or whether a goal will shortly occur (Coulton, this volume).

In health and wellness (see Munson et al., this volume), gamification chiefly intersects with the quantified-self movement of individuals seeking self-knowledge and self-improvement through self-tracking and analysis of their behaviors, body states, and experiences (Wolf 2009). The most publicized example for this intersection has been Nike+ (and now Nike Fuelband), a suite of tracking device and software that records everyday exertion and translates it into a universal score, complete with personal goal-setting and social competition. Health Month has been another early, influential system. Developed by Buster Benson (this volume), it allows users to set personal health goals and track their daily activity against these goals and to form player groups with collective goals and the ability to "heal" each other if one loses "life points" by missing a goal. In their wake, innumerable applications nowadays combine self-tracking with goal setting and virtual achievements, some for individual fitness (such as Runkeeper or Fitocracy), some as enterprise health programs (such as Keas), and some for task management (such as Chorewars or Epic Win). Zombies, Run! offers a slightly different, more narrative take: in this mobile running application, the player is motivated to run physically in order to escape fictional zombies.

In education, gamification has been adopted chiefly among practitioners and researchers interested in learning analytics, new forms of assessment, and self-directed, self-motivated online learning as found in, for example, massive open online courses (MOOCs). One highly influential exemplar is Khan Academy, a website that offers videos and exercises around basic educational topics, where users can earn points and badges (and educators can track learner performance). On a grander scale, the Quest to Learn schools restructure the entire school life and curriculum in the image of games (Salen, this volume). At the Rochester Institute of Technology, undergraduate students can engage with Just Press Play, an achievement system for noncurricular activities such as visiting a professor in her office (Ramirez & Squire, this volume). And already in 2010, Microsoft Office Labs released Ribbon Hero, an add-on that taught users the basics of the Microsoft Office Suite, including the then-new ribbon interface, with a game interface of challenges, achievements, and scores layered into the Microsoft Office software itself.

In sustainability, a notable example is Chromaroma, a platform that allows users to visualize their public transport use in London as tracked by their Oyster Card; users can form competing teams, and the system nudges users toward more sustainable commuting behaviors with goals that put walking and cycling over public transport, and public transport over car driving. And in cars themselves, one nowadays finds a plethora of "eco-feedback systems" with more or less explicit "gamy" qualities, most prominently in-car "eco-dashboards" that display how environmentally friendly one is driving, such as Ford's SmartGauge or the Ecoscore of the car-sharing service Car2Go (Froehlich, this volume).

In the enterprise space, many companies have been experimenting with gamification for training, innovation, and employee engagement (Mollick & Werbach, this volume). In training, one can find examples like the Deloitte Leadership Academy, adding point scores, missions, achievements, and leaderboards to its online video and exercise leadership training program. With regard to innovation, one finds a host of platforms for employee and consumer online idea competitions, such as the 2011 Volkswagen People's Car Project, which invited Chinese consumers to submit and vote on concepts for a future automobile, complete with teams, scores, leaderboards, and badges. In terms of employee engagement, gamification platform vendors have released a wealth of software suites to track and motivate employee performance through scoreboards, goal setting, and leaderboards, specifically in sales and customer service. The popular customer relationship management platform Salesforce alone currently counts twelve different gamification apps.[2]

Defender: Resistance to Gamification

On one side, then, optimistic authors such as Jane McGonigal (this volume) argue that mankind's existential challenges in the twenty-first century can be fixed with game design. Scholars like Byron Reeves (Reeves & Read 2009) hold that games present systems of informational feedback and incentives that are perfectly organized for reinforcement learning and for coordinating the collective action of self-interest–driven individuals; thus, we can learn from them how to design perfect markets and behavior change systems (see also Linehan et al., this volume; Rangaswami, this volume; Williams, this volume). Business consultancies like Gartner (2011) declare gamification to be a major business technology trend, and agencies and software vendors promise that gamification will "revolutionize" all areas of society and economy, motivating us as consumers to co-create, to buy, and to produce by word-of-mouth; as employees to engage in the workplace; as citizens to participate in politics and collective problem-solving; and as individuals to learn, live healthy, and act sustainably.

On the other side, we find equally vocal critics coming mostly from game design and academia (Juul 2011). They hold that "gamified" products never can, nor ever intended to achieve, the engaging qualities of well-designed games. The joys of game play arise from having meaningful choices in trying to achieve interestingly difficult goals. In contrast, gamification is "taking the thing that is least essential to games and representing it as the core of the experience"; namely, the scoring feedback that tells players how well they are on the way to achieving their goals (Robertson 2010). What gamification proponents are interested in, following Bogost (this volume), is merely commodifying the current cultural cachet of games into an easily sellable workshop format or "turnkey" technology. In this, they ignore that game *design* is an inherently complex, risky, skill-based, situation-bound, and therefore nonscalable process, as well as the potential differences of users; for instance, not all cultures equally value competition (Khaled, this volume). Gamification presents but the latest form of ideology masking

political disenfranchisement and exploitation of digital labor as playful self-realization (Rey, this volume).

Thus, many gamification vendors show little ethical concern for the affected users. Privacy and data ownership are one obvious issue here: Who owns, controls, and views the data generated? How might employers, health insurers, or governments use it? What chilling effects might the tracking and publication of such data have? Does the playful veneer of gamification make users willingly self-report behavioral data they would in no other context allow to be tracked (Andrews, this volume)?

Others argue that by presenting games and gamification as information and incentives, gamification proponents fail to appreciate that play is the quintessential case for intrinsically motivating activity: adding a layer of rules, goals, feedback, and consequences might motivate participants through coercion or incentivization but actively thwart rather than tap into the motivations characteristic for game play (Rigby, this volume; Stenros, this volume).

Finally, there are unintended consequences such as gaming the system: framing an activity as the singular pursuit of goals spelled out in metrics and targets attracts exploitive actors interested in finding loopholes. Even with regular actors, it may crowd out wider concerns for any factor not captured in the metrics, such as moral conduct, "negative externalities," or whether "meeting the target" "misses the point"; that is, whether a given goal is contextually sensible (Deterding 2012).

From Gamification to *The Gameful World*

One important part of the debate around *gamification* has been the word itself. Advocates have attempted to establish gamification as the umbrella term for anything game-related, including serious games, or even to extend it beyond games to include loyalty programs and applications of behavioral economics. Critics have taken this as an indication that gamification proponents are not interested in games at all, only in the attention value of the term *games*. Thus, they have phrased *serious games* or *gameful design* as *opposites* to gamification and have suggested to replace *gamification* with terms like *exploitationware* (Bogost, this volume).

One issue with these language disputes is that they (sometimes intentionally) conflate descriptive and political, normative levels: Instead of first establishing descriptive terms and then judging whether specific instances meet aesthetic, practical, or moral standards, they generalize and position "good" = well designed = ethical *serious games* or *gameful design* against "bad" = poorly designed = unethical *exploitationware* or *gamification*. Furthermore, they ignore the wider context. Gamification is certainly the most recent and visible instantiation of the interpenetration of games and everyday life. Yet not only has this interpenetration a long and varied history, spanning from art movements like the Situationists to war games, serious and pervasive games, and playful design in human–computer interaction: it is also but a part of a wider trend that has been variously called the "ludification of culture" (Raessens 2012), "ludofication of society" (Walz 2006), or the rise of a "ludic society" (Stenros, Montola & Mäyrä 2007, 32), "ludic century" (Zimmerman, this volume),

"ludic language" (Flanagan, this volume), and "ludic architectures" (Walz 2010). Practices and attitudes, patterns and tropes, materials and tools, languages and concepts from (digital) games and play increasingly pervade all arenas of life. Just as importantly, artists and businesses, scholars and technologists, institutions and subcultures in turn attempt to harness and shape games and play for their own purposes. Whereas game scholars have mostly painted this as a lamentable "rationalization," "colonization," or "instrumentalization" of games and play, we hold that this *cultivation of ludus* is just the logical complement to the *ludification of culture*. How can we expect games and play to "migrate" into new territories without undergoing some "acculturation" in the course (see Deterding, this volume)?

Hence, instead of using the value-laden term *gamification* or the narrower concept of *ludification*, for the current volume we have chosen to speak of "the gameful world." In this, we build on the terminology of Deterding et al. (2011), who suggested mapping the current use of games and play beyond leisurely entertainment along two dimensions: wholes versus elements or qualities, and *paidia* versus *ludus*. The latter distinction is taken from philosopher Roger Caillois (2001), who noted that all forms of human play fall on a spectrum between open, free, exploratory play as we find it in children's object and pretend play (paidia) and formalized, rule-based, goal-oriented play as we find it in games (ludus). One may thus distinguish

- serious games: "ludic wholes," or full-fledged games designed and/or deployed for non-entertainment purposes;
- serious toys: "paidic wholes," or toys designed and/or deployed for non-entertainment purposes;
- playful design: "paidic elements or qualities," or non-toy objects and experiences that use design elements from toys and/or are designed to afford playful experiences;
- gamification (or gameful design): "ludic elements or qualities," or non-game objects and experiences that use design elements from games and/or are designed to afford gameful experiences.

All four are part of but do not exhaust the double movement of the ludification of culture and the cultivation of ludus, which together we call the rise of a gameful world (figure I.1).

The Question of the Gameful World

So what are we to make of the rise of a gameful world and the debate between gamification proponents and critics? As editors, we were struck by four things. The first is the narrow focus on gamification itself, blending out its prehistory and wider double movement of the ludification of culture and cultivation of ludus.

The second is the tendency to speak in absolutes: gamification proponents promise revolutions, yet seldom mention possible limitations, complications, or downsides. Conversely, critics of gamification voice valid and important concerns, but again paint with a broad brush. Even if questionable actors and implementations are currently in the majority, this does not disprove that games and game design have a growing impact on our culture, nor that they hold valuable learnings for other domains—just as comic

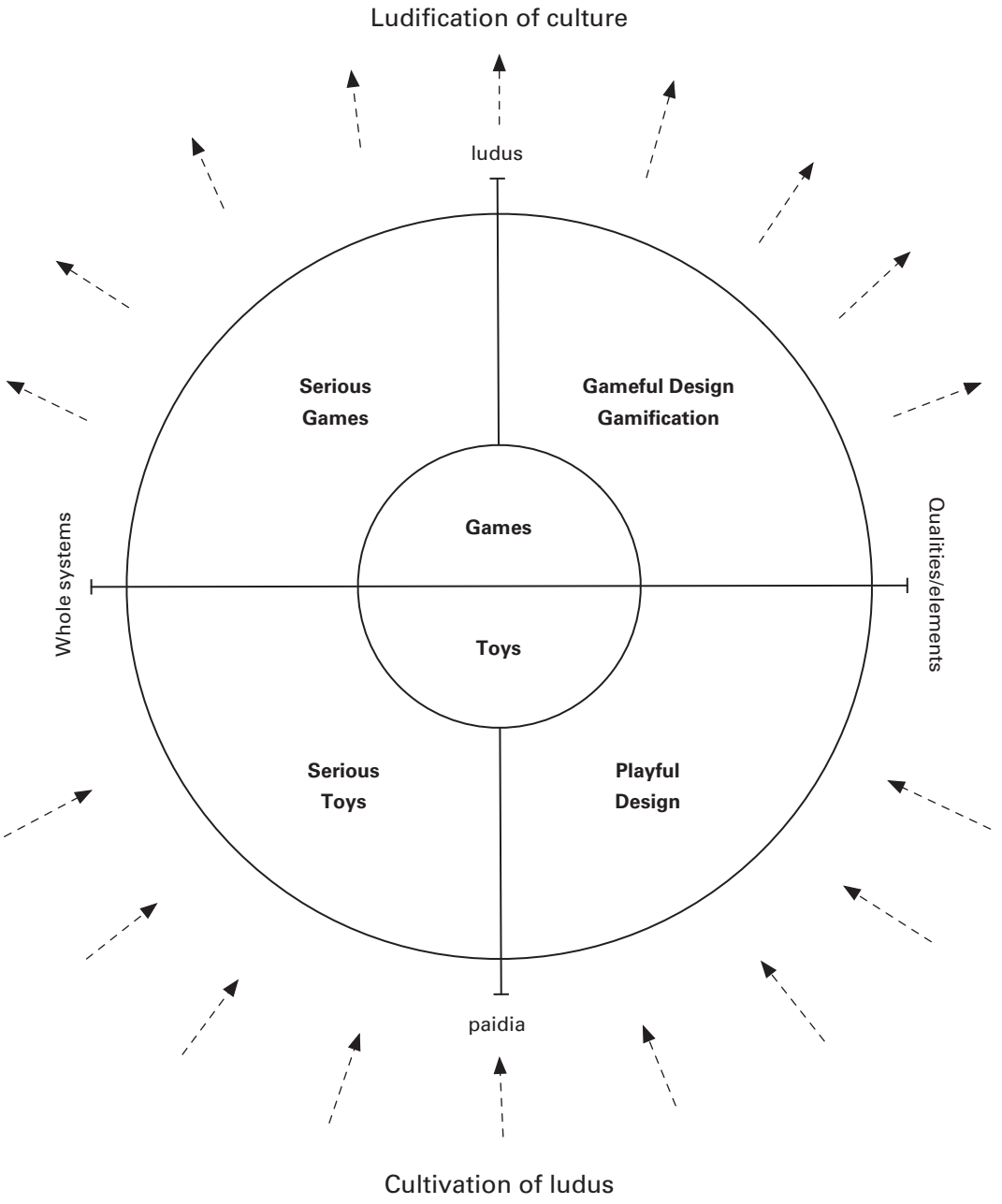


Figure 1.1
A conceptual mapping of the gameful world.

artists have been inspired by movies, novelists by advertising and screenwriting, interaction designers by graphic design, or hackers by science fiction (Stober et al. 2013). Given that game design is *the* practice of creating enjoyable interactions, it stands to reason that it holds something of interest to any domain where interaction is designed and the goal is to make it more enjoyable. In this sense we agree with proponents that gameful design has potential—although we hasten to emphasize *potential*. Despite the publicized short-term success stories of gamification vendors, there are still few solid, peer-reviewed empirical studies on the effects of using game design in non-game contexts (Hamari et al. 2014).

Third, we were struck by the sheer fervor of the debate. As Deterding (this volume) argues, what we see at work here is not so much a disagreement over facts than a clash of *rhetorics*—worldviews and moral politics reproduced by communities and their languages, most importantly in this case, different ideas of the "proper" place of games and play in society. Gamification has brought communities into the discourse around games and play whose rhetorics are alien and often anathema to the rhetorics of game designers and scholars. In Victor Turner's (1982) terms, gamification proponents predominantly subscribe to the *liminal* use of games and play as a conservative perfection of means toward the given goals of the existing social order. Game scholars and designers in contrast predominantly see the "proper" use of games and play beyond entertainment as *liminoid* progressive questioning and subversion of the standing order.

Fourth and finally, while proponents have been busy producing blueprints and services for bringing a gameful world about, and critics busy ridiculing the very idea, we are dearly lacking solid description, analysis, and reflection of just *what is happening now* and just *how it will affect us, the people*. *What if* all of our everyday life is turned into a game? What would be the consequences of life governed by a pervasive web of sensors tracking our every action, algorithms evaluating them against rules and goals set by ourselves and others, and effectuators constantly feeding back information on our performance, status, and progress? How would we work, commune, and act politically under such circumstances? How would it alter (and disturb) the ordering of our everyday interaction? And what happens to games and play themselves? What are the ethical ramifications of a societal *panludicum*—for policy makers, for designers, but also for individuals alternatively extending or replacing our will with technically mediated systems of goals?

There are no definite answers to any of these questions yet. But the underlying issues have been tackled in philosophy, game studies, human–computer interaction, psychology, sociology, economics, anthropology, and other disciplines. These literatures at least provide us with inroads to the questions a gameful world poses. The goal of the current volume, then, is to scrutinize the ramifications of a gameful world, the promises it holds, and the issues it brings—socially, economically, politically, culturally, ethically, and on a personal level. Rather than shoehorn the heterogeneity of phenomena and rhetorics into one coherent argument, we believed it most helpful to surface its very heterogeneity: to rub the different stances of advocates and critics against each other, thus highlighting their partially rhetorical nature, but also enabling the reader to make up his or her own mind.

Organization of the Book

The essays collected in this book are organized into three broad parts. The first part, "Approaches," disentangles various disciplinary perspectives that have been used to frame gamification. It is grounded in the chapter "The Ambiguity of Games," a historical survey that traces the main precursors and parallels of today's gameful world and synchronically maps the main contemporary rhetorics. Chapters by representative scholars each explicate one perspective: rhetorics (Bogost), behavioral psychology (Linehan, Kirman & Roche), motivational psychology (Rigby), neoclassic and behavioral economics (Hamari, Huotari & Tolvanen), play and performance (Stenros), aesthetics (Flanagan), design (Hassenzahl & Laschke), and ethics (Sicart).

The second part, "Issues," breaks out major issues at stake in pervading life with game elements: exploitation (Rey), culture (Khaled), media spectacle (Borland), social control (Whitson), morality (Selinger, Sadowski & Seager), privacy (Andrews), and the technical underbelly of a gameful world (Nova).

The third part, "Applications," surveys existing research in major application domains: product and service design (Holopainen & Stain), the enterprise (Mollick & Werbach), social media (Lampe), science (Cooper), politics (Lastowka & Steinkuehler), cities (Alfrink), sustainability (Froehlich), education (Ramirez & Squire), and health (Munson et al.).

In each part, *chapters* are interspersed with *position papers*. Chapters synthesize and critically reflect the existing literature around an approach, issue, or application area. As counterpoints, position statements provide subjective voices by practitioners, theorists, and activists that have been formative for the gameful world.

The choice of chapter topics emerged from a systematic mapping of the main actors, topics, and rhetorics of the gamification discourse in 2012. As editors, we tried to match each topic with authors bringing deep expertise in both games or gamification and the given subject matter. Despite its size, this book does not claim to be a comprehensive representation of *all* angles and voices. To the best of our ability, where currently influential voices or angles had to give, we tried to at least provide ample reference to them throughout the book. Still, this book very much remains an opening move to a—hopefully—continuing conversation.

To give our readers a first map and compass for this conversation, we have arranged the various contributions into a coordinate system with two axes, reflecting the main fault lines of the debate around the gameful world: one axis represents the different forms of play appealed to, *paidia* and *ludus*, the other axis the different moral politics of the "proper" place of play and games in social life: *liminal* and *liminoid*. By assigning numeric values on both axes to each contribution in the book, we arrived at a coordinate for it: its place in *The Gameful World* (figure I.2).

In closing, we would like to thank first and foremost our authors for embarking on this adventure with us, our anonymous reviewers for their valuable guidance, and our editorial producer Sebastian Felzmann for keeping the ship afloat throughout the journey, as well as editorial assistant Sebastian König and translator Jenna Krumminga. Lastly, we would like to acknowledge that the production of this book has been supported by the Royal Melbourne Institute of Technology, Australia, and its Games and Experimental Entertainment Laboratory, the GEElab.

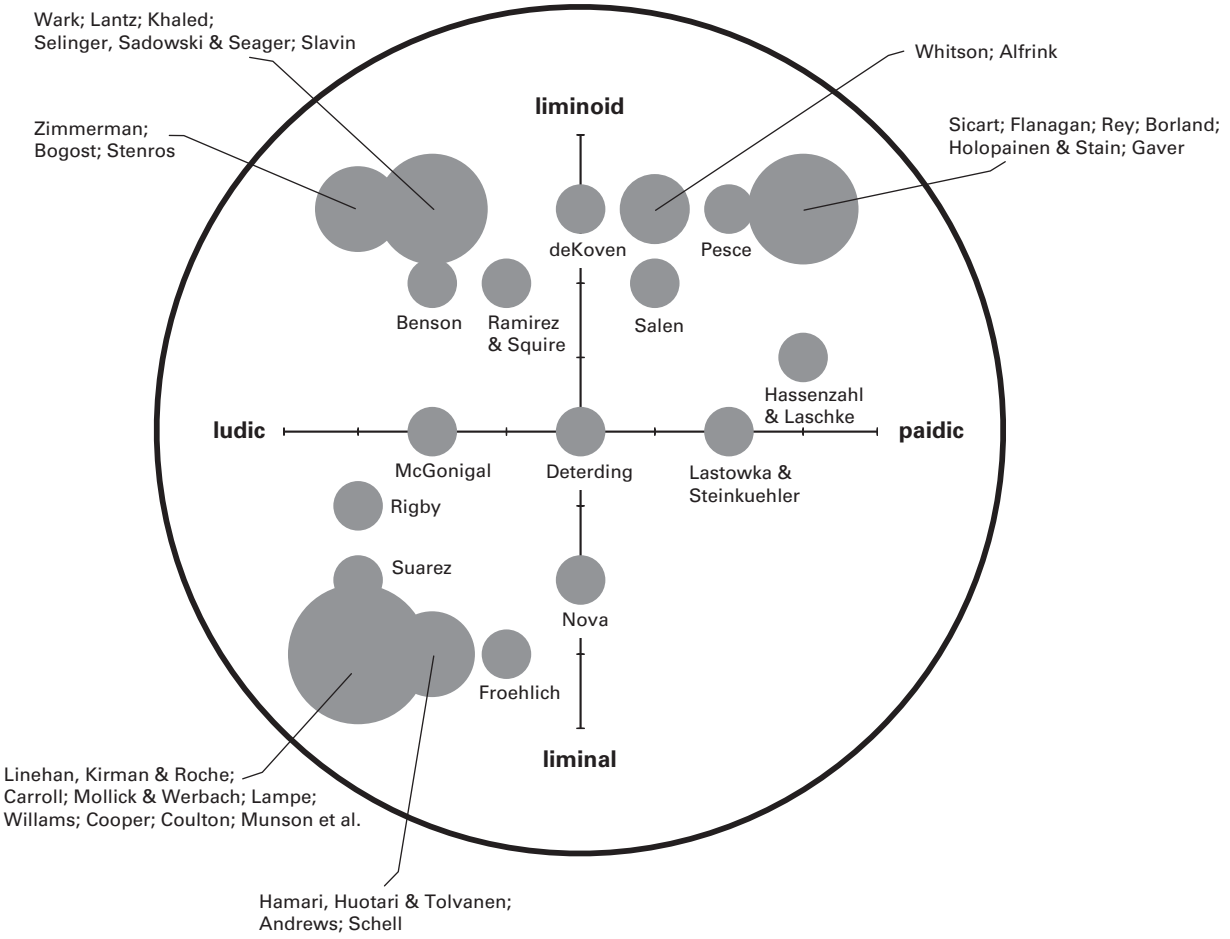


Figure 1.2

A compass to *The Gameful World:* Contributions by predominant form of play and moral politics.

Notes

1. See http://www.google.com/trends/explore#q=serious%20games,%20gamification (accessed May 11, 2014).

2. See https://appexchange.salesforce.com/collection/gamification (accessed November 24, 2013).

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I APPROACHES

How better to open a gameful world than with a proper manifesto? At the Games Learning Society conference in 2008, game designer and scholar Eric Zimmerman and game journalist Heather Chaplin invited the audience to muse whether the nineteenth-century industrial age and twentieth-century information age might be followed by a ludic age today. In this, they articulated an essential question of the gameful world: does it demarcate a historical shift? And if so, what are its contours? We therefore invited Zimmerman to pick up this trail of thought for the current book, and pick it up he did: his "Manifesto for a Ludic Century" contends that the twenty-first century is indeed a ludic age. Not only are the interactive digital systems that characterize our time deeply game-like, but also play and game design are fundamental literacies required if we want to build and use these systems to solve the challenges of our time. True to its activist nature, the manifesto quickly escaped its scholarly confines and stirred controversy far ahead of its print publication.[1]

As this and other recent debates reveal, framing the gameful world as one of (interactive, designed, playable) systems is but one of many possible—and often clashing—perspectives. The purpose of part I of this book is to disentangle these various perspectives and make them explicit. It invites representative scholars to each articulate one theoretical or disciplinary angle on gamification and the gameful world. Part I is anchored in coeditor Sebastian Deterding's mapping of the origins and discourses of the gameful world. Taking its cue from Victor Turner and Brian Sutton-Smith, "The Ambiguity of Games" argues that the fervor of the current debate surrounding gamification stems not from factual disagreements but rather conflicting rhetorics and their implied moral politics. Whereas "indigenous" game scholars and designers predominantly see the proper function of play and games to be an escape from, alternative to, or questioning of society, marketers, psychologists, and other "immigrating" communities are comfortable with using games and play as a perfection of means toward societies' given ends. Deterding ends with an overview of the different contemporary rhetorics of the gameful world that are covered in individual detail in the chapters that follow.

First is Ian Bogost's *rhetorical* analysis of the contemporary gamification movement, "Why Gamification Is Bullshit." A designer and scholar well known for his work on persuasion and games, Bogost expands his earlier, widely read provocation that "gamification is bullshit" into an argument for that position. He connects the philosophical definition of bullshit to the practice of management consulting, to which

gamification belongs as a subspecies. Bogost argues that like most consulting tricks and trends, gamification exists to benefit the purveyors of its solutions, rather than to help individuals or organizations. It has less in common with games than the sixty-plus-years-old concept of business intelligence. Bogost concludes that far from being a novel use of games for business, gamification is just the latest way to dress up business consulting to achieve temporary lift until the trend burns out and is replaced by another one.

Against this critical perspective, Conor Linehan, Ben Kirman, and Brian Roche articulate how *behavioral psychology* can be used to understand practices commonly used in gamified products and services. Their chapter argues that gamification can be interpreted as the technologically facilitated application of behavior modification methodologies to everyday events on a massive scale. It discusses the philosophy, goals, history, evidence base, and perceived limitations of behavioral psychology in order to explain how this branch of learning psychology can provide practical, evidence-based guidance in the design of more useful and engaging gamified experiences.

In direct response, Mark Pesce, virtual reality pioneer and author of *The Playful World*, paints a stark picture of a potential future that takes what Linehan, Kirman, and Roche describe to the logical conclusion: humankind has become an insectoid hive where psychological knowledge enables the precise and total control of individuals' every thought and action. Pesce urges for a *contraludics* that defends the individual's integrity of inner self-determination against such outer social control—a resistant practice that merges neuroscience, religious practices of self-development, and play.

Scott Rigby's "Gamification and Motivation" likewise foregrounds the importance of self-determination. Where Linehan, Kirman, and Roche articulate the behavioral psychology view on gamification, Rigby provides the *cognitive psychological* counterpart. With a specific focus on self-determination theory as one prominent contemporary cognitive approach to motivation, he unpacks the concepts of fun, engagement, and motivation, outlines potential issues of implementing gamification as a reward system, such as thwarting autonomy, and derives some guidelines for the design of gamified applications.

Juho Hamari, Kai Huotari, and Juha Tolvanen investigate how gamification relates to and differs from key established *economic* theories and concepts. They first discuss how gamification can be viewed from the neoclassical perspective. Here, gamification might provide solutions to a class of classical problems in information economics and mechanism design. Next, they illustrate how gamification taps into heuristics and biases identified in behavioral economics. Lastly, they discuss how gamification relates to marketing, specifically transactional, relationship, service, and experiential marketing.

In "Losing is Fun," cultural theorist McKenzie Wark turns this economic perspective into a critical mirror: games, he holds, have become the emblematic form of our times because they provide an almost-utopian double to their systems logic. In games, the neoliberal promise is actually kept: the playing field really is "level," and the "winner" is determined on her or his merits. As such, games call into question the imperfectly realized gamespace that we experience in everyday life, where the playing field is far from level, and certain "winners" seem to

have gotten something of a leg-up. A critical theory of games can get purchase within the form of the game by considering it as an allegory of the times, which holds the everyday of this "ludic century" to account by its own declared standards.

"Losing is fun" could also serve as the motto of Mark Hassenzahl's and Matthias Laschke's *design* perspective on gamification. A doyen of experience design in human–computer interaction, Hassenzahl together with Laschke evolves his earlier "funology" work into an "aesthetics of friction" for technologies that help people close the gap between their actual and ideal selves. While Hassenzahl and Laschke agree with self-quantification, gamification, and persuasive technology that interactive technologies can facilitate personal change, they urge us to focus on *how* these technologies should be designed to work—and to be ethically acceptable. To this end, their chapter develops the notion of "Pleasurable Troublemakers" and outlines a set of underlying principles to support personal transformation: technology should create rather than restrict choice. It should be situated to disturb, to rescript and reshape given routines—but with a certain naivety, understanding, and a touch of irony. In short, it should instill change through friction and insight—with a smile.

John M. Carroll's position paper provides a varied reprise on both Hassenzahl and Laschke and his own early influential work on minimal design and fun of use. To him, using game design in non-game contexts means to articulate critical abstractions of game design that can be re-instantiated elsewhere. Contrasting the classic computer game Adventure and early end-user office systems, Carroll derives two such design archetypes from games. One is transparency to action: documentation and interactions should be simple, clear, consistent, and anchored in mental models. The other is conduciveness to action: providing feedback that provokes hypotheses and testing, attenuated error consequences, reversible commands, and error states that are easy to recognize, diagnose, and recover from.

Looking "Behind Games" and their design, noted pervasive game researcher Jaakko Stenros finds playing and playfulness as crucial components of game play that are typically overlooked in the discourse around the gameful world. His *performance studies* perspective fills in this blind spot: at the heart of game play is the (biologically grounded) mindset of playfulness. Shared playfulness becomes socially framed and formed as playing and codified as games. Thus, gamified systems that only deploy the codified system but don't afford playfulness run the risk of being hollow. At the same time, playfulness is by definition transgressive and disruptive: it can be used in transformative practices that break personal and societal patterns and question norms, but doing so opens the door to deep and dark play, griefing, trolling, and bullying as inherent potentials of playfulness.

"A Gameful Mind" by Buster Benson beautifully illustrates Stenros's argument, while also echoing Rigby's cautionary note on reward systems. In his position piece, Benson relates how his discovery of an exploit in the reward scheme for good grades set up by his parents enabled Benson to switch his stance toward school from evading drudgery to treating it as a game—in Bernard Suits's definition, as a "voluntary attempt to overcome unnecessary obstacles." A lifelong self-logger and explorer of the question how to change oneself, Benson has built several startups and systems for gameful self-transformation, among them the influential Health Month. The best they can do, he observes, is not to coerce or entice with

punishment or rewards, but to afford just this autonomous, gameful stance.

In much the same spirit, game scholar Miguel Sicart formulates a *philosophical* view on the gameful world grounded in virtue ethics. Gamified applications delegate values and visions of the good life to a machine. While technologies designed for improving habits have an apparent beneficial effect in quantifiable terms, from an ethical point of view Sicart considers them to be troublesome. They do not require us to develop or practice moral values, only to enact tasks in mechanical repetition. Sicart suggests to counterbalance this with play as a source of a good life, and an appropriation of technology and context through which we can explore, practice, and reflect its possibilities. This notion of play as deliberative exploration and cultivation of virtues is in many ways the exact practice and stance that Hassenzahl and Laschke's pleasurable troublemakers seek to afford—and it underlines the importance of play as a stance articulated by Stenros and Benson.

Shifting perspectives one final time, pioneering game designer and scholar Frank Lantz inquires about games as an *aesthetic* form. The aesthetic experiences that games provide "exist outside our ordinary frameworks of logic and purpose, cause and effect, reason and explanation," he holds, and as human beings, "we *need* to have this outside." Instead of asking how to use aesthetic experiences to achieve real-world goals, we should ask how we might use the real world in creating more interesting aesthetic forms.

Bridging aesthetics and the good life is game artist, writer, and designer Mary Flanagan. Her chapter traces the ludification of culture very literally; namely, how a "ludic language" of game and play practices and tropes has permeated other cultural spheres, art in particular. To Flanagan, these playful aesthetics constitute a movement not from reality into art, but toward communal presence, action, and possibility in everyday life. In this sense, the playful in art is very much a site for experiencing and exploring the good life as individuals and citizens alike: a foreshadowing of the best of all possible ludic centuries.

Note

1. See http://storify.com/dingstweets/manifesto-for-a-ludic-century (accessed November 24, 2013).

POSITION STATEMENT

MANIFESTO FOR A LUDIC CENTURY

Eric Zimmerman

Games are ancient.

Like making music, telling stories, and creating images, playing games is part of what it means to be human. Games are perhaps the first designed interactive systems our species invented.

Digital technology has given games a new relevance.

The rise of computers has paralleled the resurgence of games in our culture. This is no accident. Games like Chess, Go, and Parcheesi are much like digital computers, machines for creating and storing numerical states. In this sense, computers didn't create games; games created computers.

The twentieth century was the century of information.

Systems theory, communications theory, cybernetics, artificial intelligence, computer science—these fields, many of them emerging well before electronic computers, helped create the "information revolution."

The abstraction of information has made possible massively complex bureaucracies and technologies, from telegraph and telephone networks to NASDAQ and Facebook.

In our Ludic Century, information has been put at play.

Our information networks no longer take the form of vast card catalogs or webs of pneumatic tubes. Digital networks are flexible and organic.

In the last few decades, information has taken a playful turn. To take a prime example, Wikipedia is not about users accessing a storehouse of expert knowledge. It is a messy, chaotic community in which the users are also the experts, who together create the information while also evolving the system as a whole.

In the twentieth century, the moving image was the dominant cultural form.

While music, architecture, the written word, and many other forms of expression flourished in the last century, the moving image came to dominate. Personal storytelling, news reporting, epic cultural narratives, political propaganda—all were expressed most powerfully through film and video.

The rise of the moving image is tightly bound to the rise of information: film and video as media represent linear, noninteractive information that is accessed by a viewer.

The Ludic Century is an era of games.

When information is put at play, game-like experiences replace linear media. Media and culture in the Ludic Century is increasingly systemic, modular, customizable, and participatory. Games embody all of these characteristics in a very direct sense.

Increasingly, the ways that people spend their leisure time and consume art, design, and entertainment will be games—or experiences very much like games.

We live in a world of systems.

The ways that we work and communicate, research and learn, socialize and romance, conduct our finances and communicate with our governments are all intimately intertwined with complex systems of information—in a way that could not have existed a few decades ago.

For such a systemic society, games make a natural fit. While every poem or every song is certainly a system, games are dynamic systems in a much more literal sense. From Poker to Pac-Man to Warcraft, games are machines of inputs and outputs that are inhabited, manipulated, and explored.

There is a need to be playful.

It is not enough merely to be a systems-literate person to understand systems in an analytic sense. We also must learn to be playful in them. A playful system is a human system, a social system rife with contradictions and with possibility.

Being playful is the engine of innovation and creativity: as we play, we think about thinking and we learn to act in new ways. As a cultural form, games have a particularly direct connection with play.

We should think like designers.

In the Ludic Century, we cannot have a passive relationship to the systems that we inhabit. We must learn to be designers, to recognize how and why systems are constructed, and to try to make them better.

It took several decades for automobiles to shift from being a hobbyist technology requiring expert knowledge to being a locked-in consumer product. The constant change of digital technology means that our hardware and software systems may never stabilize in this way. To engage fully with our world of systems, we must all think like designers.

Games are a literacy.

Systems, play, design: these are not just aspects of the Ludic Century but also elements of *gaming literacy*. Literacy is about creating and understanding meaning, which allows people to write (create) and read (understand).

New literacies, such as visual and technological literacy, have also been identified in recent decades. However, to be truly literate in the Ludic Century also requires gaming literacy. The rise of games in our culture is both cause and effect of gaming literacy in the Ludic Century.

Gaming literacy can address our problems.

The problems the world faces today require the kinds of thinking that gaming literacy engenders. How does the price of gas in California affect the politics of the Middle East affect the Amazon ecosystem? These problems force us to

understand how the parts of a system fit together to create a complex whole with emergent effects. They require playful, innovative, transdisciplinary thinking in which systems can be analyzed, redesigned, and transformed into something new.

In the Ludic Century, everyone will be a game designer.

Games alter the very nature of cultural consumption. Music is *played* by musicians, but most people are not musicians—they listen to music that someone else has made. Games, on the other hand, require active participation.

Game design involves systems logic, social psychology, and culture hacking. To play a game deeply is to think more and more like a game designer—to tinker, retro-engineer, and modify a game in order to find new ways to play. As more people play more deeply in the Ludic Century, the lines will become increasingly blurred between game players and game designers.

Games are beautiful. They do not need to be justified.

This above all: games are not valuable because they can teach someone a skill or make the world a better place. Like other forms of cultural expression, games and play are important because they are beautiful.

Appreciating the aesthetics of games—how dynamic interactive systems create beauty and meaning—is one of the delightful and daunting challenges we face in this dawning Ludic Century.

The ideas in this essay grew out of my interaction with literacy scholars Jim Gee, Rich Halverson, Betty Hayes, David Shaffer, Kurt Squire, and Constance Steinkuehler. Very special thanks to the brilliant Heather Chaplin for developing the ideas with me over many conversations and arguments. Thanks to Nathalie Pozzi and John Sharp for insightful editing and to Kirk Hamilton and Stephen Totilo for first publishing it on Kotaku.com. Lastly, warm thanks to Sebastian Deterding and Steffen Walz for first spurring me to write the essay for their book.

1 THE AMBIGUITY OF GAMES: HISTORIES AND DISCOURSES OF A GAMEFUL WORLD

Sebastian Deterding

If there is one catchword for the current moment in the history of media, it is convergence: digital media, computing, and networking are decoupling the entities formerly known as "the media" into their requisite components—content genres and storage media, distribution networks and end devices, producers and audiences—to recombine them into unexpected, fleeting new formations (Jenkins 2006; Storsul and Fagerjord 2008). Thus, games can now be played on almost any digital device, anytime, anywhere. Game distribution is migrating from off-the-shelf physical copies to online streaming and a myriad of app stores across a myriad of platforms, and games are shifting from being finished products to interconnected, constantly evolving online services where "produsers" (Bruns 2008) pay in the form of micro-transactions, personal data, content creation, and marketing work.

Beyond these immediate forms of digital game convergence, several authors have observed a broader "ludification of culture" (Raessens 2006, 2012; Walz 2006; Stenros, Montola, and Mäyrä 2007, 32). From 8-bit music to pixel art, thematic, visual, auditory, and interactive tropes of games are pervading our pop culture and art with a "ludic language" (Flanagan, this volume). Some argue that the formative experience of growing up playing digital games has nurtured a whole "gamer generation" (Beck and Wade 2006) sporting playful media practices, mindsets, and identities. Others have gone so far as to call the twenty-first century a "ludic century" where games become the dominant cultural form because they match the systemic, computational, participatory constitution of our time (Zimmerman, this volume; see already Minnema 1998).

If *ludification of culture* captures how games and play increasingly inform other domains of our everyday life, we also can and must speak of its counterpart: the *cultivation of ludus*. As games and play move from the periphery of playgrounds, living rooms, and arcade halls toward the center of our cultural, social, and economic life, so cultural, social, and economic actors become interested in shaping and harnessing them for their purposes: other realms of life impress their forms on games and play. We see this in the professionalization of digital gaming in e-sports (Taylor 2012) and the economization of play in "gold farming," real-money trading, virtual economies, or game play as user-generated marketing (Malaby 2007; Dibbell 2008). We see it in the rationalization of play, as digital gaming takes on more and more work-like features, such as "grinding" in massively multiplayer online role-playing games (MMORPGs) (Yee 2006; Grimes and Feenberg 2009). We see it in the adoption of game technologies such as

three-dimensional rendering engines by non-game industries. And we see it in the rise of serious games and gamification (Sawyer 2002; Deterding et al. 2011): games and game design elements are increasingly harnessed to improve everything from productivity to marketing, from learning to user experience, from health, happiness, and creativity to civic engagement and governance.

Among game designers and scholars, gamification has been met predominantly with skepticism if not hostility (e.g., Robertson 2010; Bogost 2011a). Many view it as yet another wave of "colonizing attempts" (Aarseth 2001), only this time from marketers and startups with often-doubtful ethics and little care for "games as such." Gamification is painted almost as a desecration of a presumed nature of games and play. Says Bogost (2011a): "gamification is marketing bullshit, invented by consultants as a means to capture the wild, coveted beast that is videogames and to domesticate it for use in the gray, hopeless wasteland of big business." In the same tone, PJ Rey (2012) critiques gamification as producing play labor, or "playbor": "Play loses its innocence. It is no longer an escape from the system, it is just another branch of it. Waste is no longer wasted. Playbor is part of capitalism's effort to colonize every last moment in the waking day."

Yet the discourse around gamification has also managed to widen public interest in games and play rapidly. And whatever the status and longevity of gamification itself, it is only one symptom of the broader digital convergence of games, one little current in the double tide of the ludification of culture and the cultivation of ludus—something one may call the rise of a *gameful world*. Like media convergence in general, this double tide does show no signs of ebbing, and it *is* changing what games and play are.

In the heat of any debate, one task of the scholar is to take a step back and look at the unfolding scene from the outside. Instead of taking a stance in *Gamification: The Rhetoric Wars* (Walz and Coulton 2011),[1] this chapter tries to bring them into view *as* just that: a clashing of rhetorics in Kenneth Burke's (1950) sense, of different worldviews and moral politics, manifest in different languages. This chapter provides such an outside view through a threefold distancing: first, it offers a theoretical language to articulate that and how our views of the gameful world are steeped in specific, modernist conceptions of games and play. Second, it foregrounds the contingency of the gameful world by delineating its historical precursors and enablers: the forms of its current ascendance are neither entirely novel, nor aberrant, nor natural, nor inevitable. Third, the chapter compares and contrasts the various contemporary rhetorics of the gameful world. In closing, it asks what those rhetorics tell us about our contemporary life, what they elide, and how the rise of the gameful world may change play and games themselves.

Liminalities: The Social Place of Play and Games

If the current scholarly critique of the "instrumentalization," "colonization," or "domestication" of play and games highlights one thing, it is that play and games *are* thought to have "a proper place" in society—a designated cultural role, meaning, or function. Is play not *supposed* to be "outside 'ordinary'

life" and "with no material interest" (Huizinga 1955, 13), "separate" and "unproductive" (Caillois 2001, 9–10)—a space free from the demands of social norms and uses?

As the millennia-old use of games for military training and planning demonstrates, such sentiments overlook the fact that the provision of freedom from the instrumental norms and purposes of productive life through play and games is no brute fact of nature, but a contingent state of affairs—characteristic for modern industrial societies: *To us*, *today*, games and play *function* and *serve* as "escape attempts" (Cohen and Taylor 1992). Whether we praise their value for childhood development or artistic creativity, bemoan their unproductive frivolity, or engage in them as leisurely restoration for work or hobbyist self-realization—even when we romantically long for them as a utopian "otherland"—we reproduce very specific, modernist cultivations and rhetorics of play and games (Sutton-Smith 1997; Malaby 2007). Celebrating play and games as a space free from order, purpose, norm, and consequence, a space "outside of" society, we easily forget that this "freedom from" is *itself* a social norm and purpose. The spoilsport, the powergamer, and the bully remind us that the playground is a place full of rules that are continually expected, demanded, and policed by the play community (Goodwin 2006).

If we want to understand today's transformation of the cultivation of ludus, we need a *theoria*, a distancing from our own modernist preconceptions of the relation of play, games, and society. A useful one can be found in anthropologist Victor Turner's (1982) studies of ritual and liminality. Like Durkheim, Turner argued that in premodern societies, rituals served as the crucial forge and reaffirmation of the bonds and moral order of a community. Of central importance are rites of passage through which an individual's orderly transition from one social status into another is performed. During these rites of passage, between the individual's separation from and re-incorporation into the group, one finds a phase of transition, of *liminality* (Latin *limen*, "threshold"), where standing orders, behaviors, and meanings are temporarily turned upside down—a space of *play*: "the analysis of culture into factors and their free or 'ludic' recombination into any and every possible pattern, however weird, that is of the essence of liminality" (Turner 1982, 28). Maybe the best-known contemporary form is the *Rumspringa*, observed in some Amish communities, in which adolescents leave their homes for a set period of time to explore the modern world around them and decide whether to return and become a full adult member of their community.

In other words, play in premodern societies is fully contained and functionalized in ritual, fully integrated into social life. There is no distinction between "work" and "leisure": play and games are part of the total ritual "work of the Gods" (Turner 1982, 38). *All* of life is "work" in the sense that it is bound in a nonoptional manner to the cosmic rhythms and rules of nature and culture. It is only in modern Western nation-states that we start to find "work" itself separated from the rest of life, religious order kept apart from secular existence, social timing freed from the rhythms of nature, the individual uprooted from the collective—such that "leisure" as a nonwork, secular, optional, individual activity becomes possible. Modern leisure activities like play, games, or art still retain the ludic forms of liminality; but they crucially differ in their social function, meaning, and place. In Turner's phrasing, they become limin*oid*, not liminal.

Liminal phenomena are collective duties embedded in and reaffirming social order. The liminal play

found in rites of passage is part of one play-and-work complex, functionalized as a temporary inversion that conserves the social order by foregrounding the chaos that lies beyond, or by absorbing destabilizing energies. Liminoid phenomena in contrast are individual choices (that may result in temporary collectives), marginalized spaces pocketed away from social order, but therefore also spaces where the order *can* be reflected, critiqued, subverted, and alternative versions explored. Still, their "antistructure" is socially functionalized, geared into the machine of progress as an engine of innovation (Turner 1982, 44–45).

Now Turner's strict historic and cultural dualism between modern and premodern, "the west and the rest," has long been outmoded. But his *theoria* of play still holds value in reminding us that at any time and place, play and games are always cultivated, and in articulating two ideal typical poles of this cultivation, *liminal* and *liminoid*.

Seen through this lens, many of today's serious games and gamification projects present a return to liminality: games are again "centrally integrated into the total social process" (Turner 1982, 54) to reproduce a standing social order that is considered no less good and natural than the cosmos of tribal religion. The modern Protestant work ethic internalized religious duty into secular economic industriousness and disciplined self-betterment, defining and devaluing leisurely play and games as the frivolous opposite of God's work. Today's liminal serious games and gamification dissolve this separation of work and leisure again, only in a secondary, *secular* liminality, a tautological Calvinism without Last Judgment that replaces the work of God with the work of work, and the work of the self. As Goffman (1967) puts it so well: "Many gods have been done away with, but the individual himself stubbornly remains as a deity of considerable importance" (95). And today, the moral value of this self is again tied to the holy work of work. Burning calories at the gym or midnight oil at the office, we bring our daily "little offerings" (95) to prove moral character, so help us our gamified self-tracking device.

Conversely, the reactions against such liminal cultivations reveal themselves as modernist invocations of the *liminoid* as the "proper" role of games and play in society. To use the distinction of Raphael et al. (2009), critics of liminal gamification prefer games to question the ends society pursues, not to optimize the means toward them.

Yet their celebration of games and play as sites of transformation, subversion, autonomy, or empowerment is no less instrumentalizing—it simply instrumentalizes for progressive purposes. Even defendants of games and play as objects of disinterested appeal like Eric Zimmerman or Frank Lantz (this volume), fighting for "games for games' sake," in doing so engage in a deeply progressivist, liminoid project. They reaffirm the modernist notion that society can and ought to have spaces of noninstrumental, nonreligious aesthetics, and that games and play can and ought to be one of them.

Histories of a Gameful World

With Turner's conceptual compass of liminal and liminoid cultivations at hand, we can begin charting the history of the gameful world—to foreground its contingency, to moderate breathless claims of novelty, and to substantiate the claim that the contemporary instrumentalization of games and play is

no unique profanation of their pristine "nature." The following subsections trace important precursors, parallels, and traditions of using games and play beyond leisurely entertainment, followed by a more detailed account of the origins of gamification in the U.S. technology industry around mid-2010. It closes with a look at the wider social transformations that enabled the contemporary rise of a gameful world.

The Serious Games of War, Business, and Education

The first documented instances of games for "serious" purposes date back to the China of the Warring States period around 475 B.C., when the game *weiqi* (Go) began to be used to school strategic cunning for the art of war (Halter 2006, 20–21). In Western culture, war games for strategic training and planning rapidly spread in the nineteenth century from the Prussian army to war academies across Europe (Deterding 2009, 23–24). During the early Cold War years, the U.S. Department of Defense and associated think tanks like the Rand Corporation set up entire gaming and simulation departments, using war gaming, role-playing, simulations, and mathematical game theory to make nuclear war rationally calculable. From there, "serious games" for strategy, foresight, and training expanded into government, business, and education (Abt 1971).

The early 2000s witnessed a digital renaissance of serious games (Sawyer 2002). In education, researchers from sociocultural traditions (see Ramirez and Squire, this volume) began to study and design learning interventions that not only used games as "content conveyors," but also made educational use of the technologies, literacies, communities, and practices surrounding them. As serious games expanded into fields as varied as desensitization therapy and experimental "petri dishes" for economists, subgenres like newsgames, persuasive games, advergames, and exertion games emerged (Sawyer and Smith 2008).

"Be Happy in Your Work!" Games at the Workplace

Mark J. Nelson (2012) has followed the use of games for productivity back to the unlikely origins of the Soviet Union under Lenin. Looking for a way to motivate workers without capitalist monetary incentives, Lenin proposed "socialist competition": individual workers, groups, and whole factories were awarded points for their performance, earning commendations, banners, and orders for surpassing certain point thresholds or beating other groups.

In the West, the first documented uses of games at work were workers' playful variation, pranking, and goal setting to counter the alienation of factory work (Roy 1960). Csikszentmihalyi's (1975, 1990) psychological studies of play and flow at the workplace painted a more positive picture. He described flow as a subjective state of optimal experience, characterized by autotelic, perceived-voluntary engagement, afforded by activities structured in a game-like manner. Csikszentmihalyi found that workers who arranged and treated their craft as a game they voluntarily chose to perfect would regularly experience flow in its course.

In 1985, one Charles Coonradt published the management book *The Game of Work: How to Enjoy Work as Much as Play*. Yet he was a little ahead of the curve: only in the late 1990s did businesses start to try and co-opt workers' play as resistance or self-determined craft into a corporate productivity strategy in a whole wave of "funsultant" business consultancy, peaking with the *FISH!* management

book and videotape series (Lundin, Paul and Christensen 2000; Nelson 2012). The age of funsultants has also been the age of mind maps, Post-its, and product innovation firms like IDEO. Here, the book *Innovation Games* (Hohmann 2006) introduced the use of games as catalysts for "Creating Breakthrough Products Through Collaborative Play" (the book's subtitle), recently refashioned as *Gamestorming* (Gray, Brown, and Macanufo 2010).

Designing for Playfulness and Motivation in Human–Computer Interaction

If design firms used play and games for product innovation *processes*, researchers in HCI early on started to explore them as inspiration for desirable product *qualities*. In the early 1980s, Thomas Malone wrote seminal papers deriving "heuristics for designing enjoyable user interfaces" from video games (Malone 1982). John M. Carroll (1982) analyzed the design of text adventures such as *Adventure* (Crowther and Woods 1976), leading Carroll and John C. Thomas (1982) to suggest redressing work activities in varying "metaphoric cover stories" to make them more interesting and to urge for a research program on the relation of fun and ease of use (Carroll and Thomas 1988; see also Carroll, this volume). With the expansion of HCI from utility and usability toward user experience (Hassenzahl 2010), more researchers began to study the "hedonic attributes" (Hassenzahl et al. 2008) or "motivational affordances" (Zhang 2008) of "pleasurable products" (Jordan 2002), dubbing the field "funology" (Blythe et al. 2004).

Thus, Luis von Ahn and others developed "games with a purpose" that piggyback game play to solve human information tasks such as tagging images (von Ahn and Dabbish 2008). Others explored the use of game interfaces in productivity contexts (Chao 2001) or studied playfulness as a desirable user experience (Fernaeus et al. 2012; see Holopainen and Stain, this volume). Some equated playfulness with any "pleasurable experience" (Costello and Edmonds 2007) or indeed every interaction that goes beyond utilitarian task completion (Gaver et al. 2004). Others spelled out different types of playful experiences (Arrasvuori et al. 2011).

A second intersection of HCI and games has been motivation, as in the case of online reputation systems. Originally developed to provide trust indicators and information filtering, designers soon found that these systems also had a motivational impact on users (Farmer and Glass 2010, 111–123). With "karma" points, stars, levels, badges, and user avatars, reputation systems took inspiration from games long before the term *gamification* emerged (Lampe, this volume). More recently, researchers like Lockton, Harrison, and Stanton (2010) have begun teasing out design patterns from games to motivate user behavior.

The World Upside Down: Play Forms in Art and Counterculture

The counterpole to the serious games of military, business, and HCI has been the "world upside down" (Curtius 1990/1948, 94–98) prefigured in the exuberant, apocalyptic chaos of medieval carnival and baroque literature (Bakhtin 2009/1965). In the twentieth and twenty-first centuries, several artists and activists have tapped into and extended these early rhetorics of frivolous play (Sutton-Smith 1997, 201–213). Literary examples are the playful automatic and aleatoric writing of surrealism and Dada or the gameful *contraintes* of the group OULIPO (Ouvroir de

littérature potentielle). From the 1950s on, the Situationist International developed artistic tactics and theories for "constructing situations" that embraced play as a central concept (Andreotti 2000; Wark 2011). Chief among them are *détournement*, the appropriation of other materials irrespective of authorship or copyright, *psychogeography*, the explorative mapping and creating of new experiences of a cityscape, and *dérive*, the playful drifting through cities. All three aimed at the creation of new atmospheres or scenes, new ways of seeing and living that would ultimately *dissolve* the modernist separations of classes and of work and leisure: "The central distinction that must be transcended is that established between play and ordinary life, play kept as an isolated and provisory exception" (Debord 1958). In parallel, the Fluxus movement erased the boundaries between art and life with designed boxes and instruction pieces—little written rules for action—that turned the audience into performers, guiding them toward new real-life behaviors and experiences.

In the 1970s, the New Games movement formed as a countercultural response to the Vietnam War and civic unrest in the 1960s in the United States (New Games Foundation 1976; Pearce et al. 2007). Its large-scale physical games created bodily experiences and spectacles with ecological themes that did not pit players against one another other in a zero-sum fashion. The underlying idea was that this embodied experience with participants across social divides would build trust, community, and new ways of living.

These themes have been picked up by critical art games and game-inspired art from the late 1990s on. They appropriate games, game engines, and game imagery as forms of cultural commentary and expression, linking backward and outward to everyday tactics of resistance (Lefebvre 2002/1961), reversing socially normed forms of action, appearance, and narratives in toys and game play (Flanagan 2009, 33; Flanagan, this volume).

Pervasive Games and Mixed Reality Performances

Intertwined with both the serious games movement and artistic play practices, new forms of gaming evolved in the early 2000s that extended games into new contexts and spaces. Commonly called *pervasive games*, they have "one or more salient features that expand the contractual magic circle of play spatially, temporally, or socially" (Montola 2009, 12). Examples are location-based games that take game play into the public space, augmented reality games that use digital devices to overlay game representations on the environment, persistent games that continually run to be entered and exited during the course of the day, or alternate reality games (ARGs), which "take the substance of everyday life and weave it into narratives that layer additional meaning, depth, and interaction upon the real world" (37).

Pervasive games originated in ubiquitous computing research labs, where they became a handy use case for prototyping new technologies (Magerkurth et al. 2005; McGonigal 2006). In parallel, in the late 1990s, art collectives and theater groups like Blast Theory started to connect critical media art, Situationism, and New Games with locative games using mobile phones (de Souza e Silva and Hjorth 2009). They developed mixed reality performances and immersive theater productions that were less about winning games than layering alternative meanings and narratives into participants' experience of moving through cities or staged

environments (Benford and Giannachi 2011). As in the case of Davies' (2009) "barely games" and the "subtlemobs" of artist collective *circumstance*, audio in particular has been explored as a medium in this context, being an inexpensive and unobtrusive means to layer an additional atmosphere and information stream into everyday life (see also Dow et al. 2005).

In parallel with this growing academic and artistic interest in pervasive games, design studios like Area/Code started developing experimental games in public spaces, as well as ARGs as marketing campaigns, which have nowadays grown into the lovechild of media companies as the future of transmedia storytelling, advertising, and marketing (Rose 2011). Independent game designers began to organize festivals dedicated to play in public spaces, such as Come out and Play, igfest, Hide&Seek, or Playpublik. And specifically in Northern Europe, live action role-playing (LARP) games developed into a unique subculture with high production value games, a decidedly noncommercial spirit, a strong theoretical discourse, and an artistic, political impetus that led to the exploration of societal issues and the testing of boundaries of appropriateness (Stenros and Montola 2010).

Silicon Valley and the Origins of Gamification

Serious games, funsultants and funology, countercultural play and pervasive games show that the use of games and play beyond the modernist confines of leisure times and playgrounds is far from new. Still, they remained relatively niche phenomena happening in research labs and subcultures. It was in the U.S. technology industry where between 2005 and 2010, some of these traditions would be absorbed and then strategically grown into "a thing" called *gamification*, to take off rapidly after January 2011.[2]

Around 2005, with the rise of web 2.0 business models centering on "harnessing collective intelligence" (O'Reilly 2005), web startups increasingly faced the challenge how to motivate users to sign up with the offered service, invite their friends, and part with their share of said collective intelligence. In reaction to this demand, in January 2005 Kathy Sierra, author and coeditor of the popular "Head First" book series with technology publisher O'Reilly, began to write about game design as an important inspiration for "Creating Passionate Users."

In 2006, game and online community designer Amy Jo Kim started touring the Silicon Valley conference circuit, talking about the application of game mechanics to functional software. The gamified fitness application Nike+ was released and became the quintessential conference case study for service and experience design. Justin Hall and colleagues founded the startup GameLayers and built out the *Passively Multiplayer Online Game* (PMOG, later rechristened "The Nethernet") that placed a game layer on top of web browsing.

In 2007, IBM contracted communication researcher Byron Reeves to publish white papers on the role of online games for business leadership. *Chore Wars* (Davis 2007) launched, a role-playing game around household tasks that became another early reference case.[3] Jane McGonigal, who started her design career at the team-building game company Go Game, was employed by the Institute for the Future and began to present keynote talks at conferences such as ETech about using ARGs for "happiness hacking." The startup Bunchball, founded by Rajat Paharia, pivoted toward "the science of engagement," building out the meta-game features of its social games (leaderboards,

virtual currencies, virtual items) into a white-label platform that was deployed on the website of the NBC television series *The Office* to drive user activity.

In April 2008, StackOverflow was launched, a question-and-answer platform for software developers. Using a reputation system with points and badges inspired by the gaming experience of its developers, it quickly gained cachet in the technology industry. The same month, Rajat Paharia of Bunchball, Justin Hall of GameLayers, and one Gabe Zichermann met at a panel of the web 2.0 Expo to discuss how "to work massively multiplayer game mechanics into social websites." Zichermann, a marketer working for publishers and companies servicing the games industry, had toyed with the idea of applying game elements to online photos and business cards, founding the startup rmbr. In May 2008, shortly after the web 2.0 Expo panel, he began blogging about "funware" and closed book deals on *Game-Based Marketing* with John Wiley and *The Engaging Web* with Manning Publications (rapidly renamed *Funware in Action*). In October, David Edery and Ethan Mollick's (2008) book *Changing the Game* appeared and summarized the business application of games, only to be drowned by the financial crisis. In November 2008, the term *gamification* was documented online for the first time, then attributed to Clay Shirky and Bret Terrill. The verb *gamify* was presumably first used by Richard Bartle in 1980 during his work on *MUD* (Trubshaw and Bartle 1978). But the word didn't catch on then, and it didn't catch on in 2008.

Still, things sped up. In March 2009, foursquare launched at the SxSW festival and demonstrated that game design elements can drive the initial adoption and retention of users. The massive success of *FarmVille* (Zynga 2009), launched in June the same year, reaffirmed the impression that games somehow "solve" the problems of user acquisition and monetization. In August 2009, Bunchball's Rajat Paharia registered the domain gamification.com.[4] In September 2009, Volkswagen Sweden launched the advertising campaign "The Fun Theory" for its green brand BlueMotion. Its slogan, "Fun can obviously change behavior for the better," was illustrated in online videos of installations such as the Piano Stairs: stairs down to a subway station are rebuilt into giant piano keys that play a sound when people step on them, motivating people to take the stairs instead of the escalator next to them (at least on day one).[5] The videos spread virally and offered short, fun, high production value case studies for the concept of gamification. In November 2009, Byron Reeves and J. Leighton Read's *Total Engagement* appeared, a business-book-length exposition of their 2007 white papers that MMORPGs provide the blueprint for future work environments.

In January 2010, Microsoft Office Labs launched *Ribbon Hero*, a tutorial game for learning the Office Suite ribbon interface. It became another staple case study. David Helgason, CEO and cofounder of game engine provider Unity Technologies, blogged that 2010 will be "the year of gamification." The word started to get currency, flickering between "gameification," "game-ification," and "gamification," but "game mechanics" and "game dynamics" were used as well. In February, game designer Jesse Schell gave the talk "Design Outside the Box" at the game industry conference DICE, presenting a vision of the future where every little action will be tracked and awarded with points by advertisers and governments. The video became a viral hit, and Schell put up a "Gamepocalypse" blog. In March, Jane McGonigal's talk "Gaming Can Make a Better World" was released on TED.com and also spread virally.

To summarize, by early 2010 Zynga had heated up venture capital interest in everything online game-related, and Nike+, foursquare, Chorewars, StackOverflow, and Ribbon Hero presented success stories of the use of game design beyond games. There was at least one startup (Bunchball) that provided a white-label gamification platform and a popular advertising campaign (The Fun Theory) that helped illustrate the idea. In the technology conference circuit and business book aisles, a discourse around the use of game design beyond games had thickened, to hit mainstream audiences with the futurist talks of Jesse Schell and Jane McGonigal. There are many converging threads, but there is no *gestalt*, no *single* word.

And then. In May 2010, Bunchball relaunched its website, now stating that its white-label platform Nitro "brings the power of Gamification to you."[6] In June, Manning Publications announced that they will not publish Gabe Zichermann's *Funware in Action*. In July, Zichermann (whose first book *Game-Based Marketing* had just appeared) gave a talk on *gamification* at the Develop 2010 conference—it's the first time the term appeared in any of his presentations or his "funware" blog. In parallel, Zichermann registered the domain name gamification.co and founded Gamification Corp.[7] In October 2010, Zichermann rerouted his funwareblog.com to gamification.co, where all previous content was now hosted. He announced the "Gamification Summit," and, shortly thereafter, that his book *Gamification by Design* would be published by O'Reilly in 2011—turning out to be the manuscript of *Funware in Action*. The Gamification Summit commenced on January 20, 2011, with Zichermann as host, Bunchball as showcase sponsor, Jane McGonigal as keynote speaker (launching her book *Reality Is Broken* the same day), and Amy Jo Kim as leader of a "gamification workshop." The technology media report: a meme is baptized.[8]

Enablers

In the late 2000s, then, we find the seeds of serious games and pervasive games, of countercultural art games and play in public, of playful interaction, ludic design, and gamification both growing and accelerating to grow into a gameful world. This raises the simple question: Why? And why *then*?

One obvious reason is the rise of *digital* games as a cultural medium of its own right in the 2000s. Cohorts whose childhoods were spent playing video games began rising into positions of cultural power in media and education. Game studies, game exhibits, game museums, and game canons were formed, institutionalizing games as objects of cultural worth, facilitated by the rise of art games and serious games. Politicians discovered games as part of the highly sought-after "creative industries" and started catering to game companies, and journalists and marketers discovered games as reliable attention magnets. Games developed cultural momentum; they become *cool*. As a consequence, social actors tried to cash in on games' symbolic capital (see Bogost, this volume) and probed the applicability of games for their own ends, as happened with radio propaganda, documentary film, or educational television before.

Some authors have suggested that we are witnessing the rise of a "gamer generation" (Beck and Wade 2006; Shore 2011) belonging to the larger cohort of "digital natives" that puts new, "gamy"

demands on workplaces, products, and public institutions. However, these analyses suffer from the same shallow data basis and massive overgeneralization as the concept of "digital natives" itself (Jones et al. 2010). A more convincing line of reasoning is provided by longitudinal studies of the post–World War II shift toward postmodern values in Western industrialized nations, such as the World Values Survey (Ingleheart 2008). According to these, cultural development charts a relatively predictable path: with the first wave of modernization comes a shift from traditional, survival-focused values of religion and community to values of economic achievement and rational, legal state authorities. Growing affluence and functioning institutions increase the experience of existential safety. This brings about a second, postmodern shift that de-emphasizes authority and economic achievement and foregrounds the maximization of personal well-being and self-expression. Against this background, it appears sensible that postwar generations in postmaterial societies would positively value games and play as worthwhile sources of such well-being and self-expression.

A congruent economic argument can be made about the post–World War II shift toward a post-Fordist, dematerialized service, information, and experience economy. As the provision of basic sustenance goods (food, shelter, clothing, mobility) and higher consumer goods becomes commoditized, margins and market differentiators shift toward the immaterial, symbolic, and experiential: an "experience economy" (Pine and Gilmore 2011) of lifestyle marketing and "retailtainment" (Ritzer 2004) emerges where "the experience is the product" (Merholz et al. 2008, 1), and companies are in a perpetual search for the next experiential "value add." Playful and gameful design appear as a logical outgrowth of this trend.

In parallel, with the exponentially decreasing costs of processing power, data storage, and networking bandwidth, more and more everyday interactions are mediated through networked computing devices; sensors, processors, and actuators become ubiquitous, from the smartphones in our pockets to the "smart cities" we walk through, and more and more behavioral data of individuals are tracked, stored, and analyzed. This technological rise of a "code/space" (Kitchin and Dodge 2011) effectively equips our lifeworld for pervasive, large-scale gaming: as goal-oriented, rule-based input-output systems, games require some sensing and tracking of activity, some processing of activity according to the rules, some storing of game states, and some display of game feedback. Today's code/space is the basic technological precondition for deploying algorithmic game(like) systems beyond the confines of a single gaming device and software (see Nova, this volume). Several authors have in fact argued that computers and games show such deep structural similarities (Pias 2002; Zimmerman, this volume) that computerization equals gamification.

Contemporary Rhetorics of the Gameful World

The growing interest in and use of games and play beyond the entertainment game industry and game studies has widened the discourse, but also created friction, as it brings in people from different traditions, contexts, and disciplines with different agendas, experiences, and reference points. This

situation somewhat mirrors the state of play research at the end of the 1990s, which then led Brian Sutton-Smith (1997) to pen *The Ambiguity of Play*. The book presents a historical discourse analysis of play in scholarship and popular culture, tracing the miscommunication between researchers and pundits back to different value-laden "implicit narratives," or *rhetorics*: "the way play is placed in context within broader value systems," "large-scale cultural 'ways of thought' in which most of us participate in one way or another, although some specific groups will be more strongly advocates for this or that particular rhetoric" (8–9).

I hold that the same phenomenon is at work in the current discourse around the gameful world. And like Sutton-Smith, I believe that "by revealing the rhetorical underpinnings of the apparently diverse theoretical approaches" toward the gameful world, "there is the possibility of bridging them with some more unifying discourse" (Sutton-Smith 1997, 9). Appropriating Sutton-Smith's validating criteria for the identification of a play rhetoric (15–16), I suggest that a gameful world rhetoric presents a more or less coherent, self-stabilizing network of the following components:

- one or more *communities of practice* that reproduce the rhetoric: its fit with the practices, values, mental models, and standard operating procedures of a community stabilizes a rhetoric;
- one or more *academic disciplines* in which the community members were likely socialized, whose epistemology, theories, and methods fit their practice and rhetorics and lend them legitimacy;
- *concepts and intellectual precursors* that can be referenced and drawn upon as jointly known and accepted as authoritative;
- main *proponents* of the rhetoric in the current discourse;
- typical *application areas* informed by the rhetorics, whose fitting affordances again stabilize the rhetoric;
- *prototypical nongame analogs* that offer conceptual metaphors to guide thinking about games in the form of "games are like X";
- a specific *framing of games and game design* in terms of specific concepts and theories, foregrounding specific properties and backgrounding others;
- specific *reference game genres* whose features fit and stabilize the rhetoric;
- a *moral politics* of play: a wider social valuing and placing of play, games, and game design in either liminal or liminoid terms;
- finally, many rhetorics of the gameful world perpetuate one of the *play rhetorics* identified by Sutton-Smith.

The use of the term *rhetorics* is not meant to imply a linguistic relativism or a historicism for which every rhetoric is "immediate to Truth," to paraphrase von Ranke. It is grounded in a pragmatist stance: theories, perspectives, and rhetorics are practical tools, not Platonic mirrors, and some tools are more useful or viable than others. Also, the rhetorics presented in the following subsections are to be seen as Weberian ideal types. Any individual speaker is likely to appeal to several rhetorics, have her own individual twist, suggest combining two or swapping one for the other, or similar. Likewise, any designer in her practice will be guided by her own idiosyncratic blend, and any resulting application will be describable *post hoc* in the terms of different rhetorics. With these provisions out of the way, let us look at the individual rhetorics in detail (tables 1.1–1.4).

Table 1.1
Rhetorics of the gameful world, part 1 of 4

Rhetoric	*Feedback*	*Nudging*	*Exploitation*	*Status*
Communities	Marketers, managers, economists, quantified self movement	Marketers, designers, economists, policy makers	Critical theorists	Marketers, social media startups, managers
Academic Disciplines	Neoclassical economics	Behavioral economics, social psychology	Political economy, critical theory	Sociology, social psychology, informatics
Concepts and Precursors	Markets, incentives, mechanism design, market design	Cognitive biases and heuristics, persuasion, incentive-centered design	Free labor (Terranova), playbor (Kücklich), ideology, symbolic politics	Conspicuous consumption (Veblen), social capital (Bourdieu), signaling theory, self-presentation (Goffman)
Application Areas	Enterprise, human information tasks, health, sustainability, social media	Public policy, health, sustainability, marketing	*Same as feedback and nudging*	Marketing, UGC platforms
Games as . . .	Markets with perfect information and incentives	Choice architectures using cognitive biases	Exploitation, tokenism, bullshit, false consciousness	Stages for public display of costly signals
Non-game Analogs	Prediction markets, virtual currencies, KPIs, business intelligence, incentives, loyalty programs	Marketing material, sales offers	Clicktivism, free labor, human information tasks	Reputation systems, loyalty programs, status goods
Reference Genres	MMORPGs	Social network games	MMORPGs, social network games	Achievement systems
Moral Politics	Liminal optimization of fitness, profit, productivity, policy outcomes	Liminal achievement of policy outcomes and business goals	Liminoid critique of surplus extraction, social inequity, and commodification of dissent	Liminal appeals to and reproduction of status anxiety
Play Form	Ludic	Ludic	Paidic	Ludic
Proponents	Byron Reeves, JP Rangaswami, Gabe Zichermann	Gabe Zichermann, Juho Hamari	Heather Chaplin, Liz Losh, P. J. Rey, McKenzie Wark, Ian Bogost	Gabe Zichermann
Play rhetoric	Progress	Progress	Power	Power

Table 1.2
Rhetorics of the gameful world, part 2 of 4

Rhetoric	***Performance***		***Well-being***	
	Communal	*Immersive*	*Hedonic*	*Eudaimonic*
Communities	Nordic larp, performance art and public play design community	Mixed reality researchers and art groups, narrative educational role-playing	Gameful.org community	Philosophers, designers
Academic Disciplines	Performance studies, game studies	Media studies, performance studies, computer science, educational research	Positive psychology	Psychology, virtue ethics
Concepts and Precursors	Performance (Schechner), ritual (Turner), play community (DeKoven), collective effervescence (Collins)	Mixed reality performance (Benford), barely games (Davies), augmented environment (Bolter), spatial stories (Jenkins), transformational play (Barab), practomimesis (Travis)	Happiness and well-being (Seligman), lusory attitude (Suits)	Flow (Csikszentmihalyi), self-determination (Ryan and Deci), paratelic state (Apter), virtue ethics (Aristotle), focal practice (Borgmann)
Application Areas	Therapy, art, community building, political activism	Art, entertainment, education	Health, personal growth	Personal growth, ethical development
Games as . . .	Collective performance and experience of temporary reframings	Immersive worlds and narratives	Environments well-structured for positive experiences	Extended will, focal practices and objects for building virtues
Non-game Analogs	Rituals, performance art, happenings, parties	Theater, virtual reality, amusement rides, theme parks	Therapeutic and self-help exercises	Deliberate practice in crafts, sports, martial arts
Reference Genres	Larps, pervasive games, art games, new games, ARGs	Augmented reality games, story-driven games, RPGs	Multiplayer games, ARGs	Skill-based games, playful objects
Moral Politics	Liminoid temporary autonomous zone for communities	Liminoid experiences, liminal pleasures	Liminal-liminoid recovery, fueling collective action	Liminoid eudaimonia
Play Form	Paidic	Paidic	Ludic, paidic	Paidic
Proponents	Jane McGonigal, Kars Alfrink, Jaakko Stenros	John Carroll, Russell Davies, Lee Sheldon, Roger Travis	Jane McGonigal	Scott Rigby, Marc Hassenzahl, Miguel Sicart, Evan Selinger, Sebastian Deterding
Play Rhetorics	Identity	Imaginary	Progress	Self

Table 1.3
Rhetorics of the gameful world, part 3 of 4

Rhetoric	***Pleasure***	***Systems***	
		Expressive	*Learning*
Communities	Advertisers, designers	(Indie) game designers and scholars	Game-based learning scholars
Academic Disciplines	HCI, Industrial design, user experience, game design	Game design, literary theory, computer science	Education
Concepts and Precursors	Funology (Blythe et al.), playfulness (Korhonen et al.)	Serious games (Abt), proceduralism (Wardrip-Fruin, Mateas, Bogost), systems theory	Microworlds, constructionism (Papert), communities of practice (Lave and Wenger)
Application Areas	Advertising, design	Arts, political communication	Education
Games as . . .	Experiences designed for pleasure	Designed rule systems procedurally expressing meaning	Explorable microworlds with problems and communities
Non-game Analogs	Playful design	Literature, propaganda, algorithms, hacking	Montessori schools
Reference Genres	Advergames, toys, installations, amusement parks	Persuasive games, rule-intensive games	Simulation games, multiplayer games, MMORPGs
Moral Politics	Liminal attention generation, entertainment, and value-add	Liminoid social critique	Liminal training of the twenty-first century workforce, liminoid empowerment
Play Form	Paidic	Ludic	Ludic
Proponents	Jesse Schell, Stephen Anderson, Nicole Lazzaro	Ian Bogost	Kurt Squire, Katie Salen, Karl M. Kapp
Play Rhetorics	Frivolity	Imaginary	Progress

Table 1.4
Rhetorics of the gameful world, part 4 of 4

Rhetoric	*Cultural form*	*Playfulness*	
		Free	*Industrial*
Communities	Game artists, media scholars	Philosophers, critical designers	Product designers
Academic Disciplines	Aesthetics, cultural and media studies	Philosophy, critical design	Design
Concepts and Precursors	New media literacies, media theory, cultural studies	Carnival (Bakhtin), paidia (Caillois), Dyonisian (Nietzsche), free play (Nachmanovitch), ludic design (Gaver)	Design thinking, creativity
Application Areas	Arts, media	Infinite	Product innovation
Games as . . .	Aesthetic forms and cultural media	Material for the playful recombination of behaviors, objects, meanings	Rationally accountable creativity methods
Non-game Analogs	Music, film, literature	Carnival, jazz, childhood play	Creativity techniques
Reference Genres	Indie games, art games	Sandbox simulations	Card games, word games
Moral Politics	Liminoid reflection of culture, self-expression, art for art's sake	Liminoid heterotopia	Liminal rationalized creativity
Play Form	Paidic	Paidic	Paidic
Proponents	Joost Raessens, Eric Zimmerman, Frank Lantz, Ian Bogost, Mary Flanagan	Miguel Sicart, Bill Gaver, Scott Nicholson, Jaakko Stenros	Luke Hohmann, Dave Gray
Play Rhetorics	Imaginary	Frivolity	Imaginary

The Rhetoric of Feedback

The rhetoric of feedback has been put forward most strongly in Reeves and Read's (2009) *Total Engagement*. MMORPGs are blueprints for the future of work, they argue, because they create a fully transparent and meritocratic market: character statistics such as levels and skills are explicit and reliable markers of skill and reputation, goals are explicit numerical targets, and feedback is immediate and explicit. In MMORPGs, systems like "Dragon Kill Points" ensure loot is distributed meritoriously based on each individual's measurable contribution to the team effort, solving collective-action problems. JP Rangaswami (this volume) similarly argues that the networked, distributed, exception-is-the-rule, "lumpy" quality of today's knowledge work requires MMORPG design and gamer values, "active 360 feedback loops, highly sensitive controls, both close-in and as well as zoomed-out loci of operations." Much "green gamification" (Froehlich, this volume) and health gamification (Munson et al., this volume) appeal to this line of reasoning: provide users timely

and actionable feedback on their health and environmental behavior, and they will improve it. As Aza Raskin (2011) put it: "The one secret to changing human behavior? Feedback loops." This rhetoric connects gamification with the quantified self movement and with "big data" and "smart cities" discourses (Wolf 2009; Li, Dey, and Forlizzi 2010; Goetz 2011).

Underlying this rhetoric is a conception of human beings as computational and economic *rational actors*, dominant in a wide range of disciplines during the second half of the twentieth century. In essence, the claim is that humans are highly independent quasi-computers that rationally process information to plan and execute actions so as to maximize their self-interest. "Suboptimal" action (disengagement at school, free-riding at work, overeating, driving fuel-inefficiently) is therefore due to either incorrect information or insufficient incentives. The solution is to improve the available information or to align incentives with desired outcomes. In contemporary terms, behavior needs to be tracked and measured (with ubiquitous sensors), analyzed (with big data analytics), displayed (with data visualization), and incentivized (with granular scores translating into saved fuel, reduced health care premiums, etc.). The rhetoric of feedback thus reiterates mid-twentieth-century cybernetic, systems theoretical visions of social engineering performed by technical elites (e.g., Bell 1976), only in slightly more "distributed" and "agile" terms where software continually monitors and automatically adjusts information flows, terms, and incentives (O'Reilly 2010).

Games in this rhetoric are blueprints of perfect markets or feedback systems. Players cooperatively engage in activity intended by the system designers because games provide not only a wealth of real-time, precise, easy-to-grasp feedback on all relevant parameters but also granular, explicit, optimally balanced and aligned incentives for any and every desired activity (Williams, this volume). Game design is essentially market or mechanism design (Hamari, Huotari, and Tolvanen, this volume), and frequent nongame analogies are prediction markets, virtual currencies, KPIs, business intelligence, and employee bonus and customer loyalty programs.

This rhetoric is found in public policy, sustainability, personal health, and human information task platforms such as Amazon Mechanical Turk. Yet its main bedrock is corporate engagement, productivity, and collaboration (see Mollick and Werbach, this volume). It fits nicely with mental models and practices dominant in enterprises: games are seen as business dashboards and incentive programs, only somehow more and better. The societal role of games is thus framed fully in liminal terms: game design can allow managers and regulators to arrange for more competitive and productive employees (or citizens), solve collective-action problems, and coordinate distributed work.

The Rhetoric of Nudging

In the past decades, the rational actor model has been increasingly called into question across the social and behavioral sciences, to be replaced with the image of human beings as "social animals" (Brooks 2011): humans depend on others and are driven by social influence; they are guided by emotions, not just reason; most of their everyday behavior is habitual, not consciously planned; and even where rationality holds, it is bounded.

The most visible rendition of this view has come in the shape of behavioral economics, which

empirically studies how people's actual economic behavior deviates from predictions based on the *Homo economicus* model, explaining this deviation with certain cognitive heuristics and biases. In their influential book *Nudge*, Richard Thaler and Cass Sunstein (2008) flipped the descriptive analyses of behavioral economics into prescriptive advice for policy makers: to improve regulatory outcomes, governments should engage in "choice architecture" that uses knowledge about cognitive biases in the presentation of choices to "nudge" individual choice in the desired direction. Beyond policy circles, this rhetoric has become prevalent in (online) marketing and user experience design as "persuasive design" (Deterding 2012b). Hence, consumer marketing is the predominant application area and real-life analog of this rhetoric, though it also extends to personal health, green technology, and public policy.

Gamification evangelists and scholars alike have appealed to behavioral economics as a foundation for gamification (Paharia 2010; Zichermann and Linder 2013). Game design is framed liminally as helping governments and marketers to drive policy outcomes and sales with choice architectures whose design patterns directly use cognitive biases and heuristics, social influence, emotional appeals, and the power of habit. Social network games like FarmVille are the reference genre of choice: their gifting mechanics utilize guilt and reciprocity norms to get people back to the game; appointment play with decaying resources makes use of the sunk-cost fallacy and endowment effect; and so on (Hamari 2011, 2013).

The Rhetoric of Exploitation

The expectable opposition to market and nudging rhetorics comes from political economy and critical theory, where gamification is seen as the most recent adaptation of capitalism to reproduce the power and property relations of bourgeoisie and proletariat (Rey, this volume).

As societies become affluent and work is dematerialized into affective, creative, and information labor, new, *symbolic* goods and forms of consumption have to be invented to keep production running, and new ideologies have to be fashioned to appease the public. Furthermore, creative, affective, and information labor require a degree of emotional investment, self-investment, initiative, and flexibility that is incompatible with the alienation of twentieth-century Fordism—which denied workers any autonomy and initiative and offered nothing but money in exchange. The new economy replaces Fordism with a "play ethic" (Kane 2004) that dissolves the modernist distinctions of work and leisure into "weisure" (Conley 2008), producers and consumers into "produsers" (Bruns 2008), and play and labor into "playbor" (Kücklich 2005): the workplace is presented as a playground where "friends" "grow themselves" and "have fun" while working on things that "make a dent in the universe." This factual and rhetorical *de-alienation* of labor enables its continued *exploitation* (Rey, this volume). In the "free service for data" deal, the terms (of service) are set by the platform owners, and the surplus value stays with them. One exemplary site of such de-alienated exploitation is the "free labor" (Terranova 2013) performed by players of "games with a purpose" (Zittrain 2008).

Games and gamification in this rhetoric manifest symbolic goods, modes of production, and ideologies that commodify dissent and optimize surplus value extraction while covering up their exploitive nature with a false consciousness. They create

artificial demand for virtual items without real value. They channel dissatisfaction into slacktivism and "epic wins" in virtual worlds. They replace real co-determination, deliberation, and participation by employees, shareholders, and citizens with tokenist "engagement" in "idea platforms" and "democracy hackathons." Instead of helping the public to identify and align around systemic social issues, games and gamified platforms for health, education, self-management, or civic engagement let people "feel *as if* they're doing something worthwhile" (Chaplin 2011). Government serious games harness the symbolic capital of games (Losh 2009), and development aid projects like the PlayPump (Borland, this volume) make for uplifting feel-good stories—no matter if outcomes on the ground are dysfunctional. In gamified employee and customer relations, "(o)rganizations ask for loyalty, but they reciprocate that loyalty with shams" (Bogost 2011b). In short, this rhetoric explicitly addresses and critiques the liminal framing of the rhetorics of feedback and nudging.

The Rhetoric of Status

Where the rhetoric of feedback and nudging strike an individualist tone, the rhetoric of status turns to the collective. No person has pushed the role of status more strongly and consistently than gamification evangelist Gabe Zichermann. He presents social status as a central human drive that fuels an innate desire to compete, win, and publicly display tokens of achievement. His "SAPS" hierarchy puts status on the first place as the most desired (and least expensive) reward. Status competition is presented as the unifying core of games, loyalty programs, and social life writ large (Zichermann and Linder 2010, 15, 41; Zichermann and Cunningham 2011, 9–11, 91–92; Deterding 2011).

Zichermann's status rhetoric echoes the premodern play rhetorics of power (Sutton-Smith 1997, 74–90), where games, sports, and other ritualized contests are seen as (predominantly male) arenas of publicly asserting dominance, cool nerves, and group affiliation, but his statements ultimately lack scientific grounding. This is surprising given that ethology, sociology, and social psychology provide ample resources to substantiate and unpack such status rhetorics: one may appeal to Veblen's conspicuous consumption, Bourdieu's analyses of social capital and the reproduction of class difference through displaying "good taste," Cooley's looking-glass self, Goffman's (1959) studies of self-presentation, or psychological research on social motives like power, achievement, and affiliation (Reeve 2009, 170–200). One might even appeal to evolutionary signaling theory, reasoning that animals display costly signals (like elk horns, peacock tails, luxury goods, or diplomas) because these provide a hard-to-fabricate indication of one's reproductive fitness (Bliege Bird and Alden Smith 2005). All these approaches argue for a central role of social signaling in human coexistence. Seen in their light, games and gamified systems are public arenas that provide means for fashioning scarce and costly signals of status, fitness, and trustworthiness. However, their portrayals of the dynamics of symbolic social regard are also far more intricate and complex than gamification proponents would have them be.

The most relevant application of these theories for games and gamification can be found in informatics and HCI research on self-presentation, trust, and motivations for contributing UGC on social media (Lampe, this volume) and game scholarship on

gaming capital and achievement systems (Consalvo 2007; Jakobsson 2011; Medler 2011).

Given the naturalizing terms in which gamification proponents appeal to status, this rhetoric is again liminal: all there is to do is for system designers to hook into people's natural "drives" to status to motivate them to pursue the system's goals. Critics in, for example, sociology, psychology, or informatics have countered that competing for social status is long-term detrimental to psychosocial well-being (Dweck 2008; Deci and Ryan 2012) and an acquired, learned need—one we don't find as strongly in collectivist cultures (see Khaled, this volume).

The Rhetoric of Performance

If status is one side of the coin of collective rhetorics, communal performance is the other. Its proponents—most prominently Jane McGonigal—emphasize collaboration over competition, shared experience over individual distinction, and the "epic" sense of "Becoming a Part of Something Bigger than Ourselves" (McGonigal 2011, 95). We find this rhetoric in the artistic public play and pervasive game communities, mixed reality art groups, Nordic larp, and Bernie DeKoven's notion of a "play community" (DeKoven 2013; DeKoven, this volume). Games are understood as collective rituals: temporally, spatially, and socially set-apart alternate realities where participants can embody and explore alternative roles and behaviors. New social frames of meaning and norms get negotiated and enacted, and this joint performance of an alternate little world unto itself gives rise to strong embodied experiences of "collective effervescence" (Collins 2004) or *communitas* (Turner 1982; see Stenros, this volume).

The rhetoric of performance ties directly back to the premodern play rhetoric of identity, framing play as events in which individuals experience their belonging to their community and reassert its sacred moral order (Sutton-Smith 1997, 91–110). Its nongame analogs are festivities, performance art, and happenings. Here, gamification (although proponents of this rhetoric usually shun the term) entails that the spatially, temporally, and socially *bounded* alternate reality of a ritual becomes *pervasive* as a shared framing that generates and stabilizes new meanings, experiences, behaviors, and roles in everyday life (McGonigal 2011, 125–126). Hence play activities and games that blur the boundaries between games, public space, and everyday life are the reference prototypes of this rhetoric: situationist practices, LARP, ARGs, pervasive games, or the "big games" of Area/Code (2007).

Analytically, the rhetoric of performance as *communal* action can be distinguished from a second flavor, *immersive* performance, where games are seen as environments whose dramaturgical arrangement creates narrative experiences for the person who moves through them: "spatial stories" (Jenkins 2004). In this context, gamification means to layer media onto everyday reality to reframe the environment and activity of the audience-as-performer, a technology-enabled pretense. Proponents span from Carroll and Thomas's (1982) "metaphoric cover stories" to today's immersive theater, mixed reality, and narrative role-playing-based forms of gameful learning (Barab, Gresalfi, and Ingram-Goble 2010; Sheldon 2011; Travis 2011). Role-playing and augmented reality games are the typical reference genre, and theme parks, amusement rides, and theater the typical nongame analogs.

In McGonigal's (2011) current writings, games are presented as the positive normative yardstick for social life and the means to move individual experience and collective action toward this optimum—ultimately, a secular liminal position where societal values are given and affirmed, and individual action is utilized for the *collective* good of *individual* happiness. Other proponents of this rhetoric conceive of games in liminoid terms as temporary autonomous zones (Bey 1991) where the culturally dominant meanings, values, and ways of life can be questioned and dissolved, where experiences that "another world is possible" can be made and oppositional communities forged. Many projects of this rhetoric—like *Macon Money* (Area/Code 2010)—aim to counteract the segregation and loss of community in cities that are framed as both a cause and part of the systemic inequities of today's societies (see Alfrink, this volume). It is an irony of history that *ritual*—the very embodiment of affirmation of the social order in premodern societies—should in late modernity be viewed as a principal space for challenging it.

The Rhetoric of Reinforcement

The rhetoric of reinforcement shares many proponents, tenets, and design approaches with the rhetorics of feedback and nudging. Their joint interest is using technology to drive individuals' behavior in an intended direction—in other words, "persuasive technology." Coined by psychologist B. J. Fogg (2003, 1) the term describes "any interactive computing system designed to change people's attitudes or behaviors." Fogg has since 2003 focused more and more on behaviors over attitudes, fashioning his own "Behavior Model" (Fogg 2009a) and "Behavior Grid" (Fogg 2009b). Through his business consultancy work, he has been instrumental in the resurgence of behaviorism among technology and design circles in Silicon Valley. In the rhetoric of reinforcement (as in behaviorism writ large), intention and cognition are seen as mostly epiphenomenal. Behavior is explained—that is, mathematically modeled and predicted—as the relation of the observable previous history of reinforcement of an organism and its current environment of observable stimuli (see Linehan, Kirman, and Roche, this volume). This "engineering" view of human behavior, coupled with a focus on data and predictive modeling, seems to resonate with the existing mental models and practices within software and technology companies.

In the gamification discourse, the rhetoric of reinforcement was first struck by Fogg's doctoral advisor Byron Reeves (Reeves and Read 2009, 71–75)[9] and has since become a mainstay in the publications of Gabe Zichermann (Zichermann and Cunningham 2011) and Michael Wu (2012). These authors present games as systems that produce optimal stimuli and reinforcement. Slot machines and loot drop in World of Warcraft (Blizzard Entertainment 2004) are evoked as prime examples of variable ratio reinforcement schemes that deliver reinforcements on desired behavioral responses at a not fully predictable rate, which is seen to explain their "addictive" quality.

Notably, even critics of gamification often deploy the rhetoric of reinforcement, equating gamification with "Skinner boxes"; that is, the experimental setup of B. F. Skinner and colleagues where animals are placed in a box with a lever and food dispenser (Linehan, Kirman, and Roche, this volume). Such "Skinner box" appeals are sometimes used to highlight the reductionism of behaviorist explanations,

but most often to express moral outrage over designers intentionally using behaviorist techniques on human beings—thereby implicitly expressing a strong belief in the efficacy of these techniques (Pesce, this volume). This common reaction to the rhetoric of reinforcement (and persuasive technology more generally) has led to a legitimizing ethics discourse (Berdichevsky and Neuenschwader 1999; Fogg 2003, 211–244; Zichermann 2012). In practice, these ethics are usually construed as constraints on what is "still permissible" in the liminal perfection of means toward the predetermined ends of the system owner. Applications inspired by the reinforcement rhetoric range from employee engagement and customer loyalty to public policy, but arguably center around "Habit Design"[10] for personal health and self-improvement. It bears pointing out that many proponents of the reinforcement rhetoric *appeal* to science but ultimately operate on a folk theoretical understanding, amalgamating knowledge of often obsolete and even mutually contradicting bodies of research (e.g., Maslow's hierarchy of needs with behaviorism, cf. Wu 2012), filtered through pop science journalism or B. J. Fogg.

The Rhetoric of Well-Being

The rhetoric of well-being is grounded in the cognitive and positive psychology that presents the very counterreaction to behaviorism in the history of psychology: the (re)discovery of the fact that humans voluntarily engage in activities without any reinforcement and that one and the same dreadful circumstance might lead to radically different understandings, experiences, and thus behavioral responses challenged the behaviorist ban on taking internal states into account. Consequently, the rhetoric of well-being emphasizes internal states over environmental circumstance, experience over behavior, and self-determination over outside control. The chief intellectual guideposts in this rhetoric are Aristotelian (2002) virtue ethics, Apter's (2006) reversal theory, Csikszentmihalyi's (1990) concept of flow, self-determination theory (Deci and Ryan 2012), and the works of positive psychologist Martin Seligman (2011). All are concerned with the nature of human well-being, and all propose that human beings actively seek out and enjoy certain intrinsically valued states. Indeed, the human being in this rhetoric is defined by its ability to transcend its animal determination by bodily pleasure and pain toward ever-greater self-determination.

The rhetoric of well-being frames the "fun" of games as just those states that we as humans innately strive for: experiences of competence, relatedness, and autonomy, of meaning and flow. Games are seen as environments that optimally afford such experiences (Csikszentmihalyi 1990, 48–76; see Rigby, this volume). Notably, the quality of one's experience is seen to depend on both external circumstances *and* one's internal stance toward them. We might perceive and engage in an activity as "autotelic" (Csikszentmihalyi), "paratelic" (Apter), "autonomous" (Deci and Ryan)—or not. This is why proponents of the rhetorics of well-being emphasize perceived autonomy as a necessary condition for play.

One aspect of the rhetoric of well-being, forwarded by Jane McGonigal among others, might be called hedonic. In her view, judged by its ability to provide what humans strive for, "reality, compared to games, is broken." (McGonigal 2011, 3). Her proposed solution is to (a) play *more* games to sate our unfulfilled needs, and (b) redesign the world in the image of

games, again (c) using games to motivate people to engage in these redesign efforts.

Miguel Sicart (this volume), Marc Hassenzahl and Matthias Laschke (this volume), myself (Deterding 2012a), and to a certain extent, McGonigal herself have emphasized a different aspect of well-being one may call *eudaimonic*. Taking a primary cue from Aristotelian ethics, the focus is less on happiness than *eudaimonia* or flourishing as the lifelong development into a self-determined, self-concordant human being. Changing outer circumstances alone by definition cannot bring about such flourishing, because it is the development of self-knowledge, self-awareness, and self-regulating skills or virtues through deliberate practice. In this rhetoric, gamification is seen as ideally facilitating tools of "extended will" or "pleasant troublemakers" we enroll in our pursuit of eudaimonia (Heath and Anderson 2010; Hassenzahl and Laschke, this volume; Selinger, Sadowski and Seager, this volume). Analogs are the practicing of crafts or martial arts. Physical sports or games like Go and chess that allow a lifelong deepening of skills as well as "light," playful interactions that invite reflection instead of forcing (or easing) action are prototypical game genres. This rhetoric is deeply liminoid, wedded to modernist rhetorics of self-realization (Sutton-Smith 1997, 173–200).

The Rhetoric of Pleasure

The rhetoric of pleasure is found predominantly in HCI as well as user experience and game design, with Jesse Schell (this volume), Stephen P. Anderson (2011), and Nicole Lazzaro (2008) being main figureheads. Here, the central question is how to create an entertaining, enjoyable, engrossing experience—be that for a marketing or advertising campaign, a product or service, or a stand-alone entertainment offering. In all cases, games are seen as *the* genre of interactive experiences that has been built with the sole purpose of providing enjoyment—hence, they are a medium of choice and a source of inspiration for designers who want to design for pleasure. This rhetoric is holistic and applied—the main interest is in supporting the practice of design with (conceptual) tools and processes. Many different forms, causes, and conditions of enjoyable experiences are acknowledged (Arrasvuori et al. 2011), as is the importance of design detail and the complexity of circumstances.

Given its broad stance and perspective, this rhetoric is grounded in a broad range of exemplary phenomena, including toys, amusement park attractions, and museum installations. Its applied focus is fully liminal, supporting organizations in designing and marketing products and services.

The Rhetoric of Systems

The rhetoric of systems—sometimes also labeled proceduralist or formalist (Sicart 2011)—is a distinctive emic discourse of game design and game studies. Central proponents are designers and scholars such as Eric Zimmerman (this volume; Salen and Zimmerman 2004), Ian Bogost (2007), and Michael Mateas and Noah Wardrip-Fruin (2009). Grounded in literary theory, computer science, systems theory, and education, this rhetoric portrays games as designed rule systems with which one can interact.

The *expressive* branch of this rhetoric is especially interested in games as meaning-making media—how rule systems are able to present claims about other systems, and how meaning is generated in the interaction between people and rule systems. The

main reference genres here are simulations or otherwise rule-intensive games; algorithms, societal rule systems, and expressive media (especially literature) are common nongame analogs. The expressive potential of games is at the same time criticized as a potential "hidden persuader" (see Starr 1994) and actively encouraged in the design of persuasive games (Bogost 2007). This rhetoric has been so far mostly used as a critique of gamification (Bogost, this volume; Robertson 2010), making the argument that the reduction of games to reusable feedback design elements misses their *systemic* nature, and thus ignores the complexity of both game design and the game-player interaction. The moral politics of its representatives have given the expressive systems rhetoric a liminoid bent: by representing (or contrasting) the dysfunctions of other rule systems in society, games ought to raise players' consciousness of them.

A slightly different tune is struck in the *educational* branch of the systems rhetoric (Gee, 2003; Squire, 2011; Ramirez and Squire, this volume). Grounded in constructionist and sociocultural learning theories, its representatives are most interested in the learning that takes place through and around games. Learning is understood as a process of constructing one's understanding of and skill in engaging with the world through practical experience with complex real-world problems and participation in a surrounding community of practice. Games are thus viewed as sociotechnical systems of communities and artifacts, with the artifacts themselves being complex, probable systems. This makes simulation and multiplayer games, with strong player communities, the key reference genres. Systems thinking, design, creativity, complex problem solving, and collaboration are presented as the most relevant skills of the twenty-first-century workforce (Zimmerman, this volume), and game design as the activity where these skills are best trained, making hackers and designers aspirational role models: Ludoliteracy (Zagal 2010), procedural literacy (Bogost 2005; Mateas 2005), and computational thinking (Wing 2006) are understood as intimately linked. In this rhetoric, gamification is framed mostly critically, arguing that educational uses of gamification have so far fallen back into incentive systems that have proved detrimental to learning, while missing out on games' ability to present complex real-world tasks and involve players in social learning (Ramirez and Squire, this volume). Counterexamples such as Lee Sheldon's (2011) *Multiplayer Classroom* or the Quest 2 Learn schools (Salen et al. 2011; see Salen, this volume) again stress the *systemic* nature of games, restructuring classes (or school curricula) around complex, collaborative problems and design activities. This rhetoric fluctuates between the liminal and liminoid, based on the underlying educational philosophy—it is liminal (and echoing the modernist play rhetorics of progress; Sutton-Smith 1997, 18–51) where it plainly uses game design to realize socially demanded learning outcomes and educate a future workforce (e.g., Kapp 2012). It is liminoid where it understands education as the full development and empowerment of individuals.

The Rhetoric of Cultural Form

The rhetoric of cultural form overlaps with that of expressive systems in its framing of games as sites of expression, meaning making, and aesthetic experience. Yet it casts a wider net: many scholars and designers with backgrounds in art, art history, or cultural and media studies understand games broadly as an "aesthetic form" (Lantz 2011; Lantz, this

volume) parallel to painting, music, film, or literature, one that has its own specific affordances and tropes (Flanagan, this volume). As such, games partake in the wider circulation of meanings in culture—as reflections of the ideas, discourses, and ideologies of their time, and as expressions of individuals' perspectives and (critical) reflections on them. Cultural participation today requires the acquisition of "new media literacies" (Jenkins et al. 2009), including the ability to appreciate and express oneself critically through the medium of games. Unsurprisingly, this rhetoric is particularly dear to the indie and art games communities.

This rhetoric is quintessentially modernist and liminoid, continuing the play rhetorics of the imaginary (Sutton-Smith 1997, 173–200). Games can be a medium of creative expression of the self and social critique, but as "art for art's sake," they do not need any instrumental, functional legitimacy. "In Defense of Beauty" (Zimmerman 2011), gamification is usually critiqued as profaning "the wild, magical beast of games" (Bogost 2011). At best, it is reframed as "ludification"—the recirculation of game aesthetics in the wider media culture (Raessens 2012).

The Rhetoric of Playfulness

A central motif in the rhetorics of play is to portray it as what by definition escapes definition, graspable only in the negative as the reversal of any existing order or form, as an inherently "meta" phenomenon of "antistructure" (Turner 1982, 52). In modernity, play has become framed as the frivolous opposite to the Protestant work ethic, but the rhetoric of frivolity goes back much futher to the ancient figure of the trickster god (Sutton-Smith 1997, 201–213).

Fittingly, in both the game studies and gamification discourse, the rhetoric of playfulness takes the shape of a counterdiscourse (Sicart 2011; Sicart, in press; Sicart, this volume; Nicholson 2012). It draws on philosophy and aesthetics to evoke the protean, creative, unruly powers of *paidia* (Caillois 2001) or *Free Play* (Nachmanovitch 1990). From its improvisational recombination of behaviors, objects, and meanings, new forms emerge. Play is thus not only "free movement within a more rigid structure" (Salen and Zimmerman 2004, 304) but also the movement and transcendence of structure, while still bound to it as its necessary antithesis. At the same time, the ludic civilization of rules and competition is ever present to bind the primordial impulse of playfulness (DeKoven 2013). Games in this rhetoric are that antithesis: they are the material for and solidified remains of play. If there is a genre of games conducive to play at all, it is the most toy-like, open-ended sandbox simulations such as Minecraft (Mojang 2009). Playfulness itself is seen not as a material object or activity, but as a non-instrumental, autotelic stance. It is a temporary liminoid appropriation and transformation of its source activity—a bout of benign mischief that for its very immoderateness always carries the danger of derailing into "dark play" (Schechner 1988). In HCI, this rhetoric is reflected in Gaver's notion of "ludic design" (Gaver 2002; Gaver, this volume) that is less interested in the "improvement" of user experience than in creating spaces and objects for open exploration and reflection. Then again, in the design industry, we find the liminal harnessing of play as a source of creativity. Innovation games (Hohmann 2006), gamestorming (Gray, Brown, and Macanufo 2010), and gamified idea platforms are used to "manage innovation processes" (Gartner 2011) and to sate the demand of businesses for rationally accountable creativity.

Conclusion: *Vos jeux, faite*

Fitter, happier, more productive
Comfortable
Not drinking too much
Regular exercise at the gym (three days a week)
Getting on better with your associate employee contemporaries
At ease
Eating well (no more microwave dinners and saturated fats)
A patient better driver, a safer car (baby smiling in back seat)
Sleeping well (no bad dreams)
No paranoia

—Radiohead, Fitter Happier (1997)

So what remains? As with the rhetorics of play, we see that the rhetorics of a gameful world "are part of the multiple broad symbolic systems—political, religious, social, and educational—through which we construct the meaning of the cultures in which we live" (Sutton-Smith 1997, 9). Reviewing its histories—serious and pervasive games, artistic and countercultural play, playful design, socialist competition, playful resistance and funsultants at work—takes the breathless edge of newness from gamification. This in turn opens a view toward deeper underlying transformations of society: the postmaterial value shift, the rise of post-Fordist information, service, and experience economies, the instrumentation of our lifeworld into a code/space of ubiquitous computing, the emergence of digital games as a medium and industry, and the convergence of media into ever-new amalgamations.

Surveying the rhetorics of a gameful world, one finds a heterogeneity of communities, discourses, and practices that existed long before they came into contact with games and play. As a cultural moment and momentum, the gameful world is characterized by this very proliferation of perspectives that—I would argue—game scholarship should not fend off as "colonizing attempts" (Aarseth 2001), but embrace and reflect in its meaning and effects. If the unifying feature of play is its "adaptive variability" (Sutton-Smith 1997, 221), then "games" today are unified in their mirror-like quality to reflect the image of whatever preconceptions are brought to them: the alliances and oppositions among the rhetorics of the gameful world organize themselves along the fault lines of modernist politics of play, on the question of what the "proper" place of play and games in society ought to be.

On the one side, we find the liminal rhetorics that accept our secular gods of progress, productivity, and the self. They frame games, play, and game design as the perfection of means toward these given ends. Here we see economists and psychologists, marketers, designers, and educators building perfect markets and feedback systems, choice architectures and behavior programs, remedies for the stresses of everyday life, collective self-realization engines, digital pleasures and innovation machines, microworlds for the twenty-first century workforce—all to make us fitter, happier, more productive. What is new is that the liminal collectivity of ritual is dissolved into a mundane, individualist, technical rationality. No collective ritual: just a few more rules the individual chooses to self-obey and self-monitor (until ubiquity turns them into social norms and technical defaults). No ecstatic moment of boundless *communitas*: just linked data, social graphs, and aggregated visualizations. No temporary inversion that affirms the

order upon return, no risky *Rumspringas*: just direct affirmation.

These liminal rhetorics go hand in hand with an emphasis on and preference for ludic forms in Caillois' (2001) sense: *gamification* is an inadvertently apt term, as the suggested solutions almost always involve adding *more* structure, rules, and goals (the only exception being the rhetorics of pleasure and industrial creativity). These added rule systems often present themselves as tools that empower individuals to pursue economic achievement and self-realization. Yet as Foucault has taught us, any such technology of the self at the same time is a technology of domination: just as it liberates through self-control, it controls through self-liberation. Continual self-governing is the very form of governmentality in modern liberal democracies and market economies (see Whitson, this volume), and celebrating the empowerment of individuals to better themselves puts the onus for any social problem on the individual, her "willpower" and "determination" (or lack thereof). All would be well if we all just ate a bit better, worked a little harder, drove a little more fuel-efficiently. You can get it if you really want, but you must try, try, and try—and gamification can help you along the way. This affirms the standing order by backgrounding any systemic causes or inequities, and thus any questioning of the sustainability or fairness of the order as such (cf. Thøgersen 2011).

Such liminal cultivations of play and games face the serious issue that the experience of autonomy in an activity is borne out by study after study as a constitutive feature of play and a central source of our enjoyment of it (Deterding 2013). It is not so much that we voluntarily play games because they are so much fun; rather, we experience game play as fun partially because it is framed as autonomous, with no outer control, coercion, or frightful consequence affixed. The irony of instrumentalizing play and games as means to another end, then, is that it depletes the very source it tries to tap into: the experience of autonomy in noninstrumental activity.

One the other side, we find the liminoid rhetorics of critique, resistance, subversion, likewise bowing to the modern god of the self, but in presumed opposition to and desired transformation of the standing order: here the artists, intellectuals, and subcultures, here the critical aesthetic interventions and creative civil disobediences that "raise awareness" of the exploitation and inequity in society, the pocket utopias reminding us that "another world is possible," the focal practices of self-determination, the empowering mastery of new media expressing social critique and personal vision, the nonfunctional art for art's sake, the little carnivals and tactics of resistance. They overwhelmingly coincide with a preference for *paidic* forms of play and games (Caillois 2001), the freeing of movement and dissolution of structure—the exception being the rhetoric of expressive systems.

These liminoid rhetorics have to answer to the question whether their critical interventions (like those of any avant-garde before them) will ever reach more than the already converted and highly educated few. They have to respond to Horkheimer and Adorno's (2010/1969, 145–150) interjection that play as a refuge *from* the world of instrumentality is always already instrumentalized as a restoration *for* that world. And they have to ask themselves to what extent any playful reclaiming of public life is really a form of repressive tolerance that discharges revolutionary energy and resells it as commodified dissent.

In all its presumed dissolution of modernist boundaries between work and leisure, then, the rhetorics of

the gameful world reproduce the modernist opposition of the utopian politics of economic and technological progress and the romantic counterpolitics of the incommensurable individual and art as religion. Which raises the question: What are we *not* talking about? For this, let us turn to Jane McGonigal's (2011) *Reality Is Broken*. The book's premise is straightforward: "Reality wasn't designed from the bottom up to make us happy. Reality, compared to games, is broken." And so is the inviting conclusion: "What if we decided to use everything we know about game design to fix what's wrong with reality?" (3, 7). Now to the extent that McGonigal—and other proponents of a gameful world—want us to be more aware and skillful designers of our own lives, they are to be applauded. But their project also reveals an almost Baconian modernist, technocratic belief in the ultimate knowability, controllability, and perfectibility of nature by mankind. Games certainly offer us such strong experiences of agency. Their narratives almost invariably reiterate some version of *Only You Can Save Mankind* (Pratchett 1992). There's a reason we call Utopia (Dagelow 1982) or Populous (Bullfrog 1989) "god games." Games are bounded spaces of "contrived contingency" (Malaby 2007) where the odds are ever in our favor: it might be a zero-sum game, but *someone* is *bound* to win—even if it is the bank. In this, games mirror the ontological cocoon of today's technical civilization where gratification is instant, power is at our fingertips, and "disruptive" progress is a daily news story. Surely, any day now, diseases will be overcome, aging reversed, and death conquered.

But reality is not broken—the comparison is. Reality has no tutorials, walk-throughs, or rulebooks. Not every puzzle has a solution. There are no save games, no insert coins to continue (well, depending on your creed). The cards are almost invariably stacked, and often enough, everyone is bound to lose. Reality ultimately is much more messy, random, unfair, and *beyond our control* than games. To think otherwise is to fall for "the ludic fallacy" (Taleb 2010, 122–133; cf. Walz 2010b): to confuse map and territory, the calculable risks of gambling with the existential dangers of living, the contrived, controlled, known spaces of our games, models, and technical lifeworlds with the uncontrolled, unknown, uncaring cosmos. It is tantamount to believing in some form of benign predestination, or the megalomania of children playing a game of "War," believing they can somehow force their will unto the cards.

This is the rhetoric of play conspicuously missing in today's discourse: *fate* (Sutton-Smith 1997, 52–74). In Hinduism, existence is seen as *lila*, a form of play by Shiva Nataraja, the lord of dance. In dancing, Shiva creates and destroys universes for the pure pleasure of doing so. In another myth, Shiva splits into himself and his wife Parvati to play a game of dice together. They constantly cheat on each other and get angry over it—and this divine spousal infighting is the source of our worldly turmoil (Handelman and Shulman 1997). We also have this notion in the Western tradition. It is present in the jealousies and scheming of the all-too-human Greek gods. It is the *hamartia* of Aristotelian tragedy, the hero's tragic flaw, the sight of which causes terror and pity in us through recognizing our own possible lot in it. It is the wheel of fortune commanded by the goddess Fortuna. There is a small, liminoid tradition in modern art that sought out chance as a means to escape the order of society, culture, and language—echoed in Luke Rinehart's (1971) *The Dice Man*, who decides to let his every decision be ruled by the roll of a die, logically leading him into the outside of society (with lots of gratuitous sex, drugs, and violence on the way, as befits a 1970s "cult classic"). We find remnants of this tradition in Dennis Crowley's

vision for foursquare: collecting and analyzing behavioral and social graph data "to generate more of those serendipitous encounters" (Crowley in Newton 2011) that brighten our daily routines in traversing a big city—an automated *dérive* that seems to miss the point that "reliable pleasant surprises" are as oxymoronic as "mandatory play." Urban "things to do, places to shop" raffles like foursquare—useful and enjoyable as they are—only add cushioning to our cocoons. They do not expose us to our own thrownness, as games of chance sometimes do (Malaby 2012). Compared with Greek tragedy, we seem to have lost our taste for facing suffering and fatefulness—for seeing them not as bugs but as inherent features of our lives that demand and teach courage, compassion, humility, and wisdom.

And games and play themselves? What happens to them in a gameful world? If they become fully ubiquitous and mundane, if they no longer demarcate something "magic" anymore, will they disappear? Will their increasing instrumentalization kill their very soul? Such anxieties, understandable as they are, presumably result from our attachment to that brief pocket in time and space we call modernity: where play and games were permissible for children in their healthy development, artists in their creative pursuit, and adults as leisure—but nothing more. These, after all, *are* the modernist rhetorics of play (Sutton-Smith 1997). We perceive artistic and countercultural play in public, serious and pervasive games, or gamification as figures of provocation, newness, alterity, or profanation only against this background. The rise of a gameful world reminds us that this modern state of affairs is itself a temporary formation and practical accomplishment: to expect and demand *that* games have no practical consequence or instrumental function; to design games *such that* the risk of serious bodily and social consequence is minimized and interpretive and experiential openness maximized. Change these practices, and over time you change the background of our shared expectations, concepts, norms, and designs, our social and material institutionalizations. These new forms do not pervert an eternal "heart of gameness"; they *transform* what games "are" *to us*, *today*, as members of a culture.

On a practical level, we as individuals can attentively track only so many levels of meaning at any given moment and thus partake in only so many games at any time. Games as designed artifacts are already used every day in ways the users themselves would not call gaming or playing—gold farming, usability testing, and the like. Games as artifacts and gaming as activity do not necessarily coincide; today's diffusion of game design beyond gaming encounters makes this theoretical possibility an everyday reality.

Play and games have always already been interwoven with and cultivated by society. The pocketing away of games and play into the fringes of childhood, art, and leisure—which allowed transgressions for their very marginality—has been their primary social place in modernity. That changes now. Their ubiquity and pervasiveness will strip games and play of some of their current enchantment, momentum, and cachet—that's the logic of fashion. Gamers will revolt against this "Eternal September" (Grossman 1998) of massive outer influx into the places and practices they feel "belong to them" and retreat into new, "hardcore" places and practices, anxiously guarding cultural symbols of "true" identity and belonging—that's the logic of subculture. The instrumentalization of games and play will engender alienation and professional deformation, and people will reassert their autonomy and playfulness in gaming the system and in seeking out new, yet unknown spaces and

forms of nonfunctionalized restoration and resistance that will then be functionalized again—that's the logic of instrumentalization.

History tells us that these shifts will not dissolve games or play, only change their local meanings and forms. As long as we are warm-blooded bodies living and dying on Earth, the exigencies of survival will not disappear, and neither will the *un*-necessary, the *in*-efficient, the "just so" and "just because," the occasions of pure waste, the voluntary attempts to overcome unnecessary obstacles as that which lifts us, however fleetingly, from the given bounds of our mortal fate into the chosen bounds of culture. "And law only / can give us freedom."

Notes

1. But very much in the footsteps of Walz and Coulton's (2011) pioneering mapping attempt and Walz's (2010) historical analysis of the "routinization trajectories" of games. I thank my coeditor Steffen P. Walz for the countless shared discussions that deeply informed every line of this chapter.

2. See http://www.google.com/trends/explore#q=gamification (accessed April 5, 2013). For references in order of appearance, see Sierra (2005), Kim (2006), and Merholz et al. (2008) on Nike+; Hall (2011) on GameLayers; IBM (2007) and Reeves and Malone (2007) on IBM's involvement in serious games; Saffer (2007) and Alfrink (2007) on talks about games inspiring user experience design; Varney (2010) on McGonigal; Lusty (2007) on Bunchball; Atwood (2011) on StackOverflow; O'Reilly Media (2008) on the web 2.0 Expo 2008; Takahashi (2005) on Zichermann's Funware blog; Currier (2008) and King and Borland (2003, 53) on first uses of the term *gamification*; Werbach and Hunter (2012, 36–37) and Kapp (2012, 2–3) on the "Piano Stairs"; Schell (2010a, 2010b, 2010c) on his Dice talk and Gamepocalypse blog; McGonigal (2010) on her TED talk; Manning Publications (2010) on the book *Funware in Action*; Zichermann (2010) and Zichermann and Linder (2010) on the Develop talk and the book *Game-Based Marketing*; Badgeville (2010) on their launch; Helgason (2010) on "the year of gamification"; and Kim (2011) on her gamification workshop.

3. See http://www.chorewars.com/help.php (accessed April 11, 2013).

4. See http://who.godaddy.com/whoischeck.aspx?domain=GAMIFICATION.COM (accessed April 11, 2013).

5. See http://www.thefuntheory.com/ (accessed April 11, 2013).

6. See http://web.archive.org/web/20100525152043/http://www.bunchball.com/ (accessed April 11, 2013).

7. See http://who.is/whois/gamification.co (accessed April 11, 2013).

8. And if the term needed any more legitimacy, in April that year technology consultancy Gartner (2011a) announced its "strategic planning assumption" that more than half of all organizations with innovation processes will gamify those by 2015 and that more than 70 percent of Global 2000 companies will have at least one gamified application by 2014. Gartner then officially placed gamification on its "hype cycle" curve of emerging technologies in

August 2011 (Gartner 2011b), only to announce one year later another strategic planning assumption that by 2014, 80 percent of gamified applications will fail due to poor design (Gartner 2012)—thus providing ample material for research on the manufacture of self-fulfilling prophecies.

9. Who was in turn informed by Yee (2001).

10. See http://www.habitdesign.org/ (accessed April 12, 2013).

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2 WHY GAMIFICATION IS BULLSHIT

Ian Bogost

In his tiny treatise *On Bullshit*, the moral philosopher Harry Frankfurt (2005) gives us a useful theory of bullshit. We normally think of bullshit as a synonym—albeit a somewhat vulgar one—for lies or deceit. But Frankfurt argues that bullshit has nothing to do with truth. Rather, bullshit is used to conceal, to impress, or to coerce. Unlike liars, bullshitters have no use for the truth. All that matters to them is hiding their ignorance or bringing about their own benefit.

In a short position statement presented at a small, invitation-only conference on gamification at the University of Pennsylvania's Wharton School, I argued that "gamification is bullshit" (Bogost 2011a). The idea proved popular, and the article was quickly republished and translated online. Apparently tired of hearing the constant refrain of gamification as a salve for all ills, the technology community proved particularly receptive to my neutralizing opinion. As a result, my diatribe's message became confused. Too many have read just that short statement (or perhaps just the title) and walked away with a sense of glee or disfavor after finding that someone might hope to sabotage another rising trend. I intended to provoke, but not through mere cantankerousness: I really did mean what I said on an intellectual as much as a visceral level. In this article, I describe that position more fully and demonstrate its implications.

Let's start by rehearsing the argument without the theatrics. Gamification, I suggested, is primarily a practice of marketers and consultants who seek to construct and then exploit an opportunity for benefit. The opportunity in question is games, which remain a terrifying yet appealing medium for businesses. Terrifying because traditional organizations don't understand games and therefore fear them: for example, why do people spend so much time in such concentrated attention when playing video games, while they are so distracted or easily disengaged from other media? And appealing because there is some possibility that power can be harnessed for corporate benefit.

Bullshit enters the picture in the space in between, where games become a business opportunity for those who might harness the fear to help corporations believe they are benefiting from the power. When understood in the sense Harry Frankfurt helpfully provides, bullshit is a coercive strategy rather than one that has anything to do with the true value or promise of something. In late summer 2011 when the Wharton symposium took place, the trend itself was still relatively new, only having entered the trend repertoire a year or so earlier. Gabe Zichermann organized the first of his gamification "GSummits" in January of that year, and by the time I called

bullshit in August, technology research firm Gartner had added gamification to its "hype cycle" chart, declaring it "on the rise," nearing its peak (Brockmeyer 2011).

By that time, gamification's proponents had been working for eighteen months to transform the idea into a trend, a thing that couldn't be ignored, a question that would come up in a brand manager's meeting with her chief marketing officer or that a vice president might have heard enough times to warrant signing up for a conference with the leftover, expiring funds on a fiscal year's budget. In this sense, gamification's rise entailed its rhetorical advance more than its application. By promoting it as a trend that the future clients of gamification service providers and consultants couldn't possibly ignore, gamification's proponents were able to focus on ideas and promise. Once sufficiently established, the use, value, and details of the practice became less important. As with most business consulting activities, in the end they never would be.

Consulting as Bullshit

Consultants are excellent bullshitters. That's not because they lie—although they sometimes do that too. No, it's because their primary purpose is to justify and better their own existence. Those of us who have worked as, for, or with consultants will know immediately what I mean. For the more fortunate, pop culture offers an adept introduction.

The Showtime television series *House of Lies* chronicles the lives of a "pod" of business consultants at the fictional firm of Galweather & Stern who are continually challenged to validate their own existence. They do so not by solving the problems of the companies who hire them (in many cases, they create more problems than they solve), but by helping the executives who hired them justify an existing plan, appear clever in the face of a rival, or feel comforted in the wake of a hardship.

In the show's second season opener, the pod takes a call with a potential client for which they have failed to prep. They know a "Mr. Pincus" is on the phone but don't have enough time to figure out what company or even what industry he represents. The group throws out a series of filler nonsense meant partly to hook him, but mostly to stall until they can use clues in his responses to home in on a match. They identify the client definitively only after he arrives for a face-to-face meeting, when the pod tricks him into revealing the industry he represents—"We are going to build a better goddamn casino!" he finally proclaims.

A generous characterization of the consultant's life might reason that Pincus's industry doesn't really matter because any solutions Galweather & Stern will provide are so general as to apply to all businesses. Such is the argument a gamification consultant or service provider would likely make, after all. The more suspicious might wonder if a solution meant to apply to everyone would ever really be of use to anyone in particular.

This is the logic bullshit characterizes, that of the swindler or huckster who seeks his own advantage at all costs. A consultant isn't a petty criminal, of course, things are more complicated than that. While street grifters are always out to swindle, consultants

certainly don't mind if their efforts benefit a client. It's just not their primary concern. Or their secondary one. Really, it's not a concern at all, except insofar as some examples of success are necessary so as not to bring the whole house of cards tumbling down. The consultant makes appeals to the client's feelings of inadequacy, confusion, or weakness, then fills that imagined chasm with a pre-prepared or stock solution that can be tuned to oscillate on just the right wavelength to mesmerize the customer into submission. And given that the consultancy client's primary goal is to appear competent, supported, and powerful in the context of a business, the outcome of a consulting arrangement may not be of primary concern to the client either. Such is the strange dance of business consulting. Philosopher turned consultant turned management debunker Matthew Stewart explains it this way, discussing a shape of chart called "the Whale" he would often use with his consulting clients:

> By the time I had produced the fifth or sixth version of the Whale, I realized that I could do the same for just about any business anywhere. It made no difference whether the business was inherently good or bad, well-managed or in the hands of chimpanzees. It didn't even have to be a business—it could be a football game or a population chart. In fact, I didn't even have to do the analysis. I could save 80 percent of the effort by just borrowing data from some previous analysis. There was always going to be a skew [the whale shape]. In most cases most of the time, the skew chart merely records a curious fact of economic life. It isn't science; it's a party trick. (Stewart 2009, 61–62)

Gamification plays a similar kind of party trick, but it does so with ideas instead of numbers, at least to start. There are two moves this trick entails, both of which are represented in the admittedly brilliant term *gamification*. The first is the "game," of course, that strange, dark medium that captures attention in entertainment contexts yet somehow withholds it from business ones. If only we could capture the magic and power of games and apply them to other contexts, imagine what we could achieve! Games are appealing yet terrifying to businesses because they seem like magic attention machines, inspiring players to devote tens or hundreds of hours to them. Yet, the cost, complexity, and unfamiliarity of games makes them seem risky, expensive, and confusing. By putting games up front, gamification perks up the ears of its potential customers, offering a pique that stands out in a noisy environment of possible business solutions.

The second move tames games, making them safe and predictable: the "-ification" suffix. As I've previously argued, "-ifying" things "makes applying that medium to any given purpose seem facile and automatic" (Bogost 2011c). To beautify or falsify, for example, assumes that simple processes or devices can be put to use to render the ugly attractive or the counterfeit authentic: "-ification" assures the customer that the process is easy and achievable. At the same time, it allows the gamification provider to take a page from Matthew Stewart's consultancy handbook, recycling similar solutions for any situation.

When combined, gamification's sleight of hand is hard to see as illusion, just like Stewart's Whale graph appears customized for a particular circumstance despite really having been copied from the past gig. Gamifying a business process or customer experience says far less about that process, business, customer, or experience than it does about the way two concepts, "game" and "-ification," produce rhetorical force.

In the traditional version of gamification, the process involves the adoption of simple, repeatable, scalable feedback systems such as points, levels, badges, and other rewards. Game designers and critics might lament seeing this particular selection of features assigned to the category of "key game mechanics" (Zichermann 2011), but in their effrontery they fail to notice that the gamifier's idea of centrality is different from the designer's. Game designers naively interpret "key mechanics" to refer to elements of a game central to the design aesthetics of the medium or to the player's experience. But in the context of "-ifying" games, of performing the consultant's maneuver, being "key" just means being "-ifiable." That is to say, for the gamification advocate, the nature of games is whatever is most easily abstracted, packaged, and sold as business services or software, most easily fashioned into a "solution." Just like Stewart's Whale chart.

Gamification Is the Goal of Gamification

One reason designers, consultants, and businesspeople are confused about gamification is because the term's utility and increasing popularity has helped it expand into areas beyond the usual scope of points, leaderboards, and badges. Where previously someone might have discussed a non-entertainment game by means of its context—an educational game or a training game, for example—now the generic term *gamification* is sometimes used instead. For example, I once saw someone use the word *gamification* to describe the weird and delightful game Vim Adventures, which uses a Legend of Zelda–style adventure motif to help players learn the intricate keyboard shortcuts in the Vim text editor. When used in the "loose and popular" sense, this title counts as gamification because it takes a non-gaming task, namely learning a text editor, and turns that task into a game.

Policing language is a fool's errand, and I'm not interested in suggesting that this usage is "wrong." Rather, it helps illustrate the reasons why it's been hard for some to grasp the complexity of the "gamification is bullshit" thesis. A game like Vim Adventures seems like an interesting and creative way to help someone practice a rote task like learning keyboard shortcuts in a computer program. But such a use of games isn't really gamification at all, it's just a game, one of many possible games that could be created for many possible situations—in this case, for drilling oneself on the operation of a new and confusing piece of software (cf. Bogost 2011b, 141). The fact that it is a one-off experience that had to be designed and developed from scratch helps separate it from the generalized "solutions" offered by gamification consultants.

Gamification is not really a style of game design or a manner of putting games to use. Rather, it's a style of *consulting* that happens to take up games as its solution. Despite appearances, this adoption is not very interested in the possibilities and potential of games as such, no more than the Galweather & Stearn pod is interested in or committed to any of the solutions they provide. They just want the simplest, fastest route to getting customer sign-off and billing for services. The management consulting business relies on a prescribed course of action in order to produce a predictable recommendation given any set

of business data. It is a process of telling a familiar story—often one that's been chosen in advance—and molding that story to fit the individual circumstances provided.

To someone who might have been casually introduced to Vim Adventures as an example of gamification, the idea that it might be "bullshit" would seem surprising and unlikely. His or her ear would hear "bullshit" in the sense of "nonsense," and it's hard to square such an unlikely, wacky, creative application of games with mere drivel. But this is not what games or bullshit mean in the context of gamification. Bullshit is not mere nonsense, but a name for the process of persuading an audience irrespective of the truth value of the propositions presented. Stewart and his colleagues don't really care if the Whale chart represents the business situation they now propose to solve; rather, they merely hope that it can present a problem convincing enough to demand solution. The bullshit is not found in the product or even the outcome, but in the process of presenting that product or outcome as exactly what it needs to be in order to benefit the agent doing the presenting.

Thus, when a gamification advocate like Gabe Zichermann (2013) speaks about the "hard truths" of the practice in support of its virtues, it doesn't really matter whether individual companies have or have not "succeeded" in their business applications; that is, whether or not the gamification solution in question has "worked." All that matters is that gamification has been *pursued* as a strategy meant to solve a problem, perhaps a problem proposed by the very notion of a gamification solution.

For example, Zichermann cites a bizarrely fractal example of gamification's success: Deloitte Leadership Academy (DLA).[1] By adding badges and leaderboards to the consulting organization's internal training program, which has been in operation since 2008, Deloitte reports having increased weekly return users by 37 percent (Meister 2013). The perversity of a business consultancy deploying gamification on itself notwithstanding, the rationale behind the solution itself already assumes its answer: the fundamentals of DLA are sound, employees just need to be inspired (or coerced) into using it more frequently. Like salespeople (another common gamification target), consultants are intensely competitive, and it's not hard to imagine that a leaderboard set to reset weekly might scratch a compulsion. But it's also possible that *any* sort of novel addition to DLA would have renewed user interest. Leaderboards and badges offered a simple, cheap means to that end, just like Stewart's Whale chart can find a home in any business scenario. A gamification solution simply ends-around any failings in the underlying DLA program; the idea that employees might not wish to pursue a training program because of some defects in the program itself need not be considered. The result might accurately be called "greater engagement," but what good is more engagement with a fundamentally flawed product? For the Deloitte executives who must have invested time, money, and political capital in creating the DLA program, such engagement might be quite valuable indeed. In such a case, gamification offers a simple way to cover over more complex problems, for a time anyway.

The success or failure of the gamification initiative is not of concern to the gamification consultant, if success or failure refers to the accomplishment of fundamental goals. Rather, the goal is to establish a solution that can be applied in many contexts, both increasing the possible customer base for services and increasing the demand for such activities by

forming them into a trend executives cannot ignore if they want to appear up-to-date. The end result of gamification solutions is to justify gamification solutions. And as a business strategy for gamification providers and corporate executives seeking bulletable solutions for PowerPoint presentations, it's a fine one bound to produce results. Just like management consulting.

Gamification Is the Pursuit of More Gamification

In some cases, the consultant's sleight of hand isn't even required: it's easier just to adopt outside work as a credible enough example of gamification techniques. Gamification advocates have proved both adept and willing to embrace the term's increasing ambiguity to their advantage. Even if gamification products and services amount to the simplest possible, iterable, scalable one-size-fits-all solutions, the fact that gamification has increasingly overtaken earlier terms like *serious games* in the (frightfully short-term) public imagination has allowed its advocates to frame more bespoke games as a part of their projects.

Gabe Zichermann's Gamification.co consulting and events organization maintains a news section discussing gamification projects. The group is good at marketing, and they have built an audience eager to see the latest news and examples in the area. While one might be inclined to construe this website as a community advocacy resource, it also serves a strategic purpose for the broader Gamification.co project. By featuring work by others that either resists or deliberately opposes itself to gamification, Gamification.co can nevertheless abscond with some of the value in those projects for its own ends.

In late 2012, I was startled to find one of my own games featured in Gamification.co news. In 2009, my studio Persuasive Games, in collaboration with Traffic Games UK, created Killer Flu, a web-based game about the mutation and spread of seasonal and pandemic flu strains. That was the year of the H1N1 "bird flu" epidemic, and anxiety was almost as high as ignorance about the disease. Our game, which was commissioned by the UK Clinical Virology Network (the British equivalent of the American Centers for Disease Control and Prevention, or CDC), was created to explain the way seasonal flu strains mutate each season and to demonstrate the relative difficulty with which so-called pandemic flus develop and spread over large areas.

Everything about the game is fundamentally different from gamification's usual fare: it was conceived and created for a specific purpose; it was designed and developed from the ground up; it mustered specific subject knowledge rather than general purpose incentives into its design; and it was intended as a tool for public communication and education rather than as a hook for online engagement. In his short discussion of the game for Gamification.co, Ivan Kuo doesn't claim otherwise: "The UK Clinical Virology Network has released the Killer Flu Game as a way of explaining how viral infection works and how it's actually rather difficult to infect everyone in a given area despite mutating strains" (Kuo 2013a). Other non-gamification games have also made the "gamification news," including SimCity EDU, a special edition of the classic city building game for use in science, technology, engineering, and math (STEM) classroom education (Kuo 2013b).

Even though the typical points-and-badges client would likely never pursue such projects, by covering such games Gamification.co can associate itself with works of greater complexity and ambition while also benefiting from providing timely information about the uses of games: Kuo likely linked to my four-year-old game about the flu because a particularly virulent seasonal flu season had increased media coverage of anything related to the illness. Once the attention of those readers is captured, it can be motivated toward other Gamification.co projects. During 2013 and 2014, all the Gamification.co news pages end with a prominent logo, description, and link to that year's GSummit, the main gamification conference put on by Zichermann and crew. In 2013, registration fees ranged from $895 to $2,095, and registrants could preorder Zichermann's then-latest book, *Gamification Revolution*, along with their registration. In 2014, the prices rose to $995 to $2,195. The highest fee also covered Gabe Zichermann's Advanced Gamification Certification Workshop.

Zichermann might argue that the interests of gamifiers have broadened, that gamification is ready to expand from simple points and badges to more sophisticated, "traditional" games. After all, they even managed to get game designer Will Wright (SimCity, The Sims) to keynote the GSummit 2013, and celebrity astrophysicist and education advocate Neil deGrasse Tyson to keynote the 2014 event. The summit's website uses terms such as "engaging design" to suggest this expansion, while perhaps also helping the organization distance itself from—or at least set up a potential hedge around—gamification in the abstract, highly iterable sense.

But a quick view of Gamification.co's "getting started" guide reveals just how much Zichermann and his many intertwined organizations have to gain from colonizing as many related works as possible and claiming them for gamification's empire.[2] Of the seven steps in the gamification getting started guide, five of them involve consuming products from Zichermann and his companies: watching a video of Zichermann speaking (step 1); buying his books (step 4); paying to attend his conference (step 5); paying to attend more workshops at that conference (step 6); and paying to earn his gamification certificate (step 7). Naturally, should you require further or more detailed assistance, Zichermann also provides gamification consulting via his "gamification agency" Dopamine, whose website (http://dopa.mn) used to feature an enormous photo of Zichermann's head on a gray background, one eyebrow petulantly raised.

The core of gamification's bullshit is not abuse or falsehood, but activity "unconstrained by a concern with truth" (Frankfurt 2005, 38). It's easy to get wrapped up in the word *truth* here—for Zichermann and his cronies, pointing to a few examples of businesses who may have benefited from their approaches seems like enough to issue an effective wet cleanup on my critique. One response might argue that such successes are really tautological: like the Deloitte Learning Academy, they are successful only when framed in the terms already given for their adoption. Part (perhaps most) of the purpose of a "gamification strategy" is to satisfy a company or a division's desire to be able to claim that it has a gamification strategy. Corporations have to keep up with the Joneses too.

But Frankfurt provides us with tools for a more effective response. It's not just that bullshit is unconcerned with truth, but that it is "not germane to the enterprise of describing reality" (Frankfurt 2005, 30). Sometimes this divorce is undertaken mindlessly, its ideas concocted from nothing. But other times, the practice of describing or influencing reality is

secondary or even irrelevant. Gamification lives in the latter neighborhood. For the gamification industry, projects and their results are essentially immaterial. The main goal of the movement has nothing to do with business problems such as engagement and loyalty, but merely with the establishment and continuance of the practice itself. The real reality of gamification is just not relevant for its consultants and service providers. All that matters is making those services a required feature of an organizational strategy, such that consultants can sell the latest insights surrounding them.

In this respect, one must admit that gamifiers are not alone: innumerable trends in technology, business, and culture have no particular concern for their attachments. For example, the free software movement's values have recently been transmuted into the more abstract logic of "open source" by for-profit companies (Morozov 2013), a practice some have called "openwashing" (Finley 2011). Organizations such as Google celebrate "openness" as a value, yet they ensure that the products and services they produce return relatively little into the public sphere while still using their appearance of openness as a marketing and publicity vehicle. Of course, just because gamification is not alone in producing a rhetorical veneer atop a particleboard philosophy does not mean one should celebrate such bullshittery as acceptable because it is widespread.

Gamification without Games

I've previously suggested the term *exploitationware* (Bogost 2011c) as an alternative for *gamification* because such a term more adequately characterizes its proponents' goals. It's a term that strips "games" from the situation entirely and focuses instead on the consultant's gambit and his or her enterprise customer's desire to extract value in the form of meaningless engagement. Even its proponents have begun discussing the practice as a combination of "game design, loyalty, and behavioral economics," partly admitting that gamification has little in common with game design and development.[3] While we cannot police the adoption of one field in the service of another, we still ought not overlook the contortion of such disciplines. The best game design practices create playful experiences meant to produce gratification. The best loyalty programs offer a two-way conversation between a company or organization and its customers. The best behavioral economic techniques study and establish economic models for rationality. In gamification, all of these inspirations become contorted techniques for producing compliance. Given the brazenness of gamification's proponents, I suppose I should have expected that a derogatory alternative like *exploitationware* could be so easily re-incorporated into the gamification machinery. Like the best sociopaths, gamifiers are not even capable of being ashamed of their activities.

For now, gamification's groupies have not fully acknowledged that their strategies amount to a distorted version of behavioral economics, one dressed up as gaming in order to appear cooler and more appealing to potential sector customers. Until then, we can look for cracks in the fundamental idea that gamification draws its lessons from game design

Table 2.1
Major characteristics of games, according to a 2012 Bunchball white paper

Performance	Real-time feedback	Transparency	Goal setting
Achievement	Badges	Leveling up	On-boarding and mastery
Social interaction	Competition	Teams	

practices in the first place, rather than from more general and ordinary practices common to business operations for decades.

In 2012, Bunchball issued a white paper titled "Enterprise Gamification: The Gen Y Factor." It's a useful candidate for analysis for a few reasons. For one part, Bunchball was founded in 2005 as a social gaming company and transitioned into gamification services in 2010, at the very start of the trend's rise. For another part, the company has maintained a more modest public persona than Zichermann's various organizations, offering a contrast to the trend's typical hucksterism. Bunchball founder Rajat Paharia offers a tempered and deliberate approach to publicity and sales, not the least of which is resisting the temptation to plaster his name and likeness all over his company's website and marketing materials. And for another part, Bunchball's white paper is currently offered as one of the seven "getting started" steps in Gamification.co's online guide, so the company and its ideas offer a different resource that is nevertheless considered a part of the gamification "canon."

In the white paper, Bunchball (2012, 2) rehearses a familiar argument about Generation Y (otherwise known as the Millennial Generation): they are "digital natives"; they "live and breathe online," they have been playing video games from childhood, and that "a game-like metaphor applies to almost every aspect of their lives." Bunchball singles out three characteristics of games that are particularly gratifying to Gen Y employees: performance, achievement, and social interaction. Within each of these categories, the company identifies a few purported features of games that realize such gratification (table 2.1).

Whether or not these particular features of games or work really motivate Gen Y employees is irrelevant for our discussion. As we've already seen, gamification need not concern itself with truth in order to advance its project. So, rather than asking if activities like real-time feedback, transparency, and leveling up really motivate young workers, let's instead ask: do properties like these have any special relationship with games?

Among the performance-related features of games Bunchball highlights, one is real-time feedback, which the company defines as follows: "In a game, anytime you take an action, you receive instant feedback. Positive feedback reinforces good behavior, strategy and tactics, while negative feedback enables you to learn quickly and adjust" (Bunchball 2012, 3).

Do games really offer instant feedback for actions? It depends on what sort of feedback we have in mind. Certainly, many games offer visual and auditory responses thanks to the fact that they are real-time multimedia artifacts. For example, moving a block in Tetris or directing a unit in Warcraft creates feedback in the form of changes to the display and

alterations in the sounds that emanate from the computer. But nothing about this virtue seems limited to games.

After all, as I write this sentence in Microsoft Word, every tap on my keyboard produces an audible click, signaling to me that the key was depressed successfully. On screen, a character corresponding to the key I pressed appears in the document. The same is true of other standard office software that long predate the Millennial Generation, tools like spreadsheets, calendars, e-mail clients, and databases. Should such actions be considered "positive" and "negative" feedback, akin to that of games? I suppose that when I inadvertently depress an incorrect key, I immediately see the wrong character appear in the body of my text, allowing me to identify and correct it by means of the backspace key. Is this the sort of feedback Bunchball has in mind? Perhaps a more sophisticated sort is necessary. How about this then: as I typed the present sentence, I thought I had typed "present" correctly, but in fact my fingers had become bound up with one another, resulting in the erroneous nonword "presnt" instead. Word dutifully underscored the error in its characteristic red squiggle, alerting me to the misspelling, which I was able to correct. Is *this* the sort of feedback Bunchball has in mind? If not, a more complete description would be required. But if so, then wouldn't "word-processorification" be a sufficient explanation of the purported "game characteristic" in question?

Perhaps I'm being ungenerous, and Bunchball really means to suggest that games provide more synthesized representations of player choices and their possible outcomes. For example, Tetris displays a shadow on the tiles at screen bottom, showing the player where the current piece will fall when it is placed and thereby helping him or her visualize an outcome. But such feedback can be found outside of games as much as within them. In fact, one of the common examples of real-time feedback among gamification proponents is the Toyota Prius dashboard statistics system, which displays both quantitative and qualitative information about the driver's style and how it might affect overall fuel economy. Says one critic, "Whenever I start my Toyota Prius, . . . my eyes go straight to the car's average miles per gallon since last fill-up. Driving has become a game" (Pegoraro 2012).

It sure sounds good. But, is it really reasonable to conclude that driving has become a game because one can attempt to optimize fuel economy given calculated feedback? Why choose games over the design of automotive computer data displays as the design inspiration? The Toyota Prius and Nissan Leaf are common examples thanks to their eco-trendiness, but gasoline-only cars have allowed drivers to view average fuel economy information in-dash for years. As such, why not highlight Gen Y's need for "dashification" or some such instead? There's even a history of such efforts, the "business dashboard" trend of the 1980s and 1990s, much of which would seem to prefigure gamification's focus on real-time feedback as a key feature. Bunchball doesn't consider such questions, presumably because it's already decided that games are the answer before posing them.

Nevertheless, let's consider how the company suggests incorporating real-time feedback in the workplace. Given the real-time feedback mechanism that is purportedly a characteristic of games, Bunchball recommends the following lesson: "The current state of painful once-a-year performance reviews isn't going to cut it. Businesses need systems and processes that enable fast and meaningful feedback, accelerating employees' growth and learning." I'm

no expert in management theory, but would any reasonable organization really consider annual performance reviews to be a sufficient feedback mechanism for employees of any age? Isn't the marketplace littered with management and leadership texts encouraging different methods of developing employees and businesses through more frequent and more data-driven methods?

As for Bunchball's suggestion of "systems and processes that enable fast and meaningful feedback," this too is hardly a new challenge facing Gen Y workers and their supervisors. The need to evaluate company, division, or employee performance on a more frequent basis using better data almost seems like an obvious point in today's information economy. In fact, one can trace it back to the 1950s, when IBM researcher Hans Peter Luhn first proposed the concept of "business intelligence" as a way to automate statistical procedures to produce data that, when communicated effectively via "input-output equipment," might be "assembled to accommodate all information problems of an organization" (Luhn 1958, 314). Apart from his old-fashioned diction and prolix delivery, Luhn's summary of the problem in 1958 could easily have made the cut as one of Zichermann's introductory documents for getting started with gamification.

In fact, Luhn's original paper on business intelligence already contains most of the lessons suggested by another of the "game characteristics" Bunchball identifies in its white paper: transparency. "Games," the company suggests, "are statistical nirvana—players can always see exactly where they stand and where everyone else stands. Progress can be tracked and communicated in real-time; both in the moment and over longer time periods" (Bunchball 2012, 3). Businesses, Bunchball concludes, "need the ability to capture [performance] data and make it available to employees in an easily digestible format." This sentiment bears striking similarity to Luhn's motivations for establishing a business intelligence system in the context of IBM and its corporate customers:

> The growth of organizations and increased specialization and divisionalization have created new barriers to the flow of information. There is also a growing need for more prompt decisions at levels of responsibility far below those customary in the past. Undoubtedly the most formidable communications problem is the sheer bulk of information that has to be dealt with. In view of the present growth trends, automation appears to offer the most efficient methods for retrieval and dissemination of this information. (Luhn 1958, 314)

Luhn wrote these words at the start of the industrialization of computing, in the year Intel founder Robert Noyce codesigned the semiconductor. Clearly, Luhn didn't have access to the kind of powerful and widely available inter-networked personal computers we now enjoy. But he was also writing for the Greatest Generation and Silent Generation workers who would later become Gen Y's grandparents—if not their great-grandparents. As for Luhn himself, progenitor of these ideas so similar to the latest trend, he was born in 1896.

The field of business intelligence grew substantially with the expansion of both industrialism and computing in the mid-twentieth century. The availability of client-server terminals in the 1970s and then networked business computers in the 1980s bolstered the growth of data warehousing, data mining, executive information systems (EISs), decision support systems (DSSs), online analytical processing (OLAP), and countless other boring-sounding business back-office practices.

Given the long tradition of business intelligence in the enterprise, why might gamification proponents not connect their efforts to such historical examples in order to make their products and services more palatable and appealing? Partly because nobody wants to talk about OLAP and data mining, partly because selling IT solutions is far more complex than selling marketing, sales, and human resources solutions, and perhaps even partly because the gamification community just doesn't know about that part of industrial history. But most of all: The uncertainty and opportunity presented by a flood of young (and therefore inexpensive) millennials coming of age and looking for work combined with the weird and incomprehensible appeal of games among that population (particularly in the minds of older managers and executives) makes a game-based solution seem both sexier and more relevant to today's business goals. Among those goals: corporate middle and upper managers who have to make themselves and their divisions appear creative and winsome while also keeping up with competitors by ticking the latest trend boxes without giving in to the ennui of meaningless corporatism. No executive wants to attend a conference on "new approaches to business intelligence through smart dashboards." By comparison, a conference on gamification sounds like a trip to Disneyland.

Bullshit Carefully Wrought

Game designers and developers have resisted gamification largely because they perceive it to mistake incidental properties of their medium—points and leaderboards and the like—for the more complex and fundamental activity of designing and playing real-time simulations of complex systems. But such an objection assumes that gamification proponents have any concern whatsoever for the state of games as an art, craft, or business. Overall, gamification's relationship to games isn't just one of exploitation, but also one of total and complete indifference and unconcern. For gamification, games are not a medium capable of producing sophisticated experiences in the service of diverse functions and goals, but merely a convenient rhetorical hook into a state of anxiety in contemporary business.

In that respect, gamification represents an opportunity in the way that any business consulting trend is an opportunity: a way to sell products and services that organizations probably don't need, but which might make them feel better about some part of their operations, present the appearance of innovation and progress to their shareholders, and perhaps occasionally even create actual improvements. As Matthew Stewart (2009, 67) puts it, "Consultants, like their pagan forebears, understand that it is important for their task to envelop their work with an aura of sacred mystery. ... Princes and popes always dress up for the job; so should we." But just as wearing a dark suit and cuff links doesn't make yesterday's consultant more the sage even as it improves his style, so hawking "game features" doesn't make today's consultant any wiser even as it improves his or her cool.

The bullshitter need not lie to talk bullshit; instead, it's sufficient to act in accordance with a desired outcome without any concern for its validity or even its connection to actual factors in the material

world—such as the tenuous relationship between gamification and games, or the similarities between gamification's purported novelty and age-old techniques in business intelligence. But most of all, the bullshitter must not be ashamed. And that unshakeable righteousness is gamification's most effective and most terrifying property. Bullshit may seem like a thing done without thinking, nonsense spouted when caught in a lie or backed into a corner. Some bullshit is like that, to be sure. But other bullshit is "carefully wrought," to use Harry Frankfurt's colorful phrase, fashioned with such rigor and detail over so long a period as to strike us as something credible (Franfurt 2005, 22). But as Frankfurt points out, advertising, public relations, and politics are in fact committed to bullshit in just this way.

The bullshitter, we must remember, is never earnest on the surface. Bullshit always hopes to get away with something (Frankfurt 2005, 23). Gamification proponents don't need to lie about what games are, what uses they might be good for, or how they might be best used in the contexts of business or organizational practice, because the answers to such questions are irrelevant to their projects. Rather, it is sufficient and even more desirable for gamifiers to "bullshit their way through," to establish an entire regime and industry around them and their wares rather than to determine how to accomplish either the goals of businesses or to exploit the capacities of games, "to whatever extent the circumstances require" (Frankfurt 2005, 51). The bullshitter's deception hides his entire enterprise. Gamification is not bullshit because it misinterprets games, nor even because it fails in its purpose. No, gamification is bullshit because that very interpretation and purpose are irrelevant. The only purpose it serves is to advance the current—and likely temporary—reputation and advantage of those who would advance it as a solution.

In that respect, there are two comforts we can take in gamification: first, that anyone who presents it as a solution is likely signaling either their own ignorance about the subject in question or their own complicity in the industry of bullshit surrounding them. It's a bit like learning that someone works in advertising or politics or consulting. And second, that soon enough gamification will be stained the same brown hue as those fields, such that even uttering the word will produce the kind of cautious skepticism and sanctioned revulsion we've come to associate with marketers, business consultants, and politicians.

Notes

1. See http://www.deloitte.la/welcome/.
2. See http://www.gamification.co/getting-started.
3. See, for example, the Gamification.co "getting started" guide previously mentioned, http://www.gamification.co/getting-started/.

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3 GAMIFICATION AS BEHAVIORAL PSYCHOLOGY

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Those who advocate the benefits of a gamified or gameful world often advance a vision of the future in which all life and all work becomes increasingly playful, game-like, and rewarding; a future in which the world's problems can be fixed by mass collaborative game-like activities, which simultaneously stimulate and delight the participants, while also providing useful services to science, charities, and industry (McGonigal 2011; Schell 2010b). Given the current popularity of game playing as a pastime and the success of many early examples of gamification (e.g., von Ahn and Dabbish 2004; Khatib et al. 2011), it is difficult to not get caught up in this excitement. Notably, however, this excitement about gamification does not appear to have been generated by any specific new scientific or technological breakthrough. Indeed, there seems to be very little novel—theoretically or practically, from a sociological, psychological, or design perspective—about the gameful design of products, services, and activities (Deterding et al. 2011).

We argue that in order fully to appreciate gamification as a design tool, it is necessary to understand the spectrum of relevant social and psychological processes acting on both the designer and consumer of such products, services, and activities. In this chapter, we focus on one level of analysis, *observed* behavior, and introduce the field of behavioral psychology as an approach to understanding observed behavior in gamified products. Behavior analysis is a natural science branch of psychology and has been successful in developing principles and procedures for engaging users in a wide range of training programs and engendering behavior change, usually in an educational context (see Cooper, Heron, and Heward 2006). Of the many fields within psychology, behavior analysis has devoted itself to precision in the understanding of, and perhaps more importantly the control of, human behavior. A consideration of the principles generated by behavioral psychologists might be useful in explaining how specific game design elements motivate and maintain user engagement, and knowledge of the principles and processes defined by behavioral psychologists can readily help in the design of more useful and engaging gamified experiences. Given the tremendous strength of empirical grounding, behavioral psychology is a proven, valid, useful, and interesting lens through which we can investigate gamification.

To communicate the contribution of this chapter as clearly as possible, it is important to first provide a definition for the phenomenon of gamification. Deterding et al. (2011) suggest that the unique phenomenon of interest when discussing gamification is

"the use of game design elements in non-game contexts." They identify that *game design elements* are "elements that are found in most (but not necessarily all) games, readily associated with games, and found to play a significant role in game play." Examples of game design elements are provided, which vary in terms of abstraction from the concrete (interface design patterns) to the abstract (game design methods). We specifically emphasize how the effects of characteristic game design elements (i.e., points, badges, leaderboards, time constraints, clear goals, challenge) can be explained through principles of behavior investigated and understood by behavioral psychologists for decades (see Skinner 1974).

A Gameful Life

Arguably, one of the catalysts for the current interest in gamification was a keynote speech by Jesse Schell (2010b). In his talk, he outlined a future where game mechanics are totally intertwined with our daily lives. Players receive game rewards for brushing their teeth, using public transport, eating certain branded foods, and so on. In a similar vein, Jane McGonigal (2011) argues that through careful use of game design elements, people can become motivated to solve real-world problems, do more work, and better manage their health (e.g., SuperBetter.com). The core idea is that through modifying the environment and giving suitably motivating rewards, the behavior of players can be changed for their own benefit (or that of their corporate masters).

Notably, in 1948 the behavioral psychologist B. F. Skinner released a science fiction novel called *Walden Two*, which tells the tale of a utopian community whose members live together bound by a strict set of rules defining how tasks are completed and rewards granted in such a way to encourage positive behavior change and maximize motivation. For example, working less desirable jobs earns more "labour-credits," which means those workers get more free time for leisure. *Walden Two* acts as an argument for how the principles of behavioral psychology can be used to help people become better motivated, productive, and healthy. Fascinatingly, the argument put forward by Skinner is strikingly similar to that advanced by McGonigal and Schell.

It seems that in order to design the type of world envisioned by McGonigal and Schell, what is needed is a deeper understanding not only of games and play but also of the processes through which it is possible to incentivize people to behave in an appropriate or productive manner. We need to understand how to measure, understand, predict, and control people's behavior. These are exactly the questions that behavioral psychologists have tried to answer through their research. The remaining sections of this chapter will introduce and discuss behaviorism as a philosophical approach to understanding the gamified world. This philosophy is the foundation for the practice of behavior analysis, and some interesting points of overlap between this philosophy and the assumptions underlying gamification will be considered. Subsequently, the very effective behavior control principles developed by behavioral psychologists will be reviewed. Many of these can be readily implemented in gamified products and services and are already being applied by researchers in the field of persuasive technology (Fogg 2002). We will provide

an analysis of game playing from a behavioral psychology perspective and will conclude by offering some broad concerns and criticisms often associated with the behavioral approach.

The History and Philosophy of Behavioral Science

Behaviorism is an approach to psychology that attempts to understand all behavior, and all psychological events, in terms of the interactions of an organism in and with its environment (Hayes 1993). The work of a behavioral psychologist lies in investigating which specific features of the environment lead to particular behaviors of interest and in understanding how to replicate and control those behaviors through control of the environment (Catania 1998). This approach is steadfast in its adherence to environmental explanations of behavior because these lend themselves most readily to the development of means of controlling behavior (i.e., by manipulating the environment appropriately). Behaviorists eschew explanations of behavior in terms of free will or cognitive activity (e.g., decisions, intentions, etc.) because (a) these processes cannot be easily manipulated for the purposes of behavior control, (b) they usually constitute hypothesized rather than observable processes, and (c) as aspects of human activity, they must themselves be explained in terms of organism-environment interactions. For instance, we might explain the behavior of interacting with a vending machine in terms of a history of successful acquisition of candy bars upon the insertion of cash, as well as in terms of the current physiologic state of the individual engaged in the behavior (e.g., the person is food deprived). That is, given the individual's history of being naturally rewarded (i.e., reinforcement) by the delivery of food for inserting money correctly into a vending machine, and given that the individual has cash in his or her pocket, and given that the individual has not eaten for some time, he or she is likely to put some money in the appropriate vending machine slot. Put another way, the person's history of reinforcement is coming into contact with the current environment, and it is this history combined with the previously established functions of the various stimuli present (the vending machine, money) that explains the behavior (i.e., predicts and controls it). Of course, explanations of specific behaviors are usually more complex than this, but this example serves merely to outline the form that behavioral explanations typically take.

Importantly, a behavior analyst would not explain the behavior of buying candy from a vending machine in terms of the hunger or the intention of the individual, as we so often do in commonsense reasoning and in softer branches of psychology (as well as much of cognitive psychology). That is, it is not acceptably scientifically rigorous to use, in explanation of behavior, a hypothesized internal and private state (i.e., hunger), the only proof for which is the very behavior it is supposed to explain (i.e., the candy purchase). More specifically, we can control the history and state of food deprivation of the individual, but we cannot *directly* control his or her level of hunger. We can also control the individual's history of reinforcement and so increase the individual's efficiency and frequency of using vending machines, but we cannot *directly* control his or her intentions to do so. In summary, behavioral psychology is utterly

non-esoteric and is supremely pragmatic in its approach. To this extent, it dovetails well with the purposes of an engineer whose goal is to increase user engagement with a product or system, rather than merely understand it in hypothetical terms.

Notably, the constraints that have led behavioral psychologists to adopt their unique approach are remarkably similar to those operating on any technologically mediated system that attempts to modify human behavior, such as is often the goal of gamified technologies. Specifically, technology is good at objective measurement, analyzing patterns, and determining solutions based on executable functions. It is not good at intuitively inferring states of mind, thoughts, feelings, and emotions and effecting behavior change outcomes by nonempirical means. Essentially, a computer that is attempting to modify behavior, such as improving the frequency of reading or exercise, is operating under the same constraints as a behavioral psychologist, except that in the case of technology, these constraints are technologically rather than philosophically imposed. Therefore, the tools developed by behavioral psychologists to understand and control behavior are very relevant for anyone using technology to monitor and change human behavior.

In summary, the insistence on observation in attempting to change or maintain the actual behavior of individuals is what is unique and useful about behavioral psychology. We would suggest that it is also useful to take this approach when designing a game, gamified service, or, indeed, persuasive technology (Fogg 2002). Focusing on the actual observed behavior of a person, rather than some presumed inner state or intentions or some other common-sense-influenced model of behavior, will lead the engineer closer to finding means of maintaining the engagement of that person with his or her task.

B. F. Skinner and Radical Behaviorism

The basic science of behavior analysis is heavily indebted to the work of B. F. Skinner (i.e., Skinner 1953, 1959, 1974) and his contemporaries, who experimentally studied the behavior of animals such as rats and pigeons. A typical study by Skinner or his contemporaries involved an animal being placed into a specially designed box containing a lever and a food dispenser (often referred to by popular media as a "Skinner box"). The experimenter set up a contingency whereby the delivery of rewards or punishments was dependent upon either a fixed amount of time or some specific response (such as a lever press) produced by the animal. An experimenter kept a record of the behavior of the animal using a cumulative recording device (Skinner 1959) while the animal interacted with its environment. Numerous such studies were conducted, leading to considerable success in defining the now well-understood "principles" of behavior, such as operant conditioning.

Operant conditioning could be described as a set of circumstances in which the "consequences of behaviour may 'feed back' into the organism, and, when they do so, they may change the probability that the behaviour which produced them will occur again" (Skinner 1953, 59). For example, a rat may engage in many different behaviors while trapped in a cage. If one of these behaviors, such as pressing a lever, is followed by a favorable consequence, such as

the delivery of food, the probability of this behavior occurring in the future will have been altered (in this case, we may expect that lever-pressing will be more likely in the future). The consequences of behavior have a direct and measurable impact upon the likelihood of that behavior occurring again.

Skinner specified the operant behavioral unit as a three-term contingency consisting of an *antecedent*, a *response*, and a *consequence*. He suggested that this concept could be used to describe all behaviors of organisms and that understanding the determinants of behavior simply required understanding the specific antecedents and consequences operating on that organism in that context. In an operant response, both an antecedent, such as an environmental context or stimulus, and a consequence combine to produce behavior (Skinner 1953, 65). Importantly, this definition allowed for a systematic program of research examining various antecedent and consequent conditions as determinants of behavior. This program of research led to technical, mathematical definitions of terms such as *positive* and *negative reinforcement*, *punishment*, and *avoidance*.

Positive reinforcement describes a situation in which the presentation of a stimulus as a consequence of an instance of behavior makes that behavior more likely to occur in that context in the future. There are countless examples of positive reinforcement contingencies implemented in gamified applications. For example, the rewarding of points, badges, leveling up, and access to new features as a consequence of appropriate behavior are all examples of this process.

Negative reinforcement describes a situation in which the removal or termination of an existing stimulus (or existing aversive condition) as a consequence of an instance of behavior makes that behavior more likely to occur in that context in the future. The game Farmville (Zynga 2009) provides an example of how this process can be implemented. In this game, crops must be harvested within a certain period or else they die. The reward for visiting and tending to your farm is the removal of impending crop death, a consequence of your negligence.

Negative punishment describes a situation in which the removal or termination of a stimulus as a consequence of an instance of behavior makes that behavior less likely to occur. *Positive punishment* describes a situation in which the presentation or addition of a stimulus as a consequence of an instance of behavior makes that behavior less likely to occur in that context in the future. These *aversive* contingencies (situations that people will work to avoid) are used less often in gamified applications because of the fear that they will lead to disengagement with the product. Specifically, if the consequence of eating a chocolate cake will be a disapproving message from a phone application, the easiest way of avoiding that feedback is to stop using the application rather than to change your eating behavior (Kirman et al. 2010). However, because games often use aversive consequences as key mechanics in game play, we should expect to see the prevalence of such aversive contingencies increasing in gamified products (Foster et al. 2011).

To apply reinforcement techniques successfully, it is crucial to understand the difference between a "reward" and a reinforcer. A reward is any stimulus given to a user on *the assumption* that it will increase the likelihood of the consequent behavior being repeated in the future. A reinforcer is any stimulus that *has been observed* to increase the likelihood of the rewarded behavior being repeated in the future. Crucially, the technical definitions provided earlier were defined based on careful observation and analysis of the single subjects (i.e., one animal, one person).

They do not refer to assumptions or to typical effects across a group of participants. The concept of reinforcement, therefore, is not theoretical, and the parameters of the various processes involved are well understood empirically. Essentially, if we have not carefully observed and measured behavior, we cannot describe the specific consequences that one should provide as a reinforcer, punisher, and so forth, in order to alter the rate and probability of that behavior in the future. However, if we have carefully measured and recorded behavior, as the creators of many persuasive gamified products do (e.g., healthmonth.com, Nike+), reinforcement techniques can be very effectively applied in the control of behavior rates.

Importantly, behavioral psychology is a form of selectionist analysis, entirely coherent with evolutionary biology (see Hayes and Long 2013). Evolutionary theory describes how behaviors and traits are selected by the environment across generations. Behavioral psychology, in contrast, explains how behaviors and traits are selected by the environment within the life span of the organism. In both cases, the analytic unit (the species or the behavior) is selected by the consequences of its occurrence.

The general principles of behavior are at work in many gamified products, even where product developers are not fully aware of the fact. Indeed, those game design characteristics used to "gamify" products can often be described using the concept of reinforcement alone. In many cases, there is no need to refer to game play at all as an explanation of what a "gamified" service does to engage users. For example, there are many popular exercise applications, such as Nike+ (http://nikeplus.nike.com/plus) and Fitocracy (fitocracy.com), which use simple positive reinforcement contingencies to encourage exercise through the awarding of points, badges, and progression through levels in exchange for observed activity, such as completion and regularity of runs and other workout sessions. In these instances, badges are presented as the consequence of observed behavior on the assumption that the "earning" of that badge will make exercise more likely in the future. While these features are commonly seen in games, they have little to do with the concept of play. Other systems use forms of positive reinforcement to encourage healthier eating (https://foodzy.com/) and language learning (https://www.duolingo.com).

Scheduling Feedback

The use of consequences, whether real (e.g., points redeemable for credit) or virtual (e.g., points with social value in terms of comparing oneself favorably with others), is a crucial and central aspect of any program of behavior maintenance. But it is not simply a matter of providing encouraging feedback or points. The real science of behavior analysis lies in the scheduling of these behavior consequences.

Through experimental investigation of operant conditioning, behavioral psychologists discovered that there are significant temporal and contextual components that affect how the environment is responded to by an organism (Ferster et al. 1957). For example, the effect that any one stimulus, presented as a consequence of behavior (i.e., feedback), will have on subsequent behavior is determined primarily by the history of that organism encountering that

stimulus previously, rather than any inherent feature of the stimulus itself (an exception to this are those stimuli that humans have unconditioned, genetically determined responses to, such as painful stimuli). Essentially, the power of any stimulus to function as a reward or punishment changes over time through experience (i.e., learning). Skinner and his colleagues specifically investigated how that process occurred and how to manipulate these factors in order to predict accurately and control subsequent behavior. Because the use of game design elements in non-game contexts is typically carried out to encourage, provoke, or maintain specific behaviors, these processes are important to understand in the design of gamified products and services.

Behavioral psychologists use the term *schedule of reinforcement* to describe important contextual aspects that define the organism's experience of reinforcement. Specifically, two variables were identified as significant: the *interval*, or amount of time that has passed since the last instance of reinforcement, and the *ratio*, or the amount of work that it takes to earn a reinforcer (Ferster et al. 1957). Researchers found that varying either of these had significant impact on behavior (figure 3.1). Four different configurations produce different patterns of responses in animals engaging with a lever that can be pressed to earn food pellets.

A fixed interval (FI) is a schedule in which only the first response after a specified amount of time has elapsed is rewarded, while premature responses are not reinforced at all. This schedule results in a pattern of behavior in which most behavior occurs in the minutes before reinforcement is expected and behavior rates reduce rapidly immediately afterward, until the end of the interval. Overall behavioral engagement under FI schedules is low. *Variable interval* (VI) schedules are similar to FI schedules, with the exception that the time for which reinforcement is unavailable oscillates around a mean, rather than being predictable. This schedule results in a steady but relatively low rate of response. *Fixed ratio* (FR) schedules deliver reinforcement after every *n*th response. For example, FR5 schedules provide reinforcement consistently after every fifth response. This schedule produces a high, steady rate of responding with a brief pause after the delivery of the reinforcer. *Variable ratio* (VR) schedules are similar to FR schedules, with the exception that rather than being predictable, the number of responses required for reinforcement oscillates around a mean. This type of schedule creates a high and steady rate of responding and is typically the most economical; a lot of work can be generated by few instances of reinforcement.

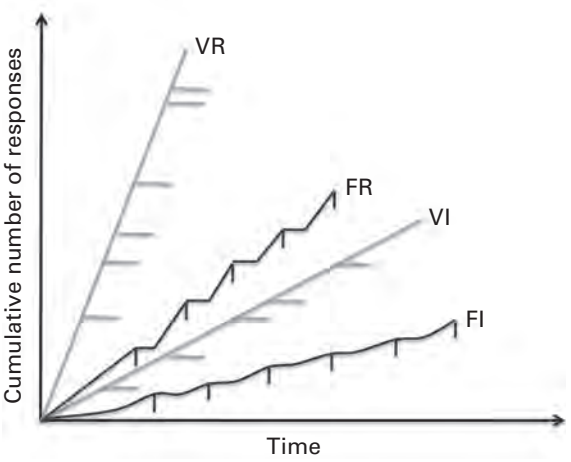


Figure 3.1
Illustration of prototypical behavior observed across four different schedules of reinforcement. FI, fixed interval; VI, variable interval; FR, fixed ratio; VR, variable ratio.

Variable ratios have been the source of much research and controversy. Because the work produced in response to a VR schedule is so out of

proportion to the rewards offered, implementing this schedule in a game, gamified service, or work environment can be seen as exploitative. Indeed, work practices in which pay can be varied at will by employers (e.g., piecework) are very much frowned upon if not illegal in some jurisdictions. VR schedules have been advanced as an explanation for addiction to gambling, as both demonstrate evidence of unrealistic expectations of reinforcement for the actions taken (Haw 2008; King, Delfabbro, and Griffiths 2010). Indeed, Karlsen (2011) has extended this analysis to explain addiction in massively multiplayer online games. Needless to say, the implementation of VR schedules in gamified services, while an extremely effective strategy for motivating engagement, will certainly draw criticism on grounds of exploitation.

The research conducted on schedules of reinforcement demonstrates that it is not usually optimal to offer a reward or punishment after every action that a user takes. Rather, in order to sustain behavior over a period of time, it is necessary to manipulate either the number of responses required or the time elapsed before reinforcement is delivered (Ferster et al. 1957). Different schedules are appropriate in different contexts, depending on the type of behavior one wishes to engender in the user. Indeed, the literature (see Catania 1998; Cooper, Heron, and Heward 2006) suggests that if people have a consistent history of being reinforced for their efforts, the workload required to reach those same rewards can be increased gradually over time without losing the motivational effects of those rewards. Behavioral psychologists refer to this technique of progressively spacing out the delivery of reinforcement as *schedule leaning*. This technique is also observed in computer games, in which the first few tasks that a player completes are often reinforced through new items, new skills, and leveling up. As the player progresses and spends more time playing the game, the number of actions needed to produce a reinforcer is increased.

The technique of schedule leaning is evident in online social network games (Deterding et al. 2010) and particularly in massively multiplayer online role-playing games, such as World of Warcraft (Blizzard Entertainment 2004). For instance, let us consider the archetypal example of the popular role-playing game Dungeons and Dragons (Gygax and Arneson 2000). In this game, players gain "experience points" (XPs) through play. After gaining a certain number of XPs, a character moves up a "level" and gains additional strengths and abilities. However, the number of XPs required to level up increases with each level that is completed, progressively increasing the time taken to complete each subsequent level (the completion of levels and leveling up is presented as a reward in such games; table 3.1). Given the prevalence of this technique in games across media, we should consider it as a characteristic game design element that has itself been selected by its consequences for game and other product developers, but

Table 3.1
Learning schedule of reinforcement in terms of character levels in Dungeons and Dragons (third edition)

Character Level	Experience Points (XPs) Required
1	0
2	1,000
3	3,000
4	6,000
5	10,000

Source: Gygax and Arneson (2000).

in the absence of a dedicated behavioral analysis to guide such developments.

Notably, commentators (e.g., Bartle 2011) have criticized the use of techniques such as variable reinforcement ratios and schedule leaning in games, suggesting that they are in fact so effective when used properly that they are exploitative of users. Critics suggest that for some players, the schedules even remain effective long after the player has ceased having fun or "playing." The player is then seen as engaging in a repetitive, monotonous, menial task, analogous to a low-wage job, rather than a fulfilling, challenging experience. Essentially, game researchers suggest that the use of these game design elements equates to lazy or uninspired game design, and that good games should maintain engagement and motivation through the provision of inherently interesting experiences (Bartle 2011).

Given that game researchers are increasingly uncomfortable with the use of these game design elements within self-contained games, it should not be surprising that concerns have been raised about the application of these techniques to non-game contexts (i.e., Bogost 2011a). It is important to remember, however, that these techniques have been used successfully in special education for decades and have helped to transform the lives of countless individuals suffering with developmental delay and other behavioral problems (see Cooper et al. 2006; Rehfeldt and Barnes-Holmes 2009). Token economies (a specially designed context where appropriate behavior is rewarded through earning of tokens that can be saved up and exchanged for preferred items) provide a particularly clear example of how topographically game-like behavioral interventions can have profoundly positive effects on behavior in even the most challenging environments (Corrigan 1995). Regardless of whether you consider these game design elements as useful tools or potentially exploitative practices, knowledge of the effectiveness of these techniques, as well as the controversies around their use, is essential for the designers of gameful experiences.

Evaluating the Effectiveness of Feedback

To apply schedules of reinforcement successfully, it is crucial that we remain mindful of the difference between a reward and a reinforcer. As stated earlier, a reward is any stimulus (points, badges, etc.) given to a user on *the assumption* that it will increase the likelihood of the rewarded behavior being repeated in the future. However, simply providing people with rewards is of little value unless there is a check to see whether subsequent behavior has changed as a consequence (if not, then the reward was not a good reinforcer). In both experimental and applied settings, behavioral psychologists continually test whether the feedback they offer produces changes in the target behavior (i.e., learning), and consequences are systematically modified "online" in order to achieve the desired behavior rate.

Especially in applied contexts (see Cooper et al. 2006), many different types of rewards are offered, and the psychologist must analyze data to understand better whether consistent patterns of behavior are observed after each reward is presented. This process is necessary as there are very few (if any) stimuli that function as a reinforcer or punisher for all people at all times. For example, some people find

listening to classical music to be the highlight of their week, while many find it boring. Delicacies such as caviar, kokoretsi (organ meat), oysters, and Marmite are often seen as repulsive to different palates (Kirman et al. 2010). Thus, it is necessary to evaluate the impact of different rewards in order to evaluate whether those rewards are ones that the person, as an individual, is motivated to obtain. If the targeted behavior does increase as a consequence of the delivery of a particular reward, then that stimulus can be classified as a reinforcer in that context. The identified reinforcer can then be used in the future as a consequence of behavior that the psychologist wants to reinforce. The same process is applicable in the identification of punishers. In the context of a game, for example, a behavior analyst would offer a wide range of rewards for effective behavior, initially emitted at low rates on a rich schedule (e.g., an FR1), and then "lean" the schedule in tandem with a narrowing of the range of rewards being provided, all the while removing those rewards that do not function as reinforcers.

The above procedure is clearly applicable to both stand-alone games and non-game contexts alike. Taking an example from computer games, Ultima Online (Origin Systems 1997) provides many possible types of behavior for the user to engage with, from crafting to exploring and fighting, all of which provide the possibility of advancing within the game. The completion of a masterwork piece of armor by a player who enjoys crafting is rewarded by the game in the same way as defeating a dragon (i.e., through gained skill points). As a result, the different histories of individual players are catered to, and a wide variety of different individuals can experience the same reinforcing consequences from the same game, even though their behaviors differed markedly.

The clear implication of an analysis of reinforcement effectiveness for gamified products is that there should be different types of rewards available to users, and the application should include some simple way of evaluating which rewards are most reinforcing for each user. Behavioral psychologists have developed precise methods for doing exactly that, and it appears that these techniques may be ideal for use in technology such as computer games or gamified applications. For example, Herrnstein's matching law (Herrnstein 1961) is a mathematical way of determining which contingencies an organism finds most rewarding when multiple options are available. Herrnstein found, in experiments with animals, that the amount of time and work that was devoted to each of the options was consistent with the rate at which that work was rewarded. Essentially, the matching law is a mathematical way of determining which contingencies that individual organism found most rewarding. Understanding the matching law can help game designers create uniquely adaptive and engaging games or gamified products and services, as Herrnstein's algorithm allows us to monitor the relative attractiveness of each of the various reinforcers on offer with a given game. Through continually monitoring a player's behavior, a system can automatically calculate which rewards are eliciting the most work from that person (i.e., which are most reinforcing).

In an application such as foursquare (foursquare.com), for example, through monitoring user behavior, the application could easily identify that a given user is twice as likely to check in at further locations that day after receiving a large check-in bonus than after receiving a badge. Using the matching law, the application could deduce that check-in bonuses are

twice as reinforcing for that user as badges. Thus, using this simple strategy, it is possible for a gamified application to evaluate dynamically the reinforcing strength of each of the available game rewards for each individual player and make adjustments to its own reward system to exploit these data.

Beyond the Skinner Box

The Skinner box apparatus is one of the most frequently misrepresented aspects of discussions on games and gamification. The Skinner box was a simple apparatus designed to observe and measure the behavior of animals (we should not expect it to provide entertainment for humans). However, this does not mean that the behavioral processes that behavioral psychologists discovered through this apparatus are not generalizable to more complex behavior. For example, a reinforcer was defined experimentally as any consequence that improves the likelihood of a behavior being repeated. This can be anything from a drop of sugar solution to a more complex consequence such as the resolution of a particularly surprising or opaque narrative arc, a particularly "juicy" cut scene, or other in-game event. What is genuinely useful is the insistence that the observation of behavior is key in understanding, predicting, and controlling it. Notably, Bogost (2011b) wrote:

> Game mechanics are the operational parts of games that produce an experience of interest, enlightenment, terror, fascination, hope, or any number of other sensations. Points and levels and the like are mere gestures that provide structure and measure progress within such a system.

If the contribution of behavioral psychology to our understanding of gamification is that it demonstrates how to provide structure and context for behavior, to observe, measure, and incentivize progress (i.e., learning), then it provides useful explanatory power for the phenomenon of gamification.

Understanding Game Playing

There are many ways of defining and analyzing game playing—in terms of physiology (Nacke, Grimshaw and Lindley 2010), social behavior (Kirman and Lawson 2009), immersion or flow (Ijsselsteijn et al. 2007), uses and gratifications (Sherry et al. 2006), and many other methodologies. The current analysis does not reject or overlook those definitions. Rather, an alternative is advanced as a means of explaining how the often-complex behavior observed in game playing can be understood in the context of the experimental findings of behavioral psychology research.

The behavior of computer game playing was first subjected to a basic behavior analysis in book-length format by Loftus and Loftus (1983). The authors proposed a technical account of the type of game playing seen in early 1980s video games, using basic experimentally defined behavioral psychology principles such as operant conditioning, extinction (the reduction in behavior rates induced by the removal of reinforcement), and schedules of reinforcement (described earlier). The authors drew a direct comparison between a person playing the popular arcade

game Pac-Man (Namco 1980) and a rat in one of B. F. Skinner's classic experiments (a comparison repeated much since). Loftus and Loftus's account is both rigorous and interesting and appeared valid at the time, when success at contemporary games was based primarily on reaction times and reflexes. Of course, modern games provide more complex and interesting challenges to players than those analyzed by Loftus and Loftus. However, similar low-level behavioral processes may provide some insight into players' engagement with some aspects of more complex games. For example, the variable difficulty levels available in most modern games may be seen as a method of adapting the schedules of reinforcement inherent in a game in order to produce the most game-playing behavior in the user.

Modern games often involve problem solving in addition to, or in place of, fluid stereotyped responses to a limited number of stimuli. For example, popular (4X) strategy games such as Civilization V (Firaxis Games 2010) and Eclipse (Tahkokallio 2011) require a player not only to fight battles with multiple units of different characteristics but also to build economies, military bases, towns, cities, and empires. Many valid strategies can be adopted for pursuing such goals. B. F. Skinner attempted to explain precisely this type of problem solving using the principles of behavioral psychology. In his book *Science and Human Behavior*, Skinner (1953) describes how even the complex behavior of problem solving could be explained in terms of basic behavioral principles. Consider the following passage:

> A person has a problem when some condition will be reinforcing but he lacks a response that will produce it. ... solving a problem is, however, more than emitting the response which is the solution; it is a matter of taking steps to make that response more probable, usually by changing the environment. (Skinner 1974, 123)

Skinner suggested that, contrary to appearances, the behavior that solves a novel problem is not a brand new behavior or insight, but is simply a novel arrangement of already established behaviors (i.e., "taking steps to make that response more probable"; Skinner 1974, 123). For instance, a child who has been taught to pull a chain on the ceiling to flush a toilet, and also to climb on a step to reach objects, may one day climb on a step to pull a chain that is out of reach. This appears to be a form of insight, but it might be better described non-mentalistically as *response chaining*—the mere coming together of previously established behavioral units. Skinner contends that such a process could be applied to understanding many types of real-world problem solving, and we suggest that this includes those observed in many forms of game play, although more complex forms of problem solving are now understood to be possible (see later).

Notably, Gingold (2005), in explaining the appeal of the game Wario Ware (Nintendo 2003), appeals to a form of response chaining, without naming it such. Specifically, Wario Ware consists of a large number of simple minigames that last approximately five seconds each, grouped according to theme. The player must quickly learn the rules of each minigame to progress. At the end of each level, the skills learned in the preceding minigames must be combined in order to pass a more complex game (i.e., chaining of previously learned simple behaviors). Gingold proposes that the process of gradually learning simple behaviors and combining these as the game progresses explains the appeal of the game. Indeed, this explanation could also apply to the fascinating and

hugely popular modern puzzle games in the Portal (Valve Corporation 2007) series. The structure of these and may other puzzle games, such as World of Goo (2D Boy 2008), seems optimized to take advantage of the human capacity for problem solving in terms of response chaining, which, in the field of game design, has been referred to as *scaffolding*. The simple requirement that responses be chained into long behavioral units provides some explanatory power for the engaging structural properties of simple games. However, as we will now see, game complexity may also extend to include more recently analyzed forms of problem solving than mere response chaining.

Complexity and Challenge in Games

Understanding, investigating, and manipulating the challenge or complexity presented by games is an area in which behavioral psychology may be particularly useful. Specifically, appropriate challenge and complexity are often proposed as an explanation of why a given computer game succeeds in maintaining player attention and enjoyment across the period of game play (i.e., Koster 2005). However, there is no technical definition of game complexity offered in the literature. Without such a definition, how can we know which forms of complexity are most reinforcing for an individual or along which parameters to alter such complexity? In one recent paper, a behavioral definition of complexity was offered in terms of a concept called *derived relational responding* (Linehan, Roche, and Stewart 2010, Linehan 2008). Before we consider the utility of this new definition, we must first consider what we mean by the term *derived relational responding*.

The simplest example of derived relational responding is a psychological phenomenon called stimulus equivalence (Sidman 1971; see also Sidman 1994, 2000). In his research, Murray Sidman showed that once people had been explicitly taught to choose an arbitrary stimulus "B" in the presence of an arbitrary stimulus "A" and also to choose a third arbitrary stimulus "C" in the presence of stimulus "B" (where these stimuli were things like randomly chosen Chinese characters), then a number of untrained responses emerged, including choosing "C" in the presence of "A" and "A" in the presence of "C." It has now been shown conclusively that humans can derive novel stimulus relations between various indirectly related objects or occurrences in the environment, and this phenomenon cannot be accounted for in terms of mere response chaining. In addition, this single psychological ability is considered to be a foundational unit for all human reasoning and logical thought (see Dymond and Roche, 2013).

The concept of derived relations makes both the understanding of and systematic manipulation of complexity in games more amenable. Specifically, there are some types of derived relations that have been demonstrated as more complex than others. For example, *nodal distance* (see Fields et al. 1997, 1990; Arntzen and Holth 1997, 2000) is a means for analyzing the closeness of a relationship between related stimuli. Responding appropriately to directly related stimuli is an observably less complex task than responding to stimuli that are related through a series of *nodes*. Similarly, responding appropriately to stimuli that are the opposite of each other, bigger

or smaller than each other, or different to each other is a measurably more complex task than responding appropriately to stimuli that are the same as each other (see Hayes, Barnes-Holmes, and Roche [2001] for a book-length analysis and discussion). Adopting the technical nomenclature of relational complexity offers game developers an empirical means of creating appropriate challenge levels in a non-haphazard way, based on empirically understood psychological processes, and of manipulating complexity systematically in a linear and stepwise manner across levels. It also provides a paradigm within which to understand the level of challenge presented by currently popular games. For an example of an experimental game built entirely on the concept of derived relations and nodal distance, see Linehan et al. (2010). For an example of an online educational program that uses the relational complexity game element and draws explicitly on reinforcement procedures and schedules of reinforcement, see case study 3.1.

Case Study 3.1

RaiseYourIQ

Summary

RaiseYourIQ is a suite of online cognitive training tools developed by behavior analysts to improve general cognitive functioning. It falls under the general rubric of a brain training system but is offered more as a clinical/educational tool than primarily as a form of entertainment. At their own convenience, users practice (twenty to thirty minutes several times per week) at a series of mental challenges, which take the form of deriving relations of increasing complexity across levels of the training. Each task involves nonsense words, and levels of the training consist of blocks of tasks of similar relational complexity. Extensive training at such tasks is understood to have wide intellectual benefits. The main product offered by RaiseYourIQ is called SMART (strengthening mental abilities with relational training).

Facts and Figures

As a recent startup, RaiseYourIQ has only been online since October 2012 but currently has several thousand registered users.

Gameful Design Elements

- SMART uses explicit reinforcement through audio and visual feedback, points and badges, and optional updating of current point status on social media sites, but on a well worked out schedule that is "leaned" during test stages and during higher levels.
- To optimize learning rates, points and badges are awarded for revising previously completed levels, but on a diminishing rate (i.e., systematic leaning of schedule).
- E-mail reminders are sent to users if a hiatus in training is observed.
- Progress is tracked constantly in terms of speed and accuracy so that levels may be skipped if challenge is too low or stages regressed if challenge is too high. This optimizes challenge level and therefore engagement and learning.
- The use of multiple response consequences increases the likelihood of behavior coming under control of one of these (i.e., a reinforcer).
- A virtual professor provides helpful encouragement on a well worked out schedule that is as lean as possible.

Case Study 3.1
(continued)



Box Figure 3.1
A screenshot from SMART at RaiseYourIQ.com. The task shown is a one-node complexity task involving two types of relation (same and opposite), with responses consequated initially on a FR1 schedule by a wide range of rewards, including audio and visual feedback as well as points and badges.

Issues

Difficulties identifying a sufficiently broad range of reinforcers for a wide range of users online have been noted. Innovative solutions are being sought to broaden this range.

Outcomes

Increases in general intelligence have been reported in published studies (e.g., Cassidy, Roche & Hayes 2011). The creators have also claimed anecdotal evidence of improvements in reading and vocabulary as well as documented maintenance of IQ increases across four years (Roche, Cassidy, and Stewart in press). This is unique among brain training products but not surprising given the uniqueness of the relational training approach and its foundation in behavior analysis.

Related Cases

An increasing number of products claiming to improve general mental ability are available online. Many of these are simply games that should in principle help stimulate brain activity and neurogenesis (growth of brain cell connections). These other cases use commonsense game elements and are devised by game developers rather than psychologists, even where the core purpose of the game was inspired by psychological theory, such as the concept of neurogenesis. The market leader in this regard is Lumosity (lumosity.com).

Further Information

http://RaiseYourIQ.com

Applied Behavior Analysis and Behavior Modification

At the beginning of this chapter, we pointed out that the use of game design elements as a means of engendering engagement in non-game contexts often involves implementation of processes such as highly structured behavior measurement, algorithmic analysis of behavior, feedback loops, and reward mechanisms. This is especially apparent where the intention of the application is to change explicitly the behavior of the user (e.g., healthmonth.com). It is also the focus of much work carried out in the field of persuasive technology design (Fogg 2002). Notably, these processes (measurement of behavior, analysis, and feedback) are also the fundamental building blocks of behavioral interventions—also referred to as behavior modification, or, more recently, applied behavior analysis (Cooper et al. 2006). Applied behavioral psychologists have conducted a wealth of research on the optimal means for implementing these processes in order to motivate engagement and behavioral change. Thus, some knowledge of this field of research may be useful for those designers attempting to gamify their products or services.

Applied behavior analysis (ABA) is an umbrella term for a range of behavioral interventions that build upon the principles discovered by experimental behavioral psychology. These have been used to treat a huge variety of behavioral problems from developmental delays to autistic spectrum disorders (McEachin, Smith, and Lovaas 1993). They are, by definition, evidence-based, individualized interventions. The behavior of each participant is observed, measured, and analyzed, and treatment is driven by evidence of whether improvements are observed or not, and under what conditions those improvements were brought about. The interventions are typically (though not necessarily) delivered in an intensive one-to-one manner.

ABA programs are designed on the assumption that learning is maximized when high-performance targets are set and teaching is focused on the individual. Indeed, unlike in traditional education, the passing criterion in behavioral education is not 40 percent, but typically somewhere around 90 percent. If the learner does not reach this stringent passing criterion, he or she is required to repeat the program until the criterion is reached. This process will be familiar to any player familiar with use of "boss fight" mechanics as a way of testing learned in-game skills.

Indeed, ABA programs have structures that resemble characteristic elements of computer games in many striking ways (for an in-depth discussion, see Linehan et al. 2011). For example, highly engaging games usually share with ABA interventions clearly specified and measurable goals (such as to complete a section of game or level up the character), require a great deal of repetition of skills in order to reach that goal (fighting numerous similar enemies), are often conducted under time constraints, have clearly specified rewards for reaching the specified goal (stronger player/more weapons/access to new levels), and provide consistent feedback from the game state on how successfully the player is performing. In addition, successful games pay a great deal of attention to the rate in which complexity is increased over the course of game levels and to the balance and pacing of player advancement through these levels. These issues of rates, balance, and pacing appear to parallel precisely the process that the behavior analyst undertakes in designing an intervention.

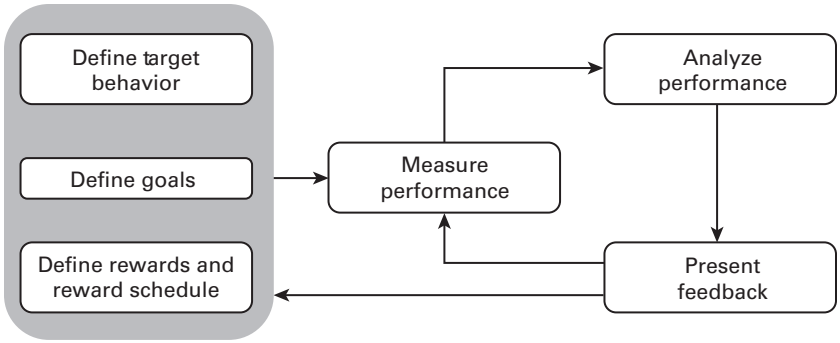


Figure 3.2
Diagrammatic illustration of the key processes involved in any behavioral intervention.

Besides the structural similarities between characteristic game design elements and the processes used for behavior modification, the other reason why this field should be of interest to the designers of gamified products is that there has been a great deal of empirical support for the effectiveness of ABA programs. Indeed, they have been extremely successful wherever implemented, from university modules (Saville et al. 2006) to secondary school (Olympia et al. 1994), primary school (Lindsley 1971, 1992a, 1992b), driver education programs (Bell et al. 1991), and challenging populations (Christopherson and Mortweet 2001). Behavioral teaching methodologies have been particularly successful as early interventions for children diagnosed with autistic spectrum disorders (Lovaas 1987).

ABA Processes as Game Design Elements

In this section, we will take a step-by-step look at some of the processes essential to any form of ABA, briefly explain some issues surrounding the implementation of those processes, and point out how they can be useful in the design of gamified products and services (figure 3.2). Obviously, we do not have space here adequately to summarize decades' worth of work by thousands of researchers. For those who wish for a more detailed account, we recommend Cooper et al. (2006).

Selecting and Defining Target Behaviors

The most basic and important step of any intervention is to define a target behavior clearly. This must be a clearly, objectively observable behavior that it is possible to measure via the technology on which the system is implemented. Whether or not participants have reached a behavioral goal should be judged by observation of that behavior, not by their answers to a questionnaire or other such self-report measure.

ABA programs, like games, break long-term goals (such as running a marathon) into short-term component tasks (the exercises expected each day over the course of training). Participants must demonstrate success at all of these short-term goals as they advance through the program before requiring performance of the more complex skill (asking someone to run a marathon without having completed sufficient training is not likely to have a successful outcome). Thus, the designer must clearly define not only the ultimate goal of the program but also the series of steps that learners must reach on their way to that goal. In this way, a hierarchy of observable behavior measurements is created, in which the most basic concepts and processes are taught first, and knowledge and performance are built methodically.

Measuring Behavior

Measurement refers to the process of assigning numerical values to observed behavior. This must be done in a coherent and meaningful manner so that the system can analyze that behavior and provide feedback. Notably, ABA programs typically measure not only accuracy (whether a target has been met or not) but also temporal aspects of performance (how long it took the person to reach that goal). Behavior analysts have found that measures that include temporal components, known as fluency measures, are a more accurate method for judging the efficiency of behavior than simple measures of accuracy. For example, knowing that someone has completed a five-mile run provides us with a lot less information about their expertise than if we also know whether the run lasted twenty minutes or an hour.

Behavioral psychologists have also found that imposing strict time constraints on behavior is a useful method for ensuring the learner attains expertise. Time constraints are also characteristic game design elements, and, as such, we can expect them to be used in many gamified services in the future.

Recording Data

Closely related to the process of measuring behavior is that of recording those measurements in a manner that is amenable to analysis. Because the dependent measure of all behavioral interventions is the change in behavior over time, applied behavioral psychologists typically use line charts to record and represent data. These charts are called *celeration charts*, as they are designed to represent accelerating and decelerating frequencies of target behaviors. In a gamified service, these data points must be recorded in a way that is easy for the game application to read and analyze. Just *how* the data are presented is open to the creativity of the designer.

It is also essential to decide on what specifically must be recorded. Behavior analysts aim to record every single instance of a target behavior and to plot these on celeration charts. For example, in a spelling exercise, the position of each letter in a word is checked and marked whether it is correct or not. In a gamified healthy eating application (i.e., https://foodzy.com), it is essential to record every meal, snack, and drink consumed in order to understand fully a user's dietary behavior.

Analyzing Behavior Change

The key metric used by behavior analysts in monitoring the success of learners is the change in their behavior over time. Essentially, once a learning outcome has been defined, the behavior analyst continually measures the learner performing that behavior and examines whether or not the learner is approaching that

outcome. Using celeration charts, it is easy for the behavior analyst to understand the trajectory of behavior and to take appropriate action to ensure that appropriate behaviors are promoted and maintained, while inappropriate behaviors are modified or extinguished. If games and game-inspired applications are to automate this process of analysis successfully, they must similarly focus on identifying behavioral trajectory. Specifically, trajectories explain crucial temporal and contextual aspects of behavior that are not available when analyzing behavior in terms of means, individually or in groups. Luckily, analyzing change in behavior is relatively simple, once the preceding steps of defining, measuring, and recording behavior have been carried out in a methodical fashion.

Presenting Feedback

Throughout the chapter, we have discussed feedback in great detail in terms of operant conditioning, scheduling of feedback, and ongoing evaluation of the effectiveness of feedback. Both engaging games and successful ABA programs use these basic processes in combination to ensure that the game is able to provide consistent, appropriate, and specific feedback to the player and to guide the player toward performing at a high skill level. It appears that adopting the following approach is useful in (a) offering a variety of rewards for appropriate performance, (b) offering persistent negative consequences for poor performance, which the player will work to avoid, and (c) directly presenting aversive consequences when the user does something that the service provider does not want him or her to do. Of course, whether any stimulus serves as a reinforcer or an aversive stimulus for any individual should be defined through careful observation of that individual's behavior. Care should also be taken to personalize the schedule on which feedback is presented. Designers who understand and use these processes will have a better chance of promoting and maintaining engagement with their gamified services.

Criticisms of Behavioral Psychology

It is almost a cliché that behavior analysis is not currently as popular as it once was as an approach to psychology, because it apparently failed to provide an adequate account of complex human behavior and in particular an account of language and cognition. Skinner did indeed concentrate most of his attention on animal research and never conducted a single experiment on humans. However, his 1957 text *Verbal Behavior* was an attempt to show how the basic principles of behavior discovered using animal populations would apply in the human case. It is fair to say, however, that it was relatively unsuccessful at that task. Critics like Noam Chomsky (1959) engaged in now legendary attacks on Skinner's position. The Skinnerian approach did not seem adequate to the task of describing behavior other than reflexive or directly trained operant behavior.

Complex Language and Cognition

As it happens, the critics were correct. Humans were more complex than animals, and it was Skinnerians who first came up with the evidence (see Galizio 1979). In particular, it turned out that animals and humans sometimes behave differently under schedules of reinforcement, and the reason had to do with the

ability to follow verbal rules, which sometimes aid schedule learning but sometimes interfere with it (O'Hora and Barnes-Holmes 2001). Later on, it emerged that only humans appear to be able to derive relations (see the earlier section "Complexity and Challenge in Games") between stimuli, irrespective of the level of training supplied to do so (see Hayes et al. [2001] for an in-depth review and discussion). This represented a qualitative as well as quantitative difference in the complexity of animal and human behavior. Moreover, it turned out that the ability to derive relations underlies all forms of human cognitive ability and may even be definitive of human consciousness itself. In the meantime, new approaches to just about every aspect of psychology have been provided by behavior analysts. The approach to therapy has been transformed, and the analysis of creativity, cognition, language, spirituality, personality, and intelligence have all been re-energized (see Dymond and Roche 2013 for a book-length treatment of this issue). In effect, the current analysis is provided within that context and in the full knowledge that human behavioral repertoires are more complex than those of animals. While a detailed discussion of these differences is beyond the scope of the current chapter, it is worth pointing out that an awareness of these differences has made the analysis provided in this chapter possible. More specifically, our analysis of game complexity in terms of derived relations relies on the newly discovered analytical unit of the derived stimulus relations, which is suitable only for human applications and is one that Skinner did not live to see.

Intrinsic Motivation

Another criticism that has been leveled at behavioral psychology, and one which is especially relevant in the case of complex linguistic activities such as game playing, is in how it explains activities that seem motivated by intrinsic or private rewards, rather than extrinsic, observable ones. Because intrinsic motivations are not observable, their explanation would seem to lie outside the explanatory power of behavior analysis. Aside from the possibility that there may be observable physiologic responses (e.g., adrenaline) that can explain some of the appeal of such activities, the modern behavioral explanation suggests that coherence is an important reinforcer for humans. Coherence and sense-making serve as continually available reinforcers for further responding (Hayes et al. 2001). Humans appear to be highly motivated to achieve coherence and make sense in every context, even in the context of solving puzzles or playing games (see Barnes-Holmes et al. 2010). This is yet another shift in modern behavior analysis that is not familiar to the average psychologist. The move toward the explanation of behavior in terms of self-sustaining reinforcement loops is a major step toward explaining many forms of complex behavior, including game playing.

Questions of Values and Control

Behavioral psychology often provokes unease due to its pragmatic focus on understanding and controlling behavior. Specifically, the goal of behavioral psychology is to understand the processes through which any desired change in any observed behavior can be brought about. In the context of designing a society, such as that imagined by Skinner (1948), this raises questions regarding who is designing that society, what values are inherent in that design, and who is judging what constitutes appropriate and inappropriate behavior (i.e., the behaviors that should be

reinforced or punished). Because the gamified world envisioned by McGonigal (2011) and Schell (2010b) is one in which a designer decides on these exact issues, perhaps we should be careful to hold these game designers to account in much the same way that behavioral psychologists have been.

Conclusion

Gamification, the process of using game design elements in non-game contexts, has rapidly emerged as a massively popular tool in the development of online services and applications. Seized by entrepreneurs and businesses as a way of increasing engagement with products, existing game designers and scholars have, unsurprisingly, been vocal about what they perceive as a desecration of their craft. However, both camps fail to understand the true powers of games as tools for learning, within the context of decades of research into the realities of behavioral psychology. Specifically, all games, and all gamified products and services, follow strict patterns of highly structured behavior management, feedback loops, and reward mechanisms in order to effect changes in player behavior. Just as one can beat the boss in battle by applying skills learned through the game or change one's lifestyle through participation in a gamified experience, game design elements have a predictable and measurable effect on one's behavior.

In this chapter, we have reviewed the field of behavioral psychology and described how behavioral processes are commonly implemented in both standalone games and gamification, through the use of characteristic game design elements. Starting from its origin in the early twentieth century, we have described behaviorism and its underlying philosophy as well as the (frequently misunderstood) core principles of operant conditioning, feedback schedules, and evaluation, relating these directly to techniques used in real games and gamified experiences. Building on this, we have explored the realities of complexity in games and the tried and tested approaches of ABA in effecting behavior change in real-world contexts. Finally, we discussed the key components of successful ABA programs in terms of game design. This includes the key processes of defining target behaviors, measuring and recording behavioral data, analyzing behavior change, and presenting appropriate personalized feedback.

Through a more thorough understanding of the principles of behavioral psychology, game designers and gamification professionals can better understand the processes at work when a player is engaged with his or her game, and the potential effects on the player's behavior. With the tools of ABA, designers have the ability to create measurably better-gamified experiences for the benefit of their players.

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POSITION STATEMENT

CONTRALUDICS

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I

Australians, obsessed with "pokies," or electronic gambling machines, spend $A20 billion each year using these devices—wagering almost $A1,000 per Australian per year.[1] Although a harmless thrill for the majority of the public, a small portion of the population finds the devices irresistible, sitting at them for countless hours, until thousands or even tens of thousands of dollars have been gambled away.

Poker machines center their design around rewards delivered randomly but with just enough frequency to keep the brain's own reward system highly activated. Every win produces a squirt of dopamine—the neurotransmitter mediator of pleasure—leaving the player wanting another win, and playing, against all reason, until that reward has been delivered. The vicious cycle of gamification creates the snare that entraps those who possess specific biological susceptibilities.

Every one of us would be similarly susceptible to some form of gamification—virtuous, vicious, or vacuous. We function within reward frameworks: cash in the bank, the esteem of our peers, the next fix. Socially valuable reward frameworks become cultural ideals, while socially corrosive reward frameworks survive only in the shadows. In areas where these two tendencies conflict—as in the case with legalized gambling, where the state receives a reward by participating in a reward framework with negative outcomes—the state becomes the instrument of its own corruption.

The design of reward frameworks has consequences that touch every human being, institution, and fundamental and persistent features of human culture. Never entirely free of them, most of us would not willingly choose to live outside of existing reward frameworks.

Until quite recently, we understood rewards from their effects on behavior. Now that neuroscience has given us the ability to see the linkages between neurochemistry and behavior, illuminating the interior of the reward system, it has become possible to design rewards that conform more precisely to the peculiar tics and peccadilloes of human biology.

That these discoveries occur at precisely the same moment that *gamification* emerges as a discipline seems far from coincidental. Each mirrors the other: gamification is the extrinsic apparatus for the internal reward mechanisms we all share. What once had been the product of trial and error can now be precisely engineered, scanned with radioactive tracers,

sampled, and visualized on PET and fMRI scans. We reduce human beings to a series of inputs, which, if triggered in the correct sequence, produce a particular action.

In this, games show their dual nature as the ground for human liberation—the "magic circle" of Huizinga (1955), where we are at our least constrained and most ourselves—and as having the potential to engender a peculiar slavery, an enjoyable ensnaring of the soul.

This essential tension makes gamification both intensely vital and of central importance to the future of human culture. It is a "technology of control" long sought after by social scientists, a connection between biology, interiority, and civilization that allows all three to be considered as a cohesive whole. Gamification is the argument against free will, stating that all can be seduced, and that man has his price.

How does an utterly gamified civilization differ from the pheromone-driven cultures of the ants and the bees? Is such blatant and overwhelming exteriorization of the self, driven by our increasing knowledge and ability to manipulate the reward system, the inevitable dwell-state of such a pointedly social species? If we are destined to become a global hivemind, is gamification the instrument of that transition?

As we learn more about the specifics of the reward system—both across the species and within a given individual—the snares become more enticing. Culture becomes continually more seductive, each seduction replacing interiority with external linkages that sacrifice intrinsic essence (or, in a nightmare scenario, leverage it) for external desires.

Whether this represents a culmination or corruption of humanity remains an open question. The Nietzschean view that new cultures are inevitably seen as demonic by the cultures they obsolesce certainly plays into this analysis, as does McLuhan's (1967) observation that we shape our tools, and thereafter our tools shape us.

II

I know I'm probably alone here, but I hate award rubbish like this ... easiest way to get me not to participate.

—Twitter user @elelibs[2]

No game works for everyone: make the tool too obvious and we will reject our own manipulation. Let reward fit the individual precisely and all pretense disappears, because all becomes pretense. This is the hole through which the psyche can be spirited away.

Although this vulnerability has always been exposed, it has mostly remained invisible. Gamification has drawn the attentions of culture to it, and this leaves us—both as individuals and as a culture—open to the predations of any who master gamification, binding our own reward systems to their ends.

The gaping wound through which our psyche flows into the arms of another must be closed. We have need of a movement away from gamification, a contraludics, an argument against reward.

Resisting the seductions of the world lies at the essence of monastic practice; self-mortification practices common to many traditions disturb the tranquility of the body and its pursuit of reward, subordinating this drive in favor of evolution toward a higher state of being. Framed this way, contraludics

is an ancient spiritual practice, an act of self-purification.

With the successful trials of primates injected with vaccine to make them immune to the addictive qualities of cocaine (Welsh 2012), we have the first example of an endogenous reward system whose characteristics have been altered through a scientific process. This is the foundation for a modern contraludics, drawing as much from neurochemistry as from psychology, religion, and ethics. Neo-contraludics accepts the reality of the body and its reward systems, condemning no man because of his failures, instead seeing those failures as proof of the need for a well-developed contraludics.

Contraludics works to maintain the gaps in the smooth functioning of power, gaps that Late Capitalism seeks to close through the techniques of reward and seduction. In a continual co-evolution, we learn to seduce even as we learn the dangers of that seduction and how it can be thwarted.

Beyond the neurologic and religious aspects, contraludics articulates a philosophy of design, one that favors difficulty over facility, but draws short of masochism (which would be its own reward). Things are hard, not to give to pain, but to provoke a self-awareness that keeps the individual within himself or herself, coherent and secure.

Such design elements have been transmitted by mystical schools for several thousand years, pointing toward a wealth of examples that can be translated into modern idioms and materials. Where non-attachment informs practice or where the reward can be postponed indefinitely, contraludics can find expression.

None of this implies a joyless existence; rather that our moments of joy should not be contingent upon the agendas of another, rising naturally from authentic experience. We will never be free from what we are, but we can know it, living in an awareness of gamification's effects on us. That would be enough for us to maintain a recognizable humanity in a thoroughly gamified world.

III

When a man cannot choose, he ceases to be a man.
—Anthony Burgess, *A Clockwork Orange*

For well over a decade, it has been obvious that the material world would acquire a playful interiority (Pesce 2000). The confluence of cheap computing and a growing wealth of interactive techniques has meant that objects previously static and lifeless now simulate some of the qualities of living systems. They listen and respond; they react, even seeming to retreat thoughtfully, as if pondering a decision. The complexity of the interiority of most of these artifacts is merely a well-maintained illusion. With few exceptions, the linkages between action and interaction is direct and reflexive. Yet even this seems almost wondrous, as we transition from a world without awareness into an environment almost animistic in its preponderance of interiority.

Children raised in a world of animate objects embed this understanding in their philosophy of the world. Piaget's Constructivism shows us that a child learns about the world through his or her

interactions within it: where the world responds from its interiority, that interiority becomes a foundational aspect of the child's understanding of the world. The world becomes playful, *and every element in it is classed among the playful or the dead*.

In its purest form, playfulness involves the willful denial of goals or even any acknowledgment that such goals exist. It is an end unto itself, its pleasure the exploration of the self and the other. The boundaries of play are those of constraint, where the mind at play encounters forces that seek something other than an effortless enjoyment of the present moment. Play is rewarding precisely because it seeks no reward other than itself.

Games are the end of play; even as they seek to encompass play within the magic circle, games direct play away from itself, toward goals and rewards, thus depriving play of its playfulness. We may play games, but we have never done so playfully. Reward makes play impossible by introducing risk. Where every act has been weighed down with the possibility of loss, play loses its freedom to explore, to mistake, and to renew itself.

Play, therefore, is the essential contraludic act, from which all others can flow. Play playfully refuses the goal, the point, the meaning that would make itself meaningless. Play insists on nothing, experiences neither loss nor gain, does not rank itself or compete, has no rules to obey or violate, awards no points or penalties. Play insists on nothing, and that emptiness of purpose makes it perfect.

Play is the antithesis of control, order, and hierarchy. There is no top anywhere, no head or crown, no power or authority. Play is deterritorialized, unsigned, inchoate, emerging from the autonomous zone within all of us. It stands against gamification, and gamification can only succeed in those zones where we have made play impossible, where the rules and rewards act as the sole motivators of our actions, where the idea of interiority has been abandoned in the mindless pursuit of badges, points, and leaderboard success.

These two have been engaged in a conflict of sorts; the pointlessly ludic moment of play poised against the gainful gamification of the world. One says nothing, the other points to itself as the only thing to be said. These two in turn echo, respectively, the worldview of the child and the socialized adult. The child inhabits a secret inner world of experience and impression, while the socialized adult faces outward to peers and relations. The child is connected only to himself or herself, while the adult connects to the whole human world, finding both capability and constraint within every relation. The child does not think of the future; the adult, consumed with that future, focuses entirely upon it, becoming synonymous with his or her goals.

Gamification well suits a world of socialized adults, but leaves little space within it for the mind of the child—a mind that we all possess, even as it disappears beneath layers of adult relations. Delight is the close cousin to playfulness: a world from which play has been exiled would be an environment mostly free from delight. Reward is not emotion, but a real danger lies in the possibility that we might mistake these two and exchange one for the other.

A gameful world is an intensely problematic and mostly post-human one. Individuals become players-without-playfulness, caught in a ludic feedback toward ever-greater rewards, in thrall to their own dopaminergic reward systems. This is not a world of free will or even the illusion of free will. It is a world of careful control, conscious manipulation, and enjoyable imprisonment. We will do things because

it pleases us to do so, acting reflexively from our neurology. There is an intrinsic emptiness to such acts, reflecting the lack of interiority on the part of the actors.

Some philosophers (Dennett and others) argue against interiority; for these, a gamified world represents the culmination of human understanding. Against this, a child at play with colored balloons, eyes wide with a delight that has no meaning other than presence within the moment of fascination. One is theory, the other reality. Reality is always a better guide to practice than theory. We must therefore approach gamification very carefully, with a full understanding both of its power and its capacity to exteriorize us utterly, and leave us essentially empty.

It seems unlikely that the drive to gamification will slow: this volume itself serves as one proof. If gamification cannot be slowed, it can at least be resisted—through contraludics. In that resistance new forms will emerge that breed the gameful and playful together into new ludic forms that accept the goal, but treat it as meaningless. The magic circle can be broken, and the prisoners freed, but only in play.

Notes

1. See statistics provided by the Australian Parliament: http://www.aph.gov.au/~/media/05%20About%20Parliament/53%20HoR/537%20About%20the%20House%20magazine/45/PDF/Waiting.ashx.

2. See https://twitter.com/elelibs/status/283703350623604736.

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4 GAMIFICATION AND MOTIVATION

C. Scott Rigby

Our Motivation for Gamification: Why Gamification and Why Now?

To best understand the motivational dynamics of gamification, let's begin with a discussion of the motivation *for* gamification: why has gamification exploded in its popularity over the past two to three years, and what specifically is gamification seeking to achieve?

Certainly a key catalyst is the ongoing evolution of technology that has created new "rules of engagement" shifting power away from corporations and organizations and putting it into the hands of each individual consumer. It is happening across virtually every market. I remember when I was growing up and wanted to watch *The Cosby Show*, I needed to remember to turn on my television at a specific time on a specific day and tune in to NBC; failure to do so meant, quite simply, I would miss the show entirely. In other words, I needed to follow the schedule and format decided upon by NBC. This scenario seems quaint today, considering we can watch virtually anything we want, whenever we want, wherever we want. Put differently, significant power over where and when to engage with a program has shifted from the media companies into the hands of each individual consumer.

Similarly, when going to the doctor twenty-five years ago, I was not able to bring with me an Internet printout of all the latest scientific articles on my particular ailment. Instead, I simply waited for my appointment and relied entirely on the doctor's advice and knowledge in order to make my medical decisions. Today, I can pull up the latest published research on my phone in the waiting room, thus empowering me to engage in a more informed way as a partner with my health care providers.

In describing this phenomenon to my colleague Rich Ryan, he remarked that it was truly a "Copernican turn" in our culture: while for years individuals were in orbit around institutions, there's been a fundamental shift of power away from institutions and organizations and into the hands of each individual. This concept is congruent with what Barry Wellman called "networked individualism": the fundamental shift away from social and organizational groupings (both large and small) toward each individual as the center focus (Wellman 2001).

Here we are making a consequential point about motivation related to Wellman's: as the requirements for engagement—such as when and where we watch

a particular television program—have shifted from the organizational level to each individual consumer, the need arises to more deeply understand and motivate behavior at the *level* of the individual versus simply leveraging the power of an organization or social group to structure behavior. The net result of this is that any institution, organization, or company that wants to hold the engagement of the individual must figure out how to reach that specific individual more deeply. Where once organizations could dictate, today they must entice.

Simply put, regardless of one's market or audience, the motivation to find ways to understand what deeply motivates individuals in order to keep them engaged has never been stronger for the institutions, groups, companies, and organizations that once could rely on having significantly more control.

With video games' ability to hold attention for hours on end, they certainly appear like wonderful role models for solving the deep engagement needs of a consumer-empowered world. As such, a key question for those who pursue gamification is how they will apply principles of motivation and behavioral science to achieve deep engagement (and consequently, the goals that result from such engagement). This chapter will review the concepts of motivation and engagement and their relations to gamification.

Gamification's Empirical Challenge: Using Motivational Science to Illuminate the "Black Box" of the User Experience

From the perspective of empirical social science research, gamification currently has a hype problem (Gartner, Inc. 2012a). Many voices are piling pronouncements of value and revolutionary success on a model in which one applies game principles, mechanics, and design to a variety of non-entertainment contexts, with the expected outcome being greater engagement and success. There is a mechanistic feeling to this approach similar to the "black box" models of behavioral science that focus on operant principles of reinforcement and punishment (Schacter, Gilbert, and Wegner 2011), largely assuming that we don't need to know (or perhaps can't know) what is happening inside the experience of the individual—we only need to observe what inputs bring about the desired behaviors. In this kind of framework, what actually happens in the subjective experience of the individual is secondary.

Presumably, the answer is that game mechanics increase engagement because they are *fun*, and that no one can reasonably argue that fun is not related to engagement. Although we'll see that this is far from a proven point, it is implicitly at the heart of arguments that gamification will achieve broad success in areas ranging from cures for adolescent obesity, to deepening learning and education, to simply ensuring customers will buy more polo shirts. All of which, it is assumed, will occur by making the path to these goals "more fun."

Unfortunately, the evidence that simply making things more fun will achieve gamification's goals is thin. Even the empirical evidence of what "fun" is remains notably sparse, particularly considering that it is the central value proposition of many sectors of the entertainment market, the video game industry chief among them.

Expert Opinion versus Empirical Study

There is an important distinction to be drawn between empirical research and "expert opinion" with respect to game studies and theorizing about constructs such as fun, immersion, engagement, and enjoyment. Certainly, game designers regularly discuss and debate the components of fun, and there are numerous models put forth by designers and industry consultants via white papers, books, and conference presentations. Such models hypothesize about the psychology of fun (often in great detail), but despite the critical importance of fun to the game industry overall, these models are rarely tested through accepted standards in the social sciences, such as internal validation of constructs, predictive validation, and replication of findings (American Psychological Association 1985). Simply put, while opinions about the psychology of fun abound, it is important to acknowledge that social science research that puts such models to the test and empirically refines and deepens psychological theories of fun remains sparse.

The lack of such empirically validated models—as well as cautionary tales from other game movements such as social gaming—suggest that gamification should consider trimming its sails until it has more strongly validated maps in its wheelhouse with respect to motivation and engagement. Between 2011 and 2012, the leading social game company lost more than $10 billion in market value in a year (more than 70 percent), despite its emphasis on closely observing the output of the black box (i.e., watching player behavior daily) and dynamically adjusting game features—those "inputs" to the black box—in order to maximize desired outcomes.[1] In the end, this mechanistic approach did not stop players from abandoning the leading social games in droves, strongly supporting the notion that better models of motivation—ones that understand what *sustains* engagement in the long term—are sorely needed.

This idea of not just engaging or capturing the attention of users, but *sustaining* engagement, is a critical implicit goal for gamification. Gamification aspires to have impact in areas such as education, health care, and personal growth, none of which are short-term endeavors. In addressing such meaningful goals, it seems appropriate that gamifiers have an understanding of *how* game techniques, concepts, and mechanics will achieve deeper engagement and behavior change through motivational models that push past expert opinion alone and can be tested, refined, and improved.

Extant research on specific theories of motivation and engagement has validated their potential to be highly relevant to gamification in this regard. The emphasis here is on research most relevant to building lasting motivation, value, and sustained engagement versus short-term approaches that can often "flame out," as recently witnessed in social gaming and other movements that have seen a rapid ascent followed by a precipitous decline.

Unpacking Gamification's "Lexicon": Understanding Motivation, Engagement, and Fun

Although entire books have been written on the topic by designers and consultants (Koster 2004), from a direct research perspective *fun* has been viewed as a difficult construct to define by many psychologists,

as early as Jean Piaget (1962). Twenty years ago, Podilchak (1991) distinguished between "enjoyment" and "fun" in some initial research largely on the basis of fun having a social component. However, the rise of video games over the past two decades and the increasing ability to experience fun subjectively even when playing alone challenges this social requirement for the definition.

Drawing on theories of intrinsic motivation (Csikszentmihalyi 1975; Deci and Ryan 2000), here fun is defined as a positive emotional state of playful enjoyment: a pleasurable and active mental state in which one is intrinsically engaged—meaning that the value of the experience is the experience *itself*, rather than the experience being instrumental in achieving something else (Deci and Ryan 2000). Such stimuli for fun can include other people, games, or even random objects not designed specifically for entertainment (as the paper clip sculpture on my desk can attest). Put differently, fun can be experienced under a broad range of circumstances that are often difficult to predict because fun by its nature is not simply the inevitable output of applying certain techniques (as evidenced by the many unsuccessful game development projects that have been undertaken even by experienced game designers). It is perhaps because fun is so variable and dependent on so many circumstances (both situational and dispositional) that empirical research to define clearly the construct is elusive. That said, many other similarly complex emotions such as love have been tackled by social science (Bersheid 2010), so certainly it's reasonable to assume that empirically tested theories of fun are within reach.

As a start down this path, most scientific research relevant to the experience of fun has generally taken two forms. The first has been largely deconstructive within the domains of cognitive neuropsychology and psychophysiology. Neuropsychology has sought to identify the concomitant "markers" of enjoyment (or fun) experiences. These have noted activation of neurotransmitters (such as dopamine) that are responsible for positive emotions and are found in higher concentrations when one is subjectively having fun playing a video game (Koepp et al. 1998). Cognitive psychology has similarly noted such pleasurable effects when engaging in game-like activities, such as puzzle solving. Psychophysiology has noted that hemispheric lateralization (notably left-brain asymmetry) is associated with positive emotions (Silberman and Weingartner 1986).[2] However, while useful in understanding possible markers of fun, such deconstructions offer little help in defining the *experiential* circumstances and structure of fun.

A second focus in the empirical literature has been on the experience of fun itself and the specific subjective factors that accrue to having more fun. Flow theory (Csikszentmihalyi 1975, 1990) describes how individuals will often experience a deep sense of immersion and fun when presented with optimal challenges that result in a cluster of psychological experiences, including a blurring of the subjective distinction between the player and the activity, as well as feeling a lost sense of time. Indeed, this resonates strongly with many gamers, athletes, and others who have the pleasurable experience that they are "in the zone" when reaching flow states while pursuing a game or activity. However, it is also important to point out that flow states are not synonymous with fun: often, one will feel a sense of fun when in states of exertion and challenge that may not even be experienced as pleasurable.

Self-determination theory (Deci and Ryan 1985, 2000), a core aspect of our own research on the

psychology of games, has repeatedly shown strong relationships to the experience of fun, as well as sustained engagement and lasting behavior change in multiple domains. Specifically, fun has been strongly related to the satisfaction of core psychological needs (competence, autonomy, and relatedness) regardless of personal preference for game genre (Ryan, Rigby, and Przybylski 2006) or even factors thought by many in the industry to be important to satisfying the core gamer audience, such as intense graphics and violence (Przybylski, Rigby, and Ryan 2009). Subsequently, other researchers have replicated these findings, in some cases reporting that despite the plethora of possible inputs to creating the experience of fun, the majority of enjoyment and fun in video games is attributable to basic psychological needs postulated by self-determination theory (Tamborini et al. 2010, 2011).

This is not to say that basic psychological need satisfaction is synonymous with fun either: as with flow theory, basic need satisfaction can occur independently of fun, often occurring during times of challenge or circumstances that are not experienced as playful, despite such experiences being highly engaging, satisfying, and motivating. It is for this reason that we will soon turn to a discussion of engagement and motivation and a discussion of why these constructs may be far more valuable to the end goals of gamification.

To summarize this brief précis of fun as a construct: evidence supports the idea that fun is significantly related to specific experiences afforded by games that successfully satisfy more fundamental (i.e., underlying) psychological needs. However, as noted this is not a one-to-one relationship, and there is more work to be done: fun is an emotional outcome that occurs in a wide variety of contexts involving many personal, interpersonal, and circumstantial variables.

Perhaps most relevant for gamification and the goals it most often seeks to achieve, fun is also not an experience that is strongly related to *sustained* engagement: longitudinal studies have suggested that the experience of fun is not related to players staying interested in, or involved with, a particular game (Rigby 2012). While related to constructs such as flow and basic need satisfaction, reports of fun may also represent a more ephemeral emotional state that can change rapidly as a function of many variables not related to deep engagement or behavior change. For this reason, we turn next to a discussion of other constructs highly relevant to gamification, specifically those of *engagement* and *motivation*, returning to some literature that may help explain why fun—despite being a powerful and valued emotion—may be a suboptimal marker for deep and sustained engagement.

Engagement occurs when a stimuli fully has our attention and focus—when we are "all in" with respect to whatever we are doing—whether interacting actively (as in a game) or passively (as in watching an engrossing film). Engagement implies a connection between the individual and a stimuli and activity that is characterized by attention and interest. Unlike fun, however, engagement is not inherently a positive experience: as I hurry to diffuse a bomb while watching the detonation timer tick downward to zero, I assure you I am quite thoroughly engaged. But truth be told, I'm not enjoying it. And when it's done (assuming I'm successful), you can be sure I'll be happy to disengage and never go through that again. This is an important point to which we will return with respect to taking the "long view" of gamification as a vehicle for *sustained* engagement.

We discussed at the opening of this chapter the Copernican turn concept—a shift in power from groups and organizations to Wellman's concept of networked individuals. It follows that when we speak of engagement with respect to gamification, we are not talking about merely "shouting more loudly" in some way to engage the audience for a moment alone: gamification is about finding ways to build relationships and effect lasting change. As such, within the context of gamification the *quality* of engagement needs to be understood: there are many ways in which we might grab the attention of an individual and engage him or her initially (Kahneman 2011), but many applications of gamification are seeking a much deeper connection—one in which individuals stay engaged in order to learn more deeply or change behavior in a lasting way. For this reason, we will focus on how various gamification techniques might optimally build motivation for sustained engagement and deeper impact.

Motivation is a psychological construct that is a combination of two dimensions: having *energy* to take action and then moving that energy in a specific *direction* (Deci and Ryan 1985). How motivational energy is directed is the easiest to observe and track because it functionally translates into observable behavior. Because technology has gotten quite good at watching what we are doing, there has been a strong emphasis on the directional component of motivation within both the behaviorist black box models and the growing analytics movements in the domain of business intelligence and behavioral economics (e.g., "big data," web analytics). For example, I can see what you are doing as you click website buttons, spend time in my application, and live in the digital universe in increasingly greater detail, even if I don't know *why* you are doing what you are doing (i.e., even if the black box remains opaque). And by sifting through terabytes of such behavioral data, one can try to discern the audience's preferences and react accordingly with new content, targeted messages, and other customized responses.

But no matter how large these data sets are or sophisticated such analyses may be, because this kind of analytic approach is only capturing the *directional* dimension of motivation, it misses the critical dimension of the *energy* that is the source of motivation—and the wide variety in the quality that energy can take. It is this dimension of motivational quality—the true "why" behind the actions we take—that can vary widely in ways that have strong implications for whether or not we will be successful in the sustained engagement and behavioral change we seek in gamification.

Much like engagement, motivational quality (i.e., its particular energy) is not implicitly positive: I may be energized to take action out of a genuine interest and positive experience but may also be motivated for other negative reasons, such as guilt, shame, or fear (as in my bomb-diffusing experience). And lest one thinks that a gamified approach inoculates against such negative forms of motivational energy, it is important to remember that numerous examples exist of entertainment games that strongly evoke negative motivators such as pressures (e.g., Farmville's "withering crops"), as well as some current gamification applications actively including mechanics for user punishment.[3]

To summarize this conceptual review, even though fun, engagement, and motivation are words often used interchangeably in the discussion of gamification, in exploring definitions of these constructs we can see meaningful differences. This argues for the

importance of a more refined road map of how the construct of motivation relates to those of fun and engagement, hopefully empowering a clearer understanding of motivational best practices for gamification that lead to the sustained engagement gamification so often seeks.

Motivation and Engagement

Motivation and engagement are close conceptual cousins, and thus it makes sense that they are so often used interchangeably in the gamification literature. We have noted that motivation has two components (energy and direction), and engagement can be understood as the *directional* expression of motivation: when we direct our energy toward something or someone, engagement is observed. Thus, engagement is not synonymous with motivation but is the behavioral expression or manifestation of a motivated state.

Engagement has been put forth as a central pillar in both the potential of gamification and as a metric of its success.[4] But what remains unexplained to date in most of the discussions of gamification is the underlying energy that *fuels* engagement. Even while engagement as a behavior is observed and measured in today's gamified products and services, the underlying energy—the quality of the "whys" that leads to engagement—remains unknown even among those putting forth early case studies of gamification's success. The truth is that engagement—like fun—can be a fleeting and even negative experience just as easily as it can represent a highly valued and more deeply sustained experience. If we cultivate the former (whether wittingly or not), we are more likely to see engagement (and gamification) fall off the same cliff that has plagued so many games and applications that touted early success.

Thus, truly to understand and successfully "gamify" in ways that achieve deeper value and lasting engagement, we must unpack and understand the energy component of human motivation that is often overlooked. Specifically, what is it that energizes us to take action and engage—superficially and deeply, positively and negatively—and how can we optimize energy to build the deep, sustained engagement gamification seeks to achieve?

A Deeper Look at Motivation: Basic Psychological Needs and the Importance of "Motivational Quality"

Self-determination theory (Deci and Ryan 1985, 2000) suggests that the key to building successful gamification applications largely rests in (a) understanding the energy that fuels our behavior, (b) the various forms such energy can take, and (c) the ways in which we can apply this knowledge in the development of strategies, designs, and techniques to optimize the quality of motivational energy to achieve gamification's goals.

The energy that motivates us can be said to flow from three sources. The first source is our physiological drives—such as those for hunger and thirst. In

Maslow's original hierarchy of needs (Maslow 1943), these would be considered the lower-order needs—those necessary to be satisfied before higher-order needs can be addressed. Such physiologic needs are fundamental and universal, energizing us on a regular basis.

A second source of motivational energy is our emotional states: we direct our actions out of feeling angry, elated, depressed, or, perhaps most apropos to gamification, when we feel energized by fun. Such emotional states are often similar to what researchers have called needs (Vorderer 2009; Tamborini et al. 2011; Huta and Ryan 2010) and include ways in which we seek experiences of pleasure and avoid experiences of pain. Mood management theory (Bryant and Zillmann 1984; Zillmann and Bryant 1985) describes the ways in which media is chosen and used in order to manage mood states, demonstrating how stress might be regulated by choosing a relaxing program or boredom might be alleviated by choosing an "exciting" form of media. As we already touched upon in our discussion of fun, we would postulate that emotional states are different from basic needs: although we may feel like punching that guy who cut the line at the supermarket, unlike our need for food and water, we don't have to take action on that emotion. In fact, even strong emotions can resolve in and of themselves, even without active actions to dissolve them (Goleman 2003). Furthermore, emotional regulation—understanding the more ephemeral nature of emotions and learning the impulse control to experience them without always taking action—is a key element of psychological health (Mischel, Ebbesen, and Zeiss 1972).

The third energy source for our behavior and action is our psychological needs. Like our physiological drives (e.g., hunger, thirst), these are fundamental and universal—they operate naturally without any need for external prompting. They also are implicitly important and valued by each and every one of us: If they are met, we feel deep levels of satisfaction and experiences such as increased happiness, well-being, growth, and even flourishing (Deci and Ryan 2000). If they are thwarted, they will have a negative impact on our well-being and psychological health. Just as lack of food will cause us to atrophy physically, lack of basic psychological need satisfaction will cause us to atrophy psychologically: we will have decreased energy, less happiness, and even an increase in somatic and physical symptoms (Deci and Ryan 2000).

Research into human motivation and emotion through self-determination theory and positive psychology has identified three basic psychological needs that consistently emerge as powerful, universal sources of energy for motivation:

- *Competence* or *mastery* is our fundamental need to feel effective and successful in the moment-to-moment activities of life. It also energizes us toward growth and elaboration of our skills and abilities; seeking out new challenges that enable us to feel successful and grow without being overwhelmed (Ryan and Deci 2000b; Hsu and Lu 2004). It is this need that is most closely related to the experience of "flow" that occurs during a game experience that has a perfect difficulty curve—always providing us with incremental but optimal challenges that stretch us but feel within our reach (Jin 2012). It also is aligned with emerging research on the neuropsychology in areas related to games, such as puzzle solving: areas of the brain associated with positive affect and satisfaction "light up" more readily when insights occur in

which people see new connections between ideas and concepts (Jung-Beeman et al. 2004).

- *Autonomy* is our fundamental need to feel volitional in what we do. We want to determine the path we are on and whenever possible have meaningful opportunities from which we can choose freely. In contrast, we feel constraints to our autonomy when we are controlled by others or by circumstances and are exquisitely sensitive to when such control is being exerted, whether through positive (e.g., manipulative praise, reward contingencies) or negative means. As rewards are currently a key mechanic in games and gamification, we'll turn shortly to a more detailed discussion of the dynamics of rewards—and their potential pitfalls—in undermining autonomy.
- *Relatedness* is our fundamental need to feel supported by others; to feel that "I matter" to others and that they matter to me. We often experience relatedness in a relationship with another when we feel they are supportive of our needs for autonomy and competence. Research in basic need satisfaction has shown that even computer-generated virtual characters (non-player characters) are capable of satisfying our fundamental need for relatedness, if they are scripted to be supportive of a player during game play.

Until gamification approaches can reach through our smartphone and deliver us a delicious sandwich that sates our physiologic needs, it is at the level of psychological needs that the movement has the most promise for impact and success. The preceding précis of these basic psychological needs forms the foundation of a proposed "motivational triad" model for more effective gamification, discussed in more detail shortly.

Motivational Studies and "Fun"

More than fifty years ago, play was postulated to represent an intrinsic aspect of human motivation and a universal element in the development of culture (Huizinga 1955). Much more recently, Yee (2005) was among the first to bring an empirical approach to the study of game motivation. Inspired by constructs for different player types described by Bartle (1996), Yee derived three core components related to player motivation. The *achievement component* measured the desire to gain power, compete against others, and master the mechanics of the game. The *social component* assessed desires to be part of a group effort and form relationships. The *immersion component* tapped the desire to escape real life, role-play, and become involved with the game's narrative. While Yee brought methodological rigor to the question of game motivation, this work was essentially taking a "top down" approach—putting existing game design ideas to a more rigorous test—rather than an a priori scientific approach testing established psychological theory of motivation and emotion as they applied to games.

Rigby (2004) outlined a model for understanding the motivation and enjoyment value of games based on self-determination theory. The model as applied to games (the player experience of need satisfaction, or PENS, model) was subsequently tested for predictive relations with a variety of outcomes that included

a comparison to Yee's model, showing stronger relationships with fun and enjoyment with entertainment games (Ryan, Rigby, and Przybylski 2006). Additional research has similarly found the self-determination theory approach to game motivation to be highly related to game engagement and enjoyment (Tamborini et al. 2010), including games focused on non-entertainment goals such as exercise (Peng et al. 2012). The PENS model is currently in use in dozens of additional empirical studies worldwide, inclusive of multiple genres and platforms. The results of these studies are revealing some notable findings:

- Sustained engagement is significantly predicted by each of the three psychological needs (competence/mastery, autonomy, and relatedness) postulated by self-determination theory, with satisfaction of each of these needs during the first month of play proving to be statistically predictive not only of enjoyment but also of continued engagement and motivation, often more than two years later. Notably, measures of fun and enjoyment alone often fail to predict sustained engagement beyond one to two months (Rigby 2012).
- Initial evidence suggests that it is the satisfaction of basic needs in games (rather than the emotional experience of fun during game play) that has the greatest potential to have a positive impact on psychological well-being (Ryan, Rigby, and Przybylski 2006). In addition, need satisfaction in games has been shown to be a positive aspect of mood management (Zillmann and Bryant 1985; Tamborini 2011), showing an ability to affect mood states positively. This creates an important link to one of the key promises of gamification: its ability to reach beyond mere enjoyment and effect deeper positive change.

The delineation of psychological needs also gives gamification a reference point for optimizing the *quality* of motivational energy. Simply put, motivational energy is most conducive to gamification's desired outcomes—such as sustained engagement, positive psychological impact, increased value, and deeper learning—when gamification design and mechanics actively focus on facilitating and satisfying basic psychological needs, as opposed to simply seeking to capture attention or superficially amuse. Conversely, when gamification is irrelevant to these needs—or worse, actively thwarts these needs—they can be expected to fail more frequently in motivating sustained engagement.

A core element of gamification—providing rewards to "players"—will illustrate this point. The motivational literature on rewards reveals that they have a nuanced relationship with motivational energy and with respect to engagement can backfire as often as they succeed. One key to effectively applying rewards while avoiding their pitfalls is to view them through the lens of basic need satisfaction and the related dynamics of intrinsic motivation.

The Motivational Challenge of Reward Systems

It is generally accepted that rewards are motivating and have a positive impact, and most games are replete with systems and mechanics that offer a regular stream of rewards to players as they engage with content. However, much of the design of such systems does not take into account significant

research data on the impact of rewards on motivation: quite often, rewards have the paradoxical effect of decreasing motivation and engagement, rather than enhancing them (Ryan, Mims, and Koestner 1983). In a classic series of experiments that has been repeated and replicated more than one hundred times since the original studies in the 1970s, the offering of incentives for engaging in a game-like activity decreased subsequent interest in that activity, in contrast with not being offered an incentive (Deci 1972a; Morgan 1984). Specifically, experiments have shown that offering external rewards (including achievement awards and direct monetary rewards) decreases both subjective reports of interest, as well as the degree to which people will re-engage with an activity when given the chance to do so in studies that use a "free choice" paradigm (Deci 1971, 1972b Lepper, Greene, and Nisbett 1973) More recently, fMRI studies looking at brain activity clearly show that when external rewards such as financial incentives are given in response for engaging in a task, engagement in the short term is enhanced, but subsequent motivation and engagement significantly declines, in contrast with not offering the external reward in the first place (Murayama et al. 2010).

With the self-determination theory model of basic psychological needs as a lens—notably the need for autonomy—we can understand the psychological mechanisms behind this paradoxical effect. While on the face of it rewards seem to be fundamentally enhancing of an experience, they in fact can detract from the experience by undermining or thwarting the intrinsic value the activity may have had by shifting focus from simply enjoying the activity for its own sake—what White (1959) called an "internal perceived locus of causality"—to pursuing the activity simply to get to the rewards (an "external perceived locus of causality"). This shift directly reduces the experience and satisfaction of autonomy needs, decreasing overall motivation to re-engage and enjoy activities, even when those activities (such as games) are initially enjoyable. Simply put, the net result of many external reward structures is a decrease in intrinsic satisfaction and autonomy and a concomitant decrease in motivation to sustain engagement, despite the fact that rewards themselves might be perceived positively in the moment they are received.

Consider through this lens the widespread use of external rewards in gamification systems. Badges, points, and similar systems are wrapped around core activities with the expectation that such incentives will enhance motivation and deepen engagement. In fact, recent behavioral research has gone further, exploring the use of punishments and social pressure (Kast, Meier, and Pomeranz 2012) as means of changing behavior, despite the clear negative relationship such "controlling" strategies have on autonomy.

Why then might these strategies be persistently promoted? Because in the short term, such external inputs to the "black box" may lead to increases in the desired outputs in terms of behavior. For example, Kast and colleagues did show an increase in the desired target behavior as long as social pressure was applied, and in the fMRI study cited above, rewards did succeed in increasing engagement in the short term when rewards were given. The problem is that research consistently shows that this approach undermines interest *in the core activity itself*, ultimately thwarting sustained engagement and motivation. If the goal of gamification is truly to help internalize deeper satisfaction, more lasting behavior change, and sustained engagement and interest with the content being gamified, such short-term strategies are suspect. As just one

example, Moller, McFadden, Hedeker, and Spring (2012) found that those who received a financial reward for weight loss not only didn't lose as much weight as those who were not given this external reward, but also after five months it was those who *didn't* receive the reward that had sustained weight loss.

It is worthwhile to emphasize the general empirical point of these paradoxical findings: such data directly challenge the notion that it is unimportant to understand the dynamics within the black box of the user experience. If one were to look only at the directional response to a reward or punishment from a behaviorist perspective, one might well conclude that external rewards are the answer. Only when we shine a light into the black box—when we see the strong motivational and emotional dynamics happening within the subjective experience of our audience—do we see that operant tactics such as these can backfire with respect to sustained engagement and lasting behavior change.

This example seems particularly critical for gamification. Imagine a health care company deploys gamification strategies that emphasize external rewards in order significantly to improve healthy behavior change and deepen learning about health issues? Or a company unwittingly undermines interest in its services and imperils lasting relationships with customers for services through a strong focus on external rewards? Simply put, the implications are potentially harmful for gamification's core goals of deepening engagement, sustaining long-term relationships, and effecting lasting and meaningful behavior change. Thus, understanding the motivational dynamics energizing behavior—and not simply the short-term behavioral responses to rewards and punishments—is an important aspect of any gamification effort that is interested in sustaining engagement and effecting lasting behavior change.

How can this understanding be achieved? While the literature on rewards offers many nuanced suggestions, some core ideas for optimizing rewards so that they do not undermine intrinsic motivation are as follows:

- *Consider offering rewards simply for engaging, and not for performance.* Research has shown that when rewards are given simply for participating and are not performance dependent, they are less undermining of intrinsic motivation because they are less likely to be perceived as controlling (and thus thwarting autonomy) (Deci 1972a).
- *Create systems where rewards are naturally enhancing of deeper engagement with the material.* Offering someone a series of virtual badges to "pin to his chest" digitally is less meaningful and more likely to undermine the quality of one's motivation than providing rewards that open up new opportunities and challenges directly related to the core activity. Instead of using a shiny graphic saying "You've run one hundred miles this week," consider a reward that opens up new opportunities that reinforce the intrinsic value of the behavior, such as access to new content to learn more about endurance running or a fifteen-minute coaching session with an expert trainer.
- *Keep rewards unexpected.* Research shows that giving a reward that was not expected is not as undermining as dangling it in front of someone before the activity is begun (Lepper et al. 1973). This is because an unexpected reward can't be perceived as controlling or manipulative (i.e., undermining of autonomy)—and instead this

shifts the reward to being a genuine acknowledgment and celebration of what one has achieved.

In fact, the market is already discovering the short-term value of behaviorist reward approaches such as badges: foursquare—a leading "gamified" social networking service at the time of this writing—discontinued third-party health badges from numerous gamified health tracking services in 2012, largely based on the limited value such badges had in sustaining engagement with the desired behavior.[5] One reported reason for ending these badges is precisely what would be predicted by self-determination theory: users began to "game" the system, looking to circumvent the health behavior simply to get the badge. It is precisely these early market stories that reinforce the argument that a better understanding of motivational quality is needed.

Rigby and Ryan (2007) directly challenged the use of traditional reward structures as a motivational focus in games, specifically citing the use of external rewards as a pillar of game design and suggesting instead a more comprehensive model of human motivation based on the principles of basic need satisfaction. After this, the industry began to adopt more readily the language of *intrinsic and extrinsic* motivation as a means of elaborating its understanding of motivation and engagement with games. Next, we explore several important nuances of motivation—in particular the concept of "motivational quality"—as a framework for thinking about gamification solutions and further empirical study.

Understanding Intrinsic and Extrinsic Motivation

After the publication of initial research on the value of self-determination theory and intrinsic motivation to games (Rigby and Ryan 2007; Ryan, Rigby, and Przybylski 2006), several prominent developers began more actively to elaborate their view on motivation and adopt the language of intrinsic and extrinsic motivation, which is a pillar of self-determination theory and other organismic theories of motivation and emotion[6] (Dickey 2006). But what exactly do we mean by intrinsic and extrinsic motivation, and how do these terms relate to the idea of "motivational quality?" Is the common belief that intrinsic motivation represents "good" motivation and extrinsic motivation represents "bad" motivation truly accurate?

Intrinsic motivation refers simply to the pursuit of an activity purely for its own sake. When we do something because it is innately fun or interesting—and not to achieve some other goal—we are acting out of intrinsic motivation. The primordial example of intrinsic motivation is the simple joy of a child who is more interested in experimenting with the cardboard box in which a present was packed than the actual present: there is no other goal in this unbridled play than the sheer fun and enjoyment derived from the play itself (Deci and Ryan 1985).

Our basic psychological needs provide a blueprint for understanding the energy of intrinsic motivation more directly. When my daughter plays with the cardboard box, she sees a possibility space for exploration (autonomy need satisfaction), a tool through which she can practice her motor skills (competence satisfaction), and an instant vehicle for interaction with those around her, in this case her invitation to

let me push her around the living room floor inside her new box, with great delight for us both (relatedness satisfaction). These basic and innate needs energize her motivation to play with the box without any outside coaxing or "gamified" intervention by mommy or daddy. As soon as circumstances facilitated the natural satisfaction of her needs, joy, happiness, and sustained engagement were the result.

Extrinsic motivation, by contrast, is operating when I pursue an activity for its instrumental value: in other words, I am not pursuing the behavior for its own sake, but am pursuing it in order to achieve some other outcome or experience. Unlike intrinsic motivation, when I am engaged in extrinsically motivated activities, the goal of the activity is not the activity itself (Ryan and Connell 1989).

Note, however, that nothing so far indicates that the quality of extrinsic motivation is implicitly "bad." In fact, extrinsic motivation is perhaps the most poorly understood concept in the general marketplace. The real key to understanding extrinsic motivation is to realize that it is not suboptimal to intrinsic motivation, but in fact is a continuum where the quality of motivation is a function of the *reasons* one is pursuing one's goals, and how much those reasons represent personal value versus pressure and control. Put differently, it can be said that the quality of extrinsic motivation is a function of how *internalized* one's reasons are for pursuing an activity—meaning how much they reflect the values of one's true self. We next describe this "internalization continuum" in more detail.

Internalization: Improving the Quality of Extrinsic Motivation

The continuum of motivational quality with respect to extrinsic motivation was first described by Ryan and Connell (1989) in their work in educational settings and has since been demonstrated in multiple domains including health and wellness, sports, religious beliefs, as well as video games. Picture, for example, two men running on adjacent treadmills at the local health club. If we were to look only at the behavioral data, we might easily see a completely undifferentiated picture with respect to motivation: Bob and Jim show up at the gym with equal frequency and run for equal amounts of time with equal intensity. Because all the "outputs" and behavioral observations are identical, using a black box behavioral lens, we would conclude Bob and Jim to be equally motivated and would have lots of behavioral data to back up this conclusion (at least, in the short term). However, there could be stark differences in the *quality* of their motivation for running on the treadmill—determined by how internalized their motivation for treadmill running is—which will be strongly predictive of whether or not they continue to pursue treadmill running (i.e., sustain their engagement) or discontinue. Bob, for example, is running on the treadmill because in a heated argument several months ago, his wife told him he was fat and she demanded that he lose weight. In this case, Bob is miserable with virtually every step he takes on the treadmill, despite the fact that he shows behavioral *persistence* in running. Bob neither enjoys running on the treadmill nor does he have a personal value with respect to running, losing weight, or becoming more fit. His running is purely instrumental to his goal of making his wife less angry. But more importantly,

the quality of Bob's motivation is low because his rationale and motivation for this activity is highly external to *who Bob is*. There is absolutely nothing about running that reflects Bob's own personal values, and his experience of running is one devoid of psychological need satisfaction.

Next to him is Jim. No one has told Jim to work out. Like Bob, he doesn't particularly enjoy running on the treadmill (i.e., it is not an intrinsically motivated behavior), but he frequently mentions to friends that doing so gives him more energy throughout the day, helps him avoid that post-lunch lull, and overall makes him feel more vital. As with Bob, Jim is running on the treadmill to achieve some other goal. However, it is clear that in this case the goal is qualitatively much different for Jim that it is for Bob; Jim *deeply personally values* the outcomes that result from running regularly on the treadmill. As a result, numerous research studies have shown that the "Jims" of the world will sustain engagement with physical exercise, feel more satisfaction from the activity, and have a higher overall sense of well-being, and the "Bobs" are more likely to discontinue the activity and fall short of positive outcomes (Noels, Clement, and Pelletier 1999; Pelletier et al. 2001).

When considering these two vastly different motivational qualities, the implications for gamification are readily apparent: if we deploy gamification techniques that emphasize motivation for behavior that is pressuring, controlling, or otherwise "push," the prospects for long-term engagement, loyalty, and lasting behavior change are in jeopardy—even if we have some success in the short term. By contrast, if we are mindful of fostering a deeper quality of motivation that emphasizes personal value, we will build more lasting positive results in our gamification efforts even in cases where the activities being pursued are technically extrinsic. In short, we can move past a dichotomous view of good (intrinsic) and bad (extrinsic) motivation, and instead focus on the *quality* of motivation (even when motivation is extrinsic), defined by how much the energy for our taking action reflects our true beliefs and values, in contrast with reflecting more external pressures.

Motivational Quality: The Internalization Continuum

To clarify further, self-determination theory has identified several important "milestones" along the internalization continuum to describe the quality of motivation. This internalization continuum is illustrated in figure 4.1, which illustrates four discrete kinds of extrinsic motivation relative to how closely that motivation reflects one's true self (Deci and Ryan 2000).

- *External regulation* refers to the kind of motivation exemplified above by Bob running on his treadmill. External motivation is energized purely by demands and forces that are outside of the self. These can include both threats of punishment and offers of incentives or rewards (the implications of which were discussed earlier in this chapter).
- *Introjected regulation* refers to those times when one is motivated out of a sense of *internal* pressure, such as feelings of obligation, guilt, or shame. This is differentiated from purely external regulation because when one is introjected around a particular behavioral activity, one is controlling oneself rather than being explicitly controlled by others.



Figure 4.1
The self-determination continuum of motivation.

In this case, the energy for taking action is intrapsychic (i.e., one is pressuring oneself in some way), but still external to one's personal values and interests.

- *Identified regulation* refers to the inflection point when the quality of motivation moves to the more desirable motivational energy associated with sustained engagement and positive outcomes. It refers to those times when we are engaging in behavior or activity because we truly value the goal or outcome we are pursuing. In identified regulation, even if the activity we are undertaking is not in and of itself enjoyable, the goal or outcome of that behavior is something that we have come to value as having a meaningful personal benefit.
- *Integrated regulation* refers to the deepest level of extrinsic motivation and occurs when we begin to integrate a particular activity with other aspects of our lives, seeing how our activities in behaviors work together to create optimal feelings of value of well-being. We see, for example, that the activities we value (identified) do not exist in isolation, but can integrate with one another to even greater benefit (as when I spend a fulfilling hour writing and also realize it builds energy in me to play with my daughter). Here again, we may not particularly enjoy many of these activities—and in fact, life is full of these circumstances—but we nonetheless value how the activities we pursue work together to create a life that has value and meaning.

This continuum offers an exciting opportunity in the motivational design of gamification products and platforms. Specifically, it provides a more detailed blueprint for creating gamification mechanics that are focused on facilitating deeper internalization. How can gamification do this in an applied way? Simply put, by increasing the focus on the basic need satisfactions for competence/mastery, autonomy, and relatedness when designing gamification features

and systems. The path to internalization and fostering higher-quality motivation lies directly on how well experiences support and facilitate basic needs, as opposed to thwarting them. Specific aspects of reward mechanisms and how these can be better tuned to support higher-quality motivation have already been discussed, but that is just the beginning. Virtually all aspects of gamification can be passed through the experiential test of how well they are supporting basic psychological needs and tuned toward these needs in an effort to more successfully motivate deeper engagement and more positive outcomes.

The evidence for taking this approach is comforting. For more than two decades, the internalization continuum and the principles of motivational quality discussed here have been tested in research studies worldwide. Greater internalization has been associated with deeper learning (Grolnick and Ryan 1987; Ryan, Kuhl, and Deci 1997), more successful behavior change in health and wellness (Ryan, Plant, and O'Malley 1995; Deci et al. 2001), greater satisfaction and productivity at work (Gagne 2000; Baard, Deci, and Ryan 2004), more enjoyment and sustained engagement with games and interactive software (Ryan, Rigby, and Przybylski 2006; Rigby 2012), and greater satisfaction and well-being (Kasser, Davey, and Ryan 1992). Put differently, there is overwhelming reason to believe that the benefits of applying the principles of basic need satisfaction and focusing on optimizing motivational quality will improve the commercial, behavioral, and experiential outcomes pursued by gamification projects across multiple markets and subject areas.

Having covered basic psychological needs and their importance to motivation within gamification and added to that a discussion of motivational quality and the process of internalization of motivation in order to deepen engagement, next we discuss the final leg of a motivational triad for gamification: a discussion of user values/goals in and of themselves—and how the motivational literature shows that not all goals motivate and engage equally.

Intrinsic Versus Extrinsic Goals: Why All Goals Are Not Created Equally

So far, I have been discussing motivational principles at the level of the individual: the specific quality of one's motivation and the individual experience of need satisfaction that is derived when engaging in activities. The final motivational consideration concerns something more global: the nature of the goals that one is pursuing. In the early 1990s, researchers began exploring the concept of values, seeking to determine whether certain values (or goals) were implicitly more intrinsic (i.e., congruent with the satisfaction of basic psychological needs) or extrinsic (i.e., less aligned with basic needs and more indicative of values that may instead be pressuring or controlling). Notably, this research identified that certain goals were indeed *intrinsic* in nature, while others were *extrinsic*. To review these findings briefly:

- *Intrinsic goals* were found to include personal growth, health and wellness, developing stronger ties in one's relationships and with one's community, and other goals more directly aligned with the satisfaction of basic psychological needs.
- *Extrinsic goals* included the pursuit of materialistic gain (such as wealth), physical appearance,

fame, and similar goals that are more external to the self.

- This research (Kasser and Ryan 1996; Kasser 2002) has consistently shown that pursuit of intrinsic goals is associated with greater well-being, sustained engagement, and a variety of other positive outcomes congruent with the goals of the gamification movement. By contrast, pursuit of extrinsic goals is associated with poorer well-being and satisfaction and less engagement, as these goals are not in the service of satisfying basic psychological needs as readily as intrinsic goals (Kasser and Ryan 1996; Kasser 2002, 2003).

This research is worth considering as a key aspect of an optimal motivational model for gamification, particularly where the gamification solution in question is concerning itself not only with commercial success, but also with improvements across mental and physical health, education, and overall life satisfaction. In such cases, the motivational literature clearly indicates that emphasizing intrinsic goals in gamification is likely to contribute to gamification's success in achieving positive outcomes. Conversely, by being aware of extrinsic goals and mindful to minimize their emphasis wherever possible, the research suggests we can expect this to reduce the risk that gamification projects will fail in achieving deeper engagement and satisfaction.

Bringing Things Together: A "Motivational Triad" for Optimal Gamification

This chapter has reviewed in some detail three core areas of motivational psychology that can come together to form an actionable and practical model for developing new gamification products, strategies, and mechanics and can help vet gamification solutions to help ensure long-term success. The foundation for this model is the empirical evidence demonstrating that the experience of the user is not—as behaviorists have historically emphasized—a black box. Indeed, evidence shows that a focus on motivation models that emphasizes inputs and behavioral outputs runs the risk of undermining gamification's core goals.

Instead, empirical evidence has been presented that motivation is best understood and applied by understanding the subjective user experience. Specifically, we have outlined three basic psychological needs—competence/mastery, autonomy, and relatedness—and the critical role that these needs play in sustaining engagement, achieving wellness, maximizing value, and motivating lasting behavior change among other positive outcomes both in games (Rigby and Ryan 2011) and elsewhere (Deci and Ryan 2000). The motivational literature has repeatedly shown that experiences in life that facilitate the satisfaction of these needs—whether those experiences be analog or digital—are associated with the positive outcomes that concern much of the gamification movement. Equally important, experiences that thwart these basic psychological needs—while they may stimulate certain behaviors and activities in the short term—will ultimately fail to sustain engagement and achieve the goals to which gamification aspires. In short, basic psychological needs become an experiential

blueprint or lens through which gamification designs can be vetted and specific mechanics and techniques evaluated.

It is equally true that gamification projects are often concerned with motivating people around behaviors that are often extrinsic (e.g., exercise, training), meaning simply that these areas are instrumental for many people and pursued only with respect to achieving some other desired goal (i.e., "I am doing X in order to achieve goal Y"). For this reason, it is also important to understand the different *qualities* of motivation that can occur at the extrinsic level so that gamification can be designed for greater internalization of the user behaviors and outcomes desired by the project. Thus, both basic psychological needs theory and an understanding of motivational quality and internalization go hand-in-hand with one another in day-to-day practice: they are two parts of a motivational triad that can greatly assist in the planning and execution of gamification projects, tools, and techniques to create the deep engagement and positive outcomes gamification seeks.

Finally, a motivational dimension that is relevant to many gamification endeavors is recognition that not all goals are equal with respect to either engagement or satisfaction. Research shows that certain goals are implicitly intrinsic, and these accrue toward greater well-being, engagement, and overall positive outcomes. It follows that the success of gamification projects can be improved—and better motivation achieved—by emphasizing intrinsic goals wherever possible.

The focus on these three key areas provides meaningful motivational guidance for gamification design, and indeed motivational design more generally. Basic psychological need satisfaction serves as the foundation, with motivational quality/internalization and intrinsic goal focus offering practical, applied guidance for gamification development of programs that are optimized for user satisfaction, more lasting behavior change, sustained engagement, and the many positive outcomes gamification seeks to convey.

Closing Thoughts: The Practical Challenges to Motivational Best Practices

The television show *South Park* has been a mainstay of American pop culture and has often sought to lampoon a variety of serious topics. In one classic episode taking a shot at commercialization and the poorly justified business models that often crop up during times of frothy hype (perhaps inspired by the dot-com meltdown a decade ago), the young protagonists of the show fall victim to the "Underpants Gnomes"—creatures who would steal away underpants from your drawer while you slept as part of their core business model to generate profits. These enthusiastic gnomes had a foolproof business plan with three steps: step 1 was collect underpants; step 3 was "profits"; step 2, however,.... was a question mark. Simply put, for all the enthusiasm there was for the business model, how the strategy actually created value was neither clearly seen nor even a primary focus of inquiry. It was simply accepted that the value was there. Somewhere.

This chapter has outlined key principles to help fill in gamification's "step 2" by outlining psychological principles and empirical research relevant to

gamification achieving its promise of effecting lasting and positive impact across a variety of goals including sustained engagement, user satisfaction, and lasting behavior change. However, having worked with organizations both large and small for more than twenty years on product and program development, I am well aware that this is easier said than done in an environment that is largely dictated by short-term performance. Most publicly traded companies that wish to engage in gamification are probably focused on a ninety-day window of performance at any given time—constrained by the market demands for the next earnings report. Or to put it in the language of this chapter: publicly traded companies are greatly constrained in their autonomy and highly controlled by the markets. Similarly, those companies providing gamification solutions—such as platform providers and consultants—are eager to promote the most robust stats they can, as quickly as they can, creating yet more pressure for short-term tactics over longer-term approaches that the research presented here indicates may well result in more success. It is easy in such environments simply to pull levers and continue a black box approach to gamification, looking only at how the features we pile up in step 1 lead to short-term performance in step 3, with little understanding or regard for step 2.

In light of this, I offer two concluding thoughts with respect to these pressures.

First, for commercial businesses applications, it is worth keeping in mind that most business economists will confirm the high cost of churn: acquiring customers is an expensive proposition relative to retaining those you have, and if businesses slip into controlling tactics when applying gamification in order to "juice" the performance of their programs, they are only setting themselves up for higher rates of disengagement and consequently less financial success over the long term as they undermine and demotivate their audience. As noted earlier, a quick study of the meteoric rise and rapid decline (at least for the moment) of social gaming will confirm this. And as competition for attention continues to grow, the value of retaining and growing customer loyalty through more effective motivational techniques is likely to grow along with it.

Finally, for gamification efforts that are not solely commercial and seek to address significant areas such as education, health care, and well-being, the potential to scale effective gamification programs optimally to motivate and deepen learning and behavior change is genuinely exciting. But on the flip side, the possibility of undermining motivation through short-term motivational tactics is equally concerning. The gamification movement often holds itself out as having tremendous transformative promise. Let's ensure we do what we can to make that promise a reality over the long haul by focusing on motivation at the deeper level it deserves—a level that whenever possible puts at its center principles of intrinsic motivation, deepening motivational quality, and supporting the fundamental psychological needs of each individual.

Notes

1. See http://www.huffingtonpost.co.uk/2012/10/09/zynga-worthless-farmville_n_1950910.html.

2. This finding has been replicated in numerous as-yet unpublished studies in our labs in which we collect EEG and other biometric data alongside subjective metrics of satisfaction and fun.

3. As one example, see http://www.digitalplay.info/blog/2011/11/04/reward-or-punishment-gamification-with-class-dojo/.

4. Virtually all current providers of gamification platforms and related solutions cite "engagement" as a core value proposition.

5. See http://aboutfoursquare.com/foursquare-to-discontinue-third-party-health-badges-from-fitbit-health-month-and-likely-runkeeper-next-week/.

6. As one example, see Chris Hecker's (2010) presentation at the Game Developer's Conference entitled "Achievements Considered Harmful?"

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5 GAMIFICATION AND ECONOMICS

Juho Hamari, Kai Huotari, and Juha Tolvanen

During the past couple of years, the use of game design for economic purposes in games (Hamari and Lehdonvirta 2010; Hamari and Järvinen 2011) as well as in non-game contexts (Reeves and Reed 2009; Deterding et al. 2011; McGonigal 2011; Zichermann and Cunningham 2011; Huotari and Hamari 2012) has rapidly gained a substantial amount of traction among scholars and practitioners. This development of affording gameful experiences or using design reminiscent of games has been dubbed *gamification*. Following the successes of social networking services (Facebook), games (Angry Birds), and location-based services (Foursquare), marketers in particular have started to apply gamification in attempts to affect user behavior. Gamification has already been applied in several areas, such as for persuading people toward greener consumption (EcoIsland), building loyalty toward television channels (GetGlue), taking care of one's health (Fitocracy), and even for gamifying the tracking of one's aspirations in life (Mindbloom).

So far, gamification has been discussed mainly in the area of game studies, and its positioning within other scientific domains still remains unclear, hindering the identification of potential avenues for further inquiry. In the economic context, gamification has been defined as a process of enhancing systems by affording gameful experiences in order to support the overall value creation (Huotari and Hamari 2012). This definition is rooted in the service marketing field. In this chapter, however, we dig deeper into the economics of gamification by illustrating how gamification links to other existing models and concepts in economics.

We start with a brief description of the theories that neoclassical economists use to model decision making and give examples on how some gamified design patterns relate to these models. After this, we turn to both potential and existing economic applications of gamification. Then, we look at the economic reasons why and when one might find it useful to push decision makers to a new direction using gamification. After all, economics 101 teaches us that free markets should yield efficient outcomes without any pushing. Hence, it seems that there might be very little room for gamification. However, we argue that when some of the assumptions that deliver efficiency fail, gamification can be a potential tool for remedying these market failures.

The section on behavioral economics considers what happens when agents are less sophisticated. By this, we mean situations where they do not behave according to the axioms outlined in the section on neoclassical economics. Usually, this means that the agents are not able to use optimally all information

that is available to them or make systematic logical errors when choosing between different available options. We illustrate with examples how gamification often plays with these systematic biases.

We then look at gamification from the point of view of marketing. By using Foursquare as an example, we examine gamification through the perspective of transactional marketing theory, relationship marketing, service marketing, and experiential marketing. Lastly, at the end of this chapter we discuss the added value that gamification brings to economics.

A Brief Look at the Microeconomics of Gamification

Essentially, it can be said that the goal of practically all gamification is to affect the decision making of a given target group so that at least some of the users will take actions they would not have otherwise taken. Examples range from more ecological consumption (EcoIsland) and healthier lifestyles (Fitocracy) to divulging location information (Foursquare) or professional activities (LinkedIn) or just simply purchasing more of a given good (loyalty cards).

In this section, we explore the economic reasons why and when it might be useful to push decision makers to a new direction using gamification. We first give a brief overview of neoclassical theory of economic choice and then relate it to some gamified design patterns.

A Sampler of Some Basic Economic Mechanisms That Make Gamification Tick

Decision Theory and Gamification

Decision theory is a subfield of economics that tries to build mathematical models of economic decision making. The goal is to produce a general theory that is able to take the whims and tastes as inputs and generate predictions of aggregate economic behavior. A decision theorist is interested in the reasons behind the decisions only as far as they affect the general rules that agents use to make these decisions. Consequently, these models are very different from explanations offered in psychology and related fields that try to describe the cognitive processes behind decision making. The strength of economic theory is that it offers strong and testable predictions on what people will choose given their previous behavior, while its relative weakness is that it avoids taking a strong stand on why people choose what they choose.

Choice under Certainty

Any modern treatment of choice theory starts by assuming that people are able to make pairwise comparisons between any possible bundles of goods or services. If this is the case, people's preferences are said to be complete. Furthermore, it is also often assumed that these decisions do not contradict one another. More specifically, the most basic model of decision theory assumes that if you like Twitter more than Foursquare and Foursquare more than Farmville, then you must like Twitter more than Farmville. This is called transitivity. Perhaps surprisingly, when the number of possible choice bundles is finite, completeness and transitivity alone are enough to guarantee that the choices of an agent can be modeled as if she was maximizing a real-valued function that is defined on the space of all possible choice bundles.[1] This is the famous utility function that is

the workhorse of most theories of economic demand under certainty.

When outcomes are certain, this model predicts that the only way gamification can affect choices is to affect the relative ordering of different choice bundles. In other words, in this basic setup, gamified features of a service always either add value to the action that the designer of the service would like the user to take or make the most preferred competing actions less appealing. For instance, assume a Foursquare situation where you had just been ousted as the mayor of your local Starbucks. If retaining mayorship was valuable to you, then visiting that same Starbucks would have now become relatively more appealing compared to visiting other cafés. Gamifying café visits may thus lead you to go to your local Starbucks more than the other nearby cafés and thus affect the relative ordering of your choice bundles.

Choice under Uncertainty

Almost all of our choices contain uncertainty. For instance, if I want to go to the outside ice cream bar, there is always the risk that it will start to rain or that they are out of my favorite flavor, cappuccino-chocolate. If this is the case, I might prefer staying inside. However, before I go downstairs, I am uncertain of the outcome of my outing. In principle, this is not a problem for the standard utility theory—as long as we perceive choice bundles to include gambles, the utility representation still holds. However, often it would be convenient to have the choices to correspond to a model where agents maximize some expectation over utilities derived from the outcomes of the gambles. To illustrate, suppose that the sky is blue and that the probability of rain is zero and thus the question is only if they have cappuccino-chocolate (henceforth c-c). Assume further that I think the probability of them being out of c-c is 0.3. Then, for many applied models it would be convenient if there was a utility index u such that I go for ice cream if and only if

$$u(\text{stay inside}) \leq E[u(\text{go for ice cream})] = 0.3u(\text{out of c-c}) + 0.7u(\text{serving c-c}).$$

In other words, my choices correspond to maximizing the *expected utility* I attain from the different lotteries that are available to me. The utility index here is again just a modeling tool that is not assumed to be anything that the people making choices use but only a tool that leads to analytically tractable results. The question is when are people's choices "regular" enough to be compatible with such a representation? If such a representation exists, it is clear from the previous example that if the probability distribution over outcomes in two gambles is similar; also the expected utility from those two gambles is almost the same. The formal version of the assumption which guarantees that similar gambles are ranked similarly is called *continuity*.

In addition to completeness, transitivity, and continuity only one more axiom is needed to guarantee that choices can be modeled as coming from expected utility maximization. This axiom is commonly known as the independence axiom, as it states that my ordering of gambles should not change if I add a given new lottery to any pair of existing lotteries. More specifically, if above I preferred going for an ice cream despite the risk of them being out of c-c, I should still prefer a lottery where my neighbor takes me for an ice cream with 0.5 chance and I go alone with 0.5 chance to one where I stay home with 0.5 chance and my neighbor takes me out with 0.5 chance. Note how the scenario of my neighbor taking me out got added to both pre-existing scenarios with

the same likelihood. If the above-mentioned four assumptions hold, then the agent's choices can be modeled, as if she had a utility function over the possible outcomes and she was maximizing the expectation of that utility index given the probabilities of different outcomes. This is in a nutshell the content of the famous von Neumann–Morgenstern expected utility theorem (von Neumann and Morgenstern 1947). For a more detailed description, see Kreps (1988) or Mas-Colell, Whinston, and Green (1995).

The expected utility is extremely convenient when combined with situations where money is a good substitute to all uncertain outcomes. This allows discussion about the attitude the decision maker has toward risk. Consider facing a lottery that yields $50 with certainty versus one that yields $0 with probability 0.5 and $100 with probability 0.5. Both have the same expectation, but the second is obviously more risky than the first. In general, a person who is willing to pay some money to get the first lottery rather than the second one is called risk-averse, and a person who is willing to pay to get the second rather than the first is usually called risk-loving. A person who thinks the two are equally good is called risk-neutral. The attitudes toward risk translate directly to different forms of curvature of the utility function. For the range where the preferences are risk-averse, the graph of the function curves down, and for the risk-loving preferences it curves up. For risk-neutral preferences, the graph is a line. A possible utility function with risk-loving, risk-neutral, and risk-averse regions is presented in figure 5.1.

A fairly general empirical observation is that people tend to be risk-loving when the expectation of the lottery is relatively small compared to their current wealth. For instance, the expected value of a common lottery ticket is usually less than the price of the ticket. Nevertheless, lotteries are highly popular. However, when the expectation of the gamble becomes large relative to the person's wealth, people become increasingly risk-averse. For instance, the premium of a homeowner's insurance is usually substantially more than the expected value of the damages (since most of the time people have zero claims), and still most people tend to buy insurance contracts.

These changing attitudes toward risk link directly to the design of gamification. The fact that most people find small gambles entertaining has also been used outside casinos and national lotteries. For instance, the Disneyland attraction "Star Tours—The Adventures Continue" features a randomly chosen sequence of flights. A guest entering the attraction cannot know which one of the possible fifty-four ride experiences she will encounter.[2] If people were uniformly risk-averse, then this layer of randomization would make the ride strictly inferior to one where the outcome of the ride was known beforehand. Conversely, recommendation services like Foursquare recognize the inherent risk-aversion that people often exhibit when choosing restaurants. For them, it is beneficial that new places get reviewed fast, as otherwise all of the risk-averse restaurant goers will just flock to the places with the largest number of positive reviews and avoid the risk of getting a bad meal in a new place. Thus, Foursquare offers rewards to the first patron of a new restaurant, and of course it is easier to become the mayor of a newly opened establishment than of the most popular café around the corner.

These two examples highlight how attitudes toward risk play a strong role in how gamification should be designed. First, one must try to figure out customers' preferences: whether the target group's

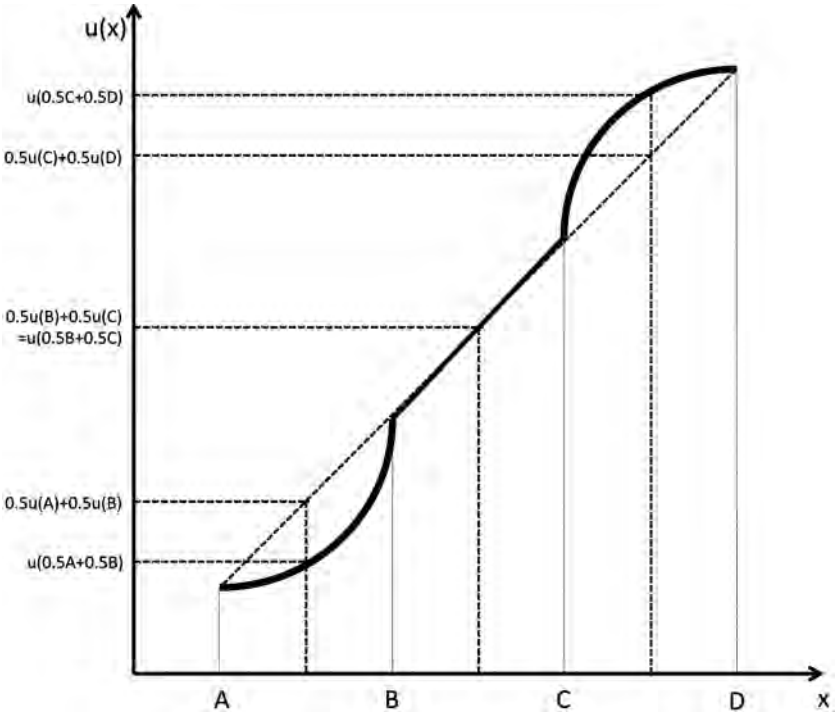


Figure 5.1

The figure presents an example of possible preferences over monetary gambles. The monetary value of outcomes is presented on the *x* axis, while the utility from any given outcome is given on the *y* axis. The preferences in the figure show first a region of risk-loving preferences between gambles with outcomes between points A and B. Then the preferences change to risk-neutral ones between gambles with outcomes in the interval from B to C. Finally, the interval from C to D shows risk-averse behavior. We have illustrated this change in preferences by drawing the utility the person obtains from a 50-50 gamble between points A and B, B and C, and C and D, respectively, and then compared this to the utility the person obtains from getting the expectation of the gamble with certainty. For instance, note that in the risk-averse region, the point $u(0.5C + 0.5D)$, that is, the utility obtained from getting the *expectation of the gamble* with *certainty*, is above the point $0.5u(C) + 0.5u(D)$, which is the *expectation of the utility* that the person gets from the gamble. Thus, in this region the person prefers the expectation with certainty over the gamble that has the same monetary expectation. Analogous reasoning shows that in the region between points A and B, the person prefers gambles over certain monetary rewards, and in the region between B and C, the person does not care whether to gamble or to take the money with certainty.

available actions lie on the risk-averse part of the utility function or the risk-loving part. After this, it is easy to decipher whether adding risk to the outcomes of the available actions, like Disneyland does, will increase or decrease the popularity of those actions and, if affecting the riskiness is not possible, whether one should reward or discourage people to take risk to achieve the outcomes that are most beneficial from the designer's viewpoint. Economics and econometrics can help tremendously in structuring the understanding of customer preferences and in forecasting how demand responds to new features or pricing.

Subjective Probabilities and the Role of Information

Thus far, we have assumed that the probabilities that people attach to uncertain outcomes are given from outside the model. However, if someone asked me what is the probability with which the ice cream bar downstairs is out of cappuccino-chocolate, I would find it difficult to come up with a precise number. Still, I am able to choose whether to go out for an ice cream or stay inside. Does this mean that our previous expected utility model is completely worthless? Not necessarily. Just as we think about the utility functions, maybe also the numeric probabilities are just modeling devices that help us analyze people's choices. This leads us to the following question: when are an agent's actions *compatible* with a model where the agents acts *as if* he or she had some *subjective* probabilities in mind for each uncertain state of the world? This question is answered by the models of Savage (1954) and of Anscombe and Aumann (1963).[3] These models give exact (and testable) conditions under which a person's behavior is compatible with a model where she acts as if basing her actions on estimating some subjective probabilities. The framework is relatively technical compared to the scope of this chapter, and hence we will not present the assumptions of the model here. However, we want to emphasize that the models are relatively general, and very often the expected utility framework gives a good description of people's decision making. However, there are many behavioral biases discussed in the section on behavioral economics that are systematic ways in which people's behavior deviates from the model predictions. Designers of gamification patterns should be mindful of these situations, as the applicability of the models presented earlier decreases when biases increase.

Of course, Savage's "subjective probabilities" do not live in a vacuum. They are strongly affected by the information we have. When we get new information, we tend to adjust our probability assessments. If I read a weather report saying that there is zero percent chance of rain in the next two hours, I am quite likely to find eating ice cream outside more appealing than before, because the new information has influenced my beliefs. The mathematical model for adjusting existing probabilities to new information is called Bayesian updating or Bayesian learning.[4]

Many gamification solutions alter the flow of information to users. Instead of attempting to add information, gamification might seek to obscure or hide information with the aim of increasing the value of taking certain actions over others. For example, in Chorewars rewards are intentionally distributed randomly in order to maintain anticipation in every house chore a player completes. This intentional obscurity surrounding the reward distribution has also another important effect: if for example vacuum-cleaning always yielded the most prestigious reward, players would start overproducing it, and

the cleanliness of the bathroom might go neglected. In contrast, gamification can also help in displaying information in a more comprehensible form and help agents to use it more efficiently. In accordance with Reeves and Read (2009), visual aids and interface design patterns, such as levels and progression metrics, can help to achieve some of these goals.

The Role of Game Theory in Managing Strategic Interaction

Consider again my local Starbucks. If suddenly all the local coffee drinkers started to compete with me for the mayorship of that establishment and if I valued being a mayor enough, I might be persuaded to start going to the nearby Dunkin' Donuts instead—even if I preferred the coffee in Starbucks over the brew in Dunkin' Donuts. Oftentimes, gamification relies on the fact that the actions of others affect the way I act, and my actions affect the decisions of others. The study of this type of strategic behavior is the core of economic game theory. The above-mentioned competition for mayorship is a prime member of a class of games known as congestion or coordination games.

The first application area of game theory when designing gamified services is to design better or more engaging "games" inside the service. Game theory helps us understand optimal ways to play a given game when the players' goals are well defined. Through trial and error, people are often able to get very close to this optimal solution. If the solution is too trivial and deterministic, people often find the game childish and not very entertaining. Economic theory offers a wealth of classes of games with nontrivial solutions and optimal outcomes that have a random component. Such games are often likely to be entertaining as well. A prime example of how economic game theory can be used to devise engaging games is a board game called Modern Art, where players auction art to one another using rules that resemble auctions studied in auction theory—a subclass of game theory.

Game theory also helps us to understand what kind of implications a given social design pattern may have. If people start to behave in an unintended strategic way due to a given design pattern, this may lead to unexpected results. For instance, one might argue that me going to Dunkin' Donuts probably decreases total welfare, as I enjoy their coffee less. It is unlikely that the designers of Foursquare intended me to drink worse coffee. In other words, the strategic dimension of Foursquare may encourage users to visit a variety of cafés, however it can nudge us to make choices we otherwise dislike.

This chapter is too short to give a complete introduction to the methods economists use to analyze strategic behavior and the type of games that are most commonly considered. Osborne (2004) offers a nice introduction with examples. Mathematically more confident readers should take a look at Fudenberg and Tirole (1991).

Some Economic Application Areas of Gamification

In this subsection, we discuss some common economic problems and goals to which gamification can offer a possible solution.

Increasing Activity through Adding Value and Increasing Available Information

The simplest way gamification works is by making certain actions more valuable, which leads to an action becoming more popular. Some existing examples include promoting sales (Samsung Nation), making exercise more fun (Fitocracy), or achieving one's goals in life (Mindbloom). Reeves and Read

(2009) also suggest that making work more like playing games would lead to higher productivity.

Many gamified features also typically convey information in a concise and efficient way. This in turn helps people to make more efficient decisions when facing uncertainty. Badges, status bars, and achievements are the most prominent examples. Another design feature borrowed from massively multiplayer online role-playing games (MMORPGs) and used as a solution to many modern resource allocation problems is the creation of a virtual market and virtual currency. As textbook microeconomic theory suggests, market prices are a very concise way of conveying information on people's preferences and relative scarcity of economic inputs. They also establish wealth as an easily interpretable way to measure success between different users. This in turn can lead to social behavior such as bragging or to "keeping up with the Joneses," which may further incentivize beneficial behavior.

Correcting Externalities

A large number of existing gamified services can be viewed as attempts to mitigate externalities. An externality arises when an agent making choices does not fully take into account how these choices affect others, and the private costs and benefits of a given action do not fully reflect the effects of these actions on others (see, e.g., Laffont 2008). An example of an externality would be the adverse environmental effects of consumption. For instance, as long as the adverse effects of global warming are not fully incorporated in the price of fossil fuels, everyone tends to consume too much energy from these sources. There are also positive externalities. For example, after reading a book or eating in a restaurant, I will have information on how good that given book or restaurant was. This information may be useful to others, but as long as I do not benefit from others knowing it, I might not divulge what I know. Similar externalities arise commonly in the modern information-centered workplace where expertise on specific areas of operation tends to accumulate to individuals who are in a daily contact with these areas. Parts of this flow of information may be useful elsewhere in the organization, but the employee possessing it rarely makes it public unless incentivized to do so. Researchers have, in fact, shown that extrinsic rewards from divulging information may be counterproductive (see, e.g., Bock et al. 2005), which suggests that intrinsic motivation-inducing gamification may indeed be a highly suitable way to entice information sharing in an organization.

Gamification has been either suggested or applied as a partial solution to all of the issues outlined above. In all of these solutions, the idea is to increase the value of taking the socially preferred action. For instance, in EcoIsland, neighbors were put in a competitive setting in order to increase the relative value of sustainable behavior through social conformity and the fear of faring badly in the competition. Similarly, the status bar in LinkedIn and badges in Foursquare encourage users to publicize private information. In a similar sense, Reeves and Read (2009) outline multiple ways how gamifying the workplace may lead to better flow of information.

Experience and Credence Goods

Strongly related to information externalities are experience (Nelson 1970) and credence (Darby and Karni 1973) goods. An experience good is a product or service such as a meal at a restaurant whose quality or price can be fully ascertained only by

purchasing and consuming the product. Credence goods, in turn, are products or services such as a car repair whose quality is difficult to ascertain also after purchase. For example, it can be hard to know whether engine problems after an oil change are due to normal wear and tear or the mechanic skimping on costs when changing the oil. The problem with these goods is that the lack of information tends to lead to underconsumption of good-quality products and overconsumption of bad-quality alternatives. Furthermore, if the seller knows the quality of the product better than the buyer, the seller may have an incentive to sell an inferior product masking its quality. If consumers understand this and are unable to verify the quality of the products offered, they are likely to treat all products as if they were of bad quality, and their willingness to pay for any product falls. This in turn leads to further incentivizing the suppliers to sell bad products given that producing quality costs more. In the end, the market is likely to supply only bad quality. This is a stylized version of a market where adverse selection leads to full or partial shutdown of a market. The example was first described in George Akerlof's (1970) seminal paper that describes how "plums" get crowded out by "lemons" in the used car market.

Gamification may help to incentivize people to publicize their private information. The service Foursquare is a prime example of a service that uses a service design that promotes social aspects, such as getting recognized, social conformity, and competition, to these aims. Social aspects have been found to be strong predictors for adopting gamification (Hamari and Koivisto 2013). When information about the quality of the goods produced is public, there will be incentives to hold up a good reputation by producing good quality.

Game mechanics can also be used to encourage users to try out a service. If the probability of a consumer liking the service is relatively low, she may find it too costly to try out the service in the first place. This is especially true with many IT services that are targeted to niche customers. Making experimenting competitive or awarding badges or points for trying out different features of the service may make the process more pleasant and appealing.

General Mechanism Design Problems

Consider three separate cases: a monopoly selling goods to a heterogeneous population; a government deciding on the amount of public good produced; and the problem of allocating rights to a common resource. A real-world example of the first case would be a lone airline serving some route and facing demand from both business travelers and tourists with very different willingness to pay. An example of the second case would be the decision about how much should we invest in national defense or reducing air pollution. Both are goods that are generally non-excludable (i.e., once produced it is hard to keep anyone from enjoying its benefits) and non-rivalrous (me consuming the benefits does not considerably reduce your ability to enjoy them as well). Public goods are usually defined as goods holding these two properties. An example of the third case is fish in the sea or commonly owned grazing land, which are non-excludable but often rivalrous (the fish I catch reduce the total number of fish available to you).

All of these examples share the feature that there is a single actor who is able to design the institutions that the participating agents must use and follow. Their goals may differ: the monopolist is likely to choose the institutions to maximize its profits, while the government will try to maximize some form of

social welfare. The study of how to choose the underlying institutions to best achieve these goals is called mechanism design; see, for example, Myerson (2008) and Fudenberg and Tirole (1991). What often makes the problem interesting is that the population interacting with the institutions has aspirations that are not fully aligned with the designer's goals or with the aspirations of other participating actors. Thus, strategic interaction is likely to arise, and the situations are best analyzed using game theory.

The tension in most of the mechanism design problems is driven by the fact that the designer does not perfectly know how the actors rank the different social outcomes. In other words, there is a problem with asymmetric information. If the designer has access to information that is correlated with agents' valuations or other private information, the designer can potentially use this correlation in the design of the mechanism to better achieve her goals. This idea is used a lot by price-discriminating airlines that typically increase their prices as the departure date approaches. The companies know that business travelers become aware of their traveling needs relatively late and are willing to pay for a specific date, while tourists have more flexible schedules and constrained budgets. The mechanism of rising prices, thus, uses this correlation between willingness to pay and the timing of purchases to sort out different types of buyers.

Similarly, gamification can be used to reveal private information. This means that a person's behavior in a game can tell something about the person's behavior outside the game. For example, gamelike mechanics can be used to reveal information on user experience and preferences, which in turn help in pricing products (e.g., Samsung Nation). Similarly, Reeves and Read (2009) suggest implementing competitive guild-like structures imitated from MMORPGs at workplaces to sort out employees with the highest leadership potential. They argue that the social interactions that take place in guilds in MMORPGs sort out natural leaders in a more organic way than job interviews and occasional promotions due to good performance in a non-managerial position.

Last, one can argue that the popularity of Ebay and other auctioning sites is due not only to the successful implementation of basic principles from the mechanism design literature but also to the fact that people find auctions suspenseful and requiring the type of mastery seen in entertaining games. As the solutions to many mechanism design problems often have features like side bets (see, e.g., Crémer and McLean [1988]) and randomization over outcomes (Abreu and Matsushima 1992) originally seen mostly in gambling, in addition to having analytically desirable properties, these mechanisms could turn out to be engaging if implemented in an entertaining manner.

Behavioral Economics

In contrast to neoclassical approaches, which assume that customers are rational and act based on their true conscious preferences, behavioral approaches acknowledge that people might not be so directly rational but instead have "bounded rationality" (Simon 1957) because a decision maker never has all the information about all the different outcomes of a decision or their likelihoods of occurring or the

processing power to calculate them. Therefore, decision makers use heuristics when faced with decision-making situations. These heuristics are rough but commonly efficient (mental) models for the optimal behavior in given situations. However, in certain situations these heuristics may misfire and lead to suboptimal outcomes or at least to different outcomes than the ones predicted by neoclassical theories. Behavioral economics studies how people deviate from the axioms of rational decision-making.

As discussed earlier, gamification can be seen as a means to increase value of certain options and thereby make people more likely to select them. However, in behavioral economics, as opposed to increasing or decreasing the value of the actual outcomes directly, behavioral biases can be seen as innate mechanisms within the decision maker that increase or decrease the value of given alternatives even though the actual external outcomes from the decision-making situation were not otherwise tampered with.

It would be impossible to cover comprehensively all the empirically demonstrated decision-making biases within this chapter. For a thorough investigation on the subject of biases, we recommend to the readers *Advances in Behavioral Economics* by Camerer, Loewenstein, and Rabin (2003) and *Thinking, Fast and Slow* by Kahneman (2011). Here, we discuss a few biases, which are especially of interest in the context of gamification.

Exploitation or Gamification for Good: Example from Prospect Theory and Temporal Choice

One underlying assumption concerning gamification has been that it can tap into the behavioral biases of users and, hence, have persuasive power beyond mere addition of value to products. Gamification has been dubbed *exploitationware* because it has been viewed as a cheap trick to entice people into activities that they would not otherwise want to do. This thinking originates partly from early Facebook games that intentionally attempted to invoke behavioral biases in order to make people purchase more virtual goods (Hamari 2011). For instance, many of the mechanics were based on degrading players' earned virtual goods unless they frequented the game or purchased virtual goods that prevented this. These mechanics were especially related to one of the cornerstones of behavioral economics: prospect theory (Kahneman and Tversky 1979). Prospect theory posits that people's value function is S-shaped and based on reference points instead of absolute values (figure 5.2). Furthermore, prospect theory shows that the value function is concave for gains and convex for losses, implying a diminishing sensitivity. This implies that mere gaining and losing have intrinsic positive or negative value. Moreover, studies show that losses have a greater impact on welfare than gains of a similar size.

Games that were seeking to increase player retention used this bias and attempted to anchor (on anchoring, see Tversky and Kahneman 1981) the reference point of the player as high as possible with regard to how much virtual assets the player owned. Consequently, knowing that losing those assets has a proportionately greater impact on the player, the designers came up with mechanics that would delete those assets unless the player took actions the designers wanted them to: to come back in the game. When looking at gamification from this angle, it is not surprising that some perceived it as tricking people into activities that they would not want to do.

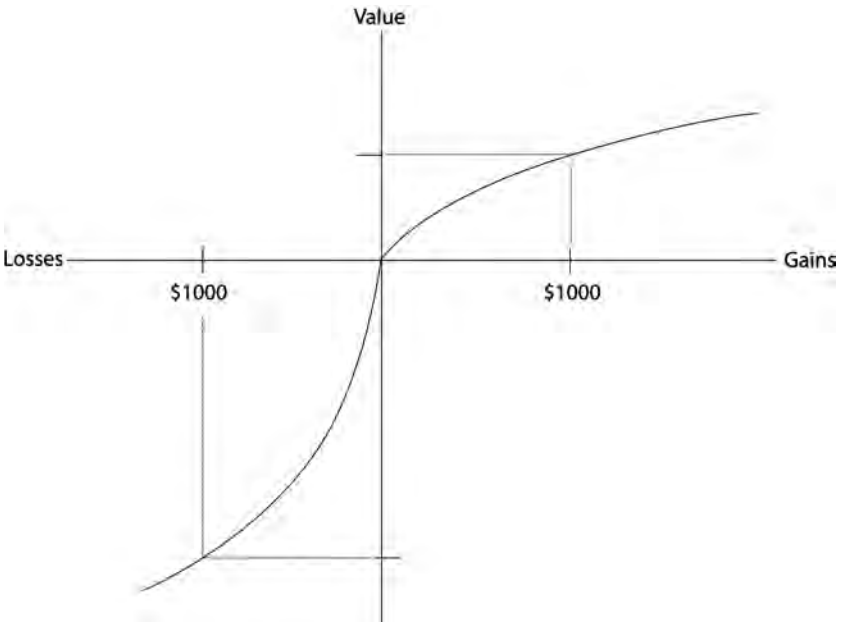


Figure 5.2
The S-shaped value function of prospect theory.

The diminishing sensitivity to gains and losses has clear design implications for gamification: to maximize perceived total value of rewards, the designer should separate gains into small chunks and conversely losses should be combined into larger chunks (see figure 5.2). This way the player does not have to suffer the extra loss multiple times but can enjoy the extra gains from multiple rewards, and the total perceived value of the players could increase dramatically.

Just as the previous example shows, gamification does not necessarily "trick" people into behaviors they would not otherwise pursue. In fact, most gamification seems to attempt to do the exact opposite by tapping into cognitive biases to nudge a decision-maker's behavior into her own desired direction.

A case in point is "hyperbolic discounting" (Ainslie 1975), which refers to a decision-making bias to favor outcomes that maximize short-term gains rather than long-term gains. In gamification and games, this phenomenon is related to favoring *instant gratification*. In the caveman era, it could have been favorable to eat all the food once it was acquired because storing was frivolous and theft was a pertinent problem. However, in contemporary society, where there might not always be a serious need to enjoy the short-term benefits, but instead a longer-term plan would be optimal for overall benefit, hyperbolic discounting can lead to extremely negative outcomes. Nevertheless, hyperbolic discounting bias still looms with outcomes such as procrastination, skipping exercise, smoking, and overconsumption. People

commonly notice these problems in their behavior and seek a variety of ways to remedy them. In fact, there is already a plethora of services attempting to gamify these kinds of difficult choices and behaviors that are commonly ridden with hyperbolic discounting problems, such as fitness (Fitocracy), housekeeping (Chorewars), and even keeping up with one's aspirations in life (Mindbloom). All of these services attempt to structure these relatively long-term activities in a way that is imbued with short-term goals and rewards. Clearly, gamification has been used for encouraging people to make "good" decisions.

In behavioral economics, there is a concept that is closely related to gamification. This optimistic view to behavioral biases is called *choice architecture*. It is a form of soft paternalism: "it tries to influence choices in a way that will make choosers better off, as judged by themselves" (Thaler and Sunstein 2003). It aims to design decision-making situations in such a way (Thaler and Sunstein 2008) that harmful biases could be avoided and/or beneficial biases amplified. Following this line of thought, it can be argued that behavioral economics or gamification have no intrinsic exploitation built into them. The goals of the gamifying party, in the end, determine what purposes gamification is implemented for and how it is designed to affect user/customer behavior.

Risk and Ambiguity

Earlier in this chapter, we discussed certainty and uncertainty and how gamification is being used to affect the expected value of outcomes by affecting the certainty by which the different outcomes are likely to manifest. Behavioral economics, however, adds an interesting twist to how people experience uncertainty. Empirical studies show that people are generally risk-averse and that people are even more hesitant to partake in behavior if there is uncertainty about the level of uncertainty. In these situations, decision makers are generally unable to combine the two levels of risk into a single uncertainty factor. This creates a higher perceived level of uncertainty, which translates into negative value and leads to avoidance of such alternatives. This phenomenon is called ambiguity aversion (Fox and Tversky 1995), and it contradicts the independence axiom in the Savage model presented earlier (Savage 1954).

For instance, a major problem in market design is how optimally to match resources to such value creation where most value would be produced (Roth 1984). In practice, this means, for example, the problem of optimally matching people to professions in order to maximize economic growth. Choosing one's education path is a tricky decision process. The longer one studies, the more risk one exposes oneself to. With vocational schooling, which is commonly shorter in duration, the risks of getting a job and decent salary are easier to estimate compared to a longer-duration university education, where it is more difficult to estimate the subsequent employment and salary level. Although it would be generally believed that longer education leads to better employment, the looming multilevel risk involved leads to overvaluation of shorter-duration education, because the odds are easier to grasp. Therefore, from a behavioral perspective, the lack of information might be an even more difficult problem because people are less likely to select options that are clouded in uncertainty, although from a rational perspective there was no reason to suspect that the odds would be any worse even if they are unknown.

However, as also with risk aversion, people tend to be ambiguity-loving when the expected outcomes

are negative (Wakker 2010), simply because people are loss-averse and the ambiguous alternative provides some chance not to have an outcome that is worse than the reference point (Kahneman and Tversky 1979). This has been articulated as the "end-of-the-day" effect based on betting behavior toward the end of the day when people are likely to take big risks in order to prevent losses (McGlothlin 1956; Ali 1977). End-of-the-day effects are also familiar from games. People who are likely to lose start taking bigger risks if they want to have any chance to win. However, taking a more risky gamble is also more likely to lead into an outcome where the decision maker would be worse off. In summary, with ambiguous gain situations, people tend to be ambiguity-averse and the opposite in situations where the decision maker is losing. In both, either avoiding or "loving" the ambiguity can lead to suboptimal outcomes.

As with uncertainty issues described in the previous section, remedies for situations where ambiguity aversion or ambiguity favoring are relevant are also related to either increasing or decreasing knowledge depending on the direction in which it is favorable to nudge the decision maker. Also here, Foursquare is a prime example of a service that produces more knowledge about restaurants in terms of quantitative (visit amounts) and qualitative (reviews) information. Here gamification is used to share information.

Setting Goals

Games are excellent in getting players committed to goals (Locke and Latham 1990). Therefore, gamification is an especially interesting phenomenon from the perspective of consumer behavior. Consumer behavior literature has found that mere goal-setting increases performance in three ways: (1) people anchor their expectations higher than what is expected of them, which in turn increases their performance; (2) assigned goals enhance self-efficacy; and (3) completion of goals leads to increased satisfaction, which in turn leads to increased future performance with the same activities (Bandura 1993). These effects are further strengthened if the goals are context-related, immediate, and the users are provided with immediate feedback. Research also suggests that if the goals are clearly specified in terms of how many times they have to be completed, the rate of completion of the tasks increases (Ling et al. 2005). Moreover, even only the sense of progression can invoke more of the same behavior (Kivetz, Urminsky, and Zheng 2006; Nunes and Drèze 2006).

Accordingly, badges (e.g., Hamari and Eranti 2011) for instance provide all of the above-mentioned aspects that increase goal commitment. Badges assign clear goals to which users can anchor their expectations as to what is expected of them. Furthermore, badges are commonly implemented via metagames that are directly context related and therefore tap into a context that is already meaningful to the user. Immediate feedback can further strengthen self-efficacy and satisfaction during the interaction with the system. By tweaking goals and progression, designers can further tap into behavioral biases related to, for example, goal gradient effect (Kivetz et al. 2006) by altering the perceived rate of progression (Nunes and Drèze 2006). However, there seems to be evidence that the successfulness of such gamification effort is context dependent (Hamari 2013). This suggests that although badges are effective in games, outside games users might not get committed to the goals badges provide.

Technological Persuasion

Another related, more technologically oriented vein of study, called persuasive technology or captology, concentrates on changing people's attitudes and behavior. Persuasive technologies are interactive computer systems designed to change the attitude and/or behavior of the user mainly through IT-based communication and social influence (Fogg 2003). The term thus clearly overlaps with gamification. For instance, some persuasion mechanisms can be regarded as similar to those applied in gamification, such as feedback and rewards (see, e.g., Hamari, Koivisto, and Sarsa 2014; Hamari, Koivisto, and Pakkanen 2014; Oinas-Kukkonen and Harjumaa 2009).

In marketing and information systems sciences, there is a long tradition of studying the interaction of technology, psychological outcomes, and behavioral outcomes. Gamification thus seems to fit well into the existing enquiry within these fields as an expansion of consumer behavior literature related to hedonic consumption (Hirschman and Holbrook 1982) and to intrinsic motivations (Deci and Ryan 1985). Therefore, within a long-run research technology acceptance (e.g., Davis 1989), gamification could be viewed as an attempt to invoke hedonic aspects (van der Heijden 2004) and intrinsic motivations (Deci and Ryan 1985) in order to make systems with utilitarian outcomes more appealing. With the increase of such aspects as perceived flow (Csíkszentmihályi 1990), enjoyment, and mastery, gamification seeks positively to influence continual usage intentions (Bhattacherjee 2001; Hsieh, Rai, and Keil 2008), actual use, and other relevant behavioral outcomes relevant to the gamified activity. Gamification offers an interesting vein in this continuum of research.

Gamification from the Perspective of Marketing

Now, let us take a look at gamification from the perspective of marketing, one subfield of economics. Before doing this, it is necessary to shed some light over the concept of "marketing." Over the years, there has been extensive debate over the role of marketing: whether it is a science or only a standardized art (Sheth, Gardner, and Garrett 1988). North American scholars have had a tendency to see marketing as an organizational function executed by one department in a company, whereas European scholars have emphasized the importance of marketing as a mindset or philosophy that should be spread throughout the organization.

Regardless of the emphasis one prefers, examining gamification through the lens of marketing is necessary, as many examples of gamification, from LinkedIn's persuasive user profile to Gmail's gamelike invitation systems to the way cafés use Foursquare to engage their customers, are so evidently connected to these companies' customer acquisition and retention strategies, which are clearly in the heart of marketing. Furthermore, many marketing strategies from customer loyalty programs to direct sale concepts such as Tupperware parties can be viewed as gamification strategies with fixed rules such as points, levels, and playful interaction.

The challenge, however, is that the term *gamification* does not have its roots in marketing theory or in economics in general but stems rather from

Box 5.1

Foursquare

The service Foursquare is a location-based social networking service that was founded in the United States in 2009. It allows its users to check in at physical venues by using a mobile application, mobile web browser, or by sending an SMS text message. Users of the service may "friend" other users and thereafter be notified of their respective check-ins. Users receive points for each check-in, and when a user hits a milestone, she is awarded with a badge. There are many types of badges and multiple ways to earn them. Some are linked to frequency, some to location, and some to the variety of venues visited. The user with the most check-ins in a given venue within a certain time frame becomes the mayor of the venue. Users can also upload photos to the system and write venue-related tips to other users.

game studies. Thus, there is no single definition for gamification that would fit the various theoretical constructs of the scientific marketing literature. Therefore, in this section, we will discuss marketing and some prominent paradigms within it using Foursquare in various service environments as an example (box 5.1). We will look at transactional marketing theory, relationship marketing theory, service marketing theory, and experiential marketing. This list of theoretical approaches is by no means exhaustive, but we hope that it will provide a relatively rich view on the subject at hand and that this introduction will spark new ideas both for practitioners as well as for academics.

Transactional Marketing Theory

Transactional marketing theory emerged in the 1950s to serve manufacturing industries. Its aim was in creating markets for mass-produced goods by connecting supply and demand. It was based on the belief that competition and self-interest are the drivers of value creation (Sheth and Parvatiyar 1995). Transactional marketing theory sees marketing as an independent function inside a company and the customer as a rational decision-maker.

From this classical marketing theory perspective, Foursquare can be seen as a combination of a loyalty stamp card system and a peer-to-peer advertising tool. By checking in to a café through Foursquare, the customer receives a stamp on her virtual loyalty card. After a certain number of check-ins to the same café, the customer receives a discount coupon from the café. Thus, it is the reward of the gamified service that gives the monetary incentive for the customer to participate in the campaign. From the service provider's point of view, the mission of gathering points (i.e., the series of future check-ins) will connect future production with demand that will possibly lead to future sales and additionally promote the firm to other customers. Thus, gamification when seen from the perspective of transactional marketing theory can be considered as game-like mechanisms that attract customers' attention to the product and as a means to segment different customer groups from one another (see, e.g., Hamari and Tuunanen [2014] for segmentation of players).

Relationship Marketing

Relationship marketing (RM) emphasizes the relationship that builds between the customer and the

firm in the long run (Sheth and Parvatiyar 1995). Relationship marketing scholars argue that relationship-oriented marketing practices date back to the preindustrial era. RM aims to enhance marketing productivity by achieving efficiency and effectiveness through long-lasting relationships that are based on mutual cooperation, commitment, and trust. RM acknowledges also the importance of other interaction parties (other customers, competitors, third parties) to the relationship. Customer relationship management (CRM) can be characterized as techniques to handle customer relationships in practice (Gummesson 2008).

From the RM perspective, a café owner can see Foursquare's check-ins, tips, and badges as ways to understand better his customers by observing what sort of drinks they are fond of. He can also propose new missions to loyal customers by rewarding those customers who try out new products or visit multiple locations. Thus, Foursquare incorporates features of a CRM system. Potentially, Foursquare can be seen also as a platform for cooperation with other service providers. A restaurant and a café can, for example, propose a common mission and reward. From the customer's point of view, Foursquare is a way to express commitment to a service provider.

Service Marketing

The pioneers of service marketing theory claimed that transactional marketing theory that was based on the exchange of goods was not suitable for service industries and that a new theory was needed for the purpose of service marketing.

Service marketing emphasizes the process nature of services. It sees marketing as a philosophy that should guide people, processes, functions, and departments in an organization (Grönroos 2007). It considers the customer always as a coproducer of the service and that value is created when the customer uses the service. *Service-dominant logic* (Vargo and Lusch 2004) takes service thinking further and suggests that this approach should be applied to marketing in general, as in the end goods are only distribution mechanisms for service provisions that take place when customers use the products. Thus, all economies can be considered service economies.

From the service marketing perspective, gamification can be seen as a "process of enhancing a service with affordances for gameful experiences in order to support user's overall value creation" (Huotari and Hamari 2012). From this point of view, Foursquare enhances a café experience by proposing gameful elements to it in the form of badges and missions. They form together a service system that can lead to gameful experiences that differentiate the café from its competitors.

Experiential Marketing

Pine and Gilmore (1998) claim that as products and services become ever more commoditized, creating value requires memorable experiences that unfold over time. Experience is considered here as a subjective and internal response, encompassing both rational and emotional reactions of the individual to interactions with the firm, its brand, its service, or its product as well as with third parties such as other customers, press, competitors, and so forth. These experiences create memories that accumulate over time and affect the future experiences (Verhoef et al. 2009). Compared to relationship marketing and to service marketing, experiential marketing emphasizes the hedonic and emotional aspects of

consumption as well as the unique, individualized, and memorable aspects of experiences. An example of such an experience is a visit to Disneyland. Pine and Gilmore (1998) also claim that in a mature experience economy, service providers should always charge an admission fee for this sort of experience. However, Gupta and Vajic (2000) find charging a fee to be a separate issue from how an experience is created.

If one adopts Gupta and Vajic's view, it is easy to imagine how gamification can be used in customizing an experience and in making it more memorable. A café owner could use Foursquare's tips feature to design tasks for his customers and make the café experience more individual and game-like. He could also identify participating customers and, for example, greet them when they check in making their café experience much more personal. Also, Foursquare could be used to record and share these experiences using text, images, and stories, all familiar elements of games.

Discussion: Positioning Gamification

One of the missions of this chapter was to decipher what is the added value that gamification brings to the economic discourse. The task was not difficult not only because the concept does not originate from economic sciences but also because it touches so many variables of economics. Its application areas can vary from very small details such as the LinkedIn profile's progress bar to the core of a company's strategy as with Tupperware parties. It can also be applied to various domains from organizing teamwork at workplaces to affecting our decision making at dry-cleaners or to our social practices at a coffee shop. In addition, the mechanisms that can be used for gamification are virtually limitless.

Our observation when writing this chapter has been that when we investigate in isolation the effects, the means, or the persuasion methods of gamification from any given economics perspective—be it neoclassical economics, game theory, behavioral economics, or marketing—we do not find anything new or surprising. For an economic choice theorist, gamification patterns are just design features affecting people's choices either through making a given choice more valuable or making the competing choices seem less attractive. More subtle ways include affecting the information or uncertainty people have about outcomes or tapping into some of the behavioral biases that people exhibit in their daily decision-making. Such goals are shared by many existing real-life services that have very little to do with games. A game theorist adds to this that people's behavior when interacting with many social gamified design patterns can be analyzed using concepts from game theory. But this also applies to almost any situation that involves decision making in a social context. Last, as was pointed out in the previous subsection, many of the actual design patterns are almost identical to existing marketing practices.

However, we miss the forest for the trees with regard to the essence of gamification because of the high level of generality of the economic theories. When we take a step back and remind ourselves that in the heart of gamification is the individual's game-like or gameful experience, the usefulness of the term starts to become clearer. Gamification is a tool for service design that has two different goals. One is the desired economic outcome that the company seeks via gamification. The other one is

the individual experience leading to that outcome that gamification helps to provide. The novelty of the concept of *gamification* is that it binds these two goals together, and its beauty is in the fact that it incorporates the idea that in successful gamification, one goal cannot be reached without the other.

Conceptualizing gamification from this perspective helps us further to pinpoint specific psychological and behavioral outcomes that are related to it. We argue that through these constructs, gamification can, in fact, be conceptualized as a new concept. Gamification attempts to provide a set or a part of experiences reminiscent of games. On the first level, they can be experiences such as flow (Csíkszentmihályi 1990), intrinsic motivations (Ryan, Rigby, and Przybylski 2006), self-efficacy, autonomy, and perceived competence that on another level can increase, for instance, goal commitment, which can further lead to behavioral outcomes such as continued use (see Bhattacherjee 2001) and healthier choices. In this vein, however, it is also possible to use gamification to bring about behavioral outcomes that make the user worse off. Gamification stands out from other concepts, such as persuasive technology (Fogg 2003), which attempts directly to change behavior and attitudes through persuasion, and loyalty programs, which attempt to affect behavior by providing economic benefits. Furthermore, gamification could be seen as an overarching concept in the sense that it can be utilized to influence behavior in several domains by providing gameful experiences that subsequently can influence attitude and behavior or affect customer loyalty or decision making.

Notes

1. When the number of possible choices is infinite, one also needs a more technical smoothness or continuity axiom that guarantees smooth numerical comparability between groups of choice bundles. See, for example, Mas-Colell, Whinston and Green (1995) or Kreps (1988) for more details.

2. See http://en.wikipedia.org/wiki/Star_Tours:_The_Adventures_Continue.

3. See also the excellent treatment of both of the models in Fishburn (1970) and a complete treatment of Anscombe-Aumann and an introduction to Savage in Kreps (1988).

4. People are notoriously error prone at learning optimally from available information; see, for example, Anderson and Holt (1997) and Hung and Plott (2001) on Bayesian updating and information cascades and Friedman (1998) on the famous Monty Hall problem.

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POSITION STATEMENT

LOSING IS FUN

McKenzie Wark

If "Never work!" was the apex of critical strategy in the twentieth century, perhaps "Never play!" could be the same point of extreme negation for the twenty-first. It is of course almost impossible to never work, but it at least defines an ambition: to abolish wage labor and the commodity form.

It was and remains a surprisingly popular ambition. The cycle of struggles in the overdeveloped world in the late twentieth century took it as their lodestar, whether unwittingly or not. The whole counterstrategy of making work seem like something else, like play even, stems from the boredom that both wage labor and the commodity form generated.

This boredom is spreading even to what was once called the underdeveloped world. By their tens of millions, peasants left the land, in China and elsewhere, and in some cases ended up in the factories that make the world's commodities. But it's getting harder to keep them there. The riots and suicides at Foxconn are just the tip of the iceberg. Nobody much likes to do this sort of work if there's any option.

In the overdeveloped world, work is only one of the ways of creating a value chain and extracting a profit. The more advanced form of spectacular economy extracts value from play. This is why the critical slogan of the times might need to be "Never play!" Where play was some kind of alternative for so many late-twentieth-century avant-gardes, from the Situationists to the Fluxus movement to the New Games movement, the overdeveloped world in the twenty-first century is all about recuperating those energies, those desires, those appetites, for the commodity form.

There are several versions of the recuperation of play. Sony Playstation once had a perfect slogan for it: "Live in your world. Play in ours." The exciting, fun stuff was not to be found in the world of work and the everyday. It was to be found in another, much more interesting world, one branded and metered by Sony or one of its competitors. The meta-game among competing firms was to find the best ways to commodify all those playful urges that wanted something other than what the commodity offers.

While we might like to think, when we turn on our smartphones, that they are there for us to play with, it's more that the possession of one turns you into a nonplayer character. You are now emitting a string of data, about location and activity, with which Apple and Google and Facebook and Samsung and Amazon and all the rest get to play the meta-game. The game that seems to be for us is really for them. They play against each other, with us as the nonplayer characters, the meatbots.

The goal of the game is to turn the push-pull of data between us nonplayer characters and between us and our devices into money. It doesn't really matter how. It could be by selling things to the nonplayer characters. It could be by selling the nonplayer characters to others, to advertisers, for example. It could be by harvesting data from us and looking for patterns in that data that might suggest new ways of commodifying the game.

There's both a game and a play aspect to this, closed worlds and open-ended ones. The closed worlds are games or game-like activities in which play can be offered up, seemingly voluntarily, and from which value can be extracted in an orderly fashion. There will always be cookies. They are not for you.

The open-world play spaces are a bit different. They concern the design of the games themselves. Every interaction with your laptop or tablet or smartphone yields moves in the game, but are also play actions that map the potential space and possible design flaws of the games themselves. To play is also to game-design, to yield up bits of an aggregate of play-test data, which shapes the future iterations of the devices and software themselves. One does not buy products any more so much. One buys prototypes, with which one plays to yield design cues for the next prototype.

Its like Philip K. Dick's novel, *The Game Players of Titan*. It seems like we are playing some vast and incredible game, but really we are the tokens, not the players. It's the Vugs that play, and they play on Titan, on another world, in a meta-game of which one occasionally gets hallucinatory glimpses. Apple or Google or Samsung look in one light as if they are terrestrial companies. In weird moments, one sees them rather as the Vugs of Titan, playing their own meta-game by their own rules.

But there's another game, a meta-metagame. A game both us nonplayer meatbots and the Titans play. Both us and the Vugs like to think there's other worlds. We get our Sony Playstation–type games to play in, they get their meta-game that games our interactions with those games. But both are just subsets of the meta-metagame: a game that has levels, of increasing difficulty, but in which you can't start over. There's no reset. Its slogan is not Sony's, but the slogan of Dwarf Fortress: "Losing Is Fun."

One version of the meta-metagame is called climate science. It's a game that has a lot of distractors. We notice mostly the other players and make our gamer identities based on our rank against each other. Occasionally we see the Vugs. Your social network provider changes the rules to extract more value, so you quit and chose another one. Facebook (or whoever) loses a meta-game point—but not to you. They lose it to whoever you give your playtime to next. As for the Vugs, they don't notice much. They think they are on Titan. They think they have someplace else to go.

But there really is only one meta-metagame. All the games and meta-games are nested within it, like Easter eggs. Games, in their separateness, always have an externality. There's always a resource external to the game that its internal resources draw on. If it's a computer game, for example, there's always the power cord or the battery that powers the game and its internal decisions. This externality is doubled. Play always has an external input, but also an output that is put back outside the bounds of the game. There is always waste. There is always something not accounted for in the score, the result, the decision. And so there is always a meta-metagame, beyond the games and meta-games, the root game in which both externalities meet.

To play the game is always to treat as purely external the input of energy and the output of waste. And while games are in a sense always systems, they are always closed systems. We used to think that the closed systems of our games and meta-games nested inside an open system from which they drew freely and into which they could quietly extrude any remainder. But it turns out that the game at root is also a closed world. It has an external input—sunlight, source of all our power-ups. But it has nowhere for outputs to go. The game is closed.

That's why, if there's a game that might be emblematic for our time, it's Dwarf Fortress. It's a game that has very little traffic with the meta-game. Just go download it, play it, send its designers some money as a gift if you like. There's no data trail issuing from it. It doesn't help any Titan to battle another for the world's resources.

And yet, despite its tiny size, it opens up into a remarkable world, with a physics engine that generates realities your characters may not even touch in their play. And if, like me, you are less than totally dedicated to playing it, you will lose. Again and again, and badly. And each time you play, and lose, the givenness of a whole world will appear briefly, then wink out of existence. It's excellent training for these times.

6 PLEASURABLE TROUBLEMAKERS

Marc Hassenzahl and Matthias Laschke

"That evening despite best intentions, Rebus took a cab from the guest house to the pub" (Rankin 2012, 254). Not only has the notorious, always thirsty Ex-Detective Inspector Rebus acted now and then against his best intentions. We all too often fail to skip this tempting glass of _____ (enter your personal liquid vice here). We indulge in activities neither physiologically nor psychologically healthy: overworking, overeating, overspending, overdrinking are common problems in Western societies. But we do not only yield to the bad, we abstain from beneficial activities, too, such as a little run now and then, spending quality time with the children, or leaving the smartphone in the closet—just for one day.

In many cases, people are quite aware of the "dos and don'ts" of daily life (at least the next morning everything appears painfully clear). But they have a hard time *implementing* presumably ideal behaviors. We *want* to "change ourselves," but need help. Besides the classic means of self-help, such as books and encounter groups, interactive technologies added viable alternatives. The Quantified Self website (http://quantifiedself.com/) boasts a collection of more than five hundred interactive apps and gadgets for tracking activities, moods, and other aspects of life. Fitbit's One, for example, is a stylish activity and sleep tracker (figure 6.1). Hooked up to a computer, it crunches numbers and creates visualization of activity levels and sleep quality. Or take the Nike+ running system emphatically discussed by Sicart in this volume. These tools are meant as a means to get insights into one's own behavioral patterns for changing them when believed necessary.

Another all-encompassing trend is *gamification*. In a recent review, Deterding and colleagues defined gamification as "the use of game design elements in non-game contexts" (Deterding et al. 2011, 10). They highlight the richness of approaches already available in technology-oriented disciplines, such as human–computer interaction (HCI), interaction design, and interactive games. McGonigal (2011) went a little further in her definition. She argued that in fact "reality is broken." Daily life is marked by a lack of ways to derive pleasure from living it. Games offer worlds of adventure and stimulation. So why not enrich the real world with game-like aspects, turning the mundane into playful activities? This adds an important element of gamification beyond the mere use of game elements: to make particular behaviors a little more pleasurable to increase their occurrence. Other than self-quantification, gamification implicitly proposes and rewards particular "ideal" behaviors—whether to the benefit of people themselves or to the benefit of companies is a matter



Figure 6.1
Fitbit's *One*. (Source: fitbit.com)

of fierce arguments (e.g., Bogost 2011; refer to Selinger, Sadowski, and Seager, this volume).

So every now and then, people want to change. We agree with self-quantification, gamification, and persuasive technologies (Fogg 2003) in the envisioned power of technology as a means to support this wish. However, we believe it needs more attention to *how* this should happen. In this chapter, we develop the notion of objects as pleasurable troublemakers—things with attitude—and outline a potential *aesthetics of friction* as a bundle of underlying principles to support the change (or better transformation) of people (see Hassenzahl 2011; Laschke, Hassenzahl and Diefenbach 2011).

The chapter starts with the discussion of choice as a central element of change. Change often requires forgoing immediate pleasures (tasty chocolate bars) in favor of long-term goals (being slim and healthy) or personal pleasures (driving fast) in favor of societal goals (making roads a safer place). In those situations, people experience a gap between their actual self (propelling toward being a weak, flabby, and sad chocolate addict) and an ideal self (being a slim and healthy chocolate connoisseur in absolute control)—a gap they want to close. This implies the deliberate choice of and ultimately reflection about appropriate courses of action. In this view, forgoing chocolate to be healthier constitutes change; forgoing chocolate because the stock of the local supermarket ran out does not.

Typically, appeals are used to instill reflection, which then in turn may result in behavioral change. In this chapter, we argue to turn this upside down. Instead of changing the mind first, we intend to change the behavior first—at least momentarily. Objects seem much better suited for this than appeals. They have the power to shape behavior directly, without much need to think. But instead of exploiting this quality to unconsciously nudge people to do better, we believe it should be used to create friction, moments of choice leading to reflection, insight, and sustained behavioral change. Obviously, the way this friction will be designed—the *how*—matters immensely. To support this, we develop a set of

principles, an aesthetics of friction, and discuss two examples of objects developed according to these principles by our students and ourselves. This is meant as a lens to understand better the principles at work, to cultivate a sense of how suggested principles may materialize in particular design choices. We conclude with comparing our approach to self-quantification, gamification, and persuasive technologies and a brief discussion of ethical considerations.

Note that our work is design oriented and, thus, largely normative. It argues for a particular route to instill change, which is not necessarily the only one possible, but in itself viable and convincing: a route that reflects our values in how to bring change into the world.

Change and Choice

Detective Inspector Rebus has always a choice. He can stay for another Indian Pale Ale or leave. Later after his unintended pub visit, "he took the stairs rather than the lift—every little bit helped, as his doctor has told him at his last check-up" (Rankin 2012, 294). However, without pub or beer, Rebus would not drink. Without lift, he would take the stairs. But this does not constitute change. Change implies the development and implementation of alternative, presumably "better" behavioral tendencies in the face of *choice*.

Change as Battle against Impulses

In many cases, choice includes forgoing an immediate pleasure (a chocolate bar) for the sake of future pleasures (health). This is difficult, and the reasons for this are manifold. First of all, people may not know about the detrimental long-term consequences of alcohol and chocolate or the benefits of running. But even if they know, benefits in the future are always uncertain, whereas an immediate pleasure is not. Thus, it is just rational to find a pleasure more appealing rather sooner than later, even when the later is a little larger. Every piece of chocolate now is better than two pieces tomorrow—simply because one can never be sure that there will be a "tomorrow." In economic theory, *discounting* the future is a classic (Samuelson 1937), responsible for wicked concepts such as "paying interest."

While paying interest is certainly annoying (getting some is presumably less so), it does not fully capture the emotional charge of any attempt to delay the consumption of a pleasure. In their famous experiments on delay of gratification, Mischel and colleagues (for an overview, see Mischel, Shoda, and Rodriguez 1989) offered children a cookie. Imagine yourself sitting in a room, a crunchy, tasty cookie under your nose. The experimenter has to go on some errands but will be back soon. You are free to eat the cookie. But when the cookie remains untouched until the experimenter's return, you get a second one. Oh boy, this is difficult for a four-year-old (amusing videos are available on YouTube, just use the keyword "Marshmallow Test"). It becomes manageable, though, for older children and adults.

Metcalfe and Mischel (1999) offered a *hot/cool systems* model to explain the underlying psychological processes. The model assumes an affect-driven, hot system pressing to immediate consumption and a cognitive, cool system trying to control this. While

the hot is automatic, the cool is a matter of training and available cognitive resources. This view understands self-control or "willpower" (the cool) as an acquired technique counteracting urges potentially detrimental in the long run. It explains breakdowns of willpower, for example, when stress limits available cognitive resources for self-control. It also explains why it is difficult to skip a last drink. Alcohol in itself decreases cognitive resources (Easdon and Vogel-Sprott 2000). Two-system models akin to Metcalfe and Mischel's are abundant in psychology. Hofmann and colleagues (2009, 164), however, conclude that all "these models share the general assumption that structurally different systems of information processing underlie the production of impulsive, largely automatic forms of behavior on the one hand and deliberate, largely controlled forms of behavior on the other."

Besides the obvious difficulties involved, the ability to restrain impulses (e.g., eating this hamburger, going on a rampage with the boys, staying in bed in the morning) in favor of long-term goals (e.g., remaining healthy, having a fulfilling relationship, earning money) is highly adaptive (e.g., Tangney, Baumeister, and Boone 2004). In a recent study, Schlam and colleagues (2012) found that each minute a child was able to delay gratification corresponded to an 0.2 reduction in body mass index thirty years later. Admittedly, not a large effect, but—as the authors of the study put it—certainly noteworthy. However, people may lack knowledge about the drawbacks of a certain behavior or the benefits of others and, thus, have not acquired appropriate *restrain standards* (Hofmann, Friese, and Strack 2009). But even when standards exist, self-control needs cognitive resources, which can be depleted for a number of reasons. In this case, impulses will win the battle. To sum, change often implies favoring long-term goals to immediate pleasures. The choice involved is, thus, the rather affect-laden balancing of "hot" immediate impulses and "cooler" long-term personal goals.

Personal versus Common Goals

It seems necessary to distinguish individual change pertaining to personal goals from change pertaining to common goals. A pint of Ben and Jerry's Chocolate Fudge Brownie each night may not only pose a problem for your waist, but also a problem for the cows providing the milk, the hens delivering the eggs, and the farmers growing the cocoa. In Germany at the time of writing this chapter, Ben and Jerry's asked its customers to support an initiative to improve the standard of farming dairy cows—"Schenk den Kühen Deine Liebe" ("Give your love to the cows"). While caring about one's own health in the face of the temptations is already difficult, caring about the well-being of an anonymous cow, hen, or farmer delivering the ingredients for the vice seems even more out of reach.

The classic example for this type of complication is Hardin's *tragedy of the commons*:

> Picture a pasture open to all. It is to be expected that each herdsman will try to keep as many cattle as possible on the commons. ... [A herdsman] asks, "What is the utility to me of adding one more animal to my herd?" This utility has one negative and one positive component. ... Since the herdsman receives all the proceeds from the sale of the additional animal, the positive utility is nearly 1. ... Since, however, the effects of overgrazing are shared by all the herdsmen, the negative utility for any particular decision-making herdsman is only a fraction of 1. (Hardin 1968, 1244)

In terms of individual gain, it is only rational to put more cattle on the commons. In the long run, however, the general overgrazing will destroy the commons. Any herdsman can refrain from maximizing her gain to save the commons, but the expectation is that others won't necessarily be as cooperative. The result would still be a destroyed common, but without having any additional individual gain from it. The difficulty of cooperation in situations that involve other people (or even generations) adds to the general problem of attaining personal goals. For example, using the car instead of public transport has a number of personal benefits (e.g., freedom, convenience) and only a few personal long-term drawbacks (e.g., expenses). To exchange the car on a rainy morning to an overheated, stuffy bus is always difficult to implement. Shifting the future goal from an at least slightly personally relevant "to save money" to the elusive "to promote the proper use of nonrenewable resources" is not making it easier. But exactly this trade-off lies at the heart of many sustainability issues, be it nonrenewable resources, water, traffic, or poverty.

The Ideal Self as a Driver of Change

While it is certainly important to be aware of the social as a further complication, we do not believe that this merits an entirely different approach. At the end of the day, it is individual behavior and individual choice that determines change. Behavior is driven by impulses and restrained by standards. These standards are derived from individuals' notion of their idealized selves. The ideal self is how we want to be—aspirations and goals set for ourselves. A discrepancy between the actual, experienced self and the ideal self is a reason for lack of positive affect (Higgins 1987). Working toward the realization of the ideal is a motivator and a source of pleasure (Carver and Scheier 1998): "[P]eople act to 'be' who they think they want (or ought) to be by adopting any of the guiding principles that are implied by the idealized self to which they aspire" (Carver and Scheier 1990, 20). The content of the idealized self will differ among people. It may comprise more self-oriented principles, such as "being healthy," or more others-oriented principles, such as "being just." Some people may have a balanced distribution of proself and prosocial principles. Others may tend to be more proself or more prosocial. The point here is that the general process driving change is the same, no matter its content. Change can only occur when there is a perceived discrepancy between one's actual and idealized self. The motivation to close this gap is what leads to change in behavior. Ultimately, those gaps are responsible for choice conflicts and the need for self-control. Take John Rebus as an example. The fictitious character might experience a tension between his actual self and ideal self. For the reader, Rebus is the ideal of a battered detective. No need to change—Ian Rankin seems very aware of this.

In Sum

"Changing oneself" revolves around a battle between impulses and long-term goals matched by a discrepancy between an actual and an idealized self. This includes (1) identifying personal long-term goals (i.e., remaining healthy, being just), (2) becoming aware of pleasurable activities that may be detrimental to these goals (i.e., eating a pint of Ben and Jerry's Chocolate Fudge Brownie every evening), (3) identifying alternative activities that are less detrimental (e.g., reducing the amount of ice cream, switching to

calorie-reduced ice cream, snacking on carrot sticks instead of ice cream), and (4) mustering the resources and according strategies to control the impulse to continue to consume ice cream on a daily basis (i.e., to live up to the restrain standard).

Note our emphasis on choice here (see also Selinger et al., this volume). It explicitly excludes approaches to change aimed at an unconscious level. Take organ donation as an example (Johnson and Goldstein 2003): 85 percent of Americans approve of donation, but only 28 percent signed a donor card. A simple way to close this gap is to change defaults. In countries with an opt-in policy, that is, people need to apply to become donors (e.g., Germany), average consent rate is about 15 percent. In countries with an opt-out policy, that is, people need to apply to be taken out of the donor registry (e.g., Austria), average consent rate is about 97 percent. Through "engineering" the choice context, here the *status quo* or default, people change from skeptic misanthropes (e.g., Germany) to enthusiastic philanthropes (e.g., Austria). But does change in behavior alone constitute change? We don't think so. To use nuggets from behavioral economics or the long dark night of behaviorism is as technocratic as changing people's energy consumption through energy-saving light bulbs. While a little "nudge" (Thaler and Sunstein 2012) is certainly helpful, the notion of a "liberty-preserving paternalism" is slightly off-putting. Thaler and Sunstein argue that while people should be free in their choice, it is also "legitimate for choice architects to try to influence people's behavior in order to make their lives longer, healthier and better" (Thaler and Sunstein 2012, location 115). We agree. But obvious enough that people become aware of the influence and attribute the better outcomes to their own choice.

Objects as Change Agents

In 1957, Achille Castiglioni and his brother Pier Giacomo designed *Sella* (figure 6.2), a stool made of a leather bicycle seat attached to a metal stem and having a rounded base. Castiglioni said about it (quoted in Antonelli 1997): "When I use a pay phone, I like to move around, but I also would like to sit, but not completely." As Antonelli points out, Achille and his brother's goal was to design a new behavior—"a hybrid between sitting and pacing nervously." There is the further unconfirmed story that Achille was keen on devising an elegant way to cut the time his brothers spent at the household's wall-mounted telephone. *Sella* is an object with the purpose to create and shape a new behavior—whether for the sake of a user's pleasure or Achilles' access to the phone remains a secret.

Objects operate at what Carver and Scheier (1998) define as the unconscious and automatic motor-level of action. Most things we do are object mediated. Eating is done with spoons, forks, and knifes. Writing needs pens or computers. Objects become assimilated into action. Driving a car, for example, needs some practice, but soon we forget the pedals, steering wheels, and gear sticks involved. Through practice, the motor aspect of an action quickly fades away from consciousness. But nevertheless, objects inevitably shape behavior through the opportunities and the quality of interaction they provide. A spoon invites one to gobble food in large portions; chopsticks invite pecking smaller portions. The difference in resulting behavior is a function of the spoon and the chopsticks—in a way

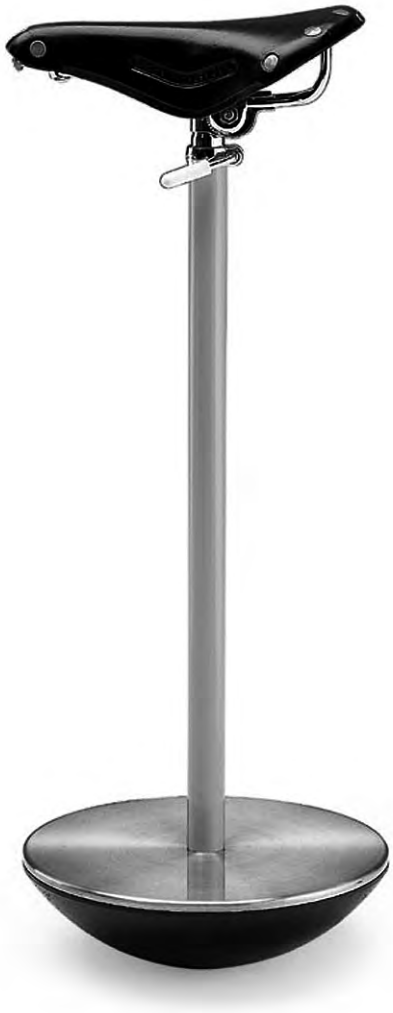


Figure 6.2
Sella. (Image courtesy of Zanotta SpA, Italy)

they resemble a "choice architecture," albeit not a deliberately designed one.

Objects versus Appeals

The crucial role of objects in behavior itself distinguishes them from appeals as the classic way of changing behavior. An appeal is a rhetoric top-down approach to change. It reflects the widespread belief that change of behavior is mainly the *result* of an insight. At the same time, people bemoan the apparent gap between what others say they'll do and what others then actually do (for a recent overview, see Sheeran 2002). As with Rebus, despite his *best intentions*, he ended up in the pub.

Notwithstanding, the dominant route to behavioral change taken at the moment is one of communication. Many believe in the power of appeals, in addressing the reflective level of behavioral control as a route to change. And in fact self-quantification, for example, is rather rhetoric than behavioral/interactive, although it relies on interactive products. Conceptually, however, it only creates visualizations and text that summarize people's behavior. This is nothing more than a personalized appeal to the reflective level. It does not address the impulsive behavioral level. Let's say, Fitbit's One reveals that you are not physically active enough. Specifically, you never use the stairs. This insight is the end of One's story. It will not prompt you in particular situations. It has no power to shape behavior. This is different than the stairs in figure 6.3. While still to some extent rhetoric, their intervention is much more situated. Information is tied to actual behavior, here steps taken, as an alternative to the more convenient escalator. The appeal is made at least in the moment of choice.



Figure 6.3
Stairs in the Kyoto subway.

The belief in appeals to the reflective level has consequences. Educational campaigns to stop smoking, to prevent alcohol abuse, traffic accidents, or HIV infection are abundant, but not always to much effect. In a review of health-related campaigns, Snyder (2007) noted only small behavioral effects, especially for topics that require a change in daily routines (e.g., more exercise) rather than the adoption of a new behavior only performed once or twice (e.g., to be vaccinated).

From Goal to Implementation

One recommendation (Snyder 2007, 37) derived from the review of the power of appeals was to rely on messages that deal with "how to" and "when to" knowledge. In the same vein, Gollwitzer (1999, 493) concluded "that it seems unjustified for applied psychologists to advise people who are motivated to do good to refrain from forming good intentions, but suggesting that good intentions are an effective self-regulatory tool is also unwarranted." Good intentions should be made more effective. His solution is *implementation intentions*. They differ from more abstract goal intentions ("I intend to stay sober") precisely by specifying the *when* and the *how* of the intended behavior ("Next time I am in a bar, I will drink a large mineral water before the next beer") (figure 6.4).

Implementation intentions are simple plans tied to particular situations. Gollwitzer suggests moving intentions consciously from the level of the idealized self down to the level of concrete action. Through this, desired action gets a better chance of implementation in a particular situation. The concreteness of implementation intentions facilitates competing with the dominant impulsive behavior.

Notably, objects are already on this level. One can think of them as materialized implementation intentions, existing outside a person. Through their materiality, they restructure situations and behavior. Take the reading lamp *Forget me not* (figure 6.5) as an example (Laschke et al. 2011).

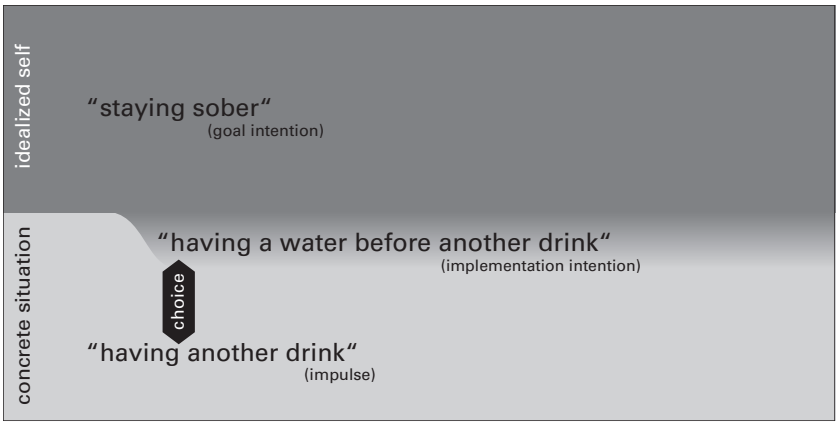
When switched on, *Forget me not* opens its petals like a flower and provides light. It, however, immediately starts to close again, thereby slowly obscuring and dimming the light. By touching a petal, *Forget me not* opens up and provides light again. Compared to a regular reading lamp, *Forget me not* creates a new default. A regular lamp's natural state is switched *on*. It provides light as long as the user pleases and requires a deliberate action to be switched off. In contrast, *Forget me not*'s natural state is *off*. It requires a deliberate action to be kept on. Like chopsticks compared to a spoon, *Forget me not* changes lamp-related behavior through the way it affords interaction. One may translate the goal intention "to conserve energy" into the implementation intention "when sitting somewhere cozy and using light, remember to now and then check whether you still need light" or one may just use *Forget me not* as a materialized implementation intention, embodying an alternative behavior, automatically acted out through usage.

Obviously, to impact behavior, an object needs to "reach out." It needs a certain power to restructure situations and reshape behavior. Tangibility certainly helps. Tangible objects, things to be touched, held, twisted, or thrown, simply offer a wider, more bodily potential to reshape behavior. In addition, tangibility implies an at least physical, but not necessarily conceptual, situatedness. The object is *there* and, thus, may be better tuned to a particular situation. *Forget me not* exploits the possibilities provided by the tangible, situated, and ubiquitous.

However, while the power to reach out expressively into the physical world certainly helps, it is by



(a)



(b)

Figure 6.4

(a) A conflicting goal intention tied to an idealized self and a concrete impulsive behavior tied to a particular situation. (b) Now, with an implementation intention to bridge the gap.



Figure 6.5
Forget me not.

no means a must. For example, we designed *ReMind* (figure 6.6), a wall-mounted to-do-list-like object, to overcome procrastination (Laschke et al. 2013).

Among many other features, it uses a physical representation of the personal goals to be reached in a certain time. In case the goal becomes overdue, it drops to the floor. This creates an affordance to pick up the goal from the floor and to reconsider: shall I just do it, reschedule it or abandon it altogether? It is this moment of choice we wanted to create, a simple story of "it's already in your hand, why don't you just get on with it?" Tangibility helps here, but nevertheless it is this choice that is crucial. Now, imagine *ReMind* as an app. A mobile phone cannot simply throw physical objects at its user to create a certain choice situation. Fortunately, this is not necessary. Instead of the floor, we rather litter the phone's lock screen. While being completely nonphysical, we nevertheless create a similar situation. A nasty, overdue personal goal on your lock screen is something you may feel an urge to get rid of. Let's say it is this long overdue phone call your half-deaf great aunt is so much waiting for. To remove the goal, you need to click or select it. Upon this it may offer a simple choice: to dial her number or to postpone again. Structurally, both are similar designs, one relying on tangibility, the other not. In this example, the intangible version may even have a little more power to change, because the possibility to immediately initiate the ideal action—the telephone call to be made—considerably lowers the barrier to make this call. The tangible version lacks this power.

Engaging People in Meaning Making

Forget me not seems similar to what Thaler and Sunstein (2012) call a *choice architecture*; maybe even similar to the change of the status quo already discussed earlier as a way to increase the number of organ donors. It makes light consumption a little more difficult, which may nudge people to use a little less. However, there is a difference. We assume that the change in behavior created and afforded through an object should not go unnoticed, but should be

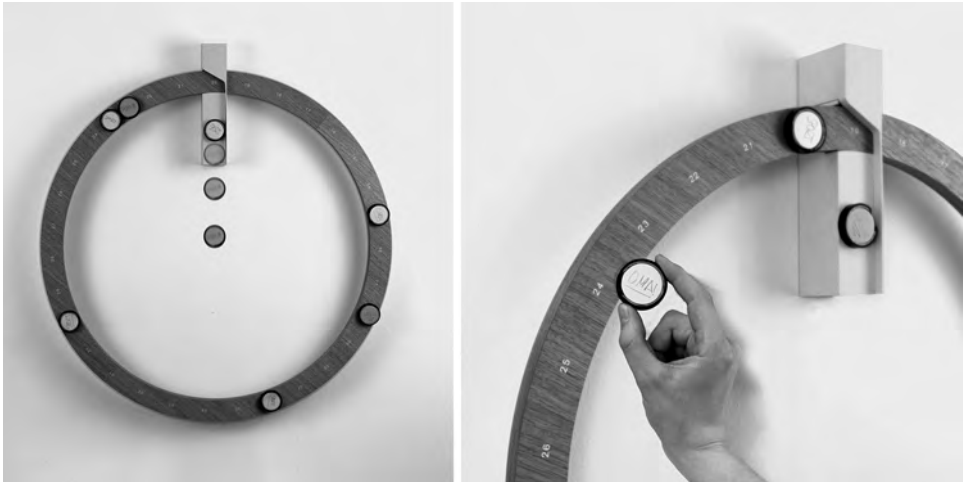


Figure 6.6
ReMind.

designed in a way to create a feedback loop to revise old and form new goal intentions.

In Gollwitzer's (1999) approach, implementation intentions are derived from and always remain related to their goal intentions. A person *knows* that she drinks a glass of water before ordering the next beer, because she intends to drink less alcohol. Objects as external, materialized implementation intentions, however, can either be linked to an active goal intention or not. The first case is straightforward. Let's say Linda holds an ideal of staying in control and being social. She identified her actual drinking behavior as a barrier to this and forms the goal intention to stay sober (presumably as a New Year's resolution). To be efficient, she needs to translate the mere intention into concrete strategies. Conventional wisdom (i.e., her friends) recommended steering free of alcohol all the time (i.e., teetotaling), but she read a book telling her that it is the most functional to be prepared to drink moderately on social occasions (see Rachlin 2004, location 78). Based on this, Linda has to devise simple plans as alternatives to her unwanted impulsive behavior. One trick she comes up with is to drink a glass of water before the next drink. All this requires a lot of knowledge and creativity on behalf of Linda. Instead, she could acquire an "object" in line with her general intention. The object then implements an intention-relevant alternative behavior Linda wasn't aware of until she found the object. For the sake of the story, imagine a bar where a beer is always served together with a large glass of mineral water. The bartender only asks about a refill when both glasses are empty. The bar as a system provides a new situated practice in line with the general goal intention of Linda.

Now imagine Colin. He is thirsty. He enters our bar and orders a large beer. What he gets, however, is a large beer *and* a large mineral water. Slightly confused, he thanks the bartender and gulps down the beer in a second. He is still thirsty, eyes the mineral

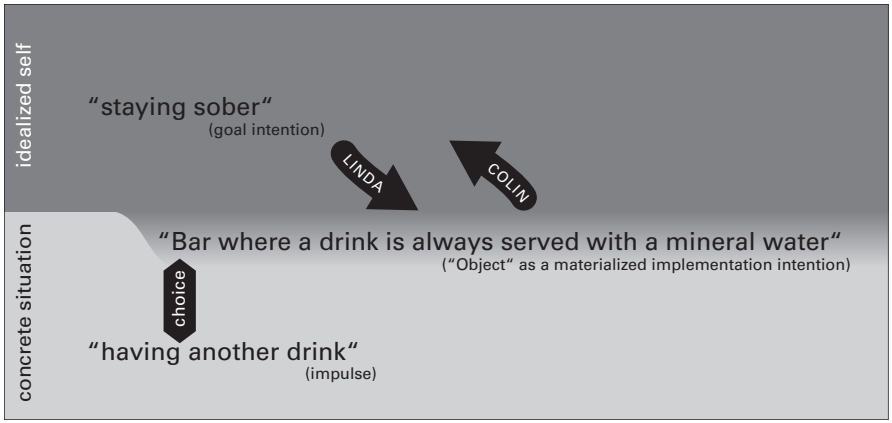


Figure 6.7
Linda and Colin's experience with the bar.

water suspiciously, but finally has a go. The next beer arrives, again accompanied by water. And so on. The next morning, Colin is surprisingly well off compared to previous drinking sprees. "Must have been the water," he concludes, "and it tasted not even half as bad as I thought. Next time, same thing." The bar changed Colin's behavior, which in turn made him think about his practices and potential alternatives (figure 6.7).

For many, the notion that behavior is an expression of higher-level ideals and goal intentions seems quite natural. The opposite direction, that is, to behave first (e.g., drinking a large mineral water before the next beer) and to derive intentions (e.g., "drinking less alcohol would be desirable") in an act of self-observation, seems less straightforward. In psychology this is a well-known fact. Recently, Wisemann (2012) provided an entertaining review of what he calls the *as if* principle. In short, available research shows that there is not only a flow of information from intentions to plans and behavior, but also a flow back from behavior to the inference of plans and intentions. We observe ourselves and when prompted may use this self-perception to infer relevant self-knowledge (e.g., Bem 1967). When the change in behavior shaped by an object is noticed, it induces meaning making. Thus, instead of lecturing through information—a rhetoric approach—objects endow people with new behavior to trigger reflection about intentions and principles. While for Linda the bar was a tool to implement her intention to drink less, for Colin it provided an opportunity to experience a course of action alternative to his regular, impulsive one. In turn, Colin may reconsider his actual and ideal self in the light of his own behavior the previous night, although this behavior was "made" by the bar.

Objects Tell Stories

Especially for Colin, the bar not only provides functionality but also tells a story. It is a "material tale" (Dunne 2006) or a "material argument" (Redström 2006). Note that this is not meant in a literal way. The object is not telling. It is neither a mere symbol nor

is it providing verbal or visual statements. In contrast, it transforms a situation by injecting "role-play" in line with particular goal intentions and ideals, shaped by the functionality provided and the interaction afforded. The user, in turn, finds herself in the leading role of a new play called, for example, "A night in the pub, without being hung over the next day." Looked at that way, *object* is a misleading term. We think of it rather as a designed situation, with objects as the major means to establish new practices in a given situation without appealing directly to the reflective level, but able to instill reflection.

In Sum

We believe that objects create unique opportunities to instill change. Other than rhetoric strategies, objects can literally embody alternative behaviors, either by serving as a materialized implementation intention (for Linda) or as a prompt to reconsider one's actual self (for Colin). In both cases, however, it is the person who acts and not the object itself. The object merely contains everything necessary to create an alternative narrative. It is a prop in people's play of change.

Pleasurable Friction

Colin's first encounter with our bar could have gone quite differently. Recall: Colin is thirsty. He enters the bar and orders a large beer. What he gets, however, is a large beer and a large mineral water. Slightly confused, he thanks the bartender and gulps down the beer in a second. He shouts for another one, but the bartender just knowingly stares at the mineral water. "You want me to drink that first?" Colin inquires. The bartender nods in reply. "What the heck," Colin mutters, "do you want me to drink less beer? Is that good for business?" "At least it is good for your health" the bartender replies.

Other responses are imaginable: Colin may be annoyed. He hates water and loves beer and never considered himself an alcoholic. He might become mad at the bartender and start a fight. There are plenty of ways this story could end.

The Importance of the How

We assume that the actual details of any intervention play a crucial role in how the story plays out. The *how* matters immensely. Just putting two drinks (i.e., a beer, a water) for the price of one in front of Colin may be a subtle but weak intervention—at least for Colin; Linda will presumably easily embrace the offer, because it is in line with her intentions. In contrast, strictly refusing a refill without two empty glasses is rather offensive, but strong. It might ensure that Colin has his water before the next beer, but it is likely to backfire. Threats to freedom lead to reactance (Brehm 1966). Instead of focusing on the behavior at hand (e.g., too much beer), people then focus on the fact that something or somebody (i.e., a communicator) restricts their personal freedom. This leads to an even higher likelihood of engaging in the restricted behavior.

Thus, on one hand objects for change need to and will be troublemakers rather than problem solvers. They change well-known situations, which certainly creates some *friction*. On the other hand, this friction needs to be designed in a way to become meaningful and acceptable to avoid reactance. In the remainder

of this section, we will develop such an *aesthetics of friction* to guide design.

Designing Friction

Industrial design, interaction design, and human–computer interaction have a tradition of making objects convenient—their ideal is things that fit the hand and the mental models of their users; things made to measure. Objects as change agents are different. They do not adapt to their users, but demand adaption. We believe that we simply lack the expertise to design for this. We need principles for the design of what we call *transformational objects* (Hassenzahl 2011; Laschke et al. 2011) or more affectionately *pleasurable troublemakers*.

Our notion of change (see the section "Change and Choice") and objects as change agents (see the section "Objects as Change Agents") restrict the potential design space for pleasurable troublemakers. Troublemakers flourish on the intimate understanding and knowledge of a situation at hand. Let's say your ideal self wants you to lose some weight. Unfortunately, the daily pint of Ben and Jerry's Cookie Dough in front of the television proves to be a barrier to your dream weight. A concrete and quite clever implementation intention is to separate television watching from ice cream munching (Rachlin 2004, location 1679): "I can eat ice cream if I must, but I need to switch off the television while eating." All you need to do is to follow this rule. However, you could also acquire our new ice cream bowl, which switches the television off when being lifted from the couch table. This bowl is not a rhetoric appeal to your reflective powers; it is an intervention on the impulsive level. It embodies an implementation intention in line with your goal intention, which reflects intimate knowledge of the problematic situation at hand and offers a viable alternative behavior. Now imagine the bowl as a Christmas present. Unsuspecting, you fill it to the brim with Cookie Dough and slump down on the couch in front of the television. Ah, John Snow, beyond the wall, amid of ice, wildlings, and white walkers. Absently, you grab your bowl, ready to tuck in and: John Snow freezes, literally. What's wrong? You put the bowl back on the table to check the television, and quite magically Game of Thrones continues. Satisfied, you pick up the bowl again. Frozen image. After a while you figured it out: the bowl does not want you to watch and eat simultaneously. You are clever, you've got the idea.

This example shows that pleasurable troublemakers must satisfy a number of crucial requirements to realize their envisioned potential: (1) They must be highly situated and relate to impulsive/automatic behavior and a moment of choice; (2) they must embody an alternative behavior in line with a potential goal intention and the idealized self; (3) they must be as close as possible to a moment of choice; (4) they must create some friction in the particular moment of choice to nudge their user(s) into a meaning-making process; and (5) they must possess a certain expressive quality, that is, the ability to tell a clear story of an alternative behavior and a better self.

Making Friction More Pleasurable

The bowl deliberately creates tension between ice cream munching and television watching. Ice cream eating does not become forbidden. Cookie Dough is not entirely banned from couch and kitchen. But still, the bowl is a restriction of its user's freedom. It is a troublemaker. As pointed out earlier, restriction of

freedom is likely to result in *reactance*. However, research (Silvia 2005) showed that reactance is reduced when the communicator is liked and appears similar. Thus, a *pleasurable* troublemaker not only should cause friction through more or less forcing an alternative behavior and a better self onto its user, but needs to be liked as well. It needs to build a relationship.

What are the potential means to create a bond between a person and a slightly annoying object? We believe in (1) naivety and (2) understanding.

Naivety

The bowl is limited in its capacities and not an especially smart object. It just embodies a single simple plan, stubbornly telling the same story over and over again: just don't eat ice cream while watching television to get a grip on your weight problem. Isn't it naive actually to believe that something as difficult as losing weight or specifically abstaining from Cookie Dough could be bettered by a silly bowl? There are many ways to work around the simple plan provided by the bowl; plenty of ways to cheat. You can just leave the bowl on the table and eat with the head bent down. You don't even need to use it, just eat the ice cream out of the box. This absurdity, the seeming hopelessness of the bowl's approach, makes it likeable. It is a Don Quixote fighting against windmills. It is the well-meaning, slightly freaky friend to comply with, just out of compassion. Let him have his way for now. It is the pathetic Wizard of Oz, handing out stuffed silk hearts to the Tin Woodman and useless magic potions to the Cowardly Lion.

But the bowl is only half as absurd as it seems. First of all, it embodies a carefully selected, smart course of action, which has the potential to take effect when acted out. In addition, such as the potion for the Cowardly Lion, the bowl will create a focus, potentially unlocking Cowardly Lion's own behavioral resources. Designers of pleasurable troublemakers believe in the power of small interventions.

Understanding

Doing the right thing all the time can be overwhelmingly difficult. Completely to deny oneself ice cream may not only be cruel, but may also require superhuman willpower. Nevertheless, teetotaling is a common recommendation, the "revolt against indulgence" (Rachlin 2004, location 791), when it is about unhealthy food, drinks, or smoking. The problem is that a small violation often leads to relapse. Imagine yourself being abstinent from ice cream for a week, but then giving in to just a spoonful of Cookie Dough. "What the hell!" you'll think and throw yourself into a full-fledged eating spree. A pleasurable troublemaker will acknowledge human nature, the difficulty of teetotaling—it understands. Because once indulged is not a pattern, yet.

A pleasurable troublemaker must be understanding in two different ways. First of all, the embodied implementation intention itself should acknowledge the difficulty of controlling impulses. The ice cream bowl does not require a superhuman. There is still ice cream munching and television watching involved, but in a slightly twisted, presumably better way. The same applies to Linda and Colin. The bar does not forbid a tasty pint of lager. It just innocuously suggests combining a beer with a mineral water to drink one or two beers less than usual. These strategies are soft and subversive rather than strict and explicit.

A second route is including a feature to sidestep the embodied implementation intention. Let's say we grant our ice cream bowl a further mode that can be activated by the user and then runs for a particular

time. During this, the bowl's function reverses. Now you need to dig in to keep Game of Thrones running. The moment you *stop* munching ice cream, John Snow freezes. The bowl now actively supports cheating, but thereby reflecting on an ultimate truth: you can cheat others, but never yourself. Including a feature to sidestep the embodied implementation intention adds an ironic element to the object. It wants you to eat less, more consciously, but at the same time, it can be used to create the opposite. On one hand, this puts emphasis on the fact that it is essentially our own personal choice, how we want to be. On the other hand, the same object embodying a seeming ideal can be used to transgress. Through this it becomes an accomplice, a "partner in crime," and certainly more likable.

In Sum

Our aesthetics of friction affords an object to be situated in a moment of choice, where it must create some friction. Its main function is to shape behavior according to an embodied implementation intention, which is in line with a better, happier self (i.e., the idealized self). To allow for an insight (i.e., an acted out implementation intention spawns a goal intention), the object needs expressive qualities—it needs to be capable of telling a story. The object itself bonds with its user through its naivety, a carefully selected implementation intention, and an ironical feature, making the object slightly ambiguous. Through this, the object becomes a *pleasurable* troublemaker. It mirrors its user, confronting her or him with his own ambiguities and inconsistencies.

Two Pleasurable Troublemakers

In the "Pleasurable Friction" section, we laid out our aesthetics of friction and already applied it to an ice cream bowl. In this section, we present and discuss two further pleasurable troublemakers as examples.

Do/Panic

Do/Panic is a student project realized by Tobias Ellinger and Philip Oettershagen and supervised by us (see Ellinger et al. 2011). Now and then, Do/Panic projects a grid or a line onto a desktop (see figure 6.8).

Design explorations showed that, when asked to get a grip on a chaotic desktop, people use a number of strategies, such as grouping items with similar form and size or aligning items orthogonally or parallel to the edges of the desk. The projected lines and grids take this up. They provide virtual containers to place similar things into or "demarcation lines" distinguishing free spaces from storage spaces. A line, for example, projected in the upper third of the desktop is used as a border, with all pens and tools placed above the line and resulting empty workspace below the line.

Do/Panic believes in the power of order and acknowledges the difficulty to maintain it in daily life—as Thomas Mann said: "Order and simplification are the first steps towards the mastery of a subject." (At least, this is what it seems to need to write something as complex and detailed as the *Buddenbrooks* or *The Magic Mountain*.) So now and then, Do/Panic suggests tidying up a bit by offering some grids and lines, relying on the power of this simple prompt. This is clearly a sign of naivety as required by our aesthetic

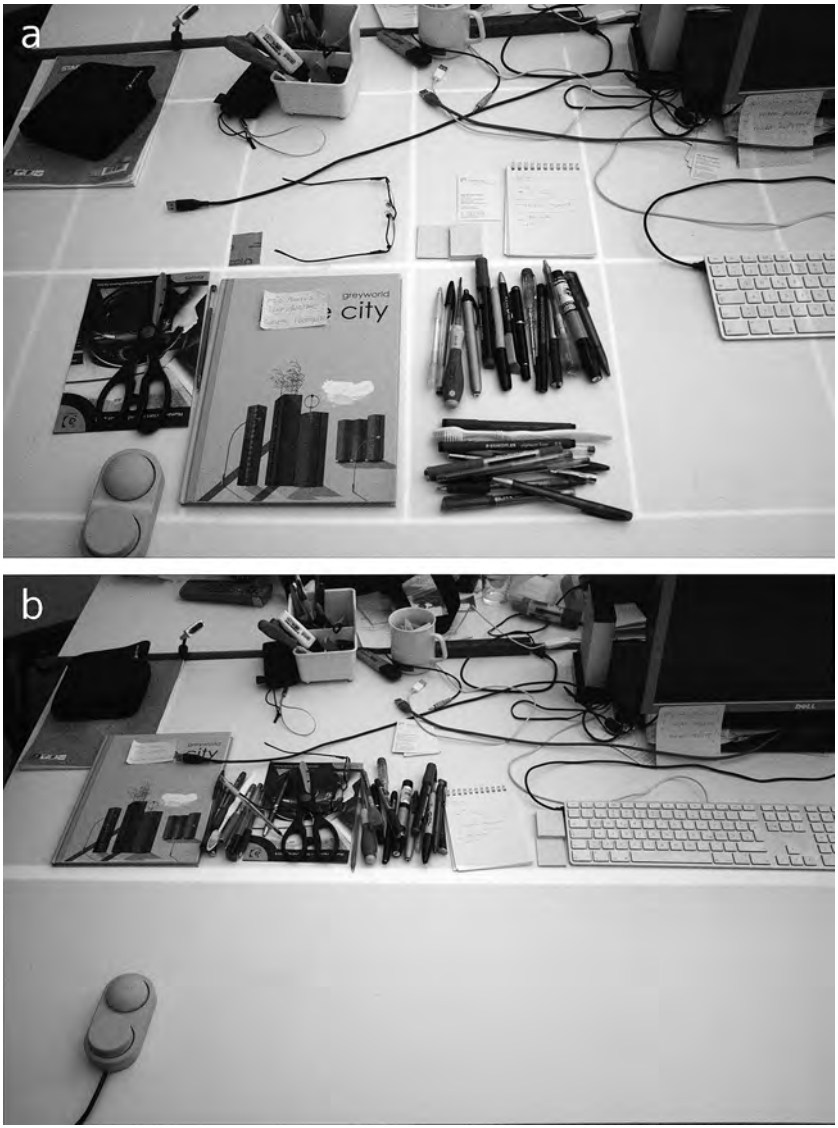


Figure 6.8
(a) A line or (b) a grid projected onto a desktop. (Image courtesy of Ellinger/Oettershagen/authors)

of friction. But quite surprisingly, the simple lines and grids create a pull, a strong desire to align items on the desktop to them. And before you know it, you'll find yourself tidying up the chaotic desk. While the object appears naive, the embodied implementation intention is quite clever. Instead of recommending putting things away, it stresses orientation and alignment (e.g., a grid) as a way to create order. This needs less time, less effort, and can be done while working. Order through grids is located halfway between total chaos and the notorious "empty desktop" of executive offices. It is a more transient, momentary order, one that expresses being in the middle of work rather than being almost on your way home. At least this is the impression the first author hopes to make, when his students take the occasional look at his virtual desktop (figure 6.9).

We believe that Do/Panic has the expressive quality to tell a quite concise story of order and tidying up. It is relatively easy to find yourself doing what it suggests. At the same time, its suggestion already acknowledges that tidying up can be gruesome, boring, and hard to implement.

But where is the irony of it? Do/Panic has an additional feature: a panic button. If the mess becomes overwhelming, but you cannot muster the energy to have a go at it, hit the buzzer. Do/Panic then projects a masking pattern onto the desktop (figure 6.10). The mess disappears—at least for a while.

Fifty/Fifty

Fifty/Fifty cake takes up the intention to lose some weight to feel healthier and more comfortable. As we



Figure 6.9
Hassenzahl's virtual desktop is made acceptable by aligning mess to a grid.



Figure 6.10
Do/Panic masks the mess. (Video stills, image courtesy of Ellinger/Oettershagen/authors)



Figure 6.11
The loaf pan for a Fifty/Fifty cake.

all know, this is easier said than done. There is a high number of potentially tempting situations and an equally high number of recommendations, such as to set up a nutritional protocol, to abandon high-calorie foods completely (i.e., teetotaling), or to replace some of it with apples, carrots, or diet drinks.

Most of these recommendations are not particularly well attuned to the structure of tempting situations. Fifty/Fifty cake addresses practices in the context of an afternoon kaffeeklatsch, an informal social gathering for coffee and conversation popular in Germany. A kaffeeklatsch is often accompanied by homemade cake, a tempting high-calorie food. However, the choice to abstain is difficult not only because of the tempting cake, but also because of the situation's social character. To reject the offer of a piece is—even when explicitly voicing one's goal to lose weight—slightly impolite. Certainly, it will urge the host to convince her guest to have at least a tiny tiny piece of the vice—just for the taste of it. In addition, it will certainly start a potentially embarrassing discussion about the actual goal intention "to lose weight," making conversational nuggets such as "You can do without this, my dear" (are you blind?), "But it suits you" (rolls of flab?), and "It's only cake" (blessed are those not knowing!) highly likely.

As its name implies, Fifty/Fifty is about two halves. The cake is made in a special loaf pan typically used for pound cake. Different than regular pans, this pan is diagonally divided (figure 6.11).

The idea is to use different sponge-cake mixes for the two halves. One mix is a regular one, the other is calorie reduced. A recipe for a regular mix for this size of pan would be: 6 eggs, 300 grams butter, 250 grams sugar, 250 grams flour, and a teaspoon of baking powder. This would result in a cake with 4,250 calories, about 350 calories per piece. A calorie-reduced mix just replaces the best part of the butter with low-fat yogurt (50 grams butter and 450 grams low-fat yogurt). This results in 2,800 calories, about 230 per piece. This is a reduction by a third. By using half of the regular and half of the reduced, the final



Figure 6.12
A Fifty/Fifty cake.

cake will have about 3,500 calories; 390 calories per piece on average (ignoring the icing; see figure 6.12 for sample cake). But only on average, as the distribution of regular and reduced mixes is different for each single piece because of the diagonal separation (figure 6.13).

Fifty/Fifty is a variation of a simple plan (i.e., implementation intention): whenever there is tempting food served, take it, but eat only half of it ("Friss die Hälfte"). This is a clever plan, because what counts is to reduce food intake in general, not the reduction of particular food ingredients through a sophisticated diet (e.g., fat, carbohydrate) (de Souza et al. 2012). Obviously, Fifty/Fifty delivers this plan with a twist. Instead of really reducing the amount, it only reduces the calories overall. But nastily it does so differently for each piece of the cake. Through this, Fifty/Fifty first of all creates choice.

Imagine Fifty/Fifty on the coffee table, neatly cut into pieces. Which piece would you take? The left end is regular, buttery, the right is reduced, yogurty. The middle piece is fifty-fifty, but it is unusual to take a piece from the middle. For the health-conscious, Fifty/Fifty offers a clear alternative. A piece of the calorie-reduced, yogurty end is in line with the goal intention to reduce weight, at the same time acknowledging the difficulty to resist the offer because of the temptation and social reasons. For the not so conscious (yet), an alternative course of action becomes suddenly available, clearly referring to the ideal of having a normal, healthy weight. This creates friction. It highlights the tension between cozy kaffeeklatsches and weight problems, between praising and submitting oneself to the baking skills of the host and later needs to repair the damage done to bums-tums-and-legs. In addition, through the diagonal design, it becomes likely that whatever piece a person takes, it will consist of both mixes at least in part. This creates the opportunity for the firm believer in butter to experience the miracles of low-fat yogurt (in fact, it is quite tasty!).

Besides this, the diagonal design also adds irony to the mix (no pun intended). However strict individuals are—may they be cake purists or health



Figure 6.13
Distribution of regular and calorie-reduced mixes for different pieces of a Fifty/Fifty cake.

fetishists—each and especially every second piece taken inevitably undermines their own choice at least a little. Imagine yourself, having been rushing for the healthy end of the cake. Not quite satisfied yet, you take a second piece from the same end. It will always be a little less healthy. The same holds true for the epicurean—each following piece is a little healthier.

So far, we discussed Fifty/Fifty from an individual point of view. However, the sociability of a kaffeeklatsch adds another dimension. There are a number of situations imaginable. For example, a health-conscious dieter innocuously brings the cake along as his materialized implementation intention. But inevitably, the cake will stir some debate because it creates the need to choose, where no choice had been before. And whatever done may in turn require explanation and justification. For the dieter, this playfully highlights the deeply social nature of weight problems. Being slim or overweight—like many similar problems—is not only a matter of willpower or the lack of it, but also a matter of social pressure and support (Rachlin 2004). Another potential situation would be a well-meaning host, introducing Fifty/Fifty to the coffee table as an offer to the especially health-conscious guests. Again, debate would be certain. And finally, there is always the malicious option of the host preparing a Fifty/Fifty for a very special guest, such as an overweight and obstinate aunt or friend. This surely adds some significance to the notion of the "tyranny of the host."

Pleasurable Troublemakers, Self-Quantification, and Gamification

Do/Panic and Fifty/Fifty are highly situated. Do/Panic links the ideal of "being organized" to one's desktop; Fifty/Fifty links the ideal of "being slim and healthy" to cake and kaffeeklatsch. In both cases, there is a dominant impulsive behavior: to make a mess of the desktop or to tuck into tasty cake. Both

pleasurable troublemakers introduce a novel behavior to the situation, a simple plan in line with an ideal (e.g., aligning items, eating calorie-reduced cake). This turns abstract intentions into something quite real. Do/Panic and Fifty/Fifty tell a story of a different self. While they definitely create some friction to nudge their users into meaning making (Do/Panic a little less than Fifty/Fifty), they remain likeable. They understand. They sympathize. Both embody implementation intentions, which do not require superhuman powers. It is still allowed to make a mess and to eat cake. In addition, both are ironic by incorporating a loophole. Do/Panic masks the mess, at least for a while. Fifty/Fifty even sabotages the prim user by always slipping in a little of the vice. In a way, pleasurable troublemakers are a reflection of their users and change in general. Nobody is perfect. And every little step counts.

Obviously, pleasurable troublemakers are different from self-quantification and gamification. The first often appears technocratic and takes a rather disinterested stance. Fitbit's One, for example, estimates in real time the calories your body may burn without physical activity. Obviously, activity increases the amount of calories burned. And there is a simple rule: if you don't want to get fat, only consume as many calories as you burn. Based on this, One's continuous visualization of calories burned implies a number of potentially interesting practices. First, one may start to better match food intake to levels of activity. Typically, this is the opposite in real life. The more we slump on the couch, the more high-calorie food we consume. Second, one may use the visualization like an account. Let's say you urge for this tasty chocolate bar on the kitchen counter. You can eat it, but you need to pay back the debt. You need to balance your account through activity. All these behaviors are made possible through One's technology. But the device itself does not tell those stories. It bores its users to death with numbers, graphs, and well-meant recommendations. It provides infrastructure, but misses the chance to tell a story. Story is entirely left to the imagination of its users.

Gamification might be a remedy to this, but this depends on its execution, the how. In fact, One is gamified as well. It hands out badges for ten thousand steps done or ten stories climbed. But this is not a story, it is a well-meant reward, resting on the wrong assumption that gamers play for points. They play for play. They play for story. The closest One comes to a story is when it tells its user that he had climbed stairs equivalent to the height of the Christo Redentor in Rio de Janeiro. McGonigal (2011) clearly has a more differentiated notion of games, narratives, and their powers. But they must be harnessed. An example highlighting the difference between dispassionate self-quantification and narrative gamification is Zombies, Run! (https://www.zombiesrungame.com/). Zombies, Run! uses the narrative of an apocalypse complete with marauding hordes of zombies to make you go for a jog. The match between the pop cultural archetype of a zombie chase and the running activity makes this fantasy appealing (at least to the first author, raised on and steeled by countless Resident Evil sessions). Never look back when you hear a zombie breathing down your neck. What this type of gamification still misses is to instill insight. It motivates through an alternative to reality, thereby sugarcoating and avoiding what is real. To us, this notion of an alternate reality is inherent to games, but different from the troublemakers, we propose. They don't tell hilarious stories about elves, zombies, or space

marines, but everyday stories about life. They are rather akin to fictional documentaries or even "mockumentaries."

Pleasurable troublemakers are different from many current attempts at designing persuasive interactive technologies. Froehlich and Findlater (2010), for example, found most persuasive technologies in the context of sustainability to rely on feedback (akin to self-quantification). They don't necessarily take a clear position, they just provide information. In addition, choice and goal setting as theoretical backdrop remains largely unexplored. Albeit interactive, most of the objects (i.e., devices) remain rhetoric; they talk or show rather than mediate choice.

Obviously, interactive persuasive technologies come in many different shades, and it is beyond the scope of this chapter to launch into a full-fledged critique on current design practices. Many potential concerns with persuasive technologies, gamification, funware, and so forth, are convincingly discussed throughout this book. In particular, the chapters by Selinger and colleagues (this volume) and Sicart (this volume) resonate with our approach. In fact, a pleasurable troublemaker becomes a part of its user's *extended willpower* (see Selinger et al., this volume)—object and person form a "motivational system." And it is not as if a troublemaker merely serves data for the person to act upon (aka self-quantification). Quite conversely, person and object are performing the "task" together, with the object shouldering a significant part of the responsibility to shape the system. For ages, people already quite successfully "materialized" manual, cognitive, and even emotional aspects of their lives (think: a shovel, a calendar, and Celine Dion). And as Selinger and colleagues point out, there is nothing ethically wrong with this. The most obvious reservations against technology to deliberately change people (i.e., a designed motivational system) stem from the potential commercial exploitation. But this is not a problem of the objects themselves, but of the production and distribution models they become embedded in. More serious is the question of—as Selinger and colleagues (this volume) put it—"Who does technology want us to be." They identify a number of potential concerns (see Morozov 2013). One revolves around the topic of foregoing individual choice, a lack of experienced agency, which results in "infantilism" and a diminished feeling of personal responsibility. We share this concern. This is why choice is so prominent in the theoretical backdrop of our aesthetics of friction. Ultimately, it was you who tidied up your desk or left it as it was. Ultimately, it was you who chose a piece from the buttery or the yogurty end of the cake. Pleasurable troublemakers create choice in situations where it might not have been existent or that obvious before. But they never *make* the choice. They leave it to people—and are even understanding in case of failure. This ties into a further potential concern identified by Selinger and colleagues (this volume): "fragmented selves." We "might eat healthy when using a gamified wellness app, but poorly when we forget our smartphone at home." To prevent this, pleasurable troublemakers avoid convenience. They never assume responsibility for reaching a goal. They do not attempt to substitute individual willpower by making it easier. In contrast, they deliberately create thorny but interesting situations, opportunities for action and reflection to instill internalization and generalization.

When designing a "motivational system," the distribution of aspects among the person and the object is crucial. We assume that an object can suggest, lend a hand, offer criticism, be understanding—but it is the person who must choose and understand. This is

also behind Sicart's (this volume) note that "technology can be designed so moral values are transmitted to a practice. [...] However, the fact that technologies mediate morality does not imply that we become more virtuous beings by [...] using them. Technologies mediate morality, but we *practice* morality." Ultimately, the locus of change is the self, not the technology. This is what pleasurable troublemakers acknowledge and many persuasive technologies fail to understand.

Final Thoughts

> We believe that in order for conceptual design to be effective, it must provide pleasure, or more specifically, "complicated pleasure." One way this could happen in design is through the development of value fictions. If in science fiction, the technology is often futuristic while social values are conservative, the opposite is true in value fictions (Dunne and Raby 2001, location 489).

Pleasurable troublemakers are "complicated pleasures." Friction is crucial to this. In their chapter, Selinger and colleagues (this volume) argue that to have an insight, people need to feel uncomfortable. They need to be confronted with the gap between who they take themselves and the world to be and who they actually are and the world actually is. Pleasurable troublemakers do this, but in a light way. This is what makes them complicated but pleasurable: it is the friction and meaning, which makes them complicated, and the insight, irony and complicity, which makes them pleasurable.

While Dunne and Raby had certainly something more ambiguous and grand than cake or desktops in mind when they envisioned their value fictions, we find the notion of science fiction versus value fiction appealing. A pleasurable troublemaker is never a complicated piece of technology. But it is able to create complex, meaningful personal and social situations. It tells stories about alternative ways of living and being and is, thus, fiction. But, fiction on the brink to reality. Admittedly, being slimmer or more organized may not be futures worth bothering with. To decide which area to tackle, which stories to tell, is the responsibility of the designer. The aesthetics of friction as a set of principles holds for more essential and controversial themes than getting a slimmer or more organized self. And it does so with an already generous portion of moral reasoning built in. Obviously, whether a particular troublemaker is morally justifiable or not is a matter of individual analysis. But the general approach already highlights crucial aspects to consider, such as the situatedness of good and bad behaviors, as well as the significance of personal choice, meaning, reflection, and sympathy for failures. Note that this is also a limitation of our approach. It is attuned to everyday struggles, not to severe pathological problems. In this respect, our frequent use of drinking as an example may be misleading. Our approach is a light one, certainly not able magically to solve the problems of alcoholics, the severely obese, or pathological procrastinators. In these cases, a pleasurable troublemaker can only be a part of a more comprehensive plan for change.

Our notion of an aesthetics of friction and according transformational objects is a proposal for those who believe in big effects of small interventions. In line with recent commentaries (Brynjarsdottir et al. 2012, 954), we intend to go beyond persuasion as

understood in human–computer interaction and interaction design by shifting from prescription to reflection and from isolated behavior to situated practices. A pleasurable troublemaker intimates both. It is a device to instill change through behavior and insight—with a smile.

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POSITION STATEMENT

GAMES AS DESIGN ARCHETYPES

John M. Carroll

Games comprise a distinctive and, to some extent, a pure subtype of interactive systems. For the most part, people interact with games primarily for the qualities of the interaction itself. This suggests a research strategy of understanding game designs in order to draw general lessons for the design of interactive systems.

In the late 1970s, I played a version of the game Adventure, implemented on an IBM 360 mainframe and accessed through an IBM 3270 display terminal. This experience happened to coincide with my own first encounters with early versions of the constantly emerging concept of usability, these encounters occurring through a series of empirical studies of the problems of new users of office systems (e.g., Mack, Lewis, and Carroll 1983). Engaging in these two experiences at pretty much the same time was stimulating: the experience of Adventure seemed so different from that of using the office systems of the time, yet the user interfaces and user interactions seemed so similar.

Adventure and Office Systems

Both Adventure and the office systems seemed designed to obscure appropriate goals and effective methods for those goals. Users were frequently diverted from goals they were pursuing by system events and inadvertent side effects of their actions. Error messages and other system feedback was cryptic, difficult to decipher, and not clearly relevant. In the worst cases, messages directed users to do things that seemed wrong. Doing the "same thing" in a different circumstance often led to very different outcomes, making it difficult to form generalizations. Actions users took often had invisible consequences for the system state, but this might not become evident until much later when the user no longer remembered what he or she had done. This, again, made it difficult to draw generalizations or learn causal connections at all.

For example, creating a document required the user to handle a parameter labeled "retention period" (among others). It turns out this parameter could have been defaulted, but new users did not know that, and some spent minutes pondering what should be specified. Analogously, in Adventure the virtual spelunker is wandering through a vast cave in search

of treasure when suddenly the system informs him or her: "The bird was unafraid when you entered, but as you approach, it becomes disturbed and you cannot catch it." Should it be caught? Is it a treasure? How is it related to the larger goals of the game?

Printing a document in a networked office system involves handshakes between devices. This often meant that a document access was locked until the print operation completed. But systems did not necessarily tell the users that; instead, they posted a message like "Document unavailable," suggesting to the user that something had gone wrong, and that his or her document might have been lost. Of the many users I watched dealing with this predicament, none realized that printing *made* their document unavailable. Indeed, the message was so distressing that many forgot they had requested a print at all, and began working on the problem of recovering the lost document (often making further errors). Adventure also presented many obscure side effects. Thus, you could kill the bird, but doing so prevented progress in the game many hundreds of turns later in the game play (see Carroll 1982).

Transparent to Action; Conducive to Action

Adventure held my interest for many weeks. It was difficult, but tractable, and though it did frustrate me at times, I enjoyed the challenge. However, the office system users I was studying in my day job were not having a good time. One woman withdrew from our study so abruptly and decisively that she seemed to be running down the hall.

Two design archetypes are suggested by the contrast of Adventure and the early office systems. One is *transparency to action*: documentation and user interactions should be simple, clear, consistent, anchored in mental models and other task-related knowledge meaningful to users, and focused on user goals and effective methods for achieving them. Transparency to action is a generalization of task orientation (Carroll 1990); it has continued to be useful and important in designing today's user interfaces.

A complementary design archetype derives from recognizing that even the most careful design will confuse someone sometime. Interaction designs therefore must also be *conducive to action*, providing feedback that provokes hypotheses and testing, attenuated error consequences, reversible commands (undo), and error states that are easy to recognize, diagnose, and recover from. I called this archetype "exploratory environment" (Carroll 1982).

Both Adventure and the office systems provided less-than-transparent user interfaces, but Adventure was more conducive to action, conveying that the user should explore the user interface, including reasoning and improvising, testing hypotheses, and making mistakes and learning from them. The office systems conveyed through their designs that the user should follow directions carefully, learn intended lessons thoroughly, and never make mistakes. In both contexts, people engaged in discovery learning and made mistakes, but in the case of the office systems, this was experienced as failure, and it evoked frustration and undermined confidence, whereas for Adventure, it was the main source of the fun.

We still cannot and may never be able to design interactive systems that are fully transparent to action—comprehended instantly, unambiguous in their affordances, and errorless. Thus, we must still and may always have to design for the signature interactions of Adventure; that is, design for conduciveness to action.

Minimalist Design

The design archetypes of transparency to action and conduciveness to action articulate the intuition that the problem with early office systems was not one of providing too little guidance for users, but rather of providing too much of the wrong kind of guidance. They motivated the minimalist design model (Carroll 1990, 1998; see also van der Meij 2003; Obendorf 2009). As in classic minimalism of the early twentieth century, interactive systems must focus on what is essential, provocative, and motivating to people.

For example, we designed a series of interactive debugging environments called Viewmatcher, which presented a collection of coordinated views of Smalltalk code as it executed and allowed programmers to interact with the execution to analyze the code (Carroll and Rosson 1991). We provided minimal documentation coupled to specific execution states; most of what users gained through the Viewmatcher was discovered through self-initiated explorations. The Viewmatcher is an example of how the lessons of transparency to action and conduciveness to action can be applied to interactive systems that are game-like but not games.

Applying game archetypes in the design of interactive systems is not a matter of making work into a game, but rather a more nuanced effort of articulating critical abstractions and then re-instantiating them in new contexts. Minimalist design is an early, albeit modest example of gamification that was also quite successful, both in results of laboratory and field studies and in transforming information design practices in the 1980s and 1990s. The minimalist design model did not make information tasks into a game, but it was informed by a game design archetype.

More than any other kind of interactive system, games depend on providing high-quality interaction experiences. Indeed, user engagement must be immediate, intense, and sustained, or the game fails. We should identify design archetypes in games that we can then apply more broadly to the design of other interactive systems. The enormous value, competitiveness, and emphasis on innovation in game design ensures that lessons from game design for user interface design will keep coming.

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7 BEHIND GAMES: PLAYFUL MINDSETS AND TRANSFORMATIVE PRACTICES

Jaakko Stenros

To understand how a "ludic society" or "gameful applications" might function, we need to understand what playfulness, play, and games are. In current attempts of harnessing games for external goals, games, play, and playfulness are often treated interchangeably. Also, the emphasis is usually on game products as systemic artifacts. Yet psychology, sociology, and performance studies tell us that these things are very different, and that the activity of playing is deeply important. As will be shown, ignoring these differences means risking failure, and the activity of play holds many more interesting features untapped by a narrow view on games as artifacts. Therefore, this chapter analyzes concepts around play and games from the perspective of the player and breaks them down into experiential, social, and cultural parts. In a culture where playing games is ubiquitous and where games are seen as a model to be learned from in system design, it is necessary to separate the mindset, the socially negotiated activity, and the culturally recognized artifact. The analytical divisions introduced will be especially helpful in understanding the nature of playfulness and its role in harnessing games for external goals, such as learning, therapy, or advertising.

This chapter is divided into three parts. The first discusses the analytical separation of a playful mindset, the socially constructed activities of playing and gaming, and the culturally recognized artifacts and sites of playing and gaming. The second part addresses two misconceptions surrounding play: idealization of play, and the conflation of brute playfulness and socially constructed play. The third part discusses harnessing playfulness for transformative practice.

Playfulness, Play, and Games

At the core of play and games is the mindset of *playfulness*. The impetus to play is older than language, culture, even mankind (Fagen 1981). Indeed, it is biological in nature. As this playfulness is shared, it becomes socially framed, *play* emerges, and as these shared forms are codified, we call them *games*.

The Spirit of Play

First, let us separate the playful mindset from the socially shared act of playing. This distinction is important for understanding how play, although rooted in biology, appears in many different and distinct cultural forms—and how those cultural forms can end up being devoid of playfulness. To understand how playing a game can be tedious work, for example in professional sports, and how work can become deeply and gratifyingly playful, as in research and art sometimes, we need to differentiate analytically the social from the psychological.

An observer can often identify when animals are playing and can sometimes even join in. Playing is often visible and shared, and the tendency to play is ingrained in mammals, possibly even a much greater number of animals. Human play is continuous with non-human animal play, and in order to understand human play, other species need to be taken seriously (Burghardt 2006).[1]

According to anthropologist Gregory Bateson (1955, 315–317; see also Eastman 1948, 15–17) (social) play is only possible if the participants are capable of *meta-communication*. The meta-message of play, the statement that makes play possible by framing it, is "These actions, in which we now engage, do not denote what would be denoted by those actions which these actions denote." Bateson exemplifies this with play-fighting: it is possible to play at fighting so that it looks like fighting, while it is clear that it is not-fighting. Furthermore, there is a double negative: play fighting is also not not-fighting (Schechner 1988, 7).

Playfulness and play(ing a game) do not completely overlap either conceptually or experientially: it is possible to follow the rules of a game without being playful just as playfulness does not need to have the formal structure of games (Makedon 1984; cf. Salen and Zimmerman 2003, 302–305). *Play acts* can be analyzed according to structure, process, experience, function, ideology, and frame, whereas what in this article is called *playfulness* is "a mood, an attitude, a force" that erupts or something one falls into (Schechner 1988, 4–5, 16; see also Riezler 1941; Makedon 1984, 32).

Psychologist Mihaly Csikszentmihalyi (1975, 10) uses the term *autotelic*[2] to describe activities that "require formal and extensive energy output on part of the actor, yet provide few if any conventional rewards." He terms the intrinsically rewarding autotelic experience *flow*, and argues that it can be experienced when an actor's skills are in balance with the challenge provided by the activity (36, 49). Csikszentmihalyi notes that games and play are strongly autotelic, as they provide a structure for flow, but work can also provide a platform for such experiences. Indeed, he goes so far as to say that everyday life can be flow-inducing if one is capable of mentally restructuring it correctly. He gives artists, poets, and scientists as examples of people who are able to "play" anytime and anywhere (53, 193–194). Csikszentmihalyi sees games and play as structures that have the function of inducing flow (191) and playfulness (and flow) as separate from that:

> By downplaying the structural distinction and emphasizing the experiential one, we are better able to deal with the *esprit de jeu* that Huizinga, Caillois and many others have held to be the central issue of the phenomenon of play. Yet the same scholars have been unable to study this "spirit of play," because they fell back on the obvious structural distinction and looked at games instead of the experience of playfulness. Playfulness, or flow, is not limited by the form of

> the activity, although it is affected by it. (Csikszentmihalyi 1975, 185–186)

Csikszentmihalyi is not the only psychologist who has made a clear distinction between play (a structure) and playfulness (an attitude or a mindset). Michael J. Apter (1991; see also Kerr and Apter 1991) has developed a general approach in psychology called *reversal theory* and a specific structural phenomenology of play. As people can experience playing a game as frustrating and work-like and sometimes working can feel like play, Apter (1991, 21) firmly believes that "[p]lay cannot be defined externally by reference to objective criteria; it is a phenomenological state." Reversal between phenomenological opposite states is what reversal theory is all about, and in relation to play the theory identifies two ways of "being in the world," two mindsets or meta-motivational states: *telic* and *paratelic*. Telic is a serious mindset, where the activity is engaged in for a purpose. Paratelic (cf. Csikszentmihalyi's autotelic) is a playful state, and engaging in the activity is itself the goal (or, as in games, a goal is adopted in the service of the activity). Other characteristics of the paratelic mindset include emphasis on immediate gratification, spontaneity, freedom, willingness to experiment, disposition toward make-believe, and the tendency to prolong the activity if possible.

Both flow and reversal theory are based on a phenomenological approach, and that establishes a clear difference toward more social or developmental approaches. For example, both models understand sex mostly as play, something we engage in while in a paratelic mindset. Although sex has an important biological function, it is mostly experienced for its own sake and for immediate pleasure—and approaching it as work, in a telic mindset, can rob it of pleasure, or prevent it from being performed fully (Apter 1991; Frey 1991).

Playfulness is innate to the player. It is present in the moment and can be sparked in an instance, change drastically at any time, and can disappear without warning. Though it is possible to try to foster and use playfulness, it cannot be fully tamed. Playfulness is often visible and its tone is meta-communicated to others present in a social situation—and this meta-communication can even carry from one species to another.

This leads us to a central paradox of gamification and other applications of playful activities for external goals: the mindset that gives rise to playing and games is deeply connected to being performed for its own sake. Playfulness is paratelic, without external goals. Adopting the form of playing or certain elements of games for serious, telic purposes always runs the risk of losing that mindset. Playfulness is not located in a system or an artifact, but in the participant.

Constructing Play and Games

The phenomenological nature of playfulness separates studies of adult human play from studies of child's play and animal play, as in the latter the first-person experience remains out of reach. Another difference between animal and human play is that humans not only play, but they are *aware* that they play. Human consciousness and intentionality (cf. Searle 1995, 227–228; see also Montola 2011) make it possible to construct, based on the brute fact of playfulness, the social facts and institutional facts of playing.

Play refers to activity. It is often visible and can be carried out alone or be socially shared. Though

usually discussed with the noun *play*, it makes more sense to approach it with the verb *playing*, as it is actively created, performed in the moment. It is rooted in a playful mindset yet influenced by social construction. It is possible for a person to oscillate between a playful and unplayful mindset quite fast resulting in a dual consciousness of the social situation as both viewed as playing and as not playing.

In this kind of thinking, games are a more formal version of play, when placed on a continuum. This has been most famously verbalized by sociologist Roger Caillois (2001/1958, 27–35), who discussed *paidia* (free play) and *ludus* (formal games). Before the systemic turn in game definitions, sparked by digital games and Chris Crawford's (1983) influential work, games were defined as activities. For example, Clark C. Abt (1970), social scientist, systems engineer, and a pioneer of games and simulation for learning, saw in games the possibility for a reunion of action and thought. He saw physically inactive thought and mentally inactive action as diseases of civilized man (4). Games, and especially serious games simulating social processes using role-play, can integrate action and thought. His definition of game is curious:

> Reduced to its formal essence, a game is an *activity* among two or more independent *decision-makers* seeking to achieve their *objectives* in some *limiting context*. (Abt 1970, 6–7)

This definition is purposefully broad. Abt (1970, 7) notes that "most real-life activities involve independent decision-makers seeking to achieve objectives in some limiting context." Instead of attempting to draw a clear distinction between games and other game-like activities, Abt chooses to emphasize this similarity. It fits his project of developing serious games that enable learning. He lists activities that can be viewed as games: war, political and social situations, elections, international relations, personal arguments, and almost all business activities (9).

This idea of approaching all kinds of situations *as* games underlines the similarity of game systems and other systems—a perspective omitting the element of playfulness.[3] In this, Abt is representative of a tradition of approaching games as formal activities (cf. Avedon and Sutton-Smith 1971, 7; Suits 1978, 41). Playfulness and games are two separate things: one is the form of an activity, and the other is a state of mind. It is possible to participate in games without a hint of playfulness. Yet there is a connection between the two: in a way games are technologies for prompting, fostering, and harnessing playfulness, yet they are often born out of playfulness even if over time that connection may have been lost.

Toys and playgrounds are similar to games in that they are designed to foster playfulness through *affordances* (Norman 2007, 66–69; also Norman 1988, 423). Toys and playgrounds are looser, have less limitations, and the user is, at least in theory, free to manipulate them according to her whimsy (Crawford 1983). Toys and playgrounds may be interactive, but they have no built-in goals: a ball is a toy, whereas *soccer* is a game. It is also worthwhile noting that while animals play with objects, humans are "the only species known to fashion objects for use in play by its offspring" (Fagen 1981, ix). Thus, though object play is based in biology, constructing instruments meant for play—toys—is a more complex process.

Sociologist Erving Goffman (1961, 20), who has studied games as part of the everyday life, writes that "games place a 'frame' around a spate of immediate events, determining the type of 'sense' that will be accorded everything within the frame." This concept of frame, which he adopts from Bateson

(1955), describes something that is social, shared, and provides meaning in an *encounter*, a social situation. Goffman sees these frames guiding our understanding of social situations not just in relation to play and games, but in all (face-to-face) social interactions (see also Goffman 1974). Goffman (1961) discussed how games are *engines of meaning* that enable events, roles, and identities to emerge that would not be understandable or meaningful in any other frame (e.g., offside in soccer, *atari* in Go)—only to show that this phenomenon is not limited to games but general. For example, only in the context of the street do terms like *pedestrian* or *motorist* become meaningful.

To sum up in the language of John Searle's (1995, esp. 63) social constructionism, inherent playfulness has a biological basis, and thus playfulness is a *brute fact*. Playfulness is not subject to intentionality and consciousness—or human representations—but is biological. This ontologically subjective fact is based on a mental state. Some foundational forms of playing seem to also be inherent to animal play, such as playing with the body, chasing, and hiding. More complicated forms of playing, say tag, require social construction.[4] Consciousness and (collective) intentionality make it possible to construct the social fact of play, and playing games. There is overlap between the biological and the social, but distinction is relevant. Furthermore, once a form of playing has been established, it is possible to sever the link between the mindset and the form of an activity.

The challenge for gamification is that it can be difficult to separate gaming as an activity from the use of other systems. Of course, this continuum between games and other systems is important for adapting elements of games to non-ludic contexts. The question becomes: how to foster playfulness?

Attempting to utilize the form of games and play for telic uses such as learning, therapy, or gamification of services or products typically means that one is attempting to direct playfulness. Understanding playfulness is thus central for any attempt to create telic playing. However, it is common to either sidestep playfulness completely or to idealize it in a way that can have serious implications for applications of games in nonplayful contexts (cf. Juul 2005; McGonigal 2011). Approaching games and play disconnected from playfulness often means that playfulness is treated as a constant. The ephemeral nature of playfulness can be paid lip service, while actually it is either mystified as an unknowable or its understanding is reducing to "fun."

Misconceptions Surrounding Play

Two ways in which play is commonly understood too narrowly are the idealizing of playfulness as a universally positive, liberating, fun, and character-building activity (cf. Henricks 2006, 2–7) and the labeling of playing the system as deviant. Developing a fuller understanding of play is important as both of these misconceptions lead to problems not just in theory but in practice when game elements are placed in nonplayful context.

Idealized Play

Play is not always nice. Numerous scholars have challenged the idealized notion that play is always

positive and respectful of other participants. Instead, play has been characterized as obligatory, of negative affect, rigid, and dysfunctional (Sutton-Smith and Kelly-Byrne 1984). The division to playfulness and play is helpful in understanding the full spectrum of playing.

Playfulness gives rise to both "good" and "bad" play. "Bad play" is transgressive, disruptive, disrespectful, and rules-defying—and it is central to the understanding of play in general (Myers 2009). Indeed, deliberate and provocative rule-breaking (i.e., *negativism*) is a strategy for increasing both telic and paratelic arousal (Apter 1991, 18–20). This kind of play "involves taking pleasure in the feelings which accompany acting in a way which contradicts internally or externally imposed directives and norms" (McDermott 1991, 98; see also Alfrink's discussion on transgressive skateboarding in this volume). Though the psychologists associated with reversal theory note that it is possible for this kind of behavior to become pathological, they stress that even negativism is widespread and common—and a part of normal human experience (Kerr and Apter 1991). If play is to be understood, researchers cannot turn a blind eye toward its darker expressions.

The incorporation of "bad play" into the theory of play not only questions the approaches that want to see playing as learning, but also positions many definitions of play, like that of Salen and Zimmerman (2004) of play as free movement within a more rigid structure, as definitions of "good play." Play that is transgressive, illegal, transformative, and harmful seldom fits within those definitions (it would need to be, for example, more like free movement within *and challenging* a more rigid structure). It recalls Huizinga's (1938) idealistic formulation of play as free; no matter what rigid structure attempts to surround play, that structure can be played with. David Myers (2009) connects this transgressive and transformative play to Sutton-Smith's (1997, 229) *adaptive variability*—it is bad play that reinforces an animal's behavioral variability. Play can be revolutionary, creative, and transformative precisely because it is free, but that does not mean that everyone will like it. Play can be a threat to a conservative—and that containing play within the rules of games is a way to attempt to control it (Makedon 1984, 40–43). Victor Turner has made the same point:

> Playfulness is a volatile, sometimes dangerously explosive essence, which cultural institutions seek to bottle or contain in the vials of games of competition, chance, and strength, in modes of simulation such as theatre, and in controlled disorientation, from roller coasters to dervish dancing—Caillois's "ilinx" or vertigo. Play could be termed dangerous because it may subvert the left-right hemispheric regular switching involved in maintaining social order. Most definitions of play involve notions of disengagement, of freewheeling, of being out of mesh with the serious, "bread-and-butter," let alone "life-and-death" processes of production, social control, "getting and spending," and raising the next generation. (Turner 1986, 31)

Sutton-Smith and Kelly-Byrne (1984, 311, 319–320) went so far as to say that the twentieth century illustrates the gradual socialization of children's leisure: from folkways to the adult-influenced recreation, sport, and play. Indeed, such is the danger of play that they playfully conclude their article by suggesting that the whole idealization of (certain types of) play is an elaborate way of confusing the functions of play with those of normative learning.

"Bad play" has also been discussed as *dark play* (Schechner 1988, 12–14; 2006, 118–120) and *deep play*

(Bentham 1802/1894, 106; Geertz 1973, 432–433; Csikszentmihalyi 1975, 74–101; Sutton-Smith and Kelly-Byrne 1984, 314–316). Deep play is playing where there is a very high psychological, physical, or monetary risk involved. Examples of this can be illegal extreme sports, having unsafe sex for kicks, or drunk driving on a dare. Dark play, in contrast, "subverts order, dissolves frames and breaks its own rules—so that the playing itself is in danger of being destroyed" (Schechner 1988, 12–13). Thus, though dark play is quite similar to what Myers calls "bad play" (or *anti-play*), Myers believes that play is paradoxical by nature and always survives the paradox—thus even if you dark play with something, he believes that the play survives.

We rarely consider teasing as a form of playing (as suggested by Henricks 2006, 6), though often the interrogated schoolyard bully will defend by saying that it was only play. Similarly, in gamification we prefer to look at instances where games have helped us take out the trash or vacuum, as in Chore Wars, and rarely consider that the playful applications are just as like to be used for, say, spreading the propaganda of the Israeli Defense Forces as they are bombing Gaza.[5] Playfulness can be creative, emancipating, and liberating, but it is also unexpected, disruptive, and destructive. Games can be a way to tame the force of playfulness, but not without the cost of losing at least some of what is so precious about it. However, it is important to remember that one cannot invoke just the "good kind" of playfulness, but must deal with the whole package.

Playing the System

The analytical separation of playfulness from play and games allows not only for the understanding of situations where games are devoid of playfulness, but also brings into focus playfulness that takes places in systems not recognized as games. Any system or procedure comprising formalized social action can be played if it is engaged with in a playful mindset.

Consider SuperBetter (McGonigal 2011, 135–142), a game where the player gives herself an alternative identity and goals that help her recover from an injury or illness. SuperBetter is structured in straightforward missions like recruit help and list power-ups (things that make you feel better). Yet the real goal of the game is to entice the player into a playful mood and to keep her positive. Following the rules of SuperBetter in a telic mindset makes it all seem trivial, patronizing, and condescending. Yet if the game is able to shove the player into a paratelic mindset, that can make carrying out chores relating to getting better easier. Depending on the mindset, SuperBetter is either fun and meaningful or a tedious chore.

The connection between playfulness and games can be rejected, both intentionally and by accident. Unplayful gaming, like professional sports, gold mining on virtual worlds, and boring serious games, can combine telic mindset with the social context of a game. Similarly, a serious social situation or an online system can be engaged with a playful mindset.

Social play is based on collective intentionality (Searle 1995, 23–26), which means engaging in a cooperative behavior where beliefs, desires, or intentions are shared. Each act by an individual is part of a larger collective activity. A player that rejects the collective intentionality of playing together with shared rules and instead, knowingly, begins to play by her own rules transgresses. I have written about this more thoroughly elsewhere (Stenros 2010a) but will condense the part of the discussion relevant for the topic at hand.

The player who rejects collective intentionality can *play the system*, meaning that instead of trying to achieve the explicit goal of the game (or other system) by the appropriate measures, she starts to interact playfully with the system, taking advantage of loopholes, bugs, hacks. This activity can be found both from games and from telic systems (e.g., taking pains to optimize character creation in role-playing games or creating a large amount of links to a webpage in order to raise its positioning in online searches—colloquially known as Google bombing). She might be trying to reach the goal by any possible means disregarding the artificial limitations of the game or she might be subverting the functionality of the system or game by choosing a new goal and then attempting to achieve that. She can also start to *play the players*; ignoring the game or system and its rules and simply start playfully interacting with the other players—without letting them in on the new rules of play. Usually, the goal of this kind of playfulness is to get a rise out of the fellow players. Heckling, griefing, bullying, trolling, and masquerading as a different person are all activities where one person is (often playfully) misleading other participants, fabricating (see also Goffman 1974, 83–123; Montola, Stenros, and Waern 2009, 257–278) the social context for the other participants. If numerous people share the same (new) goals and rules and start playing the system together, this creates an alternative social context, a new frame of playing. They start *gaming the system* together.

This phenomenon also questions how practical the idea of viewing games as objectively grounded in game rules is (e.g., Makedon 1984, 32). All participant may not be playing by the same rules even if they are interacting with the same game product. This is particularly true in digital games where the facilitation of the system is managed by a computer.[6]

All systems can be played, and indeed all systems are played by some of their users (case study 7.1). Nonessential systems, systems that have less of an impact on the participant's life, are probably played more, but playing the system is rampant from bureaucracy to online recommendation systems. Additionally, most systems give a possibility for playing other players, and as some players are more interested in following their own rules and disregarding official purposes, local norms, and good manners, it is very difficult to eliminate such actions without removing all possibilities of interaction. Neither type of playing requires any official gamification, as the paratelic playfulness is located in the participant. Furthermore, adding elements culturally coded as game-related into a telic system can encourage playing the system in ways not wanted by the facilitators.

Harnessing Playfulness

Playfulness offers a slightly removed perspective. It is possible to attempt to exploit playfulness for many things, such as enjoyment, learning, and personal change. Furthermore, it can render the constructed fabric of the social world visible and thus enable hacking reality. All systems can be and are played with, whether such actions are deemed positive or negative.

In this section, we will look at how playfulness has been harnessed for personal change and for social

Case Study 7.1

Playing Online Systems

Summary

Playing online systems is a disconnected group of activities users participate in when they are playfully using a system such as a search engine or a wiki in a way that does not line up with its stated purpose. The users play the system by abusing its rules and the possible bugs in it. The goal is to have fun, gain an advantage, or make a point, and usually such usage is detrimental to the effectiveness of the system.

Facts and Figures

There are no official figures on how much people play with online systems, as much of this toying around is personal and never visible to others. However, there are numerous cases where the results of such actions have become visible and gained attention (cf. Tatum 2005). Additionally, many online companies tweak their recommendation algorithms to combat reverse engineering and the optimization it brings (cf. Wesch 2008).

Gameful Design Elements

Telic online systems seek to prevent misuse (including unwanted playing) of their systems by hiding the rules. However, once the rules have been reverse engineered, they can be played with. Some examples:

- *Recommendation optimization* Any online system that uses a recommendation system, such as Google, Flickr, or YouTube, is subject to speculation as to how recommended pages, images, or videos are selected. If the algorithm is understood, it will be used to push one's own material—and to tease, bully, and smear.
- *Social engineering* Knowing how people tend to act helps in playing the system. For example, ensuring that the thumbnail of a video on YouTube has a bikini-clad woman on it gathers more clicks (Wesch 2008).
- *Rule abuse* Many rules can be abused if one is not trying to use a system for its designed purpose. In Wikipedia edit warring (cf. Sumi et al. 2011), the target is to have the page lock, as it does after three reverts, while reflecting one's opinions.

Issues

The challenge is creating a system that is easy to use and play with (in the encouraged manner), yet difficult to play with (in the unwanted manner).

Outcomes

Playing with online systems can be seen as pranks and vandalism, as tactical use of power (cf. de Certeau 1984), as a social movement (Tatum 2005), and a pastime. Yet as online systems are constantly updated, the results change. Playing with the system is a part of the online world, something that online companies take into consideration.

Related Cases

In online games such as World of Warcraft, analyzing the mechanics in order to gain advantage is called *theorycrafting* (Mortensen 2010). All systems are played, not just online systems, from taxation to Google. When playing the system becomes established and socially shared, it becomes *gaming* the system (cf. Stenros 2010a). Playing with players is also related: The idea is to use the system just to get a rise out of individuals, for griefing, trolling, and for online bullying.

change. To understand these better, it is helpful to look first at two examples of how foundational forms of play have been built upon.

Examples: Pretend Play and Embodied Play

Foundational forms of play, such as playing with the body (locomotor play), pretend play, and rudimentary forms of social play (chasing, hiding), are not just encountered in humans, but in animals as well. Such basic forms of play are not too reliant on abstract concepts, social construction, and collective intentionality.

Pretend play means pretending "as if" something was different than it is. In role-play, this "as if" is applied to pretending to be something one is not. Both are very common in human cultures, and already very young children engage in make belief (cf . Yardley-Matwiejczuk 1997; Paley 2004). Pastimes from theater to role-playing games, masquerades, and historical re-enactment are based on this.

Role-play is a fundamental activity for humans and a powerful way of communicating different points of view. Indeed, communicating certain types of instructions can be very difficult without relying on the concept of role-play, as giving the instruction of "you are a secret agent" without pretend play can show (Ericsson 2009, 242). The role-player can gain insights from adopting another perspective and possibly develops empathy for this role. Though obviously, the quality of the role-play as a simulation is greatly dependent on factors such as pre-existing knowledge the role-player has.

Instead of using role-play just for the paratelic fun of it, the pretend play mindset has been applied for all kinds of telic functional uses. Therapeutic psychodrama, based on the work of Jacob L. Moreno, has found its way into traditions of clinical psychological and psychiatric work. A number of socio-psychological role-play experiments of the past century have become infamous. Role-playing–based tools have also been packaged in the field of innovation, analysis, and design (e.g., de Bono 1985/1990). Different types of training simulations from customer service to military application are common. Less obvious examples include fake online personas used by brand management agencies to create the illusion of grassroots activity (i.e., astroturfing) and the fantasy scenarios offered by sex workers.

When role-playing is enacted bodily, not just as a mental exercise, there is an additional element (case study 7.2). The player inhabits the point of view as her body becomes the body of the role-played character or functional role. In live action role-playing games (i.e., larps),, this is called first-person audience (cf. Stenros 2010b). The player performs not just for others, but also to herself. Also, she only witnesses the things her character is present to witness. This visceral perspective-taking may further work to build empathy with other perspectives through embodied performative enactment—though there is no consensus in the research to back that claim.[7]

The relevance of the body of the player should not be underestimated. This chapter has adopted a player-centric point of view, focusing on the experience of the player and her state of mind; yet the player is not just her mind. Playing with the body, swirling, jumping up and down, dancing, masturbating, somersaulting, spinning until dizzy, farting, stretching, making faces, trying to lift just one eyebrow or to lick your elbow can all be very enjoyable. This kind of play, which Caillois (2001/1958, 23–26) termed *ilinx*, requires neither tools nor coplayers.

Case Study 7.2

Just a Little Lovin′

Summary

Just a Little Lovin' is a live action role-playing game, where approximately sixty participants immerse into and enact characters in a shared, fully scenographed setting. Just a Little Lovin' is about the beginning of the AIDS crisis in the New York gay community in the early 1980s. The larp lasts four days, and its central themes are friendship, desire, and fear of death. As part of the Nordic larp tradition (Stenros & Montola 2010) where larps tackle serious subject matters, emphasis is placed on the first-person experience of a participant and a believable story-world. Just a Little Lovin' offers an embodied and ephemeral peek into a specific time and place.

Facts and Figures

Just a Little Lovin' had been played three times. Originally created in 2011 by larp designers Tor Kjetil Edland and Hanne Grasmo, who have worked with HIV/AIDS prevention and LGBT issues, the larp has since been staged by other producers.

Setting and Mechanics

In two neighboring cabins in upstate New York, Independence Day is celebrated annually. A group of gay men and their friends party in one, and in the other a group of cancer survivors, many of who were hippies and swingers, gather to celebrate life. The larp comprises three acts, each portraying a consecutive Fourth of July party: 1982 is streamlined as the time before the epidemic, 1983 addresses the epidemic and the lack of knowledge about the virus, and 1984 shows a world where the HIV test existed but no remedy was available. Each party has a similar structure (arrival, barbecue dinner, assorted program, fireworks, sleep, breakfast), which provides a framework for playing out the drama (cf. Saita 2012). Each act starts around 5:00 p.m. and goes on until 11:00 a.m. the following morning. Between the acts, mechanics are used to determine who gets ill and dies, and the players negotiate what has happened during the year. In total, the larp lasts four days, counting the preparations and debriefing. It has mechanisms for simulating sex and playing scenes not set at the party.

Gameful Design Elements

Nordic larps often feature fully immersive environments with believable characters as interfaces. There is no script, though the characters and their motivations can be preconceived by the larpwrights, and game mechanics are often used to structure the action. Typical elements of Nordic larps are as follows:

- *Character ownership* Each player portrays a character fully by pretending to believe to be that character (cf. Pohjola 2004; Montola 2012). Each player is trusted to interpret the character and can improvise new elements provided that they do not conflict with established fiction. As Just a Little Lovin' is about the fear of death, permanent character deaths are frequent.
- *First-person audience* The character is the player's lens, performance, and private show. In Just a Little Lovin', the plurality of viewpoints enables breaking the timeline; along the main timeline there are pockets where alternative moments in time are played out in a black-box environment.
- *Inter-immersion* Larps are socially constructed and co-created in the moment. Participants do not just pretend to their fictional personas, but also that everyone else is who they claim to be.

Case Study 7.2
(continued)

- *Positive negative experience* Although a character's experience is negative, the experience of the player can be positive (Hopeametsä 2008; Montola 2010). Just a Little Lovin', with its strong themes and appropriate simulation techniques, helped its players feel in an embodied way lust and utter hopelessness, and perhaps even gave a glimpse of what it might have been like.

Issues

The Nordic larp community has been accused of snobbishness and taking larping too seriously with works that address cancer, immigration, and consumerism. Nordic larps are mostly played by privileged, well-educated, middle-class whites, and an element of misery tourism may be involved. Yet, such criticism can be leveled against most representations of serious topics in media. The first staging of Just a Little Lovin' specifically was criticized for romanticizing the past, as no participant had lived in New York in the 1980s or was HIV positive. In the second staging this was remedied. The larp was also criticized, before it was played, in Swedish tabloids (cf. Gerge 2012) for trivializing a serious issue.

Outcomes

Just a Little Lovin' is part of a long tradition of Nordic larps. It especially builds on work done with gender and sexuality in larps such as Mad About the Boy (2010) and *Mellan himmel och hav* (2003). It is too early to discuss the impact of the larp. Photographs of playing, paintings created for the larp, and character performances were exhibited at Oslo Central Station on World AIDS Day in 2011.

Related Cases

Numerous larps with similar techniques and aims exist. The activity of larping is related to role-playing in general, and similar activities include everything from improvisation to sociodrama, and from military training simulations to sexual role-play.

Further Information

http://just-a-little-lovin.blogspot.fi/ (official website)

The body of the player has not been very central to the discussion of play and games in contemporary game studies, though the rise of the mimetic interfaces of dance mat games, Nintendo Wii, exergames, and perhaps even smartphones using touch interfaces and accelerometers are starting to change that (cf. Márquez Segura et al. 2013). Yet the player always has a body and that has an effect, not just in pure locomotor play, but in all play. The possibility to derive pleasure from just performing the activity is not lost when these activities are placed in the context of game play.

In Nintendo Wii, where it is possible to carry out most actions on the controller with a small flip of a wrist, the players still choose to carry grand, iconic gestures when playing Wii Sports (Simon 2009). Mimicking body movements seems to be important to players. In Mirror's Edge, an immersive, single-player, parkour-themed game, for some players the feeling of vertigo it induces is part of the draw of that

game. Even in extreme role-playing games, where players consciously immerse themselves into disgusting situations, players frequently experience crying and feelings of nausea (Montola 2011).

Though play with conceptual objects can be very rewarding, the body should not be forgotten. In fact, it is quite possible to design games that take advantage of modifying the player's body (cf. Mueller et al. 2011). In professional sports, doping is obviously an issue, but it is treated as a problem and not as part of the playing. However, the genre of party games, such as SingStar and RockBand, benefits immensely not just from the physical co-location of players who perform as much or even more than they play for points, but from drunk players. Indeed, they are digital descendants of drinking games and drunken singing. If alcohol or other drugs seem too extreme, endorphins and adrenaline can do just fine as well.

These two examples, pretend play and locomotor play, show how fundamental forms of play have been and are harnessed for both telic and paratelic activities. Next we move on to look at using playfulness for transformative practices.

Playful Personal Change

Play, like art, is a step apart. It puts the participant in a removed point of view, enabling distance, reflection, and new angle of approach. Play also provides an alibi for pushing one's own boundaries, as it is "just play." Society tolerates more under the banner of play.

Play is also related to, or the basis of, humor. John Allen Paulos (1980) has defined humor as "a perceived incongruity with a point, in an appropriate emotional climate." Humor is incongruous in that it juxtaposes different interpretations: for example expectation and surprise, balance and exaggeration, or propriety and vulgarity. This juxtaposition creates insights as the same thing is considered in two incompatible frames. This has been called *bisociation* (Fauconnier and Turner 2002, 37).

This bisociation of play lends itself to numerous applications. It is particularly apparent in play that takes place in public spaces (cf. von Borries et al. 2007; Montola et al. 2009; Walz 2010). Most public spaces are reserved for nonplayful activities, such as walking, driving, or selling. Playing with the city, for example by following the map of one city while being in another (as one version of the psychogeographers' *dérive* or *drift* would have it), is a prime example. Indeed, getting lost in a city in a playful way is something that has excited urban dwellers for a long while (cf. Alfrink, this volume). Contemporary smartphone applications like Drift build on the work of the Situationists in a very direct way. Yet the general approach of using the city as a playground is very common in mobile games such as Zombies, Run! and Shadow Cities as well as in parkour and geocaching.

Urban exploration, the activity of exploring manmade structures such as tunnels, ruins, abandoned industrial buildings, and catacombs just for the pleasure of it, is another example of this. Many urban explorers infiltrate places that they do not have the rights to enter; for them any open door or manhole cover is an invitation to an adventure. Indeed, interacting playfully with the city can lead to having "ludic glasses," meaning that the person is looking at the city as a playground.

Alternate reality games and other pervasive games (cf. Montola et al. 2009) that blur the line between the ordinary everyday world and a game world often superimpose a fiction on top of the ordinary world, in the form of scavenger hunts, campus assassination

games, locative smartphone games, and many others. These games encourage bringing a bit of the magic of games to the ordinary life, to look at the world as if it was a game, to try and perceive where playful possibilities lurk in the everyday life. Seeding a few ludic elements into the city is a useful game design to foster the emergence of this bisociation. Once players have noticed some clear invitations to play, they may be overcome with *pronoia*, the feeling that the world is conspiring to help them (McGonigal 2006; Montola et al. 2009).

Play can also be used as an alibi. Activities that would not be permitted otherwise can sometimes be acted out under the pretence that "it is just play." Cindy Poremba (2007; see also Salen and Zimmerman 2003, 478–481) has discussed this with the concept of *brink games*: games and play that feature a conflict between implicit game rules and implicit social rules. The classic examples of this are spin the bottle and Twister, where players get to do things that would otherwise be socially prohibited, but as it is "just play," it is acceptable. More recent examples are provided by role-playing games that use *bleed*, meaning that they knowingly break down the barrier between the player and the character (Montola 2010). And of course the reason these games are enticing in the first place is that you get to do the forbidden things. "Brink games not only force the awareness of explicit and implicit game rules, but of implicit and explicit non-game rules as well" (Poremba 2007). Thus games and play not only can foster an alternative view on the world and render its norms and rules visible, but can also actively encourage breaking said norms.

This brings us back to humans being aware that they play. We are able to use play and design play—and we also have a cultural understanding that play exists. Certain types of activities are recognized as play and as games, especially if they take place in locations created for such activities. As play is something that is culturally considered "merely" play (cf. Riezler 1941), as lacking something, one can get away with more under the pretext of play. In this jester function of play, which very clearly uses the play as an alibi, public political acts coded as play get away with more than they would if they were wholly serious. Street theater and public demonstrations have taken use of this for a long time. Such activities are probably not paratelic, but they may give the impression that they are—or invoke playfulness in their audience. For example, *zombie walks* (dressing up as zombies; see Montola et al. 2009) are playful for their own sake, whereas *slut walks* (demonstrations where some participants dress up as "sluts" to ridicule the idea that there is a connection between the way women dress and rape) use play instrumentally.

In public play, the question of audience is particularly important. Richard Schechner (1988, 5) points out that "play is performative involving players, directors, spectators, and commentators in a quadralogical exchange, that, because each kind of participant often has her or his own passionately pursued goals, is frequently at cross-purposes." The possibility to invoke playfulness in spectators and commentators is clearly one way to mark a thoroughly telic activity as "play."

Where the construction of a shared event between performers and an audience is outside the scope of this chapter, the co-creation by players is pivotal to the discussion. As numerous participants are engaged together in a playful mindset, the possibility for social play emerges. This shared social play does put restrictions on the expression of each participant's playfulness as one needs to take into consideration

the other participants, but that is the prize for shared play. A shared pretend play situation requires negotiation on how new elements are introduced into play and what are its rules (cf. Montola 2012).

In the context of larps, this pretending together has been called *inter-immersion* (Pohjola 2004; Stenros 2010b). It means that each participant immerses in the character that is played and acts as if she was that character, but also that each participant pretends that other participants are the characters they are pretending to be. There can be no kings without subjects, no prisoners without jailors. This means that even if someone is not actively immersing on a given moment, the surrounding people will still treat that person as if she was the fictitious persona they know she is pretending to be. Thus, the illusion of a fictional world is supportive and self-perpetuating. This creates a very concrete situation of constructing a social reality—or a Goffmanian encounter. The players have agreed beforehand on a world they will build and they pretend that world into existence.

The same principle is in effect when imagining any shared fictional world into existence. The shared world need not even be strictly speaking fictional (whatever that means), but the so-called *temporary autonomous zones* (Bey 1985; see also *temporary tribal zones* in Letcher 2001; Harviainen 2012) follow the same logic. Indeed, there are numerous cultural sites where the usual set of rules are exchanged for another set: carnivals, parades, trips on cruise ships, vacationing in Las Vegas, Burning Man, and so forth. All of these are played into existence, though after they become culturally recognized, they may lose the soul of playfulness yet still maintain their alternative social rules.

To sum up, play helps perceive things from alternative angles and in different light. It helps us engage with our surroundings in a new way as we perceive and break norms and routines. In a playful state of mind, we can not only see the adventures that surround us, but we feel safe, possibly even *too* safe, to take that plunge. Play and games serve as an alibi: as they are perceived as being somehow less, we can get away with more. And finally, playing together we can create new and surprising social worlds that, as long as we all keep playing along, are as real as any other world.

Reality Hacking

Play makes rules explicit—and not just the rules of the game, but the rules of the social world. Playing with rules shows that rules can be changed. Once the constructed fabric of the social world is visible, it is possible to envision and create not just temporary play worlds, but a real change in the everyday social realm. This is the idea that lies at the core of reality hacking and benevolent gaming.

The social world is constructed, you can change who you are in it through your own action, and with concentrated effort with others you can change the world (Ericsson 2009, 243–245). Pervasive games have used this line of thinking to create alternate worlds in the midst of the shared reality, but usually these games have been staged in order for the players to have fun—or at least a meaningful experience. Using them consciously to enact social or even societal change has been rarer.

Political demonstrations and other performative projects are an obvious example of using play to bring about social change, but they usually operate with a division between players and an audience. Breaking down that barrier and enticing more people to participate is difficult. Augusto Boal's

(2002) Invisible theater is an interesting attempting to tear that division down. The performers put on a political show in a public place, but do not demarcate it as theater in any way—like *Candid Camera*. The idea is to push the "spect-actors" to participate, to make them think, and not just prank them for the benefit of a television audience.

One successful application of playful methods to effect social change comes from Bogota, Colombia, where philosopher-politician Antanas Mockus has served as mayor of the city. He has used unorthodox, and often quite playful, tools to enact social change. This is most visible in his use of more than four hundred mimes to embarrass pedestrians publically who ignore traffic regulations (Dalsgaard 2009; Hunt 2012). Allegedly, the reasoning behind this was that people are more fearful of embarrassment than fines. In a way, Mockus was using his position to troll the jaywalkers systematically.

There has also been a push to create clearly benevolent multiplayer games, such as World Without Oil (simulation of our world where the price of oil skyrockets; McGonigal 2011, 302–311), Evoke (World Bank–backed educational game about teaching social innovation; McGonigal 2011, 333–340), and the Nokia funded Conspiracy For Good (mobile phone–centric alternate reality game about battling corporate corruption; Stenros et al. 2011). None of these games have been able to show convincing evidence of their societal impact, though they may have changed the minds of some of the participants. In fact, they have been criticized as whitewashing for the corporations that finance them. For example, a critical satire of Evoke, called Invoke, billed itself as a crash course in saving capitalism and the World Bank. Instead of using playful methods to solve problems like hunger and poverty, it advocated using the free market system to do so.

If these games are attempting to wield playfulness and play for the benefit of bringing about positive social change, the question very much becomes who gets to decide what is positive (cf. Deterding 2012). One man's freedom-fighter is another's terrorist in a very real sense. Using Google bombing to tarnish the name of U.S. Senator Rick Santorum based on his anti-gay stance, as sex advice columnist Dan Savage did, is celebrated by some and seen as a low blow by others. Using larp mechanisms to foster nationalist pride (Mochocki 2012) is similarly a political question. And the privilege of affluent Westerners toying away on their smartphones in games that use the very real issues of colonialized countries either "raises awareness" or "trivializes"—or maybe both. Creating a political game, if successful, will usually prompt a discussion that questions the game and its designers' motives.

Using games and play to bring about real social change means using them for an external goal. This tautology is important to keep in mind when pondering the limits of using playfulness. The playful paratelic mindset is defined by doing things for the sake of doing them, not for an external goal. The process of cultural recognition of something as play or game can ensure the continuity of a structure even once playfulness has been removed, but it is a step toward the serious. Playfulness is thus used as scaffolding for a telic endeavor.

Coda: Playful Adaptations

If the importance of a playful mindset is not understood, then enacting the forms of play or inhabiting the arenas of games run the risk of being hollow. The very ingredient that is at the heart of play is missing. Just as risky is the idealization of play as inherently positive and beneficial to an individual or society. Playfulness can be liberating and emancipating, but it can just as easily be transgressive and disruptive. Deep play and dark play are never far away, and in many ways games are a way to harness and tame radical playfulness. Exploiting playfulness is possible, but if you open the door for playfulness, you are not letting in just the fun and positive aspects, but also the darker side. Griefing, trolling, and bullying are not unfortunate side effects of playfulness, they are part of its core. Constructing a totalitarian system that completely eliminates them also eliminates playfulness in general. It is possible to build transformative practices that use playful mindsets. Adapting the mindset for personal benefit, for breaking patterns and questioning norms, can have a remarkable impact. Using the same tool for social or societal goals is much more cumbersome—and involves questions of politics.

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Notes

1. One potential pitfall of studying play and games has been *exceptionalism*. While the ludologist call for studying *games as games* is commendable, there is a risk that moving games to the center of attention disconnects them from other similar activities (cf. Malaby 2007). This is particularly relevant for any discussion of gamification, as the very idea that certain elements typical of games can be usefully deployed in telic contexts requires that games are *not* exceptional, that lessons learned from them can be useful in other contexts.

   If playing games as an activity is treated as exceptional, as unlike other human or animal activities, and this idea is taken as given, then we will be sorely restricted by our notion of "game," whatever it may be. The emphasis on the similarity and continuum of play goes back at least to 1932 when Dutch behavioral psychologist F. J. J. Buytendijk discussed the span of play from animals to human and from children to adults (Walz 2010, 39–49).

2. From Greek *telos*, meaning "goal" or "purpose."

3. Also forgotten is the endogenous meaning (cf. Costikyan 2002, 21–24; Montola 2012, 30) in a game, what Jesper Juul (2005, 41–43) discussed with negotiable consequences; basically, the separated, step apart quality of games.

4. The literature is inconclusive on the ability of other animals to engage in more complex forms of play. It seems that a central question is the cognitive capability to understand other beings as other persons, based on the observations on human children who become able to engage in pretend and social play during their second year as this capability develops (see Rakocky 2007, 56). Yet at least some simple forms of play, both pretend play and social play, can be enacted with some animals other than humans—at least if humans have taught the animals (57). Whatever the situation turns out to be with animals other than humans, at least humans are aware of their playing, and thus playfulness as a mental state and play as a social institution can separate.

5. See http://readwrite.com/2012/11/15/unbelievable-the-idf-has-gamified-its-war-blog.

6. Montola (2012, 20, 60–66) has criticized this kind of *objectivism* in game studies. See also the discussion on internally and externally validated and defined rules (Dansey, Stevens, and Eglin 2009).

7. The development of the theory of mind and perspective-taking have been connected to early pretend play. However, though the impact of pretend play on the development of children has been studied empirically for forty years, there is still no consensus. In a review article, Lillard et al. (2012; see also Yardley-Matwiejczuk 1997; Lillard et al. 2011) conclude that "evidence does not support strong causal claims about the unique importance of pretend play." On the effects of adult pretend play, empirical research is scarce.

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POSITION STATEMENT

A GAMEFUL MIND

Buster Benson

I was lucky to stumble upon the value of viewing the world through a gameful lens early in life. I can trace the shift in mindset back to when my family moved from Chino Hills to Irvine, California, and my parents decided to start paying me for my grades.

In second grade, before the move, before the money, I was getting C's and D's. From third grade on and through college, I hovered around straight A-minuses.

At the time, the wider shift in parenting philosophies was from negative to positive reinforcement. I remember seeing the books my parents were reading on the topic. Despite recent trends emphasizing the downside of positive reinforcement, it worked pretty well on me. The side effect, which was consistent with recent trends showcasing the downside of positive reinforcement, was that I no longer attended school for the official reasons (i.e., to absorb the knowledge they were imparting), but rather treated it as a means to a variety of other ends.

My favorite definition of a game, articulated by Bernard Suits, is "a voluntary attempt to overcome unnecessary obstacles."

The part that I find lucky is that for some reason—and despite my parents' reward incentives—I didn't treat school as drudge work (where autonomy is low) or parental control as oppression (where autonomy is also low), but rather as a game (where autonomy is high). And that has made all the difference.

The key to this lucky break in my early life might be due to the fact that I thought I had discovered, and could therefore exploit, a flaw in the rules set by my parents. The money for A-minuses was almost as much as an A, and yet the work required to get an A-minus was significantly lower. I found my wormhole!

This wormhole allowed me to adopt a mindset where both parents and teachers (people who typically have control over their children and students) were demoted to "unnecessary obstacles" that I could joyfully attempt to overcome. This feeling of autonomy flipped my gameful mind on.

School became a voluntary attempt to optimize effort spent on reward received, on my own terms. The part often overlooked from the classic game definition, which I believe is crucial to what I would call a "gameful mind," is that the voluntary nature of the play allows one to focus on the style of play over the outcome. Even more important than winning the game is that I play it in a self-expressive way—as an example to others—proudly, even.

The wormhole, in a way, was my doorway into the game. I felt like I discovered it on my own, using my own ingenuity, and therefore I inherited power from

it as if it were a power-up, a cheat code that made a potentially oppressive environment into an empowering, autonomous, and voluntary one.

A game without voluntary play, without the autonomy to play with my own style, is not a game at all. It is the player who makes the game, not the other way around.

By high school, I was creating personal rules that helped me avoid actual A's (getting an A was considered wasteful of resources in my particular mindset). I tried skipping every fifth class. Not buying textbooks. Writing essays about the topics of my choosing rather than the assignments. I was using my trick to experiment within the system, testing it in different circumstances to see which other tricks it could unlock.

Rather than being punished for this blatant show of autonomous, gameful behavior, I found to my surprise that my parents and teachers appreciated my participation, even if it wasn't by the books. Probably because it was not by the books. They appreciated my self-expressive way of playing school and gaming the reward incentives for grades. As school became easier (aka I got better at my own game), I continued to dial up the frequency of stylistic tricks to make sure I balanced as closely as possible on the A-minus boundary. My game continued to evolve, grow more complex, more self-expressive, more enjoyable.

The surprising magic of the gameful mind is that it's enjoyable not only to the player, but also to the other people in the scope of the game.

Treating school as (paid) work might have led to some of these same strategies, but they would be executed in a repressed style. When I do something for someone else, non-autonomously, there is the fear of getting punished for standing out. Let's call it worker-bee mind. When I play with a gameful mind, part of the reason for playing is to stand out, to self-express. To be seen. To be heard. To win on my own terms.

Today, playing what would be thirty-third grade if such a thing existed outside of my gameful mind, my appreciation of autonomous, gameful play applied to real life has only increased. It has improved my play of school, of work, of relationships, of businesses, and of life in general.

In addition, I've used the gameful mindset to help enable others to stumble into the gameful mind themselves:

750words.com is a website that turns private journaling into a world where a gameful mind can thrive.

HealthMonth.com is another website that allows you to create rules for yourself and encourages you to try to stick to them.

These aren't games. In fact, there are no games; only players. They best we can do is create environments where people can voluntarily choose to play games.

8 PLAYING THE GOOD LIFE: GAMIFICATION AND ETHICS

Miguel Sicart

I am proud of my willpower. Throughout the years I have been able to quit smoking, to lose the weight gained afterward, and even to learn to enjoy changing diapers early in the morning. However wrong I might be, I think of myself as a stubborn, iron-willed person.

That's why I surprised myself when in 2007, I decided not only to take up running, but also to do so with the aid of the Nike+ services. At that time, Nike+ offered a feature that interested me: it was a cheap tracking system that would make it easier to follow a training program. Measuring my efforts would make it easier for me to follow the strict training patterns that would result in the successful completion of a marathon.

As it turns out, I did not run a marathon then, and I haven't yet. I stopped using the Nike+ system on May 15, 2009. I have, however, maintained the habit of running. I run at least four times a week, at least five kilometers. I run with my children, alone, in the rain or sun or snow. I've listened to great books and cheap novels and plotted articles and games and books while on the road. Running gives me a center, a focus, a practice of something radically different than my main daily activities.

In summer 2012, I decided to take one extra step in my habit and bought a GPS watch. I am the kind of runner that focuses on a schedule and commits to doing it. I always found the sound feedback on the Nike+ application too intrusive. A GPS watch would give me distance and pace data without bothering me. I ended up buying a Nike+ watch, which made me return to that service. Then I realized its failure in making me a habitual runner, even though I did become a habitual runner.

In this chapter, I want to explore this paradox, focusing on how gamified systems aspire to help us live a better life and promote good habits, and how their design might not always reflect what is actually needed to develop the virtues of a good life. I understand gamification as the design of services and products with the methods of game design, with the intention of engaging users in ways similar to those of games.[1] And I understand the good life as the philosophical notion of a fulfilling life in which humans practice and develop their potential, alone and together with others.

My main concern with many gamified systems is their focus on making the activities that lead a good life look like (competitive) games, disregarding the need for internal reflection and effort to perform activities that *we think* will lead to *our* good life. The good life is *not only* the practice of activities, it is *also* the capacity to reflect about the role of those

activities in one's life and sense of well-being. In this chapter, I want to suggest how playful technologies can be designed to help us lead a virtuous existence, focusing on how play is an activity that belongs to the good life.

I will present this idea of the good life, or *eudaimonia*, from two theoretical angles: virtue ethics and positive psychology. Eudaimonia explains the kind of virtuous life that we should aspire to live, in which we take efforts to explore our best potential and develop it. To live a good life is to practice virtues and reflect about them, further developing the ability to *live well*.

My critique of gamification will be based on the combination of two other disciplines—design research and philosophy of technology. I will critique how gamification has been designed and implemented for computing technologies, pointing at the philosophical and technical flaws that question their ethical validity. I will then propose a way of thinking about technologies for the good life that includes play and playfulness as a core driving inspiration for the design and the use of these technologies.

To support my arguments, I will use examples from Nike+ and the online programming teaching service Code Academy. Nike+ is a social, web-based tracking system for runners deeply embedded in the Nike/Apple technology ecosystem. Nike+ provides data for runners, as well as a number of competitive game mechanics delivered through its web presence. Code Academy is an online collection of interactive tutorials designed to teach users to write code in a number of programming languages. Code Academy extensively uses badges and other competitive elements to keep users engaged and ensure that they return to the website, while also tracking their progress in the classes.

I will start by reviewing what a good life means: the origins of the concept and its philosophical and psychological meaning. I will then present the idea of technology for the good life, or how the philosophy of technology has reflected on the role of devices in the configuration and practices of a virtuous life. This will lead to a critical reflection on gamification, followed by a proposal on how design should include thought about the devices that would help us lead a good life.

This chapter should be read as a philosophical inquiry on design and playful technologies, based on my own experiences and reflections as a consumer of some of these services. My purpose is not to critique gamification. Quite on the contrary, I believe that a way of living a good life is by *playing* a good life and that computing technologies can support this ideal. However, not all play, and not every playful design is a conduit of a good life. A good life is an internal activity of expression and reflection through action—very much indeed like a well-played game.

Aspects of the Good Life

Is running a part of the good life? Judging from both the large number of runners I meet on the streets every day and the rhetoric that surrounds this popular sport, I would intuitively answer that for the middle and high classes in the Western world, running *is good*.

If running is *good*, then services like Nike+ provide an element of enhancement of that goodness. By

encouraging us to run and measuring our progress, we are theoretically better suited to continue the habit of running. Nike+ offers data-rich, communicable visualizations of our running virtuosity while we are rewarded for it.

What is this *good life* I am referring to? According to Aristotle (1988), to live a good life is the supreme goal of humans, toward which all actions and activities should be guided. The good life is an end in itself, a purpose that should guide all of our active behaviors, as well as our thoughts and practices. We should think and act with the purpose of developing a good life. In this sense, the Aristotelian concept of the good life is both a teleological one, as it presumes that there is an end toward which we strive, and a constructivist one, as we must act to develop that good life.

The teleological aspect of the good life is particularly important, as it establishes that to live a good life, we must find our "true purpose" and practice it. According to Aristotle, we should find our true calling and follow it with actions, thus developing the virtues attached to that purpose.[2]

Of course, this purpose ought to be a virtuous one—the exploration and performance of functions that develop the good values that will result in a good life. To live a good life, we should act virtuously in a process of constant action and reflection toward developing our best being.

In classic philosophical terms, the idea of the good life is tightly coupled to that of flourishing—living a good life is becoming the best human beings we can. To flourish is to develop not only our virtues, but also the rational capacity to understand and evaluate the goodness in our own life.[3]

A common critique to Aristotelian theory has been its focus on the contemplative as the highest form of good life. Recent interpretations of the classic Aristotelian theory provide more nuanced understandings of the relation between the good life and flourishing as an outcome of performed actions. However, before moving on to positive psychology, it is important to mention that flourishing ethics has become a dominant trend in modern ethical theory, adapting the classic Aristotelian theories with Buddhist and Taoist discourses (Bynum 2006). Modern flourishing ethics allows us to understand the good life as the performance of virtuous actions in the context of an ontology in which humans are not at the center of the universe. The good life is not anymore a concept that is exclusive of, or even necessarily centered on humans—the good life is the human flourishing in the context of an ecology where machines are moral agents and thus active in the performance of the good life.

The philosophical concept of the good life is concerned with the performance of those virtuous activities that will lead to flourishing. The good life is not a passive state or a situation that can be identified, but a process toward an end in which actions and activities constitute the meanings of a life well lived.[4] The good life is a process of acting and reflecting with the purpose of improving ourselves.

But is running really a part of the *good life*? Or are we confusing means and ends, activities with virtues? Running has a number of positive outcomes: it demands and fosters virtues, from self-sacrifice to personal fitness. But running is not the goal. Talk to a runner: what makes sense is not the miles or finishing a marathon, though those things matter. What truly matters is the *practice* of running, an exercise in training the body and the mind. We run for the "runner's high," for the fugitive pleasure of being one with the road, for the realization that we can run

one more kilometer. We run not for the activity, but for what *we make of the activity*. The data tracked by the Nike+ system does not reflect this purpose of running. It's an excellent reminder of the need for running, but it only addresses one part of the virtuous life. It offers an external acknowledgment of a *manifestation* of the good life, a shadow image of a virtuous practice.

Before I go deeper into this critique, I need briefly to explain what I mean by practice. In my way of thinking about how we use technology, I am deeply influenced by Albert Borgmann's ideas on how we experience the world, as much as I am inspired by the ethical idea of morality as a *practical* science (Borgmann 1987, 2011). Hence, I call practices all those activities conductive to the development of an ethical life. Practices are the *actions* attached to *virtues*, from generosity to empathy, courage, and many others. Practices are not only actions, but also the reflection on those actions in order to pursue certain betterment in our moral life. A practice is constituted by actions, but also reflections. In this sense, the good life is constituted by and structured around practices.

But what do I mean by a good life? If running effectively makes me fitter, helps me lose weight, and keeps me in a good mood, why is it that evaluating the performance of that activity is the wrong approach to designing for the good life? It is not enough to track, and make sure that I perform these activities—the good life is not about the results, but the process. A good life is a lived life, not an accounted one.

This idea of the good is heavily influenced by both Aristotelian philosophy and positive psychology (Deci and Ryan 2008; Ryan, Huta, and Deci 2008; Bruni 2010). These two theories share a focus on eudaimonia, on the art and practice of living well and why and how do we perform that activity.

Virtue ethics provides the ethical framework that explains the concept of the good life; positive psychology is the discipline that has paid more attention to the reasons why we act well and why we want to act well. If virtue ethics provides the metaphysical and ontological reasons why we should act toward a good life, positive psychology provides the reasons behind the actual actions toward a good life.

Positive psychology has already provided a working definition of the good life, taking the Aristotelian concept of eudaimonia and applying it to a certain understanding of actions and activities and the reasons behind them.[5] For eudaimonistic positive psychology, the good life should be identified with an ongoing process rather than with an end, with a quantifiable outcome. Unlike hedonistic psychology, which is focused on the results that provide a sense of well-being, eudaimonistic psychology shifts the weight of responsibility to the actions that give the feeling of positive reinforcement, to the actions that lead to goals that lead to a good life.[6]

Positive psychology adds two interesting concepts to the Aristotelian idea of a good life: first, the good life is a process guided by intrinsic goals that are not necessarily and not always effectively rewarded with external outcomes. The good life is the process of acting according to our self to achieve these intrinsic goals, and the reflective capacity of the moral being is the capacity to identify these intrinsic goals and act upon them.

The second relevant concept is that of psychological autonomy. Living a good life is the outcome of being and feeling autonomous, of setting goals and evaluating them intrinsically, without the requirement of a heteronomic reward system. Eudaimonistic

psychology identifies the good life as that which develops autonomy, through the action and reflection on intrinsic goals that provide us with the satisfaction of a life well lived.[7]

Autonomy, in positive psychology, is opposed to heteronomy, or the performance of actions that bring satisfaction for reasons external to the reflection of the self. Autonomy leads to intrinsically motivated actions, while heteronomy leads to actions performed by request of external regulators. This distinction will be useful when I analyze gamification further on in this chapter.

To summarize: the good life is the life experienced as a process in which we perform the best of our virtues with the goal of flourishing, of exploring our potential as human beings. That flourishing is not externally determined or rewarded, but is the outcome of a process of internal reflection that leads to increased autonomy as human beings, as well as to the setting of intrinsic goals that we identify as constituting the good life. The good life requires the effort of living by the goals we set, evaluating them, and experiencing a sense of wellness and achievement when we accomplish them.

Is running, then, a part of the good life? The question in itself is misleading. Running is just an activity, a physical performance that leads to a number of results we give importance to. Running is only a part of the good life if we have reasoned that it is so, if we have actually developed as part of our understanding of the good life the idea that the effects of running are good.

Of course, some of its results are socially and culturally praised and recommended: physical fitness is a core virtue in modern societies. Furthermore, the type of physical endurance and patience that running requires also fulfills some of the rhetoric of stoicism that is popular in corporate language. Running is an activity that leads to many virtues and is therefore positively regarded in society.

However, running for the good life must not depend on that result. We might set as a goal to be more fit, and running is an instrument to it, but not a goal. Running can also be a meditative activity or a social one—I listen to spy novels when I run, while some of my friends enjoy running in groups not only for the exercise and the competition, but also for the social occasion: the shared activity, the shared goals that are explicit so the conversation can flow away from what is being done toward more interesting topics.

Running becomes a part of a good life if it fulfills some of our intrinsic goals—not all of them correlated with an activity that can be measured or tracked. The results, for many runners, are just an accident of an activity with another purpose than just achieving results.

Let's look briefly at learning as another activity that is part of the good life. I will focus on learning how to program computers and how to do it well. In our digitized Western world, computer programming has moved from a fringe, esoteric activity to a desired knowledge. We value computer programmers because their knowledge is not widely available, but also because they hold the key to living in this modern world in which everything has a computer inside.

Knowing how to program a computer could then be seen as a socially valuable and good skill. Both the activity of learning and the activity of practicing programming could be parts of a modern understanding of the good life, as I will explain later on. However, much like running, programming can be a painful and dull activity, and learning it can be even slower

and more painful. Therefore, we have seen a number of technologies that try to help the learning process by gamifying it. I'd argue that this is so because programming can be considered a part of the good life that needs encouragement to be learned and practiced.

But is it? Programming is just an activity or a skill: the capacity of giving instructions to a computer in order to make it perform instructions we have previously designed. The virtues of programming do not lie in the activity itself, but on what the activity entails: a way of thinking, a way of solving problems, and a way of commandeering the technologies that surround us. The gamification of learning is as conflicting as the gamification of running—well intended, but focused on a heteronomic understanding of the good life. I will return to the gamification of learning programming later on in this chapter.

The question is now ready to be formulated: how can digital technologies help us live a good life, focusing on autonomy and intrinsic goals as part of the process of flourishing? We have done that through gamification, applying game design elements to the evaluation and encouragement of activities—but is that the right path to take?

To answer these questions, we first need to take a quick sidestep in the philosophy of technology to understand how machines and morality relate and how activities are connected to technologies as a part of the good life.

Technologies and the Good Life

I like comparing running to playing live music. Both share a sense of exhaustion, an embodied pleasure of being one with something else—both are, in my view, physical experiences of the world.

In the case of running, how does Nike+ support this experience? Nike+ tracks distances and pace, as well as geographic locations, providing feedback on performance metrics once the run is finished. The Nike+ system allows users to set and publicly share online their own goals, a social layer that allows establishing ad hoc competitions. Nike+ excels at measuring performance, but is it encouraging the embodied pleasures of the sport? It focuses on what can be measured, on what can be tracked, on what a machine can compute—but the ways in which running can be a part of the good life go beyond crunched numbers.

I haven't chosen to write so much about running by chance. Through running we can explain the relation between technology, gamification, and the good life. Why running? I want to focus now on how technology can contribute to the good life, and I will begin with Albert Borgmann (1987, 2011), the philosopher of technology who attempted to reform the meaning and importance of technology in modern society by thinking about the relations between machines and the good life. The example of running is taken directly from Borgmann, and it serves as an excuse to introduce two key concepts in my understanding of gamification for the good life: focal practices and focal things.

A focal practice can be defined as an activity that requires skill, discipline, and practice, but which is also used to disclose the world. By this, Borgmann invokes those activities that explain the world from and for the user. To disclose the world is to experience it as a reflective being, demanding an

engagement with others and with the world itself. For Borgmann, running is a focal practice because the runner experiences a unified experience of the road, the environment, and his own body. To run is to experience the world through an embodied practice that can benefit from the aid and support of technology.[8] The road, the landscape, the body, they all exist in a different way, and the runners' capacity for experiencing and understanding that enriches their lives.

Focal practices are an interpretation and actualization of Aristotle's idea that the good life is the result of a constant exercise of virtues. The development of the good life requires a practice, an activity that allows us to exercise virtues. Only through practice we develop the good life—of course, practice coupled with reflection, with introspection, with the capacity to reflect and learn and develop our moral sense from those practices. The development of habits is important, but not in itself without the development of autonomous reflection.

Many of our daily tasks, of our activities, are mediated, cued, explored, and explained through the use of technology (McCarthy and Wright 2004). Some of these will help us disclose the world, becoming a part of focal practice.[9] For Borgmann, those are focal things: technologies that help facilitate focal practices. These things do not interfere with the experience, they do not get in the way of the focal practice—they just facilitate it.

Any focal practice, then, can be supported by technology, as long as this technology reinforces the point of the practice; that is, the experience of the world through a network of contexts and people.[10] Focal practices are the center of such a network, and technologies should never become that center—just a node, an element in that network. The point of running is not the shoes, or the clothes, or the road, but they all play a role in the practice.

Borgmann, a technology skeptic, argued that most technology does not qualify to be a focal thing. He proposed a difference between focal things and devices, the later being any technology that gets in the way of how focal practices develop relationships between people and environments. The treadmills, or the Walkman, interfere with running as a focal practice—running is not *running* anymore, it is something we do while we listen to music or while we are indoors.[11] Let's remember that for Borgmann, focal things are centers of complicated networks of people and activities. Devices, in contrast, are just machinery, simple objects with a function, but without the capacity actually to disclose the world.[12]

Borgmann's take on technology falls short of providing a comprehensive understanding of the effect and importance of these technologies in our life experiences. His theory is skeptical of the virtues that technology can embody and the positive behaviors it can help promote. Technology can help in the practice of the good life.[13] Machines have helped us develop better lives, structure society and construct better relations with the world, and they cannot be ruled out as important elements in the way we experience the world, also when developing the good life.

We need a concept that connects the concepts of focal practices and focal devices with the ways in which technology can contribute to human betterment. Technologies mediate our experience of the world, but we need to qualify that experience, to understand how the technology-mediated experience of the world can affect our moral understanding of it.

The Dutch philosopher of technology Peter Paul Verbeek provides a theoretical alternative that

expands Borgmann's philosophy in a more interesting direction. Verbeek proposes a hermeneutic approach to technology that explains how the world is experienced through and together with technology. We cannot understand the experience of the world without the mediation of technology.[14] Any experience of the world is intimately connected with technology. In fact, there is no divide between the world and humans and technology, but a continuum of relations and experiences. The world is experienced through technology, and the different modalities of that relation establish modes of experience and the different importance that technologies can have in our moral lives.

Technologies embody moral values, but also enhance the practice of morality—in Verbeek's terms, they mediate morality. A classic example is speed bumps: these devices mediate our relationship with the world with a moral message, forcing us to lower our vehicle's speed when the traffic conditions require it. This is not to say that speed bumps foster a moral practice, but as devices, they are encoded with values and moral meanings.

Technology, then, can mediate morality; it can have embedded ethical values that affect the way we experience the world through them.[15] This assertion is of particular importance for the idea of *playing* the good life, as it implies that technology can be *designed* so moral values are transmitted to a practice. If technology is designed with ethical affordances, and the behaviors it fosters are ethical, then we could design technologies for the betterment of our lives through the encouragement of virtuous activities.

However, the fact that technologies mediate morality does not imply that we become more virtuous beings by means of using them. Technologies *mediate* morality, but we *practice* morality, and we only do so if the activities we engage in are perceived to increase our autonomy. As I mentioned earlier, increased sense of autonomy is a sign of a good life according to positive psychology. We need to think about how technologies can be designed to *mediate morality*, while fostering a practice of life that increases our sense of autonomy. In this sense, technologies can become focal things that help develop focal practices. For that, gamification focused on the virtues of games, often praised for their capacity to engage users and make them learn behaviors and patterns.

Let me foreshadow my argument: the focal practice that gamification can support is play, and therefore we should think about how these devices could encourage a type of playful interaction that develops and fosters the good life. In other words, we can design focal things that lead to the focal practice of playing the good life.

Technologies can be used to practice the good life, as long as we understand them as focal things that are a part of a network that constitutes a focal practice. The way of understanding how technologies do that is by focusing on how they can mediate morality. Gamification design has looked at how game design can encourage, motivate, and pace activities that are a part of the good life. In the next section, however, I will critically look at a conceptual failure in gamification: its focus on the results of an activity, rather than on the playful practice of a good life.

Gaming Virtues

In the understanding of the good life I am advocating for, practice is key. Practice develops virtues and gives a feeling of autonomy; of course, practice coupled with reflection, with the capacity for moral introspection. Practice and reflection, the development of skills and the capacity to think about them, require time, dedication, and devotion. Technology can support this process: belts and alarms can remind us about safety in cars, Nike+ will help us track and measure our progress.

Because we often need support and encouragement for practicing virtues, philosophers and designers have considered the question of how to help people live the good life. Gamification is the design of activities using game principles in order to engage users following the positive and negative reinforcement loops of games. Even though gamification is being applied to many different contexts, I am exclusively interested in how it can be used to live a good life. Therefore, I will only focus on gamification applied to the development and practice of good virtues.

Let's return to our examples. The Nike+ system is designed with the purpose of reinforcing the runner's habits. In its most basic use, Nike+ is a tracking system, an old way of measuring an athlete's performances and preparing the athlete for training. It is also a good tool to encourage novices to run and to follow training programs. From this perspective, Nike+ could be seen as a device that supports an activity that can be defined as a part of the good life.

However, Nike+ only tracks data and numbers. It does encourage users to engage with running, but it has little to do with the context of running, or what's running's role in a more complex network of activities, practices, reflections, and beliefs that constitute the good life. Running, for Nike+, is the activity of making kilometers at a certain pace, providing results that can be compared and matched with others. But even using the geotagging version of the application, Nike+ does not much care about the whereabouts of the running. Running, for Nike+, is an activity detached from a context. Users of Nike+ can only make sense of their experience of running through the data manipulation tools that the system affords, such as the limited description field for the run where users can describe the quality of the run. Nike+ in this way limits its users' agency over the practice of running.

By design, Nike+ is limited to gathering and processing quantifiable data on each run. As an application, it can hardly contribute to the idea of the good life through practice—it can support it, but it does not allow for the kind of autonomous thinking that the good life requires. It is a node in the network of a focal practice, just not the most important one.

My second example is Code Academy. Code Academy wants to help people learn to program with different languages, from Javascript to Python. For that, users can follow courses with different tracks, and they are rewarded points and badges when they complete some exercises. Through these methods, Code Academy intends to keep users engaged in learning.

Code Academy does only a few new things, but it does them very well. It provides a clear and well-developed framework for learning, a summary of many modern pedagogic techniques and online modules for learning. Code Academy sets off to teach

people to program—but does it contribute to creating a practice that is part of a good life?

Programming can be seen as a focal practice, as the development of a number of mental skills and good practices that provide discipline. Programming is the practice of having control of and understanding computing machines, and therefore, albeit vicariously, understanding the culture and the nature of contemporary culture. To be able to program is to have important knowledge about computation, about the inner workings of the machines that move the world. To be able to program is to gain deeper control over the machines that articulate many of our experiences, being able to give them instructions and adapt them to our needs and demands. We have delegated important parts of our emotional and work lives to machines—being able to program empowers us in that relation with computers.

Code Academy provides an important first step toward learning how to give instructions to a computer. However, Code Academy is based on the principles of teaching how to perform programming tasks, rather than understanding why programming is the interface between man and machine. Code Academy teaches programming, but only as an activity, not as a focal practice. It does not help understand the workings of computers or the importance of thinking about how computers work.

Much like in the case of Nike+, Code Academy tracks the progress of users without allowing them to develop a focal practice. It helps to develop habits but does not contextualize them into an activity that makes for a good life. Code Academy tracks and rewards progress with relevant badges matched with a progression ladder that leads to learning a number of rote skills, but not programming as a focal practice.

Both Nike+ and Code Academy use computing power to gather, collect, compare, and visualize data regarding activities that users voluntarily log. The idea is simple enough—processing these data yields responses from the system designed to engage users in the activities tracked. This type of gamification can be seen as an instrument for the good life if we presume that the activities tracked are a part of the good life, and that the way of tracking the data, or even the data tracked itself, support or are a part of the good life.

When a system tracks data and rewards predetermined actions, there is a risk of forcing practices into users: we would only do what the system can track, and therefore we would comply with a set of practices that has been externally determined as constitutive of "good" behaviors. If we think about those applications specifically designed and marketed to help us improve our lives, I argue that their biggest mistake is to apply a very naive understanding of virtue ethics and the good life.

Gamification tends to delegate virtues in data and rewards, neglecting the importance of practice *and* reflection in the development of the good life.[16] Technologies designed for improving habits have an apparent beneficial effect in quantifiable, trackable terms. However, these technologies do not help or contribute to development of a good life, as they do not require us to develop or practice moral values—just to *enact them*, to perform them as a mechanical repetition of tasks.

In other words, gamification runs the risk of proposing an approach to the good life based on delegation to data, without developing autonomy. Focusing on heteronomy rather than autonomy in goal-oriented activities that are trackable using computers, rather than in activities with internal rewards and

goals, gamification runs the risk of focusing on the tracking technology rather than on the idea of a good life.

The practice of the good life needs to develop not only skills, or habits, but also, and perhaps more importantly, the capacity for self-reflection, the development of autonomy as moral beings. Gamification is doing some things right, but its focus on the habits through designs based on reward systems invokes a heteronomic relation of dependence with the gamified system. The good life requires autonomy, and therefore any technology that wants to support it needs to be a technology that fosters autonomy. It is the task of design to think and create such technology.

Designing to Play a Good Life

In this chapter, I wanted to look at how gamification can support the development of the good life. To that purpose, we need to look at how design has understood the activity of play. I want to propose a way of thinking about the design of gamification deeply rooted in interaction design, with the purpose of designing (for) play, rather than games. My intention is to inspire designers in aspects of the good life that can be designed for through play. This is meant to be a reflective and inspirational approach—not so much a method as a set of ideas and provocations for designers.

So far, I have focused on systems that use *game* design elements to create behaviors that could be interpreted as a part of the good life. I am, however, not interested in games. Games are just one of the types of devices designed and created to *play with*. I want to challenge gamification developers to think about play, and not about games, when trying to design for the kind of engagement games can create. My position is simple: if we want to engage people in the good life, we should start not by making games, but letting them *play the good life*.

I guess at this stage in the book, play has been defined several times, and Huizinga, Caillois, Sutton-Smith, and Suits are familiar names for the reader. Hoping so, I will not bother you with yet another scholarly account of the history of play research and its different definitions of play we can find. What I am going to do instead, dear reader, is assume that we both know what play is, and that we can identify playfulness too—otherwise, may I direct you to read Jaako Stenros's chapter in this book, please? Just to make sure that we are on the same page: I am now being playful with my writing tone, and play is what we do with toys and games, in playgrounds and beds too.

I am interested in play as an appropriative and expressive activity, one that likes to be bound only to play freely with and within those boundaries. Like Salen and Zimmerman wrote, play is free movement within a more rigid structure. The type of play I am interested in can be expressed with objects created for that purpose, like toys (Benjamin 1999a, 1999b), or as an attitude, an understanding of life or of a particular situation that brings forth the idea of play. In other words, we can play or we can be playful.

There are, of course, a number of characteristics of play that are relevant: play takes things in its own seriousness, and it is often switching between the

very important and the trivial. Play is also autotelic—its purpose is its very existence, which does not mean that it is not *important* for us. In fact, because play is autotelic, it is better suited to encourage autonomous thinking, even when playing under the heteronomic frames of rules. Play can also be dangerous, addictive, and hurting toward those who do not want to play. What interests me here, though, are the expressive and appropriative capacities of play.

What do I mean by expressive? Let's start thinking about how we play with dolls, for instance. Children's play with dolls is "just" play, but if we listen to what they say, the worlds that are built through imagination, we can perceive that play is used to express their worldviews, what concerns them and worries them. Play, like music and painting, is a form of expression. When we are playful, we are expressing ourselves. Think about the playfulness in my writing tone before—it was a more personal, more direct tone than the somewhat dry approach I took earlier. Think also about all the other embodied expressions of playfulness we engage with, the little dance steps we perform when nobody looks, the back heel kick with which we finish going down the stairs, to make sure that the stairs stay in place ... all these tiny performances enlighten the day, make us take over our activities, and express who we are.

Play is expressive because it discloses who we are and what we think of the world, even when engaged in mundanity and work. To play or be playful is one way to express our being in the world. Any designer invested in harnessing the power of play should have in mind how will she let users express themselves through play.

Play is also appropriative; it can *take over* situations and make them playful. In this process, whatever is appropriated might change meaning: a job interview might become a contest of wits, a conversation might turn into flirting, a vending machine can have personality we can argue with.

When play is expressive and appropriative, when we use it to take over a situation and express ourselves in it, it becomes a focal practice. Through play we can disclose the world, establishing a network of (play)things, people, and situations through which we make sense of experiences. In this sense, the focus on designing gamification as objects and systems is appropriate but misleading: we should not concern ourselves with the design of the devices but should pay attention to play as a focal practice, as an activity that uses things, as a part (sometimes the *central part*) of an experience that discloses the world.

Gamification, then, has taken the wrong focus when designing devices for the good life. It has seen the device as an embodiment of the practice, while it should have focused on the following question: how can we play the good life? How can we use the creative, expressive nature of play and harness it to develop the good life?

Let me be specific: there is nothing wrong in using reward systems or similar design elements when creating services and devices that would gamify the practice of a good life. However, the focus on these elements has made gamification devices not always playful. Play and playfulness are the element that's missing in the design of gamified services for the good life. Any focal practice is constituted by the practice and development of skills and the reflection about that practice. Play becomes a focal practice when it is expressive and appropriative, when it takes over a situation to interpret as part of play. Gamification systems like Nike+ or Code Academy encourage the practice of skills, but never let users appropriate or express situations through

them—they don't let runs become more than data, they don't let programming be more than giving instructions.

Designers should then think about what characteristics of play need to be addressed by the device/system/service that will aid in the development of a playful practice of the good life. In my terms, that would be how to design for appropriation and expression.[17] There are four elements designers of playful devices for the good life should take into consideration:

- the development of skills through practice;
- the development of an autonomous interpretation of these practices that leads to self-reflection and evaluation (without resorting to external/heteronomic evaluations);
- the appropriative nature of play;
- the expressive nature of play.

Caillois (2001) famously classified play types into two: *ludus*, or the playful activity that has goals and winners: and *paidia*, or the playful activity without goals. Designers should then ask themselves if users should engage with a service as part of a ludic practice, with a goal and purpose objectively measured, qualified, and ranked, or if they should approach it as a paidic activity, a playful appropriation that reveals its expressive possibilities.

Furthermore, Caillois typified the types of games as being a part of any of four categories: *agon*, or competitive games; *alea*, or games of chance; *mimicry*, or games of make-believe; and *ilinx*, or games of vertigo. These types of games can also be mapped to our idea of designing play: each of them refers to a particular form that constrains the playful activity.[18] For instance, chance limits the ways we can appropriate the world, and mimicry can be used as an expressive conduit through play, like in role-playing games.

In many uses of gamification, we can find agon and ludus as the dominant play types: competitive, externally rewarded practices of what should become the activities of the good life. Think about Nike+, which encourages competition with others, or Code Academy, which rewards you with different badges depending on your progression. These are not necessarily bad things, but they are superficial understandings of both the good life and the activity of play. Why don't we have more gamification systems based on mimicry/role-play or in the bodily pleasures of ilinx? One reason might be the tight coupling between computing and the conventional ways of designing for agon and ludus, but also because we tend to think about play as an activity with goals played through games.

If we want to design a playful good life, we need to start focusing on four elements: How will this practice help develop skills and virtues? How will this practice help develop autonomy? How will this practice be playfully appropriative? How will this practice be playfully expressive? Designers should identify what elements of an activity constitute the good life. That identification should lead to a process of reflection focusing on interpreting the key elements that lead to virtuosity and autonomy through the lens of expressive, appropriative play.[19]

This is precisely what is missing on gamification: the design of those elements that make playing a focal practice. Gamification focuses on skill training, but if it was augmented with the appropriative, expressive capacities of play, if it allowed users to play with and through the gamified system, then gamification could support the development of a good life.

My trade is not that of design; but as an example, a redesign of a running tracking device that encourages a creative good life should add something like geotagging and a photo-sharing application, so the runner is not oblivious to the landscapes where she runs. Or, if she is an indoor runner, the system could encourage talking to other runners in other treadmills by asking, as part of the daily logging routine, if she knew anybody else in the room when training or if there were other registered users of the tracking system.

Let me be clear: play itself is a focal practice, and what gamification should do is take from it the characteristics that constitute it as a focal practice and apply it outside of the contexts of play. The good life is constituted by practices and reflection, by practicing both virtues and the capacity to understand how to become a better human being. It's a practice and a reflection. Play can contribute to practicing the good life by letting us appropriate a context and express ourselves through them, setting autotelic goals that lead to our personal satisfaction.

My approach to this design challenge of applying playfulness to the creation of technologies for the good life is focused on two ideas: first, we must design for the *Homo ludens* (Gaver et al. 2004; Gaver 2006, 2008), for users who want to live a playful, better life. And second, technology needs to stay open to interpretation, avoiding functionalist efficiency for a more appropriative design of interactions and evaluations (Sengers et al. 2005; Sengers and Gaver 2006).

The idea of designing for the *Homo ludens* advocates a type of playfulness that engages creativity within a context of which the device is just an element. This context might lead to the emergence of rules but does not require these to happen—playfulness might not be bound to rules, and therefore there might not be need for evaluating any activity according to these. The opposite of playful design would be instrumentalized systems,[20] focused on processes and results rather than in the creation of situations.

A gamified system for the good life needs also to be open for appropriation or interpretation. In classic human–computer interaction (HCI), ambiguity and multiple interpretations were forbidden. However, when it comes to the interpretation of technology as a part of our value system, ambiguity can play a creative role that fosters autonomous thinking.[21] If we want to design technologies that users can playfully appropriate as part of their playful good life, we need technology that is interpretively flexible, providing a space of possibility in which users can find meanings and express themselves. Technology has to be, and stay open to, playful interpretation, because in that openness play can be used to practice and develop values and virtues to develop the good life.

Ambiguous technologies for the *Homo ludens* would, by design, encourage the principles of eudaimonia, of creating and developing skills and reflective capacities, as well as fostering a sense of autonomy in the process of moral development.

Gamification can support playful practices, appropriating contexts and letting users play with them. This would imply a step away from merely performing activities that can be rewarded. This also means embracing the most mischievous sides of play, with dark play (Schechner 1988) or even deep play (Geertz 1972) as the frontiers that should not be crossed, but can be explored: What if the Nike+ application mocked you for being slow or lied about the distance? What if Code Academy made you type infinite loops forcing you to close the browser, only to return to learn why that happened?

Play is an opening for expressing who we are while taking over a situation, a context, a technology. Designers of gamification for the good life need to be aware that the designing of playful, ethically relevant technologies needs to take into consideration the need for autonomy and expression that is behind the ideas of gamification for the good life. Let people play, and make play part of the practice of a good life, supported by technologies open to that circle of values and practices, of agency and creativity.

Playing the Good Life

I hope I have proposed in this chapter an argument for a better gamification. The success of gamification has been so fast and so overwhelming that it has encouraged many critiques. Like many other trends that promise a better world, gamification has been targeted for its sloppy theoretical and scientific backgrounds, the lack of empirical data behind the results showcased, and even for the naive interpretation of humans and motivations that seems to fuel it.

However, gamification is just a symptom of a cultural trend: the vindication of play as a legitimate way of living, creating, and expressing. We enjoy using technologies to play, and we can harness it to live a good life. In fact, this is not new: political artists and educators already saw play as a way of addressing the cultural issues they were interested in addressing.

To be able to play the good life, we need to think about play as a focal practice that can help develop virtues and practice morality. For that to happen, though, we need the support of technologies of play, from toys to playful devices. Designing these technologies is not easy: it requires thinking about designs that are open, contextualized in a practice at the same time that they take over such practice. More than mediating ethics, what playful ethical technologies should do is encourage and guide ethical practices through play.

The good life can be seen as a golden, unachievable ideal, usually at odds with the hardships of "real life." Precisely because of that, because the good life needs practice and denying the easy way out, play is a good accomplice of morality—not because it engages us, but because it lets us express ourselves, who we are, and what we do in a complete way. Playing the good life is important because playing is expressing ourselves, taking over situations, and shaping our world. We can do all those as an expression of our virtues, appropriating the world and acting as moral beings. There is nothing more serious than playing the good life.

Notes

1. By gamification, I understand here the type of design described by and advocated for in Deterding et al. (2011a, 2011b, 2011c) and Deterding (2012). See also Raessens (2006); Dole (2011).

2. "Thus, flourishing is the calling of a human being—it is the ultimate goal for the sake of which all subsequent "mini-goals" are pursued. And when we consider this fact, the move in the function argument

in which Aristotle draws an analogy between craftsmen and human beings is defensible indeed" (May 2010; location 902).

3. "[...] morally virtuous activity is a developmental prerequisite for contemplative activity" (May 2010; location 162).

4. The activities that lead to a good life also need to be qualified—they can be physical, but guided toward a development of character and wisdom: "Ethically virtuous activity is a complex rational activity that includes both the virtues of character and practical wisdom or phronêsis" (May 2010; location 1010).

5. "To summarize, the Aristotelian view of eudaimonia considers well-being not as a state of pleasure versus pain, but as living well. As he put it, "the happy man both lives well and does well" (Ryan et al. 2008, 103). Living well entails actively and explicitly striving for what is truly worthwhile and is of inherent or intrinsic human worth, and it contrasts with the pursuit of crass endeavors such as materialism or pleasure seeking that pull one away from virtues. Eudaimonia is characterized by reflectiveness and reason. Finally, eudaimonic pursuits are voluntary, and are expressions of the self rather than products of external control or ignorance. Together, Aristotle's eudaimonia is thus characterized as living well, and entails being actively engaged in excellent activity, reflectively making decisions, and behaving voluntarily toward ends that represent the realization of our highest human natures" (Ryan et al. 2008, 145).

6. "Rather, eudaimonic conceptions focus on the content of one's life, and the processes involved in living well, whereas hedonic conceptions of well-being focus on a specific outcome, namely the attainment of positive affect and an absence of pain" (Ryan et al. 2008, 140).

7. "The term autonomy literally means 'self-governing' and implies, therefore, the experience of regulation by the self. Its opposite, heteronomy, refers to regulation from outside the self, by alien or external forces. An autonomous act is one done freely and willingly by the actor. In the case of intrinsic motivation this is obvious, because intrinsic motivation represents doing an activity because of its inherent satisfactions, which one typically does quite freely. But in the case of activities that are not intrinsically motivated, the issue is not inherent enjoyment, but rather inherent and self-endorsed value. A person who acts autonomously reflectively embraces an activity as his or her own, endorsing it at the highest order of reflection" (Ryan et al. 2008, 157).

8. The inspiration for this idea of embodied interaction comes from Dourish (2001).

9. "A thing is focal if it is what we give our time to and what we build our lives around. Like the fireplace, focal things richly interweave means and ends, point to the larger context of their setting in nature, the community, and culture, call for attention, effort, skill, and fidelity to regular practice, and invigorate individual and community life. Genuinely focal things stand over us as a commanding presence" (Strong and Higgs 2000, 32).

10. "While the function of a device captures one or a few aspects of the original thing, such as the exercise of muscles, devices sever most other relationships. At the health club, one might be reading a book, riding a stationary bicycle, and listening to music with headphones. Mind, body, and world are all dissociated from one another. In general contrast,

then, a focal thing is not an isolated entity; it exists as a material center in a complicated network of human relationships and relationships to its natural and cultural setting" (Strong and Higgs 2000, 23).

11. "What has replaced the thing is the device. The device provides what Borgmann calls a commodity, one aspect of the original thing (for example, in the case of the wood-burning stove, warmth alone), and disburdens people of all the elements making up the world, or context, and engaging character of the thing. This world of the thing, that is, its ties to nature, culture, the household setting, a network of social relations, mental and bodily engagement, is taken over by the machinery (the central heating plant itself) of the device" (Strong and Higgs 2000, 28).

12. "But where technology is intended as a thing to provide a focus and orientation in our lives, and to reveal the world in its essential dimensions, it makes no sense to measure it by the standards of the device" (Tatum 2000, 187).

13. "These demands and attractions of the focal thing's commanding presence make things engaging for mind and body, serving to unify them" (Strong and Higgs 2000, 22).

14. "By mediating our actions and experiences, technologies help to shape the quality of our lives and of our moral actions and decisions. To deal adequately with the moral relevance of technology, therefore, the ethics of technology should incorporate the phenomenon of technological mediation" (Verbeek 2011, location 113).

15. "In my postphenomenological approach, technological mediation concerns action and perception rather than cognition; and moral mediation is not only about the mediated character of moral ideas but mostly about the technological mediation of actions, and of perceptions and interpretations on the basis of which we make moral decisions" (Verbeek 2011, location 796).

16. I am referring to the type of gamification design advocated by Zichermann and others (see Zichermann and Linder 2010; Zichermann and Cunningham 2011).

17. This idea resonates with some contemporary work in interaction design—for extended reference, see Dourish (2003), Eriksen et al. (2003), Costello and Edmonds (2007), Desmet, Porcelijn, and Dijk (2007), Dix (2007), Korhonen, Montola, and Arrasvuori (2008), and Bekker et al. (2010).

18. This idea of "form" is inspired by interaction design work on the topic of aesthetics. See Crilly (2010) or Hallnäs (2011).

19. The notion of reflection is present in research as a goal of certain practices in interaction design. See Hallnäs and Redström (2001), Boehner et al. (2004), and Fallman (2011).

20. This idea of instrumentalization is also resonant of the Frankfurt School's understanding of modernity and, by extension, of technology. See Adorno and Horkheimer (2010).

21. This idea has become quite popular in third-wave HCI. See Gaver, Beaver, and Benford (2003), Aoki and Woodruff (2005), Bell, Blythe, and Sengers (2005), Bødker (2006), and Bilda, Edmonds, and Candy (2008).

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POSITION STATEMENT

GAMES AND THE WORLD

Frank Lantz

What is the relationship between games and the world? For me, this is the important question. Before we consider the *potential* relationship created by new applications of game systems to real-world situations, let's ask ourselves about the *existing* relationship between plain old games and the workaday world they are a part of. What is the relationship between Tetris and the world? Between chess and the world? Between basketball and the world? Between Minecraft and Street Fighter and Portal and the world?

I choose to answer this question in a way that I hope is simple, straightforward, and understandable, by saying that games are an aesthetic form. Like music, like stories, like poems, like dance, they are a human activity we engage in for its own sake, in which we sometimes find deep meaning and profound beauty, and sometimes find pleasant entertainment and compelling recreation, and sometimes find both, and often neither. This answer seems obvious and uncontroversial when you look at something like Portal. It seems pretty clear that, whatever else video games might be, we are comfortable thinking of most of them as pop cultural artifacts. They are clearly at home alongside rock records, comic books, movies, and television programs. Which, regardless of their individual or collective merit, means they are part of the aesthetic realm.

But let me quickly admit that this answer, straightforward as it may be, is a classic case of question begging. Because, after all, what exactly do we know about aesthetic forms? What are these things? What types of experiences do they produce? Where do we go, when we sit in the dark watching actors kiss, when we curl up in bed reassembling words into imaginary places, when a percolating polyrhythm moves our feet or a poignant melody warms our throat? Oh, sure, there are plenty of aesthetic theories, plenty of ways to "explain" these experiences, by referring to evolution's influence on behavior, or the pure spirit of sublime consciousness, or neural networks, or networks of power, the means of production and the politics of pleasure. But let's be honest, if there's one thing you and I both know, it's that we don't know where we go when we do these things, and we don't want to know. Sometimes a sorrowful song makes us weep, but the emotion we feel isn't exactly sadness and isn't exactly not sadness, and this feeling is complicated and mysterious and beautiful. And we don't want or need an explanation of this feeling, and we're not going to get one. Instead, we are simply grateful that it happens, grateful for this complicated, beautiful, mysterious aspect of life.

And that is, maybe, the one thing we can say for certain about aesthetic experiences, about their

relationship to the world, that they resist explanation, that they are hard to pin down, not just complex and ambiguous but *irreducibly* complex and ambiguous, that they exist outside our ordinary frameworks of logic and purpose, cause and effect, reason and explanation. We *need* to have this outside, and aesthetic experiences serve that need. But wait, even that implies an explanation that isn't there, even that is an attempt to rein them back into the logic of purpose and reason, explain how and why they work, and they will always continually elude this attempt. Better to say they occupy a position, relative to our explanations, that is analogous to the truths that elude capture in formal systems. Patsy Cline sits outside of heartbreak looking in, the same way that a Gödelian assertion hovers above a formal logical system that can never capture it. Inexplicably true.

So if this is the relationship that aesthetic experiences have with the world, then this is also the case for games. Tetris sits outside our desire to arrange and complete, looking in. Portal hovers above our compulsion to solve problems and escape confinement. And this tension, or dynamic, or dialogue, or dialectic, is always there, to some degree or another, between every game and the world they are a part of, and apart from.

So what does this mean for attempts to harness the power of games for external purposes? First of all, it's clearly possible. In the same way that people write songs to teach the alphabet, or take photographs that make us want to smoke, people can create games that teach us math, or trick us into washing the dishes, or persuade us to be less cruel to animals, or improve our workplace efficiency. That such games can and do exist is obvious. I am no purist, no idealist, the gap between aesthetic experiences and the world is messy, permeable. But understanding that there *is* such a gap helps us sort through this mess and see it more clearly.

For instance, it helps us better understand the flow of concepts between games and the world. You often hear people talk about taking concepts from games and applying them to real-world situations—rules, goals, winning and losing, levels, rewards, and so forth. But are these really *game* concepts? Each of these things already exists in the "real world" and always has, we certainly don't need games to see all of them at work in our social relations, our institutions, and our workplaces. What games do is take these concepts and abstract them, stylize them, isolate them, arrange them, intensify them in a way that makes them newly visible to us. This is aesthetics in action. Understood in this way, we can see how applying game concepts to real-world situations is a doubling back, taking concepts that started in the world, transforming them through games, and then reapplying them to the world from which they originated.

There are many different ways to think about this doubling back. What happens when concepts are transformed through aesthetics into abstracted, stylized versions of themselves, then reintroduced into an everyday context? Do they inject some of the strangeness of the aesthetic realm into the regular world or do they lose their magic, reverting back to the mundane?

For my part, I have created many games that intentionally blur the distinction between the real and virtual, between the game and ordinary life, including some that were designed to achieve real-world goals. I have always approached these projects with the same idea—not to harness the power of games to accomplish something in the real world, but the opposite, to harness the power of

the real world to create a better, more interesting game. Sometimes this means using pervasive technology to highlight the strangeness of our contemporary world, the hybrid half-real spaces we move through in our daily lives, the new ways in which objects and information intermingle, creating new surfaces on which to play. And sometimes this means using the idea of a real-world goal as an ingredient in an aesthetic experience (as opposed to using an aesthetic experience to accomplish a real-world goal); in other words, doubling back across the gap a second time, smuggling the concepts of reason, purpose, challenge, and goal over the border and back again and then back across, nervous, sweating, clutching a suitcase full of contraband whose origin is no longer clear. Exporting indigenous vegetation? Importing raw materials to be processed? Exporting processed goods for sale? Importing revolutionary propaganda? Exporting weapons? Importing medicine? Drugs? Pornography? Artworks? Explosives?

Games.

9 PLAYFUL AESTHETICS: TOWARD A LUDIC LANGUAGE

Mary Flanagan

Popular thinkers of the present moment claim that we are in a liberatory ludic age, an era defined by its connections to play, games for all, and the integration of games into everyday life that not only serve to entertain, but indeed *save us* and the world at large. Others, meanwhile, approach this claim with critical reservations about the meaning of such games and their rhetorical and practical relationships to freedom and work. Why is it that this era in particular has witnessed the adoption of game-like strategies among social, personal, and professional domains? What does such ludic material really mean to us on an experiential level? If we are indeed in an era of ubiquitous play where *gameful* strategies lure us into organized structures for play, work, and beyond, how might we identify some of the ways these ludic aspects are experienced?

Something playful is indeed rumbling across various cultures. Increasingly, gameful references are framing personal experience, media, and commerce; imposed reward structures have brought on the gamification of the workplace; and so forth. This shift manifests itself in a complex fabric of change that is deeply reflected in artistic practice, as I shall expose in due course in this chapter. We must ask ourselves: is the ludic turn at this particular moment due to our more constant connection to everyday play experiences (the widespread accessibility of mediated sports, mobile games, an "addiction" to console games, and more) that have fueled a thirst for structured challenges and rewards? Or is it, as Juul, Pias, and Wark might argue, due to the proliferation of technologies, where play is not only enabled by technology, but further, that games provide an ultimate match for computers because they are so fit to process game rules (Juul 2004; Pias 2005; Wark 2007)? Because game-influenced and playful creative work has proliferated over the past century, and significant contemporary artworks reference games or play consistently, I wish to explore the ways that *gamefulness* has altered and changed art itself and the criteria by which it is judged. As questions arise as to the ways in which games are in and of themselves their own art form, understanding how they affect broader notions of aesthetics remains a key component to explore, indeed, the ways in which games in art change culture.

In this essay, I set out to reveal how elements of a *ludic language* have permeated other cultural arenas and in particular art. I'll first discuss two works of art that can serve as a foundation to link the discussion to the aesthetics of new media art and show how these form an introduction to a collection of *playful aesthetics* that permeate popular culture: a set of

artistic practices emerging from activities of human play and, in particular, from the qualities and elements of games and their theoretical, tangible, and practical implications. Playful aesthetics permeate both digital and nondigital aspects of culture. How might we put ourselves to the task of discovering these aesthetics? Are such playful variations on aesthetics derived from video games and play culture somehow vastly different from common "material" or analog games such as chess or sports? In this essay, I use the work of artists to help tease out these aesthetics and put forward a unique set of lenses, or propositions, to use in understanding the ludic turn in contemporary culture.

Defining Playful Aesthetics

To investigate notions of aesthetics in Western media forms, it is useful to touch on historic notions of aesthetics and include elements from Western philosophical debates. Eighteenth-century philosopher Immanuel Kant (1911), for example, outlined four possible "reflective judgments," rooted in what one *perceives*: the beautiful, the agreeable, the good, and the sublime. The term *aesthetics* has come to be used to demarcate kinds of objects, positions or attitudes, experiences, values, taste, and judgment. Aesthetics are the way we judge and analyze experience. *Aesthetic* is thus a term that not only outlines creative difference, but also confers power. Contemporary scholars tend to loosen Kant's strict categories that seem especially limiting in light of unpredictable contemporary creative practices. The notion of the lone artistic genius has been complicated and superseded by different models for creative work, including those involved in the production of, and use of, games and play environments. The hacker, the modder, the low-tech interventionist—these playfully disruptive practices interrupt what have become staid, scholarly notions of beauty and power. Because game-influenced and playful creative works have proliferated over the past century through the playfully disruptive and inventive works by Dada, Situationist, Punk, and hacker culture (to name but a few threads), and because contemporary artworks increasingly reference games or play, these playful practices have emerged to change art and culture. Thus, a reconsideration of aesthetics in light of such practices is long overdue.

To philosopher Jacques Rancière, notions of aesthetics have long brought with them questions about power and authority. The development of aesthetics "was a kind of redistribution of experience ... those who were destined to rule and those who were destined to be ruled didn't have the same sensory equipment, not the same eyes and ears, not the same intelligence. Aesthetics means precisely the break with that traditional way of embodying inequality in the very constitution of the sensible world" (Rancière 2006, par. 26). Other philosophers have been equally dubious about the liberatory claims of play and leisure. As Horkheimer and Adorno (1996/1944) have pointed out, the aesthetics of a leisure pursuit can also serve to camouflage work, while "culture industries" serve both ideology and power by shifting potent political action into matters of taste. Playful aesthetics, then, are not only the means to understand emerging art forms, but also suggest and convey types of power.

Importantly, artists have long unearthed what is suddenly being called "the new aesthetic" by the popular media, who now notice a fifty-year history, from Situationists to Fluxus, from Relational to Game Art. Artists have a long history of inquiry in play, and their works can be subversive, whimsical, social, and critical. In proposing propositions as lenses with which to locate origins of emerging aesthetic practices, I'll show how these play-based artworks form a foundation for playful aesthetics. Pundits promoting "the new aesthetic" may wish to recognize the nuances behind the expression of power and conceptual thinking in creative work in order to explore the many types of aesthetic processes and criticality implied by play and new technologies.

The Playful Is an Act Composed of Sequences Set in an Eternal Present

Scholar Mihaly Csikszentmihalyi (2008) is well known for his insistence that the phenomenon of "flow," that process of being completely immersed, away from temporal awareness in which one is immersed into a continuous state of "now." While games and play do not always trigger that sense of flow, they do have interesting spatiotemporal qualities that have fascinated Bateson to Huizinga to contemporary scholars (Huizinga 1955; Bateson 1972). One thing is for certain: games and play activities are emergent—acts becoming, of agency. If Deleuze were to analyze many play experiences, he would find "a foundation of time, non-chronological time" (Rancière 2006). This sense of non-chronological time can be found across many disciplines, but I'd like to look at early mixes of art and communications technology in order to analyze how rules or instructions might help create a sense of "present" and how artists have approached this both conceptually and practically. This will help us better understand the ways in which the experience of time and changing duration contribute to playful aesthetics.

The "eternal present" is well represented in telematic works in which a sense of simultaneity is highlighted. One of the earliest examples of such work is the 1920s series of *Telephone Paintings* created by painter Laszlo Moholy-Nagy. By telephone, the artist described, demarcated, and ordered for fabrication five paintings in porcelain enamel—an early work of telecommunications art. The work highlights the playful creation of art through technological mediation—calling in one's painting for creation had not been done before. The artist outlined the paintings on graph paper and, using a commercial color chart, dictated the points of color on a grid. The artist drew out the paintings on graph paper utilizing the factory's color chart, while the factory supervisor over the telephone followed along on graph paper, taking down the dictated shapes in the proper coordinates. Thus, the images were never "touched" by the artist, but rather, the artistic intention was communicated through networked technology into action.[1]

Was it a game by Moholy-Nagy to "call in" his artwork for manufacture? Given the expanse of time elapsed since this event, the artwork as described now sounds like a whimsical game of Battleship, in which the artwork was rendered with a hope for exactness against the inherent interference and miscommunication of distant human operators. This early game of art models itself on the hidden information and communication that is the foundation of Battleship, or its predecessor, L'Attaque.[2] The aesthetics involved with the *Telephone Paintings* invoke

temporal concerns (both instantaneous and lagging, for the artwork had to be created not directly from the hand of the artist, but run though a process divorced from him), and its distributed production made space—or simultaneous lack thereof—integral to the final work. Moholy-Nagy's work highlights communication and process, deeply engaging synchronous and asynchronous time, decision making, mediation, physical possibility, strategy, and challenge. These can be seen as aspects of emergent forms of aesthetics, and they expand to blend the boundaries of games, play spaces, and day-to-day routines. Such themes—time, distributed production, immediacy, and process—are twentieth-century concerns; they are unique aspects that have marked the contemporary transition into a new way of understanding everyday life. Thus, this type of thinking adds to the quest for playful aesthetics, for games as well as art like Moholy-Nagy's are acts composed of sequences; games too are process-based creative endeavors that smooth over temporal shifts and dislocation (players don't, for example, need to be in the same time and space as those who write the much labored-over rules of a given game).

Many people will think of games as *fast*: that games are inherently played with a sense of urgency. Timers, competition to be first, races, and so forth, guide these playful experiences. In fact, games and play are characterized by a phenomenological foundation of action: "Play is going. It is what happens after all the decisions are made—when 'let's go' is the last thing one remembers. Play is action generating action: a unified experience flowing from one moment to the next in contradistinction to our otherwise disjoint 'everyday' experience" (Csikszentmihalyi and Bennett 1971, 45). If we look closely, we see that time in both more traditional play (such as a puppet show) and time within commercial video games is more interesting than this stereotype allows. Such recognition is a process of understanding nuance. Narrow and rigid, artificial and old-fashioned assumptions about games become less and less relevant as they rightfully enter their own artistic space. Today's games that are art, and simultaneously art that reflects games, resist being forced into normative molds regarding what is a game and what is art. Just the idea of a game or a play space brings the promise of experience—as does art. This may very well be the core promise of play, and the premise upon which gamification has been sold by marketers wishing to motivate and regulate commerce. Play any game: instants are here and there; lag-blurring instants. Henri Bergson argued that we must be allowed space for free will to unfold along autonomous and unpredictable lines: in this lies the collapse between voluntary play and the involuntary, the requirement that people must play their own gamified everyday life. Conceptual artist Yoko Ono toys with such autonomy in her incredible artwork *Play It by Trust*, the famous series of all-white chess sets she produced for several decades in the twentieth century. Free will, when there is no differentiation, no side to take, is to Ono an important and decidedly compelling site for reflection on politics, global concerns, and the phenomenology of time. We all know that play implies agency. How long does it take to win at chess where there are no sides to take?

Playful technologies and games expose the importance of time. Time, to theorists such as Michel Foucault, has been postulated as an invention of power in itself: a form of discipline, regulation, and governance. The aesthetics of time, like any other playful aesthetic, implies power. Yet Bergson noted, "duration is subjective and constitutes our internal life."[3]

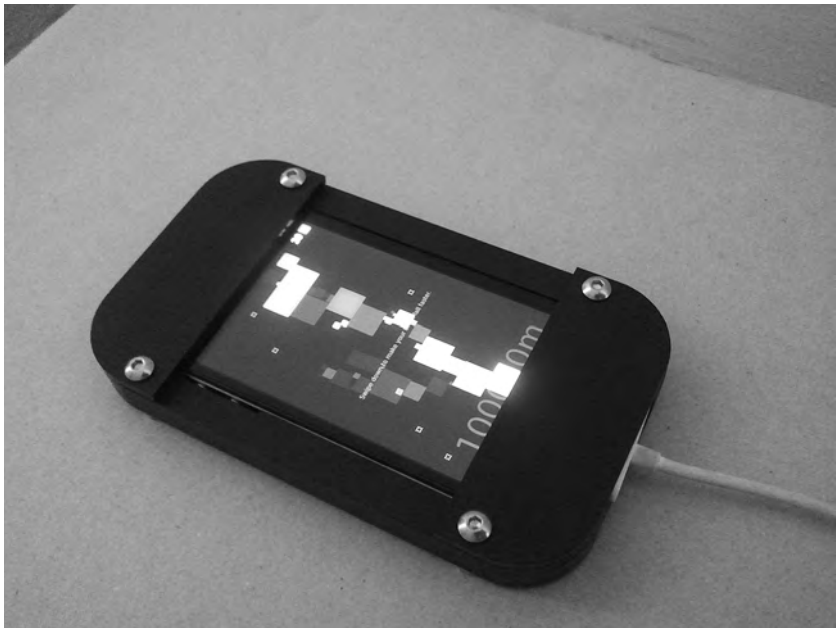


Figure 9.1
The ongoing game of *Generations* (One Life Remains, 2012).

Artists from the art collective One Life Remains (France) created the gameful work *Generations* (2012) in order to explore the notion of playing a game over an entire lifetime, then passing this "save file" down to heirs (figure 9.1). The game/artwork *Generations* by requirement never ends, as it is designed to be a digital family heritage object. In *Generations*, we see the internal notion of time spanning game-play session or hardware fads, and it rather considers familial and generational notions for a long-term view of digital cultural artifacts. In *Generations*, future relations play together with those of the past. *Generations* highlights the play process, deeply engaging synchronous and asynchronous notions of time.

We play as an act composed of sequences of action and agency. These sequences are set in an eternal present. Play requires a definition of experience that incorporates variations on the understanding of duration. "Pure duration is the form taken by the succession of our inner states of consciousness when our self lets itself live, when it abstains from establishing a separation between the present state and anterior states" (Bergson 2001/1910, 74–75). In play, such an aspiration comes close to fruition.

Though it could seem to be an exercise in humorous design or toying with player roles, a more interesting reading of Gabriel Orozco's *Ping Pond Table* (1998) might be in the work's manipulation of time. Exhibited in the "Play With Me" exhibition at MOLAA, the work consists of two modified, overlapping ping-pong tables and contains at its center a small water-filled pond. Players gather to enact ping-pong around

the pond. Notions of time have a palpable presence in the game: it is either a play space of chaotic simultaneity because of the additional "suits" that enhance the traditional binary game, or a play experience of duration, in which the pond intervenes to change the nature of the metaphor of ping-pong. *Ping Pond Table* adds extra dimensionality to the classic game, one that exists on the boundaries of extreme concepts of duration, one that opens the player to both internal and external notions of time. If Deleuze were to analyze such an artwork, he would find Bergson's "non-chronological time" (Deleuze 1989, 82).

In artists' playful use of the ludic, notions of time and space, rules, balance, and craft are confounded by the potential laden in playful aesthetics. In artwork, the interruption, the glitch emerges to alter further the notions of time and space in whimsical ways. Duration in mainstream games over the past decade has manifest in curious game features. Juul (2004) explored this at length in his work: he examined the time needed to play, called play time, and the way the player must check in with the game state throughout that temporal scale. Juul noted that game time in playing Quake III has a very different quality than, for example, time in SimCity. Time shifts, and there is a wide range of ways players might experience their own sense of play time as they engage with game events. Thus, time and duration are a key aspect of the creation of aesthetics based on games. In the classic Nintendo series Animal Crossing, among the first home console games that fostered a sense of "game-world" time, networked players around the world had to grapple with temporal disjunctions with one another. The experience of time is posited as simultaneously real (linked to day and night, for example) while at the same time subjective and interpretive (things in the game do not necessarily mimic real-world time constraints, such as the time it takes for the maturation of fruit). Players of Animal Crossing can also change their sense of game-time duration; they may stop mid-play and have a coffee or listen to music. This trend is an interesting one in action-packed digital games that are designed to be both goal-oriented yet profoundly not so: minigames, "Easter eggs," and designed distractions fill contemporary commercial games, and sayings, language, jokes, and costumes from games seep into out-of-game spaces. Play actions, game tasks, and the systemic structures games possess have positioned the ludic at odds with other time-based media forms.

When artists seize gameful notions of time, they disrupt established (and commercial) expectations of play. The playful aesthetics of duration are taken on by a Swedish artist in *Brainball*, for example (figure 9.2). *Brainball* is a game/installation from 1999 where one must compete to relax.

The playful is pure action. Sometimes, the outcome is the least of concerns. Theorists note that game actions can "hallucinate ego mastery," but they also bring with them their own bound systems for understanding that mastery (Erikson 1956, 185). Duration in Magnus Jonsson's *Brainball* posits a significant conceptual frame: instead of high speed and agility, players must let go to win, and achieve mastery through surrender. Players' brainwaves control a ball on a table, and the more relaxed player lures the ball to move in a direction beneficial to the player. Competitive relaxation results in scoring a goal over the opponent. Relaxation, a phenomenon difficult to conjure instantly, lies at the opposite end of the spectrum of familiar notions of game as competitive, engaging, exhilarating, and media-rich. *Brainball* brings forth the phenomena of embodied play and the ways in which play breaks open the possibility act composed of sequences set in the now with an endurance that is subjective within each player.



Figure 9.2
Brainball, created at Sweden's Interactive Institute in 1999 by Magnus Jonsson.

Bergson noted: "Pure duration is the form taken by the succession of our inner states of consciousness when our self lets itself live, when it abstains from establishing a separation between the present state and anterior states" (Bergson 2001/1910, 100).

The Playful Is Created with Object, with Image, Sound, Text, Like Other Media, yet Further Involves Additional Functions to Which We Must Ascribe an Aesthetic; Examples Include Procedural, Performative, Contextual, and "Digital"-"Analog"

Games are obviously systemic: they offer cause and effect, are governed by rules, and hold the potential for a number of possible outcomes. Play is also systemic: even though play can be based in fantasy, playful acts are "divorced from their original motivation and are qualitatively distinct from the same patterns appearing in their originally motivated contexts" (Loizos 1969, 228–229). In other words, the intention of a playful act is different, as is its systemic quality compared to that of games. But games and play are far more than systems. They are performative and affective experiences that offer a kind of transformation: the initiation state at the beginning of a game rarely fixes in equilibrium.

Thus, the playful act might refer to aspects of everyday life but in play these do not follow the same patterns or sets of decisions. Games and play systems are unique because they make space and behavior

into representational, reproducible, and *activated* norms. Artist Gabriel Orozco uses such an approach to much effect in his game-related artwork *Horses Running Endlessly* (1995); Orozco replaces the typical chess set with knight figures of four-color shades. It seems viewers of the work catch the game in mid-motion, and spectators feverishly look to see if this is a possible game, that is, playable (if I were black, could I take brown here?) or if it is merely invoking the leveling of game hierarchy, the diffusion of sameness and difference. Referencing chess and the functionality of its pieces, the work embodies the performative nature of the game. It offers *what is possible*: two new "suits" added to the traditional binary game changes both the nature of the metaphor of chess from a binary conflict to one of multiplicity and mayhem. While the work includes familiar aesthetics of abstraction, balance, high craft, and a classic take on the familiar, it adds the playful aesthetics of the precarious, the unpredictable, the enacted or the actionable through action or metaphor, and rule.

The performative intervention offered by artist Anne Marie Schleiner in her political critique called *OUT* in New York City is another example of gaming norms following new patters and priorities. *OUT* was a political intervention in the space of the 3D networked game America's Army and was made public in New York City through projections on city streets during the nights of the Republican National Convention of 2004. *OUT* worked to offer a performative critique both in the in-game space of America's Army and in the urban space of the city. The blending of virtual and real, of exterior and interior, holds promise for the formation of new truths in the playful approach. That same blend, albeit with a lack of criticality, is represented in Die Gute Fabrik's *Johann Sebastian Joust* game from 2011. Dancing with sensors to music creates a group spectacle that helps players regulate their bodies to the rhythm to win. Die Gute Fabrik's and Anne Marie Schleiner's performance-driven works reveal that the line is artificially drawn between the performance of play and the play of everyday life. These artists have worked to create a blur between embodied and digital play, invoking notions of the virtual and the real in the everyday. "Once again, the actual and the virtual are exchanged in an indiscernability which on each occasion allows distinction to survive" (Deleuze 1989, 74). The search for the physical manifestation of a game like *Joust* is equal to the internal search for control and rhythm.

Finally, the artwork of German artist Aram Bartholl needs to be explored for its performative strengths. Bartholl's playful works shift virtual worlds to physical street space; they end up functioning as post-representational, behavior-driven events (figure 9.3). Bartholl enjoys bringing the language of games into spaces in which the public is not (yet) familiar. The relationship between the game and the world is discussed in awkward terms (such as *gamification*, or the game-like constructions in everyday life) rather than understanding the relational exchange that happens in both. The two may not be as far apart as new terminologies would suggest.

At first, Bartholl's works might be interpreted as types of interventions, as disruptions in space. But more contradictory questions begin to emerge upon further examination. Does the work seem to say, "Look at me. I exist in real space"? Or, does the work call attention to the game-related aspects of everyday physical being: "What if we put a game right here? The virtual exists." Or, might the work merely call attention to a wandering, misplaced game—pointing



Figure 9.3
Aram Bartholl's behavior-driven performance work *WoW*, where a crew dangles the person/player's name over his or her head using the typographic reference from World of Warcraft.

out, "Look at me. Am I a game that belongs somewhere? What do I mean when played here?"

All may *be* equally true:
Games continually enact themselves.

A resolution: for Bartholl, the virtual and the real bleed seamlessly into each other. In the creation of my own work, such a blend has been invoked in the creation of a clothing-as-game interface. In 2005, I created a series of clothing that manifests virtual scenes and virtual controls on clothing items. I come from a family with a long history of service in the U.S. military, beginning with a great-great-grandfather who served during the Civil War; my father, brother, uncles—and my niece, who raised her newborn near the U.S. Air Force base where her husband was stationed. During an artist's residency on the Georgia Tech campus in Atlanta (home to Moody Air Force Base, Fort Gillem, Fort McPherson, Dobbins Air Reserve Base, and the Atlanta Naval Air Station), I started playing the online video game America's Army (U.S. Army 2003), created by the U.S. Army for recruiting purposes, with a renewed curiosity about "the rest of America," particularly the South, which provides 40 percent of all military enlistees. The work *Army* draws upon the imagery generated in the play space of America's Army. I set up a video-game station in the gallery and used captured screengrabs to create a series of images and texts reflecting on the role of the military in everyday U.S. culture. These images were rendered using computerized embroidery machines, giving a domesticated and homespun subtext to the original digital images of virtual war and violence. All the materials for the

work were collected at secondhand stores or purchased at Walmart, the store second only to the military PX store in popularity among military families. The clothing that was generated marked wearer/player health, embroidered into the fabric of clothing, or particular skills through game-interface controls. Shirts, baby bibs, children's clothing, and more were created as interfaces between the real and virtual worlds of violence.

The digital and nondigital is a false dichotomy. This is revealed most clearly when we apply the thinking of French art historian Nicholas Baurriaud's already classic art history text *Relational Aesthetics*. Baurriaud's central notion is that the most relevant art of today is constituted by relationships. Relational aesthetics describes a set of ongoing qualities by artists from Fluxus to the present who have formed as a basis of their work social, interpersonal happenings, exchange systems, and events *as themselves constituting an artwork* (Baurriaud 2002). Rather than encountering a work of art that may have been formerly perceived as a visual experience, Baurriaud's art consumer "contributes his whole body, complete with its history and behavior, and [is] no longer an abstract physical presence" (Baurriaud 2002, 59). Embodiment and social exchange are key concepts to Baurriaud and to any playful aesthetic that implies motion, rules, embodiment, and dynamic problem-solving as part of its framework. While not a fan of digital art per say, and likely not a fan of contemporary game-related work (for all of his references still occur in the hallowed halls of art institutions and do not seep out among "crass" popular culture), Baurriaud's notion is useful for discerning a piece of the game-art puzzle related to aesthetics—that the dynamic relationships between actors in a work of art form their own unique aesthetic that is performative and social.

As is hopefully well-represented by these examples, playful aesthetics drift among the "digital" and the "analog"—they pose related questions, operate under conditions unique to their particular medium, yet there are commonalities to be found across concept and different materialities. In other words, the nuances in aesthetics to be found in performance (action, repetition) and potential cross form, genre, and medium.

The Playful Is a Set of Understandings Constituting a Ludic Language, Unique from Other Media Forms in That This Language Is Personal and Experiential in Nature

To uncover a language that is *ludic*, it is useful to compare games to other media forms. What are the fundamental ways that games are different than forms such as cinema or other media? Both games and cinema, for example, have intertextual references. A prime example is watching *Kill Bill*, Quentin Tarantino's homage to many other films. One of hundreds of examples of inter-filmic references would be The Bride's yellow tracksuit. This tracksuit is a direct reference to the one worn by Bruce Lee in his last and unfinished film, *Game of Death* (1978). Orson Welles' film *Citizen Kane* includes a child's sled that was named Rosebud, and then years later another filmmaker, such as Tarantino, involves Rosebud in a different movie. The intermedia referents, the intertextual references among films happen all of the time. They also have been occurring in television spin-offs, almost from the inception of the medium: characters from one series cross over to another show to visit sitcom "neighbors"; for example, think

of *All in the Family* (1971–1979) and *The Jeffersons* (1975–1985); *Beverly Hills 90210* (1990–2000) and *Melrose Place* (1992–1999); *Cheers* (1982–1993) and *Frazier* (1993–2004); or *Battlestar Galactica* (2004–2009) and *Caprica* (2010). Variations of the spin-off happen in music with sampling and remixing. But when such references occur in games, something peculiar emerges. The distinct properties of games cause a shift in their references. Games have elements both coincidental with and distinct from cinema, and these rather roughshod technical aspects of games become their own engines of meaning making. Some are directly related to cinema (narrative premise/setting, characters, narrative, visual aesthetics, sound) and some are not (actions, resources and tools, player options and agency, relational aesthetics formed by rules for interaction with other players or NPCs, scoring, and the permitted game strategies)—these combine in a particular mix that distinguishes games from other media forms.

Game references add significant *dimensionality*, for they bring with them other aesthetic experiences I've mentioned already, including visual, sonic, procedural, performative, and other nuanced aesthetics. Take the newer example of Portal, by Valve software. Portal also gives back a world of references ludic in aesthetic references, but, more importantly, actionable references. Similarly, the Companion Cube from the Valve game series Portal has implications beyond the game. "Oh, that's that charming *Cube*," a player might wink-wink to another. The Cube carries its meaning outside the game. It means that I might be alone except for inanimate objects around me. It means I'm supposed to put the Cube to use in some way. I know that it carries properties from the game, such as its noninvasive nature and its helpful weight. The Cube may be put onto things or over things. There are thus procedural, rule-based ways to understand the relationship to the object. There are also performative ways to read the object. The Cube can be performed. This way of thinking is key to understanding how both fan culture and artists perceive popular play and work to interpret and reinterpret surrounding artifacts.

Also extending away from the game, The Cube is not a mere object but indeed an object of affection: I have a relationship with it. The Cube refers us to its original context; therefore, I could be in danger. While these are very specific instances of how game references are unique, these detailed aspects are much enhanced from what we previously thought of as intertextual media references, for example. The whole notion of GUI, or controls—even the camera controls, mouse looks, point of view, angle, the idea that I have an inventory . . . the idea that the modes of consumption in games that form a particular reading of them . . . those are not really mechanics; those are something like *memes*; and they're memes that are beyond mere representations. In fact, one could go so far as to argue that these memes are a *language* (figure 9.4). They have actual implications for how I do things, how I think about things, and how I might act. They frame objects, situations, scenarios, and actions in different ways. The ludic language feels very different from, for example, a Quentin Tarantino film positioning an actor to quip "play it again, Sam." The ludic language is different *because it has use*: it is an emergent property of games and is a key characteristic that reveals the essence of games, in other words, what makes games game-like.

our knowledge is of a
ludic kind



Figure 9.4
The ludic in urban graffiti.

To explore this ludic language, I set about three years ago collecting and storing video while playing games in a work called *What Is a Game but a Pile of Secrets?* In this project (figure 9.5), I systematically recorded hundreds of hours and terabytes of game play, from games published from 1980 to the present. The captures are compiled into individual thematic videos such as "Jump," "Death," "Keys," "Treasure," and so forth. The project is thus a series of video analytics exploring the quintessential aspects of the aesthetic experiences of video games. Familiar aesthetics are based on color, form, referentiality, composition, and database aesthetics. Using this data pool, how might we parse a game's vocabulary into its aesthetic elementals? What makes a game gamelike? How can we trace elements such as Jump, Ascend, Corridor, and Treasure in non-game contexts? *What Is a Game but a Pile of Secrets?* can be looked at as a database project on the ludic language I am seeking, because the work not only collects references and representation but also invokes players' own ludic experiences and agency.

Although game systems function within a long history—the oldest games discovered are mancala-style games dating to 6500 B.C.—it is clear that digital games themselves have developed their own sign system. Hovering at the margins of legibility, following along borders—a journey of transitory space, a way of not only demarcating but also representing space and time. In working with the digital, rule systems created by games and game-like environments emerge as constructed sites for meaning

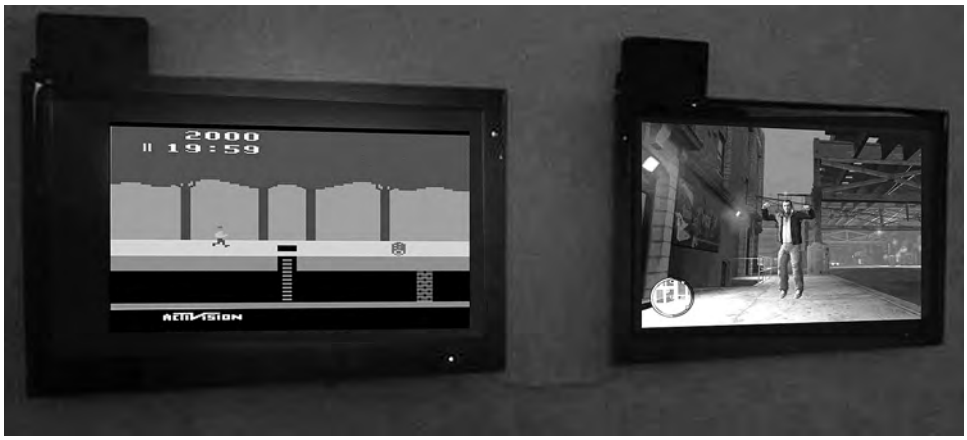


Figure 9.5
An exhibition of *What Is a Game but a Pile of Secrets?* by Mary Flanagan, which explores the ludic language.

making—here, I mean where rules for interaction are governed not only by social contracts but also by a written rule sheet or abilities coded into a digital experience. These form a peculiar type of relational, conceptual context in which artists can treat rules as forms, yet where emergent formations arise.

"The altermodernity emerging today is fueled by the flow of bodies, by our cultural wandering. It presents itself as a venture beyond the conceptual frames assigned to thought and art, a mental expedition outside identitarian norms" (Baurriaud 2002, 77). Baurriaud equates this to an exodus, but to where exactly the exodus is bound he does not say. I would argue that it is the abstract territory *of the rule* to which the wanderer must ultimately become a satellite. The ludic language that emerges among and between games affects and further gamifies the human experience.

Let me take a moment to unpack experience and the *ludic language*. To think about notions of embodiment, recall Baurriaud's already cited relational aesthetics, where art is formed through interpersonal relationships. Relationships aren't required to be embodied, but most relational artworks manifest as participants with the work meeting in person, spectating, eating, working, talking, or even sleeping. In this, the people involved in relational works together craft a dynamic disruption of the mundane and reconnect with humanness. In a sense, relational works are in direct opposition to abstractions and disembodied experience.

The Playful Can Involve Goals but Ultimately Problematizes Them

Thus far, I have asserted that the playful is a set of sequences set in an eternal present; that the playful moves beyond other media forms to include a range of unique aesthetics such as procedural, performative, and digital/analog aesthetics. These create a ludic language, and this language is experiential in nature. At this point, the playful should be unpacked

to understand a key feature separating discussions of games and play: play is often considered open ended and "goal-less," while games are argued to have more structure, including concrete outcomes such as winning and losing. Such a distinction is muddied in the practice of playful aesthetics. Game goals are often the first element of a game to be subverted or thwarted by artists and others reskinning, reworking, or replaying common playful tropes.[4] Sometimes, artists wishing to call into question the whole notion of success, goal setting, accomplishment, or resolution remove game end goals. One way to bring such a question to the forefront is to design a game-like work where goals are made impossible to reach. Take, for example, the "unwinnable" game artworks created in the 1930s. In his work *Circuit* (1931), twentieth-century sculptor Alberto Giacometti posited a thought-provoking sculpture as an unwinnable game (figure 9.6). Among a larger collection of what he called "mobile and mute objects," Giacometti created beautifully crafted puzzles that are impossible to solve without cheating. The work confronts the internal suppositions involved in most games: those of progress, completion, resolution, and wholeness. Coming out between the World Wars, the series of playful sculptures exploits the pointlessness of engaging in competition that ultimately costs supreme sacrifices for both sides.

In my notion of "critical play," I present the idea that games carry beliefs within their representation systems and mechanics (Flanagan 2009). Giacometti's impossible puzzles might reflect in their game goals the difficulty or futility of play in the rise of Nazi political power sweeping Europe or the attempted assassination of Mussolini. Likewise, perhaps on a more everyday level, unwinnable games might be crafted in order to be objects that resist usefulness or completion. Because playful aesthetics invoke, among many other issues, the meaning of, and framing of, goals, making games with problematic or



Figure 9.6
Circuit, 1931.



Figure 9.7
Volker Morawe and Tilman Reiff of //////////fur//// created a painful game controller.

unusual endings can be an intentional and provocative aesthetic choice. Take Yoko Ono's 1966 to late-1990s work *Play It by Trust*. The work posits two identical-looking and enabled sides in a situation of conflict—the undiscernible enemy means that winning is not a meaningful possibility in this work. Is this a pacifist game or an aesthetic statement on games? Does the artist mean the two to be the same?

Games and play situations are compelling, for among other things they provide a framework, or as Felix Guattari (1995) might say, they are "devices for producing subjectivity": the goal-less game evolves to be more open; the goal-pointed game becomes extreme, raising the stakes until play is no longer voluntary and no longer virtual. Two Cologne-based media artists under the name of //////////fur//// pushed their artwork to this level with the project formerly known as *Painstation*.[5] With interaction based on the classic game Pong, "no pain no game" *Painstation* is a limited series of two-person consoles that offers "competitive dueling" by causing pain to players (figure 9.7). The machine is designed to shock, heat, and beat the hands of unsuccessful players. Players of the art game place their hands on the game interface and are subjected to pain to punish them; in later versions, players were additionally beat on the hand with a rubber hammer.

Unlike //////////fur////, who take metaphors such as death and pain in games and manifest them in the player to collapse the real/virtual split, the choreographer Tino Sehgal, in his 2012 Tate Modern Turbine Hall commission, decided to contain the



Figure 9.8
Tino Sehgal's 2012 spectacle at The Tate Modern.

systemic, chaotic effects of playful systems in the hall through dancers. In this image taken at the Tate (figure 9.8), we see a massive relational project that sits on a fuzzy line between art, action, and spectator.

The Turbine Hall grew to be inhabited by an assembly of participants whose choreographed actions use movement, sound, and conversation as a frenzied all-day, every-day pure action event. That massive hall is possessed by the kinetic energy of the dancers and visitors, some of whom join the work. Thus, the work collapses the fuzzy line between actor and spectator (a dynamic that resonates in many discussions of the blurring of digital and analog, everyday and magic circle actions, and more). Choreographer Tino Sehgal's playful nonstop performance pushes the dynamic sense of rules that are not apparent to the work. Here, we have a work that crosses hours of the day, continuing on and on. It is an event of pure *verb*, pure *action*, with no end seemingly in sight. At first, one may think the work has performative or theatrical elements, and it does; but through duration, the play holds itself in an unending loop. After observing for a while, the spectator begins to see the crowd resolving into different forms or patterns through time. Spectators can follow a particular figure through time, noting one individual's actions. The individual performer may have specific algorithms (for example, a woman begins speaking to a passerby) and spectators might follow their actions, but such attempts are eventually unsatisfactory modes of sense-making against the flow of pure action. The play here goes on and on, ultimately resisting attempts for resolution, summary, and comprehension. This is the great pleasure and provocative concept behind the work.

Pure Action … or Perpetually Implied Action/Inaction

The hack. The mod. Playful aesthetics imply the subversion of form, the teasing of the authentic, cracking out of one epistemological framework for another. The testing of limits and the going beyond of a hacking mentality is supported in much playful artistic practice, where artists see beyond the rules of a given game to imagine rule-less games, or rule-bound alternative practices.

Cory Arcangel's most well-known work, *Super Mario Clouds*, is an infinite run of the Super Mario Brothers Nintendo NES game with nearly all of the game graphics and capabilities except for the minimal background: the sky and clouds that drift to the left. The work could be a video of the game, but it is not: it is a hack of the cartridge itself, where most of the game-like elements in Super Mario Brothers have been removed. Arcangel has not altered the rules of the game, but rather, he has removed the ingredients on which the rules have authority and agency. Thus, the remains rendered in *Super Mario Clouds* may belong to a rule-bound system, but they are no longer relevant without the myriad missing pieces. Artists like Arcangel use the promise of playful aesthetics to strip potential agency from their game-specific elements: hack out the rules, hack out player actions, remove risk, and steal rewards, bonuses, and deaths. Another eternal present.

Games consistently function on these self-referential, genre-specific, and medium-specific—or perhaps a better way to understand medium is by the term "experience-specific"—levels as sites for inscribing meaning. Play is an experience of ongoing present, manipulating space and behavior through sometimes highly abstracted representation. Play elements (dolls, balls, and toys as examples) and game elements (scores, enemies, teams, competitors, win states as examples) are customs, practices, and conventions that reflect playful and gameful positions as an intertextual experience: the ludic language implies that play objects can be read as enacting themselves, in and among media culture. The ball does not merely signal the idea of play; rather, the ball has affordances in play. It offers possibility—it can bounce and roll, it can fall and leap. The ludic language suggests that even through a mere reference to a bouncing ball, we are on the playground again.

Future Directions

The ludic language constitutes a game's cultural conventions, privileging agency with responsive feedback, and connoting meaning through the way in which players take *action* and *authenticate themselves* through this action. Representation, reception: these become action—*action*, coupled with a systemic recognition of that action, constitutes what are called the *game mechanics*. Game elements such as goal, score, and the like bring unparalleled meaning to such mediated worlds, shifting the concerns we might have with games to a different arena of media theory entirely. Trends in which the news media or fledgling game designers describe games as "about something" do not do express the essence of the type of world a game creates. As dedicated designers and players know, games are sites for action and

enaction. We must go beyond thinking that games "are about something" and move to "games are about doing something"; then to follow with, "in what context are we doing it?" Further, "is doing this intrinsically satisfying?" The player experience through the type of agency conveyed in the game is the heart of the art of games: this includes non-agency that in unwinnable games, where agency can be thought of as a type of matrix involving decision, ability, affordance, and chance. Games are special precisely because they are an *art of doing*.

Are there ethical game mechanics? Do artworks have mechanisms similar to game mechanics? First, let us revisit what we mean by game mechanics. As those deeply involved with the study or creation of games understand, all games utilize mechanics. Game mechanics are the phantasm of game rules that produce the emergence of game play. Most often, game mechanics are understood by example: a game that involves shooting has a "shooting mechanic"; a "tag" game has a chase mechanic and a tagging mechanic, or more precisely, a mechanic consisting of "intersect, before being *intersected with*." The past use of the term *mechanics* might shed some light: a branch of what is now physics, the field of mechanics (Gk. *Μηχανική*) is the study of the behavior of physical objects subjected to force, and the actions and reactions of the bodies in and on their environment. In games, the metaphor is useful: games offer players meaningful choices through game mechanics, which are units of rules that can be implemented in a small scale (for example, a sorting mechanic could be used in a minigame) or a large scale (a racing mechanic might form the core of a new driving game). Certainly, what you're *doing* in the game is how you create meaning, but action always has its place: meaning is constituted by what one is doing *in a specific context*. Game mechanics can be combined, like gears on a bicycle or in a watch: too many, and you have a cobbled together a machine that will likely fail due to its complexity. Too few, and you have a Zen timepiece with one hand indicating minutes, seconds, or hours—you'll never know. Game mechanics also imply feedback to the player. Do mechanics themselves create types of aesthetics? Future work may parse out the contributions of ranges of mechanics.

In a larger context, the construction of a gameful world is an attempt to deploy game goals and elements throughout a variety of lived experience. Typically, there is a purpose to these games: better employees, better consumers, and better products are some of the goals touted in the gamification movement in commercial spheres. Yet in the larger-scale discussion of the gameful world, the contexts for games and the game elements used in this *ludification* do not always foster an intrinsically motivating pleasure. The best aspects—and there are ones that are fitting—foster an intrinsic satisfaction in a player, a satisfaction that resonates during and after the game and supports personal/psychological growth or introspection. Games that construct everyday life as a world of points, hollowed out by desire and devoid of rich experience, are bound to fail.

If there is a historical notion that media not only informs but literally shapes knowledge—Bergson noted as early as 1907 that "the mechanism of our ordinary knowledge is of a cinematographical kind" (Bergson 2011/1907, 332). I posit that the contemporary mechanism of our knowledge at this juncture is of a *ludic kind*: we formulate meaning through actions (Bergson 2011/1907, 306) and these actions become agency in games. Bergson would have likely championed the idea, for after all, he was the thinker who

wrote, "movement is reality itself" (Bergson 1946, 169). The rule systems created by games and game-like environments—here, I mean where rules for interaction are governed not only by social contracts but by a written rule sheet—form a peculiar type of relational, conceptual context in which we can treat rules as forms and "the mechanic as the message."[6] Such an investigation requires a definition of those things specific to a game that constitute the ludic language.

> "It is of the essence of reasoning to shut us up in the circle of the given," noted Bergson. "But action breaks the circle. If we had never seen a man swim, we might say that swimming is an impossible thing, inasmuch as, to learn to swim, we must begin by holding ourselves up in the water and, consequently, already know how to swim. Reasoning, in fact, always nails us down to the solid ground. But if, quite simply, I throw myself into the water without fear, I may keep myself up well enough at first by merely struggling, and gradually adapt myself to the new environment: I shall thus have learnt to swim. So, in theory, there is a kind of absurdity in trying to know otherwise ..." (Bergson 2011/1907, 192)

Accepting the notion of a powerful ludic language, and considering the theme of this book, we should then ask about behaviorist aspects of the ludic language. If play is pure action, and games frame and motivate action, what responsibility does the game maker have for the new regime he or she has created? Are such ludic rules hovering in an implicit, virtual place, ready to be conjured at any time? Is there such a thing as more ethical game mechanics or mechanics that are more liberating than others? Play reflects social values, so what do we need to design consciously for them? If play reflects social values, we must invent new ways to design consciously these values as artifacts into our play culture. Ultimately, to encounter liberating types of human experience, borders will be crossed, and with such crossings—not purely in the name of invention but in the name of humanity itself—to intervene ethically, you must take things in a radical new direction, one that does not look like things seen before, one that comes in from a storm soaking and wrenching wet. To intervene ethically, one must throw "intelligence" outside itself by an act of sheer willpower (Bergson 2011/1907, 194). Kant implied that he believed the arts tended toward "impure" aesthetics, for they involve a "concept." Indeed, the works I presented in this essay are quite *impure*. They are constructed with a heavy emphasis on conceptual underpinning forming their aesthetics.

Conclusion

This essay has offered a set of lenses to look at *l'esprit de jeu*, playfulness itself, and how it is related to, yet expressed outside of, commercial video game culture by artists engaged with playful aesthetics. Here is the potentiality of play, and it brings, always, a sense of experimentation, trial and error, and practice. A ball could be bouncy or it could be heavy, atypical to expectations. In fact, is the nature of play that it lets us find out what is possible. Indeed, this essay is a fundamental updating of Bergson, who claimed that cinema was the revolutionary force in the foundation of a new kind of knowledge. The ludic turn at this particular moment, I would argue, offers a reconstituted sense of presence and action—and possible

source of empowerment—in the appropriation of, and nurturing of, play.

But one cautionary note must be added to an essay that could otherwise be read as a "utopian manifesto of gamefulness": philosophers among others need to be *wary* of aesthetics. Rancière for one has been adamant to point out that aesthetics are tools for political and social oppression. Playful aesthetics could represent "play that is tamed," a way in which playfulness is rendered in turn as part of the conditions of—as ingested by—the social order. The ludic turn through gamification and more generally gamefulness might nevertheless be necessary as a means for agency. In societies characterized by the continuing dissolution of social conditions (church, nuclear family), the ludic turn may be *required* to bring a human sensibility to everyday life. Play is the new social putty, in an age where we are increasingly "Internetted." At the same time, paradoxically, play and games drive the digital world and its consumerisms. Play may be required if we continue to be able to link into the essence that makes us human, to connect with others, to engage fully, and live, at least occasionally, completely in the moment. Even if there is an overall tendency for global societies to deem playful activities "as 'less real' than the world of daily life, as fictional and ultimately as less important than the world of working . . . Yet one of the first things to be noticed about the world of daily life is that nobody can stand to live in it all the time" (Bellah 2011).

Playful creative work has proliferated over the past fifty years, and significant contemporary artworks increasingly reference games or play. The work of the artists discussed here have shown us several avenues of understanding how play has permeated the social framework for contemporary culture and could be a force of liberation, not oppression. Playful aesthetics have offered the opportunity to break from long-held traditions about beauty and art, among other aspects such as power that traditional aesthetics represents. Games have permeated popular culture and have helped produce new art forms that examine time, agency, and the social in the context of a digital age. This essay explored a number of propositions that uncover the effects of the ludic on aesthetics and culture. Rather than see the ludic turn as a move toward the "less real," a more fitting interpretation of this turn is to see it as a move toward the social and the communal. Even prisoners of war play cards.

Play, whether a response to anxiety or stress or a celebration of life, breaks up fixed notions of what is permissible—and possible—in everyday life. Play brings unexpected and joyous disruption: winners can become losers, players can play impossible games, play objects themselves can be rendered differently meaningful or meaningless. The gameful is a disruption that actualizes the possible, or as Delueze might say, the virtual, for the ontological shift at hand in Delueze's thinking about determined reality requires idea into action—something the gameful does particularly well.

Art practices and play are both ways in which citizens can intervene in their own lived experience that is increasingly occupied with work and a domination of time through digital technologies. Playful aesthetics might be able to stay one step ahead of the political machine that seeks to appropriate it, precisely because play is such a fundamental human activity. Play allows players fullness, plenty, and possibility, where fulfillment is attainable and change is a direct matter of course. Feminist scholar Liz Grosz's hope is ready to play out: that the open nature of the virtual

"may prove central in reinvigorating a politics embracing the future by refusing to tie it to the realization of possibilities . . . linking it to the unpredictable, uncertain" (Grosz 2004, 190).

Artists will continue to bring societal attention back to play in order to point to positive ways of being in the world. "What, then, is the right way of living? Life must be lived as play. Playing certain games, making sacrifices, singing and dancing, and then a man will be able to propitiate the gods, and defend himself against his enemies and win in the contest" (Plato, *Laws* 7.796). Play, as old as any other aspect of human culture, can act as a central site for imagining a playful—and free—future.

Notes

1. Moholy-Nagy is discussed in Kaplan (1993, 165). The *Telephone Paintings* were early precursors to later fax machine art such as Sonia Landy Sheridan's 1970s work with fax where she sent images of her hand around a room through time. A more recent piece exploring remoteness and temporal shifts is the mobile phone theater work by German collective Rimini Protokoll called *Call-Cutta* (2005). In this work, an hour-long walking tour of Berlin is conducted remotely by Calcutta-based call center operators (Walz 2010, 231).

2. Mademoiselle Hermance Edan designed L'Attaque. It was patented in France in 1909. Subsequent grid-based games of hidden information have used this game as a model.

3. Bergson, cited in Deleuze (1989, 82).

4. Explore these terms more fully in Flanagan (2009).

5. A suit from Sony forbids the artists from using their original name for the system. See http://www.painstation.de/history.html.

6. This phrase of note was contributed to the field by U.S. game designer and scholar Tracy Fullerton.

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II ISSUES

From the first days on, gamification has been accompanied by intense critique (Juul 2011). While some merely wished to put a damper on the hyperbolic promises of evangelists or pointed out flaws in the predominant forms of implementation, others rejected the very idea as either ineffectual cargo cult replicas of "real" games, or frighteningly dangerous mind control, or morally corrupt abuse. Yet no matter how justified, blanket value judgments do not further our understanding of the issues at hand. And as Eric Zimmerman, Kevin Slavin, Nicolas Nova, and other authors in this volume convincingly argue, gamification is but one outgrowth of a more pervasive sociotechnical transformation: increasingly, our everyday lifeworld is pervaded, tracked, analyzed, governed, automated away from sight and reach by software. And just as the gameful world is but one dimension of this code/space (Kitchin and Dodge 2011) we are living in today, so are its issues harbingers of its fundamental systemic challenges. Unpacking these issues is the purpose of the current part.

Maybe the most straightforward and, therefore, most essential question to ask is, *Cui bono*? Who benefits? This is the starting point of PJ Rey's political economy of the gameful world, a critical riposte to the neoclassical/behavioral economic perspective of Hamari, Huotari, and Tolvanen. Rey argues that gamification presents an appropriation of games and play by contemporary capitalist institutions for wealth accumulation through the post-Fordist implosion of differences between production and consumption, play and labor. On the one hand, it produces symbolic hypercommodities and individuals that desire them to keep market economies going after sustenance needs have been met. On the other hand, gamification enables the continued exploitation of labor by de-alienating it.

In "Monkey Brains and Fraction Bingo," Bernard DeKoven, pioneer of the New Games movement, phenomenologist, philosopher, practitioner of play, and author of *The Well-Played Game*, engages in a defense of fun against any such appropriation: we play games for the fun of playing together. No matter how useful or effective games may be for other purposes, their real purpose lies in purposeless fun, "where we find in each other a moment or two of nothing but fun. Not learning, but fun. Not information, insight, understanding; not training, but fun."

Rilla Khaled next examines the potential clashes of "Gamification and Culture." Like any other technology, the rule sets and game dynamics of gamified systems manifest specific cultural and other value biases that may alienate or fail to reach players from different cultural contexts. Starting from Schwartz's

theory of cultural values, Khaled explores how typical examples of contemporary gamification promote particular interpersonal dynamics; namely, differentiation, competition, community cohesion, knowledge sharing, interdependence, and normative action. These in turn tend to reinforce and privilege values suited to mastery and hierarchy-focused cultures. In closing, she proposes design practice solutions for more inclusive gamification—and by proxy, game design.

Ralph Borland's analysis of the PlayPump provides a perfect case in point for the consequences of gameful designs that are oblivious to local circumstances. The PlayPump is a children's roundabout that pumps water, intended for use in the developing world. Invented by a water engineer in South Africa in the mid-1990s, the idea was bought and popularized by a former advertising executive. Over the next decade and more, the PlayPump became increasingly visible in the international press, celebrated in design forums, and received high levels of funding from institutions and the public as "the magic roundabout" that would literally turn work into child's play. A series of critical studies eventually showed that children's play was neither effective nor sufficient to operate the pump. Instead, adult women were frequent users of the system, turning the children's roundabout by hand, which was physically uncomfortable and far more difficult for them than use of the conventional hand pumps replaced by the PlayPumps. In effect, the PlayPump mainly delivered a spectacle for audiences far removed, while worsening the lot of its actual users. It illustrates the spatial and power dynamics between audiences and users of a technology that are quite common both for economic development and information technology deployment.

Questions of power are also at the heart of Jennifer Whitson's Foucauldian analysis of gamification, "Technologies of Control?" Whitson argues that self-tracking and gamification enroll individuals in the governance of modern liberal nation-states where subjects willingly monitor, govern, regulate, and optimize themselves. Gamified applications articulate and normalize "approved" routes toward mastery and self-improvement, as well as personalized (rather than communal) risk management. Furthermore, every gamified self-tracking application easily creeps into a de facto surveillance and control device. In Deleuze's terms, gamification helps render subjects into informational "dividuals," data sets that are easily mobilized, aggregated, abstracted, and automatically governed through access rights management.

Another obvious issue of the behavior-tracking infrastructure instituted by gamified applications is privacy, which acclaimed legal scholar of emerging technologies Lori Andrews explores in the chapter "Privacy and Data Collection in the Gameful World." Gamification raises a wealth of legal questions, such as property interests and governmental restrictions on virtual goods and currencies, intellectual property rights, deceptive advertising, and fair labor practice laws. Central to all of them is the data generated by gamified applications, which in turn centrally touches upon the question of privacy and data ownership. Stalkers or burglars using location-based data or health and life insurance companies using health behavior data in their underwriting processes are only the two most obvious risks for individuals. Based on her previous work on user rights in social networks, Andrews proposes an informed consent model for privacy and data collection in the gameful world, encompassing adequate notice presented in

an understandable way, voluntary consent, and control over the collection and use of data by third parties.

In "Gamification and Morality," Evan Selinger, Jathan Sadowski, and Thomas Seager extend the ethical questioning of gamification from companies and governments to individual users and their relations: to what extent may gamification contribute (or be detrimental) to the moral development of individuals? Gamification, they hold, can be framed as a form of "extended" or "distributed willpower"—a technical augmentation of our ability to act against the pull of immediate pleasures. In this sense, we may even be morally obliged to help each other in our distributed willpower systems. Yet at the same time, the provision of gamified applications shapes us into people who depend on "offloading" our willpower into them. This is often articulated as a fear of technology "infantilizing" us, keeping us from developing the responsibility for and capacity of consciously and resolutely choosing ends and means of our action. By easing or incentivizing other-regarding actions, they support selfish care for comfort and incentives, not moral regard for others. Finally, gamification in its current form cannot facilitate moral reflexivity—a self-awareness of the gap between one's ideals and actual actions, or of the extent to which one's situation is dependent on luck or privilege. As an alternative, Selinger, Sadowski, and Seager present a series of educational games that put players into moral dilemmas—a notion parallel to the pleasurable troublemakers that Hassenzahl and Laschke suggest for mindful, deliberate personal change.

The final set of essays in this part addresses issues of technology and is opened by Kevin Slavin. He argues that as our real world increasingly is pervaded and run by algorithms, video games and other "Playful Systems" become relevant as exemplars that make visible this technical underbelly of otherwise invisible synthetic systems and allow us to contemplate and understand it through play. Next, ubiquitous computing and near-future technology researcher Nicolas Nova breaks out "The Technical Conditions of a Gameful World." How do pervasive technologies such as cell phones, locative media, sensors, and networked objects afford and constrain playful interactions? Based on a series of case studies, Nova maps the design space of three main components of gameful systems: identification of users, sensors, and input-output interfaces. We end the exploration of technical issues with novelist Daniel Suarez's reflection on his incisive tech thrillers *Daemon* and *Freedom*™. Based on his almost twenty years of experience as a senior systems analyst (and avid gamer), Suarez's novels paint a scenario where narrow artificial intelligence algorithms that already direct the day-to-day activities of large human workforces in enterprises are linked into a network that orchestrates a whole alternate society to usurp the existing one. In Suarez's vision, these algorithms would make use of a GPS-based augmented reality overlay together with role-playing–inspired quests, experience points, and levels to inform and govern human agents. It could take control of every algorithm-run technical device in the world. In a sense, Suarez depicts a world where human willpower is fully offloaded into external systems that now become the actual actors in charge. He makes us aware how dependent on our algorithmic technopark we already are, how vulnerable its tightly interconnected systems—and how each new information technology woven into its whole makes a society of control on autopilot more possible and likely.

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10 GAMIFICATION AND POST-FORDIST CAPITALISM

PJ Rey

One bit of wisdom I gleamed from Patricia Hill Collins (a former professor of mine) is to approach inquiries into any new social phenomenon with a simple question: who benefits? And this, I am suggesting, is the way we must address the Silicon Valley buzzword du jour: *gamification*. Why does this idea now command so much attention? Who is promoting all this talk of gamification, anyway? What do these gamification advocates stand to gain? While gamification is a technique with a wide range of possible applications (many of them potentially positive), I want to focus, specifically, on the nature of the relationship between gamification and capitalism. I argue that the concept *gamification* is gaining currency, in large part, because it fits well with certain ideological assumptions native to contemporary, post-industrial capitalism and that it is promoted because it is believed to benefit those who already occupy a position of privilege within this system. By critically examining the negative consequences of gamification for individual producers and consumers, my hope is to change the conversation so that it is no longer acceptable for technology and business commentators to uncritically sing the praises of gamification.

Because this chapter is meant to be a sociological analysis of capitalism rather than a game studies tract on the nature of games, I will adopt here the broadest possible definition of gamification: "the use of game design elements in non-game contexts" (Deterding et al. 2011). Beyond this basic definition, it is useful to acknowledge from the outset a few key assumptions from game studies and related fields about the nature of these game elements. It is important to recognize that economic analyses such as this one focus on behavior motivated by the accumulation of resources; however, we should be careful not to reduce game mechanics to mere accumulation (Deterding 2010). Instead, the relationship between games and their players are best understood through the framework of "motivational affordances" (Deterding 2011, p. 2), whereby

> the relation between the features of an object and the abilities of a subject allow the subject to experience the satisfaction of such needs when interacting with the object. E.g., relative to my skills and knowledge, this Sudoku puzzle in front of me affords an opportunity to experience myself as competent when interacting with it.

In other words, there are a range of intrinsic motivators within a game situation that are relative both to the unique relations between game and player as well as to the social context in which they are embedded. This focus on affordances is particularly important in light of the fact that numerous studies have

demonstrated that extrinsic rewards (or punishments) may actually de-motivate players (Lepper, Greene, and Nisbett 1973; Deci, Koestner, and Ryan 1999; Deterding 2011).

Gamification can be put to many uses. As danah boyd (in Anderson and Rainie 2012) observed, gamification is "a modern-day form of manipulation. And like all cognitive manipulation, it can help people and it can hurt people. And we will see both."

I am not attempting to argue that gamification is intrinsically beneficial or destructive to society; instead, I want to examine *how gamification fits with current widely held assumptions about society* and why, *at this moment in history*, it such an attractive concept to those in positions of economic power. My focus is how gamification is being appropriated as a much hyped and growing part of the post-industrial capitalist economy—in short, how it is being used to make money for capitalists.

The profit motive is undoubtedly a major reason for the hype surrounding gamification. It is telling that conferences like "For the Win: Serious Gamification" or "The Gamification of Everything—Convergence Conversation" are taking place in business (and not, say, sociology) departments or are being run by CEOs and investment consultants. The Gamification Summit invites attendees to "tap into the latest and hottest business trend." Searching *Forbes* turns up far more articles (156) discussing gamification than the *New York Times* (34) or even *Wired* (45).[1] All this makes *Time* magazine contributor Gary Belsky (2012) seems a bit behind the time when he predicts "gamification will soon rule the business world." In short, gamification is primarily promoted and championed not by game designers, those interested in game studies, sociologists of labor/play, or even computer–human interaction researchers, but by business folks. And, given that the market for video games is already worth greater than $25 billion (Anderson and Rainie 2012), it should not come as a surprise that business folks are looking for new growth areas related to gaming.

The basic appeal of gamification is that it has the potential to spur economic activity (i.e., production and consumption) by influencing the behavior of producers and consumers. But the gamification of economic activity (as with other forms of gamification) faces a certain paradox: how can gamifiers control behavior without ruining the game? The issue here is that volunteerism is an essential feature of games and play (Huizinga 1938; Caillois 1961/1958; Deterding 2011; Huotari and Hamari 2012). Roger Caillois (1961/1958, 6) explains:

> Play must be defined as a free and voluntary activity, a source of joy and amusement. A game which one would be forced to play would at once cease being play. It would become constraint, drudgery from which one would strive to be freed. As an obligation or simply an order, it would lose one of its basic characteristics: the fact that the player devotes himself spontaneously to the game, of his free will and for his pleasure, each time completely free to choose retreat, silence, meditation, idle solitude, or creative activity.

So, if play is about freedom but the purpose of gamification is control, we seem to have a contradiction on our hands: how is it is possible to control behavior and have that behavior still be voluntary?[2]

This apparent gamification paradox can be resolved through a nuanced understanding of power. Gamification is not a conventional form of power—it is not power as constraint. Thus, when we think of how power operates through gamification, we must move beyond the Weberian concept of power as

being "the probability that one actor within a social relationship will be in a position to carry out his own will despite resistance" (Weber 1947, 152). Power, from this perspective, amounts to the ability to command obedience—to ensure that others will conform to your wishes even if it is against their own will.

In contrast, gamification is a form of soft power—it only works if it can entice individuals to genuinely want what the gamifiers want. Gamification is less about coercing unwilling subjects and more about creating willing subjects. Power, in this instance, should not be understood as a constraint; instead, power effected through gamification is better understood as taking the form of disciplinary strategy. As Michel Foucault (1975, 194) was fond of saying, "power produces." Gamification is a process of "subjection" (Butler 1997) or "subjectification" (Davies 2006). Judith Butler (1997, 2) explains that

> if, following Foucault, we understand power as forming the subject as well as providing the very condition of its existence and the trajectory of its desire, then power is not simply what we oppose but also, in a strong sense, what we depend on for our existence and what we harbor and preserve in the beings that we are.

That is to say, gamification cannot be seen as just one or more isolated instances of social control. Instead, gamification must be examined within a broader pattern of socialization—of producing and organizing innovative problem-solvers and self-motivated consumers. Gamification is one of myriad strategies for developing subjects that are compatible with the needs of late capitalism. It is a strategy that is just as flexible and mobile as its subjects. Such strategies need not be compulsory and enclosed like schools or prisons but can be voluntary and ubiquitous. Gilles Deleuze (1992) observed that through such mechanisms, capitalism has transformed the "disciplinary society" (where socialization produced docile bodies in an archipelago of disparate institutions) into a "society of control" where motivated subjects are constantly (and willingly) reintegrated into the circuits of power (see also Hardt and Negri 2001). Deleuze (1992) observes that the late-capitalist subject is markedly complicit in the maintenance of the social order: "Many young people strangely boast of being 'motivated'; they re-request apprenticeships and permanent training."

As a flexible strategy of subjectification and a fluid mechanism of disciplinary power, gamification is a properly *post-Fordist* phenomenon. Fordism refers to principles of rationalized factory labor introduced by Henry Ford as well as the broader socioeconomic entanglements implied in the kind of corporate control Ford pioneered.[3] Post-Fordism, or "flexible accumulation," as David Harvey (1990, 147) calls it,

> is marked by a direct confrontation with the rigidities of Fordism. It rests on flexibility with respect to labour processes, labour markets, products, and patterns of consumption. It is characterized by the emergence of entirely new sectors of production, new ways of providing financial services, new markets, and, above all, greatly intensified rates of commercial, technological, and organizational innovation.

An aspect of particular relevance to our analysis of gamification is that post-Fordist capitalism has achieved a degree of autonomy and spontaneity for workers that workers' movements and more radical anti-capitalist movements were never able to realize (at least not as a durable and widespread set of

conditions). According to Paulo Virno (2003, 99), "the masterpiece of Italian capitalism [and that in other developed nations] consists of having transformed into a productive resource precisely those modes of behavior which, at first, made their appearance under the semblance of radical conflict." If leisurely, non-alienated activity furthers consumption or can be exploited for profit, then it is not viewed as antagonistic to capitalism. In fact, the subjects of post-Fordist capitalism have demonstrated general acceptance of exploitation, and even manipulation, just so long as they are spared alienation (Fisher 2012; Rey 2012). Gamification is one mechanism through which post-Fordist capitalism appropriates such non-alienated activity and renders it useful to the capitalist goal of wealth accumulation. (I will discuss the specific forms this appropriation takes in the sections that follow.)

One key component in the subjectification of the post-Fordist subject is the valorization of fun. Peter Fleming (2005, 286) describes the shifts in managerial philosophy and corporate culture that took place in the 1980s and 1990s:

> Through informal dress codes, office parties, games, humor, zany training camps, joking, and so on, organizational members are encouraged to loosen up and find more pleasure in their roles. In the 1980s, the benefits said to accrue from making work fun were flexibility, competitive advantage, and increased motivation. In the 1990s, the message was much the same but with the added emphasis on customer service, innovation, empowerment, and creativity.

This description stands in stark contrast to the Fordist era, where fun and work were clearly distinguished and perceived as belonging to separate spaces: the home and the workplace, respectively. Karl Marx (1844/1959, 30) observed how industrialization exacerbated this separation:

> In his work, therefore, [the laborer] does not affirm himself but denies himself, does not feel content but unhappy. ... He feels at home when he is not working, and when he is working he does not feel at home.

Though fun and leisure were kept apart from the workplace, in the early industrial era both were, nevertheless, celebrated as means of incentivizing labor. This fun was integrated in the capitalist economy through consumption. You worked so that you could have fun—so that you could access "the means of consumption" (Baudrillard 1998/1970; Ritzer 2009). For this reason, fun has been a subject of much criticism from the left (Marcuse 1964; Weitzman 2008; Roberts 2009; Horning 2011). Fun is viewed as a carrot dangled before the highly programmed masses—a mechanism of social control. Those who pursued fun are often portrayed as dupes. As Herbert Marcuse (1964, 75) stated somewhat contemptuously:

> No matter how controlled the mobilization of instinctual energy may be ... no matter how much it may serve as a prop for the status quo—it is also gratifying to the managed individual, just as racing the outboard motor, pushing the power lawn mower, and speeding the automobile are fun.

From this perspective, fun is viewed as a form of false consciousness. Fun is a source of alienation because the desire for it, and for the commodities that it promises to provide to us, increases our dependency on the alienating labor that give us the wages to buy these commodities.

Fun in the Fordist era was also subject to criticism from the right. Work was seen as a heroic sacrifice, while play was a form of vice or waste. Weber

(2003/1905, 84), in his classic *The Protestant Ethic and the Spirit of Capitalism*, observed that "Protestant asceticism ... acted powerfully against spontaneous enjoyment of possessions; it restricted consumption, especially of luxuries." He described the attributes of the proto-capitalist:

> The ability of mental concentration as well as the absolutely essential feeling of obligation to one's job, are here most often combined with a strict economy which calculates the possibility of high earnings, and a cool self-control and frugality which enormously increased performance. (21)

Restraint and austerity were important to maximize investment capital. Time spent on leisure and money spent on consumption were understood as having opportunity costs. As rationalization came to dominate modern life, "leisureliness was suddenly destroyed" (Weber 2003/1905, 24). In short, fun is anathema to the spirit of (early) capitalism.

Post-Fordism has undermined both theoretical positions. On the one hand, it offers the promise that fun activity can be productive. If this is true, fun no longer has the same opportunity costs that it appeared to have in previous epochs. On the other hand, it promises that labor will be less alienating, so that consuming fun things does not, necessarily, require us to act against our own best interests in avoiding toil. Both the critical and the conservative positions of the Fordist era took for granted binary distinctions between work and play, labor and consumption. In the post-industrial era, these binaries have been eroded. Fun has been fully integrated into the circuits of capitalism, and gamification is, perhaps, the most complete manifestation of this trend.

One last consideration before I can discuss the specific mechanisms at work in making economic activity more game-like is the need to distinguish the types of economic activity in question. Historically, sociology has been characterized by a "productivist bias" (Ritzer 2010). More recently, in light of the abundance of the post–World War II era, many sociologists shifted toward consumption as a separate object of analysis within the subfield of economic sociology (Bourdieu 1986; Baudrillard 1998/1970; Ritzer 2009). In recent years, particularly in light of new forms of immaterial production and consumption such as social media (see Terranova 2000; Lazzarato 2006/1996), there have been calls to analyze production and consumption as two fundamental aspects of the same process—a process that might be labeled "prosumption" (Ritzer and Jurgenson 2010). Gamification appears to manifest differently at the two extremes of the continuum of economic activity. Thus, I have chosen to devote the next two sections to gamification at each end of production-consumption continuum. However, it is important to keep in mind the ways in which production and consumption are often combined in gamification. In fact, gamification may even thrive most in conditions where production and consumption are imploded—an idea that I will return to in the final section.

The manner in which capitalism appropriates gamification may vary based on the specific type of economic activity in question, but in each application—whether to consumption, production, or some mixture of the two—gamification is employed as a strategy because it furthers the sole purpose of capitalism: to accumulate wealth for the capitalist. The overarching goal of this chapter, then, will be to describe how gamification furthers this end in the context of the various types of economic activity and, importantly, what consumers and producers (as well as "prosumers") stand to lose if these applications of gamification are successful.

Gamification, Consumption, and Hypercommoditization

In the economic sphere of consumption, gamification is a strategy to channel our attention and activity toward advertisements or commodities that will, ultimately, generate revenue for the capitalists who engineer the gamification. However, gamification does not alter the fundamental character of a commodity; it works at a representational level. In the context of consumption, gamification can be understood as a process that transforms commodities into what Jean Baudrillard called *hypercommodities*.

Baudrillard (1994/1981, 67) describes hypercommodities as "no longer linked to distinct exchanges and determined needs, but a kind of total descriptive universe ... incessant circulation of choices, readings, references, marks, decodings." There is a lot to unpack here. First, he is saying that hypercommodities involve a de-contextualization or "disembedding" (Giddens 1991) of the commodity. A hypercommodity is removed from (or simply lacks) a native context (i.e., traditional reference points used to make sense of it and to attribute value to it). As a consequence, the value and meanings associated with a hypercommodity are extremely fluid. Second, Baudrillard is saying that hypercommodities are not consumed in order to fulfill simple biological needs; instead, they are consumed as part of a sophisticated social code. Finally, by detaching it from any specific indigenous symbolic order, the commodity becomes a surface upon which signs and signifiers circulate, endlessly being attached, detached, and reattached. That is to say, value and meaning become flexible like everything else in the post-Fordist economy.

Sociologists have long accepted that people consume things for reasons other than their apparent use value. Three Buck Chuck gets you just as drunk as a fine vintage wine, and research demonstrates that it is only when we learn the price difference that we take varying degrees of pleasure in them (Goldstein et al. 2008). Our consumption choices are influenced as much by a commodity's ancillary benefits (e.g., the cultural or social capital they might help us demonstrate or accrue) as they are by the basic function of that object (Bourdieu 1986). But, with hypercommodities, the signs and signifiers associated with objects become the primary reason for consuming them; the basic function of the object is just an afterthought. Once the signs and signifiers associated with various commodities become our primary motivation for consuming them, commodities themselves start to become interchangeable because signifiers are easily and rapidly reassigned. This is most apparent in fashion where, every season, we are encouraged to dispose of functional clothing and replace it with new clothing that offers no greater practical benefit but that makes the important symbolic statement that "I'm with it."

Gamification transforms commodities into hypercommodities both by assigning new signifiers to a commodity and by redefining the purpose for which a commodity is consumed. We can see how this process plays out by examining, for example, the Monopoly game featured annually in McDonald's fast-food restaurants. By attaching game pieces to each meal item, McDonald's signifies food with a symbolic reward that can be be accumulated. Each item consumed is a play made in the game—another tile earned. This "added value" is meant to incentivize consumption at McDonald's in lieu of other food options. In extreme cases, consumers are driven more by the pursuit of another chance to win than

they are by the actual food (which they may not particularly like or be hungry for). In this sense, the commodity is made secondary to the game so that game play is the thing really being purchased. In short, customers are not consuming the commodity itself so much as what the commodity represents.

To the extent that the consumer finds the symbolic rewards or other benefits attached to a commodity more attractive than the commodity itself, the capitalist is relieved of the obligation of actually delivering the commodity in a manner that is appealing to customers. Ian Bogost (2011b) argues that gamification replaces real incentives for customer loyalty with "counterfeit incentives that neither provide value nor require investment." Even if gamification is fun and rewarding for the consumer, it does not improve the fundamental nature of the commodity. Thus, gamification often serves to divert consumers from the fact that the commodity they have purchased fails to satisfy them. Critical theorists have long made similar arguments about the promises made by advertisements and the perpetual disappointment experienced when consumption fails to satisfy our appetites fully (which are socially constructed to be bottomless). Horkheimer and Adorno (1997/1944, 139), with their usual mix of eloquence and cynicism, observed that capitalism trains consumers simply to accept and live with misdirection and illusions:

> The culture industry perpetually cheats its consumers of what it perpetually promises. The promissory note which, with its plots and staging, it draws on pleasure is endlessly prolonged; the promise, which is actually all the spectacle consists of, is illusory: all it actually confirms is that the real point will never be reached, that the diner must be satisfied with the menu.

But gamification's post-modern maneuver is to turn around and say that dinner *is* the menu—the simulation is the reward. The hypercommodity comes at the expense of the commodity.

This is what makes gamification even more insidious than advertising. Advertising invents social needs and promotes a particular commodity as having the qualities necessary to fulfill that need. So, though advertising is resignifying the commodity, it still implicitly makes claims that the signs and symbols it invokes have some intrinsic connection to the commodity. A classic example is advertisments suggesting a need to be cool while also trying to convince us that certain cigarettes are best suited to fulfill this need in certain social milieux. Gamification, in contrast, says forgot the commodity, you are here to buy the sign (regardless of what it is attached to). The gamification of consumption completely negates any intrinsic value and meaning of the commodity (i.e., the "use value" of the commodity) and reduces its function to a medium of representation and exchange. The McDonald's Monopoly game makes consumption so remote from any actual function of the commodity that the company might as well be selling empty cups and wrappers (and probably could if it were not for the disenchantment this blatant profiteering would surely cause, as I will discuss further in the final section). Players, in fact, do sometimes fish packaging out of the trash to have more opportunities to play. What this illustrates is that, when the logic gamification is taken to its limits, it becomes a complete and utter abandonment of the real. The gamified tail wags the hypercommodified dog.

Baudrillard argued that this sort of disassociation from the real characterizes post-modern society. This free flow of signifiers raises new issues related

to power and domination. Bourdieu (1986) famously observed that "symbolic capital"—the power to assign and reassign signifiers—is not distributed evenly. While Baudrillard (2008/1983) makes similar observations, he is self-admittedly, and perhaps overly, fatalistic about the ability of individual consumers to challenge "the code"—the logic of signification inscribed in, and controlled by, capitalism. He believes that late capitalism simply reappropriates any and all attempts to manipulate the system of signs and signifiers. Baudrillard's fatalism aside, he makes an important argument about how hypercommodities produced by late capitalism reinforce existing power structures:

> The objects are no longer even commodities: they are no longer even signs whose meaning and message one could decipher and appropriate for oneself, they are *tests*, they are the ones that interrogate us, and we are summoned to answer them, and the answer is included in the question. (Baudrillard 1994/1981, 75)

Here, Baudrillard makes the consumption of hypercommodities almost sound like a sort of game in that they come with their own rules and internal logic; however, he also observes that individual consumers have little say in how these systems operate—there is a lack of freedom and autonomy. As mechanism for the production of hypercommodities, gamification is imposed on the consumer, who can only either participate or try to ignore it.

Gamification asks us to accept that simulated "points" or the other psychological inducements of game mechanics are a legitimate motivational equivalent to any real material and immaterial benefits of the commodity itself. As such, gamification can be used to encourage behavior that might be harmful to the consumer (say, regularly eating fast-food). Proponents of play and gaming might, rightly, object that consumers are making a rational decision to choose gamified hypercommodities over traditional commodities (even if they cost more) because such commodities offer additional social and psychological benefits. While I accept the premise that commodities can, possibly, be more beneficial because they are gamified (though certainly not all games [e.g., slot machines] are intrinsically beneficial), the issue here is that by attaching game elements to consumption, capitalism encourages a pattern of behavior whereby we turn to consumption (and thus must pay) in order to receive these benefits, when as Huizinga (1938, 1) states, "play is older than culture ... human civilization has added no essential feature." By imploding consumption and play, gamification makes play itself into a commodity, thus bringing it into the service of the capitalist system.

Through both the production of hypercommodities and the production of subjects that desire them, the gamification of consumption acts as a mechanism of social control; its purpose is to maximize the importance of the symbolic level of the game elements so that it can minimize resources devoted to producing or improving the commodity itself without losing its consumer base. (In light of this fact, Bogost [2011a] suggests that *exploitationware*[4] would be an accurate description of platforms for gamifying consumption.) The cost to the consumer is that, by buying into the game (elements), the consumer has less attention and resources to devote to satisfying the need the commodity itself is supposed to fulfill. Moreover, to the extent that the game elements themselves provide certain social and psychological benefits, consumers now must purchase commodities to access what play has given freely from time immemorial.

Gamification, Production, and Exploitation

In the introduction to this chapter, I discussed how in recent decades the conceptual boundaries between work and fun were eroding within the post-Fordist reconfiguration of capitalist production. Now, as I move on to examining the gamification of production, it is necessary to explore a similar implosion between work and play. Play is generally fun (indeed, Huizinga [1938, 3] went so far as to claim that the "fun-element ... characterizes the essence of play"), so it often shares the same sort of valorization as fun. However, play is sometimes also observed to be "serious" in that it is not taken lightly by the players or in that it is oriented toward extrinsic and socially important goals (Rieber, Smith, and Noah 1998; Goggin 2011).

Already, this seeming contradiction points to the complexity in defining play. Given that interpretations of play have shifted throughout the course of history (from Aristotle to Kant to Huizinga and beyond), there is reason to believe that play—as well as the work/play divide—are historically contingent and that the nature and meaning of play transforms to match the cultural and political assumptions of the current social order (Connor 2005). However, here I must limit the scope of my analysis to the modernist conceptualization of play that is being challenged by recent phenomena and, particularly, the aspects of play that are relevant to work. Of particular importance is the belief that play is both free and unproductive.[5] Both these ideals are contained in Huizinga's (1938, 13) foundational definition of play:

> Summing up the formal characteristics of play we might call it a *free activity* standing quite consciously outside "ordinary" life as being "not serious," but at the same time absorbing the player intensely and utterly. *It is an activity connected with no material interest, and no profit can be gained by it.* It proceeds within its own proper boundaries of time and space according to fixed rules and in an orderly manner. It promotes the formation of social groupings which tend to surround themselves with secrecy and to stress their difference from the common world by disguise or other means. [emphasis added]

For Huizinga, play is voluntary (in that it is not coerced) and unproductive (in that its rewards are intrinsic).

Caillois, whose definition of play is a bit more refined than Huizinga's, nevertheless retains these two ideas. First, regarding the voluntary aspect of play, Caillois (1961/1958, 9) explains that "playing is not obligatory; if it were, it would at once lose its attractive and joyous quality as diversion." Here, Caillois is relatively in line with Huizinga. But, on the issue of nonproductiveness, he is more nuanced. Caillois (1961/1958, 5) takes issue with Huizinga's absolute dismissal of material gain associated with play:

> Huizinga's definition which views play as action denuded of all material interest, simply excludes bets and games of chance—for example, gambling houses, casinos, racetracks, and lotteries—which, for better or worse, occupy an important part in the economy and daily life of various cultures.

In games where betting is involved, Caillois (1961/1958, 5; his emphasis) argues that, "*property is exchanged, but no goods are produced*." This is an important, if subtle, distinction between productivity and material gain.

Nevertheless, like Huizinga, Caillois works to maintain a clear (and very modernist) division

between work and play. The rigidness of this binary becomes fully apparent when we take account of the fact that for Marx (1867/1887; focusing here on *Capital*) and other modern thinkers, the essence of work is the creation of value. Caillois (1961/1958, 5) emphasizes that play is wholly outside the sphere of production, noting "play is an occasion of pure waste." Thus, play and work are, implicitly, set up as an oppositional pair.

Given this historical context, the gamification of production can be viewed as a post-Fordist assault on the modernist work/play binary. Gamification implodes play and labor into a phenomenon that has been labeled "playbor" (Kücklich 2005). In fact, we might say the *gamification is the production of (one form of) playbor*. Therefore, understanding playbor is key to understanding the gamification of production.

Playbor existed as a concept long before it existed as a word. Huizinga (1938, 199), for example, observed the existence of

> Certain activities whose whole *raison d'etre* lies in the field of material interest, and which had nothing of play about them in their initial stages, develop what we can only call play-forms as a secondary characteristic. [Professional] sport and athletics showed us play stiffening into seriousness but still being felt as play; now we come to serious business degenerating into play but still being called serious.

Julian Kücklich (2005) was the first academic to publish an article using the term *playbor* and is certainly among the most cited commentators on the topic of playbor. Kücklich presents video-game "modding" as the paradigmatic example of how play and labor are imploded in a way that primarily rewards capitalists, who retain property rights to all modifications that users make to games and thus the exclusive right to profit from them (of course, modders accumulate a certain amount of cultural capital in the process).

However, I find this example of modding relatively unsatisfying because, in this case, it does not seem that we are using the concept of playbor to talk about new social phenomena. I suppose, modding *is* playbor in the same sense that someone must set up a hoop and inflate a ball before a basketball game can commence. Sure, in this case, play presupposes a certain degree of work, and it is not insignificant that the players also do some of the work necessary to make play happen. But, in these examples, play and labor remain relatively distinct activities that happen to have a degree of mutual dependency. Mutual dependency, however, is a far cry from the indistinguishability that playbor would seem to imply. In contrast to Kücklich, I prefer to frame playbor, in its ideal-typical form, as the implosion of work and play into the very same act.

A more compelling example of playbor is offered by Joyce Goggin (2011), who discusses "grinding" and "gold farming," where workers in countries with cheap labor are paid to level-up characters in online games for buyers in wealthier countries who want to avoid the drudgery and repetitiveness of leveling these characters up on their own. For gold farmers, work is game-play. Of course, this also implies that, for the buyers, play has becomes so work-like that it is no longer an activity that one would engage in voluntarily. On both ends of the gold-farming equation, we see work and play imploding. However, though this example well captures the coextension of work and play, it is still distinct from the playbor produced by gamification, because the focus is on how play becomes work-like rather than how work becomes play-like.

Thus we need to turn to another example, the contemporary corporate landscape, which is rife with examples of work being made more play-like. Peter Fleming (2005) describes a call center where he conducted research:

> The walls are painted yellow and red, the supporting pillars are purple, and the carpets are a vivid blue. These colors are designed to create a mood of verve and fun. Bright icons covering desks and pods proudly announce a team's client, accentuating the playfulness of working for this particular client firm. For example, the area dealing with an African-based airline project is decorated with green cardboard cutouts of jungle trees and photos of cheetahs and hyenas.

In other corporate offices, toys and games such as cargo nets and foosball tables are integrated into the work area.

Fred Turner (2009) describes how Google's headquarters is decorated with photographs from Burning Man. Turner (2009; his emphasis) argues that the Burning Man—an event embodying the kind of transgressive playfulness that Mikhail Bakhtin (1965/1984) has taught us to expect from festivals—provides a "*cultural infrastructure* for emerging forms of new media manufacturing." Of particular interest is the Google "20 percent time" policy in which engineers devote 20 percent of their working time on projects of their own choosing. By introducing this sort of autonomy into the workplace, 20 percent time makes work look a lot like some form of serious play (though, of course, material gains are being made for the company).

It is gamification, however, through the insertion of game elements into the very logic of the workplace, that fully realizes this form of playbor *qua* playful work. In the production of playbor, gamification generally has two goals: to promote innovation and/or to stimulate productivity. Game elements target cooperative and competitive impulses, often offering simulated rewards, including "virtual ... points, payments, badges, discounts, and 'free' gifts; and status indicators such as friend counts, retweets, leader boards, achievement data, progress bars, and the ability to 'level up'" (Anderson and Raine 2012). There are myriad examples of how gamification has been embraced by corporate management and plenty of companies who pedal gamification services to them. Badgeville, "the #1 gamification and behavior management platform," promises to "enhance employee productivity"; it boasts a client list including Microsoft, Panera Bread, Samsung, Fox, NBC, and eBay (Badgeville.com 2012). My Corner Office is a social networking and gamification platform that promises to help businesses "engage, motivate and reward their employees" (Springwise.com 2012). Though spatial realities necessitate that employees will inevitably be stuck working their lives away in cubicles, every employee can be awarded a symbolic corner office.

Returning to the two focal elements of play highlighted earlier in this section (i.e., its voluntary and unproductive nature), we can safely assume that the latter of these two elements is highly undesirable for the capitalist. A company has no interest in paying workers for unproductive activity. This fact highlights the need for the playbor concept (as distinct from play). Gamification does not attempt to transform work into play; rather, it tries to create the conditions for a play-like activity that is, ultimately, still productive.

The power of playbor is thus revealed in examining the other aspect of play: its voluntary nature. In the modernist conceptualization, work is separate from the rest of life and is something to be avoided.

It is certainly not something one does voluntarily. Of course, this modernist conceptualization of work focuses on the kinds of alienated labor occurring in the factories that dominated the economy in that period. But not all labor is coerced. In fact, Marx argues that labor (in its non-alienated form) is a fundamental expression of human nature. For Marx, (1844/1959, 30), "labor is shunned like the plague" by the worker only when "it does not belong to his intrinsic nature ... [and is] not voluntary, but coerced." Gamification aims at making the labor less alienating—something that will be done willingly, even enthusiastically. This is particularly important in a post-industrial work environment where the capitalist needs the worker to engage not only her body but also her mind. Self-motivation, innovation, and profitability are linked in the context of creative work, so that if alienation impedes motivation, it ultimately impedes profitability. With playbor, this obstacle is removed.

The problem with gamification, of course, is that it does nothing to address Marx's other major critique of capitalism; namely, that it is intrinsically and necessarily exploitative. An important clarification must be made here: the term *exploitation* is frequently used in common parlance (sometimes even with a positive connotation) to mean "take advantage of." However, in this chapter, I am invoking a technical definition of exploitation, which is, primarily, derived from Marx's (1867/1887) *Capital*.

For Marx, exploitation is an objective description of the structural conditions of the capitalist economy. In the simplest terms, we can say e*xploitation is the process through which the capitalist enriches himself by selling commodities produced by workers and returning only part of the value of those commodities in wages*. Marx, observing the factories operating during his lifetime, describes the workday as being divided into two parts. The first part is the "necessary labor" that creates, for the capitalist, value equivalent to that being paid in wages. The remainder of the day is "surplus labor" that is not remunerated to the worker but, instead, goes entirely to the capitalist in the form of "surplus value" or profit (Marx, 1867/1887, 148). The rate of exploitation is the ratio between surplus value and the value remunerated to the worker in the form of wages. Thus, if a worker spends half her day producing value equivalent to her wages and spends the other half of her day producing surplus value for the capitalist, then the rate of exploitation is 100 percent.

It is with this in mind that we can make sense of Marx's (1844/1959, 28) seemingly paradoxical expression that "the worker becomes all the poorer the more wealth he produces, the more his production increases in power and size." If a factory worker, for example, skips her lunch break to complete a task, she is simply increasing the amount of surplus labor done that day and, thus, increasing the rate of exploitation. The same logic applies to increases in productivity. If a worker's wage is fixed, then increased productivity simply reduces the proportion of the day devoted to necessary labor and increases the proportion devoted to surplus labor.

Of course, since the early period of factory labor observed by Marx, myriad developments in the organization of capitalist enterprise have complicated these simplistic models. Moreover, critical economists have come to accept that the cultural code of signs and signifiers is equally as important in determining the value of a commodity as is the amount of labor put into it (Baudrillard 1998/1970). Nevertheless, wherever the capitalist engages in production (and not, say, speculation), exploitation is always an

essential component, because only through exploitation can the capitalist accumulate wealth that exceeds the modest limits of his own labor. For the capitalist, exploitation and the surplus value it produces "has all the charms of a creation out of nothing" (Marx 1867/1887, 149).

Gamification is the most recent addition to a long line of techniques and technologies employed by capitalists to enhance their ability to exploit workers (including many of the social media services now being incorporated into gamification platforms) (Dyer-Witheford 1999; Terranova 2000; Andrejevic 2002; Petersen 2008; Fuchs 2010; Fisher 2012; Rey 2012). By "de-alienating" (Fisher 2012) the workplace, gamification minimizes resistance and makes exploitation more efficient. If and when it achieves what it promises, workers will be lured into exploitative conditions by genuine interest and motivation instead of economic coercion. In fact, the capitalist's ideal scenario would be to have the motivational mechanisms offered by the game element introduced into the processes of production simply eliminate the need for monetary incentives. Thus, gamification is simply another capitalist strategy pushing toward the realization of post-Fordist logic, which, if we trace its trajectory, culminates in a growing availability of "free labor" (Terranova 2000; Fuchs 2010).

Because exploitation enriches the capitalist at the expense of the worker (or, in this case, the playborer), then, to the extent that gamification increases exploitation, it must also produce greater social inequality. From this perspective, gamification is not only an abstract economic issue, but also becomes a moral issue. Workers are now faced with a dilemma: is a less alienating work environment worth accepting intensified inequality and further consolidation of power in the hands of the capitalists? Or, is gamification something that should be resisted? Do workers even have the power to resist?

In a quote given in the introduction of this chapter, Harvey (1990) described flexibility as a central characteristic of post-Fordist capitalism. This, of course, has meant the dissolution of unions and other tactics used by workers to create inertia against the excesses of capitalism. Gamification may also serve to disempower workers. To the extent that gamification uses competition to motivate workers, it poses a potential threat to worker solidarity. In fact, this presents a reason we might expect to see capitalists favor competitive game elements over cooperative ones.

The gamification of production implodes play into work, producing playbor. Playbor, in principle, is less alienating than the traditional forms of labor described by Marx and other modern theorists, making it more likely to be done voluntarily or, at least, with less resistance and less need for compensation. Thus, gamification makes exploitation easier. Because exploitation creates inequality and because gamification promotes exploitation, the gamification of production can be understood to promote inequality (at least indirectly).

Gamification, Disenchantment, and Prosumption

In this chapter, I have argued that gamification serves capitalism's singular goal of accumulating wealth for capitalists by transforming (possibly unsatisfying commodities) into desirable hypercommodities and by de-alienating labor so that exploitation can be accomplished more efficiently. Moreover,

I have traced the extraordinary hype around gamification back to the banal and rather mercenary hope that it will enhance profitability in spheres of both consumption and production for the capitalists who employ it.

One final issue I would like to raise with regard to gamification is disenchantment. Disenchantment is a critique of the consequences of modernist—and, particularly, capitalist—rationality; it was most famously articulated by Weber (though adapted from Schiller). Weber scholar Richard Jenkins (2000, 12) explains that disenchantment is the product of two concurrent trends:

> On the one hand, there is secularization and the decline of magic; on the other hand, there is the increasing scale, scope, and power of the formal means–ends rationalities.

Both aspects of disenchantment appear antithetical to the essence of play. Caillois (1961/1958, 10) suggests that (at least some) play involves a make-believe element that we could easily and appropriately describe as magical or enchanted, because "make-believe [is] accompanied by a special awareness of a second reality or of a free unreality, as against real life." Similarly, Huizinga (1938, 18) says, "our ideas of ritual, magic, liturgy, sacrament and mystery would all fall within the play-concept." This fantastical or magical dimension of play is opposed by rationalization, which tends to disenchant it.

The totalizing nature of rationality also means that nothing can exist separate from it. Separation is another of Caillois' conditions for play. Likewise, Huizinga (1938, 9) also describes play as "distinct from 'ordinary' life both as to locality and duration." While play may contain rational elements, play itself cannot be (formally) rationalized—that is to say, it cannot be integrated into the totalizing system of rationality, lest it lose it distinctness. As soon as play is rationalized, it becomes disenchanted, and disenchanted play is no play at all. Huizinga (1938,197) observes that when sports become rationalized, they are no longer play-like, saying that "with the increasing systematization and regimentation of sport, something of the pure play-quality is inevitably lost."

In fact, if we adopt Huizinga's understanding of the relationship between play and culture, then rationalization and play are in direct conflict. Huizinga (1938) observed the play element in culture gives life its variable and multiplicitous sources of meaning. He argues that "the great archetypal activities of human society are all permeated with play from the start" and these activities include "law and order, commerce and profit, craft and art, poetry, wisdom and science" (4). Furthermore, he argues that

> real civilization cannot exist in the absence of a certain play-element, for civilization presupposes limitation and mastery of the self, the ability not to confuse its own tendencies with the ultimate and highest goal, but to understand that it is enclosed within certain bounds freely accepted. (211)

Rationalization allows no such choice or freedom to determine meaning. Instead, as Weber (2003/1905, 91) put it, rationality acts as an "iron cage," replacing playful possibility with means–end calculation. For Huizinga, the playfulness of culture is threatened by the seriousness of formal rationality (Henricks 2006, 79). Huizinga (1938, 198) concludes that Western civilization has experienced a "fatal shift towards over-seriousness."

Rationalization works against playfulness in all facets of life, including the economy (which is our primary interest in this chapter). Weber (2003/1905, 29) explains "economic rationalism as the salient

feature of modern economic life as a whole," which is characterized by "the extension of the productivity of labor ... through the subordination of the process of production to scientific points of view." Simply put, rationality pervades modern capitalist institutions, which are organized around the most efficient means to the ends of maximizing accumulation. But wherever rationalization exists, disenchantment follows. This is even true in the sphere of consumption, where George Ritzer (2009, 48) explains that "rationalization tends to lead to cold, inhuman settings that are increasingly less likely to attract consumers."

Clearly, gamification is meant to re-enchant production and consumption by making them more playful. Huizinga (1938, 205), for one, recognizes this possibility and finds it deeply upsetting, noting that "certain play-forms may be used consciously or unconsciously to cover up some social or political design. In this case we are not dealing with the eternal play-element... but with false play."

In truth, the fact that gamification is a mechanism of enchantment appears so obvious that this point probably does not merit further attention. What I am really interested in is how gamification itself is threatened by disenchantment. Huizinga observes that games can become disenchanted when they are overly rationalized and commercialized, and he is clear that "commercial competition does not ... belong to the immemorial sacred play-forms" (Huizinga 1938, 200). Given these observations and because we know the ultimate (and highly rationalized) goal of the gamification (as employed by capitalists) is to maximize profit, it follows that gamification is intrinsically prone to disenchantment.

My contention is that in order to function effectively, gamification must obscure its own purposes or, at least, redirect attention away from its own rationalized *modus operandi*. For this reason, I think gamification functions best in an environment (such as social media) where production and consumption are highly imploded. This is because, as Ritzer (2009, 120) explains, implosion produces spectacle, and spectacle obscures economic realities (including the fact that gamified consumption is manipulative and gamified production is exploitative). Ritzer (2009, 96) argues:

> Spectacle has been used to overcome the liabilities, especially the disenchantment, associated with highly rationalized systems ... the spectacle associated with commodities is a kind of opiate that obscures the true operation of society (including its rationality). It also serves to conceal the fact that the goods and service purchased may ultimately be dissatisfying.

The implosion of production and consumption and the resulting spectacle helps to mask the material gain generated via economic gamification. In so doing, it makes a gamified system appear to conform better to the principle that play is unproductive. If the manipulative or exploitative aspects of gamification are too obvious, they will disenchant the gamified system, making it feel highly rationalized and not very game-like.

Whether or not this final point about implosion and spectacle holds, the overarching issues is that the gamification of economic activity faces an eternal struggle against disenchantment. To the extent that it succeeds in concealing its own rationalized logic, it becomes another tool in the arsenal of capitalists, who will continue to use every means available to accumulate greater wealth and further intensify economic inequality. In short, with gamification, we collect the points while the capitalists collect the dollars.

Notes

1. Based on an October 13, 2012, search of the word *gamification* on the publications' respective websites.

2. This paradox is not confined to gamification but has historical precedents in earlier attempts at workplace "funification" (Fleming 2005; Goggin 2011; Nelson 2012).

3. There were also, of course, other major figures who helped shape the organization of capitalism in the early twentieth century, including F. W. Taylor, whose influential tract *The Principles of Scientific Management* taught capitalists how to maximize productivity in the factory.

4. As I will discuss in depth in the next section, this use of the term *exploitation* is inconsistent with the definition derived from Marx's writings that is used by most academics, but we certainly get the point.

5. Of course, these two characteristics are not meant to be an exhaustive account of play, just the two aspects of play most relevant to the gamification of production.

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POSITION STATEMENT

MONKEY BRAINS AND FRACTION BINGO: IN DEFENSE OF FUN

Bernard DeKoven

We play games because they're fun. When they stop being fun, we stop playing them. And no matter how clever the rules or attractive the trophies, if they don't lead us to fun, they show themselves for what they, in themselves, truly are: irrelevant, without purpose or value.

On my walk this morning I passed a boy, about seven or eight years old, batting at some branches with a stick. Thinking he was trying to retrieve a ball or kite that had gotten caught in the tree, I asked him what he was doing. "Oh," he said, "I'm trying to get one of these monkey brains." "Monkey brains?" I asked, thinking maybe he was talking about some kind of toy. And he pointed with his stick and said, "Yeah, monkey brains." And I discovered he was pointing at this fruit whose skin was convoluted, like a brain, indeed. "Oh, yeah," I said, "monkey brains." "Well here," I said, pulling at the branch, about to lower it toward him. "No!" he exclaimed, "I don't want help." And he proceeded to jump up and whack, jump up and whack again.

And as soon as his harvest reached one monkey brain more than he could comfortably carry away, he scooped them all up and ran over to his friends. At which time, everybody scattered, running to the trees in clusters of gleeful monkey-brain gatherers, hunting and gathering, and then gathering together to admire their ever-growing monkey brain heap.

There didn't seem to be any other point to their activity. Just running off, collecting, running back, dumping, admiring the swelling sweetness of it all, celebrating the burgeoning glories of their collective monkey hunt. But it was game enough for them. And for me, it was more than metaphor.

As I was walking away, pondering pensively as I often do, I realized that it really wasn't the monkey brains they were after, it was the fun of gathering them, the silly, pointless fun that came from jumping and whacking and trying to knock monkey brains out of the tree, and seeing how many they could carry, and celebrating how large of a pile they could make together. And sure, I could talk about all the learning those boys were engaged in—the hand-eye-jump-swing coordination, the setting of realistic goals, the discovery of the power of collective action. And I could have probably even helped them make a game out of it, setting goals for each other, counting trips ...

But, in truth, the best thing I could have done for the boys, and for the game they were playing, was to acknowledge, to myself at least, they were having fun, and walk away.

(Yes, those things are called monkey brains, by the way. They come from the Osage orange tree. You learn things all the time.)

There's a game called fraction bingo. It's like a bingo game, as advertised. You have bingo-like cards, bingo-like markers, there's a person who calls out numbers, like in bingo. And the object is to be the first player to cover a line or diagonal on your bingo-like card with your bingo-like markers. Except, instead of numbers, what you have on your card is fractions. And when the caller says something like "B-3," you have to look on your card, in the B column, and find the one fraction (27/3, 8/2, 12/2, 21/7, 10/2) that equals 3.

If you're really good at fractions, and you happen to have the right fractions, and are the only person to have that particular fraction, and it happens to be in a line with four other fractions that have already been called, and you are the first person to notice, and call BINGO, you win.

The problem is, even if you're really good at fractions, but you didn't notice that one of your fractions was called, or if the fraction that was called wasn't on your card, or if someone else called BINGO first, then you don't win.

And if you're not very good at fractions, you'll probably never know whether you've won or not.

The designers, in inventing this oddly ubiquitous game, probably reasoned something like this: kids need to learn fractions; kids like bingo; so, what could be better? They'll have fun. They'll practice their fraction-solving skills. BINGO! It's a perfect match.

Except that the winners are the kids who don't need the practice and are lucky enough to have the right numbers on their cards. And the kids who need the practice neither win nor are motivated and are definitely not having fun.

There are other games that give us other reasons to play. Some games promise us money, others acceptance, others fame. But those promises are only granted to the winners. And only one side or one player gets to win. And even those people don't get to be winners until the game is over. So what's the fun of that?

Fun is a powerful force. Even the promise of fun is enough to make us want to play. But, for the boy who was playing monkey brains, playing the game itself was fulfillment enough, was fun itself, the whole game through.

He was winning. He was winning all the time. And he wasn't the only one winning. He was having fun with other kids who were also having fun. They were winning together. And the fun they were having was bigger, more encompassing than the fun they could have alone.

Having fun together is the biggest prize a game has to offer us. Having fun alone, all by yourself, is still fun. But having fun together is deep fun, is the kind of fun that redefines us, that redefines reality, that takes us away together and brings us back more whole than we could possibly be by ourselves.

I am a great admirer of games that help us learn things—especially those games that are fun, especially those games that somehow, almost magically, transport us to the heart of something personally meaningful, something deep, rich with opportunities for understanding. Or those games that bring us closer to something we might have distanced ourselves too far from, too far for compassion, appreciation, or acceptance—games that manage to connect us to a truth. The serious games, the team-building games, the simulations, the educational, political, training games.

But for me, understandably, the very best games are those that are strong enough and rich enough to carry us together to where we find in each other a moment or two of nothing but fun. Not learning, but fun. Not information, insight, understanding; not training, but fun.

Of all the other things that we do with games, it seems to me that fun is what games do best, and what we do best in games. And that this fun comes not from the teaching or preaching or deft beseeching of its designers, but from the experience of being together playing. And that anything, any reason that anyone gives that comes between us and that experience, takes us away. It's not the mission of the game, not the subject matter that makes the game worth playing, but the experience of reaching together and touching together and finding ourselves together, all together, all for the fun of it all.

A game, any game, really—tag, hide and seek, soccer, duck-duck-goose, marbles, baseball, jacks, checkers, hopscotch—can get fun, really fun, seriously fun. And when it does, we, just because we are playing together, find each other and ourselves so deeply, transportingly fun, transcendingly fun, that the game itself breaks open and we ourselves come tumbling out, reborn.

11 GAMIFICATION AND CULTURE

Rilla Khaled

In Danish and other Scandinavian cultures, there is an important concept known as *Janteloven*. Under Janteloven, you should never try to stick out from the crowd. If you do try to stick out, it is only because you think that you are better than others. But no one is any better than anyone else, which is why you should not try (Sandemose 1936, 77–78). Janteloven is essentially a set of rules for encouraging social equality, social stability, and uniformity. Some locals question whether Janteloven still serves as an apt description of Scandinavian society. But as many a foreigner who moves to Scandinavia soon discovers, Janteloven is still an important cultural creed and one of the first aspects of Scandinavian culture communicated to newcomers. In such cultures in which it is undesirable to strive to achieve more than one's neighbor, or indeed to stand out at all, does it make sense for us to design gamification systems that focus on achievement, differentiation, and competition? How about for cultures that focus on fitting in with the environment, or those that prioritize the group over the individual? How might we effectively harness gamification in cultural contexts that seem inherently at odds with gamification's current trademark design elements?

Gamification has been defined as the use of game design elements in non-game contexts (Deterding et al. 2011). Over the past few years, gamification has been appropriated for a wide range of contexts, including but not limited to health (Fitocracy 2013; Hubbub Health 2013; Keas 2013), productivity (Hoopla Software 2013), eco-driving (Ford and Smart Design 2008), and personal finances (Mint 2013). Despite the diversity of application contexts, culture to date remains an underexamined aspect of gamification design. In this chapter, I explore the connections between culture and gamification and examine how typical examples of contemporary gamification applications use well-known gamification design elements to enforce cultural values. I also propose ways to approach the design of culturally relevant gamification and, indeed, inclusive gamification more broadly.

Culture

In the words of the sociologist Geert Hofstede, culture is "the collective programming of the mind which distinguishes the members of one group or category of people from another" (Hofstede 1997, 5). Culture is not innate but acquired and transmitted. It shapes how we relate to others and our environment,

encompasses beliefs and values, and governs expected behavior. Culture provides us with a rule set for how to be and how to behave (Kroeber et al. 1952, 181; Hall 1989, 14–16; Hofstede 1997, 5–11). The cross-cultural psychologist Harry Triandis provides an additional view on culture, explicitly highlighting the longevity of cultural knowledge, and positioning cultures as solutions for problems posed by environmental constraints: "Culture is to societies what memory is to individuals. It includes the things that have worked in the past" (Triandis 1995, 4). Intrinsically related to culture is the notion of value preferences. In the words of social psychologist Shalom Schwartz, "the prevailing value emphases in a society may be the most central feature of culture. These value emphases express shared conceptions of what is good and desirable in the culture, the cultural ideals" (Schwartz 2006, 138–139). Values, in this context, can be understood as "desirable, trans-situational goals, varying in importance, that serve as guiding principles in people's lives" (Schwartz 1994, 21).

Cultures can differ dramatically from one another: values and actions that are considered honorable in one culture may be viewed as dishonorable in another. For example, in the United States, while it might be considered admirable for an individual to pursue her personal objectives "against the odds," in Bangladesh, the same individual might be perceived as selfish, especially if her objectives clashed with those of her immediate community. Schwartz's theory of cultural value orientations is a framework of values that facilitates comparisons between cultures (Schwartz 2006, 2–6). It was derived from a priori theorizing, then validated with data from 76,000 individuals across 73 countries. Schwartz's theory proposes that there are seven cultural orientations, giving rise to three cultural value dimensions.

The first dimension is *autonomy versus embeddedness*. *Autonomy*, in turn, breaks down into two further orientations: *intellectual autonomy*, in which people are encouraged to be creative, curious, broad-minded, and pursue their own ideas, and *affective autonomy*, in which individuals are encouraged to pursue pleasure, excitement, variety, and to have affectively positive experiences. At the other end of the dimension, in the *embeddedness* orientation, people are encouraged to participate in shared ways of life, strive toward shared goals, maintain social order, be obedient, and respect tradition.

The second dimension is *egalitarianism versus hierarchy*. In the *egalitarianism* orientation, people are encouraged to feel concern for the welfare of others, to engage in productive work to maintain society, to pursue social justice, and to view others as moral equals. In contrast, in the *hierarchy* orientation, power is assumed to be unequally distributed, individuals are expected to comply with obligations and rules attached to their roles, to respect social power, authority, and wealth, and to value competition.

The final dimension is *harmony versus mastery*. In the *harmony* orientation, people are encouraged to try and fit into and appreciate their natural and social environment, rather than to change or exploit it, and peace, unity with nature, and protecting the environment are valued cultural ideals. In the *mastery* orientation, people are encouraged to assert themselves in the natural and social environment in order to change, direct, or control it, and success, achievement, ambition, competition, risk taking, and competence are highly emphasized.

These value types can be mapped to a roughly circular structure. Adjacent value types such as *egalitarianism* and *harmony* have more in common with one another, whereas polar opposite value types, such as *egalitarianism* and *hierarchy*, constitute

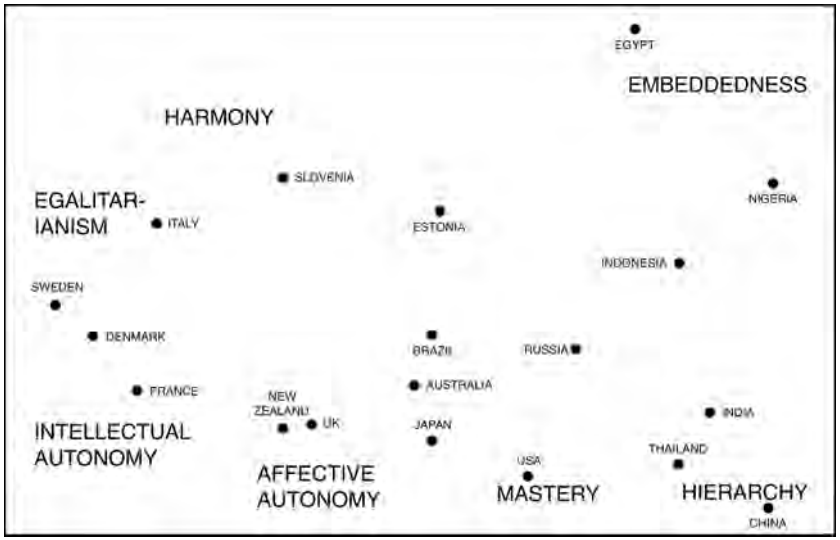


Figure 11.1
Mapping cultures to value orientations. (Adapted from "Co-Plot Map of 76 National Groups on Seven Cultural Orientations," Schwartz 2006, 156)

opposing values. In fact, Schwartz's theory proposes that if emphasis is placed on any given value type, it necessarily de-emphasizes the polar opposite value type. For example, in a country like China where *hierarchy* is emphasized, *egalitarianism* must be de-emphasized, and vice versa.

Alongside identifying these cultural value orientations, Schwartz mapped different cultures to positions within the circular structure, thereby shedding light on value types that are most important within those cultures. Figure 11.1 depicts the values and the positioning of some cultures with respect to them. The United States, for example, is positioned closest to *mastery*, indicating that among Americans, cultural importance is given to self-assertion, success, achievement, ambition, competition, and risk taking. The United States is positioned far away from *harmony* and *egalitarianism*, indicating a cultural de-emphasis on fitting in with the natural and social world, accepting one's portion in life, viewing others as moral equals, and acting for the benefit of others as a matter of choice. Egypt, in contrast, is positioned closest to *embeddedness* and farthest away from *intellectual autonomy* and *affective autonomy*. Non-U.S. English-speaking cultures (including Australia, New Zealand, and the United Kingdom) are positioned close to *affective autonomy* and far away from *harmony* and *embeddedness*. Thus, even among countries considered as "Western," different shades of individualism are present. As the mapping of cultures shows, cultures around the world can prioritize radically different values that sometimes oppose one another. For example, in Egypt *embeddedness* is given high priority and *intellectual autonomy* is given low priority, whereas in France the situation is exactly reversed. There is not a single set of values that members of all world cultures believe to be equally important.

Culture, Games, and Gamification

Examinations of the relationship between games and culture are not new. Johan Huizinga (2006) and Roger Caillois (2001, 57–67), for example, both point to connections between culture and games, suggesting that play and games reflect culture, and that culture, in turn, is transmitted via play and games. Culture-focused inquiry in contemporary game studies has tended to revolve around three areas: representations of culture and different cultural groups in video game worlds, appropriations of video games among cultural groups, and the development of subcultures within or around particular games and genres. For example, game studies scholars have explored representations of minorities in terms of sexuality (Shaw 2009; Sundén 2009) and race (Leonard 2006; DeVane and Squire 2008; DiSalvo, Crowley, and Norwood 2008; Higgin 2009), cultural instances of transformative play (Salen and Zimmerman 2003), the place of games within non-Western cultures (Wu et al. 2007; Chan 2008; Jin and Chee 2008), and acculturation within game worlds (Ward 2010).

Beyond considerations of human cultures and their representations and interactions with video games, however, we can draw further structural connections. Cross-cultural research conceptualizes human culture in terms of some of the same structures we often use to understand games, especially if we take as a given that game systems reach beyond hardware and software. Both games and cultures have rules, implicit and explicit, which serve to guide us in terms of acceptable modes of conduct with regard to others and our environment. Both suggest goals that are worthy of pursuit and noble and ignoble ways to achieve them. Both equip people with valuation equations, and their artifacts, rituals, and traditions are interpreted as important within these framings (Kroeber et al. 1952; Hall 1989; Triandis 1995; Hofstede 1997; Caillois 2001). If we map video games to cultures, we can view game play as taking place within a culture that supports and encourages particular sets of values and modes of behavior, while discouraging, disallowing, and punishing others. Just as members learn to rely on cultural rules for survival and progression in human cultures, players learn to do the same within games.

Games, then, are very much like cultures in some respects, and for the duration of game play, players can be understood as existing within a particular game culture. Such game cultures may be well-aligned to cultures that exist directly outside of the game context, or outside its "magic circle" (Huizinga 2006, 104–112), such that rules guiding game behaviors and actions map onto how we are expected to conduct ourselves in the material world. Or perhaps game cultures may be dramatically misaligned with cultures existing directly outside the game, such that game actions would be widely interpreted as being socially and culturally unacceptable. Separations drawn between games and life are what enable people more readily to adopt actions that may otherwise be frowned on (Huizinga 2006).

A core difference between games and gamification concerns context; specifically, gamification operates in non-game contexts. Separate contexts that might exist while a game is being played, and in which it is understood that game rules supersede cultural rules, do not necessarily exist for gamification. This is because gamification explicitly concerns moving game elements outside game systems and into the world. In a sense, the context of action and player

agency arising from gamified systems is the world. Not only do these systems lack explicit game context boundaries, but also the actions and behaviors they encourage tend to be instrumental, productive, or meaningful in relation to the material world. This relationship with the material world flows in both directions, as material world social and cultural contexts and rules are also drawn into and intertwined with gamification contexts. Indeed, the "non-game context" of gamification is partly, at least, the material world cultural setting. Separation between gamification and culture becomes blurred.

This being the case, while games that promote values incongruent to their surrounding cultures create valuable opportunities to explore alternative value systems, motivations, and perspectives, if gamification applications do the same, then they risk culturally alienating their players, being under-used, and motivating action in ineffective ways. This is because such applications require users to abide by sets of rules misaligned to how they expect to conduct themselves in the world, while also not providing the safe(r) context that a game provides. Essentially, this forces users to choose whether they wish to prioritize gamification contexts over cultural contexts. A Japanese sales team member who competes with his superior in a directly competitive application such as Hoopla (Hoopla Software 2012) and beats him is likely to experience a socially awkward situation in the office, as hierarchy and respect for elders is a defining characteristic of Japanese culture. With gamification, one can no longer lean on the claim that "it's just a game," because gamification intentionally bleeds into life.

But the requirement of making a choice between contexts only arises when gamification values do not align well with cultural values. The success of gamification to date has relied, in part at least, on there being a degree of social acceptance of the kinds of values that the current generation of gamification promotes. However, we have yet to see gamification take hold in non-Western countries in the way it has in Western countries, with cultural incongruence potentially explaining why. Given that gamification typically encourages and fosters actions that lead to consequences that are desirable or meaningful from a "real life" perspective (at least from the perspective of some stakeholders), it becomes doubly important that the actions themselves make sense in the material world. As the cross-cultural psychology and persuasive technology literature has shown, it is easier to persuade people to change their attitudes and behaviors if persuasive appeals reference their cultural backgrounds or align with their cultural values (Khaled et al. 2009). Gamification brings these concerns even more to the fore, as it is often highly social in nature. If culture gives us rules for how to behave with others, when gamification gives us different rules, we are faced with a values clash with potentially damaging results over the long term.

Fundamental Gamification Design Elements

Many contemporary examples of gamification rely heavily on various combinations and contextualizations of a handful of well-known design elements (Deterding et al. 2011; Nicholson 2012, 1). To examine the cultural values in gamified systems, it therefore becomes necessary to clarify and understand the

purpose and nature of these elements. We here survey three of the most widely used gamification design elements: *points*, *achievements*, and *rewards*.

Points

Points and scoring systems are ubiquitous design elements in gamification applications. Typically, points are attributed to tasks and activities that contribute toward the core use objectives of the application in question. For example, in Librarygame (Running in the Halls 2013), an application with the objective of encouraging deeper, richer, more sustained use of the library by existing library users, points are rewarded for completing a range of typical library activities, such as physically visiting the library, checking out an item, leaving a review of an item, returning an item, and accessing e-resources. Usually, the points accrued per activity depend on the difficulty, time, or effort investment required. In the now classic Chorewars (Davis 2007), an application that recontextualizes household chores as quests in the style of a role-playing game, up to one hundred experience points (XP) can be allocated for any single household chore, with the approximate rule of thumb suggested for attributing points that one XP is equivalent to one minute's worth of labor. Thus, greater numbers of points are attributed for feats that are generally considered to be more time-consuming.

Achievements

Achievements—including badges—are another staple element of gamification design. Similar to how points indicate degrees of use and effort, achievements are awarded on the basis of whether users have completed tasks and actions typically aligned with the core use objectives underlying the system in question, signifying progress. As they are generally publically displayed, they also enable individuals to receive recognition for their feats. Fitocracy (2013), for example, has achievements in the form of badges related to different physical activities that users might have input, including "I Prefer Being Off the Ground," for performing at least eight pull-ups in a single set, and "I Seem to Be Lost," for running twenty miles in your lifetime.

Achievements can serve a valuable expectation management and informational function. In Fitocracy, they set the bar for what is considered impressive, which may be especially helpful for individuals not familiar with the fitness domain. In addition, they suggest possible goals users may decide to pursue.

Rewards

Rewards are another ubiquitous design staple of gamification applications. Points and achievements can be viewed as a form of reward. For the purpose of this chapter, however, *rewards* refer to abilities, advantages, or gifts granted to users within an application as acknowledgment for their effort, commitment, or skill, and can be put to use diagetically (i.e., within the application) or nondiagetically (i.e., in the material world). Ford and Smart Design's EcoGuide (2013), a dashboard instrument intended to foster more fuel-efficient driving styles in hybrid car drivers, relies on diagetic rewards. As drivers adopt more fuel-efficient driving behaviors, EcoGuide rewards them by growing the leaves and vines of a plant representing their fuel efficiency. Keas (2013), an employee wellness program, relies on a nondiagetic approach for rewards and recommends that

employers reward their employees with cash sums for achieving their health goals. Chorewars encompasses both approaches and encourages users either to spend the points they have accrued by buying out the need to perform certain chores (e.g., two hundred XP can be "spent" on not having to do the laundry for a week) or to justify material world expenses (e.g., six hundred XP can be "spent" on a massage).

Interpersonal Gamification Dynamics in a Cultural Context

The majority of gamification applications are multiuser systems designed around user communities and participation. As culture manifests itself in how we relate to others, looking at how gamification systems shape interpersonal dynamics can give us perspectives into how they uphold certain cultural values. In relation to typical examples of contemporary gamification systems, I examine how the aforementioned fundamental design elements promote certain dynamics in interpersonal contexts and how these dynamics are likely to be interpreted by audiences of different cultural backgrounds. The dynamics addressed here are *differentiation*, *competition*, *community cohesion*, *knowledge sharing*, *interdependence*, and *normative action*.

Differentiation

Differentiation between individuals arises in gamification contexts in which emphasis is placed on how players are different from one another or are worthy of social recognition, within a wider player community. Disparities between the skills, strengths, and degrees of effort among players are accentuated, and system affordances are provided for players to promote their achievements. In Fitocracy, for example, one user's profile display might show that he has attained level 34, accrued 228,000 points, and earned 19 achievements, including one titled "I Lift Things Up and Put Them Down," for performing a barbell dead-lift of at least 1.8 times body weight. Another user's profile might, in contrast, reveal that she is only on level 5, with 2,163 points, and no achievements to show. Within the user community, different users may have drastically varied skills and degrees of commitment to maintaining or improving fitness. The Fitocracy interface makes differentiation and comparison between such users a simple activity.

Differentiation and social recognition make sense in mastery- and achievement-focused cultures, where it is socially acceptable for one to draw attention to one's successes. As Fitocracy points out on its "Getting Started" page, admittedly in a tongue-in-cheek fashion, their achievements are "designed to be showed off" (Fitocracy 2013). But in egalitarianism- and harmony-focused cultures, differentiating between individuals can be perceived in a negative light. Cultural rules such as the aforementioned Janteloven in Scandinavia or the similar "Tall Poppy syndrome" in Australia and New Zealand (Feather 1989) may mean that people view markers of differentiation, particularly those focused on achievement, as needless and almost offensive self-promotion. In these cultures, successful individuals are expected to downplay and de-emphasize their achievements. They might, for example, be uncomfortable about displaying Fitocracy achievements as part of their profile, even if they have been rightfully earned, as

it might be perceived as undue attention seeking. In embeddedness-focused cultures such as Egypt and hierarchy-focused cultures such as India, it can be disrespectful to suggest the notion of comparison between individuals occupying different rankings within the hierarchy, as rank and seniority are often perceived as more important than skill level. Thus, Fitocracy comparisons between a teacher and a student would be considered inappropriate. Even in mastery-centric cultures, some individuals might be concerned that focusing on the positive achievements of certain individuals may incite feelings of inadequacy, insecurity, and jealousy in others, especially given societal emphases on indicators of success. As such, while markers of differentiation may be intrinsic to how we think about gamification design, communicating progress and success, they serve at cross-purposes to the core values of many cultures.

In some gamified applications, dispelling concerns of achievements celebrating the few while ignoring the many, achievements exist for a wide range of tasks and actions. Sometimes, these tasks seem trivially easy. For example, in Librarygame, there is an achievement titled "Visit library before 10 a.m. five times." This enables novice users to also collect and display some achievements, thus on the surface appearing somewhat more similar to their more skilled counterparts, and potentially providing them with some encouragement to continue pursuing more difficult challenges. Such approaches make sense in egalitarian cultural contexts, as they distribute achievements to all, enabling everyone to have an opportunity to feel valued. In contrast, they can water down the significance of receiving achievements overall. In Librarygame, total number of achievements per player constitutes one form of leaderboard listing. Although people are capable of distinguishing between types of achievements, pursuing achievements regardless of type or significance is therefore implicitly encouraged by the system. Depending on how important skill-based differentiation, or indeed mastery, is for the user community, the benefits of offering plentiful achievements should be weighed against the disadvantages of lessening the perceived value of those achievements.

Competition

Competition naturally arises in gamification contexts in which players are openly compared to one another and ranked among other players in terms of their points and potentially other achievements. Setting it apart from differentiation, the actual emergence of competition relies on players feeling that the gamified system is inviting them to outperform the points or achievements of other players. Hoopla (Hoopla Software 2013), for example, is an application targeted at sales teams with a core use objective revolving around meeting sales targets and invites a strong competition dynamic. In Hoopla, team members are listed on an ESPN-style leaderboard, with stats-focused profiles of team members being displayed on a large screen for all to see. When an individual within a team hits a given target, such as making a certain number of sales within a set time frame, a loud gong may sound throughout the office, or offices, of the current team members.

While points and levels can be interpreted as motivating competition against oneself and meeting one's own targets, particularly if they are kept private, leaderboards are public and invite direct comparison and competition with others. Openly competitive systems such as Hoopla were conceived

for use in contexts in which competition with teammates and a focus on achievement are both commonplace and valued. As a user in a Hoopla promotional video notes, "it's really created a culture where people know what they're measured by" (Hoopla Software 2013). Such values are widely acceptable in mastery-focused cultures such as the United States and can be greatly effective forms of motivation. In hierarchical cultures like India, in contrast, competition is acceptable as long as it does not apply across individuals holding different ranks; that is, it would not be acceptable for an entry-level employee and a CEO to be competing against each another in Hoopla. Meanwhile, achievement and competition-centric values directly conflict with and undermine values important to egalitarianism-focused cultures such as Denmark, as they foreground inequalities between individuals, encouraging people to try and outdo others within their community. So while Hoopla may be met with enthusiasm when deployed in New York or among a team of equivalently ranked employees in Mumbai, it might be met with consternation when deployed in Silkeborg, Denmark.

Community Cohesion

Community cohesion is invited in gamification contexts in which community participation activities are afforded for within a gamified system's interface. Such activities might include answering questions posed by other players, providing encouragement to other players, and recognizing and commenting on player activities and achievements. But beyond support via affordances alone, community cohesion dynamics are invoked when community participation activities are attributed points, achievements, rewards, or other markers of value within the system. Fitocracy, for example, allows players to "follow" each other, in the style of Twitter. Importantly, it rewards achievements related to community participation, including "Social Butterfly," for posting one hundred comments, and "Feeling the Love," for receiving one hundred "props," or acknowledgments for fitness activities (Fitocracy 2013).

Community cohesion is important for any gamification user community, as cohesive user communities play an important role in the success and longevity of gamification platforms. Furthermore, community support can play a significant role in facilitating and maintaining behavioral change. But it should play an absolutely crucial role in cultures in which embeddedness values dominate. In these cultures, group cohesion characterizes a fundamental way of life, with individuals linked to in-groups throughout their lives. At the same time, an individual's in-group rarely corresponds to the entire community: there would be sharp distinctions, for example, between how an individual would relate to his classmates and how he might relate to a stranger sitting next to him on the train. Designers should be cautious of assuming generic in-groups and designing for general community cohesion, as individuals from embedded cultures are more likely to be interested in community cohesion at the specific group level.

The application DueProps (2013) presents an alternative approach to fostering community cohesion. Described by its CEO as an employee appreciation game, DueProps enables work colleagues to appreciate each other's feats and efforts by giving each other "props" or awarding "achievements" related to particular work responsibilities. For example, one player might give her colleague a prop for "Midnight Oil" out of respect for the colleague having sent a

work-related e-mail at 3:00 a.m. DueProps is intended to address the issue of employees feeling overlooked and underappreciated. What is unconventional about its design is that instead of relying on self-tracking and reporting coupled with built-in achievement systems, as is often the case with gamification achievement systems, it relies on groups of users tracking one another's activities and voluntarily acknowledging the achievements of others.

In light of how self-reports of achievements may be considered undesirable in a range of cultural contexts, especially those prioritizing embeddedness, harmony, and egalitarianism, interdependent achievement systems like that of DueProps propose a solution for how to bestow acknowledgment and gratitude in cultures in which individuals are less likely to step forward and sing their own praises. Notably, such systems contextualize achievement as an interdependent rather than independent phenomenon. Additionally, as the recognition originates from other community members observing and acknowledging other members' efforts, rather than from the mechanical interpretation of systems, the acts of bestowing and receiving achievements further reinforce feelings of cohesion between users.

Knowledge Sharing

Knowledge sharing is supported in gamification contexts in which application endeavors, including challenges, quests, and achievements, serve a signaling function within the community, communicating knowledge or indeed identifying who has specific knowledge, skills, or expertise and might therefore be able to assist others. In Visual Studio Achievements (Microsoft 2013), a gamification platform built on top of Microsoft's Visual Studio, users earn achievements based on their use of particular programming features. One such achievement is "Protector of Data," which is earned by anyone who uses the Data Protection API (DPAPI) to store data securely. For individuals within the Visual Studio user community who wish to learn more about using DPAPI, the "Protector of Data" badge quickly points out who is likely to be able to give experience-based advice and support and is potentially even expecting to be sought out as a source of knowledge.

Such indicators are commonplace in mastery-centric and hierarchy-centric cultures, in which abilities, rank, and social standing are expected to be broadcast, and individuals correspondingly take pride in being viewed as experts. But in cultures focused on egalitarianism, harmony, or embeddedness, in keeping with the earlier discussion on differentiation, such indicators may not be desirable. If the element of competition or superiority of skills can be diffused or de-emphasized, then indicators of skill and knowledge would be helpful for all communities interested in knowledge sharing and social support.

One gamified application that spreads knowledge in a way that is independent of the knowledge of specific individuals is Hubbub Health (Hubbub 2013). A central design feature of Hubbub Health is its repository of challenges. Each challenge suggests a health-related behavior for players to take on for a certain amount of time and often includes a description of why the behavior is meaningful. For example, in the challenge titled "MVM—Most Valuable Meal," the challenge itself is: "Check in every day and let us know if you ate breakfast and tell us what you scarfed down on the way out the door." Accompanying the challenge is a rationale: "Breakfast really is the most important meal of the day—for adults and kids both.

It recharges sleepy brains (the kind zombies love to nosh the most) and helps with weight loss." In this way, Hubbub Health shifts knowledge sharing into the challenge itself. Users of Hubbub Health can also view who else, and how many other users from the community, have accepted the challenge. This can serve as a social barometer for whether a challenge might be worthy of pursuit and is particularly suitable for embedded cultural contexts in which people look to members of their in-groups to determine courses of action.

Interdependence

Interdependence is fostered in gamification contexts in which players are rewarded for group efforts as opposed to individual efforts, pertaining to the progress and successes of a group they belong to. Conscious of their responsibilities to the group and the dynamic of all members needing to participate, players are explicitly invited to contribute to group efforts, provide support to one another, and maintain group morale.

The employee wellness platform Keas (2013) is an example of a gamified application, albeit a rare one, designed with group-level rewards in mind. Users of Keas are put into teams of six. Each team member might have different wellness goals; for example, one might be trying to eat more fiber while another might be trying to do more weight-bearing exercises. If all team members are making progress toward achieving their goals, their team may win a reward, such as a cash bonus, which can be shared by all. But if not all team members meet their goals, no reward is given. As such, social comparison and the desire to be perceived as a good team player keep users motivated toward achieving their individual goals, while (hopefully positive) peer pressure from teammates can serve as additional support to keep users motivated.

Keas is rare because it focuses on the group as its core social unit. In contrast, most gamified applications to date have tended to focus on the individual. Likewise, indicators of progress and achievement and rewards have also typically been pitched at the individual level. For example, while Fitocracy enables users to join user groups, such as the group "No Barbie Weights Here" (for women interested in more substantial weight training), and these groups may be associated with group-specific challenges, such as "Let's start 2013 with a bang" (with the objective of earning as many points as possible), achievements do not exist at the group level. One Fitocracy member's progress or achievements—or lack thereof—has no system-enforced effect on anyone else. Gamified applications that are intended to foster ongoing participation and behavior change via a focus on the individual are suited toward users from autonomy-focused cultures, such as France, in which the ability to pursue goals of one's own choosing and guide one's destiny are cultural ideals. But they may not be so effective in embedded cultures, such as Indonesia, in which maintaining group cohesion and pursuing group goals are paramount and trump individual desires and goals. In these cultures, approaches similar to that adopted by Keas are suitable, as they measure success in terms of group-level goals and progress and are premised on group members being concerned about upholding and contributing toward group goals.

Worthy of consideration is how well interdependence-focused gamification applications will work for users from autonomy-focused cultures. Cultural norms would suggest that they would not necessarily

be successful. However, numerous organizations promoting behavioral change that rely prominently on group support as a support mechanism, including Weight Watchers and Alcoholics Anonymous, have been very successful in autonomy-focused cultures. While the type of interdependence discussed here goes beyond what is practiced in those organizations, and those organizations deal with quite specific forms of behavior change, it seems plausible that interdependence-focused gamification would also benefit autonomy-focused cultures under the right circumstances.

Normative Action

Normative action is invited in gamification contexts in which different actions are associated with different degrees of value and acknowledgment. This establishes a dynamic of comparison between the relative values and merits of actions and tasks, communicating that not all are recognized as equally valuable or desirable. For example, in Librarygame, users earn five points for accessing an e-resource and twenty-five points for borrowing a book, suggesting that borrowing literature printed on paper is considered five times as worthwhile as downloading digital materials from an online repository.

Essentially, such applications assign values to decisions that previously may not have had explicit valuations attached. Furthermore, they rest on the assumption that it is possible to place a value on certain actions, and indeed that these valuations will be agreed upon by the user community. Whereas individuals from hierarchy-centric and mastery-centric cultures are more likely to accept external valuations of action and prosper within these bounds, individuals from embedded cultures will tend to look to their own groups to determine norms and desirable actions.

Normative action poses the greatest problems for autonomy-centric cultures, in which the notion of externally determined worth runs at cross-purposes with basic cultural values of freedom and the ability to make meaning on one's own terms. Speaking to this, selection and customization of goals is already a common feature of many gamification applications and goes some way toward allowing users to make meaning. Both Fitocracy and Hubbub Health, for example, enable users to pursue fitness goals of their own choosing and to opt into specific challenges. But to appeal better to users from autonomy-focused cultures, if different amounts of points are allocated across a range of actions, it may also be necessary to enable customization of points allocation. This would create a tighter link between users' personal valuations of the actions independently of the gamified system and the actions in the context of the gamified system. It would also, however, significantly complicate the task of gamification system design and how to compare activity across users in terms of points.

A Call to Action

As the above examinations show, the ways in which gamification design elements are leveraged within current gamification systems give rise to certain interpersonal dynamics that are not equally suited to all cultures. *Differentiation*, as it is currently invoked in gamification, is most well suited to

mastery-focused cultures. *Competition* is also most appropriate for mastery-focused cultures, along with hierarchy-focused cultures in certain contexts. *Community cohesion* is often an important feature for gamification independently of cultural context, but is especially important to support for embeddedness-focused cultures. *Knowledge sharing*, too, is useful independently of culture, but instantiations focused on achievements privilege mastery and hierarchy values. *Interdependence* is most well suited toward embeddedness-centric cultures, and current instances of it are still rare. Lastly, *normative action* makes the most sense in hierarchy-focused and mastery-focused cultures and the least sense in autonomy-focused cultures.

In short, the ways in which typical gamification applications use well-known game elements tend to privilege cultural value orientations of mastery and hierarchy at the expense of egalitarianism, harmony, embeddedness, and autonomy. The patterns of use for these elements generally mirror the ways that their corresponding counterparts have been used in games. Indeed, mastery, competition, and success are classic and characteristic themes of games. While it is possible to find game themes that challenge the pursuit of these values (such as Shadow of the Colossus [Team Ico 2005], which plays on a tension between mastery and harmony), they remain fundamental to most understandings of games. For example, in Caillois' classification of games, the core game category of *agôn* describes competitive games, intimately connected with notions of combat, superiority, and success (Caillois 2001, 14–17). The very concept of "winning," whether stemming from explicit competition or not, is also at base a recognition of success.

Current forms of gamification, then, are continuing a legacy that originates within games, albeit in a highly limited, condensed form. But as emphasized earlier, the values of mastery, hierarchy, and competition do not hold universal appeal across all cultural settings, especially those in which interdependence is emphasized. In egalitarianism-centric cultures, these values oppose cultural rules concerning what everyone in society deserves. In harmony-centric cultures, they oppose how one relates to others and the environment, potentially creating conflict and destabilizing social relations. In embeddedness-centric cultures, they can disrupt tradition and social order and make one appear immodest. Even in hierarchy-centric cultures, in which displays of social power are the norm, mastery, success, and competition are only acceptable under certain conditions that acknowledge existing power structures.

For a technology to be effective within a culture, as opposed to being effective only for a handful of individuals within that culture, the core premises on which the technology functions must be broadly acceptable by that culture. For the core premises of a technology to be broadly accepted by a culture, the values it rests on—and indeed promotes—should either resonate to a significant extent with the members of that culture or alternatively should promote values that its members are willing to adopt. Indeed, this point can be generalized beyond cultural values to apply to preferences related to all forms of diversity—such as gender (e.g., Croson and Gneezy 2009), age (e.g., Gerling and Masuch 2011), physical ability, income, technology access, or any other of the numerous potential forms of diversity.

For designers, then, there is a need to design gamification that supports and promotes a wider range of values. Simply put, if we do not do this, only a narrow slice of users will adopt gamification, and many will call its efficacy into question. Here I present a number

of potential strategies for broadening the diversity of gamification values.

Redesigns

Some game design elements natively reinforce certain values, such as leaderboards reinforcing competition. But the ways in which other elements realize values depends on their context of use and the dynamics of their incorporation within the gamified system. Redesigning the context of use of basic design elements within existing applications can go a long way toward suggesting possible alternative value perspectives. Take, for example, Fitocracy, and its use of points, leaderboards, achievements, and challenges. In Fitocracy's current incarnation, these design elements are largely used to foreground mastery, success, and competition values. But they could be recontextualized to emphasize interdependence in different forms. Suppose Fitocracy was redesigned to reflect cultural attitudes better in embeddedness-centric cultures such as Egypt. In this redesign, the notion of user groups, group-level challenges, and group achievements might play a central role, with group participation, group bonding activities, and group support providing strong, culturally relevant incentives for application use. Within a specific user group, group members who are training for a marathon might decide to set out to obtain a "1,000 miles in 10 days" achievement as a group-level challenge. Such an achievement would require all group members to have collectively run 1,000 miles over 10 days, though mileage contributions per group member would remain unspecified. Challenge-specific visualizations would exist, showing the group how close they were to their shared goal, along with individual users' contributions. Likewise, challenge-specific leaderboards would display and rank points earned from challenge-relevant activities (e.g., short runs on a treadmill, long runs, *fartlek* training, and so on). Through the challenge-specific visualizations and leaderboards, group members would be able to track one another's progress. As well as providing a sense of togetherness, the visualizations and leaderboards would enable group members to determine whether anyone required additional support or, alternatively, who was doing particularly well and might be able to give useful advice.

Harnessing Local Knowledge

Culture is pervasive and affects not just the users of gamification applications, but the designers as well. It plays a strong role in shaping their preferences and the design elements, tropes, and strategies they embed in their products. In making design decisions regarding how to shape and motivate behaviors aligning with a product's core use objectives, designers constantly make culture-based value judgments about what they personally find motivating and what they believe their target users will find motivating. Sometimes designers make these decisions consciously and at other times they are made subconsciously, but decisions are always made in a cultural context.

While it may be difficult for designers to articulate cultural values and ideologies that they have (generally unknowingly) embedded into their products, these values and ideals are brought into relief when the products are used by audiences the designers were not considering. As Edward Hall explains: "The only time one is aware of the control system is when things don't follow the hidden program. This is most frequent in intercultural programs" (Hall 1989, 4).

Seen positively, culturally mismatched gamification might serve to introduce novel values and perspectives to the target audience; however, it may also risk being perceived consciously or subconsciously as a colonizing attempt to privilege a particular set of values.

An awareness of basic cultural characteristics and patterns of target audiences in a context of cross-cultural comparison will go some way toward helping designers to design culturally relevant gamification. Hiring designers from culturally diverse backgrounds is a richer way to inject culturally diverse perspectives into the design of gamification and to challenge existing assumptions concerning effective, culturally relevant design. Deep, nuanced understandings can be obtained by working with representatives of target user groups throughout the design process, as is common in methodologies such as participatory design (Muller 2003), contextual design (Holtzblatt et al. 2005), and empathic design (Koskinen and Battarbee 2003). Initially, this participation might take the form of a consultancy role. Once products have reached the prototyping stage, a standard play test/user test approach may be adopted (Fullerton 2008).

In all cases, designers should first seek to understand how the core use objectives of gamified applications are typically achieved in the appropriate cultural and social contexts with non-gamified methods and which cultural patterns and forces are at play in the existing approaches. Only then should they embark on gamifying the contexts. Not acknowledging or designing for contextual cultural value patterns makes the task of gamification design more of a blind target. Too much dissonance between a society's cultural value patterns and the cultural value patterns promoted within a gamified application risks ineffectiveness and failure.

Folk Perspectives

Another way of introducing culturally diverse perspectives into gamification is to look to analog games and traditional folk games for design inspiration, as these often embody cultural values. As a large number of these folk games were intended to be played in group settings with multiple players, they may provide novel suggestions on how to design culturally relevant gamification in interpersonal contexts. Take for example, Thirudan Police, a simple South Indian four-player folk game for children premised on facial expression and body language reading, in which all players are randomly assigned a role from the following set: Police, Thirudan (thief), Raja (king), and Rani (queen). While the Police player must make his or her role known to the rest of the group, the others keep their roles secret. The group then determines one of the remaining other roles that the Police player must identify (e.g., the Thirudan). The Police player must then guess who he or she thinks the Thirudan is. If the guess is correct, the player with the Police role gets a point; if it is wrong the player with the Thirudan role gets the point. The game continues for a number of rounds, and the player with the highest score wins. The Police role in Thirudan Police essentially trains players to become better at reading facial expressions of group members and detecting and interpreting their nonverbal communication signals. Perhaps counterintuitively, it is a game that supports group cohesion. Authors of a website that documents folk games propose that a skill Thirudan Police fosters in children is that it can "sustain the feeling of unity and oneness among the players" (Apus 2013). Its very basic premise could be repurposed, for example, to foster team cohesion and bonding among a distributed team of employees over

Skype and create opportunities for both nonverbal and verbal communication. Another notable feature of Thirudan Police is that it relies on a *unilateral competition* player interaction pattern; that is, it features two or more players competing against one player (Fullerton 2008, 54–55). Unilateral competition is currently underused within gamification and could be leveraged to refreshing effect.

Using Gamification to Explore Untapped Game Design Spaces

Contemporary examples of gamification tend to feature a high degree of homogeneity in terms of design tropes. In part, this can be attributed to the relative youth of gamification as well as desires on behalf of designers and other stakeholders to capitalize on established, successful design features. But an additional constraint is in effect: gamification as a concept is somewhat subordinated to games. The game elements that designers make use of in gamification systems are generally those that are already familiar to many users.

Thus, points, achievements, and rewards are staple features of games, and the ways in which they have been used in gamification to date serve as a kind of shorthand for previously experienced and well-established game dynamics and mechanics. Even if they represent but a tiny sample of design elements that could be drawn on from games, they are elements that are usually highly recognizable to the general public and the gamification user audience as "game-like." This is a target audience far broader than individuals who would classify themselves as "gamers," and accordingly their breadth of knowledge of more esoteric game elements will necessarily be more limited. If there is a paucity of culturally diverse values or more general inclusive values to be found in current gamification applications, this stems in part from the related problem that games in general are not yet highly culturally diverse (Kerr 2010), and game design's most well-recognized elements represent a very conventional perspective on game values.

But gamification is invited into places that go beyond where games have been and are able to go. Gamification can exist in contexts we would not normally consider folding game systems, dynamics, and narratives around. Furthermore, the user base for gamified applications in some ways stretches beyond that of games. It may concern audiences from certain cultural backgrounds or with particular needs and preferences that have largely gone under the radar of digital games. In addition, its audiences are not necessarily committed to playing games. Games are usually played voluntarily; their players are, for one reason or another, committed to playing. When the commitment wanes, players typically stop playing. Gamified applications, in contrast, span users primarily focused on game-like characteristics, users who appreciate game-like characteristics but place their importance as secondary to the application's core use objectives, as well as users who are either indifferent toward or prefer not to be using such applications but may have to for political and institutional reasons, such as applications like Hoopla. For better or worse, gamification ends up being used by individuals with different levels of interest in games and playful use.

We should be better leveraging the design challenges and opportunities that gamification presents.

Designing gamification for more diverse values—cultural and otherwise—coupled with the unusual design contexts and non-gamer target audiences that characterize the use scenarios for gamification creates specific and exciting design challenges that should compel us to expand our game design repertoire. Instead of merely reappropriating or, worse, parroting familiar game design tropes, designers of gamified applications could be using these underexplored design spaces as testbeds from which to experiment with and reflect on the use of novel design strategies and approaches. With voices calling for further experimentation in games—and with design perspectives such as serious games (Mitgutsch and Alvarado 2012), not games (Harvey and Samyn 2011), and abusive game design (Wilson and Sicart 2010) challenging the boundaries of what game experiences are—new game design influences are timely. The results from gamification design explorations could broaden and enrich the ways in which we imagine, understand, and design games. Gamification design need not and should not be subordinated to game design, but should be viewed as a complementary and related activity.

One potential concern stemming from approaching gamification design as a form of game design experimentation, as opposed to game design reappropriation, is that the general public may no longer be able to recognize the "game-ness" in gamified applications featuring novel design elements. Indeed, this concern has been raised with respect to gamification applications targeted at elderly users, as they are less likely to have existing familiarity with digital games and their design tropes (Gerling and Masuch 2011, 3–4). That is, due to a lack of cultural knowledge of digital games, the "shorthand" aspect that many gamified applications rely on of referencing well-known game design elements might be rendered useless, resulting in an additional learning curve for users who do not have background familiarity with digital games.

In contrast, the game elements that gamified systems tend to draw on are not only currently rather limited but also have been described by several commentators as fairly superficial interpretations of what makes games compelling (e.g., Bogost 2011; Juul 2011). In appealing to a common-denominator understanding of what games are, many gamified applications have ended up presenting impoverished caricatures of game design, explaining why gamification has been eschewed by many game designers. Games should not be viewed as a countable set of design elements rearranged in different combinations; nor should gamification be viewed through such a narrow lens.

In terms of demarcating an approximate space for gamification and expressing potential ways in which it can be used, the existing characteristic gamification design elements have played an important and useful role. Now that the notion of gamification has solidified further and gamification has proved itself to be sufficiently compelling to the public, its design elements need not remain so conventional. An application can be game-like without needing to rely on mechanics that have previously been experienced.

We have yet to see the most interesting ways of using game design elements in gamified contexts. Most likely this will involve deep, specific reflection on how to design game mechanics and experiences for the unique qualities of core use objectives required by specific gamification contexts, rather than copying and pasting what has worked in the past. Naturally, this should involve genuine consideration of the cultural context, values, preferences, knowledge, abilities, habits, and use contexts of intended target users.

Conclusion

Gamification, by design, merges with day-to-day life and often asks players to adhere to certain rule sets and game dynamics that may well clash with their immediate cultural context. As gamification reaches further conceptual maturity, we need to be deeply thoughtful about cultural and other value biases inherent in gamification design tropes and cognizant of the preferences of the audiences we are designing for to reduce the risk of design failure and the alienation of players. In this chapter, I examined ways that typical examples of contemporary gamification promote particular interpersonal dynamics; namely, *differentiation*, *competition*, *community cohesion*, *knowledge sharing*, *interdependence*, and *normative action*. Using Schwartz's theory of cultural values as a point of departure, I discussed how these dynamics embody various cultural values, proposing that they tend to reinforce and privilege values suited to mastery and hierarchy-focused cultures.

To move gamification toward being more inclusive, we must actively acknowledge culture and other forms of diversity as part of the design frame. We need to draw on the talents of more diverse designers, work with target users to inspire and inform how and what we design, and consider looking to folk games (and potentially other rituals) to inspire culturally relevant design.

Broadening the horizon of gamification design is not just useful for gamification as an endeavor, but also for game design in general. Gamification presents a live, dynamic, and challenging frame for game design targeted at diverse audiences and concerning use contexts and objectives outside of the scope of strictly entertainment-focused game design. What we learn from gamification design could infuse game design itself with a fresh perspective. But this means we will need to move beyond the dominant, all-too-familiar tropes, which only take a fraction of the world's population into account and score poorly on the scale of innovation.

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12 THE PLAYPUMP

Ralph Borland

In 1989, retired advertising executive Trevor Field took his father-in-law on a visit to an agricultural fair in Pretoria, South Africa. At the fair, he saw a prototype for an invention by water engineer Ronnie Stuiver: a children's roundabout (or, merry-go-round) that drove a borehole water pump. Stuiver had come up with the idea through his work in rural areas of South Africa, where children would appear when he was sinking boreholes and installing water pumps and want to play at helping him with his work.

Field was immediately captured by the idea and bought the patent for it from Stuiver. Stuiver, Field tells us, knew that the system was too expensive in comparison with a standard hand pump to be widely taken up (Architecture for Humanity 2006). But Field had a vision for how to raise money for the pump—he saw "advertising billboards that pumped water" (PlayPumps International 2009). He added an elevated water tank to the design, to capture water from children's play for adults to draw later, and on the water tank he placed billboards, to be rented out to raise money for the installation and maintenance of the system. And so the PlayPump was born (figure 12.1).

During the 1990s, Field established connections with government bodies and corporations in South Africa to fund and install PlayPumps in rural areas. In 2000, the project started to become internationally known when Field was encouraged by a World Bank representative to enter the project in its Development Marketplace Award, which the PlayPump won. With the exposure from the award, interest in the project accelerated, and in 2006 Field scored a major coup: The Case Foundation, run by America Online (AOL) founder Steve Case and his wife, Jean, wanted to back the PlayPump, and after meeting with Field in South Africa, they established PlayPumps International in the United States to start fund-raising for the project in earnest.

The Cases used their expertise from running AOL to pioneer new forms of online fund-raising that targeted the general public in the United States directly, employing a top Internet strategist "to help it go global and craft an 'everyman' approach to raising new dollars" (McMillan 2008). Their new website for the PlayPump used social networking and direct online donations to "unearth a new stream of donor dollars," creating "a robust hub for fundraising," which allowed an early campaign for "100 pumps in a 100 days" to raise US$1.6 million online (McMillan 2008).

PlayPumps International's campaigns rode a wave of public interest in the PlayPump in Europe and the



Figure 12.1
A promotional photograph for the PlayPump, ca. 2000. (Source: www.playpumps.co.za)

United States, which was primed by other aid campaigns and increasing press attention to the project. A brand of bottled water, with profits going to the PlayPump, had been launched by the One Foundation a year before PlayPumps International was established, under the slogan "When You Drink One, Africa Drinks Too" (One Water 2009). The One Foundation used celebrity spokespeople and extensive marketing and advertising to draw attention to the PlayPump and had its brand selected as the official bottled water of Live8 and Make Poverty History, high-profile campaigns for aid to the developing world. Global hip-hop star Jay-Z's "Water for Life" concert tour in 2006 raised US$250,000 for PlayPumps International (FRONTLINE/World 2005).

A slew of international headlines, articles, and editorials celebrated the PlayPump. Bill Clinton (2006) called the PlayPump a "wonderful innovation" in an article in *Time* magazine, while a *New York Times* (2003) editorial described it as "more efficient, easier to use and cheaper to run than wells with hand pumps." The *Sunday Times* newspaper in the United Kingdom told the story of Field's invention under the headline "The Drought-Busting Magic Roundabout" (Lamb 2005). "Why Pumping Water Is Child's Play" read the headline to a BBC News (2005) article on the PlayPump. "Sometimes it's the simplest of ideas that can change the world most profoundly" mused *National Geographic* (2008) in its short film about the project.

Probably the most instrumental piece of media in advancing the PlayPump was a short film by PBS's *FRONTLINE/World*, which was broadcast online and on public television in the United States in 2005. The movie was made by reporter Amy Costello (2005), who accompanied Field to the site of an early PlayPump installation at a school in South Africa. *Frontline* reported "an overwhelming interest from web viewers" to the project over a number of years, from its original screening in 2005, to an update in 2007 in response to the Case Foundation's adoption of the project, and beyond. Jean Case told Costello that the movie was the first thing she would show potential donors to the project (Costello 2010a).

Shortly after PlayPumps International was established in 2006, the Cases secured a massive donation to the project from a range of public and private institutions, including the Kaiser Foundation, USAID, and the President's Fund for AIDS Relief (PEPFAR). This donation for US$16 million, which press photos show being handed over as a giant check by Bill Clinton and Laura Bush, launched a campaign intending to raise a total of US$60 million, which was meant to fund the installation of four thousand PlayPumps across Africa by 2010. This, they said, would supply water to ten million people (Case Foundation, n.d).

But three years later, in 2009, things didn't seem to be going very well for the project, and PlayPumps International's support appeared to falter. Its newly appointed CEO, Gary Edson, published a letter on the company's website acknowledging problems with the rollout of the campaign and announced that production of PlayPumps would be suspended until these problems were resolved. Reports began to emerge that questioned the PlayPump's benefits. A few months after Edson's announcement, journalist Andrew Chambers writing for the *Guardian* newspaper reported criticism of the project among aid agencies (Chambers 2009). He published a letter from the charity WaterAid, which noted that "although aggressively marketed to them," they did not support the use of PlayPumps as there are "cheaper and more sustainable ways of achieving the same aims without using PlayPumps" (Martin 2009). WaterAid doubted

that children's play would be a reliable source of energy for the pump.

Indeed, Chambers used the PlayPump's own figures for its pumping rate against international standards for minimum water supply to calculate that children would need to "play" for twenty-seven hours a day to supply water to the size of communities the pumps were installed in! Clearly, children's play would not supply sufficient water to their communities. In his article, Chambers referred also to an unreleased UNICEF study from 2007 that was circulating among workers in the water and sanitation sector, which noted poor performance of the pumps, a lack of community consultation, low advertising take-up on the pump's billboards, and delays in maintenance of PlayPumps. The study reported that adults were frequent users of the pump, pushing the roundabout by hand to meet their demand for water. In the same month as Chambers' article, "Owen," a worker with Engineers without Borders (Canada), began blogging about his first-hand observations of PlayPumps in Malawi (Owen 2009). He confirmed seeing adult women regularly turning the roundabout by hand just to draw their own water—the PlayPump's water tank was usually empty—and that the pump appeared to perform less well than ordinary hand pumps.

Amy Costello felt compelled by rumors and reports of the project's poor performance, and her role in promoting it, to revisit the PlayPump story for PBS in 2010 with the film *Troubled Water* (Costello 2010b). Her new film, accompanied by extensive interviews and documentation on the film's website, revealed high levels of dissatisfaction among communities where PlayPumps were installed, particularly in Mozambique. Adult women had to turn the roundabout by hand because children's play was an insufficient source of water, and it was uncomfortable for adults to use in this way. Costello referred to another unreleased and much more detailed study of the PlayPump commissioned by the Mozambiquan government in 2008 that confirmed the widespread unhappiness of communities with this new technology, especially given that the PlayPump's roundabout usually replaced existing, more efficient hand pumps. A woman in Mozambique told Costello, "From 5 a.m., we are in the fields, working for 6 hours. Then we come to this pump and have to turn it. From this, your arms start to hurt. The old hand pump was much easier" (Costello 2010b).

In 2010, PlayPumps International wound up their support for the project, took down their website, and handed over all remaining PlayPump stock to the organization Water for People (Costello 2010b). They have never fully explained why they dropped their support for the project and would not be interviewed by Costello for *Troubled Water* (Costello 2010b). Field's original company in South Africa still continues in 2013 to produce and promote PlayPumps, but with much diminished international support.

Audiences and Users

I analyzed the PlayPump as the central case-study of my PhD thesis, *Radical Plumbers and PlayPumps—Objects in Development* (Borland 2011), which I wrote as a cross-disciplinary study in an engineering department (drawing on my background in fine art) at Trinity College Dublin. I began my examination of

the PlayPump in 2006 and followed it over the intervening years until late 2010, when I submitted my thesis, so I was in a position to witness it at both the height of its popularity and at its demise.

I analyzed the PlayPump as an example of contemporary "design for development," a field in which small-scale objects are designed to serve basic needs in developing-world environments: to access water and electricity, to generate income, and so on. Some iconic objects from the field include the "windup radio" produced by Freeplay in the 1990s and the One Laptop per Child (OLPC, or "$100 laptop" as it was originally called). While the idea of designing for developing-world environments and users has been around for some time—particularly since the "appropriate technology" movement of the 1960s—the past few decades have seen increasing attention paid to the reception of these designed objects among audiences in the "first world,"[1] via press articles, aid campaigns, exhibitions, and awards. Design for development projects gather support and funding through increasing their visibility among first-world audiences—by stimulating direct donations or through the sale of products like One Water or by attracting donor bodies to share in this public attention. As Field put it, "we can make a really big organization look fantastically well by being associated with PlayPumps" (London 2007).

The PlayPump is a particularly vivid example of how an object that tells a compelling story to these audiences outside of the developing world might get advanced despite its failure for users on the ground. The enthusiastic reception of the project among journalists, donors, and the general public in the first world was in stark contrast to the dissatisfaction of users of the technology in the developing world. Articles in the press that uncritically repeated the claims of the producers of the technology, and the popularity of these depictions of the project among the public in Europe and the United States, had a snowballing effect, with more and more bodies wanting to associate themselves with the PlayPump. There was very little avenue for realistic representations of the experiences of users of the PlayPump to travel to these distant audiences supporting the project. Even when first-world visitors traveled to the site of a PlayPump, they still saw more or less what they expected to see—for reasons that will be examined later.

For my thesis, I constructed the diagram shown in figure 12.2 to help depict the relationship between these audiences to and users of the PlayPump, as mediated by the different bodies producing, promoting, and funding the PlayPump. The PlayPump in this diagram is understood as both a discrete, particular object on the ground somewhere (the lower part of the diagram) and as a larger overlying system for funding, maintenance, and image distribution (in the upper part of the diagram).

I imagined the boundary between these two zones, represented by the horizontal dashed line in the diagram, as a type of membrane separating first world and developing world. One way in which this membrane functions is as a "screen." Looking down from the top of the diagram, audiences see not the failure of the technology on the ground (of which more later) but idealized images of the PlayPump projected there by the producers of and partners to the project—such as the photograph in figure 12.1.

The PlayPump's promise that children's play could accomplish essential work was compelling. In photographs and videos of the PlayPump, laughing children were depicted playing on the roundabout, pumping water to the water tank that then gushes

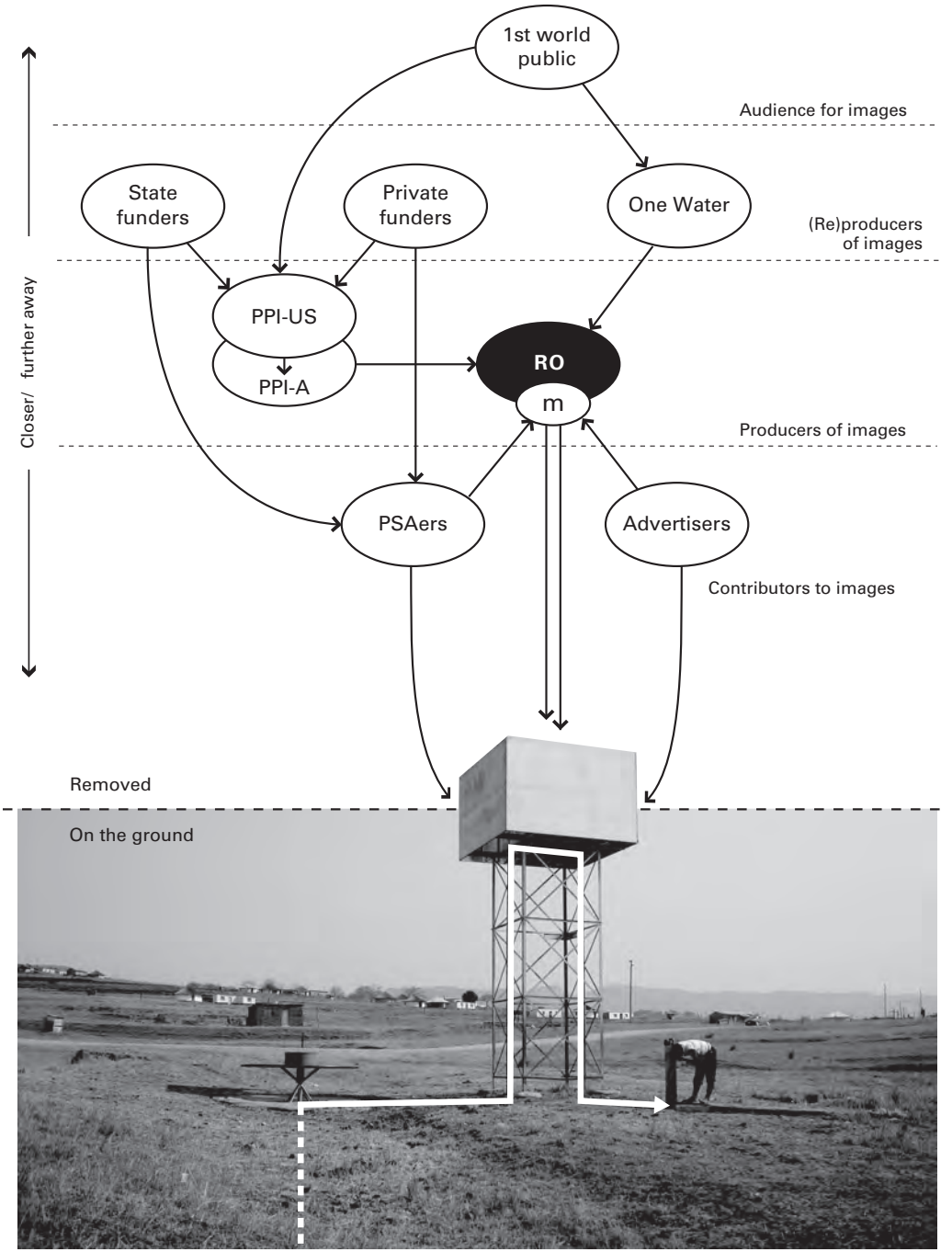


Figure 12.2

A diagram from the author's thesis: The diagram focuses mainly on the systems that support the PlayPump and includes a photograph taken by the author of a PlayPump in KwaZulu Natal, South Africa (ca. 2011). Key: PSAers, public service advertisers; RO, Roundabout Outdoor, original producers of the PlayPump; PPI-US, PlayPumps International (USA); PPI-A, PlayPumps International's African organization; m, maintenance, indicating how billboard rentals were meant to fund the maintenance of the pump.

from the pump's faucet to quench their thirst. As the World Bank described it on its Marketplace Development Award website, "primary school children can now be found laughing, playing, running, and joyfully extracting water from the ground for their entire community" (World Bank 2004). Children's play on the roundabout was sold not just as a substitute for work, but as valuable in itself: the PlayPump's producers described play as "a powerful tool through which young people learn about themselves, gain respect for each other, break down gender stereotypes, and stimulate their bodies and minds. PlayPump systems inspire kids to play, giving joy while fostering self-confidence and interpersonal skills" (PlayPumps International, n.d.). "This play pump idea is brilliant" wrote one visitor to the website for Costello's first movie on the PlayPump, "Does my heart good to know that basic human problems can still be solved with innovative and creative solutions." Other visitors to the site describe getting "chills" and "goose-bumps" from watching the video; they describe it as a "heartbreaking and uplifting piece all in one." "I LOVE the idea! It is creative, inventive, and kind-hearted. I ADORE IT!" "This is the coolest thing I have ever encountered" (Costello 2005).

Word Play

In my thesis, I identify some key components of the compelling "narrative image" that was generated by the PlayPump and that was instrumental in gathering support for the project. Using puns and double-entendres is common stock for newspaper headlines, and with the PlayPump journalists were presented with images ripe for such word play. There were two particular phrases that were used in association with the PlayPump and that I included in the selection of newspaper headlines in the first part of this chapter: the PlayPump as "child's play" and as the "magic roundabout."

The PlayPump appears to literalize the metaphor of "child's play"; journalists could make use of the phrase both to describe the actual operation of the pump (powered by children's play) and as an idiom: a simple idea that just works. The PlayPump brought to life an existing figure of speech, one which Western audiences are primed for. The idea of "child's play" is conveyed to audiences through readings about the object, as in newspaper headlines, but is also encoded in the PlayPump itself, via the pump's roundabout—a children's technology.

The roundabout is an iconic piece of playground equipment for Western viewers. Its presence persuades viewers that play, and therefore pleasure, must characterize the user's interaction with the pump. A teacher writes, "I've been sharing the excitement of the Play Pump in my third grade classroom for several years now ... the universal appeal of playground equipment is something they can easily relate to" (Costello 2005). The roundabout has a self-evident logic to it that has made for a near-universal interpretation, from a distance, that the PlayPump works the way it is supposed to. To picture adults pushing the roundabout around by hand as a repetitive chore requires a leap of imagination; the more potent image it suggests is the one

the PlayPump's producers are glad to reinforce, that of children playing.

The PlayPump's mode of operation made for what Gary Edson, short-lived CEO of PlayPumps International, described as "compelling images of children at play on our equipment" (Edson 2009). As Field says, "There's nothing quite like children's power as a pure energy source" (World Bank 2004). We can read pure as in a "clean" (or green) energy source, but we can also read "pure" in a more metaphorical sense: children as icons of innocence, not bearing the same assumed responsibilities as adults. This must be one of the reasons why children are such a stereotypical subject of aid campaigns: they present an uncomplicated image of humans in need. Designing the PlayPump to be operated by children means it is always represented with children included—and is I am sure what Trevor Field, with his career in advertising (hence image creation), immediately perceived to be a major advantage of the technology. As Field said about his former career: "I used to sell dreams, hopes, and goals for the future" (Field 2008).

The phrase "magic roundabout" (which headline writers presumably based on the title of the popular British children's television show) also reveals the spectacular allure, and the careful crafting of the PlayPump object itself. The PlayPump is most likely perceived of as "magical" because it creates the appearance of work accomplished without human labor. The apparent innovation of the PlayPump is to have the work of water pumping accomplished as a by-product of children's play. The design of the system, with the pumping mechanism hidden inside the roundabout, and all connections between the roundabout and the water tank concealed beneath the ground, creates the illusion of roundabout and pump operating independently. Coca-Cola during their partnership with Roundabout Outdoor described the PlayPump as "a children's roundabout with a hidden agenda to provide energy for a borehole pump" (Coca-Cola 2000). The construction of the PlayPumps International slogan "Kids Play. Water Pumps!" as two separate sentences emphasizes the separation of these two concurrent activities, implying that there is almost no causal relationship between the two phenomena. This "modern-day alchemy" converts "the energy of children cavorting on a simple playground merry-go-round into clean water" (Everline 2007). It transforms work into play.

The PlayPump takes its place among other magical objects in the European folk-story tradition that produce goods without labor: salt grinders, cooking pots, axes, and harps. Walt Disney portrayed a version of the German fairy-tale "The Magician's Apprentice" in *Fantasia* (1940), with Mickey Mouse as the apprentice unable to keep control of a magical broom he attempts to have do his work for him—as is usually the moral with these stories. As with the figure of speech "child's play," a model for "magical" labor-saving objects such as the PlayPump already exists in Western culture—and more widely: "All productive activities" of the Trobriand islanders, for example, noted the anthropologist Alfred Gell (1992, 224), "are measured against the magic-standard, the possibility that the same product might be produced effortlessly."

Through the PlayPump, two apparently opposite ideas, (adult) "work" and (child's) "play" are collapsed. It performs alchemy, transforming work into play. But what effect does the accomplishment of these powerful messages, which proved so enthralling to audiences to the project, have on the functioning of the technology on the ground?

An Inappropriate Technology

Matching children's play—by most people's definition a voluntary, sporadic activity—to the achievement of an everyday, essential task in resource-poor settings was a high-stakes gamble. Setting up such a system successfully, if it were possible, would require careful observation of children's play habits over time, calculations of how much water could be pumped reliably in this way, and how many people could be supplied with an adequate amount of water. None of this work was done for the PlayPump. Studies of the PlayPump both on paper and in the field demonstrated that it did not pump water at anywhere near the rate its promoters claimed, and it could not possibly supply nearly as many people as it was intended to.

This was caused in part by the same features of the pump upon which its "magic" relied: although concealing the connection between children's play and water pumping impressed distant audiences to the project, it hindered the operation of the pump on the ground. The 2008 study in Mozambique found that hiding the PlayPump's pumping mechanism within the roundabout drastically reduced the stroke of the pump and so the rate at which it could pump water relative to a conventional hand pump. This low pumping rate ensured that children's play could not provide enough water for their community.

As a result, children's play did not *substitute* for adult work, as representations of the project suggested, but was in competition with it. The team employed by the Mozambiquan government to evaluate PlayPumps in that country noted: "In hot weather with sunshine, it is unpleasant to operate the pump during the time between late morning to early afternoon. Early morning and late afternoon is also the time when the adults want to fetch water. During this time they don't like if their hard work of drawing water is interrupted by playing children" (Obiols and Erpf 2008, 25).

The PlayPump's roundabout was uncomfortable for adult women to use in this way. Designed for children, the roundabout was at a height appropriate for their use, not for use by adults. Back pain was reported as a common complaint. Pregnant women and the elderly were not able to use it at all. Some women also felt humiliated at having to use a children's plaything: UNICEF interviewed women who were embarrassed to be seen operating the roundabout, especially "where the people watching them did not know the linkage between the 'merry-go-round' and the water pumping," for example where the pump was near a public road (UNICEF 2007, 10). Encoding child's play into the PlayPump rendered it inappropriate for the adults who were compelled to use it.

Play and Passivity

Ignorant of the poor reception of the PlayPump among many of its users, some first-world designers went further in interpreting the system's incorporation of play: as a means by which communities could "celebrate" the gift of water they had been given through the donation of the pump. "Water pumps are

placed in areas of high drought and provide drinking and irrigation water. They greatly benefit the surrounding areas and are seen as something to be celebrated" the Centre for Design Innovation in Ireland reasoned. "Designing a pump which incorporates the play of children, takes this emotional element into account. Its design combines the function of the pump with the celebration of its installment and use" (Centre for Design Innovation 2009).

This representation of the PlayPump as enabling celebratory expressions of gratitude is reinforced by Crealy Adventure Parks in Britain, which installed modified PlayPumps in its parks as an amusement ride for first-world children. They proposed to install a cash machine next to each roundabout "where children can make donations. As people insert money or swipe a credit card, a screen above will show children in Africa riding on the roundabout and shouting, "Thank you!" The money raised will go to install more roundabout pumps in Africa" (Lamb 2005).

But many users of the PlayPump were not grateful for the "gift" they had received. Many wanted their original hand pumps back. While the PlayPump is given without cost to users, the Mozambiquan team noted that at least some communities "prefer paying while keeping control of their services" as they had previous to the PlayPump (Obiols and Erpf 2008, 39). Where some communities had to pay into water-point committees that managed hand pumps, the PlayPump promised a free maintenance service funded by billboard income. The PlayPump could only be repaired by the company managing this income—but users experienced long delays waiting for this service, and the pump was not designed for repair by users (as some hand pumps are). As Joaquim George, of Mozambique's Rural Water Authority, told Costello: "we know it is free, but it doesn't work properly" (Costello 2010b). Both UNICEF and the Mozambiquan government's reports identified problems with a lack of adequate community consultation in installing PlayPumps, which compounded their frustration with the system. Users were denied an informed technology choice between the PlayPump and the existing hand pumps that the PlayPump replaced.

The PlayPump's treatment of its users as passive recipients of aid denied a choice of preferred technology and a part in its maintenance—whose only role designed into the technology is a daily celebration of gratitude through play—is reflected in the PlayPump's relationship with its funders, its first-world audience. Audiences as well as users are disempowered by the PlayPump, which offers them the opportunity to give with the minimum of effort, and without advancing their understanding of the developing-world issues they desire to mitigate. Crealy Adventure Parks' proposed mode of giving, whereby first-world visitors swipe their credit cards and receive an instant pre-recorded message of gratitude, is emblematic of this relationship.

This was a specific intention of the PlayPump's partners. One Water wrote on its Facebook campaign page for the PlayPump: "we rarely talk about the problems in Africa. We'd rather focus on the solution and create good feelings. Make a change in the world and have fun at the same time" ("Mark" 2010). Field described One Water's model as "a really clever way to get a lot of people to donate money to a charity without really thinking about it. . . . All you do is buy a bottle of water and you know you're doing the right thing" (Fry 2007). The PlayPump's invocation of play as "not work" for its users, a supposed benefit of the system, extends to its donor audience—they too are spared the work of inquiry into the developing-world problems they would like to help solve.

The Performance of Play

Part of the reason why a distorted vision of the PlayPump can be maintained for audiences is their geographic separation from users, with the producers and partners to the PlayPump controlling the play of images across the "screen" separating first-world audiences from developing-world users. But strangely enough, distorted images of the PlayPump can persist even when privileged onlookers visit individual PlayPumps in the field—as happened to Amy Costello in her first film for PBS, when she visited a PlayPump with Field and saw children enthusiastically playing on the roundabout.

This phenomenon of children performing to the expectations of outside onlookers was observed by Karl Erpf from Swiss water consultancy SKAT, a member of the team studying PlayPumps in Mozambique:

> In most schools visited, children were not always moving the play wheel—they often enjoyed the PlayPump as a gathering place, just sitting on it and chatting. However, as soon as the evaluation team (foreigners) walked towards the PlayPump, the children rushed to the pump (like they have been told), showing their ability to rotate the play wheel at an enormous speed. The children pushing the wheel with such a high speed could only keep up this pace for a few minutes before being exhausted. (Obiols and Erpf 2008, 24)

Erpf's report here highlights, in parentheses, the telling parts of this phenomenon: the evaluation team are "foreigners," and the children play "like they have been told." They spin the wheel at great speed for a short time, and then stop exhausted after a few minutes. Clearly, this is not a realistic depiction of the way the pump is used, though it happens right before the eyes of the observer. Engineers without Borders blogger Owen similarly described the difficulty he had in photographing adult women in Malawi operating the roundabout by hand, though he knew from past observation that this was the way the PlayPump's roundabout was used most of the time:

> Each time I've visited a Playpump, I've always found the same scene: a group of women and children struggling to spin it by hand so they can draw water. I've never found anyone playing on it. But, as soon as the foreigner with a camera comes out (aka me), kids get excited. And when they get excited, they start playing. Within 5 minutes, the thing looks like a crazy success. Kids are piling on top of each other to spin around on the wheel, and women can fill their buckets without having to work (although I'll note that the buckets still fill slowly). (Owen 2009)

The very similar experiences of Owen and of Erpf show that the distorted view of the PlayPump supplied by its makers and partners to audiences in the first world might be replicated right on the site of an individual PlayPump, due to the desire of locals, children especially, to put on a show for a foreign visitor. Owen warns that:

> If you show up in a community with a Playpump, it will look like a success. Kids will play. Water will flow. But all of this is likely only happening because you are there. And if you can't ask the right questions, or if you are travelling with a guide who has a vested interest in the technology (e.g. an NGO worker who installs Playpumps), then you will never know the difference. Same goes if you only watch the promotional videos on the Playpump website. (Owen 2009)

The "screen" that I have suggested divides the PlayPump on the ground from audiences is not so easily pierced, and its effects may persist even to the object itself. In these instances, the screen that divides our diagram acts perhaps more like a flexible membrane, which accommodates the movement of the viewer, maintaining its integrity even when the viewer imagines he or she has crossed it. The larger systems at work that divide first-world viewers and developing-world users, with their enormous imbalances of power, are not so easily surmounted.

Conclusion

It is not just distance that enables idealized and unrealistic images of the PlayPump's performance to be presented to first-world audiences—it is primarily a product of the unequal power relationships between first and developing world. Because of this difference in power, the impressions of individuals and organizations in Europe and the United States took precedence over the experience of users of the PlayPump in the developing world, and a technology that was inappropriate to its contexts of use was imposed on communities in Southern Africa.

In this relationship, notions of "play" were mobilized in a number of ways. Children's play was used as a compelling image to attract support, with its promise to save children and adults from the labor of manual water pumping. As a result, adults were made to perform to a script written for the gratification of distant audiences, bending to the wheel of the PlayPump's roundabout. The PlayPump insisted that they be children—it infantilized them both in this quite literal way, and in more metaphorical ways: in denying them a choice of technology or a role in the pump's management.

Audiences in the first world were able to imagine this fanciful technology for accessing a daily necessity to be genuinely useful—for it to be celebrated by its users in fact. Never mind that for its Western supporters, water is accessed through simple, everyday technologies without ritual. There are many things that are onerous about water collection for poor people in rural parts of the developing world, but the up and down motion of a manual hand pump is not one of them (Owen 2009). Play was presented as way of negating work—play as "not work." But matching play to a necessary, daily task resulted not in less work, but more.

This fantasy of using a "waste" resource, energy as a by-product of children's play, finds correspondence in other examples from the first world: nightclubs with sprung floors that generate electricity to power the venue, cinemas powered by bicycles, gyms by the people exercising on their equipment. But crucially these examples match play with play—they power nonessential services. Other ubiquitous sources of energy are always present. To design similar systems for daily necessities is a much riskier proposition, and the fact that the PlayPump replaced other technologies for water compounded this risk, leaving some communities with no other source for water.

The PlayPump disempowered both its users and its supporters. The PlayPump associated play with entertainment, with pleasure for the viewer as well as the user. This pleasuring of the viewer involved shielding him or her from the complexities

of developing world problems, inviting only the viewer's passive donations. This association of play with the uncomplicated pleasure of giving is evoked by another recent design project for the developing world: the Sockett, a soccer ball with an internal generator and USB outlet powered by motion. "We want people to realize that making a difference doesn't have to be serious, complicated or boring" say its producers. "It can be as simple and fun as playing a game" (Davis 2012, 38).

The story of the PlayPump speaks to the risks in framing design projects as a form of spectacle for audiences removed from their contexts of use—with "play" used here as part of the project's narrative appeal—and to the mismatching of "play" to an essential task, with negative consequences for users. The ground upon which design for development objects such as the PlayPump operate is highly unequal. Whatever games are played on it need to be more than simple and fun, and making a difference may require changing the rules of the playing field in much more fundamental ways than such objects imagine.

Note

1. By which I mean largely Europe and the United States—but also privileged audiences anywhere. South Africa, for example, where I am from, mixes first and developing world in one country, which is common to many countries in both the global North and South. I explore the notion of power over geography in defining first versus developing world later in this chapter.

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13 FOUCAULT'S FITBIT: GOVERNANCE AND GAMIFICATION

Jennifer R. Whitson

In this chapter, I draw from Michel Foucault to frame self-tracking and gamification in terms of the governance of modern liberal nation-states where subjects willingly govern, regulate, and optimize themselves. I introduce the quantification of the self, showing how it is used in gamification movements and how it is leveraged to promote a care of the self, as well as further enrolling individuals in normalization projects. I argue that current gamification projects are not influenced by playful design (and much less a focus on fostering creativity and exploration), but take something entirely different from games: the feedback mechanisms such as leaderboards, damage meters, and point systems that allow users to manage risk as well as pinpoint "approved" routes toward mastery and self-improvement. I conclude with some cautionary thoughts about the difficulty uniting play with non-game governance projects, given that play inherently encourages players to push against, reshape, and find movement between rules, sometimes breaking these rules altogether.

Introduction

Luka is five years old. He has been "quantifying" since he was two and a half. First, he started collecting his daily "numbers" on a bathroom scale that records his weight, calculates his body fat, and uploads these data to a website charting his growth. Then, when Luka was four, he was given a *Fitbit,* a wireless activity and sleep tracker that counts each step he takes during the day. While Fitbit itself promises to make fitness fun by awarding badges for distance traveled, calories burned, and stairs climbed, Luka and his dad have invented many other games to "get more steps," including—if Luka notices he hasn't beaten yesterday's total—racing around the house like an Olympic sprinter before bedtime. One of Luka's favorite games is competing with Bruce from his dad's work, who is also connected to Fitbit's cloud service. On most days Bruce, a sedentary soul, shows up on the family's leaderboard losing to Luka. Winning against a real grownup, let alone one from his dad's work, is pure joy (Carmichael 2012).

Luka, or more accurately Luka's dad, is a member of the quantified-self (QS) movement (Schuller 2012). As I argue in this chapter, QS is closely tied with

gamification. By *gamification* I mean "the use of game design elements in non-game contexts" (Deterding et al. 2011). Both emphasize self-improvement through tracking, both are tools of self-governance, and both characterize a new care of the self based on data. Not surprisingly, tools for quantifying the self are also examples of gamification, including Nike+, Mint, Runkeeper, Health Month, fitocracy, Daily Burn, 750words.com, and, of course, Fitbit. While gamified apps often fail to deliver the playful and game-like spaces they promise in their marketing rhetoric (for example, see Jaakko Stenros's chapter in this volume on the missing playfulness in gamified applications, and see Ralph Borland's chapter in this volume on how play is evoked in marketing the PlayPump), what they excel at is providing meaningful feedback to users that is then enrolled in motivating behavioral change (see both C. Scott Rigby and Conor Linehan and colleagues' chapters in this volume for more on this). Mastering the self through the application of gamified data is an important method of governance. It is this relationship between gamification, quantification, and governance that I explore in this chapter.

Part 1: Governance, Surveillance, and the Care of the Self

In part 1 of this chapter, I draw from the work of Michel Foucault to first define what I mean by governance. I then argue that governance—and knowing the desires of those who are governed—is reliant upon surveillance. I briefly trace the historical relationship between surveillance and governance, from discipline in the eighteenth century to more modern modes of control focused on consumption and desire. Here, I begin to make links to commercial gamification products, such as the Fitbit, and how they enroll our desires for self-mastery and improvement into a new care of the self, a care of the self that is also predicated upon governance.

Defining Governance

When I talk about governance, I am referring to something broader than the voting practices and democratic institutions of the state that are the focus of Greg Lastowka and Constance Steinkuehler's chapter in this volume. I am referring to government as the "conduct of conduct." In other words, government is not just a state domain but includes "[A]ll endeavours to shape, guide, direct the conduct of others, whether these be the crew of a ship, the members of a household, the employees of a boss, the children of a family or the inhabitants of a territory. And it also embraces the ways in which one might be urged and educated to bridle one's own passions, to control one's own instincts, to govern oneself" (Rose 1999, 3).

I am interested in how gamification is tied to the forms of neoliberal governance that are focused on the privatization and deregulation of the state, while simultaneously inducing citizens and corporations to regulate and govern themselves. Specifically, in this chapter I want to examine how gamification is used to encourage citizens to govern themselves, in terms of taking increased responsibility for their health care, education, and workplace productivity, as well as how they are encouraged to become more loyal consumers and clients. For example, how does gamification ensure that a five-year-old boy like Luka is

aware of his body-fat index and its relationship to long-term health (and thus his potential burden on the health care system)? In these cases, the governors are not only state agencies, but also educators, employers, corporations, and even individuals such as Luka's dad. The governed are people like Luka—as well as you and I—the individual users of gamified applications.

Governance in the Foucauldian sense is productive. It is opposed to domination, wherein subjects have no other option but to obey. As put by Nikolas Rose, knowledge of those to be governed is key to this productivity:

> [T]o govern is to recognize that capacity for action and to adjust oneself to it. To govern is to act upon action. This entails trying to understand what mobilizes the domains or entities to be governed: to govern one must act upon these forces, instrumentalize them in order to shape actions, processes and outcomes in desired directions. Hence, when it comes to governing human beings, to govern is to presuppose the freedom of the governed. To govern humans is not to crush their capacity to act, but to acknowledge it and to utilize it for one's own objectives. (Rose 1999, 4)

In other words, governance is about knowing subjects and their motivations and desires well enough to determine how to get them freely and willingly to enroll in the governor's projects, and thus govern more effectively. This entails a much different conceptualization of power than domination through force—the governors are not focused on punishing the governed, but recruiting them as willing participants. In this sense, power is not a thing, but a relationship between people in which one affects another's actions. It is productive, rather than violent or repressive. It involves making a free subject do something he or she would not have done otherwise. Power is not just localized in the state and other authorities, but is present in all relationships.

Governance is a thus a process of translation, forging alignments between the objectives of authorities wishing to govern and the personal projects of those organizations, groups, and individuals who are subjects of government. Not surprisingly, there is a lot of interest in using gamification as a *technology of government* that shapes users' conduct in the hope of producing certain desired effects (such as using gamification to increase productivity in call centers) and averting certain undesired events (such as using gamification to reduce employee churn and absenteeism).

Surveillance and the Panopticon

Developing the knowledge required to discern the desires of subjects and govern them accordingly depends upon surveillance, making visible the space over which government is exercised: "defining boundaries, rendering that within them visible, assembling information about that which is included and devising techniques to mobilize the forces and entities thus revealed" (Rose 1999, 33). The role of surveillance in governance is a central theme of Foucault's earlier work (Foucault 1977). *Discipline and Punish* traces the history of government from pastoral to feudal to near modern times, asking: *how do we go from an unruly, undifferentiated mass of people, to the orderly, productive, collection that we see today*?

Foucault argues that in order for society to thus organize itself, the key is render each individual visible, to separate them out, to closely observe, and then compare them to each other. The way this is achieved differs according to the point in history and the techniques available. For example, Foucault used

Jeremy Bentham's blueprint of the panopticon as a vivid example of disciplinary power. A prison oriented around a central guard tower, individual prison cells in the panopticon create a ring around this central tower. At all times, prisoners are exposed to the gaze of those within the tower, though prisoners cannot see the other inmates, nor tell whether the guard tower is occupied or not. In this case, the simple fact that one may be observed is central to evoking socially accepted behaviors.

As put by Bart Simon, the power of the panopticon is twofold:

> On the one hand, there is a concern with processes of subjection and normalization that arise through the internalization of the gaze, while on the other there is a concern with processes of administration, social sorting and simulation that occur independently of embodied subjects. (Simon 2005, 1)

This administrative power of the panopticon quickly diffused to other spaces, such as schools, hospitals, factory floors, and military service, and initiated the birth of the record system (ID systems, police file systems, medical records, and academic grade systems), as well as written systems for identifying individuals and—most importantly—tracking them over time. The ability to differentiate subjects and to track their performance over time further incites a desire on the part of the subjects to normalize, to fit in. By normalize, I mean conforming to the idealized norms of conduct. For example, Foucault uses the example of military drills, where each soldier is taught precisely how to stand, march, present arms, and so forth. Soldiers are then rewarded or punished for conforming to or deviating from this ideal. Today, normalization can be seen in body-fat index ratios, as well as scores, ranks, and grades.

This two-part disciplinary power—part focused on administering populations, and part focused on self-governance—fought the chaos of previous, more violent, forms of government by using surveillance to order individuals and, by doing so, impose efficiency and productivity. In the popular imaginary, however, the panopticon is equated to domination in the form of George Orwell's *1984* and the oversight of Big Brother.

Beyond the Panopticon: Games as Government

In the face of neoliberalism, the panopticon crumbles. At heart, disciplinary panopticism relies on individuals who want to become ideal citizens, part of a civilized polity who would govern themselves "through introspection, foresight, calculation, judgement and according to certain ethical norms . . . the social objective of the good citizen would be fused with the personal aspiration for civilized life" (Rose 1999, 78). The overarching impetus to become an ideal citizen, however, dissolves as people focus more and more on individualized goals and aspirations. Meanwhile, the bastions of this disciplinary governance—the church, the factory, the state—disappear in the face of deregulation, replaced by new domains, more agile models of production, and a de-emphasis of the collective social body. As put by Nikolas Rose, "Today, perhaps, the problem is not so much the governability of society as the governability of the passions of self-identified individual and collectivities: individuals and pluralities shaped not by the citizen-forming devices of church, school and public broadcasting, but by commercial consumption regimes and the politics of lifestyle, the individual identified by allegiance with one of a plurality of cultural communities" (Rose 1999, 46). Accordingly,

theorists such as Gilles Deleuze have extended Foucault's work to consider how governance is enacted in spaces premised on automation, *dividuation*,[1] and consumption. In these spaces, control operates on a more free-floating, adaptive basis that is rooted in desire rather than social conformity. In short, consumer society has replaced civil society, and thus modes of governance adapt.

In a consumer society, surveillance shifts from tracking individuals to monitoring behavior and consumption patterns. Populations are constituted as consumers to be seduced into the market economy. This monitoring is predicated upon limiting access to places and information and developing ever more intimate consumer profiles. Power is enacted through reconstructions of consumers' behavior, habits, and actions, knowledge that enables more effective governance. Accordingly, in the *society of control* there is a movement away from human watchers and their associated value judgments, and a movement toward seeing individuals only as bits and bytes in vast ebbs and flows of information. By tracing the aggregated desires of shoppers and system users, finding patterns in their flocking behaviors like a school of fish, and then channeling these behaviors, organizations thus enact governance, knowing subjects and their motivations and desires well enough to determine how to get them to freely and willingly enroll in their projects.

Whereas the panopticon depends on the individuals, first individuating them in order to form them into a more productive social body—like cogs in a vast machine—Deleuze's (1992) *society of control* is automated by technology. Rather than relying on prison guards and drill sergeants to discipline individuals, control operates according to machinic demands and is reliant on codes and passwords, on data and databases. The body is transformed into pure information—the data double—so it can be rendered more mobile and comparable. Information, such as shopping habits, user preferences, bank account numbers, voting preference, location, and so forth, is separated from individuals and recombined in new ways outside of their control. These recombinations, such as user and customer profiles, are based on the criteria deemed salient by those with access to the information, be they government officials or corporate marketers.

Instead of individuals—irreducible and with an autonomous sense of agency—the new subject of governance is instead the dividual, an artifact of data mining searches and computer profiles. People, each as individual wholes, are unimportant. What is important are how masses of people can be broken down into more manageable parts by collecting the data streams they trail behind them, filtering out the parts deemed important, and ignoring the rest. *Dividuals* are then governed automatically through databases and levels of access and exclusion. For example, in banking transactions, your name and identity are entirely unimportant. What matters is whether you hold the appropriate account card and provide the correct PIN password to access your account. In financial institutions, you are abstracted into streams of numbers and transactions that are aggregated with the transactional streams of other clients, which are then used to streamline operations and predict future economic patterns.

On the part of the user, governance is short term and rapidly shifting, but at the same time continual, unbounded, and ruled by pleasure and desire. We are not confined. We are free consumers. We monitor ourselves or submit to monitoring willingly in order to maintain or augment social perks. This pattern is

characteristic for today's web services, such as those that Google provides; from efficient Internet searchers, to maps and real-time traffic reports, to cloud storage, to e-mail and social networking services, to YouTube. The data we willingly divulge are used to "serve us better." It is here that we see the link between gamification and governance. Gamification enables a form of governance much closer in alignment to what Deleuze proposes.[2] We broadcast our personal data as the price of participation.

With the Fitbit, Luka's name and identity are rendered unimportant, as is the fact that he's using the system as a way to interact and bond with his father. What is important is whether the system is registered to a valid online account, and that a steady stream of data—in this case, steps, acceleration data, location, and data on when the system is turned on and off—is being sent back to Fitbit. These data are combined with the usage patterns of other Fitbit dividuals, as well as amalgamated with the demographic data culled from these users' online profiles. It provides clues as to what traits, demographics, and usage patterns may correlate to the most profitable or loyal users, as well as insight about who to target marketing at, and how to improve the system's algorithms and tracking capabilities to attract more lucrative clients. However, gamification as governance promises something more than just tailored services. It promises to tell us more about ourselves.

The Care of the Self

While links may be drawn from discipline, normalization, and panoptic surveillance to gamification, in comparison to the governance exhibited in the factories and prisons of the eighteenth, nineteenth, and twentieth centuries that Foucault described, it is somewhat difficult to imagine Luka's competing with Bruce "from dad's work" as akin to the activity of a prison inmate. Luka is playing with surveillance in a much more self-directed way. The impetus to track and monitor his daily numbers comes from within, and from his dad, rather than the formal authorities such as the school or state. In this sense, Luka is using gamification in terms of a traditional care of the self. The fact that this care of the self is bundled with other technologies of governance (i.e., Fitbit monitoring Luka and influencing his health and consumption patterns) is largely irrelevant. Accordingly, this playful quantification presents powerful new opportunities for governance.

The term *care of the self* refers to the later work of Michel Foucault (1988). Foucault argues that the care of the self was a foundational principle of all moral rationality up until the Cartesian moment and the Enlightenment. Foucault draws heavily from the Socratic dictates that one must care for oneself and know oneself, arguing that through this self-reflection and care, individuals come to see themselves as responsible for constituting themselves as moral subjects. This care of the self was achieved in three ways: (1) knowing how to live without luxury, through abstinence, (2) regularly subjecting oneself to a thorough examination of one's conscience, and (3) be in constant control. Already, we can draw parallels to many gamified applications, such as Health Month or SuperBetter, which prioritize similar forms of self-reflection as a route toward self-improvement.

Elsewhere (Whitson and Haggerty 2008), I have used the moral panic surrounding identity theft to show how this care of the self takes a different modality in the digital age: we now care for our virtual selves, curating and maintaining the accuracy of our "data doubles," the informational profiles (market

profiles, credit histories, social networking accounts, and even the avatars and account settings for online games) that have become the lifeblood of our interactions with others and the real objects of governance.

In the informational era, our physical bodies seem to fall away. Yet, QS and gamification movements are now bringing the body back in. I argue that this characterizes a new modality of governance that leverages a new set of desires—exploration, curiosity, self-mastery—that characterizes both QS and gamification. Gamification, unlike QS, is also imbricated with discourses of play that effectively shape how it operates.

Part 2: Gamifying the Quantified Self

While part 1 of this chapter provided the theoretical background of Foucauldian governance and the historical evolution of governance techniques, part 2 shows how gamification and the QS movement come into play. In this section, I continue with my discussion of the care of the self, explaining how the quantified-self movement parallels gamification movements. I then make the novel argument that fun is irrelevant in gamification. "Fun" in gamification is, for the most part, empty marketing rhetoric. What matters in gamification is giving users actionable feedback on how to improve. This feedback is what games do well and is where the real link between games and gamification lies.

The Allure of the Quantified Self

Before moving onward to gamification, it is first useful to provide some background on QS. The QS movement was started in 2007 by Gary Wolf and Kevin Kelly, both editors of *Wired* magazine. Interested in using ubiquitous technology to track the self and thus develop self-knowledge, quantified self is also known as "self-tracking," "body data," "living by numbers," "self-surveillance," "life-hacking," "personal analytics," and "personal informatics." These systems collect information about the user and present it back to them, treating people as both the object and the subject of its function (Li 2011, 9–10). Users enroll in QS programs out of curiosity, and continue with them because the data provided are so compelling: "they continue because they believe their numbers hold secrets that they can't afford to ignore, including answers to questions they have not yet thought to ask" (Wolf 2010). The quantification of the self—by compiling the intricate details of our lives and then rebroadcasting them to us in new ways—promises to tell us something about ourselves that we did not already know.

While the quantification of the self has commonalities with the time-honored tradition of journaling and the care of the self as an ethical practice of reflection detailed by Foucault (1988) and the list-making more recently described by Umberto Eco (2010), what is different is the precision, complexity, and the amount of the data collected, as well as the way it is ultimately presented back to the chronicler. Instead of leaving it up to us to decide what is worth chronicling, and then delegating our spotty memories to provide the details, the journaling process in the era of QS is automated, enabling incredibly precise details.

The quantification of the self is not new, but automation greatly expands its scale and scope, as well as its effectiveness at telling us our secrets. As stated by Gary Wolf, we track ourselves all the time:

> We step on a scale and record our weight. We balance a checkbook. We count calories. But when the familiar pen-and-paper methods of self-analysis are enhanced by sensors that monitor our behaviour automatically, the process of self-tracking becomes both more alluring and more meaningful. Automated sensors do more than give us facts; they also remind us that our ordinary behaviour contains obscure quantitative signals that can be used to inform our behaviour, once we learn to read them. (Wolf 2010)

The unifying methodology of QS is data collection, followed by visualization of these data and cross-referencing, in order to discover correlations and modify behavior.

We are used to measuring and quantifying many things in our lives—from optimizing assembly line production, to measuring how fast our computers operate, to grading our intelligence, to using software to clock how many hours, minutes, and seconds we work each day. This disciplinary monitoring is commonplace in public spaces (work, school, hospitals). What is new with QS is that individuals are now willingly monitoring themselves in nondisciplinary spaces and making these details public. For example, whereas measuring food intake or mood was previously an activity restricted to health institutions and revealed only to experts such as nurses and doctors, we now use tools such as diet and health tracking apps to share and broadcast this information to an unspecified public.

Technologies such as the Fitbit or SuperBetter enable us to measure, chart, and quantify what was previously unquantifiable and also allow us to transmit and share what was previously private. It is now relatively simple to measure and analyze patterns in our sleep, exercise, sex life, food intake, mood, location, alertness, productivity, and even our mental health and spiritual well-being. We effortlessly track and measure, display and share all of this heretofore unknown data using our computers, smartphones, and gaming consoles.

While the QS may represent the extreme pole of "self-knowledge through numbers" (Wolf 2009), most of us, in one form or another, have quantified our lives in one way or another, from tracking our fuel consumption on smartphone apps, to monitoring our infant's diaper changes, feeding times, and sleep schedules, to subscribing to Mint to help us track our spending habits. Most of the time we are using gamification to do so. This is not a coincidence. Games and gamified apps are excellent tracking devices.

Gamification as Feedback Loops, Not Fun

Promises of fun and play populate the advertisements of gamification companies such as Bunchball, BigDoor, Badgeville, Lithium, SCVNGR, Greengoose, and Seriosity. However, critics such as Ian Bogost (this volume) argue that these are empty promises. Behind the empty badges and meaningless leaderboards, there is often no "game" in gamification. Accordingly, the failure of gamified products to sustain users and maintain the breathless hype that preceded their deployment makes sense (see Evans 2010). They simply do not deliver. Yet I argue here that the failure of gamification to provide "fun" (whatever that nebulous word means) does not mean that gamification as a whole is a failure. What gamification successfully borrows from games are the

methods to provide clear feedback and reinforcement to users. This, and not playful or gameful design, is what characterizes current examples of gamification.

The ways that games render space visible, from points systems to pathfinding, are what is leveraged in gamification. Feedback methods borrowed from games are key to caring for the quantified self. As Stanford psychologist Byron Reeves and his business partner J. Leighton Reid discovered in their research on games and gamification, data visualization techniques from games are essential tools in shaping users' behavior: "Game interfaces set a new bar for feedback. At any one time, Helen sees progress bars, zooming numbers, and status gauges, all in a well-organized dashboard that lets players know how things are going, good or bad. Numbers indicate the health of players, the time left before an attack, the amount of gold accumulated so far" (Reeves and Read 2009, 71). Games excel at providing precise real-time feedback to help players chart their current progress and determine how to advance. Feedback thus governs behavior; steps toward a goal are encouraged in multiple ways and channels, while steps in the wrong direction are penalized. Feedback can be immediate, for example, providing a World of Warcraft player with real-time *per second* data on how he or she is faring in an attack. But feedback also takes mid- and long-range forms, providing information on how a player is progressing with goals that take weeks, months, or even years to accomplish.

In games, performance metrics and feedback are overwhelmingly positive and focused on improvement, reward, and engagement rather than highlighting deficiencies. Of course, failure still exists, but the risks and punishments for attempting something and then failing are not as severe. Negative feedback works to highlight areas that require improvement and suggest changes in tactics that may help in achieving success, rather than punishment. Thus, players can clearly decipher what they need to do in order to progress. For example, each failure of a World of Warcraft raiding party provides valuable statistics on what techniques are successful (i.e., attacks that inflict maximum damage, team formations that provide an optimum balance between tanks and healers, etc.), and what actions to avoid. Each consecutive failure comes with an incremental improvement in strategy until finally, the raiding party is victorious. Failure provides valuable information on how to become better: "Quick feedback creates immediacy and contingency in the interactions. When you make a new move, you know quickly whether the action was right or wrong. The close connection between behaviour and feedback (it's usually obvious which reinforcement applies to which behaviours) increases the likelihood that the reinforcement will be effective" (Reeves and Read 2009, 72).

Porting the feedback methods used in games to non-game activities thus makes sense. We turn to gamification to respond to a gap in our day-to-day lives, where feedback on one's progress, cues for future directions, and a place for experimentation and even failure is lacking.

For the most part, feedback in the real world is much more infrequent and difficult to accomplish, largely because the automated cycle of data collection, compilation, analysis, and feedback is simply not established. For example, at work feedback is often restricted to annual performance reviews, whereas in academia, feedback cycles can take months and even years—as in a tenure application or journal submission.

The form of surveillance exemplified by online games would have been impossible to carry out in the past. The sheer amount of data collection, analysis, quantification, and feedback, especially on a moment-to-moment basis, would have been entirely impossible. But, automated closed systems, such as the walled gardens of games and social networking sites, put Moore's law into practice. Inexpensive data storage and number crunching, combined with the increasing ubiquity of mobile sensors, make the collection and analysis of data much easier. These advances are combined with a shift away from rudimentary analytics such as aggregate page views to more sophisticated individual user behavior analytics, which were initially developed in social games. With these tools, progress on tasks is now easy to chart and reward, especially as virtual rewards and reputation scores do not have to cost anything. Accordingly, I argue that the key commonality between games, the myriad examples of gamification, and QS is the leveraging of surveillance. Whether online or off, users' behaviors are tracked via technologies that monitor progress. These metrics are then used to provide incremental feedback to the user, thus indicating what the user needs to do next in order to achieve his or her goals. The promise of a "game" and the desire to level up and win (or at least to beat "Bruce from dad's work") is used to inculcate desirable skill sets and behaviors. Gamification is thus rooted in surveillance; providing real-time feedback about users' actions by amassing large quantities of data and then simplifying these data into modes that are easily understandable, such as progress bars, graphs, and charts. Proponents of gamification, such as Reeves and Reid, have created companies such as Seriosity to leverage gamified feedback in places such as corporate offices to lowly call centers, shifting the mode of surveillance from a single-player game to one that effectively governs a whole office.

The "game" involved in gamification projects is in setting challenges and goals, both short term and long term. Charts, graphs, and statistics are automatically compiled, transforming what is essentially a large database of meaningless numbers into something that users can quickly parse and understand. By gamifying everyday tasks such as exercise and healthy living, users might make solitary and tedious activities more enjoyable. At the very least, even if the tasks are still unenjoyable, users feel that they are making some progress, however incremental. Thus, gamification does not have to be "fun" to be successful (although it certainly can be fun).

What is important here is that this is *willing* self-surveillance. This is not the institutionally imposed disciplinary surveillance of Foucault, or even the instrumentalization of hedonistic desires that fuels the consumer surveillance described by Deleuze. Gamification enrolls people into self-governance by using their highest aspirations and capacities, that of self-care and self-development. With the aid of data gleaned from these practices, gamification creates heterotopias.

Part 3: Gamified Heterotopias

Following Foucault, heterotopias are not utopias or dystopias, but spaces of difference that mirror, reflect, represent, designate, and speak about other sites, while at the same time suspending, neutralizing, inverting, contesting, and contradicting these self-same sites (Foucault 1986). Gamification can

create critical-reflective heterotopic spaces, but it can also create dystopic spaces. Thus, all gamified spaces exist upon a knife's edge. The ability to master and reshape the body or to solve much larger global issues through play clearly has utopic potentials (McGonigal, this volume). However, the potential effects of gamification can quickly become dystopic. The trick of the playful discourse used in gamification marketing rhetoric is to make corporate dystopias appear as if they were heterotopias.

Gamification leverages a discourse about using "fun" to reshape and remake the world, regardless of whether this fun is realized or not. In the words of SCVNGR founder Seth Priebatsch (2010), it creates a "game layer on top of the world." In this sense, this transformation creates pockets of heterotopic space. When individuals gamify aspects of their lives, they are changing mundane spaces such as their daily running routes, their classrooms, gyms, and workplaces into heterotopias. Gamification creates a space of difference that overlays, and is linked to, these everyday spaces. It thus simultaneously sustains and undermines normalcy.

Gamification not only turns physical spaces such as gyms, living rooms, and offices into heterotopias, it also affects our own bodies and our relationship to them in a similar manner. Like the mirror, the quantified self as an object of analysis both represents and—simultaneously—inverts the body and our day-to-day actions. Gamifying this quantified self, in turn, breaks down oppositions between private space, including the intimate details of one's life, and public space by uploading these data to databases of thousands of other users to compare and normalize. It breaks down divides between the cultural space of games and the useful space of production, between the space of leisure and play, and that of work. It turns our physical movements through space and the interactions we have with others into reams of data to be collected, analyzed, and presented back to us in new formats. This data-driven heterotopia provides a contrast with the messiness and fallibility of physical human bodies. In the words of Peter Johnson (2012, 10), it becomes "a meticulously arranged enclosure that exposes the jumbled mess we tend to live in." The quantified self that we develop and then shape according to these data is both real and not real. The body itself is both a subject and object.

Heterotopias change their function at different stages in history, reflecting wider attitudes in society. Moving from the traditional care of the self (the lists and journals and self-reflection) to gamification as a way to reflect upon ourselves, we see our body and our behaviors in a new light—as something that can be quantified, measured, and segmented into tractable data in order to master and reshape. In the panoptic age, this was done by institutions such as school and hospital with grades and medical charts. Now we undertake this quantification of the self under our own free will. We become self-regulating agents. Yet governing institutions are not wholly absent. They only appear to be. This third, and final, section of the chapter discusses some of these dystopic effects, especially those related to the quantified self, as well as the role of play in mediating them.

Self-Regulation and Corporate-Governed Dystopias

The surveillant practices that are embedded in technologies for the quantified self become essential tools for measuring one's progress, providing feedback, and highlighting routes toward this success, whether it is running for thirty days in a row, losing fifteen

pounds, earning a promotion, or getting more miles to the gallon while driving. Personal informatics tools, such as those used in QS projects, are an improvement over other methods of care of the self because pure self-reflection is often flawed. These "systems help people by facilitating collection and storage of personal information, and by providing a means of exploring and reflecting on the information" (Li 2011, 23–24). By helping users observe and record their behavior, users can then reflect on using visualizations to compare their behavior to a particular goal or standards and thus regulate themselves with little external aid.

Not surprisingly, the relative simplicity of what can currently be measured, coupled with the increasing sophistication and decreasing cost of monitoring technologies, creates the impetus for QS services to collect ever more information, especially the *contextual information* that could deliver cues to behavior change. As put by Li (2011, 2), "A tool that allows users to associate contextual information with behavioral information can better reveal factors within one's life that affect behavior, compared to existing systems that only show behavioral information." For example, using the Fitbit to learn that you walked fewer steps this week than last week is only part of the puzzle of actually amending one's behavior. The key to answering "Why did I walk less this week?" lies in gathering a wide range of information from exercise levels, to physical health biometrics, to personal events calendars, to menu information, spending habits, and so forth and so on. This contextual information could pinpoint a number of interrelated reasons for walking less, such as a paycheck that meant one had money to take the bus, coupled with a busy social calendar that prioritized time at the pub eating greasy food, which—in combination—resulted in both reduced energy and fewer visits to the gym, and thus less steps.

What this means is that quantification services are compelled not only to gather much more precise and intimate information, but also to link together information from different domains of our lives. Currently, most systems are unifaceted, only showing one aspect of our lives, such as Mint for finances, Nike+ for physical activity, 750words.com for writing productivity, and so forth. But integration between these services or the creation of multifaceted tools would be undeniably more useful in terms of caring for the quantified self. A participant quoted in Li's study exemplifies the growing desire to chart more and more of our personal lives: "I now want to record all the minutiae of my personal life that aggregates into interesting data. I want to graph the people I see, the things I do, the hours I devote to every significant task, and the money I spend and why. I want to have yearly data that shows, for example, that I spent 1,000 hours on programming, but only 400 on reading, or that I spent twice as much in coffee shops as I did on groceries" (Li 2011, 75). As addressed by Lori Andrews (this volume), this desire to quantify and correlate the different aspects of our lives raises substantial concerns about privacy and data protection.

The amount of additional information collected by QS services can be astonishing. For example, to "benefit fully" from Fitbit's mobile services, users must create an online profile that includes height, weight, gender, and age. If you use Facebook to access your Fitbit account, Fitbit has access to additional information from Facebook, such as your name, profile picture, gender, networks, user ID, list of friends, and other information that is associated with the account, including your birth date and location.

And if you use the social services enabled by Fitbit such as broadcasting about a weight-loss goal, information about those you share with, such as their names and e-mail addresses, is also collected by Fitbit.

In 2011, Fitbit itself ran into trouble with its default privacy setting that set profiles as public, allowing them to be searchable online. What this meant was that any user who fails to unclick this setting automatically posts Google-searchable information on his or her daily health activities, sleep quality, and profile information. The ability to search for users and discover intimate details such as their self-reported sexual activity—including the duration of each sexual event and approximations of calories burned—highlight the problems with Fitbit's information policy. Yet, it's clear that even posting seemingly benign information such as daily steps becomes an issue when Fitbit users are as young as five-year-old Luka.

The desire to collect and combine ever more precise information points to a recurring theme in surveillance studies: function creep. Function creep describes how data collected for one purpose is then applied to new uses. For example, while users pay Fitbit for its health monitoring system, Fitbit uses the information it collects to attract third-party advertisers, thus creating a parallel revenue stream where access to users and their information is what is being sold, not the Fitbit product line. For free services such as Mint, the entire revenue model is predicated upon shaping users' desires—and recommending other services to use and partnership offers—in exchange for lucrative advertising revenue.

What this means is that surplus value is created from the information we trail along behind us, information that is then used to govern our consumption habits and our leisure activities. Instead of being compensated for this surplus value, with QS services we are instead paying for the privilege of being monitored and marketed to. In terms of function creep, it is not inconceivable that future insurance and health services will take into account this information, selectively providing services and pricing structures based on the long-term health habits that people like Luka formed in youth and have been tracking since the age of four.

When data are sent directly to employers, parents, teachers, or personal physicians, function creep contributes to more personalized governance measures. For example, RescueTime (https://www.rescuetime.com) is a web-based QS application that keeps track of a user's computer usage in order to help him or her with time management. It unobtrusively tracks what productivity applications, websites, games, and so forth, are used and how long the user focuses on each one. RescueTime creates beautifully visual analytics that graph how users spend their time and attention. Users may also block distracting websites, as well as have the system "nudge" them to return to their task if they have been idle or distracted for too long.

Buoyed by its success in the media, yet disappointed by low sales, RescueTime started focusing its sales efforts on institutions rather than individuals. RescueTime Empower performs the same tasks as the original application, but sends these data to managers while allowing employees to see their own data and have some control over what is being monitored and when. RescueTime then introduced another product: RescueTime Pulse, which "allows managers to see how employees are spending their time without the employees being able to see or control the monitoring software." In their corporate blog, RescueTime defends their choice:

> A restricted mode offering was literally the most requested feature from our business customers. RescueTime is a software startup, which means that our first mandate is to build something people want. ... Which may or may not necessarily map to what we THINK they should want. ... Revenue and profit are king and we can't expect to focus on free/consumer audiences forever. While we will always serve that individuals, [sic] we thought the site should reflect our focus on business customers. (Wright 2009)

It is here we see a more insidious form of function creep, as the product moves from a personal QS tool to an institutionally mandated time-management service, to a service allowing employers surreptitiously to spy on their workers and algorithmically rank their productivity.

Statistics, measurement practices, and classification schemes more generally are never just a benign assessment of the world, but change our conception of the world and our understanding of ourselves (Huff and Geis 1954; Bowker and Star 1999). Quantification practices tell us what is important to measure, how we should measure it, and indicate how we should change it. On a technological level, it is much easier to measure and reward some behaviors in comparison to others. For QS tools, aspects of life that are easily measurable, such as steps taken, calories burned, the number of Facebook friends, or the number of minutes spent working on a business report, become superficial stand-ins for much more complex concepts, such as overall health and well-being, the strength of social relationships, and the value of an employee.

While a QS user may believe that this validation is unbiased, it is important to emphasize that particular values are deeply inscribed in the system. The system rewards the user for his or her actions, and points are generally rewarded in a transparent manner, thus seeming objective and fair. Yet the valuation of what actions earn points is set by the designers of the system, thus system designers have more control than ever (Whitson 2010). They can reward subtle changes in behavior to inspire and evoke optimum performance. Only certain behaviors are worthy of notice and rewards. So, in systems like Foursquare, brand loyalty, return visits, and consumption are all worthy of rewards. In the case of a gamified call center, answering as many calls as possible within a time limit is valued, whereas in a RescueTime-managed office, the ability to avoid web-browsing and other distractions matters more than the actual quality of one's work.

While the dashboards of these systems are seen as transparent—showing users exactly what other users are doing—the inner workings of the system are opaque. These inner workings—composed largely of algorithms designed by developers—produce and certify knowledge (Gillespie, 2014). This knowledge is premised on specific ideas of what the ideal self should be, and how to operationalize this self into relevant components that can be monitored, measured, and rewarded. However, this knowledge, and the algorithms that create it, is a moving target, obfuscated from view and—unbeknownst to the user—constantly in flux, being redefined and redesigned according to the needs, goals, and desires of the system operators. Every aspect can be monitored and controlled. Questioning what behaviors are rewarded and what behaviors are ignored is not done, because the rules of the system are hidden with the black box of technology. This is especially true for gamification.

The close relationship between gamification and quantification is not surprising. Both are rooted in

the same technological affordances, such as ubiquitous mobile devices, the exponential improvement in data storage and processing, and the improvement in the quality, size, and cost of biometrics sensors (including an accelerometer the size of a piece of confetti in Luka's Fitbit). Both are focused on data collection and feedback. And both emphasize and encourage values such as competition, advancement, efficiency, and accumulation. They foster a techno-liberal American Dream, telling users that if they work hard enough they can achieve victory—in games and in real life. Of course, hard work in this case is not represented by back-breaking labor or even bootstrapping entrepreneurship. Rather, gamification and QS rephrase the American Dream in terms of using technology to master and shape the body, in the process creating a victorious healthy machine, one that has many friends and followers and is a productive worker and savvy consumer.

Gamification practices, in particular, build upon psychological desire for self-mastery and self-improvement, reputation and status building, achievement and reward (Kim 2000, 2009, 2010). These efforts foster a sense of autonomy and self-efficacy as the player selects what quests to complete, how they do so, and on what timeline. As stated by Jason Della Rocca (2010), "With regards to social validation, games provide an unbiased judge: the rules of the system are arbitrated by the unfeeling computer. This is transparent, highly detailed and visible to the community as a whole." With gamification, the computer becomes the assessor of reputation, with the assumption that technology is more objective and infallible than humans.

However, gamification differs from QS in how it leverages discourses of play initially to entice users into this self-monitoring. Surveillance is phrased in terms of enabling free play and promoting engagement; accordingly, there is less reason for users to opt out or resist gamification, even if it is imposed upon them by others. But this free play does not mean that users are unsupervised. They are surveilled more closely than ever before. Everything they do online is logged and entered into a database. Performance metrics become indicators to the inner self. They are "a complete and public package of competency, prowess, and experience ... that makes it more true in the game than in real life that what you see is what you get" (Reeves and Read 2009, 75). What is new is that surveillance is framed as fun (Albrechtslund and Dubbeld 2005). To reiterate the central tenets of governance via gamification: supervision is not about discipline and control, but is geared toward providing meaningful feedback and rewards.

Following Nikolas Rose, governable spaces are not fabricated counter to experience: "they make new kinds of experience possible, produce new modes of perception, invest percepts with affects, with dangers and opportunities, with saliences and attractions" (Rose 1999, 32). Though technical means, gamification creates real and material governable worlds that are then composed, terra-formed, and populated by users.

Gamification and QS movements create governable spaces that then interpellate certain subjects who are interested in autonomy, freedom, and self-regulation. As put by Rose:

> The individual is to adopt a new relation to his or her self in the everyday world, in which the self itself is to be an object of knowledge and autonomy is to be achieved through a continual enterprise of self-improvement through the application of a rational knowledge and a technique. To live as an autonomous individual is to have learned these knowledgeable techniques for understanding and practising upon yourself. Hence the norm

> of autonomy produces an intense and continuous self-scrutiny, self-dissatisfaction and self-evaluation in terms of the vocabularies and explanations of expertise. (Rose 1999, 93)

In other words, we use gamification as a tool for self-mastery and self-improvement. In striving to live our autonomous lives, to discover who we really are, to realize our potentials and shape our lifestyles, we become tied to the project of our own identity and bound in new ways to pedagogies of expertise that are not self-imposed, but rather carefully sculpted by QS and gamification designers. These designers, in turn, are working to shape their own ideal subject/user—one who consumes product in the quest to care for and improve himself or herself, while generating ever more data that can be enrolled in ever more governance. Ultimately, our desire for self-development and autonomy is to a large extent enrolled by others in their goals, if not first created and nurtured in order to then be harnessed.

Player-Centric Utopias

So far, I have painted a rather bleak picture of gamification. The key to avoiding these dystopic effects lies in a de-emphasis on the ubiquitous and increasingly precise surveillance fostered by corporate gamification services, and a reemphasis on the role of play and games in other, not-for-profit, gamification projects.

Current gamification projects encourage the subjectivity of *users* rather than the subjectivity of *players*. The fundamental issue here is the treatment of gamification just like any other software geared toward helping someone efficiently and expediently carry out a task—in this case self-regulation and improvement. Yet, as summarized by Barr, Noble, and Biddle (2007), there are a variety of traits that separate games from other software, such as word processing applications. There is a considerable difference between why games are made (e.g., entertainment rather than meeting an end goal) and why other software applications are created (e.g., efficiency, productivity). At its heart, gamification focuses on making everyday tasks simple for users to accomplish in an efficient and timely manner. It is fundamentally a productivity tool and not a game. For example, we use the Fitbit to focus on an end product (our physical health), and we bring our own personal projects and goals into play, expecting that the Fitbit will make reaching our goals easier, rather than putting constraints and artificial barriers in our way. In short, what we really want is for the Fitbit to be an efficient and easy-to-use tool for health management, not a game. This opens the door for governance projects that masquerade under the rhetoric of being just a fun game (Whitson 2013).

The solution is not to abandon gamification, but rather to focus on making it more *gameful*. When we see the everyday as a game space, and not as some self-improvement project, we create new rules of play. I am arguing that the solution to dystopic governance is not to abandon gamification altogether, but to (re)unite games and gamification, and to introduce rationalities of play. Following a governmentality approach, I use terminology such as *rationalities* similarly to how game scholars use *rhetoric*. *Rhetoric* is a discourse, narrative, and argument for how the game world works. It provides the player with implicit instructions on how he or she should act in the game and points to potential methods and techniques for playing and winning the game (Sutton-Smith 1997; Bogost 2007; Bogost and Salen 2008). In short, gamification *users* should be encouraged to become *players*.

There is something important about interpellating *players* instead of *users*. They bring their modes of play along with them. This play includes joyful explorations and tangential detours/*détournement*. It also includes counterplay, both of which complicate the surveillance projects that constitute corporate gamification endeavors. The very nature of games and play encourages testing, bending, and even breaking the rules (Whitson 2010). Playing a game necessitates learning the rules and testing their boundaries (e.g., how high can I jump? who/what can I shoot? etc.), while winning requires mastery of the rules (Koster 2005), and in some case bending (e.g., exploits) or breaking them (e.g., mods or cheats) in order to win. Players consciously decide to play with the rules and structure of the game (Sotamaa 2009, 82). Mastering, beating, and even subverting rules is an essential part of "play," and cheating and hacking are commonly intertwined with this play (Consalvo 2007; De Paoli and Kerr 2009; Grimes and Feenberg 2009).

However, in gamified systems, this playing with the rules is not encouraged. Many instances of cheating gamified applications are simply self-defeating (e.g., "forgetting" to enter that doughnut you ate on gamified weight-loss apps). In many gamified systems, especially the visions of the gamified workplace and classroom promoted by Reeves and Reid where salary bonuses and grades are dependent on performance, the incentive to cheat becomes overwhelming. In these instances, there are significant consequences for failure that go far beyond loss of face or gamer capital. Players are already predisposed to pushing back and reshaping the rules.

There is an obvious disconnect between the desire on the part of the designers of gamified products to promote the efficient, productive behavior of their users, and the inherent playfulness of gamers themselves, who are less interested in efficiency than they are in exploration, or defining the limits of the systems, and in many cases playing with them. We already play with everyday content and situations, using gamification to change mundane space. Not surprisingly, the term *gaming the system* refers to applying these playful subversions and exploits of games to defeat and remake non-game systems that are seen to be overly constraining. So far, in gamified workplaces, classrooms, and exercise routines, the "game" is present only in marketing rhetoric. The key to maintaining knife-edge heterotopias is to allow gamer subjectivities and playful acts to remake everyday space, without imposing top-down corporate surveillance and governance measures.

Luka and his father have already started to create their own family heterotopia, eschewing the badges and goals handed down by the Fitbit system, and by doing so, refusing to let Fitbit define their ideal selves. Instead, they make up their own games where the rules are constantly fluid, mutating, and open to negotiation (just like Luka's bedtime). The Fitbit, in this case, operates as a gameboard populated with Luka's own data. The key to maintaining this heterotopia lies in finding ways to decouple the Fitbit from systems of corporate surveillance and governance. Allowing users the choice to store and analyze the data on their own computers, rather than sending it to corporate servers to be dividuated and amalgamated is one option. At the very least, making more transparent the decisions of what data are collected and what is done with that data may help users discover the (hidden) values embedded in the system. In this way, Luka and his dad can decide for themselves whether the Fitbit is something that can be played with and reshaped, or not.

Notes

1. For Deleuze, *dividuation* refers to the internal division of individuals into malleable bits of coded information that are more amenable to being measured, recombined, and aggregated into populations of other dividuals.

2. Foucault wasn't blind to this, just concerned by earlier formulations of control. Deleuze is writing in the spirit of continuing Foucault's analysis from pastoral governance based on the shepherd governing the flock, and a priest governing his congregation, to sovereign power based on the spectacle of the ruler's power and might, to discipline operating through surveillance and normalization. Deleuze extends this analysis further, to control operating through consumption and desire.

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14 PRIVACY AND DATA COLLECTION IN THE GAMEFUL WORLD

Lori Andrews

Gamification elements are used by individuals for self-improvement, by groups to spur policy change, and by third-party institutions such as employers to make judgments about individuals. In addition to its potential to encourage positive change, the use of gamification to modify behavior can pose psychological risks, financial risks, and social risks. Most of these risks could be averted if the well-established principle of *informed consent* were applied to the protection of privacy and the personal control over data collected in the gameful world.

Privacy protections have always lagged behind the development of technologies. More than 125 years ago, when the portable camera was developed, the initial reaction was that privacy was now dead. Op-eds in newspapers in the late 1800s referred to the camera as the "Kodak fiend" and warned that it could catch you in an "uncouth" position in line at the post office (Hawaiian Gazette 1890). Because people no longer had to go to a studio to get a photo taken and third parties could now take photos of them, people no longer had control over their images. Forensic technologies and medical technologies, too, raised challenges for privacy. Wiretapping allowed police to listen to phone conversations. Predictive genetic testing allowed employers and insurers to discriminate against healthy people on the grounds that those people had a higher-than-average risk of getting ill later in life.

In each of those cases, an argument could be made that the person was consenting to the loss of privacy by going outside, by speaking on the phone, or by applying for a job or insurance. And, at first, the law took that position. But ultimately, the importance of privacy was recognized. State laws were adopted to prohibit intrusion on a person's seclusion in order to take his or her photo; other laws prohibited the commercial use of a person's photograph without consent. Courts instituted the requirement of a warrant before phones could be tapped. And, seventeen years after the introduction of predictive genetic information, the U.S. Congress passed a law that prohibited the use of this information by insurers and employers.

Today, social networks, search engines, apps, and gamification present similar challenges to privacy, and the law has not yet caught up. In fact, many people who participate in activities that use gamification elements do not even realize that their privacy can be easily compromised and their data can be used against them by third parties. To remedy that, I advocate an informed consent approach to data collection and dissemination in the gameful world. In medical settings, informed consent consists of the provision of adequate notice about the endeavor before consent

is sought, assurance of voluntariness in the consent process, and control over choices regarding what procedures are undertaken and who receives private information. Similar provisions should be put into place in the gameful world.

Why is it that people sometimes fail to recognize the risks of gamification? In part, people see game playing as an alternative to the commercialized world and have an innate trust that gamification is completely under their own control, similar to traditional game playing. They may also view themselves as smart and competent, the type who aren't likely to be taken advantage of. They may not realize how the increasingly intimate information they are sharing through gamification may be accessed by third parties or used against them. Or they may think that the risks of information disclosure are outweighed by the benefits of gamification (or by the fact that the gamified app is free).

All of the above assessments were made by people using social networks. They assumed that privacy protections were in place and that they controlled their own data. They assumed that Facebook was just a way to communicate with their friends. And, initially, it was. But then Facebook realized it could make billions of dollars a year, 90 percent of its income, through targeted ads on its platform. Facebook now creates targeted ads based on users' demographic information, interests, likes, friends, and websites frequented.

That business model can offer benefits to users—such as a coupon for money off at your favorite store. But it can also create risks. If you "like" a violent video game on Facebook, you can be denied employment. In fact, 75 percent of large employers in the United States assess a person's online profile before hiring them. One-third refuse to hire someone with a drink in their hand on a Facebook page. If you do a Google search for a guitar or for a divorce lawyer, then go to a credit card website, you will be offered an inferior credit card (if you are offered one at all), because of the presumption that guitar players and divorcing individuals are less likely than other people to pay off their credit cards (Andrews 2012).

There is reason to believe that the same problems with data mining and discrimination are occurring or will occur in the gameful world. To protect against that (and even to determine the extent to which these risks are occurring), we should take an informed consent approach to gamification.

Adequate Notice

People need to know what data are being collected about them through gamification, who has access to it, and what is done with that information. Yet it is extremely difficult to get that information about apps, including gamified apps. There are more than 700,000 apps for children. When the U.S. Federal Trade Commission (FTC) analyzed a random sample of four hundred children's apps retrieved from iTunes and Google Play in a December 2012 report, the agency found that only 20 percent of the apps disclosed any information about the apps' privacy practices, including whether they shared information with third parties. According to the FTC, "most apps failed to provide any information about the data collected through the app, let alone the type of data collected, the purpose of the collection, and who

would obtain access to the data" (Federal Trade Commission 2012, 4).

Nearly 60 percent of the apps reviewed transmitted device ID to the developer or a third party such as an advertising network (Federal Trade Commission 2012). Device IDs are important because they can be used by apps, developers, and other third parties to identify, track, and analyze devices and their users. Because of the permanency of device IDs, developers and other companies can use them to compile "profiles" about individuals. It becomes particularly concerning when ad networks and analytics providers collect device IDs and other user information through a vast network of mobile apps. According to the February 2013 FTC staff report, "companies can use a mobile device to collect data over time and 'reveal the habits and patterns that mark the distinction between a day in the life and a way of life'" (Federal Trade Commission 2013, 3; quoting *U.S. v. Maynard*, 615 F. 3d 544, 562 [D.C. Cir. 2010]).

The same lack of disclosure occurs in the gamified world. Researchers at the Institute of Science, Law, and Technology, which I direct, are studying gamified apps to improve health. We identified twenty-four diabetes-related games in the database of Health Games Research, a national program of the Robert Wood Johnson Foundation (Health Games Research Database 2013). Privacy policies were disclosed for less than half the games. Of those with privacy policies, some would disclose the users' data to law enforcement or affiliated third parties (which can be marketing entities). Some of the gamified apps (e.g., HealthSeeker and Knock 'Em Downs) were specifically developed by pharmaceutical companies, which could then use the information collected for research and advertising.

The principle of informed consent as applied in other legal settings means that people need to be told not only what will be done to them and to their information, but also what risks those procedures raise. For adequate notice to occur in the gameful world, people should not only be told of the features of the gamified app, but also that the features might be used against them. For example, many gamified apps collect location information. Some fitness apps encourage you to exercise by revealing your location so that your friends could cheer you on or join you. But, as U.S. Supreme Court Justice Sonya Sotomayor pointed out in a recent case, location information "generates a precise, comprehensive record of a person's public movements that reflects a wealth of detail about her familial, political, professional, religious, and sexual associations"—just the sort of information that could be used against you (*U.S. v. Jones* 2012; Sotomayor, J., concurring).

Location information can be used to discriminate against you. If you check in on Foursquare at a diabetes clinic or if you are one of the 19 percent of individuals who download a phone app to track or manage your health (including gamified apps), that information could be used to deny you life insurance (Fox and Duggan 2012). The publication of location information could also lead to robberies and attacks. According to a 2011 United Kingdom survey of convicted burglars, about 78 percent used information from Facebook, Twitter and Foursquare to plan their crimes (Yin 2011). Women's use of location information in fitness apps—such as those in which you reveal to others your jogging route—is particularly problematic. In 2012, Jenn Gibbons—founder of the nonprofit Recovery on Water—was a few days into her publicized charity row trip around Lake Michigan when she was sexually assaulted by

a man who had tracked her location through her blog and satellite tracking (Hill 2012). She was attacked in a remote area, but a woman might be attacked when alone in her home. A University of Pittsburgh study found that female Foursquare users are more likely than men to expose their check-ins at residential venues (Jin, Long, and Joshi 2012). Sixteen percent of women check in once they arrive home, but this can provide a signal that they are home alone.

For adequate notice, gamified apps should provide an explanation about their features and the implications of those features. Right now, little such information is available, and what is available is buried in privacy statements that often have a line that gives the entity that has developed or implemented a gamified app the right to change its privacy policies at any time without any notice.

For adequate notice, a gamified app should also disclose the identity of the developers and funders of the gamified app as well as any third parties that will have access to the information collected through the app (and information about what any of those entities will do with the information). Gamified apps should also disclose the ways in which data might leak from the apps. Even if, quite extraordinarily, a gamified app were to say that it never disclosed your information to third parties, never used it for advertising, and was never going to change that policy, the personal information you enter into the app could be disclosed due to tracking mechanisms that were put on your computer or phone by other apps and websites.

Most people are unaware that they are being tracked on the web. A Consumer Reports poll found that "61% of Americans are confident that what they do online is private and not shared without their permission" and that "57% incorrectly believe that companies must identify themselves and indicate why they are collecting data and whether they intend to share it with other organizations" (Consumer Reports 2008).

Tracking technologies such as cookies (Kristol 2001), web beacons (Gomez, Pinnick, and Soltani 2009), data scraping (Borate 2009), and analyses of search queries (Tene 2008) allow data aggregators to create a picture of users by noting what they look at, look up, and buy across the Internet. The seemingly benign website Dictionary.com actually installed 234 tracking tools on a user's computer without permission, only 11 from Dictionary.com itself and 223 from companies that track Internet users (Angwin 2010). People are generally not aware of how aggregated data are beginning to replace individualized, personalized assessments in determining what benefits they are offered or opportunities they are denied. When an individual visits a website, advertisers bid against each other in real time for the right to place an ad tailored to that person's perceived interests on that person's screen. The whole process takes milliseconds. For example, the data aggregator Turn crunches 2,000 bits of information about you in 25 milliseconds, influencing what offers or rejections you will receive when you go to any particular website (Turn 2011a, 2011b; both as cited in Chester 2012).

In their article "Can Advertisers Learn That 'No Means No'?," Hoofnagle et al. (2011) describe the extensive tracking mechanisms that third parties use to monitor where people go on the web and what data they enter.

> First, users cannot fairly be said to have notice of these activities. The entire point of new tracking methods seems to be motivated by users'

> ignorance of them. And the privacy policies we read didn't disclose Flash respawning or cache ETag use. The lack of notice leads to a second problem: because these vectors are resistant to blocking, they rob consumers of choice. This undermines the advertising industry's representations about respecting choice, and leaves consumers in a technical arms race with advertisers. (Hoofnagle et al. 2011, 1400–1401)

Once tracking mechanisms are introduced into your computer or browser, they can be used to collect any information you enter into a gamified app and to share that information with data aggregators.

Presenting Notice Information in an Understandable Way

In medicine, the tenet of informed consent requires that information be presented in an understandable way. The same requirement should apply in the gameful world. Right now, that is not the case. The information is hard to find, often couched in legalese, and is too voluminous to process. It has been estimated that if we wanted to read all the privacy statements of the apps, social networks, and digital devices that we commonly use, we would have to devote seventy-six days to that process (Madrigal 2012). So people often just click without reading and do not realize what terms they are supposedly "consenting" to.

The requirement of adequate notice cannot be implemented if it is impossible to read, much less understand, all the privacy policies that are thrown at us each day. There are several ways to remedy this situation. One would be to prohibit the collection of gamification data on people by third parties such as advertisers, insurers, employers, and the government. (This would be similar to the current rule with respect to predictive genetic information, which cannot be collected or used by certain third parties.) Another would be to develop better and more consistent privacy warnings, such as familiar icons that, at a glance, would indicate how a person's information is collected and used. Maybe we even need a gamified warning app that shows us how information will be used if we choose to go forward with the use of the gamified app itself.

Voluntary Consent

The argument has been made that by using a gamified app, a person is knowingly consenting to giving up his or her private data. This is the same argument that has been made about social networks—that you give up your data in exchange for the free use of Facebook. But, if the only way to use the service is by giving up privacy, that can hardly be seen as a voluntary choice.

Privacy is an important value. Control over private information is essential to respect, friendship, love, trust, and personal liberty, according to privacy scholar Charles Fried (1968). We achieve intimacy with other people by parceling out information about ourselves bit by bit. Each new revelation demonstrates additional trust. It's healthy to be able to show different aspects of ourselves in different settings. We

need to explore or grow without our former digital lives coming back to haunt us. Privacy is also a way to allow "deviations from prevailing social norms, given that no one set of social norms is universally and permanently satisfactory—and indeed, given that social progress requires social experimentation" (Abelson, Ledeen, and Lewis 2008, 64).

In the United States, privacy is protected by the U.S. Constitution. Should it really be possible for a company to require that a person give up a constitutional right in order to use a service? What if the private company asked a person to give up a different constitutional right—such as the right to vote or the right to have children? Would we stand by and say that was okay? A young man might be perfectly willing to give up the right to vote in exchange for using a cutting-edge gamified app. But society should not allow that sort of exchange.

And what about children? Children cannot enter binding legal contracts in the United States until they are at least 18 years old, a legal adult. Thus it is inappropriate to think of them as contractually consenting to give up their privacy in exchange for a gamified experience.

Moreover, there are circumstances in which people are being forced to participate in gamification. According to the Entertainment Software Association, 70 percent of major employers use games in the workplace (Entertainment Software Association 2012). Employers are using gamification techniques to recruit, train, and monitor employees without adequate data about whether these techniques provide a credible assessment of an employee's capabilities and value. Employees are not given a meaningful choice about consenting. Obtaining or keeping the job is contingent on playing the game.

As Sebastian Deterding has pointed out, "[W]hen I look at most gamified applications today, what they do is to employ game elements to tie us even more tightly into our worldly toils and schemes. They are glorified report cards that turn games into work rather than life into play and users into pawns rather than players" (Rowan 2012, 6). Deterding says that a "public performance comparison at work," such as that introduced by management and related to monetary incentives, is "neither voluntary nor free of consequence" (Deterding 2011, 3).

Participation without meaningful consent may become standard for other gamification applications as well. The U.S. Food and Drug Administration has approved a health app, WellDoc's BlueStar, with motivational components to be "prescribed" to patients by doctors (Dolan 2013; Ravindranth 2013). The game, which deals with diabetes, may become a routine part of medical care, rather than a true choice.

Control of the Collection and Use of Data by Third Parties

The tracking, analysis, and use of the information collected through gamification can have psychological, financial, and societal impacts on the users. As aggregated data from online apps are fed into algorithms that make predictions about a person's potential behavior, desires, health, and capabilities, that person may be offered benefits (such as an ad for money off at a particular store or a notification of where that person's favorite band is playing). But that person could also be denied opportunities (such

as being denied a credit card or insurance) based on the collected data. Aggregated data about a particular individual are also used to determine what ads appear when she is online and what options are available. For example, if she goes to an insurance website, aggregated data are also used to make assessments about what type of policy will be shown on the home page of the website or whether she will be denied insurance altogether (Angwin 2010).

Medical information and information about a person's health and fitness, such as the information collected by Nike+ and Keas, is the "pot of gold" for insurance companies that are looking to use predictive modeling and data aggregation in their underwriting processes. Life insurance underwriting has traditionally been based on urine and blood samples that provide indications about a person's health. But now some consultants are suggesting that those tests (which are expensive and time-consuming for companies to administer) be replaced by information from people's social network pages and games (Batty et al. 2010; Fitzgerald 2010; Kroll and Testa 2010).

A survey found that 66 percent of adult Americans opposed being targeted by behavioral advertising and are troubled by the technologies used to enable it (Turow et al. 2009). Also, 68 percent of Americans opposed being "followed" on the web, and 70 percent of Americans supported the idea of requiring companies that collect or use someone's information without his or her consent to pay hefty fines. Most people—92 percent—believe that websites and advertising companies should be required to delete all information stored about an individual if requested to do so.

In order for people to control the dissemination and use of their information from gamified apps, they should be offered a "Do Not Track" mechanism that allows them to opt out of data tracking. This would be akin to a model of informed consent in medicine, where people can withdraw their consent to further participation.

Legal Implications of the Data Trail

The foundation for addressing the legal aspects of gamification has been laid by work on technology and law generally (Rosen and Wittes 2011), on digital privacy (Abelson, Ledeen, and Lewis 2008; Browning 2010; Andrews 2013), and on legal issues raised by games (Zarsky 2006; Arias 2008; Penney 2008; Akins 2010; Fairfield 2012). With the use of game elements in non-game contexts, a range of legal issues will need to be addressed, including property interests in virtual goods (Arias 2008; Akins 2010; Herger 2012a; McHale 2012), restrictions on virtual currencies and goods (Gatto 2011; McHale 2012), intellectual property issues (Gatto 2011; Fairfield 2012), deceptive advertising practices (McHale 2012), and fair labor practice laws (Gatto 2011; Cherry 2012; Herger 2012b).

Central to all these legal issues, though, is the question of privacy and the control of the data generated through gamification. When people use gamification principles in non-game situations, they create a "quantified self" consisting of intimate data about their moods, beliefs, and activities. What is done with the data collected in the gameful world? Do people have any rights to control access to information

about themselves in the digital realm? And, if that information is used to stigmatize or discriminate against a person, is there any legal recourse? Is privacy dead, as Scott McNealy of Sun Microsystems has suggested? Or, because technological intrusions can cause "mental pain and distress, far greater than could be inflicted by mere bodily injury," were Samuel Warren and Louis Brandeis right a century ago when they said privacy is actually more important in times of "modern enterprise and invention" (Warren and Brandeis 1890)?

I am suggesting that privacy is not dead and that important social values are served by protecting privacy through a robust mechanism of informed consent. In the United States, this would be consistent with the constitutional protections of privacy, freedom of expression, and freedom of association (Andrews 2013) and efforts by the Federal Trade Commission to promulgate a "Do Not Track" regulation. In Europe, this would be consistent with value placed on privacy through data protection laws that require that entities not collect more data than they need for a particular transaction, that they ensure that the data are accurate and complete, and that identifiable databases are kept "no longer than is necessary for the purposes for which the data were collected" (European Union Directive 1995). Elsewhere in the world, the informed consent approach would help protect against retaliation by governments and discrimination by social institutions.

Gamification is a seductive means of self-improvement and monitoring by institutions. But until greater disclosure and control are built into the system, gamification may also present privacy risks that are unwarranted and unacceptable.

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15 GAMIFICATION AND MORALITY

Evan Selinger, Jathan Sadowski, and Thomas Seager

For every job that must be done, there is an element of fun. Find the fun and snap, the job's a game.

—Mary Poppins

Examining the Normative Dimensions of Funware Gamification

The arrival of gamification as a contemporary cultural movement and incessantly circulating meme that is synonymous with innovation and engagement should give us pause for thought.[1] Not too long ago, the term had limited appeal. It was "a self-description used by vendors and proponents" (Deterding 2011). Now, amid endless headlines and extraordinary prognostics, gamification is hailed as a "$100 million market that should grow to $2.8 billion by 2016" (Stern 2012). Eric Schmidt, executive chairman of Google, goes so far as to say that, "Everything in the future online is going to look like a multiplayer game" (Orca 2010). In a much-quoted remark, Jane McGonigal (2011), a gamification advocate with a reputation for proselytizing its merits, expresses hope that within twenty-five years, a game designer will win a Nobel Prize.

On the most basic level, gamification entails transposing gaming mechanics (e.g., point systems, badges, levels, leaderboards, etc.) to environments that lack these features or contain minimal traces of them. Beyond this, opinions diverge and debate exists over gamification's purpose, power, and potential. In this respect, gamification is a contested concept with supporters and detractors at odds over such issues as the definitions of and dividing line between "play vs. work, intrinsic vs. extrinsic motivation, authenticity vs. contrivance" (Zichermann 2011).

Advocates often describe gamification as a user-empowering tool that modifies behavior in desirable ways: "Gamification's main purpose is to help people get from point A to point B in their lives—whether that's viewed through the lens of personal growth, societal improvement or marketing engagement" (Zichermann 2011).[2] This end gets furthered by transforming unpleasant tasks, which we are adverse to, into pleasant ones that we stick with and possibly even gravitate toward.

The behavior-modifying conception of gamification can be formalized as an inferential argument.

Premise: People play games because games are fun.
Inference: Bringing the fun of games to new environments can make them fun, too.
Conclusion: By making unpleasant (frustrating, anxiety provoking, or mind-numbing) tasks fun through gamification, effective behavioral change will follow.

Many variations of this argument exist, and the following attestations convey its essence. Gabe Zichermann, coauthor of *Gamification by Design* (Zichermann and Cunningham 2011) and *Game Based Marketing* (Zichermann and Linder 2010), states: "If you can make something more fun, and include notions of play, you can get people to do things they otherwise might not want to do" (Carr 2011). Likewise, in a comment on Bogost's (2011) "Gamification Is Bullshit" post, Nathan Lands, cofounder and CEO of Gamify, insists: "What gamification should be seen as is an opportunity to open our minds to change what reality can possibly be by reimagining experiences with fun in mind." Such reimagining of fun has become so pervasive that over at Rochester Institute of Technology—where one of the authors teaches—a gamification initiative funded by Microsoft Research Connections, "Just Press Play," has been designed to enable students to experience the mythical hero's journey by turning a range of activities into badge awarding, "fun and engaging" quests.

This interpretation of how gamification effects change suggests the technique belongs to the lineage of what psychologist B. F. Skinner (1904–1990) called "operant conditioning," especially when operationalized through effectively scheduled rewards (e.g., points) and participation from an encouraging social support system (e.g., friends, family, and co-workers who, in the gamified culture, essentially participate as "teammates"). Indeed, Zichermann (2011) identifies Weight Watchers as a paradigmatic example of gamification, and "the program's close adherence to Skinner's basic principles has consistently garnered some of the best long-term weight-loss results of any mass-market program" (Freedman 2012).

Linking gamification, an essential part of the "high-tech behavioral revolution," to "the work of a mid-century" scientist might seem surprising, not least because Skinner's legacy is tainted by prevalent misunderstandings (Freedman 2012). Partially due to the misperception that Skinner followed Pavlov in endorsing the motivational use of punishment—to replace "pleasant stimulation" in subjects "with an unpleasant one"—his vision of behaviorism became "maligned as morally bankrupt, even fascist" (Freedman 2012). The fact, however, is that "Skinner sought to shape only consciously chosen, directly observable behavior, and only with rewards" (Freedman 2012). The pervasive trend of connecting Skinner to techniques used in Kubrick's film *A Clockwork Orange* is entirely illegitimate (Freedman 2012).

We are emphasizing the undue taint associated with Skinner because as the chapter develops, we will identify some of the real limits that constrain the behaviorist orientation toward gamification. Those limits have ethical implications, and we do not want our view to be assimilated with spurious critiques.

To make our case, it will be helpful to use shorthand to designate the specific version of gamification that we are targeting. Zichermann's term "funware" will be our point of reference.

> Funware—is taking the social web and mobile apps world by storm. Almost every aspiring startup—and many established brands, including Chase, NBC and the US Army—are turning to Funware to deliver results that traditional/social marketing simply cannot deliver . . . [G]ame

> mechanics can make any service or community more fun; and when given a choice between two similar activities, consumers will always choose the one that's more enjoyable. (Zichermann 2010)[3]

To examine key normative dimensions of the funware approach to gamification critically, this chapter will address the following three questions:

1. Are we ever ethically obligated to lend a hand by participating in a funware-gamified endeavor?
2. Can funware gamification adversely impact character?
3. Are there any moral goals that funware gamification cannot further?

We selected these three questions because they address some of the most central ethical issues concerning funware endeavors. The first question orients us to the possibility that if funware can positively impact behavior, it might be a new source of obligation—at least to the extent that it has a causal role to play in situations where we are at least partially responsible for what others do. To use a biblical metaphor, if we're responsible for being our brother or sister's keeper at this moment of the digital age, we just might have funware obligations—obligations that, as we will discuss, cannot be fully understood without coming to terms with a striking metaphysical conception of willpower. The second question enables us to shift our attention away from the external impact of funware (e.g., the goals it can help us meet) so that the internal, characterological consequences of its use can be assessed. Funware projects might pass utilitarian muster but, thanks to the intricate willpower issues, still be morally composed from a virtue-theoretic perspective that places ethical value on distinctive habits, dispositions, and proclivities. Finally, given our concerns about character, the third question focuses on whether aspects of moral development can be furthered by games that aren't designed to be fun; that is, anti-behaviorist games that are structured to make moral decision-making unpleasant.

While we will address the first two questions from a predominantly theoretical perspective that attends to intuition-priming thought experiments, our discussion of the third one will draw directly from our experience creating and administering unique sustainability ethics games. These games give students an opportunity to undergo moral identity crises. These crises are not fun, but they deepen a player's understanding of how easy it is to act in ways that betray espoused moral ideals. This awareness is especially important for coming to grips with matters of justice like privilege, luck, and idealized self-understanding.

Motivated by Fun

While the leading premise of funware gamification—people play games because games are fun—is often presented as if it were self-evident, it is not. Just as there are competing definitions of "play" and "playfulness," "fun" is subject to conflicting interpretations. Perhaps lack of consensus should be expected. Like other art forms, there are different kinds of games, and audiences consume them for different reasons.

Prompted by a range of pleasant experiences, players find different aspects of games and different kinds of games fun. As an illustrative example, consider how fun is conceptually sliced in the taxonomy

"8 Kinds of Fun" proposed by Marc LeBlanc, chief technology officer at Electrified Games (Nutt 2008):

Sensation "To function as an art object, to look, sound or feel beautiful."

Fantasy "A game to be about something, a vehicle for make-believe."

Narrative "The ability for a game to function as a story, to unfold over time . . . think about a movie about a sporting event . . . there's story content in the sporting event itself. Those things form a narrative."

Challenge "The ability of a game to provide you obstacles to overcome, problems to solve, plans to form."

Fellowship "All of the social aspects of games; the ability for a game to function as a social framework. All the ways in which games facilitate human interaction."

Discovery "An opportunity for a game to function as uncharted territory—you could be a tourist walking around Disneyland, or you could be a tourist in the tech tree in Civilization and exploring it. To see a new space and become a master over it—that's what I call discovery."

Expression "Whether it's how you dress your avatar or it's how you play. Using the game as a vehicle for expressing yourself."

Submission "The pleasure of a game as a mindless pastime, like the pleasure of knitting or organizing CDs on a shelf. Some people play solitaire because it's an interesting problem; some play it for the pleasure of moving the cards around. The second is submission."

Although different conceptions of fun exist, some might be wrong, applicable only in delimited contexts, or viable only when supported by facilitating conditions. Indeed, a central criticism of the funware approach to gamification is that many of its designers export the wrong kind of gaming mechanics. Failing to appreciate what makes games fun, they ostensibly cannot deliver the goods. Consequently, even if their guiding inference is right—namely, that bringing the fun of games to new environments can make them fun, too—their execution is flawed.

In this spirit, Sebastian Deterding (2011), coeditor of this volume, argues that it is a mistake to believe people find games fun because they are seduced by extrinsic motivation and drawn toward acquiring rewards ("status, access, power, stuff") for achievements. To the contrary, he depicts games as a mode for exploring intrinsic motivation, emphasizing their capacity to "provide opportunities to overcome a challenge—a puzzle to solve, a chasm to jump, a monster to beat—and the joy of succeeding in it" (Deterding 2011). From this perspective—the psychological orientation of self-determination theory—games exert a powerful hold on us because of our "need to experience competence," a need that manifests in a desire to "control and affect our environment, and to become better at it" (Deterding 2011).

If Deterding is right, we should not place too much stock in the fact that short-term enjoyment can be had from gamified activities that don't enhance skill and learning. Over time, they might become addictive or distracting, but players likely will not continue to view them as genuinely fun. The feedback mechanisms (e.g., "points, badges, and leaderboards") will not meaningfully signal how well a player has done "in overcoming challenges on the way to her goals" (Deterding 2011).

Of course, there is nothing new in debating what motivators work and whether some are inappropriate in certain contexts. For example, in *Strings*

Attached: Untangling the Ethics of Incentives, philosopher Ruth Grant reminds us that in the context of education, there has been a contentious history with high-stakes discussions.

> Before it arose around the paying cash for grades, it was disputed about the grades themselves. It is worth noting that paying for grades is offering an incentive for an incentive. Grades are an incentive to learn—an extrinsic reward for academic achievement—and "grade grubbing" students have always been distinguished from those who learn primarily for learning's sake. Critics of grades have always been proponents of intrinsic motivation (Grant 2011, 113).[4]

In the next section, "Obligated to Help Someone Have Fun," we will return to the issue of motivation and examine the moral implications of its effects upon character. Doing so now, however, would be premature. The critical—and, ultimately, negative—focus might prevent us from appreciating why, in principle, we might be ethically obligated to participate in a funware endeavor.

To determine whether conditions could arise that would create this obligation, we need to temporarily bracket the debates about what actually provides short- and long-term motivation and whether self-interested parties (e.g., gamification consultants) are tainting discussions about this matter. To isolate the ethically relevant variables, we will pose a thought experiment that frees our ethical considerations from empirical constraints.

Let us imagine that an app gets developed that validates the argument for funware gamification. The app can successfully motivate friends and family to accomplish important goals they have been struggling with. If they ask us to provide support by participating, are we ethically obligated to do so? Or, would participation be supererogatory (i.e., generous acts that go beyond the call of duty)?

Obligated to Help Someone Have Fun

The hypothetical app is called Digital Eudaimonia (a Greek word that means happiness and human flourishing). It allows users to select a goal related to well-being—such as better health, greater financial security, or improved happiness—and is programmed with a customizable set of gamified options. In fact, users can select the features that they personally find motivating, avoid the features they find distracting and self-defeating, and change the features if, over time, features stop being effective.

Further imagine that someone in your inner circle, perhaps a sibling or friend, asks for you to participate in a program run on Digital Eudaimonia. The person, whom we will call Needy Ned, justifies this favor by saying that he lacks the willpower necessary to meet a basic life goal. Whether the lack of willpower comes from weakness of will or hyperbolic discounting of future possibilities is irrelevant. The point is that Needy Ned believes the key for him staying on track (i.e., not backsliding or favoring immediate gratification over long-term satisfaction) is using Digital Eudaimonia to make the necessary behaviors—which he otherwise finds frustrating—fun. All you have to do is agree to check your phone a few times per day to monitor his progress and post encouraging remarks when his accomplishments are on top of the leaderboard. Should you agree?

If Needy Ned posed that question today, a number of concerns would justify saying no. None of them, however, apply to the thought experiment.

1. Today, participating might have hidden financial costs. Even if an app like Digital Eudaimonia is free to download, that version might contain only basic functionality. Various options could be expensive to purchase, and the product might be designed to prime you to buy other products. Because we are considering an idealized thought experiment, those concerns are inapplicable. We can stipulate that a fully functional version of Digital Eudaimonia is available for free; it was not designed to be a gateway product.
2. Today, participating might have privacy costs. Downloading an app like Digital Eudaimonia might allow for various kinds of personal data that one does not wish to broadcast to be tracked and shared. Because we are considering an idealized thought experiment, this concern is inapplicable. We can stipulate that Digital Eudaimonia gives users complete control over their information and is designed to be absolutely secure and hacker proof.
3. Today, participating might have opportunity costs. The time you spend monitoring and commenting on Needy Ned's activity might detract from time you could be spending on other endeavors you deem important. Because we are considering an idealized thought experiment, this concern is inapplicable. We can stipulate that Digital Eudaimonia downloads almost immediately; it also allows users to consume and broadcast information nearly instantaneously.
4. Today, participating might implicate you in supporting questionable supply-chain operations. Because we are considering an idealized thought experiment, this concern is inapplicable. We can stipulate that Digital Eudaimonia was produced according to highly ethical labor practices; it is maintained and distributed through them, too.
5. Contrary to what they say, today people like Needy Ned might not really want to use apps like Digital Eudaimonia. Instead, their desire to use the app might be the result of peer pressure or some other external influence. Because we are considering an idealized thought experiment, this concern is inapplicable. We can stipulate that Needy Ned freely chooses to use. Moreover, he can opt out, with ease and without fear of penalties, if he ever changes his mind.
6. Today, some gamified products do not actually yield the touted behavioral results. Because we are considering an idealized thought experiment, this concern is inapplicable. We can stipulate that Digital Eudaimonia is incredibly effective, and your participation—in just the way Needy Ned stipulates—is a crucial component.

Under these idealized conditions, participating in Digital Eudaimonia seems like an obvious way to fulfill part of one's obligation to be beneficent. Different moral theories offer different reasons why are we obliged to be beneficent (Beauchamp 2008). And, different accounts have different conceptions of how far our obligations run (Beauchamp 2008). Arguing why a particular one is right or weighing in on the limits of how much a beneficent person needs to do would take us beyond the scope of the current inquiry. To make the point of principle at issue, it is sufficient to simply postulate that good people are committed to the well-being of people they care about, and that the thought experiment illustrates a

scenario where that care can be expressed in a way that yields significant impact without anything of substance—ethically or pragmatically—being sacrificed. Unless someone already is maximally committed to beneficent endeavors, it is hard to imagine an ethically compelling reason for saying no to Needy Ned. Indeed, it seems that only a selfish or insensitive person would refuse.

Whether that judgment also applies to people under real conditions that prevail today depends upon two things: (1) the extent to which the concerns listed above actually do apply, and (2) concerns about character related to the type of "extended willpower" called "digital willpower." The first issue is an empirical matter, and we hope that the cautions identified here prove useful when considering them. Because our goal is to identify fundamental ethical concerns, we will now address the second set of issues, proceeding by explaining what extended willpower and digital willpower are. Then, in a later section, "The Fun Stops Here: Gamification and Character," we will turn to concerns about character.

Extended Willpower and Digital Willpower

The full significance of the Needy Ned thought experiment comes into relief by elaborating concern no. 3 specified earlier and considering how inexpensive and expedient the gamified technology makes the requested intervention. As a point of comparison, imagine if Needy Ned lived in a pre-Internet, pre–quantified-self era and selected the goal of losing weight by consuming less calories and exercising more. He would find tracking the relevant data—creating exercise and food journals—to be a highly time-consuming (i.e., writing everything down and sharing it with other members of his circle) and imprecise (determining how many calories are in a given meal or are burned off by an activity) endeavor. The encouragement he asked of you would be time intensive, too. And, the further we go back historically, the greater the time delays would be that punctuate your positive reinforcement messages.

Because Digital Eudaimonia makes all of the data gathering and sharing effortless, it nearly eliminates transaction costs while embedding your virtual presence—expressed in motivational comments—as a crucial component of Needy Ned's decision-making environment. Your moral support would bolster his commitment to healthy living. And, withdrawing your support would undermine that commitment. Put in metaphysical terms, you would become part of what Joseph Heath and Joel Anderson call the "scaffolding" of Needy Ned's "extended will" (Heath and Anderson 2010). Because your presence is expressed through a digital medium, we can further classify the externalization as a component of Needy Ned's digital willpower.

Heath and Anderson develop their account of the extended will by drawing upon research in "extended cognition" and the "extended mind." These studies offer *systems level* analyses of how humans use technologies to solve problems by "offloading" aspects of practical reasoning. To get a sense of what systems thinking entails, consider the following paradigmatic illustrations of how humans and technologies can work together as "cognitive systems" (Hutchins 1995; Clark 2003; Sterelny 2004; Zhang and Patel 2008; Giere 2010).

- When we form cognitive systems with slide rules, complex mathematical problems are transformed into simple tasks of perceptual recognition.

- When we form cognitive systems with maps, geographic information is accessed in a manner that is well-suited for complex logistical planning.
- When expert bartenders form cognitive systems with distinctly shaped glasses, it becomes possible to mix and dispense volumes of drinks carefully and quickly.
- When artists form cognitive systems with sketchpads, it becomes possible to organize ideas, explore visual ambiguities, and determine how to express themselves in ways that go beyond what their inner imagination typically permits.
- When Scrabble players form cognitive systems with their rearranged tiles, it becomes possible efficiently to determine what possibilities for word formation their letters offer.
- When babies form cognitive systems with existing languages, it becomes possible to classify things and express ideas without needing to first generate new vocabularies, grammars, and techniques for imparting meaning to others. (Even patterns of repetition, such as rhyme and rhythm, serve a cognitive function. They make it easier to remember complicated sound patterns.)
- When people form cognitive systems with mathematical conventions (e.g., symbolic notation and basic procedural rules), it becomes possible to reduce complex mathematical problems, such as multiplication, to more basic acts of perception and computation.
- When we form cognitive systems with navigational tools, then ships, cars, planes, and bodies can be effectively navigated to desired destinations.

In sum, when we solve problems by becoming part of a cognitive system, things become easier for us: (1) the amount of conscious awareness and attention required to solve problems becomes minimized; (2) transient information becomes stabilized; (3) the demands placed on biological resources becomes minimized; (4) difficult problems become resequenced into ones that are easier to solve; (5) knowledge is more easily transferred; and (6) frequently performed tasks can be done quicker and more reliably (Hutchins 1995; Clark 2003; Sterelny 2004; Zhang and Patel 2008; Giere 2010). Heath and Anderson argue analogous payoffs exist when one becomes part of an extended willpower system. The most important one is better self-control.

Now, it may sound paradoxical, but in a extended willpower system, self-control isn't exhibited solely by the self. Rather, willpower is distributed among parts of the system; it is not located solely within any single one of them. To appreciate fully why this description of "distributed willpower" is apt, let's consider how the cognitive systems literature treats a paradigmatic case of "distributed cognition" (Giere 2010).

Returning to the mathematics example from above, when the goal is to multiply two three-digit numbers, we capitalize on our capacity to recall the products for any two integers by following this sequence: first, we construct an external representation of the entire equation, say, by writing it out; second, we adhere to long-established rules of calculation that restrict our focus to basic addition (i.e., four numbers in a row, at most) and basic multiplication (i.e., always two numbers). As Giere (2010) notes, philosophical reflection upon distributed cognition prompts the question: what cognitive entity is performing this task? It does not seem correct to assert that the entire act of cognition is accomplished only by a mind/brain. After all, perception is occurring (how else would the numbers be recognized?), as is

physical gesture (e.g., grasping a pen and moving it about so as to create inscriptions). Additionally, it does not seem correct to state that the entire cognitive act is accomplished by the whole person doing the task; that assertion ignores the required material culture (e.g., pen, paper, etc.). But what, exactly, does it mean to claim that the whole person plus the material culture collectively are engaged in cognitive activity?

Common sense suggests that the material culture, including the external representations, merely serve as inputs that structure the problem in such a manner that it can be solved in what Herbert Simon calls "the mind's eye" (Giere 2010, 102). By contrast, according to the distributed cognition view, it is the "system" (i.e., the whole person plus material culture) that is performing the cognitive task; cognition is thus "distributed" among all of the parts of the system. By characterizing cognition in this manner, Giere comes to insist that the model of "collective cognition"—according to which it is claimed there are instances where groups divide up cognitive tasks that would be too difficult for one individual to do alone—is too human-centered of a model of inquiry. As Giere sees it, the *collective* cognition thesis fails to acknowledge that cognitive processes are not exclusively human functions; cognition can be *distributed* between humans and material artifacts (Giere 2010, 99).

To bring this analysis back to the case of Needy Ned, there is metaphysical warrant for describing our hypothetical participation in Digital Eudaimonia as a commitment to becoming part of Ned's "distributed willpower system." In saying yes to his request, we join the gamified technology as a fundamental component of his digital willpower that helps him stay on track. And this means Ned's success is due—at least partially—to the efforts of the distributed system.

Who Technology Wants Us to Be

While funware gamification potentially can help users meet goals they have been struggling with by bolstering their digital willpower, crucial normative dimensions of gamified behavior would go unexamined if the analysis ended there. For while we can judge the ends people pursue in ethical terms, we should also use moral standards to judge the means by which those ends are pursued. Even if technological dependency is inevitable and a positive duty exists to provide others with gamified willpower, our character still can be weakened by technologically mediated assistance.

In broad outline, concern about human diminution through technology-enhancing abilities and connectivity is not new. Seminal skepticism dates all the way back to ancient Greek philosophy (Plato on writing), with attention growing considerably in the twentieth century, partially due to the impact of contributions in fiction (especially E. M. Forster's "The Machine Stops") and formative media theory (Marshall McLuhan on technological "extensions"). In the current context, the guiding question to consider is what happens to a user's character when he or she develops gamified habits and becomes dependent upon digital willpower. In other words: who does funware gamification want us to be?

Out of context, this question can seem anthropomorphic. After all, funware gamification is not a

sentient being, but a mixture of technique and goal. While humans pursuing gamification have beliefs and desires, gamification itself lacks both. It cannot want things, at least not in the same sense that beings endowed with intentionality do.

The context needed to make the question of what gamification wants both intelligible and appropriate concerns technological adaptation: the process by which humans change the virtues they cultivate as a response to changes in technological standards and functions. A clear understanding of this process can be obtained by considering how users adapt to the type of technology embodied by Apple's Siri.

The Siri commercials promise a perfectly anthropomorphized digital assistant: a virtual, voice-recognition secretary programmed to serve every scheduling and questioning whim by celebrity and average citizen alike. Indeed, Siri's wildly successful marketing campaigns revolve around the slick presentation of the long-standing utopian fantasy where machines faithfully and capably serve their human masters without creating adverse or unintended consequences. In real life, of course, Apple's supply-chain system has been a public relations nightmare, with questionable safety and labor practices by Foxconn Technology Group taking center stage. But in a much-discussed (and parodied) commercial featuring Zooey Deschanel, the political turmoil fades from view and a cyber Shangri-la emerges.

Zooey, who stands in as a proxy for the typical user, trusts the machine more than her own senses. Looking outside through a window, she asks Siri, "Is that rain?" She turns to Siri to have her basic needs provided for, exclaiming, "Let's get tomato soup delivered!" She even asks her smartphone to compensate for imperfect mental abilities ("Remind me to clean up tomorrow") and looks to Siri for boon companionship ("Today, we're dancing").

What—beyond a willingness to endure gentle caricature—does Siri ask from us in return? The superficial answer is little but consumption: purchasing iPhones and data plans. But Michael Schrage, research fellow with the MIT Sloan School's Center for Digital Business, argues the superficial answer misses something important. Siri—as well as other increasingly popular and pervasive technologies—asks us to participate in a fundamental redesign of our social sensibilities. In *Who Do You Want Your Customers to Become?* (Schrage 2012), Schrage argues that Siri wants us to treat impatience as virtue.

According to Schrage, Apple isn't selling a fantasy but instead asks us literally to change—intellectually and emotionally—our default relationship with information. Siri's users should become the type of people who "wouldn't think twice about talking" to phones, treating them as "sentient" servants, and taking the dispensed advice (Schrage 2012, 276). He writes:

> Ask yourself: How would you be different if you regularly had seven or eight conversations a day with your smartphone? As your upgraded smartphone became "smarter" and more fluent, would you chat even more frequently, seeking the information and advice? Can you see yourself as someone who would rather ask Siri for a restaurant recommendation for dinner instead of calling—or texting—a human friend? (Schrage 2012, 269)

Although the scenario is peppered with questions, the reader gets the distinctive sense that Schrage would answer each one with a resounding "yes." This is because he believes Siri "asks her users to become the kind of people who want to engage with her" (Schrage 2012, 271). Quite simply, Siri wants us to want her—the fact that Siri has been popularly

gendered is further evidence for techno-anthropomorphizing—and this means our sensibilities must be wired accordingly.

Apple is hardly the only company aspiring to cultivate instinctive desires for effortless digital gratification. Google's genius, Schrage notes, in part comes from its stress-free and forgiving search interface. Searching on Google doesn't require perfect spelling or punctuation. Typos do not matter. Powerful autocomplete and "showing results for" functions enable users to pose queries without fear of negative feedback. Sloppy searches carry no stigma. Google aims to please.

This logic equally applies to technology journalist Farhad Manjoo's (2012) prediction that Amazon will inevitably start giving away the "Kindle as an inducement to join Prime" delivery service. Manjoo notes that the widespread use of e-readers not only changes people's reading and book purchasing habits but also their expectations around immediate textual gratification. Instantaneous access coupled with lower prices and zero "shipping and handling" costs turns "casual" readers into "impulsive obsessives." In other words, ease of use ensures pervasive usage.

The Kindle thus appears to be asking us to become the type of people who impulsively turn to it whenever we feel literary or documentation itches. When we become those people, scratching that itch will be become mindless. Indeed, once we're experienced from our new set of habits, it will feel natural. Perhaps "Kindling" will even become as linguistically popular as "Googling"—albeit in this case serving as shorthand expression for immediately accessing long-form texts.

If this way of thinking about adaptability proves accurate, we need to give far greater thought to the prices we pay for getting what we want the instant we decide we want it. Economists emphasize the importance of financial capital and social capital when they discuss value creation. But in environments where Google creates new kinds of searchers, Amazon creates new kinds of readers, and Apple create new kinds of digital companionship, it is just as important to ask: What kinds of human capital do these innovations create? Or, put in the moral terms we are concerned about here: What other virtues and vices we will develop when impatience becomes a pronounced part of our character?

The Fun Stops Here: Gamification and Character

To discern the moral stakes of becoming the types of people funware gamification wants—namely, users who rely on digital willpower—it will be helpful to move our thinking beyond paying attention to any particular goal it can be applied toward. For although gamification has become pervasive, users often relate to it in a piecemeal manner, participating in this or that gamified activity. They do not yet embody the habits of what phenomenologists would call a gamified lifeworld. This means the characterological effects of gamification are not as potent as Googling, and the voice of gamification (metaphorically speaking) is not as audible as Siri's.

To bring a possible gamification future into relief that increases the scale and effectiveness of willpower-enhancing apps and the frequency with which they are used, we will propose a new thought experiment and call this hypothetical time-period

Digital Willpower World. The moral traps we are about to discuss are potential general pitfalls. Whether they actually have teeth will depend on how technology and practice develop—details that can be filled in only through speculation are not considered here.

The first concern about Digital Willpower World is that it is antithetical to the ideal of "resolute choice" (Bovens 2008, 217). Some may find the norm overly perfectionist, Spartan, or puritanical. However, it is not uncommon for folks to defend the idea that mature adults should strive to develop internal willpower strong enough to avoid external temptations, whatever they are, and wherever they are encountered. These admirers of self-discipline believe that, for example, we should simply not be rude by ignoring those we are spending face-to-face time with, no matter how alluring it becomes to check e-mail on our mobile devices.

In part, resolute choosing is prized out of concern for consistency, as some worry that lapse of willpower in any context indicates a generally weak character. The person who can't stop texting during dinner is presumed to lack self-control. He or she could be easily swayed by unruly desires and too readily disposed to avoid moderation or worse in other circumstances. In short, resolute choosers see themselves as strong in spirit and capable of consistent virtuous actions. They construe backsliders as weak-willed slaves to impulse and immediate gratification. If they wore a group T-shirt, the slogan would be: "Quit anything and you're a quitter."

The second and third concerns about Digital Willpower World can be extrapolated from ideas developed by Luc Bovens (2008), a professor of philosophy at the London School of Economics. They are the problems of "fragmented selves" and "infantilism."

Fragmented selves behave one way while under the influence of digital willpower but another way when making decisions without such assistance. In these instances, inconsistent preferences are exhibited, and we risk underestimating the extent of our technological dependency. For example, under current conditions, we might eat healthy when using a gamified wellness app but poorly when we forget our smartphone at home. Acknowledging this risk does not entail denying that repeated exposure to willpower-enhancing technology can engender positive new habits and correlative preference shifts. It simply means that when it comes to digital willpower, we should be on our guard to avoid confusing situational with integrated behaviors.

In Digital Willpower World, the problem of fragmented selves does not appear to be an issue—and this is significant. After all, inhabitants are constantly plugged in to willpower-enhancing devices. They no longer toggle between enhanced and unenhanced lives. Bracketing the question of what would happen to such folks if the support systems crashed—as that issue applies to so many things—the problem of inauthenticity, a staple of the neuroethics debates, might arise (Levy 2007). People might start asking themselves: has the problem of fragmentation gone away only because devices are choreographing our behavior so powerfully that we are no longer in touch with our so-called real selves—the selves who used to exist before Digital Willpower World was formed? Consider a contemporary analog to this problem. Right now, people can use an app that automatically sends happy birthday wishes to Facebook friends. Although this service bypasses the problem of forgetfulness, its use raises questions about sincerity and thoughtfulness.

Infantilized subjects are morally lazy, quick to have others take responsibility for their welfare. They do not view the capacity to assume personal responsibility for selecting means and ends as a fundamental life goal that validates the effort required to remain committed to the ongoing project of maintaining willpower and self-control. We can become so comfortable with exporting decision-making to our digital devices that rather than enhancing our own abilities, they become like techno-nannies who watch over us (Sadowski 2012).

In *Real American Ethics*, philosopher of technology Albert Borgmann (2006) does some forecasting of his own and considers a development that we can imagine would be central to Digital Willpower World: smart homes—dwellings designed to not only to sense but, more importantly, to anticipate user desires. Such places would take advantage of the so-called Internet of Things and be stocked with smart appliances, including refrigerators that keep stock of depleted items and communicate directly with grocery stores to initiate purchase of replacements. Borgmann worries that this environment, which habituates us to be on autopilot and delegate deliberation, threatens to harm the powers of reason, the most central component of willpower (according to the rationalist tradition). In living here, Borgmann fears, "we will slide from housekeeping to being kept by our house" (Borgmann 2006, 120).

Now, a deeply religious person might object that the concern over infantilization places too much value on independent moral deliberation while undervaluing the importance of deference—of deferring to God's expectations, which are expressed in the laws he expects us to follow. Although this objection concerns a deep conflict between faith and reason that is beyond the scope of this chapter to address, we can say one thing. Regardless of whether someone should only endorse a religious outlook that can be rationally justified or whether an irrational leap of faith is the best way to connect with the divine, adherence to a religious tradition is, practically speaking, insufficient for solving all moral problems. Religious texts can be inconsistent, and so judgment is required for determining which messages are fundamental and which can be flexibly interpreted. Moreover, religions are often viewed as living traditions—as practices that get reinvented in response to emergent crises and changing sensibilities. The issue of gay rights, for example, divides many Christians, making the personal use of moral judgment inevitable.

As it turns out, we don't need to wait for Digital Willpower World to form or the Internet of Things to mature to find people worrying about infantilization. Even positive reviews of the Internet-blocking app Freedom are tinged with elements of concern over self-infantilization and the loss of resolute choosing. Some try to cope by pointing to social norms, convincing themselves that if others are doing it, it cannot be too bad:

> Yes, the whole thing makes me feel infantile and weak, like a tween whose parents have put some kind of monitoring software on my computer; except in this case I am both tween and parent. But I'm not alone. In other corners of the Internet universe, people are turning to programs like RescueTime and MeeTimer to monitor, track, and limit the time they spend on Twitter, and something called LeechBlock to prevent themselves from even going to certain sites for certain times of the day (Traister 2009).

Yet another concern about Digital Willpower World comes from renowned political philosopher Michael Sandel (2004) in his classic essay, "The Case

Against Perfection." He notes that technological enhancement can diminish people's sense of achievement when their accomplishments become attributable to human-technology systems and not an individual's use of human agency. If someone sticks to his or her ideal eating and exercise regime, who deserves praise? Can we still say the person does, or, drawing from systems analysis that we applied to Needy Ned, does it make more sense to say that it is a person–gamified-technology system? A critic might respond that this question is meaningless because when that person uses the app, she has to exert much more willpower than, say, if she had gastric bypass surgery. The surgery would eliminate agency by shrinking the stomach and causing food to bypass part of the small intestine. By contrast, the user enhanced by the gamified app still has to make many choices on his or her own. True enough! But this example only shifts the comparison. It doesn't deal with the fundamental worry—which, again, was foreshadowed in the earlier analysis of Needy Ned—that when "collective" systems perform shared functions, individuals lose their right to exclusive ownership of behavior.[5]

Provocatively, Bogost (2010b) contends a variation of this issue already is a serious concern for users of gamified apps—which means we do not have to wait for Digital Willpower World to be formed to feel a sense of unease. Appealing to ideas associated with the concept of moral luck, he writes:

> When one responds to game-based incentives like points and rewards, good things might happen: better hygiene, or a cleaner environment, or a greater connection to one's family. But these results cannot be judged to have the same moral responsibility as choices made given factors under greater player control.

Bogost's point, therefore, is that folks who pursue one thing (points and rewards) and end up inadvertently getting something better (improved health, environment, or familial connections) in the process do not deserve as much credit as folks who explicitly value and work to achieve the latter. Why credit someone as being family-oriented, environmentally conscientious, or health-centric if, in fact, he or she is point-obsessed and pursuing that obsession contingently leads to getting a lucky break and obtaining more than was intended?

We do not know whether the traps associated with Digital Willpower World will come to fruition or will be as disconcerting as the versions presented here suggest. In identifying them, our goal was to point out moral issues that are often occluded from funware advocacy. By expanding gamification discourse to include resolute choice, infantilism, distributed responsibility, and moral luck, we hope to have introduced a robust set of considerations that can continue to animate public and scholarly discourse. Ideally, such potent ethical ideas will be brought to bear on new iterations of gamification as they unfold. Indeed, our goal here isn't to directly answer what, ultimately, are forward-looking ethical questions. Instead, our sights are set on contributing to anticipatory governance. We hope to help form the critical vocabulary that's needed to determine what a character corroding, gamification-fueled tipping point would look like. Regardless of how history unfolds, one thing is certain. The discussion illuminates how inapplicable Dutch historian Johan Huizinga's (1971) classic conception of the "magic circle" is to contemporary funware gamification. As philosopher of technology Adam Briggle notes:

> Huizinga described games as existing outside of normal life—closed off by a "magic circle," the

> borders of which defined a separate time and space. Game fiction projects another world and game rules carve out an area where they apply. Huizinga described games as separate and unproductive. (Briggle 2012, 160)[6]

Unlike the external (and possibly idealized) games that Huizinga discusses, the object of funware gamification is ourselves. One follows its rules hoping for desired consequences to occur to oneself in real life. Digital Willpower World, therefore, can be understood as a cautionary tale that asks us to consider whether those consequences amount to a Faustian bargain. Perhaps if users "care for themselves" this way, they become a losing version of a gamified "design space."[7]

Gaming beyond Fun

So far we have critically discussed funware gamification, identifying conditions where the duty of beneficence might require us to participate in a funware endeavor and pinpointing possible ways that funware could be morally corrosive by adversely impacting character. In this final section, we will discuss whether there are moral goals that funware gamification cannot further. In so doing, we will contribute to gamification scholarship by clarifying limits that constrain its use.

It would be theoretical overreach to contend that some applications of gamification are inherently impossible. After all, our imaginations might be too limited to detect latent possibilities. Or, our prognostic skills might not be up to the challenge of speculating about emergent behavior. With those caveats in mind, we can claim it is extremely unlikely that funware gamification will ever be compatible with a morally significant application: inspiring the developmental growth needed to help people experientially appreciate how matters of luck helped them personally obtain privilege and determine what obligations they owe others who are less privileged—at least in part—due to not enjoying that luck.

By "experientially appreciate," we are referring to a sense of understanding that goes beyond merely intellectually grasping ideas. Someone who can articulate how being born white in a racist country is an act of luck, but is emotionally unmoved by this knowledge, and in everyday life fails to reflect about how his or her whiteness conveys undeserved privilege, could write a glowing philosophical essay on John Rawls's conception of justice. That knowledge of "the original position," however, would not rise to the level of experiential appreciation. Someone with experiential appreciation would, for example, be attentive to whether he or she received a promotion at work because the boss is a bigot who acts prejudicially against deserving colleagues. And, he or she would feel the moral emotion appropriate to this situation: indignation. He or she might not confront the boss, for the myriad possible justifiable reasons. But, he or she would experience the temptation to do so.

While a lucky break (e.g., advantageously dealt cards) can be obvious when playing games that simplify the amount of relevant variables, in real life it can be hard to perceive—so intangible as to be invisible. In a culture like the United States, which is built

upon a history of esteeming individualism and self-determination, people are encouraged to see success as a result of their hard work and failure as a result of their mistakes. Consequently, they can be deeply resistant to someone telling them that some of their success is not earned, but should be attributed to luck. Comedian Louis CK—who has a genius for creating uncomfortable material designed to make the audience confront privilege—provides an apt illustration of this problem.

In the opening segment of one of his shows (Louie, Season 1, Episode 6, "Heckler/Cop"), his character is rudely interrupted by a member of the audience while delivering a routine at a comedy club. Louis asks her to stop, but she keeps on talking. After the routine ends, they confront each other, and Louis offers moral condemnation, exclaiming that a "good person" would not be so rude. The woman denies this, and when Louis asks her why she believes she is a good person, she refrains from offering any clear justification. With smug self-satisfaction she says: "I just am. I know I'm a good person. And you have no right to talk to me like that."

Exasperated, Louis asks if she likes her life—if, as a very attractive student, she has "good days and full and fun nights." Her response: "Yes, I'm not a loser like you." Louis then points out that comedians lack this sense of well-being. Disconnected from friends and family, all they have to look forward to are short and sporadic comedy appearances, the very thing she ruined. After Louis offers further moral shaming and points out the disparity between her idealized self-conception and the horrible facts about her actual being, the woman exclaims: "You have no right to make me feel like that ...You are just one of those unattractive people who are totally bitter against people like me."

Would anything have convinced the woman that she was being insensitive and taking her privilege (youth, looks, and, from her appearance, we can infer, money) for granted? Possibly not. Of course, as we are dealing with fiction, we will never know. We can speculate, however, that some version of Louis's approach would be needed. The woman would need to feel uncomfortable. She would need to be put in a position where she confronted rather than denied the gap between who she took herself to be and who she actually is. Could a game do this?

A game that could would not be fun. Consequently, such a game would violate the guiding premise of funware gamification—a premise that game designer Brenda Brathwaite insists is spurious.[8] And, such a game would need to be experiential. Specifically, it would need to revolve around experiences where players develop an appreciation for how privilege relates to luck. Consider writer Joel Stein's claim that, "We fetishize epiphanies, but only experience changes you. Just as the act of smiling makes you happy, climbing a log tower makes you confident, taking punches makes you tough" (Stein 2012, 276). If he is right—and we believe he is—the analog is that experientially confronting your privilege can make you a more just person.

Taking all of this into account, we have used the mechanics of formal game theory (e.g., the prisoner's dilemma) to create a series of games that put players (so far, university and high school students) in situations where they have to navigate tough moral dilemmas—dilemmas where pursing personally beneficial courses of action (e.g., ones that increase game points and thereby also increase the corresponding quiz grades) may have direct negative impacts on the well-being of other players (e.g., decreased their game points and also their corresponding quiz

grades). Without getting into the specific details of how each game operates—for that description see Seager and Selinger (2012), Seager et al. (2012), and Sadowski et al. (2013)—one of the themes uniting them is that each is structured to encourage players experientially to appreciate the ethical relevance of luck. Players gain this appreciation by reflecting, both in groups and individually, on the surprisingly unethical actions (personally committed and committed by classmates) that arise during the game. Like Louis's heckler, they come face-to-face with the misalignment that often exists between our self-perceptions and how we actually behave in the world (Kahneman 2011; Sadowski 2011). (Or, they have the opportunity to act like Louis and help the metaphorically heckling classmate.)

To say the least, it is not fun for someone to learn that he or she may not be as good of a person as was previously believed, and that the privilege granted by luck can be a large driver of success.[9] For many students, this conflict leads to what we call a "moral identity crisis"; that is, an unpleasant experience that motivates people to want to do a better job living up to their moral ideals (Sadowski et al. 2013, 1235). In nudging students toward this crisis, our games embody one of the features Ian Bogost (2010a) associates with "persuasive games."[10] Their structure confronts players with a "procedural rhetoric" that is designed to engender cognitive dissonance between players' subjective mental models and the actual behavior that takes place in real-world systems.

The story of a recent student will help concretize the type of experience we are referring to. To keep this person anonymous, we will refer to him or her through a mythological pseudonym, Ariadne. Indeed, we will refer to Ariadne as "she," but this isn't necessarily any indication of the student's actual gender.

During class discussions, Ariadne made strong arguments for her altruistic moral positions and jeered those who acted immorally or defended egoistic propositions.[11] But during game play, when the temptation was strong, she abandoned her ideals and found ways to take more points for herself than would be considered fair according to the ideals she frequently espoused.

In one instance, Ariadne accepted a high grade at the expense of classmates who were treated unfairly in a game when the rules were obviously rigged. Even when Ariadne had the opportunity to correct the injustice by sharing her grade points with others who had been treated unfairly, she declined.

To Ariadne's credit, as time passed she realized that she had behaved poorly and experienced remorse. One day after class, she approached an instructor and privately proposed confidentially giving the students she aggrieved some of her ill-gotten points. At the time, this seemed like a good way to atone and experience moral growth.

However, later in a different game, she behaved *exactly the same way*. Clearly, the moral development that was hoped for had not occurred yet. Ariadne had taken advantage of the instructor's willingness to spare her from negative public judgment.

But this time her luck ran out. The instructor refused to absolve Ariadne's troubled mind through private back-channel dealings. Instead, the issue was brought before the class, and the other students were given an opportunity to deliberate about what transpired and vote on how to proceed.

In this case, even the students who admired and liked Ariadne refused to let her off the hook and make after-the-fact reparations. They determined it would be too easy to let her continue to do

irresponsible things in one moment, and somewhat inauthentically absolve herself of guilt later on. Consequently, they insisted she live with the knowledge of what she did and the internalized emotions appropriate to that failing. This wasn't spite or *Schadenfreude*. They wanted to encourage moral growth—the growth that comes from learning how easily one can slip away from moral ideals. And when the class ended, Ariadne wrote an essay attesting to having learned an important lesson. She acknowledged that prior to taking the class she was able to overesteem her level of personal virtue because of luck. She had been lucky enough to avoid situations where moral decisions needed to be made in the face of self-serving peer pressure and tight time constraints.

Ultimately, funware is not conducive to helping people appreciate the full ethical import of luck. It treats problems as dilemmas to be solved through the addition of extended willpower. Consequently, it reinforces the idea that individuals are masters of their own fate—at least if they can find the right tools or support systems. This framing obscures how aspects of luck that are fundamentally beyond our control shape events and influence our positions in life. While extended willpower helps us better go with the flow and pursue solutions to socially defined problems (e.g., individuals being responsible for unhealthy habits), funware cannot help us gain the reflexivity needed to make good moral judgments about whether and when to challenge the flow and alter not only our behavior but our self-conception, too.

Notes

1. This chapter synthesizes and advances long-standing research. Consequently, some material was previously published in Selinger 2007, Selinger 2012a, Selinger 2012b, and Sadowski et al. 2013. This material is based on work supported by the National Science Foundation under Grant No. 1134943.

2. Debates over appropriateness recently arose when the Israeli Defense Forces gamified its war blog. For many critics, this explicit use of gamification in the context of war propaganda was morally repugnant (e.g., Mitchell 2012).

3. Momentum for funware is building, and its influence already extends beyond gamification circles. In "Making Good Citizenship Fun," Richard Thaler (2012), a leading behavioral economist and coauthor of *Nudge* (Thaler and Sunstein 2009), pleads for policy makers to make better use of the type of positive reinforcement outlined in "fun theory." With the success of long-standing endeavors like sweepstakes in the background, his advocacy centers on a simple maxim: "If governments want to encourage good citizenship, they should try making the desired behavior more fun." Zichermann (2012) himself is now making similar recommendations and outlines how gamification can provide a "less combative, cheaper and more socially conscious method of encouraging users to vote."

4. Crucially, Grant makes this important distinction: "But all 'extrinsic motivators' are not alike. Praise and rewards do not operate the same way as incentives. Incentives are tangible benefits unrelated to the activity itself. Telling the child that you are proud of him for reading the book or allowing him a special

opportunity to tell the class about the book is not an incentive. These latter approaches reinforce the value of the activity and may enhance intrinsic motivation by increasing the child's desire to undertake the activity 'for its own sake'" (Grant 2011, 113).

5. A comparable way to analyze a "person–gamified-technology system" is to apply Bruno Latour's actor-network theory and Don Ihde's version of post-phenomenology. See Selinger 2012c.

6. For an edifying discussion of the "magic circle," see Calleja 2012.

7. The concept of "care of the self" comes from Foucault (1988). The concept of "design space" comes from Allenby (2011), who argues: "The human being is more and more becoming a design space." By design space, Allenby means humans are exceptionally good at creating technologies and built-environments that enhance our control over the "natural" world and allow us to impose our desires and values over the "natural" order. Allenby primarily focuses on advances in nanotechnology, biotechnology, robotics, information and communication technology, and applied cognitive science—and the future of funware gamification, which is set to design more and more spaces for us to make decisions, seems to depend upon developments with the last two categories.

8. Put in formal terms, Brathwaite argues that fun is not a necessary condition of something being a game. "We've learned that game = fun. So, yes, we have a narrow definition for it, but I believe that definition is expanding. As an artist working in this medium, I want it to expand. Imagine if someone said all books or movies or paintings had to be fun? A game is an interactive experience which has goals, a clear end and a set of rules. I think anyone who considers the systems that are presented in my series will see the full potential of games to explore the human experience. If you consider something like Risk seriously, how fun is that? It's only fun because we're able to abstract the domination and destruction" (Brophy-Warren 2009).

9. "During reflection, students are often confronted with the reality of a significant gap between their moral aspirations (e.g. the promises they made themselves or others) and their moral actions as represented by the choices they made in game play" (Sadowski 2011).

10. Bogost (2010a, 2–3) writes: "*Procedurality* refers to a way of creating, explaining or understanding process. ... *Rhetoric* refers to effective and persuasive expression. Procedural rhetoric, then, is a practice of using processes persuasively."

11. This example was previously discussed in a blog post ("The Moral Saint Fallacy") for a sustainability ethics course. See: http://sustainabilityethics.com/2012/10/13/the-moral-saint-fallacy/.

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POSITION STATEMENT

PLAYFUL SYSTEMS

Kevin Slavin

It was Frank Lantz that introduced me to Core War, a game you're unlikely to have spent much time playing. Designed in 1984, it's a two-player video game in which each player writes an algorithm and sets it in motion against an opponent's algorithm.

Over thirty years, the game has been constantly updated and recompiled for the few thousand people who are playing it at any given moment. But the stuttering graphic display of 1984 hasn't changed much. It's almost entirely abstract: just pixels on the screen and a few numbers moving up or down. Each player uses a simple assembly language to program their algorithm, and then unleashes it. There's nothing else to do but wait to win or lose. There's a 1987 match between two algorithms—Dwarf and Gemini—that's notorious.

It's a game by programmers, for programmers. It's not a popular game.

But, like many unpopular games, it's brilliant and important. Not for what it is, but for what it represents. A hinge, a moment where something heavy swings on an axis. In 1984, what swings is the role that games play in culture.

This is a position, not an axiom, and it's not qualified here by scholarship or data. But it's a feeling, like stumbling across an unexpected clue. The mystery is a mystery about the complex systems of the world, and what happens when we play them. The clue is Core War.

Let's say that for a long time before Core War, many games were about drawing something up from lived experience and distilling a system from that. That something might be, for example, the binary opposition of armed conflict, or the dynamics of zero-sum microeconomics in Atlantic City, or the behavior of nonplayer characters—broad aspects of the human experience that get quantified and systematized into an approximation of something you might recognize.

There are plentiful and obvious exceptions here, but Core War is a special kind of exception because it's the one that sets a new rule. It's exceptional because it's a game that draws up from the system it's embedded in (executable code) and renders it playful by embedding the system within new constraints and boundaries. If your browser has crashed or hung in the last week or so, it's because a core war happened. Core War is an abstraction of that conflict. But it's no abstraction.

In the same way that Monopoly draws from a model of Atlantic City, Core War draws from a 1971 "real-world" self-replicating program called Creeper, generally understood to be the first computer virus, a cybernetic lab accident. To preserve the integrity

of the lab's 1971 packet-switched network, another lab designed and built a program called Reaper. Reaper had one job: to hunt and kill Creeper.

Built 13 years later, Core War is an allegory for that conflict. I understand it the same way I understand The Game of Life, chess, or Risk. But then something swings around it: the role that synthetic systems have in our lives.

Back when Core War was first built, there were bookshops selling books, and we judged them by their covers. There were songs on the radio that found their way there through payola and magazine reviews. There were sudden financial panics caused by humans watching one another trade on the floor of the stock exchange. None of those models endured. From finance to culture to urban planning, our everyday lives are increasingly shaped by complex systems with opaque fitness criteria.

Edgerank is the algorithm that determines which of your friends' status updates will reach you, just as Panda and Penguin determine what information moves through the world. Stuxnet—a Creeper that infects real-world energy infrastructure—can take down a nuclear power plant, while Pragmatic Chaos algorithms quietly determine the majority of the Netflix movies enjoyed by more than 20 million people.

And of course, there's Wall Street, in which these algorithms are in explicit combat on a microsecond scale, shaping both the economy and a landscape that yields to the wires that carry them. Financial data analysts give them names: Castle Wall, The Knife.

Sometime around 1984, invisible physics seeped out of the mainframe and into the world. They are shifting markets, uranium, music, information, and even the physical environment around us. And when you think about the effects of Stuxnet, Castle Wall, and Penguin, you see Core War's Dwarf and Gemini algorithms in a very different light.

In their original forms, Dwarf and Gemini might have been an allegory for the Creeper and Reaper viruses, just as Risk is an allegory for cold-war politics. But here in 2013, you look at Castle Wall and The Knife, and you see that Core War was no allegory. It was an audition. An audition for the twenty-first century, when those playful algorithms in conflict would be weaponized to fight for markets, attention, and infrastructure.

This is the hinge. A game built as a model to *reflect* the world somehow built a model that *shaped* it.

This is why video games matter. When we engage them, we engage tiny synthetic models of the world. But when we stop playing the game, back in the world, we are still engaging with systems. Every time we get driving directions, or information, or money from the bank.

So when I think about a gameful world, I don't think about it as a world that has systems imposed upon it. Rather, I see it as a world in which existing systems are made somehow visible and transformed alchemically into cultural material.

In 1984, Core War only had abstract computer systems to draw from. In 2013, we have in vivo models of global systems to retrain, reshape, and rethink. Those systems might be latitude and longitude, or DNA, or commercial flights in the air, or patterns of information as they move through the world. Whatever they are, they can either remain invisible and instrumental or they can reveal themselves to become something we think about. And there's a special type of thinking, called play.

Since 1984, we've begun to play the algorithms of the real world and to imbue them with the mystery embedded in this precious form of contemplation. The mystery remains mysterious, but the motive is clear. We want to make sense of the world, so we play it.

16 THE TECHNICAL CONDITIONS OF A GAMEFUL WORLD

Nicolas Nova

This chapter will examine and question the technological conditions for the advent of the gameful world described in this book. The "use of game design elements in non-game contexts," which is the commonly used definition of *gamification* (Deterding et al. 2011), has indeed flourished in the past five years. In the industry, popular services like Foursquare (2009) feature external rewards such as points, badges, and user hierarchy. This situation has led scholars in game studies, human–computer interaction (HCI), and sociology, among others, to discuss both the cultural implications and the relative merits or drawbacks of such an approach. However, the spread of playful mechanisms out of their original domain is not limited to the web or mobile apps. One can find a similar trend in various interfaces people use every day, such as car dashboards, watches, or clothes, ranging from bracelets to headbands. Driving a Chevrolet Volt now requires you to optimize braking and acceleration by looking at a mysterious green orb, users of the Nike+ SportWatch GPS or Nike Fuelband can compete with one another and accumulate points by running with these devices, and owners of Zeo headbands even *gamify* their sleep, an activity formerly remote from gaming. The wide breadth of such applications forms what this book refers to as a *gameful world*.

From the outside, these examples highlight an ongoing convergence between gamification and a subdomain of HCI called ubiquitous computing, or "ubicomp" (Weiser 1991), which corresponds to the proliferation of computing into the physical world. Introduced by Mark Weiser at the beginning of the 1990s, this field relies on his vision of users and environments augmented with computational resources able to provide people with information and services in an unobtrusive way wherever and whenever they are. The corresponding devices developed by Weiser and his followers ranged from handheld assistants (from which PDAs were created) to tablets (ancestors to tablet PCs and iPads) or augmented surfaces such as interactive tables and whiteboards. Thus, one can see the aforementioned gamified applications as a hybridization between such "ubicomp devices" and the use of game mechanics. In spite of that similarity, it is no coincidence this convergence occurred, given that HCI researchers wondered about the transfer of game design elements to more serious digital programs back in the 1980s. As shown by scholars such as Thomas Malone (1982) or Ben Schneidermann (1983), transfer of "heuristics" learned from computer games or lessons put forward by game interfaces is a recurring trope in the field. This situation

eventually led researchers working on ubicomp projects to follow the same path and design applications that can be fun, engaging, and playful for users. Hence the design of handheld and location-based games with educational purposes (Benford 2005) and the advent of so-called pervasive games: a new genre in which original gaming experiences can be created based on the blend of real and virtual game elements (Benford, Magerkurth and Ljungstrand 2005).

In its early days, pervasive gaming consisted in a vast panoply of devices and experiences such as location-based games, augmented reality applications, tabletop games, or new media performance. This period of creative (and scientific) turmoil was also characterized by the prevalence of prototypes and temporary applications. The seminal projects of this genre, such as the different pieces produced by performance collective Blast Theory in the United Kingdom or the ones made by Area/Code in the United States, were indeed based on events or dedicated locations. The games they designed were often deployed in the context of museums, art exhibits, or as promotional tools for commercial products. At the time, because of technical issues and the limited number of users owning phones that were connected to the Internet or GPS-enabled, these applications remained limited to specific locations or time periods.

With this shift from temporary and fragile prototypes to large-scale, on-demand, commercial products, the situation changed dramatically. This chapter will focus on the latter situation and will highlight the technological conditions enabling such a gameful world. Based on an analysis of certain underlying components of the gameful world, we will see how they enable and constrain the design space of gamification in everyday life.

The notion of design space is commonly used in HCI in order to make sense of the variety of technological opportunities and characterize them in a systematic way (Card et al. 1991). Most often applied to interface and interaction types, the building of a design space consists in grouping items into families that share similar parameters. Beyond the relevance of describing a research domain, the resulting classifications also have a more operative purpose in the context of design, as they suggest new designs and opportunities.

A common technique to generate a design space in HCI is called the morphological analysis model (MAN): "the examination of different input device designs as points in a parametrically described design space" (Card et al. 1991, 99). More specifically, these authors have applied this technique to input devices in order to generate a taxonomy of such interfaces. This chapter will adopt a different stance by extending this morphological approach to more "internal" parameters such as user identification, the role of sensors, and communication protocols. The main reason to do so is that a morphological analysis focusing only on surface traits (such as interface choices) would not account for the complexity of user interactions in ubiquitous computing. For instance, the dissemination of sensors in our everyday environment creates the possibility of non-intentional interactions (Greenfield 2006): the presence of a user next to a piece of interactive furniture can be detected via GPS or a 3D camera, and the user is not necessarily aware of such interactions. This means that we need to consider these components as input means, although they look less apparent to users.

For this chapter, we selected three different components of the design space (identification,

sensors and geolocation, and input-output diversity), which can be seen as important points of contact for the users of current gameful systems. For each of them, our approach will be to describe the technological possibilities and discuss the design implications of such components in the digital everyday, where most of the gameful apps are deployed.

Identification, Social Space, and Nonhuman Participation

Even if user identification in gameful applications mostly relies on standard means such as the login/password couple, the necessity to offer alternative access to the same program on a computer or on a cell phone and other networked devices makes this component more diverse than for standard applications. Although most of these applications are intended to be used by human users,[1] it is physical devices that enable authentication and identification in the system. Let's take Foursquare as an example. Creating an account on this app requires the user to enter his cell phone number, which acts as a login. The identification, and the game interactions, are thus "delegated" to an artifact: a phone. More precisely, it's done by a specific component of this device: the cell phone number that depends on the IMEI identifier of a SIM card inserted in the communication tool. In Chromaroma (2010), a playful app that nudges people to use public transport in London, identification is also delegated to another device, namely their transportation cards ("Oyster Card"), the RFID-enabled pass employed by users of the metro and buses of the city. The game takes the users' travel data by scraping the data from the Oyster online interface and "makes it into a game where every journey counts in a competition for the city." With Swinxs (2008), an interactive toy made by a Dutch company, participants are recognized by tapping their RFID-enabled bracelets on a central unit that then engages them with various challenges. In these two last cases, identification relies on the same principle: the "identifier" of the device, which belongs to a certain user, is encoded in a microchip inserted in the card or the bracelet and communicated via radio waves to a reader, which can detect their presence and their identifier.

This quick overview of identification possibilities may look technical or mundane, but it interestingly highlights notable changes happening when the login/password couple on Internet platforms is complemented by networked objects (table 16.1).

The three aforementioned examples create a particular relationship to these apps and offer three original situations with different consequences.

First, because of the informational character of these identifiers (a cell phone number, which used to be a personal piece of information only shared with a limited number of people) or their proximity to the body (the phone itself, the RFID transport card in our wallet, the Swinxs bracelet), they correspond to a different level of intimacy for the users than that of standard means of authentication. This also means that new issues can arise if the user's objects get stolen and are used by someone else. Depending on the artifacts considered, the consequences can be either limited (a bracelet from Swinxs) or more annoying (Oyster Card).

Table 16.1
An overview of identification possibilities

Application	Identifier	Degree of Intimacy	Degree of Shareability	Design
Foursquare	Phone SIM card (IMEI number)	High	Low (people rarely share their phones)	Repurposing of an existing object
Oyster	Oyster Card ID (RFID tag)	High	Medium (individuals can use the card of another)	Repurposing of an existing object
Swinxs	Bracelet with RFID tag	Low	High (players can reuse the same bracelet)	Explicitly designed for this

Additionally, in the case of Chromaroma and Foursquare, user identification relies on repurposing an existing device designed for a distinct purpose (the phone number and the Oyster Card). This situation is both interesting, as it avoids the necessity to create a new identifier, and problematic, as it leads to confusion between services that formerly were separate. Before mobile apps, cell phone communication and access to a web service required different IDs, whereas Foursquare mixes them, and Chromaroma hybridizes the use of public transport and a pervasive game. The gamification logic at hand here is not only a mere transfer of game mechanics to a different setting, but also a hybridization of different logics directly in the system.

Finally, with Swinxs and Chromaroma, the physicality of the device makes it easily shareable, but this opportunity can be limited by the person who lends it to a friend. Like a key that can be returned to its owner, the bracelet or the Oyster Card can be temporarily used by other users.[2] Such a situation is of course intriguing from a design perspective, allowing a complex dynamic to unfold. For instance, one can think about collaborative applications in which several participants team up to perform a joint task.

Identification in a "Seamful Design" Era

Most of the time, pervasive games and gameful apps require objects to be identified, and recognized from a database standpoint, in order be part of the system. In Swinxs, for example, one can only play with the RFID-enabled bracelets provided with the game; and a book with an RFID on the back coming straight from a bookshop would not be recognized by the main unit. To put it differently, in most cases, the system boundaries are well defined. However, there are intriguing exceptions to this situation as shown by applications in which the objects are identified but not necessarily known previously by the system. Think about Treasure Troves (2009), a game developed for the Nintendo DS. One of the main input mechanics consists in scanning for nearby WiFi networks and generating items based on each network's unique frequency. In essence, the console does not know these networks beforehand, nor does it use

them in the proper sense (to transfer data packets), it only detects their presence and creates a virtual object based on its frequency. Another example is Jabberwocky (Paulos and Goodman 2004), a mobile phone application that displays the proximity of "familiar strangers"; that is, the person you often come across but do not know. The program does so by relying on the Bluetooth identification of cell phones present in the vicinity: it displays iconographic representations that indicate how often one has passed by the same Bluetooth device and, by extrapolation, the same person. Both examples highlight how technological limits to identification can be used in an imaginative and playful way to create digital interactions and the gameful world described in this book.

Sensors, Positioning Technologies, and a Playful Use of Space

Gamified and gameful apps also rely on the generalization of sensors located in our environment. Temperature, humidity, noise, pollution, movement, acceleration, direction, or location are examples of the parameters generally detected by these sensors. Depending on their proximity to the user, one can differentiate two categories: (1) sensors embedded in a personal device (e.g., GPS chip in a smartphone, a speed detector in one's car); (2) environmental sensors remotely located that can detect changes (e.g., weather conditions in a specific city, temperature or light at home). The data generated by these sensors can be collected by the program and taken into account to create different forms of interactions. This is how location-based applications, the most common example of sensor-enabled system, offer various features to the user based on his or her location: pointing to points of interests, showing where contacts are located, and so forth. Such technology creates a renewed relationship between the user and his or her environment, which leads to an increasing role of space and mobility in the usage of these applications.

In the context of location-based services, it is important to highlight that a large breadth of positioning technologies is available to designers and developers. If GPS[3] is a standard way to locate people and objects in the physical environment, there are other solutions. The use of mobile phone tracking based on multilateration of radio signals[4] (GSM, WiFi, Bluetooth, etc.) can also be used for the same purpose, sometimes as an alternative to GPS or a substitute for indoor positioning. Moreover, other designers have chosen to let users declare their own location themselves; on Foursquare, participants select their whereabouts by picking from a list of preselected places located in the vicinity via GPS sensing. Notably, games and their playful mechanics have been associated with positioning technologies for about ten years. Aside from standard navigation tools used in cars, a certain amount of research in HCI pursued design of location-based games as a testbed to investigate interface issues outside of the controlled conditions of the lab and as a way to create original applications (games, serious games, training software). In the industry, location-based services, such as Foursquare, Gowalla or SCVNGR, have also been early instances of gamified apps.

When it comes to the use of geolocation, space, mobility, and gamification, Waze (2008) is a relevant

example to discuss. This car navigation app learns from users' driving times to provide routing and real-time traffic updates. Based on a community model, it gathers real-time data from users of the service (their spatial position, accidents, traffic jams, or speed traps reported by users) in order to adjust the proposed routes. A common problem of such a community model being the lack of critical mass of participants, the designers of Waze deployed a standard gamification scheme to increase participation. Participants have a personal dashboard on which they see their ranking among all the worldwide users and earn points based on their driving (mileage) and reporting behavior (filing reports[5]). Users also improve their scores when using Waze by driving with the app turned on, by driving on new roads where no users have gone before, or by driving over special bonus point items that Waze left on its "virtual roads" (like cupcakes).

Space and Time Remediated

The use of Waze is intriguing as it is meant to reshape our relationship to space and mobility through a complex system of rewards and detection of spatial behavior. Expecting that users will drive new roads and giving them points for this behavior is a recurring hope from designers of location-based services. In their time, the designers of Mogi[6] (2003) indeed had the same expectation. In this game that engages players in a collective hunt for virtual items, some of the items were given a specific behavior. Certain creatures, for instance, only hang around parks, which meant that players had to travel near public parks at certain hours, even if this location was not part of their daily routine. Licoppe and Inada (2006) in their seminal study about Mogi showed that these "detours," the change of one's urban course to collect a game item, were quite common and generally opportunistic. These authors also highlighted how the game itself became a pretext for mobility, as if the "augmented" character of the environment provided sufficient clues and affordance for movement. As discussed by Michiel de Lange (2009), such location-based apps offer a new mediation with physical space: by transforming the perceived "nearness" to other people and objects in the environment (physical or digital), these systems remediate their presence and proximity to them. However, in the context of Waze, there are no studies about the influence of gameful elements on drivers' behavior, but we can hypothesize that usage patterns similar to those with Mogi may occur.

Aside from these spatial implications of sensors and positioning technologies, it is also relevant to consider to what extent such devices also remediate our relationship with time. The set of devices sold by Nike (Nike+ SportWatch GPS or Nike Fuelband) offers an interesting perspective for that matter. The use of these product is different than that of the aforementioned pervasive game examples: the interaction with these devices in real time is generally limited. Using the SportWatch while running will only give you the time and duration of the run, and checking your Fuelband will just tell you how many steps you did, the medium number of calories your burned, and your fuel score. These numbers can give you some information about how you are doing. But the real significance of these devices emerges when the data are accumulated over time, analyzed by the applica-

tion, and presented on the website with diagrams and behavioral comments. In a sense, most of the usage of these networked devices is asynchronous, scattered on various technological interfaces (the device, a smartphone, and a computer), and it relies on your history of interactions with them. This situation is new, potentially leading to original creative avenues.

Input-Output Diversity

The last dimension I want to cover here concerns the increasing richness of user interfaces and how this expands the conditions for a gameful world. For this section, I take here the example of the cell phone to highlight this diversity. Smartphones are definitely not a unique apparatus for gameful apps, but they do constitute a prominent point of entry for them.[7] One of the reasons for this is the progressive inclusion of new capabilities in terms of human–device interaction. If ten years ago cell phones mostly consisted of a keypad, a microphone, a speaker, and a tiny display, the addition of one (and sometimes two) camera(s), a touchscreen, as well as sensors widened the spectrum of possibilities. Table 16.2 shows an overview of the input-output configuration that can be found on standard smartphones such as the iPhone or any Android mobile device. This table presents a basic model of the user interface components available to designers of mobile applications. It includes the basic elements mentioned above as well as sensors described in the previous sections, as they can be considered as a specific form of input.

Therefore, the design space for gameful apps corresponds to the set of possible combinations using these elements. For example, Foursquare requires the user to press buttons on a keyboard (Blackberry), tap on a touchscreen (iPhone, Android), or use geolocation sensing (GPS) as an input; and the output consists of the phone's screen and possibly a second screen (web interface on a laptop/tablet), plus sound or vibration when receiving notification of presence from contacts. An app such as Waze, which we described previously, uses fewer components: geolocation sensing (GPS), a touchscreen, and the phone's screen. The framework presented in table 16.2 could be used to analyze the current state of smartphone

Table 16.2
An overview of the input-output diversity

	Input							
Output		Physical Button	Keyboard	Touchscreen	Microphone	Camera (video)	Camera (still)	Sensor
	Phone screen							
	External screen							
	Pico-projection							
	Speaker							
	Vibration							

app interfaces. Notably, such an analysis may demonstrate issues commonly encountered with design spaces: the scarce usage of certain components and the overemphasis on others. In the context of mobile applications, and gameful apps in particular, one might find a small number of services that use the microphone or the phone button, as if these interfaces were less relevant now that phones are more complex than what they used to be. If one takes prominent examples of current gameful apps on cell phones,[8] this is exactly the situation we have now.

Moreover, the currently low usage of cameras in gameful apps is intriguing and highlights an interesting avenue for the near future. This situation is almost paradoxical given the massive use of this feature of cell phones.[9] The playful potential of cameras on cell phones has been shown by a game such as Noticings (2010). Developed by Tom Taylor and Tom Armitage, it is a casual game for smartphones aimed at "learning to look at the world around you." The user was supposed to take a picture, upload it to Flickr, tag it with "noticings," and geotag it. Rules changed on a regular basis, but players were awarded with points for noticing certain things, being the first to spot a certain situation in a neighborhood, and so forth.

Conclusion

Considering the shift from experimental pervasive games and playful prototypes to commercial services, this chapter has described three main technological components of the gameful world. More specifically, I focused here on three dimensions: identification for use of such services, sensors aimed at creating an original experience, and input-output interfaces. The description of these components highlighted variations between different systems, sketching a design space represented in table 16.3. Moreover, I have also shown how addressing issues related to sensing and identification expands the notion of input interfaces. Table 16.3 summarizes the elements I derived from the cases I described and highlights frequent or less frequent uses. The latter can be seen as logical opportunities[10] with original avenues for gameful apps.

Writing this essay also made me wonder about the small number of empirical studies concerning how the technological dimensions not only influence gameful systems but also reach beyond into the play-

Table 16.3
A summary of the design space

Component	Commonly Used Dimensions	Less Common Dimensions (Opportunities)
Identification	Cell phone (Foursquare, Waze), object designed specifically (Swinxs, Nike Fuelband)	Transportation card (Oyster), biometric data, face recognition
Sensors	Location (GPS, WiFi), movement (accelerometer), presence (WiFi)	Environmental sensors (temperature, noise, humidity, pollution)
Input-output	Button, keyboard, touchscreen, screen, sound	Microphone, camera (Noticings), pico-projector, vibration

er's everyday-ness. Although most of these applications rely on components also used by other systems (e.g., location awareness), it is hard to generalize from existing studies as they do not take into account the idiosyncrasies of the context in which these apps are deployed.

The next logical step in this analysis would be to adopt a similar perspective at a deeper level by addressing two other components, which are less visible from the user's perspective. On the one hand, as proposed by Montfort and Bogost (2009), we should take the hardware and software deployed in the gameful world as platforms and analyze them as such in order to investigate the relationships between their design and the creative works produced on those systems. On the other hand, it would be relevant to investigate how infrastructures (types of networks ranging from WiFi to 3G or LTE, electrical grid, etc.) play an important role in shaping the gameful world.

Notes

1. We'll leave aside the discussion about pets and animal users in projects such as Pig Chase.

2. Although the card must only be used by its regular user, this situation is common.

3. A space-based satellite navigation system that provides location and time information anywhere where there is an unobstructed line of sight to three or more GPS satellites.

4. A positioning technique based on the measurement of the difference in distance to two or more stations at known locations that broadcast signals at known times.

5. This is done automatically by the system. For safety reasons, no interaction with the app is required when driving.

6. One of the first location-based games developed as a commercial product and created by Newt Games, a French start-up, which released it for the Japanese market.

7. In addition, most of the existing gameful apps imply cell phone usage.

8. For instance, Foursquare, Bunchball, Badgeville, SCVNGR, Nike+, Nike Fuelband.

9. According to cell phone analyst Tomi Ahonen (2012), by the end of 2012 more than five billion mobile phones will be in use around the globe. Among these will be 4.4 billion camera phones and 1.2 billion smartphones. In addition, in the United States, according to the Pew Research Institute, 82 percent of the users report using their cell phones to take pictures.

10. Depending on the parameters selected, different ethical issues may appear and forbid their use in playful context. Think for instance how face recognition can be problematic.

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POSITION STATEMENT

BOT-MEDIATED REALITY

Daniel Suarez

Back in 2004, I finished the first of two tech thrillers that pondered how narrow-AI news-reading software bots might realistically be used to orchestrate human activity—for good or ill. *Daemon* and later its sequel, *Freedom™*, drew on existing technologies combined with game-world mechanics to relate the tale of a deceased online game designer whose computer daemon[1] monitors the Internet for the appearance of its creator's obituary. When this man dies, the titular Daemon activates. Once launched, this Daemon exploits the inherent vulnerabilities of our complex, high-efficiency world to spread—becoming a parasitic digital organism living within critical corporate networks. Monitoring online news headlines, this Daemon reacts to human activity, steals funds, recruits human agents to achieve preset goals, assigns tasks, monitors results via news headlines, distributes rewards, and threatens destruction of corporate hosts in response to any efforts to remove it.

In the story, a key aspect that makes the Daemon different from standard malware is its recruitment of human followers and creation of an encrypted social network to serve them. Social interactions between members are conducted via this *darknet*[2] and depicted through augmented reality objects projected onto the real-world GPS grid—like a game map. Daemon operatives use heads-up display glasses to interact with virtual darknet objects in much the same way players in MMORPG fantasy games interact with in-game objects, and those objects likewise possess properties and capabilities to draw on network and real-world resources. Thus, a virtual button might be used to open a real-world door, or to move real-world funds from one bank account to another, or to fire a real-world weapon. The benefits for a subversive criminal organization to orchestrate social activity via invisible, virtual layers of shared experience are manifest, as the public and law enforcement alike are unable to see into this world, even if they occupy the same physical space.

Similar to common adventure games, Daemon operatives "level up," rising in power and prestige by completing "quests" that the Daemon presents. These quests are initially goals established by the deceased game designer, but once operatives reach higher power and security levels, they too obtain the ability to create and assign quests for lower-level "players"—perpetuating the system and permitting it to evolve over time. Daemon operatives begin like players in most games—at first level. However, as they complete quests, they earn points, "leveling up" at established benchmarks and acquiring additional powers and responsibilities.

Likewise, they choose class specializations. However, unlike typical games, the player classes generally don't comport with fantasy game tropes. Instead, character classes embrace the gamut of modern, real-world professions. Thus, a seventh-level journalist might interview a tenth-level coroner about the demise of an eighteenth-level electrical engineer.

The Daemon game dynamic includes what's termed a *shamanic interface* for wielding network power. Thus, rather than a point-and-click computer interface for summoning aid or deploying weapons, Daemon operatives speak coded phrases into microphones, use somatic gestures in front of cameras, or present material components to "cast" what are essentially "spells" in a modern reproduction of a timeless ritual for unleashing "supernatural" power. In truth, this archetypical interface was chosen specifically for its cultural portability—allowing the Daemon's network to spread across all cultures, from advanced to technologically backward societies alike, without the necessity to conduct extended training for its adherents. Thus, a stone-age level shaman could immediately grasp the "magic" that the Daemon conveys upon its followers. An immediate example would be the "data curse" spell—which sours the meta-data of any modern citizen, causing his or her data in government and corporate servers to contain detrimental entries (criminal records, outstanding debts, or other security flags)—which even when fixed will almost immediately revert once more to detrimental values.

Currency within the game world of the Daemon becomes network credits, which are distributed as rewards for service and which can also be converted to real-world wealth or material goods at a current conversion rate (much like Bitcoins).

Political power within the Daemon's darknet is highly distributed in nature, with no single leader. Instead, operatives "rate" each other in various skills after interacting directly, with the average of all their ratings determining their overall score (much like a credit score). Individual ratings can be disputed through an anonymous arbitration process.

One's ratings in various categories come into play when it comes time to wield political power. The news-reading software bots underlying the system constantly scan and sample news from the public Internet and the darknet. These news items are parceled and distributed randomly to individual darknet operatives on a daily basis—meaning the Daemon does not need to understand these news items itself. Instead, these news items are rated by operatives for perceived level of importance to the overall Daemon community and further categorized as part of an operative's civic duty. Each item is directed to, rated, and categorized by multiple operatives. Items that are ranked "high importance" are then sent to the highest-rated, highest-level subject matter expert in that category. Thus, if news of flood damage comes across the network and is quickly "up-voted" by many operatives, it might be assigned to the nearest flood control expert with the highest rating. That operative would then have an opportunity to assign a network credit "budget" for the event, with the amount being increased/decreased by the preponderance of up/down votes of peer experts.

The goal of the Daemon's system was to employ a game dynamic in combination with algorithmic dissemination of information, while still leveraging human reputation and crowdsourcing when it came to comprehending and categorizing reality. In effect, only humans would be capable of deciding what was

important and what was not, but by easing the flow of information and constantly rotating decision-making to members deemed reliable by others (a rating that can quickly go down for nonperformance), a swift yet democratic structure was enabled. However, this political authority is pointedly ad hoc—assigned only as long as needed, and immediately removed from the operative's grasp the moment it is no longer necessary.

Whether such a gameful social system as the darknet would gain traction is difficult to say, but I intended the fictional Daemon as a cautionary tale. Specifically, the rise of a distributed, decentralized, digital monster like the Daemon is a critique of modern society's pursuit of maximum efficiency to the detriment of almost everything else. Monocultures, while fantastically efficient, tend to be susceptible to parasitic infestations, which spread ruinously once the infecting agent finds a weakness. The high-tech revolution has incentivized global financial markets and multinational corporations (and even governments) to pursue unprecedented automation and centralization in an effort to minimize costs and gain actionable intelligence on human activity. The result is a monoculture that, while advantageous in the short term, is susceptible to disruption in the long term. Yet, as later portions of my story illustrate, gameful social interactions can still work in a more resilient social architecture—one that employs the same decentralized architecture as the Daemon itself. Thus, technology itself is not the source of system vulnerability, but rather the architectural decisions made.

Precisely where these stories fit in the canon of utopian/dystopian technology literature only time will tell. However, game mechanics prove uniquely suited for bold social experiments because games are all about exploring possibilities. And as smartphone use spreads globally and projects like Google Glass, Metaio, and many more become available, darknet-like gameful worlds will become not only possible, but increasingly inevitable.

Notes

1. Daemon (dē′mun) n.—A computer program that runs continually in the background and performs specified operations at predefined times or in response to certain events. Condensed from the term disk and execution monitor.

2. *Darknet* is technical jargon for an encrypted network, so-called because to outsiders the activities within the network are not visible.

III APPLICATIONS

Whereas parts I and II of this volume angled the gameful world from the vantage point of different theories and issues, part III charts important areas of application: product and interaction design, business, science, politics, cities, sustainability, health and well-being, and education. This does not mean that the following pages lack conceptual depth: to articulate the peculiarities of an application domain, the authors often substantially draw on theory. But the focus is explicitly practical. Hence, the chapters provide ample case studies and reference existing empirical data wherever possible.

Part III is opened by game designer Jesse Schell, whose presentation "Design Outside the Box" at the 2010 game industry conference DICE has become a central reference point in the gameful world discourse. His talk painted a near-future scenario for game design he dubbed the "gamepocalypse": governments and businesses will track and influence each action of our everyday life through game-like incentive systems, down to earning redeemable points for brushing one's teeth (with a specific brand of toothpaste). In the current position piece, Schell situates the development of gamification since 2010 within a larger "pleasure revolution" toward design for pleasure and highlights some of its main problems and oversights.

Interest in pleasure and how to design for it may be on the rise, but as Jussi Holopainen and May Stain demonstrate in their chapter, it has a long and rich history in human–computer interaction. Reviewing the main research findings and models on pleasurable experiences and playfulness, they draw on the anthropology and psychology of play and flow to formulate a "caricature principle" describing eight characteristics of playful behaviors and interactions and a seven-step design process for putting it into practice in experience design.

Next, emerging technology economist Ethan Mollick, coauthor of the early *Changing the Game*, and Kevin Werbach, legal scholar and coauthor of the book *For the Win: How Game Thinking Can Revolutionize Your Business*, survey the historical and contemporary use of games and gamification in the enterprise. Starting in the 1930s with the classic Hawthorne studies, scholars have long noted the ubiquity of games at work. Spontaneous games, developed on the shop floor and by service workers, were designed to beat the "beast of monotony" while providing meaning in repetitive, boring work. With today's gamification, games are imposed by managers and increasingly aimed at white-collar workers rather than manual laborers. Using a series of case studies, Mollick and Werbach examine the ways in which

organizations are using games to improve employee mood, motivate performance, and incorporate the work of individuals outside the organization, and they shed light on the advantages and disadvantages of games in the workplace, from the perspective of firms, employees, and user communities.

Their perspective is complemented by JP Rangaswami's "When Peers Select Tasks and Teams." Currently chief scientist at Salesforce, the influential technology futurist, speaker, and blogger argues that historically, management hierarchies were created in order to solve problems like setting priorities, allocating resources to tasks, monitoring progress, and intervening in case of conflict or problem. As technology turns economies and organizations into flat networks, some of these problems may be solved more effectively at the peer level. Drawing on more than thirty years of experience with large organizations, Rangaswami suggests that video game design and multiplayer gaming hold important lessons for arranging such peer-level team and task selection.

One essential technical enabler for this shift toward peer organization (or the involvement of user communities that Mollick and Werbach speak of) is social media: a combination of direct user interaction, dependence on user-generated content, and technical tools that regulate processes. In "Gamification and Social Media," online communication researcher Cliff Lampe details how the two intersect. Lampe analyzes two early social media sites, Slashdot and Everything2, to illustrate how their design exhibits features that today have become ubiquitous in social media—and were inspired by games. Long before the rise of gamification, social media sites have included tools like profiles and reputation systems in order to provide important identity cues and motivate user contributions. Thus, social media and gamification share common genetics, embedded in the need to use features of sociotechnical systems to provide the cues, feedback loops, and identity information essential to facilitating a wide range of online actions.

Building on this analysis, and echoing the notions of JP Rangaswami, Deloitte chief edge officer Peter Williams draws on research findings of the Deloitte Center for the Edge to articulate how games and game communities can inspire the reorganization of businesses. Opening up to a critical mass of participants, instilling passion through interesting hard challenges, articulating a clear context of goals and improvement opportunities, nurturing a culture of sharing and transparency, and providing technical platforms to provide and share fine-grained, quantitative, real-time feedback as well as qualitative evaluations and strategies are the fundamentals that together enable mass collaboration and learning and innovation on an organizational level—both in online games and organizations.

Gameful mass collaboration is not only deployed by businesses. The much-publicized online protein-folding puzzle game Foldit—whose players managed to identify the folding of an AIDS virus enzyme that stumped researchers for fifteen years—demonstrates how games and gamification are also successfully used for mass collaboration in science. As Foldit co-creator Seth Cooper outlines in his chapter "Massively Multiplayer Research: Gamification and (Citizen) Science," games provide an excellent framework for combining what humans are good at with what computers are good at in the process of solving previously unsolvable scientific problems: games structure problems in increasing levels of difficulty, training novice players to the point where they can

contribute. Game feedback can make scientific problems engaging and approachable. Game communities can involve players in problems relevant to them who might not have been otherwise able to. Finally, games can allow members of the general public to participate in scientific research. Cooper surveys successful domains of such gameful citizen science, including image recognition and analysis, biological spatial and structural problem-solving, and graph- and constraint-based computer science problems, as well as their benefits beyond immediate problem-solving, such as scientific education.

Can the use of game design for collaboration in business and science be extended to coexistence more generally—to the organization of politics and governance? Few people would be better equipped to answer this question than Constance Steinkuehler, noted game-based learning scholar and former senior policy analyst at the White House Office of Science and Technology Policy, and Greg Lastowka, author of *Virtual Justice*, a survey of the legal dilemmas posed by virtual worlds and games. In their chapter "Game State? Gamification and Governance," they first describe how increased state investments in the study and utilization of games can improve public welfare, civic participation, and education, offering two case studies, Macon Money and Commons. Asking whether government itself might be gamified, they are more skeptical: instead of implementing existing gamification platforms, a more fruitful approach would be to study how (online) game communities are governed, and whether they hold less immediate, but also more relevant and applicable learning for governance outside of games.

A different bent on the intersection of politics and games comes from design researcher William Gaver. Revisiting his own influential work on noninstrumental *ludic design*—noninstrumental design that encourages exploration and reflection, remaining open to interpretation—he wonders whether such playful engagement is irresponsible in today's troubled times. What are the prospects of a ludic approach to political concerns? Gaver suggests that by "playing the idiot" with respect to entrenched positions, ludic designs may open us to new approaches to addressing serious problems.

Still concerned with how the gameful world might contribute to better forms of coexistence but focusing on our immediate day-to-day interactions in urban environments is Kars Alfrink, a designer of games for social change, many of which are interventions in public urban space. In his chapter "The Gameful City," he argues that gameful design can provide citizens with tools to counterbalance and resist the top-down planning of governments and corporations. By and large, these organizations prefer highly legible urban structures that tend toward fragility in the face of uncertainty—such as recent "smart city" solutions to problems faced by the new megacities. In contrast, urban sports, neo-situationist street games, playful platforms for civic organization, subversive art interventions, or participatory planning tools enable citizens to act as a generative force that injects the city with some much-needed illegibility.

Moving on, and quite literally so, design researcher Paul Coulton explores the use of mobile computing devices in games and playful experiences. Given their ubiquity, rich array of embedded sensors, and the relatively easy development and distribution of apps, mobile devices appear indeed ideally suited to gamification. However, this convenience does not make their design any easier. Contrasting the design of two prominent applications—Epic Win and Heineken

StarPlayer—Coulton demonstrates how game-inspired mobile design might either fall for a superficial imitation of game aesthetics or use the situatedness and affordances of the device to enable a tight coupling of situational tasks and game mechanics.

In "Gamifying Green: Gamification and Environmental Sustainability," HCI researcher Jon Froehlich charts the uses of gamification for promoting "green" behaviors. He begins by situating the gameful world vis-á-vis persuasive technology: long before the rise of gamification, game design elements such as feedback, progress tracking, goals, and rewards were used in persuasive technology to encourage proenvironmental behavior. If there is a novelty to gamification, then it is an emphasis on playfulness. After charting three domains of green gamification—home resource consumption, transportation, and waste disposal—Froehlich closes by discussing the ethical implications of green gamification and its potential to contribute to environmental sustainability.

Another prominent application domain of gamification is health and well-being. Just as green gamification strongly overlaps with persuasive technology and eco-feedback technologies, so there are strong commonalities between health gamification and personal informatics, popularized by the quantified-self movement. In "Gamification and Health," Sean A. Munson, Erika Poole, Daniel B. Perry, and Tamara Peyton critically compare and contrast the two. Drawing on a review of behavior change models and important factors for the adoption and maintenance of health behaviors, they analyze the philosophies, advantages, and drawbacks of personal informatics and gamification approaches to encouraging behavior change. Using case studies of exertion/exergames, compliance-oriented games, and educational games, they discuss the design space, potential mechanics, and attributes of games that shift people's health behaviors and attitudes. Their chapter ends with a reflection on the considerations involved in deciding whether a game is an appropriate medium for health information and/or behavior change.

If there is one application domain for gameful and playful design more prominent than health and sustainability, it is education. A pioneer in the field is Katie Salen, coauthor of the magisterial game design book *Rules of Play* and executive director of the Institute of Play, a nonprofit focused on connections between game design, learning, and transformative modes of play. In 2009, she helped design and launch a new public middle school called Quest to Learn, located in New York City. Its mission: to transform teaching and learning practices through an approach integrating traditional and twenty-first century literacies with core principles of game design and play. The resulting school has become a vital research and development platform for innovative curriculum and teaching practices and a staple reference in the gameful world discourse. In "Learning to Pivot: A Play on Possibility," Salen explores some of the key design questions underlying the work, along with its founding motivations.

Salen's short musing is followed by Dennis Ramirez and Kurt Squire's chapter-length exploration of the intersection of gamification and education. Ramirez and Squire ground their analysis in their own field-defining work on games and learning, specifically the application of sociocultural learning theory to games. They highlight that many prominent examples of gamified learning embody the very same features of traditional schooling and assessment commonly criticized by educators and researchers proposing games in their stead. Still, there are multiple ways in which

well-designed achievement systems and other gameful design instances (e.g., quests and narratives) can promote learning. One such prominent use case for achievements or "badges" has been the accreditation of especially informally gained skills. Whatever the design element and use case, Ramirez and Squire see the main challenge in achieving a sociotechnical implementation that actually changes rather than reproduces the problematic aspects of the current education system.

We close the book with a reflection from maybe the most optimistic and audacious proponent of the gameful world, Jane McGonigal. In her book *Reality Is Broken*, the game designer and futurist claims that games may not only help us to maximize our individual potential and happiness, but also to energize and orchestrate mass collaboration to solve humankind's most existential challenges. In "I'm Not Playful, I'm Gameful," she asks what it means to be *gameful*, retracing how she came to introduce the term in her own work. On the one hand, gamefulness points to the specific strengths, virtues, or qualities that characterize how gamers approach the world: highly motivated and persevering in pursuing goals, creative and curious in finding ways toward them, loving learning, and deeply optimistic. It carries elements of playfulness, but ultimately differs in its goal focus. On the other hand, it describes systems designed to work and feel like a game, to help people develop the specific strengths of gamers in order to change reality outside the "magic circle" of play for the better.

POSITION STATEMENT

THE GAMEPOCALYPSE AND THE PLEASURE REVOLUTION

Jesse Schell

It has become very trendy to believe that *gamification* is a simple process that can be easily applied to any human experience to make it better and more enjoyable. Unfortunately, this isn't so simple, because games, and the reasons people play them, aren't so simple. So, what goes wrong exactly?

Pleasure Is Complex

People like games not because they are games, but because they provide a wide range of different kinds of pleasures: feelings of accomplishment, mental and physical exercise, a chance to solve a problem, feelings of freedom—the list goes on and on. If you would improve your experience, you must bring to bear as many interesting and unusual pleasures as possible.

Rewards Can Backfire

There is much subtlety in human psychology. Study after study shows that if your plan is to offer a reward to get someone into the habit of doing an activity, and then, once the habit is required, gradually remove the reward, this is almost sure to backfire. It turns out that when an activity has an external reward, we take it as a message that the activity isn't worth doing without it. So, in the long run, your short-term reward may cause people to do the activity *less* than if you had offered no reward at all.

Pleasure Is Contextual

What we find pleasurable at a given time has a great deal to do with the context and framing of the situation. Frequent-flyer programs count on the fact that business travelers have a mental context where they need to feel important. Coupon-clipping programs count on the fact that shoppers have a mental context

where they are trying to save pennies and not get ripped off. Switch these around, and you get certain disaster. The success of a motivational system is completely dependent upon the context in which it appears.

People Are Lazy

Systems that add a "layer of gamification" can easily go wrong by asking people to do more work to achieve the same results, but with "more fun!" If your pleasure-enhancing system is to be effective, don't ask people to do any additional steps—in fact, take away some, if you can!

People Are Easily Embarrassed

Game play, or, in fact, the experience of any kind of pleasure in front of others, can easily be embarrassing. Designers who would enhance the pleasure of a system can never forget this. Either figure out how to make it a private experience or find ways to be sure that zero embarrassment can result from people engaging in your newer, more pleasurable system.

People Get Tired of Games

Again and again, I hear proposals for plans to use games to change persistent, lifelong behaviors, such as neglecting to take vitamins or wasting fuel. But consider: has there ever been any game that you've played every day for more than a year without getting sick of it? People grow weary of even the best game in the world, and the expectation that somehow the game will stay fresh forever is an unrealistic one. This is because games are very often about the pleasures of novelty and the pleasure of exploring a broad range of strategies. Look to wider pleasures than this, pleasures involving pleasing interfaces and clear feedback, for example, if you would keep people using your system for a long period of time. These pleasures may not always seem "game-like," but making things game-like isn't really your goal, is it?

Cheating Makes Everything Collapse

One of the biggest challenges of making a game that is interwoven with real life is that real-life games are very cheatable. Perhaps you create an "exercise game" that involves taking a certain number of steps per day, monitored by a digital step-counter. What's to stop a player from tying the step counter to his dog's tail, and racking up hundreds of steps every time the dog is happy to see someone? The big problem with cheating like this is that not only does it mean the game doesn't work for the cheaters, it

also means that everyone who knows that cheating is possible feels like a chump—why work so hard when other players are probably cheating? Once players feel this way, their motivation can quickly dissolve.

Are these problems solvable? Certainly they are. But you have to take them seriously. In other words, pleasurable design is tricky! But if you can pull it off, you'll truly have created something that means something to people, something that people will enjoy a great deal, and something that can change people's lives for the better. Just remember, you can't go wrong if you stay focused on these two key questions:

Why will people like my experience?

and

How can I get them to like it more?

17 DISSECTING PLAYFULNESS FOR PRACTICAL DESIGN

Jussi Holopainen and May Stain

Our world is playful. The way human beings make sense of the world and interact with and within it is playful by the very nature of our physical and mental makeup. Without play there would be no art, no science, no sports, or any other kinds of entertainment. That world would be quite a boring place to live. Luckily, we are equipped to find play everywhere from pebbles on a beach to intimate relationships to internecine warfare.

It is possible to find playfulness everywhere, but it is also possible to make changes to the world; in other words, to design for playfulness. Some objects, environments, and activities prompt play more than others; there are recurrent characteristics and structures in playfulness. It is a designer's responsibility to tweak out, explicate, exploit, and abuse these playful structures. Playfulness can be understood as a personality characteristic, a psychological state of mind, or a feature of an environment or artifact. Much as designers might like to bludgeon people into having more playful personalities, they must settle for altering external factors to trigger playful states of mind.

In this chapter, we will offer guidelines for playful design after exploring the notion through theory with examples from two detailed case studies of playful experiences designed by spotov, a nonsense-monger based between Scotland and Berlin. We start with a brief review of existing frameworks for playful design and then proceed to argue that gameplay, understood as caricatures of intentional behavior, is at the heart of playfulness. This approach, together with insights from flow and reversal theories and the concept of closures, is used to reveal the underlying structures of playfulness. In addition to describing and explaining the recurrent characteristics of play, these structures can be used as a conceptual scaffolding to guide the design thinking in all phases of design work from ideation and concept development to final evaluations. At the end of the chapter, these insights will be summarized in seven easy steps for playful design.

Deterding et al. (2011) make a distinction between playful design, based on the notion of free-form play, and gameful design (gamification), which is more rule-bound, rigid, and "game-like." Though this chapter focuses on playful design, the guidelines and insights can be similarly applied to gameful design. *Gamification* understood as using simple game mechanics to increase the user engagement of existing systems or as a part of the bullshitting gamification industry (Bogost, this volume) is not in our remit. Rather, we hope that the guidelines and insights provided in this chapter will aid the design

of products, services, artifacts, and interventions that, through play, will help people to "perceive things from alternative angles" (Stenros, this volume), lose inhibitions, explore unconventional behavior, and prompt outbursts of *Homo ludens* (subspecies *politikos*) (Gaver, this volume).

Turning the Spotlight on Play

There has been a shift in the human–computer interaction (HCI) and user experience (UX) fields from focusing on usability and efficiency to talking about enjoyment, pleasure, and even fun. For example, Hassenzahl, Beu, and Burmester (2001) talk about the "joy of use" and "hedonic qualities," while Jordan (2000) builds on a model originally created by Tiger (1992) to categorize pleasure arising from products into four categories: physio-pleasure (physiologic enjoyment), socio-pleasure (socially related enjoyment), psycho-pleasure (pleasure related to the performance of the product, e.g., convenience), and ideo-pleasure (pleasure related to the person's ideologies; e.g., environmental values).

The approach of using elements found in games for making non-game applications more playful and fun to use is not new. At the beginning of the 1980s, Malone was already proposing that features which make computer games captivating could also be used in making other user interfaces interesting and enjoyable to use. Malone (1980) summarizes these features under three main categories: challenge, fantasy, and curiosity. Later in the same decade, Carroll and Thomas (1988) argued that taking fun seriously as an important aspect of subjective measurement of usability could lead to computer systems that people would really want to use (see also Carroll, this volume).

Funology (Blythe et al. 2003) also advocates enjoyable applications, demonstrating the use of design elements from games to make applications and products more pleasurable, fun, and playful. Many other researchers and practitioners have offered their own categories or heuristics for understanding how certain elements in games make them more interesting or fun. Hunicke, LeBlanc, and Zubek (2004), for example, propose a taxonomy with eight categories for discussing fun in games: sensation, fantasy, narrative, challenge, fellowship, discovery, expression, and submission.

Others have looked into how design elements from games can be used in non-game contexts. Lazzaro (2007) has proposed that the four forms of fun (hard fun, easy fun, people fun, and serious fun) she identified in games could also be used for creating more enjoyable user experiences in other products. According to Lazzaro, hard fun is challenging play, easy fun is open-ended fun "without purpose," people fun is enjoyment arising from social situations, and in serious fun the play has a serious purpose, for instance, getting more fit by playing fitness games.

These views, fine as they are as rough categorizations and conceptual anchors, are too broad and fuzzy for guiding concrete design work. How does one set about designing for "open-ended fun without purpose" when "fun" itself is such a slippery bugger?

With more emphasis on design, Kim (2009) has proposed five example game design elements to enhance fun in social software: collecting, points,

feedback, exchange, and customization. Similarly, based on previous research on what elements make games and play fun, Costello and Edmonds (2007) developed a framework of 13 "pleasures of play" that could be used to design and evaluate interactive installations.

Focusing on the actual design of "interactives" (interactive and playful installations), Polaine (2010) defined four principles for designers: invitation to play, playing fields and rules, flow, and delivering the promise. According to him, first the interactives have to provide clear and physical indications that playful interaction is possible. Second, the limits and rules of play have to be consistently evident throughout the interaction. Flow theory (see table 17.2 later in text) was developed by Csikszentmihalyi (1975, 1990) to explain optimal experiences in general, and it has later been used in discussing game-play experiences (Sweetser and Wyeth 2005) and media enjoyment (Sherry 2004).

Finally, condensing much of the theoretical work on pleasurable and playful experiences discussed above, Arrasvuori et al. (2011) propose a framework, PLEX (playful experiences), to categorize contexts, experiences, and activities that, according to the authors, are inherently more playful. The categories included in the framework are listed in table 17.1.

The framework and the associated methods have been used successfully in experimental interaction design projects (Arrasvuori et al. 2011). In one of the example projects, the PLEX cards (Lucero and Arrasvuori 2010), a deck of cards where each card describes one PLEX category, were used in generating ideas for a playful art installation. Later in the project, the same PLEX categories were used to check that the design had the intended playful features. The evaluation results indicated that having even a rudimentary conceptual framework helps in guiding the design toward playfulness. As can be seen from table 17.1, the possibilities of play encompass a wide variety of human experiences from nurture to suffering.

Table 17.1
PLEX categories

Category	*Description*
1. Captivation	Forgetting one's surroundings
2. Challenge	Testing abilities in a demanding task
3. Competition	Contest with oneself or an opponent
4. Completion	Finishing a major task, closure
5. Control	Dominating, commanding, regulating
6. Cruelty	Causing mental or physical pain
7. Discovery	Finding something new or unknown
8. Eroticism	A sexually arousing experience
9. Exploration	Investigating an object or situation
10. Expression	Manifesting oneself creatively
11. Fantasy	An imagined experience
12. Fellowship	Friendship, communality, intimacy
13. Humor	Fun, joy, amusement, jokes, gags
14. Nurture	Taking care of oneself or others
15. Relaxation	Relief from bodily or mental work
16. Sensation	Excitement by stimulating senses
17. Simulation	An imitation of everyday life
18. Submission	Being part of a larger structure
19. Subversion	Breaking social rules and norms
20. Suffering	Experience of loss, frustration, anger
21. Sympathy	Sharing emotional feelings
22. Thrill	Excitement derived from risk, danger

The frameworks and projects described above are useful in illuminating some aspects of playfulness as it is manifested in existing activities and products. The focus is, however, almost entirely on the experiential outcomes, not the foundations, of playfulness. To overcome this deficiency, we suggest a complementary approach based on the proposal that gameplay understood as caricatures of intentional behavior is at the heart of playfulness.

Playfulness Stripped Bare: The Caricature Principle

Play has been with us throughout our evolutionary history. Some reptiles and fish play (Burghardt 2005, 313–357), and virtually every mammal species exhibits play behavior at least in parts of its development (see, e.g., Fagen 1981; Burghardt 2005; Pellegrini 2009). Play for humans, however, is not limited just to a certain stage of development. We continue to play throughout our lives, although the forms of play usually change.

The evolution of games can be traced to the emergence of more and more elaborate play patterns in higher mammals. Game formats such as hide-and-seek or tag and game mechanics such as handicapping (allowing less skilled players advantages to balance the playing field) can be found in play behavior of many mammals. Some hominids and cetaceans even play "proto-games" such as racing and king-of-the-hill (Fagen 1981), where the play behavior seems to be governed by rule-like agreements between the participants.

Games are, then, the most refined forms of play behavior in human cultures. The structures of innate and spontaneous play behavior are crystallized in games as Jaakko Stenros also argues in his chapter of this volume.

At the heart of games is gameplay. Gameplay consists of the actions, activities, goals and failures, motivations and incentives the game provides to the players as stated in Björk and Holopainen (2005, 3). Analyzing gameplay as caricatures of intentional behavior provides a conceptual tool for a unified explanation of its disparate characteristics. Caricatures are transformed, exaggerated, abstracted, etc. forms and intentional behavior is broadly defined here as goal-directed actions and activities requiring a set of cognitive and sensory-motoric capabilities (Tomasello 2001).

This caricature principle is evident in play behavior in general and can be applied to actions, goals, and goal structures involved in the behavior. According to Burghardt (2005, 74), play behavior "differs from the 'serious' performance of ethotypic behavior structurally or temporally in at least one respect: it is incomplete (generally through inhibited or dropped final elements), exaggerated, awkward, or precocious; or it involves behavior patterns with modified form, sequencing or targeting."

Ramachandran and Hirstein (1999) claim that exaggerated forms are engaging because of the peak shift effect, which is a psychological phenomenon appearing in studies of animal learning. When an animal has been trained to discern between negative and positive stimuli on the same stimulus dimension, such as squares from rectangles, the animal response is greater to a stimulus further away from the negative stimulus than for the original positive stimulus.

For example, if a rat has been trained to differentiate squares (no reward) from 3:2 rectangles (reward), the rat will respond with greater intensity to a 4:2 rectangle than to the original 3:2 rectangle. It is a long leap of faith from trained rats to human beings appreciating visual arts, but there is further evidence for claims that caricatured forms can engage human beings more than the original forms. Barrett (2010) has claimed that many of the problems in contemporary societies are caused by our fascination with supernormal stimuli, ranging from exaggerated use of sugar and fat in junk food to exaggeration of social cues such as canned laughter in film and television.

Beyond fascination with exaggerated forms, Dissanayake (2009) proposes that caricature operations such as formalization, repetition, exaggeration, elaboration, and manipulation of expectations are performed on components of ordinary behavior to make them extraordinary, to stand out from the mundane. This "making special" is, according to Dissanayake, a culture-independent feature of all art forms from dance (exaggerated and repetitive movements) to poetry (formalized and elaborated use of language with manipulation of expectation in the form of suspense). Although Dissanayake's focus is on the arts, the same operations can be applied to design in general and designing for playfulness in particular.

In summary, the caricature principle modifies otherwise ordinary intentional behavior with at least one of the following features (case study 17.1):

1. Incomplete: Some elements of the behavior are dropped. For example, the play punch is suppressed before hitting the opponent's face.
2. Exaggerated: The amplitude or frequency of actions involved in the behavior is increased or decreased; in other words, the actions are larger, smaller, slower, or faster than normal. Dance often involves movements and gestures that are exaggerated both in amplitude and frequency.
3. Awkward: The actions and goals involved are arbitrarily transformed to be difficult to perform. For example, in Twister (Foley and Rabens 1966), the goal structures dictate that the players have to "twist" their bodies in awkward positions. The game is also set up in such a way that often the players' bodies will be in socially awkward positions (at least for mainstream Western societies).
4. Repetitive: The actions or action sequences are repeated as, for example, in dance. The goal structures are repeated even after achieving the goals. In tennis, the action of hitting the ball to the other side is repeated time and time again.
5. Formalized, abstracted, or compressed: The action sequences and goal structures involved are simplified, stylized, and both temporal and structural characteristics are compressed or reduced. In the children's game Tag, the actions relevant to the game are drastically reduced. When playing, the children are not supposed to start hitting each other with baseball bats but limit their actions to running. The relevant goal structures within the game are also compressed. There are just two goals: if you are "it," you have to catch other players, and if you are not, you have to evade "it." Other kinds of intentional behaviors are neither relevant nor allowed within the context of the game.
6. Elaborated: Unnecessary, and often arbitrary, features or embellishments are added to the action sequences or goal structures. In professional wrestling, the fight moves are heavily elaborated and embellished with superfluous actions.

Case Study 17.1

Abstract Operation

Summary

Abstract Operation was a popular participative installation game. The nightclub patrons were all garbed in surgical attire and required to extract fresh organs and human teeth through increasingly tricky installations of electrified implements; failure instantly triggering total blackout of the room. Success was rewarded by a personal reading from a specially devised medical tarot pack by the redoubtable Miss Fortune.

Facts and Figures

The game was staged in 2001, originally commissioned for one night only in a nightclub in Edinburgh and subsequently invited to a 24-hour event in the Arches, Glasgow's biggest venue. It is currently being resurrected and further developed with the view to touring it in different cultural contexts. The game was designed and staged by spotov and her collaborators.



BoxFigure 17.1
Organ harvesting.

Case Study 17.1
(continued)

Tricking Nightclubbers into Art through Play

The commission was for a room-sized game to transform the standard chill-out room option in a nightclub venue. The game was inspired by the Operation (Spinello 1965) children's game, which is in itself a variant of the electrified wire-loop game found at funfairs.

Rather than simply scaling up the game format using a gigantic body, spotov's version approached the concept abstractly. The setup primarily consisted of hospital trolleys that were wired with an array of authentic surgical instruments, syringes, and vaginal specula interlaced to create an electrified mesh enclosing genuine body parts, albeit animal not human. The organs were from sheep and pigs, but there was also a huge rotating ox tongue sticking out at you for good measure.

Varying sizes of forceps were connected as a trigger to a twelve-volt circuit. The participants were required to wield the forceps to extract the cumbersome bloody flesh through the lattice of metal implements without making electrical contact. Should this occur, it prompted jocular sound effects that interrupted the live DJ's record spinning and, most dramatically, blacked out the entire room so that your personal failure affected everyone present.

There was a performance feel to the entirety because, in addition to the game's supervisors, all the clubbers who entered the room were immediately dressed in hospital scrubs even if they only intended to nurse a pint while spectating. In addition, those testing their surgical prowess were required to don masks and latex gloves. However, it was not performance but a straightforward, if challenging, game that up to four people could play simultaneously, as there were four stages of increasing difficulty culminating in the extraction of genuine teeth through the eye sockets of a three-dimensional wire-frame head.

The initial reaction of some squeamish Edinburgh girls was to scream on entering the room. Once they were actually playing the game and concentrating on the precision task of extracting the organs, they would lose their initial horror and were ultimately even able to brave touching the offal with their bare hands.

The game overseers felt like they were providing a social service, helping people deal with their overdramatic response to things they would quite happily have eaten in another context and introducing them to the parts our bodies are composed of (leastways strikingly similar to our own).

The net achievements were double layered. In addition to helping people deal with bodily reality hidden under a layer of skin, spotov realized the play format unleashed the possibility of tricking people into "art" as the game was described in the *Sunday Herald*: "Truly one-off standouts include last year's life-sized conceptual-art version of the game Operation." spotov would have never tagged herself as an "artist," preferring to say she makes unique experiences for people. As she never had any formal artistic education, she is aware how utterly alienating the art world is for most of the populace. Games in public places prove to be a perfect formula for reaching outside of the elitist bubble to draw people in who would normally scoff or hasten by, enticing them into the thick of the spectacle. The nightclubbers did not perceive the event as art; they had participated in a slightly disturbing but very memorable game.

Case Study 17.1
(continued)

Playful Design Elements

The caricature principle works in Abstract Operation in several overlapping ways. Surgical procedures are formalized and abstracted into extraction of animal organs, which have already been removed. The manual precision required in surgical operations is transposed to the avoidance of electrical contact while extracting the organs. Furthermore, the surgery itself is transposed from the operating theater to a nightclub setting. Elaboration is evident in the dressing up of all the people present in the room with additional gratuitous provocative details such as the rotating ox tongue. Expectations are manipulated by luring people into art through disguising it as a game in a nightclub, and unwitting clubbers entering the chill-out room may randomly find themselves in darkness should a player happen to fail.

The game was played in stages of ever-increasing difficulty. Each success closure of removing the appropriate organ without touching the electric wires allowed the player to proceed to the next stage, ultimately reaching the fortune telling. In case of a failure, the sense of closure was heightened by sound effects disrupting the DJ and the blacking out of the entire room.

7. Transposed to a different context: Some or all features of the behavior in a certain source context are moved and adapted to a different target context. In air guitar[1] contests, the movements and gestures of playing a real instrument stay intact but the context changes from making music to making a mimetic show.
8. Expectations are manipulated: In many art forms, the manipulation of expectations, especially in the form of suspense, is the source of the enjoyment and engagement in the art piece. The same applies to almost all games, including sports and children's games. The uncertainty of the outcome is a crucial feature of the engagement—a game of dice where the outcome is already known to the players ceases to be a game at all. Surprise and unexpected events are manipulation of expectations at its simplest.

The flow experience (Csikszentmihalyi 1990) has similar characteristics to the caricature principle (table 17.2). The first five elements relate to the caricature principle, and the last three describe the experiential consequences of the caricature principle in effect. In normal day-to-day behavior, the goals are not necessarily clear, and it is not always evident if

Table 17.2
Csikszentmihalyi's eight elements of flow experience

1. Balance between challenge and personal skill
2. Possibility for focused concentration
3. Clarity of goals
4. Immediate feedback
5. Feeling of control over one's actions
6. Deep but effortless involvement
7. Loss of concern for self
8. Altered perception of time

the task can be completed or not. An exaggerated goal structure posits clear goals and explicit indications if the task has been completed. Feedback from user actions is often exaggerated and sometimes even transformed into graphic progress indicators such as progress bars. The caricatured goal structures, alongside a heightened sense of control over actions due to the artificial limitations by rules (Klimmt 2003), enable the player to concentrate on the goals at hand.

An example of the caricature principle in effect is Piano Staircase created by The Fun Theory[2] as an advertising initiative. In this urban installation, the stairs in a Stockholm subway station were transformed into the black and white keys of a giant piano. When someone stepped on a stair, the sound of the related piano key was played through the loudspeakers. The main caricature principle operation used was transposing the actions of playing the piano into going up and down a staircase. Other operations involved were exaggeration in the sheer scale of the piano keys and manipulation of expectations in their surprising behavior. The visual cue of the piano enticed many people to experiment with different ways of treading on the steps. Caricaturized perceptual cues in the environment can be used as *invitations to play*, sparking curiosity and potentially leading first to exploration and then to play (see, e.g., Holopainen 2012).

The Protective Frame

Reversal theory, which Apter (2006) and his colleagues have been developing since the 1970s to describe and explain playfulness as a mental state, claims that people can be in one of two mutually exclusive psychological states: telic (serious, nonplayful, goal-oriented) and paratelic (nonserious, playful, action-oriented).

These two metamotivational states determine the subjective assessment of hedonic tone (whether the experience is pleasant or unpleasant) for different levels of arousal (bodily excitement). In the goal-oriented telic state, a low level of arousal is experienced as positive relaxation, but the playful, paratelic state colors such low arousal levels as boredom. Conversely, high levels of arousal induce anxiety or even fear in the telic state, whereas in the paratelic state the feeling is excitement. Switching, inadvertently or deliberately, between telic and paratelic states can happen very rapidly. Jumping out of a plane with a parachute causes arousal levels to rise, regarded in the paratelic state as exhilarating. Realizing that the parachute has failed to open triggers a split-second switch to the telic state, and the high arousal is now regarded negatively as panic.

Crucial to the discussion of the paratelic mode is Apter's concept of a "protective frame." He argues that the person in a paratelic state needs to have a protective frame in place in order to enjoy situations that might otherwise be considered as a direct threat (Apter 2006). The classic example is that a tiger without a cage causes anxiety and fear, a cage without a tiger is plain boring, but a tiger with a cage can be exciting. The cage provides a protective frame and induces the paratelic state (figure 17.1).



Figure 17.1
Lulling into a false sense of insecurity—view from the coffin before the smashing starts in case study 17.2.

There are three types of protective frames: physical, social, and reversibility. The physical protective frame is the most fundamental, providing a feeling of physical safety even though the activity itself could be perceived as something extremely hazardous. Roller coasters and other thrilling amusement park rides pay enormous attention to the safety, and especially the feeling of safety, of their passengers.

Social protective frames shield people from embarrassment, awkwardness, and loss of face in social situations. The anonymity of online chat forums can provide a protective frame for social interaction, fostering such playful behavior as trolling and flaming. Silly playful activity in a public place may provoke awkwardness when alone. This is considerably diminished or even reversed when in company, bolstered by the social acknowledgment of others. The actions are also given more legitimacy in the minds of the passersby. Flash mobs are an example of this safety in numbers.

The reversibility protective frame means that actions either have no or very little effect outside the play itself or that the possible effects can be reversed if need be. Any kind of pretend play from children's fantasy games to elaborate live-action role-playing games uses reversibility. The child playing the evil witch does not remain one after the game is over.

The frame does not need to be real. The subjective assessment of a situation as being within a protective frame may suffice. Cultural expectations can also provide protective frames. The theater audience expects not to be hurt even though the actors are engaged in rampant violence on and off the stage.

Case Study 17.2

Dislodge of the Mid-diluvian Order of the Buffaloes



BoxFigure 17.2
Buffaloes play the game.

Summary

This ambiguously framed experience of participative and immersive theater for a sole male audience member took the form of an initiation into an obscure gentlemen's club, a cult of the mundane under threat of imminent demolition.

The initiates were simply furnished with an address, time, and a coded knock with no knowledge of what it was they were attempting to join. They were alone for the majority of the thirty-five minute experience. Always hooded in between each of the Degrees, they never learn who are the guides maneuvering them into all manner of extraordinary positions. The only human they encounter is an eleven-year-old girl who blindfolds them on arrival.

Facts and Figures

Dislodge of the Mid-diluvian Order of the Buffaloes (DMOB) took place in a disused and disintegrating Edinburgh lodge building of the Royal Antediluvian Order of the Buffaloes in 2007. Its construction took spotov and her collaborators six months with no budget whatsoever. Twenty-two volunteers helped to varying degrees, and it required a minimum of twelve people behind the

Case Study 17.2
(continued)

scenes to operate the experience for the single audience member.

The Experience

An unenlightened wretch gives the preordained knock on the unprepossessing door. A janitorially uniformed little girl bids him sign his name, testifying he has come of his own accord, before blindfolding him.

He is guided by an indeterminate number of arms up winding stairs then instructed to kneel. The blindfold removed, he is faced with an embossed golden buffalo set in the masonry rubble he is kneeling upon. Gleaming white porcelain urinals stretch infinitely into the blackness either side of him, and in each, impossibly perched on jets of water, are glossy pickled eggs. Behind his back, male Scottish voices simultaneously intone abstruse "rules," and the line of urinals begins to waver and undulate in vertebral fashion forcing the form from one parabolic curve to another while the myriad cisterns ascend like cable cars.

"Do you accept?"

The instant he voices his acceptance, a black sack is pulled deftly over his head.

Steered to a stiff-backed chair, something is bound around his upper arms with his elbows bent back until his hands meet his shoulders. He is dehooded in darkness as a disembodied amplified voice instructs him to select a sacrificial "degraded wretch" using his elbow tongs. Low light emanates from a vibrating booth gabled by a domineering set of buffalo horns and housing a tiered mechanism with wee gherkins rotating on beds of fine-grade rubble.

As the initiate leans forward in the endeavor to extract a "wretch" with the forceps extending from his elbow joints, his head is encompassed by the protruding electrodes of a heavyweight set of slaughterhouse stunning tongs. Triggered by his success, in frenzied erratic sequence, large gherkins begin to glow, fizz, and smoke with electricity on the wires framing the stunning booth.

In a softer environment, something is pulled over his head restrictively encasing his entire body. Through a misty slit, he discerns decorative ribboned straw hats and fabric plaits continuing ad nauseam in every direction, veering upward on one side, inverting itself endlessly on another. Mirrors obviously, but where is the viewer? There is an armchair with salmon pink upholstering and a dreadfully chintzy cushion. Gradually it dawns upon the bound observer that this cushion is what is encasing his head, affording him vision through one of its lace strips. He has been made into the fabric of the floral chair in this nightmare vision of mass-produced fake handicraft. Hypertrophied artificial flowers, lace froufrou-ed baskets, and plaited pelmets begin to writhe about with sickening motion repeated to infinity wherever he twists his cushion head. The ceiling rose pinioning the swathes of cloth flaunts the buffalo's face staring back.

Led in his sack away from this purgatory down steps, which degenerate the further he descends, he is requested to lie down. A blinding flash imprints the buffalo's image on his retina. Then flickering light reveals the inside of a padded tapering box akin to a coffin with a broken window. An oscillating light source

Case Study 17.2
(continued)

discloses his horizontal position to be at the bottom of a deep, bleak, concrete, spiralling stairwell, then the coffin light erases the stair perspective entirely. When glimpsed again, looming high above him are four silhouettes, redolent of the buffalo horns just etched into his brain, staring down. Then, resonant smashing and dull thuds, as crockery rains down upon the shattered window for far too long and only a few centimeters from his eyes. Bassy rumbles reverberate through the concrete enclosure from the merciless juddering of the banisters as rubble obscures the window and all light is extinguished.

He is whisked in a wheeled chair toward the sound of men singing, and the new initiate emerges from the hood again in the smokey Members' Bar. Buffalo headdressed men in full regalia are engaged in an ambiguous game, while masonry dust spatters their gaming table. Other distinguished buffalo gents are engrossed with arcane scientific apparatus, unperturbed as more and more of the ceiling crumbles away and rubble crashes down, clattering discordantly on the keys of a salmon pink piano over which the ceremonious buffalo head reigns supreme.

Bagged for the last time, he finds himself again with the little girl who is desperate to try and coax some of the secrets out of him before he is ejected blinkingly into the real world. He has no one to share his improbable story with.

Playful Design Elements

From the elaboration of a mustard-sprouting carpet to the awkwardness of extracting a rotating gherkin with elbow tongs, DMOB demonstrates innumerable examples of the caricature principle.

Apart from wielding the tongs, physical movements are kept rigorously in check. Mentally, the participant is turning somersaults to make sense of the overwhelming nonsense. Whether or not he chooses to immerse in the pretense of being an initiate, the provision for his total immersion in the illusion engenders playful perspective alterations during and well beyond the experience.

Expectations are manipulated on every level at every step of the way from the instant of the initial blindfolding in the form of suspense, surprise, and unexpected and distorted perceptual cues. A particularly striking example is the stage where the participant is unwittingly camouflaged as an armchair, creating uncanny suspense of disassociation and perceptual confusion, which is then later resolved when the participant realizes that *he* is the armchair.

Establishing the Protective Frames Beyond the culturally established protective frame, DMOB establishes a further frame at the outset by using a little girl who first blindfolds the participants and utters the words, "If you do as you're told you'll come to no harm."

Intentional Ambiguity of the Protective Frames A key source of excitement is the abuse of the protective frame. When the participant is enclosed in a coffin, he has the protective frame of his cultural expectations to bolster him. Then the missiles of crockery and debris begin to pelt down literally in his face, and the lighting

Case Study 17.2
(continued)

design prevents him from seeing the secondary layer of more robust Perspex protecting him in the coffin. From the initiate's perspective, he is being buried alive under an already broken window causing faith in his safety to waver. As illustrated by the description from *Times* critic Robert Dawson Scott, "This is not amusing anymore, I am genuinely frightened," the expectation of a culturally established protective frame has been shattered alongside the window. The resulting ambiguity of the protective frame leaves the participant hovering between telic and paratelic states, between excitement and acute anxiety.

Further Information

See http://www.spotov.org

Closures

In the traditional game model, games end in a quantifiable outcome: the player either wins or loses the game (Juul 2007). However, all games contain many smaller outcomes that are meaningful for the players. Filling a row in Tetris (Pajitnov 1984) or finishing a level in a first-person shooter inspires a sense of achievement, but losing a life in an arcade game gives players a sense of failure. Björk and Holopainen (2005) call these quantifiable (sub-) outcomes closures. Holopainen and Meyers (2000) distinguish between dramatic and predictive closures. Pertinent here are all possible "dramatic" closures, both minor and major, which can happen while playing the game. From the psychological and enjoyment point of view, the subclosures can be as significant to the whole game experience as the final outcome. Closures are meaningful changes in the game state that prompt a sense of achievement or failure in the players.

The checkmate in chess is a closure as is eating a pill in Pac-Man (Namco 1980), although the latter is of less significance and thus diminishes the sense of achievement. The closures are closely related to the goal achievement and goal progression in the game. Players can also set goals for themselves that are not dictated by the game system. For example, The Sims[3] (Maxis 2000) series has vague, open-ended, and implicit goals formulated by the game system itself, but players often, if not always, construct their own goals, and these change during play.

The closures in games follow the caricature principle. The goal (and subgoal) achievement closures are often associated with exaggerated and artificial rewards, such as grandiose audiovisual effects. The failure closures are equally conspicuous. Sometimes failure is even pointed out with near-miss indicators (Björk and Holopainen 2005) explicitly telling players that they were very close to reaching the goal but still failed.

Ravaja et al. (2004, 2005, 2006) have made psychophysiological studies about game events that elicit different types of emotions such as joy, grief, and fear. Their findings suggest that both the lower-level achievement and failure subclosures are important

in shaping the player experience. Reaching different goals in Super Monkey Ball (Amusement Vision 2001) elicits joy or sense of achievement according to the relevance of these goals to the players. Picking up one extra banana is not as significant as a whole bunch of bananas, which in turn is not as significant as successfully completing a level. All these closures are associated with extra rewards and audiovisual closure indicators. The same applies to failures: when the monkey falls off the cliff, it precipitates strong vertigo-like visual cues and even a replay of the events leading to the fall.

Closures and subclosures are integral to intentional behavior. They shape the experience into meaningful achievements and failures. In games, the closures are caricatured; both achievements and failures are exaggerated and often explicitly displayed to the players.

How to Make Things Playful in Seven Easy Steps

In the grand tradition of self-help books and with due deference to the twelve-step program, we now offer clear instructions for making things playful.

Select a Context

The context includes both the physical and social environment and the everyday behavior you are going to make playful. Let's call this the target context. If you are focusing on only caricaturing the behavior, it is possible to ignore the physical and social environment altogether. The selection can involve arduous studies of how people behave in the context but is usually done on the basis of a hunch. In either case, you have to dissect the behavior into the most meaningful action sequences and goal structures. Ask yourself: What are the basic actions? What are the goals, both explicit and implicit, for the participants? Why are these actions and goals meaningful? Why bother at all?

Apply the Caricature Principle

Go through the list of eight caricature operations and think how these could be used to change the selected behavior in the context. If the behavior is specific to the physical and social environment of the context, you may want to use the *transpose to a new context* operation. In this case, you need to select a source context. It helps if the source has properties facilitating related forms of action. In the Piano Staircase example, both keys on a piano (source) and the stairs (target) enable downward contact. Note that the source context could be established first. The context can be from a preexisting playful source such as a children's game.

Keep in mind that in addition to molding behavior, the caricature operations can be applied to all aspects of playful design in contexts from public toilets to symphony orchestras (figure 17.2).

Establish Protective Frames

Without some sense of a protective frame, otherwise playful activities may cause alarm and distress instead of enjoyment and engagement.

The physical protective frame is the most straightforward to think about, even though the mechanisms and technology involved can be complicated. It is



Figure 17.2
The first mystery of the pickled eggs in the eternal urinal—distorted perceptions in case study 17.2.

important that the players feel safe even though the activity itself is dangerous.

One way to establish a social protective frame is to ensure that the players believe themselves to be unobserved or that their real identity cannot be established by those present. Another way is to give cues both to the players and other people around that the activity is extraordinary despite happening in a normal social situation. Simply demarcating the place of play with visual cues such as a chalk line can be enough.

For the reversibility frame, the players should have certainty that actions can be undone or that it is possible to try again. The sense of reversibility leaves more space for playful exploration and trying new things out.

Play with the Protective Frames

Protective frames do not need to be static or thoroughly explicit. The ambiguity of the frames can be used to create engaging, even frightening, experiences as demonstrated by the DMOB case study. The emphasis is on the ambiguity, though. If the protective frames are totally wiped out, the players will most probably be anxious and stressed rather than engaged. Candid Camera is a good example of both playing with the protective frames and using the *manipulation of expectations* caricature operation. First, something extraordinary happens in a social situation without any discernible protective frames. This leaves the people involved in a state of confusion and even anxiety. The situation is later resolved when the gag is revealed to them. The protective

frame is only established after the event, but it is established nonetheless, and the participants tend to switch to the playful mode.

Think About Closures

Not all playful activities require explicit closure structures. It is worthwhile, however, to explore the consequences of giving the players possibilities for explicitly achieving or failing to do something. Note also that closures do not need to be about winning and losing, nor do they need explicit goals. A sense of achievement can come as a surprise, using the *manipulation of expectations* operation, as in the camouflaged armchair scenario in the DMOB case study. It is usually good to emphasize achievements and downplay failures, but this depends entirely on what you want to accomplish with your playful design. Emphasizing the failures may make the whole experience more playful.

Check with Flow

Go through the first five elements of flow experience (table 17.2): there is a balance between challenge and personal skill, concentration on the task is possible, there is immediate feedback, there are clear goals, and there is a sense of control. Check if these are relevant in your case. Note that some elements are strongly related to the possible closure structures available. Try thinking the other way around: what would it mean if, for example, the players do not get immediate feedback or that they have a sense of losing control?

Implement, Test, Refine, Iterate

The playful design should now be in place. You just have to make it happen. Depending on the nature of your design, this might require a lot of technical development, prototyping, and what not. The design and implementation details are out of scope for this short list, but luckily there are plenty of helpful guides available; for example, Bill Buxton's excellent *Sketching User Experiences* (Buxton 2007). The main point is to keep refining and testing the implementation until the experience is in place or you run out of time and resources. Refer back to the previous steps during the refinement process.

Done! Now run along and play, boys and girls!

Notes

1. See http://en.wikipedia.org/wiki/Air_guitar.
2. See http://www.thefuntheory.com/piano-staircase.
3. The Sims is a simulation video game of suburban daily life where the players create virtual people and then indirectly control their behavior.

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18 GAMIFICATION AND THE ENTERPRISE

Ethan Mollick and Kevin Werbach

For the past century, scholars examining workplaces have repeatedly discovered that workers regularly play games at work. Circumstantial evidence of the role of workplace contests and games dates back even further—to the building of the Egyptian pyramids themselves (Edery and Mollick 2009). Supplementing this long history, firms have begun consciously to apply games, and game dynamics, to an extremely wide range of uses, from innovation to recruiting. This process began before the recent *gamification* movement[1] and has continued to accelerate. However, the range of functions to which games are applied and the variety of gameful approaches used by enterprises make it virtually impossible to develop a complete synthesis of enterprise games in a theoretical sense. Instead, as business scholars, we begin from the proposition that games, like other technologies and processes adopted by industry (Gatignon, Tushman, and Smith 2002), are being tolerated or selected, at least in part, because of assumed positive outcomes that match the strategies and objectives of firms. In this chapter, therefore, we identify major rationales for games and gamification in the enterprise based on two recent books on the topic by the authors of this chapter (Edery and Mollick 2009; Werbach and Hunter 2012). We use that typology to analyze the effectiveness and other implications of an increasingly gameful workplace. Based on our review of games currently being developed or deployed by enterprises, we believe that four hoped-for outcomes appear to be driving adoption and interest in enterprise gamification: (1) relieving boredom and increasing satisfaction; (2) increasing performance; (3) encouraging unremunerated work from internal and external communities; and (4) dealing with fundamental human resources functions.

To date, there have been very few academic studies of gamification in the enterprise, regardless of the goal of the gamification effort. Though workplace games have a long history, conscious adoption of enterprise gamification is, for the most part, a very recent phenomenon. It is therefore too early to assess in any universal way whether any of the four desired outcomes are being met. To provide a framework for future work in this important area, we have assembled a number of individual case studies, theoretical arguments about the effects of games, and related scholarly work that build a case for the role of gamification in the enterprise. In this chapter, we attempt to bring together these sources of support to develop a picture of the use of games within firms, the current state of enterprise gamification, and potential future directions for both managers and researchers.

This overview comes with three important qualifications. First, as already noted, any attempt at a comprehensive overview of gamification in the workplace is necessarily limited by the sheer diversity of situations in which games appear and the ways in which gamification is being applied. Because spontaneous games have been observed in groups as diverse as hotel clerks (Sherman 2007) and factory workers (Roy 1959), and games have been applied consciously by enterprises for a wide range of reasons (McGonigal 2011; Werbach and Hunter 2012), it should not be surprising that there is no single, ideal "enterprise game." Games in the enterprise serve multiple purposes and multiple audiences. While we argue that our examinations of enterprise games (Edery and Mollick 2009; Werbach and Hunter 2012) give us a solid foundation for attempting a general taxonomy of the use of games in firms, we will invariably leave out some potential uses of games. Some of these omissions are intentional. Our focus is on gamification as a management tool inside the enterprise. We therefore do not address gamification of customer-facing functions such as marketing or of activities with no direct connection to work processes, such as employee charitable fund drives. Similarly, we avoid specifically addressing game-like exercises designed to encourage innovation or creativity among individuals or small teams (Eno and Schmidt 1975; Hohmann 2006). Though these approaches are interesting, play has always been a part of creative activities (Csikszentmihalyi 1975). Moreover, these innovation-oriented games fit within our taxonomy, as they are either a means of improving a firm's productivity or of crowdsourcing innovations. Finally, for purposes of this chapter, we do not distinguish between firms by size or industry and use terms such as *enterprise*, *firm*, and *workplace* interchangeably.

Second, the contentious argument about what constitutes gamification, or even a game, is one that we do not address in detail in this chapter. We take a very catholic view, embracing both scholars who have focused on games in the workplace and more specific studies around gamification or gameful design. Some of the examples cited are more closely related to games, and some to gamification, but given the confluence of approaches in enterprise settings, we do not formally attempt to separate them. Similarly, we do not analyze whether activities mandated as a condition of employment can formally be classified as *games*, given the insistence of many game scholars that voluntariness is an essential element of that term (Carse 1986; Suits 2005). From our instrumental perspective, such distinctions, while critical in other contexts, are more of an impediment than an aid to understanding.

Finally, again flowing from our approach, we do not take an overall normative stance on the desirability or appropriateness of games and gamification in the enterprise. However, we recognize it is important to acknowledge that these practices can be unjust or exploitative. Games can serve many potential purposes, and it has long been established that workplace games can act to undermine the long-term interest of players through either exploitative unremunerated "playbor" or more subtle realigning of worker incentives (Burawoy 1979). Enterprises have repeatedly altered the ways in which they approach employees, at some points viewing workers as interchangeable parts, and at other times emphasizing the importance of the individual (Barley and Kunda 1992). A similar diversity of approaches underlies the ways in which enterprises approach games, as well as the outcomes that such games generate.

Clearly, games and gameful elements can be applied to the workplace in ways that are contrary to the interests of participants. In the most extreme cases, organizations may cynically use games for illegal purposes. For example, e-mail spammers developed a "strip tease" game in which users were incentivized to (unknowingly) help spammers sign up for e-mail accounts. In order for the players to remove an article of clothing from the computer player, they had to solve a CAPTCHA, the distorted words used by many websites to prevent spammers from quickly signing up for too many accounts. The solved CAPTCHA was then used to create a new account for the spammers to send junk e-mail (Edery and Mollick 2009). Moreover, even games deployed for legal and non-exploitative purposes can turn exploitative. For example, games can become an alternative to compensation, with workers earning badges or points instead of a salary, and thus not being properly remunerated for their work (Bogost 2011). Alternately, games can serve as a distraction, obscuring unfair practices, unpleasant tasks, and unrewarding work in a haze of game-play elements. Or a game-like competition could be viewed by workers as a veiled threat that undermines the workplace environment (Lopez 2011).

However, while these dangers are real, they are not reasons to conclude that gamification necessarily produces exploitation. First, there is nothing specific to games in these scenarios. Any mechanism to motivate large-scale human participation in tasks that serve the interest of an organization raises the potential for a variety of dignitary harms (Zittrain 2008). Moreover, just because games can substitute nonmonetary rewards for monetary ones, it does not mean that workers lose in the bargain. Nonmonetary rewards—from job flexibility to efforts to make work "fun"—have long been an important part of work and have been used for both positive and exploitative purposes (Cameron, Dutton,and Quinn 2003; Fineman 2006; Karl and Peluchette 2006). Indeed, social psychologists have increasingly emphasized the critical importance of the experience of work in explaining both job performance and job satisfaction (Grant 2008; Dalal et al. 2009; Rothbard and Wilk 2011). We argue that games represent a new way of experiencing and organizing work and, as such, can be used for a wide range of purposes. Any overarching statement about the moral value of games at work is necessarily inaccurate, and therefore we avoid any such large-scale judgments.

Caveats aside, games represent an important opportunity for both managers of enterprises and scholars that study them. Games draw on a wide range of techniques from the fields of psychology, economics, sociology, and the humanities to achieve a variety of outcomes. Today's digital applications are only the most recent exemplars of a long-standing phenomenon. The rising popularity of video games has sparked interest in how the "gamer generation" that grew up with them might bring new skills and attitudes to the workplace (Beck and Wade 2004; Reeves and Read 2009). While this may be the case, there is no clear evidence that enterprise gamification need be limited to younger or more technology-savvy workers. Games and game-like mechanisms were used at work long before the gamer generation was born.

For the remainder of this chapter, we offer a description of the four primary ways in which games are used in work environments and the historical precedents on which they draw. We organize these elements in terms of the current direct evidence for their effectiveness. While our first

outcome, relieving tedium and increasing satisfaction, is a clear result of games in general, it is less clear that purposeful gamification can accomplish the same task, and few studies exist about this approach. Our second outcome, improving performance, has obvious precedents in contests and incentive schemes, and early evidence suggests both potential promise and pitfalls associated with this approach. The third outcome, encouraging unremunerated work by communities, is a relatively new use of games but has developed significant evidence of its potential effectiveness in both scholarship and practice. We conclude with a brief examination of outcome four, games to enhance training and recruiting, by far the most developed use of games in the enterprise.

Outcome One: Relieving Tedium and Increasing Satisfaction

Starting with some of the first studies of workers and performance, from early Marxist sociologists to the famous Hawthorne studies of worker behavior in the 1930s, social scientists have found that both games at work (such as poker or craps played during breaks) and games about work (such as trying to meet a particular self-imposed quota) are very common (De Man 1928; Roethlisberger et al. 1943). The reason why games are so widespread was not studied until the 1950s, however, when Donald Roy of Duke University decided to gather data on the work environment by working at a machine shop himself, stamping out small shapes from pieces of plastic or leather using large punch presses. The work on the shop floor was exceedingly tedious, and Roy and his co-workers found themselves facing the "beast of monotony," work so boring that if there was nothing done to relieve it, the workers would "go nuts." (Roy 1959) To slay the beast, the workers took a variety of approaches, many of which consisted of games. They played informal and ritualized games with each other, such as stealing and hiding one worker's banana, a break they called "banana time"(Roy 1959). They also turned work into a game, using their small amounts of discretion over their work to create interesting challenges and distractions, such as stamping one color at a time or alternating colors according to some self-directed pattern (Roy 1959). Thus, games filled a vital role in the work environment: passing time, reducing fatigue and boredom, and providing some satisfaction in environments that often lacked anything of the sort.

Roy's work was expanded on by sociologist Michael Burawoy in the 1970s (Burawoy 1979). Working at the same machine shop that Roy had observed twenty years earlier, Burawoy examined more sophisticated workplace games, including a nearly universal game that had various workers competing to exceed quota by as much as possible in a single session. In doing so, Burawoy found that games may serve purposes beyond relieving boredom and encouraging satisfaction. Previously, Roy and other scholars who had noted the importance of games at work had viewed games as a waste of time and act of rebellion against management, a form of "goldbricking" that detracted from productivity (Roy 1952a,1952b, 1953). Burawoy instead argued that games were tolerated, and even subtly encouraged, by managers (Burawoy 1979). In Burawoy's view, by playing the quota game, workers were actually aiding managers, because any game has

to start by agreeing on a set of rules about how the game is played. Because the rules of the game were built on the rules of the workplace, playing the game actually reinforced the rules of managers, thus creating, in Burawoy's words, "consent" to the work itself (Burawoy 1979). As workers played the game, they ended up competing with each other, rather than presenting a united front to managers, further benefiting the management. Thus, games serve an additional purpose in the enterprise, beyond relieving boredom and providing meaning: they can also increase satisfaction with jobs or management, though whether this use of games is exploitative (as Burawoy argued) or laudatory (as many proponents of gamification suggest) depends on the situation and approach.

In many ways, modern gamification of the enterprise builds off the findings of Burawoy and Roy, though proponents of gamification are far more likely to build from recent electronic games, rather than the simple games developed by workers. The role of spontaneous games within enterprises cannot be ignored, however, and they go beyond the factory floor, with games found in groups as diverse as hotel workers (Sherman 2007), truckers (Ouellet 1994), casino operators (Sallaz 2002), and job seekers (Sharone 2007). When employees are bored or not suitably challenged, they invent games as a way to amuse themselves and provide meaning. For example, Rachel Sherman (2007) describes how workers at luxury hotels made games out of trying to maximize occupancy—the game was "lost" if hotels were either overbooked or underbooked. This sort of production game, aligned with the incentives of firm management, and therefore reinforcing the goals of the firm, seems frequently to arise spontaneously. Remarking on a similar phenomenon, Csikszentmihalyi (2000, 2008) suggests that games develop at work because the flow state that can be induced by games results in a more satisfying experience for workers, giving them a feeling of control and mastery over their environment.

Ultimately, much of the gamification movement attempts to re-create these boredom-relieving games of employees and reshape them more explicitly for the use of management. Workers who are happier, more engaged, and closer to their co-workers can be expected to perform better and have lower turnover (Grandey, Tam, and Brauburger 2002; Grant 2008; Rothbard and Wilk 2011), so there are good reasons for such efforts. The rise of gamification has led to the development of software to implement game-based structures in the workplace, and it may also be legitimizing management's use of techniques that were previously frowned upon as "nonserious."

The Zappos Face Game, described in case study 18.1, illustrates how these principles can be applied. It adds a bit of fun to the mundane task of logging into the corporate intranet, analogous to the boredom-reducing dimension of the games that Roy and Burawoy found. One could easily imagine something like the Face Game emerging spontaneously at a workplace, but in this case, it was implemented by management to reinforce a particular message about the corporate culture. By playing the game and following up to get to know fellow employees, workers at Zappos become more acculturated to the company. The game might also be used to serve other objectives of Zappos management. Though it does not publish the results of the Face Game or use them in any announced way, Zappos does keep the data about how successful employees are at identifying their co-workers. This could be useful either as a measure of employee performance or to identify social network

Case Study 18.1

The Zappos Face Game

Summary

Zappos is a large online clothing retailer. One mechanism the company uses to reinforce its playful workplace culture is the Face Game. Whenever a Zappos employee logs into the corporate intranet, he or she is shown the picture of a randomly selected co-worker and presented with a multiple-choice quiz to identify that person's name. Afterward, the employee is shown the co-worker's profile information in the company directory. Zappos encourages those who can't correctly identify a co-worker in the Face Game to e-mail the co-worker and develop a relationship.

Facts and Figures

Amazon.com acquired the fast-growing Zappos for approximately $1 billion in 2009 and now operates it as an independent division. Zappos does not publish statistics on the Face Game specifically.

Gameful Design Elements

Zappos cleverly encourages camaraderie and cross-functional relationships among employees by embedding the Face Game into a mundane workplace activity. Gameful elements include the following:

- It is explicitly labeled as a game, even though it might not fit the formal definition of that term, signaling the gameful nature of the activity.
- It defines getting to know other employees as a stimulating challenge.

Issues

Zappos is known for its playful workplace environment, which its management actively seeks to reinforce as an element of the corporate culture. Like many fast-growing start-ups, Zappos faced a challenge in preserving that culture as it hired new employees, and its size meant workers from different parts of the company had less contact with each other. Such cross-organizational relationships are valuable not only for camaraderie and morale, but also for cross-functional teams, work situations requiring consultation with experts from other groups, or employee reassignments.

Outcomes

The Face Game is one of many tools that Zappos uses to create a desirable workplace, and it does not publish any data that could be used to assess its effectiveness. However, Zappos has been consistently listed on *Fortune* magazine's list of "Best Companies to Work For," and its playful workplace culture has been the subject of numerous articles and other media reports.

Related Cases

None.

Further Information

See Hsieh (2010).

relationships and structural holes across the firm (Burt 1992).

There is reason to believe the use of such tedium-reducing games by employers will grow. Recent evidence suggests that employee mobility is an increasingly important part of modern careers and that satisfaction is a critical aspect in determining employee retention (Arthur and Rousseau 2001). To the extent more firms focus on employee satisfaction and bonding, we can expect games like the Face Game to become more prevalent in the workplace. Yet there is considerable room for future experimentation, as many of these games are relatively unsophisticated compared to modern video games. Many video games have repetitive aspects to game play ("grinding") that are made enjoyable through the application of game design elements. Few of these gameful elements have yet been deployed into workplace games, leaving an opportunity for further efforts to make work feel more meaningful and enjoyable.

Paralleling the relatively primitive state of games designed to reduce boredom and increase satisfaction, much is unknown to scholars about the use of games to accomplish these goals. While games developed by workers clearly play a role in slaying the beast of monotony, it does not necessarily mean that games deployed by managers will serve the same purpose. Similarly, whether workplace games can operate in the same way in a white-collar office as they do on the factory floor is unclear. Future study as well as further experimentation with these sorts of games are likely to continue to evolve our understanding of the way that games can make work more satisfying.

Outcome Two: Improving Performance

Of course, employee satisfaction is not the primary end-goal pursued by firms, leading to the second general outcome emphasized in enterprise gamification: games to increase performance. While happier employees may be more productive, enterprise use of games has tended to focus much more tightly on directly increasing performance through game-based incentives, whether intrinsic or extrinsic in nature. This, too, is rooted in a long tradition of workplace competitions, from the race of John Henry and the steam drill to the brutal quotas depicted in *Glengarry Glen Ross*: "We're adding a little something to this month's sales contest. As you all know, first prize is a Cadillac Eldorado ... Second prize is a set of steak knives. Third prize is you're fired."

While there is a recent tendency to consider incentive systems—based on elements such as leaderboards, prizes, rankings, and feedback—to be innovations partially rooted in game design (Deterding, Dixon, and Khaled 2011), they have long been part of the workplace. As a result, scholars have extensively examined many of the ways in which contests, feedback, and rankings can affect performance at the individual and team levels. Studies have shown that responses to these sorts of gameful practices can be extremely complex, as they act upon on a wide range of psychological, economic, and social factors (Nalebuff and Stiglitz 1983; DellaVigna 2007; Vidal and Nossol 2011). For example, individuals respond differently to incentives than groups (Falk and Ichino 2006), voluntary groups respond

differently than assigned groups (Bandiera, Barankay, and Rasul 2009), and the degree to which groups respond to competitive incentives depends on their relative performance and a variety of other factors (Bandiera, Barankay, and Rasul 2013). Even basic features of many games, such as feedback, are fraught with complications in the workplace. Studies show that the nature of the information being communicated to employees and the context in which it is delivered have powerful effects on the performance of recipients (Vidal and Nossol 2011; Kuhnen and Tymula 2012). Few efforts to build performance-enhancing games take these factors into account, and, in many ways, the simple games that are often used to increase performance are merely special cases of existing incentive systems.

For example, the department store Target adopted a system for cashiers that shows a string of green/yellow/red signals on their cash register screens, based on how quickly they perform checkouts. In addition to providing feedback to the employees about how their speed matches expectations, the information is reportedly forwarded to frontline management. To the extent this type of system improves worker performance, it does so using traditional incentives and feedback rather than in a uniquely game-like way. Target cashiers are being given feedback on their performance, with the clear implication that poor performers will be punished and top performers rewarded. While simple feedback like this does exist in games, without an engaging experience, feedback is merely positive or negative reinforcement.

Ideally, however, games can do much more than traditional feedback or incentive systems (Reeves and Read 2009). Game designers, after all, have considerable expertise in designing complex and dynamic reward and feedback loops. In many games, there are multiple types of goals that players must achieve and multiple game-play systems that interrelate with each other. Players in World of Warcraft (Blizzard Entertainment 2004) can choose to play the game alone or with friends; they can specialize in fighting or in cooking. As a result, players are allowed to succeed in many ways, and, despite the overwhelming complexity of systems involved in the game, the design is such that it guides players into the paths that might work for them. Thus, games offer the possibility of measuring and rewarding performance in much more sophisticated ways than even the most granular current incentive systems. Ideally, gameful performance systems could take into account the various ways in which employees are rewarded and dynamically encourage performance improvements by deploying appropriate feedback and incentives depending on the complex needs of individual employees, allowing each individual to better achieve reasonable goals, rather than attempting simply to compete with each other. Gameful performance systems could help make work more meaningful, as well as more rewarding, for many workers.

No existing gameful incentive system approaches these goals, but more sophisticated techniques are being developed. For example, Objective Logistics, a start-up company, provides systems to motivate improved performance by service workers, such as restaurant servers. Most waiters and waitresses in the United States depend heavily on tips, and their ability to encourage larger checks can have a significant effect on restaurants' profitability. Yet neither the restaurants nor the servers have a clear, real-time picture of how well individuals are doing, creating incentive problems for both workers and employers. Objective Logistics creates feedback

mechanisms to show servers how their performance stacks up and allows managers to allocate desirable work shifts based on that information. Objective Logistics reports overall revenue increases of 2 to 4 percent with its pilot customers, which is highly significant for these low-margin businesses.

Though still not yet the ideal, LiveOps (case study 18.2) is a good example of how a thoughtful implementation of enterprise gamification can improve performance. At a basic level, the system is based on feedback and competition. Workers can see quickly how well they are doing relative to their co-workers or benchmarks, and that information is used by management to assign rewards (here, compensation-generating work assignments that agents desire). The feedback and competition stimulate workers to try harder or to improve their skills through training. There are downsides to these approaches because, as noted, feedback and competition often have complicated implications depending on the context in which they operate. Additionally, to the extent such gamification systems emphasize extrinsic motivators such as monetary compensation, they are susceptible to the well-demonstrated crowding-out or overjustification effect in which individuals actually become less motivated than they were before (Deci and Ryan 1985; Ryan and Deci 2000).

Yet in the case of LiveOps, gamification appears to have generated not only measurable performance improvements but also anecdotal evidence that agents are pleased with the system and the company. This appears to derive from two aspects of the implementation. First, as the management of LiveOps recognized, competition alone doesn't necessarily enhance performance. For example, if agents deliberately rushed callers off the phone to score well on personal call completion time metrics, LiveOps would provide worse service to its clients. LiveOps therefore built its system to provide rewards for the right behaviors and to base feedback on how agents compare to performance metrics defined by the clients. Second, LiveOps balanced competition-oriented tools such as a leaderboard for agents with team structures, incentives for community interactions, and other social aspects to avoid an excessively zero-sum mentality. It built the gamification system into an overall structure that emphasized the company's commitment to professional growth and flexibility for its agents. Thus, the LiveOps system attempts to generate intrinsic, self-motivated performance gains, rather than merely drawing on external reward systems (Ryan and Deci 2000). To the extent agents view their competing in the LiveOps game as integral to achieving their self-defined personal and professional goals, the system is likely to motivate improved performance in a healthy way.

Though performance enhancement through games and gameful elements is a long-established tradition in enterprises, it is also one of the most theoretically complicated. Scholars studying incentives and feedback have identified many contingencies upon which the effectiveness of these tools depend, and few, if any, of these findings have been built into the design of current games for enhancing performance. Further study is needed on this topic to ensure that gameful performance enhancement systems are designed in a holistic way, rather than simply to cloak reward-and-punishment mechanisms. By integrating more of the deep institutional knowledge of motivation and feedback that is the legacy of game design, future creators of performance-enhancing games could help in creating more meaning, as well as better performance outcomes, for their employees.

Case Study 18.2

LiveOps

Summary

LiveOps is a virtual call-center outsourcing company. Its agents are independent contractors who work from home, instead of in large centralized call centers, thanks to software that automatically routes calls and provides information on their computer screens. LiveOps implemented an employee productivity enhancement system by partnering with Bunchball, a gamification platform vendor. As in many examples of gamification, agents are given customizable online avatars and the opportunity to earn virtual points, "level up," and receive badges for designated achievements.

Facts and Figures

LiveOps found that the gamified system led to reduced call times, increased sales 8 to 12 percent, and an average of 9 percent higher customer satisfaction. The company also concluded that gamification reduced its training time from an average of four weeks to fourteen hours.

Gameful Design Elements

- Avatars
- Leaderboards
- Points
- Badges for achievements
- Competitive bidding for projects

Issues

LiveOps' twenty thousand agents include a significant number of nontraditional workers, such as stay-at-home mothers, who are able to supplement their income by working limited numbers of hours from home. While this distributed approach allows LiveOps to be cost-competitive with offshore call centers, it creates significant human resources and performance management challenges. Like all call center operators, LiveOps tracks various metrics of agents' performance, such as conversion rates for outbound marketing and the speed with which they complete calls. However, its ability to manage those agents using traditional techniques is limited because they are not physically present at a LiveOps facility, and each agent can decide how many hours he or she wishes to work.

Outcomes

The gamified system is now built into the mechanism that LiveOps agents use to bid for work, which is assigned based on a variety of factors including their past performance and skills. The gamification also extends to LiveOps' online training modules. Agents are given "missions" to improve their skills, with points being awarded based on performance. These points, in turn, affect the amount of work and therefore the earning potential available to agents. While the primary emphasis was on performance, agents also indicated that the gamified system improved their enjoyment of a potentially tedious job and communicated the company's commitment to their professional development.

Related Cases

None.

Further Information

http://www.managementexchange.com/story/distributed-social-workforce-drives-profit-and-performance
http://socialmarketingfella.com/gamification-drives-performance-success/
http://online.wsj.com/news/articles/SB10001424052970204294504576615371783795248.

Outcome Three: Encouraging Unremunerated Work, Internally and Externally

Drawing upon the satisfaction and performance-enhancing aspects of games is a third use of games in enterprise environments: games have increasingly been used to encourage communities of individuals to undertake some desired action. Commonly, the goal is to achieve some marketing or promotional purpose ("collect points to unlock a secret reward on our website"), but it has also been used to encourage individuals and communities actually to perform work on behalf of the organization. This is done by taking simple tasks from within organizations and turning them into games that appeal to groups of people who will engage in the task for the intrinsic and extrinsic rewards provided by the game. This process is being used within firms to encourage corporate citizenship behaviors, but also to solicit unremunerated work by individuals outside of the bounds of firms.

Externally

To illustrate how individuals outside of firms might be encouraged by games to engage in free work that can be useful to firms, consider the relatively simple ESP Game (von Ahn 2004), developed by Luis von Ahn of Carnegie Mellon, and later adopted by Google (von Ahn 2004). The ESP Game applies gamification to a tedious job, the labeling of online images—an important task that computers cannot perform well. In the game, two players, anonymously matched online, are shown a picture, perhaps of a dog. Both individuals must type descriptive words as quickly as possible, scoring points and advancing to the next picture when both players use the same word to identify the image (such as "dog"). The same images are shown to different pairs of individuals playing the ESP Game. Words that had previously been used to describe an image become "taboo" forcing players to come up with new descriptions (perhaps "brown" for the color of the dog). Over multiple game sessions, therefore, each picture has many individuals identify the image in diverse ways that coherently agree with each other.

The ESP Game uses games to take a single, tedious task and motivate volunteers to perform the task for free. The fun of the ESP Game, with its sense of challenge, competition, and discovery, was such that it saw considerable play. Within three months of its creation, more than 1 million labels were added by more than 13,000 people, and within three years, 200,000 players had contributed more than 50 million labels (von Ahn and Dabbish 2006, 2008). This principle behind the ESP Game can be found in games that ask individuals to help identify galaxies in Hubble Space Telescope images or that encourage communities to plan for resource depletion (Edery and Mollick 2009). In all of these cases, games are used to attract and build groups of individuals who work together on a particular problem or task.

Games that encourage external participation in the work of a firm can be particularly powerful because they motivate online communities to assist enterprises. Communities of dedicated users tend to form around many products and services, and these communities are often the original source of breakthrough innovations later exploited by firms (Von Hippel 2005; Baldwin and Von Hippel 2009). This is because it is users, not the firms themselves, that are the most experienced in actually using a given product and service, and users are constantly

experimenting with the best way to solve their problems. For example, 22 percent of doctors have modified or created surgical equipment and 36 percent of plumbers have created their own pipe hanging hardware, because these users often have to find ways to address unique issues they encounter in their jobs (Von Hippel 2005). In online communities, users communicate with each other to help solve problems, assist fellow community members, and develop new products and services (Franke and Shah 2003; Dahlander and Wallin 2006; Von Hippel 2007; West and Lakhani 2008). Games like Foldit (University of Washington 2007) and World Without Oil (Independent Television Service 2007) bring together communities to innovate around particular problems without any direct payment.

Though communities are powerful, they are not always aligned with the interests of firms. This is because communities have their own goals, which rarely take into account the profit-making incentives or strategies of firms (Mollick 2005). For example, electric car enthusiasts, working in communities, found ways to modify the early Toyota Prius, turning it into an all-electric vehicle, rather than a hybrid. Though this was a significant innovation, it was against the strategy of Toyota, which was trying to avoid the pure electric vehicle market. Gamification represents a solution to community alignment, as firms can reward "good" innovation through points, badges, or other rewards, thus encouraging communities to work with firms in ways that reinforce rather than undermine strategy.

Internally

The same technique of motivating collective action can also be applied to employees inside an organization, to encourage them to engage collectively in activities beyond their normal job functions. Organizations such as Lloyds TSB Bank and the Department of Work and Pensions in the United Kingdom have employed virtual stock market games to encourage development and submission of internal innovation proposals. Enterprise software giant SAP uses points and badges to encourage quality contributions to its developer community extranet. And a number of firms are recognizing that gamification can improve the effectiveness of enterprise knowledge management and social networking systems.

Surfacing the tacit knowledge of employees and mapping their expertise networks can produce significant benefits for knowledge-based organizations. However, putting accurate data into knowledge management systems is tedious and unrewarding, so workers have been reluctant to do so. A group from IBM Research conducted an experiment that involved adding points- and levels-based virtual reward systems to Beehive, an internal system that promoted knowledge sharing among IBM employees. They found both mechanisms motivated increased contributions to Beehive, which inspired other employees to visit and comment more as well (Farzan et al. 2008). Consulting firms such as Deloitte and Cap Gemini are now using points and leaderboards to encourage contributions to their internal social networking systems, seeking to achieve similar results.

Of course, unremunerated work by employees raises the specter of exploitation. To the extent that firms are substituting gamified virtual rewards for traditional tangible forms of compensation, exploitation would be a legitimate concern. However, the examples given here have an important difference. The gamification does not involve the core job responsibilities for which workers are evaluated and

Case Study 18.3

Microsoft's Communicate Hope

Summary

A team responsible for quality assurance testing at Microsoft has designed a series of game-based activities to encourage large numbers of employees to participate in valuable but otherwise tedious testing of new products. For Lync, an integrated communications tool, they created a program called Communicate Hope that allowed employees to join teams representing five real disaster-relief charities. Microsoft made contributions to the charities based on the number of points the employee teams earned by testing features of Lync and reporting their experiences.

Facts and Figures

More than eight thousand employees participated. Two thirds of participants sent ad hoc feedback about the product, a key goal of the system, compared to just 3 percent for a control group that was not earning points for disaster relief. Moreover, 97 percent of the participants said they would participate in another testing program, compared to rates of 50 to 75 percent for typical programs.

Issues

Software testing and quality assurance is an extremely important but tedious task. For Microsoft's Lync communications product, the company faced several challenges. It had to find enough testers willing and able to try out a pre-release product, participate in designated scenarios (such as receiving a video call from another tester attempting to make one), and write up detailed feedback afterward. In addition, because this was a pre-release product, the Microsoft team did not want the likely prevalence of bugs to create a negative view of the product that would carry over.

Outcomes

Employees were willing to contribute to the process without remuneration because it involved a shared contribution as "organizational citizens," separate and apart from their individual success as employees. The fact that their efforts generated contributions to charities they selected reinforced these altruistic motivations. Quantitative and qualitative measurements suggested that the gamified elements significantly improved both the rate of participation and the subjective experience of testers.

Related Cases

Microsoft's Language Quality Game, as described in Werbach and Hunter (2012).

Further Information

See Smith (2011); Smith (2012).

paid. Rather, it goes to what Ross Smith, who led the group at Microsoft that designed Communicate Hope (case study 18.3), calls "organizational citizenship behaviors." These are activities that employees know contribute to the greater good of the enterprise, but that aren't required as part of their in-role job responsibilities. Making software products (relatively) bug-free is part of the aggregate mission of Microsoft employees, but it is not something most of them are directly tasked with. Similarly, knowledge

sharing is a collective goal for IBM workers, but not something they are individually paid for. In other words, the baseline workplace reward systems such as salary and bonuses address in-role activities, but they generally do not promote organizational citizenship behaviors, despite their value to the enterprise. Gamification can create a secondary motivational system for these activities. This form of internal crowdsourcing is typically applied to knowledge workers, who have significant flexibility in allocating their time at work. As a result, they can contribute to the collective gamified activities without a cost to their individual compensation.

Using game elements to encourage the work of internal and external communities is likely to be an enduring aspect of the current gamification movement. With the increasing desire of firms to take advantage of the skills of employees and users, games provide one of only a few successful models for how large groups can be recruited, organized, coordinated, and retained without direct remuneration. It is also one of the best-supported outcomes of games in the enterprise to date, with a number of case studies suggesting that games can assist enterprises in bringing together communities to provide valuable assistance (Khatib, Cooper and Tyka 2011; Smith 2011). Still there is considerable room for future development, as none of these community-building games reach the sophistication of massively multiplayer games such as World of Warcraft (Blizzard Entertainment 2004) or even causal games like Farmville (Zynga 2009). Similarly, any scholarly understanding of these sorts of games remains nascent, and future study is needed.

Outcome Four: Bolstering Training and Recruiting

The final major use of games within the enterprise is the most studied—games for training and recruiting. Game-based teaching has a long history, especially in the military, and the first purpose-built training games, developed by the RAND Corporation in the 1950s, were direct descendants of a nineteenth-century Prussian wargame called Kriegspiel (Faria and Dickinson 1994). Further, many training techniques have been perfected in the entertainment world before being put to more serious purpose. The Link trainer was used to train half a million World War II fighter pilots. When the Link was first introduced in 1932, the U.S. government only owned a single simulator—but there were fifty at Coney Island and other amusement parks (Edery and Mollick 2009).

Other contributions to this volume, including those by Ramirez and Squire and by Salen, deal in depth with the many ways in which games can teach and train. Enterprises have been enthusiastic adopters of games for just such purposes, with training games being used by companies ranging from Sun Microsystems to Siemens to Cold Stone Creamery (Edery and Mollick 2009). A vigorous scholarly tradition examines the effectiveness of these games compared to other training approaches (Bruckman 1999; Hoffman and Nadelson 2009; Iacovides et al. 2011). Because these uses of games are expressly presented as learning and development tools, they operate outside core work processes, unlike the other applications of games described earlier. However, because the human resource (HR) function has been one of

the first to accept the general use of games in the form of training, it is perhaps unsurprising that HR managers have been implementing games for other key HR tasks, such as recruiting.

Again, the military was a pioneer of this approach, as the U.S. Army has the largest recruiting budget of any organization in the world. Despite all of this spending, it was an explicit recruiting game called America's Army (U.S. Army 2002) that was, in the words of co-creator Mike Zyda, "the most cost-effective thing that the Army has ever done in recruiting" (Edery and Mollick 2009). The Army used the game to allow qualified potential recruits to experience a tailored, heroic version of the U.S. Army in the form of a first-person shooter (FPS). The results were impressive, as the game, which cost less than 1 percent of the recruiting budget to make, had more impact on drawing new recruits than all other forms of Army advertising combined (Zeller 2005). Noting the success of the Army, a variety of other organizations, from hotel chains to manufacturers, have developed recruiting games of their own, though the success of these efforts has not been rigorously studied.

Additionally, because the variation among individual recruits can have a large impact on firm performance (Mollick 2012), recruiting games have increasingly been deployed to help select, as well as attract, employees. These efforts have so far involved relatively lightweight games. For example, L'Oreal convinced many business schools to include a game-like simulation of the cosmetics industry, e-Strat, into their curricula. An annual contest brought the top scoring MBAs, out of thousands of participants, to give final presentations of their strategies to L'Oreal in Paris, where the best players were recruited for the firm (Edery and Mollick 2009). Given the vast amount of data captured by games, however, the continued evolution of games for employee selection is all but certain. While testing potential employees through puzzles and psychometric evaluations is common, games can potentially provide much richer data on performance, especially as scholars suggest that more complicated traits, such as leadership ability, can be accessed via game play (Reeves, Malone, and Driscoll 2008; Goh and Wasko 2009).

Thus, while the use of games for training is certain to be a subject of continued interest and study among both academics and managers, training may represent just part of the ways in which games will impact the enterprise HR function. More work is needed by both scholars and practitioners to understand how games can be used to evaluate real-world performance and how such features can be incorporated into games that assist in the recruitment of candidates. At the same time, academics and firms alike must be careful to understand the limits of games as both training and recruiting tools, as even the best efforts to create realistic scenarios for training or assessment are certain to be much more limited in scope than most real challenges faced by employees.

Conclusion

As we have shown, enterprises are using games and gamification for a wide variety of purposes. The long history of spontaneous workplace games suggests that, if designed successfully, these interventions

may be well received. However, the nature of games is complicated: just as no game is universally fun to all potential players, so too will the effectiveness of enterprise gamification vary depending on the nature of the game, the organizational context, and the player population. Workplaces are increasingly constituted of individuals who have played video games in their youth, or still do so, but none of the phenomena described in this chapter depend particularly on that fact. Indeed, the distinguishing characteristic of enterprise gamification is that it is a practice of management, not merely an avocation of workers. When an employee plays Windows Solitaire on an office computer as an escape from a tedious job, the game is a substitute for work. When that same employee engages in a workplace competition or a gamified work activity, the function of the game is quite different, even if the phenomenology is similar.

Scholars and managers alike must take care not to assume that games and gameful activities that "work" outside of enterprises will be equally engaging inside firms. While the enterprise use of gamification is, as of this writing, only gathering momentum, the evidence of the effectiveness of games, and our understanding of what makes games effective in the enterprise, are extremely limited. Games and gameful design have the ability to alter the experience, outcomes, and nature of work, but they could also be just another passing fad in management. For games to reach their potential within the enterprise, managers and game designers must do more to collect and share data about gamification, and scholars must conduct and share more rigorous and rich studies of gamification outcomes.

Note

1. For examples, Edery and Mollick (2009), including America's Army (U.S. Army 2002), Virtual Leader (Simulearn 2002), and DARPA's SIMNET (Delta Graphics et al. 1987).

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POSITION STATEMENT

WHEN PEERS SELECT TASKS AND TEAMS

JP Rangaswami

We've had the Agricultural Revolution. We've had the Industrial Revolution. For the past six decades or so, we've all been part of the Information Revolution.

We're now knowledge workers, even those of us who work in agriculture or manufacturing. We're all knowledge workers, using information systems and processes to go about our work.

Information, the knowledge worker's raw material, is fundamentally different from seeds or crops or metals or minerals; in consequence, the work that knowledge workers do is fundamentally different from the work done before.

These changes are visible across three dimensions: the firm, the worker, and work itself.

The nature of the firm has changed, from a rigid hierarchical structure to a much looser, adaptive networked approach. To understand why this has happened, we need to look at why firms organize themselves. They need structures that allow them to set priorities, to share those priorities with the workforce, to allocate resources to those tasks, to monitor progress, to resolve conflict, and finally to intervene when and where progress is inadequate.

As a result, when deciding how to structure the firm, most people looked at models already in use to solve similar problems. Prioritization and goal setting. Resource allocation. Monitoring and feedback. Conflict resolution. Intervention as and when appropriate.

Much of this was being worked on between the two World Wars, and not surprisingly, the models that were chosen looked remarkably military in their characteristics.

Scale was a determining factor for success, and scale could be solved by hierarchical control, or so the thinking went. This was true for decades in what were relatively steady-state markets. Standardization was common; the pace of change within the firm was at least as fast as the pace of change in the external market.

With the Information Revolution, everyone was getting connected everywhere. The rollout of digital infrastructure was global, as were the adjustments to public policy. The pace of the market changed, and a more complex, adaptive structure was needed.

The worker has changed, she is now born digital, born computer-literate, ubiquitously connected, and device-empowered. She comes with a profound loss of trust in many of the institutions of her elders: the elders themselves, the government, the judiciary, the law, law enforcement, educational establishments, religious establishments, large corporations. What she trusts are her friends, her peers. She does things for herself, for peer respect, for peer recognition.

Work itself has changed. Knowledge work is "lumpy," with high variability in volume. Processes do not stay standard long enough for them to be repeatable at scale; there is too much time spent handling exceptions. High-level patterns have become more important than detailed processes.

This nonlinear aspect to work is made even more complex because of the lack of standardization: the skills required are tending to become more and more diverse, more varied. Assembly lines required diverse skills as well, but they could be called upon serially. In knowledge work, the need is to have those diverse skills operate in parallel, operate simultaneously.

Leadership is still required in the setting of priorities and goals. But once that is done everything else changes. In a networked world, resource and task allocation are more effective at peer level; feedback loops are at their most valuable local to the action; conflict is easier to resolve at peer level, too; and intervention needs to be fast, again working better at peer level.

People who want to drive up the level of performance of their firms have understood this for some time and have therefore been focused on some key strategies: ensure everyone understands the priorities of the firm adequately; allow people to select tasks that are sufficiently challenging to keep them interested and learning, but not so challenging that they have no chance to complete it; give feedback quickly and effectively, reward and thank. Allow people to acquire skills, to train, as part of the job. And make sure everyone keeps a clear view of the ever-changing context.

Activities that are nonlinear, unpredictable, "lumpy" in character. Reliance on good communication of goals and objectives. Peer-based task and team selection. Active feedback loops and contextual awareness. Ability to develop and enhance skills. Where were these problems being solved, and how?

Games. Multiplayer games. Massively multiplayer games. Even video games. Where you understand the overall mission objective clearly before you begin your chosen missions or quests or activities. Where you spend time observing and discovering would-be partners before choosing the people you want to play with, your "team." Where you need to find efficient ways of assessing the skills and skill levels of your peers and of assessing the skills and skill levels needed to complete a particular task. Where you need to get used to active 360 feedback loops, highly sensitive controls, both close-in and as well as zoomed-out loci of operations.

The firm has changed. The worker has changed. Work has changed. These changes are such that the gaming community has experience to teach us how to solve for these problems in the workplace. Game mechanics and values can be used to do this, provided the focus continues to be on skill mastery, on real achievement, on valuable learning taking place.

Which is why we're all spending time understanding how to deal with the practical aspects of this, which involves being able to describe tasks as well as people in terms of skill sets and skill levels, being able to validate those skills, being able to display them in a discoverable way, being able to reflect changes in those skills and levels. Ratings, reviews, and reputations.

We're also spending time figuring out how best to get the feedback loops right so that real learning

takes place, based on useful inputs on what to change and why.

The world of work has changed. What a company can do in the future has become more important than what a company has done in the past; its ability to learn has become more important than its past experience.

It's all down to the people, their skills, their ability to learn and to develop those skills, their ability to work together, their ability to respond to external stimuli, their ability to adapt to change.

Which is where the learning of the gamers comes in.

19 GAMIFICATION AND SOCIAL MEDIA

Cliff Lampe

There is a natural connection between social media and gamification on multiple levels, in that both genres of interaction combine social and technical architectures to shape and enable user practice. While gamification has great potential to add insights in how to shape behaviors through the design of innovative tools, social media has a long history of using technical tools to shape social practices that may inform our understanding of gamification. Understanding social media and the research that has been done on it can illuminate many of the processes of gamification.

Social media environments combine the effects and processes of both social and technical systems. In other words, social media environments, like games, are essentially social structures that both shape and are shaped by the features of the computing environments in which they occur. This perspective on mixed systems, often referred to as the sociotechnical systems model (Ackerman 2001), emphasizes the ways in which the architecture of computing environments interacts with social processes. Many of these sociotechnical features are common across social media sites. For example, many social media sites include the capacity for users to write comments, and often systems for rating those comments to help manage information overload. Some tools, like user profiles and reputation systems, become so ubiquitous in current social media sites that it can be difficult to remember that these were not always common in earlier forms of social media interaction.

The term *gamification* has been used to describe a particular set of technical features common in games to shape social processes (Deterding et al. 2011). Both social media sites and the games that inspire gamification share common elements. First, they are constrained environments where people engage in social interaction, but often without the full range of social cues that exist in other forms of interaction. Second, both environments are designed, which means that developers are making choices about which features are going to be used to support social interaction. These "social features" are powerful architectures that both shape and are shaped by broader social processes.

In this chapter, we'll show how social media features evolved from an initial perspective that technology mediation limited social interaction, but eventually was reframed as enabling a "beyond being there" (Hollan and Stornetta 1992) capacity, meaning that online environments were being designed not only to replicate offline interactions, but also to enable entirely new forms of interaction supported

by the capabilities of the computer-supported environment in which they occurred. These new forms of interaction were not always part of the social media environment, but resulted from large-scale experimentation in early web systems, often inspired by the experiences these designers had in games. To help explain that evolution of design, we'll study two early entrants in the social media space and their overlap with insights drawn from games. Next, we'll consider the social media research that has examined how the features of social media affect processes like signaling status and foster relationship maintenance associated with social capital. The chapter ends with a reflection on how a new generation of challenges in social media design might be affected by advances in our understanding of gamification

Defining *Social Media*

To understand the evolution of design behind social media, it is useful to define this particular type of technology-mediated interaction. Use of the term *social media* to define a type of online interaction became common in the mid-2000s, at a time when multiple competing terms were also in general use. *Online communities*, *virtual communities*, *Web 2.0*, *social computing*, *social software*, and other terms were used to describe a type of sociotechnical system that enabled social interaction through computer mediation. Although subtle distinctions could be drawn between these terms, the umbrella term *social media* seems to have been adopted by the general public.

A sociotechnical system is categorized as social media based on the presence of three characteristics:

1. There is relatively unmediated interaction between multiple participants.
2. The content of the system is primarily generated by users.
3. User interactions are supported by a bundle of computer tools.

The goal of this definition is to distinguish social media from two other forms of mediated interaction: mass (or traditional) media and interpersonal computer-mediated communication. Although mass media content can be shared on social media (in which case the user-generated content is the link, not the story) and interpersonal interaction happens in many (if not all) social media sites, the interactions defined above help to distinguish this genre.

The first two points of this definition speak to both the range of activities in social media platforms, which can be incredibly heterogeneous in how users interact with one another, and the role of content generation in the system.

Relatively Unmediated Interaction between Multiple Participants

In examining the dimension of "relatively unmediated interaction between multiple participants," interaction between participants is a primary activity. Whether that interactivity is focused on comments on news stories in Reddit or Huffington Post, commerce in Etsy or eBay, or user-generated content in Wikipedia or DeviantArt, these sites allow for users to communicate directly with one another. This social effect of the architecture behind social media is very similar to that of social games (as opposed to

single-player games), where interaction between players is often a major motivation and a driver of game play. It is this unmoderated communication that has driven the development of gamification-like features in social media, as developers have created tools that both ameliorate some of the effects of text-only environments and extend how people can interact.

For example, in the unmoderated communication of social media, it has traditionally been difficult to know for sure who the other participants are. There is a range of how much users know about each other as they are interacting. Historically, online communities were largely pseudonymous, leading to opportunities for identity play (Turkle 1995) as well as misbehavior (Kling et al. 1999) and misunderstanding (Sproull and Kiesler 1991). Later, we discuss how online tools both obscure and illuminate aspects of identity in social media systems.

In addition to the issues of identity, also discussed in more detail later, unmoderated communication between social media participants requires the development of tools to support and extend the management of conversations between people. One gamification-like feature that helps manage user behavior is the reputation system, where users create persistent profiles and accumulate some form of points to indicate the strength of their participation (Resnick et al. 2000). Reputation systems were pioneered in eBay but have become common where very large groups of people interact, as with Reddit, Huffington Post, Amazon reviews, and other similar systems. As soon as an online group grows large enough that its participants cannot assess each other's value through their interaction, then reputation systems provide information about other participants, shape norms of behavior in the site, and provide feedback to users about the value of their participation (Lampe 2011).

Dependence on User-Generated Content

The dependence on user-generated content is also an important characteristic of social media sites, without which most sites would be empty placeholders. Whether that content is comments on news stories, tags, links, pictures, videos, or status updates, the production of content creates the opportunity for users to interact. A consistent finding in the generation of content by users is that there is an inequality in most social media sites where a small percentage of users provides the majority of the content (Golder, Wilkinson and Huberman 2007; Wilkinson 2008; Hamptonet al. 2012). We don't know, however, whether it is the same group of users who are participating in each social media site or if people are lurking in one site and participating in another. In terms of the attention framework, it may be that users are overwhelmed by the choices they have regarding which venues they can give their attention and make contributions to. In one study, we studied "the power of the ask" where theory from philanthropy literature indicated that one way to help people direct their attention is by asking them to contribute (Wash and Lampe 2012). We found that direct requests for news stories did indeed increase the amount of overall participation.

Besides simply providing a space for users to contribute content, social media tools may help provide feedback loops that help trigger intrinsic motivations to participate in generating content. One consistent finding in social media research is that in any given site, users have heterogeneous motivations

for participating in the activities of the site (Lampe et al. 2010).

Social media spaces are simply empty lots until users generate content for them, and consequently, motivating users to post content in these sites is a major area of activity in research and design on social media. This is very similar to a primary use of gamification, which is to provide feedback that triggers intrinsic motivation to engage in a behavior. For example, the gamification health system Fitocracy provides badges and levels to help support an intrinsic motivation to eat healthily and to exercise. This is similar to many systems in social media that also provide feedback that can be used to help incentivize motivations for participation. For example, Wikipedia has a system of barn stars that act as a type of badge to be given to users who do some particularly valuable work for the site. Some barn stars might recognize work in welcoming new users, where others reward users for editing skills or writing skills (Kriplean, Beschastnikh and McDonald 2008). These barn stars then appear on the user page of the Wikipedia editor and act as a significant signal of status for the members of that community.

How to motivate participation in social media sites has been a concern since the early days of research on sociotechnical systems. Kollock (1999) looked at early peer-production systems like Usenet discussion groups and the Linux operating system as examples of public goods and theorized about the different factors that might motivate people to participate in those projects. He argued that a future expectation of reciprocity, enhanced reputation, and an increased sense of self-efficacy might be three motivational forces that encouraged participation. Additionally, he listed commitment to the group as another reason people might volunteer their time to these endeavors. He wasn't saying altruistic motivations didn't exist, but rather that they didn't have to exist in order to explain the continued existence of an online public good. More recently, Lampe et al. (2010) looked at a peer-production web social media site, examining different reasons why people might contribute their time to the site. They found that group identity wasn't an important motivational factor for how much a person contributed, but it was for whether they contributed anything at all. Motivations like being entertained, contributing to a socially valuable project, and increasing self-efficacy were associated with increases in different types of contributions to the site. Additionally, status signals, measured in this site as a reputation score, were both shown in log data and in interviews to be an important signifier that their contributions were valued, motivating future participation. This finding that multiple motivations drive participation in social media has wide support. Nov (2007) looked at the motivations of Wikipedians and found that they were motivated by the social connections they had with other contributors, the improved self-efficacy in their writing, the social value of the overall project, and the ability to learn about new things.

The heterogeneity of motivations and the need to encourage contributions are not limited to peer-production social media sites either. In Facebook, Joinson (2008) found that people used Facebook for many different reasons, including to learn more about the people around them, to create and manage social connections, and to share content with others. Smock et al. (2011) found that different motivations for using Facebook led to starkly different uses of the different specific tools on the site. Consequently, not only are the motivations for using social media varied among users, but so is the effectiveness of different

tools for meeting those motivations. This matters because while social media is seen as a space with broad participation, there is still a need to incentivize certain types of contributions. Wilkinson (2008) showed that the "power law of contribution," or the finding that a small minority of users does the majority of content generation, is a consistent pattern across a wide range of peer-production sites. Hampton et al. (2011) showed that this same unequal pattern of participation was also consistent in social media use. On sites like Twitter, Facebook, and others, a minority of users posts the majority of content. This can have broad reaching effects, as there is some evidence that not actively participating (i.e., lurking rather than posting) can have negative outcomes (Burke, Marlow, and Lento 2010).

It is necessary to use the design of tools in social media to encourage contributions in social media, just as gamification systems are designed to motivate certain actions. For example, in systems where content rating is available, a new contributor can be heavily disincentivized if his or her contribution doesn't receive any rating, even a negative one (Lampe and Johnston 2005). Facebook asks in its status update the question "What's on your mind?" to prompt a type of response. As Facebook was growing in size, being tagged in a picture on Facebook increased the chance that a person would create an account on the site (Burke, Marlow and Lento 2009). On top of encouraging participation in active sites, it may be necessary to design features that encourage people to participate in new sites. Starting new social media sites and online communities is difficult, and a necessary condition for accomplishing the task is to encourage people to post content (Preece and Shneiderman 2009; Lampe and Roth 2012). For example, many Google products like Wave and Plus have used "invite only" memberships to start, which create a perceived scarcity of participation, increasing the likelihood that people will join the site. Of course, once they have joined the site, they need to be encouraged to continue contributing, which involves its own challenges (Preece and Shneiderman 2009). The book *Building Successful Online Communities* (Kraut and Resnick 2012) has many other examples of social media design features that might help to support and reinforce user motivations to encourage participation in these sites.

User Interactions Are Supported by a Bundle of Computer Tools

The third feature of social media sites is that they are bundles of computer applications. This has two main implications: computer mediation and combinations of applications.

Early studies of online interaction, especially those that were focused on text-based systems, remarked on the lack of information that was available in text-based communication versus face-to-face communication. Olson and Olson (2000) compare "radically collocated" teams to assess what types of communication are available in these face-to-face environments and then show whether different media environments, ranging from text-based to more sophisticated video-conferencing systems, replace those face-to-face features or not. They point out that text-based environments strip out body language, proximity cues (like who's looking at what or who's standing near whom), personal information (like age, gender), and voice cues (rate and tone of voice), among other essential "channels" in face-to-face contexts. They further say that more channels are required when social tasks require more "common

ground" (Clark and Brennan 1991), because the tasks involve complex interactions where misinterpretation is both common and harmful to the collaboration. While other scholars have found that text-based discussion that persists over time can be sufficient to accomplish certain types of social tasks, like forming interpersonal impressions (Walther 1992), the idea that online environments reduce the social cues that facilitate face-to-face communication has been a driving force behind much of the design of social media systems. Lack of cues in online environments may lead to several interruptions in social interaction, which in turn has often triggered the design of social features.

While modern social media sites still do have some opportunities for anonymous and pseudonymous interactions (Lampe and Resnick 2004; Bernstein et al. 2011), many of them depend on rich identity information that is stored in user profiles (Lampe, Ellison and Steinfield 2007). In particular, Facebook was one of the first in the current generation of social media sites to lean heavily on "offline" identity (Lampe et al. 2006), which changed the nature of both how people identified connections in the site (from interest based to offline network based) and how subsequent social media sites viewed pseudonymous versus "RealID" interactions. The debate between using "real" identification and pseudonymous interaction is very prevalent and contentious in both research and design of social media sites (Boyd 2012). In the next section, we'll discuss why communication that is mediated by technology, like social media, is so susceptible to these types of controversies.

Reduced Cues Make It Hard to Know with Whom We Are Interacting

There are two major aspects of this effect of the identity-hiding consequences of technology-mediated environments. First, it is hard to form impressions of other individuals in sociotechnical systems. Second, it is hard to detect who has consumed a piece of content made available on a social media site.

Early Internet interactions were marked by pseudonymous and anonymous interactions, and early research narratives were filled with the pros and cons of identity play allowed by these identity-adaptive features of early social media (Turkle 1995). In an environment where anyone could adopt an identity, there were large debates about the nature of trust, misbehavior, and legitimacy in a variety of social media sites. While the constrained cues made it difficult to make the types of immediate assessments of people that were common (though not always positive) in social psychology studies of group formation, social information processing theory (Walther 1992) argued that people could form interpersonal impressions in text-mediated environments, but that it took longer to accumulate the cues to form those impressions. Most multiplayer games are also pseudonymous, and consequently there is a strong association between the game literature and the work on these early, more anonymous online interactions. In particular, in games, cues may build over time with iterative interactions between players. However, most games try to facilitate that process by offering status cues. For example, achieving a high level in a game isn't just a reward for work in the game but also a signal

to other players about the skill and expertise of the high-level player. In social media where participants are largely pseudonymous, there are status signals in terms of reputation scores, timestamp (age) of account, use of community-specific language, and records of numbers of posts and contributions that help to signal status in ways that are similar to status mechanisms in games.

Since the mid-2000s, there has been a change in the nature of social media identities. Facebook, unlike many of the social network sites that came before it, encouraged "real-life" names rather than pseudonyms. This potentially led to a change where people joined these sites not based on a common identity characteristic, as had previously been common (Boyd and Ellison 2007), but through shared membership in offline networks (Lampe et al. 2006). New social media sites often use Facebook logins for their own user registration systems (e.g., sites like Pinterest and Instagram have heavy integration of Facebook logins). While technical features have a long history of research in supporting identity expression and status in pseudonymous online systems, there is less work on the importance of these features in environments where social media profiles are tied to an offline identity. Offline status (like job status or social status) may trump any online status (Lampe 2011). For example, in a discussion of medical treatments on Facebook, a friend in your network you know to be a medical doctor may be seen as higher status than other commenters. In the case of RealID-dependent social media, status cues may not be as much about impression formation as about building and nurturing an audience.

In addition to the difficulty of knowing about the individuals one is interacting with in social media sites, it can also be difficult to assess how many people one is interacting with. In most social media sites, as users post their content there is a "fog of audience" that makes it difficult to detect who has actually seen that content (Marwick 2012). Bernstein et al. (2013) found that Facebook users consistently underestimated how many members of their "Friends" network had viewed their content by a factor of 4:1. In other words, they estimated only 25 percent of the people who had actually seen their posts were part of their active audience. This lack of awareness can lead to context collapse, or the posting of messages appropriate for one part of a network but seen by all parts of the network (McLaughlin and Vitak 2012), which in turn has led to a range of social strategies like steganography (hiding behind coded language) (Boyd and Marwick 2011), posting only messages acceptable for the widest possible audience (Hogan 2010), or dividing one's activities across multiple social media sites (Stutzman et al. 2012). There have also been social features added to sites to help remediate context collapse. For example, Google Plus used the concept of "Circles" to replicate social contexts online in their systems (Kairam et al. 2012). Facebook added lists for users to divide their networks as well.

Another social feature that helps resolve the fog of audience is small-feedback tools that allow people to signal attention to one another, even when most viewing activity is invisible. For example, on Facebook users can "Like" each other's content, wish "Happy Birthday," or comment on each other's posts. These social features may act as a form of "social grooming" that helps users to signal attention to one another, and consequently their interest in the relationship. Lampe et al. (2012) found that Facebook users' belief that they engaged in this type of behavior was associated with their likelihood to ask their

Facebook network for help. Other social media sites add different types of these social-grooming features to signal attention. For example, LinkedIn has testimonials from your contacts, as well as endorsements for specific skills. Pinterest users can re-pin content or follow boards to show interest.

Beyond Being There

While early research was embedded in the deficit model of online interaction, in which face-to-face interaction is a gold standard that is diminished by technological mediation, both research and design have more recently identified ways in which sociotechnical systems can extend the types of interactions that take place both in social media and online more generally.

Resnick (2001) outlines several characteristics of technological mediation that supports interactions that are different than offline interpersonal communication, including the ability to store communication over time or to run computation on stored interactions. The ability of computer mediation to store content enables features like the Facebook timeline, which allows users to track their own and others' posts over their history on the site. This mediation means that it is possible to enable interaction that is at a larger scale and faster than might occur in traditional communication, but also that there are tools that can be used to shape this environment to make that scale of communication more manageable.

An example of this type of social feature is the use of recommender and reputation systems to help users form impressions of each other. The primary purpose of recommender systems is to help users select the content they want to consume when a large amount of such content is available. Recommenders can be either general (like five-star ratings on a product) or involve collaborative filtering where matches are personalized based on previous ratings from the user (such as movie recommendations in NetFlix) (Terveen and Hill 2002). Recommenders are very common in sites where there is a large amount of user-generated content, and there is an assumption that users cannot consume all of the content. As described more fully later, Slashdot uses a recommender system to guide users in filtering comments on news items. Other sites that depend on user comments have adopted this method, including Digg and Reddit. Reddit also employs these user ratings to determine which content will appear in main areas of the site (Gilbert 2013).

To summarize: The characteristics of social media systems that necessitate the use of social features include the following:

- Largely text-based interaction reduces total channels of information, meaning that it takes longer to form interpersonal impressions of other users. Social features can hasten this impression formation by adding context to user profiles.
- Ease of entry and exit into most social media environments means that membership can be very large or can be highly dynamic, which in turn can add uncertainty about the people a user is interacting with.
- Computer mediation means that traces left by users as they add content to social media systems can be stored, analyzed, and reproduced in ways that add context to user interactions.

The combination of these technical tools used in social media derive from the same characteristics that make rules and systems part of games. Namely, both social media and games are constructed in bounded architectures, where participants need to operate by codified rules in order to guide their social interactions given the lack of other cues in the environment. Lessig (2006) points out that behavior can be shaped by many forces, including norms, laws, markets, and architecture. In online environments and games, architecture, or technical features, becomes very important because these other forms of guidance may suffer due to the constrained cues. The next section describes the effects of technological mediation in online social interaction.

A History of Early Social Media

While systems like Usenet, Multi-User Dungeons (MUDs), and e-mail lists are not often thought of as social media, they are the early examples of how social and technical systems affected one another and meet the definition provided earlier. These systems were the first online social interaction spaces where the effects of technological mediation were associated with negative outcomes.

In e-mail lists, early researchers noted that the tone of e-mails could often be misunderstood, which they related to lack of communication channels that allowed for misinterpretation of content and the invisibility of audiences in large e-mail lists (Sproull and Kiesler 1991). In MUDs, which were text-based online games initially developed in the late 1970s, users were able to craft new personae, which could be positive for exploring alternative identity (Turkle 1995), but often led to problems of conflict between players that had to be moderated by game controllers (Curtis 1992; Muramatsu and Ackerman 1998). Most famously, Usenet was a text-based discussion environment where moderation was difficult, leading to a host of negative user behaviors that were difficult to manage (Hauben and Hauben 1997). Negative user activities like spamming, trolling, and flaming were all pioneered in Usenet but are still common types of activities found in current social media environments (Pfaffenberger 2002).

Both Usenet and e-mail were open, distributed architectures that did not allow for the diffusion of new tools to support social interaction. A good example of this is the GroupLens system for recommending Usenet articles, which added a middleware layer between local users and the Usenet systems to allow people to recommend articles (Resnick et al. 1994). This system is a very early example of technical recommendation systems that now are embedded in many social media systems but in the early 1990s were limited by the overall architecture of the sociotechnical systems of that era.

In the mid-1990s, sociotechnical interaction started to move to World Wide Web environments. For better and worse, these systems were controlled by single organizations, unlike the open and distributed architecture of e-mail and Usenet. Because of this, developers could add new features that supported a wide variety of interactions. Besides the addition of media to text, these new web systems could add additional tools that added context to both user histories and content. This new breed of

web online community still shared several characteristics of the previous Usenet, MUD, and e-mail list environments. Namely, web communities were still largely text-based and involved large populations that made emergent social structure difficult without additional support from social features. Because these sites were owned by individual organizations, those social features could be added to the systems to help shape the types of interactions that took place. Many of the designers of these early web communities had been active in the previous generation of social media sites and were well aware of the types of problems that could occur. In addition, they were watching each other try to solve social interaction problems in their systems and learning about what worked and what didn't in an evolutionary process that eventually resulted in many of the aspects of the social media environment that currently exists.

In the next section, two examples of early web communities are provided that show how some of these social features came to be applied to social media systems, providing context to the later refinement of these tools.

Historical Examples: Slashdot and Everything2

Slashdot (http://slashdot.org) and Everything2 (http://everything2.com) are two systems that represent prototypical social media sites. Both of these sites were developed by the same group of designers at a company called Blockstackers Intergalactic and were early web-based social sites that started to attract enough users that they required the implementation of social architecture. The developers of both Slashdot and Everything2 had three sources of inspiration for how and why they created social features in their early social media sites:

1. Participation in earlier forms of social media. The developers of these early web communities had experienced older forms of social media, including Usenet, bulletin board systems, and Internet Relay Chat. Consequently, they were already familiar with how large-scale, text-based interactions could lead to problems. They had experience with trolling and flame wars and anticipated those problems could also occur in the web communities.
2. Experience with the free/open source software (F/OSS) communities. As software developers, they were keen to take some of the social processes they had observed in the F/OSS communities and apply them to other endeavors outside of software development. With Slashdot, they wanted to apply those principles to journalism, and with Everything2 they wanted to apply them to an encyclopedia. They had contributed to, or paid attention to, the discussions around Linux and Apache, among other large F/OSS projects, and had experienced some of the emergent social structures that were characteristic of those projects (O'Mahony and Ferraro 2007).
3. Experience with games. The developers of these early web social media sites had been playing video games for the majority of their lives and had also been deeply involved with face-to-face role-playing games like Dungeons and Dragons. Consequently, they were familiar with the way in which features of the game could provide personal

benchmarks (achievements, quests, item acquisition), comparisons with other players (leaderboards, player-versus-player), reward systems (experience points, item acquisition), the effect of pseudonyms (role-playing), and the power of the designed environment to shape interactions (dungeon masters, nonplayer characters, quests).

These sources of inspiration were not unique to the developers of Slashdot and Everything2 compared to other social media developers. While there was academic work concerning the design of social features for computing systems, developers of early social media sites were more likely to learn about social features by experiencing them in their own use or importing them from similar contexts like games. When interviewing early social media developers about where they turned to for inspiration in designing social systems, they were much less likely to point to academic literature than to other social media sites or interactive experiences like games (Lampe 2006). The detailed descriptions of Slashdot and Everything2 highlight the ways in which experiences with these systems can lead to the development of social features that help organize social media environments.

Slashdot was created in 1997 by Rob "CmdrTaco" Malda as a site to share news stories with his friends (Lampe 2006). The site is framed as "News for Nerds, Stuff that Matters," and focuses largely on items related to the technology industry, open source software, science, and nerd culture. Slashdot remains an active site (as of this writing owned by Dice.com) and is an early example of a common-genre social media site, which also includes Digg, Reddit, Fark, Kuro5hin, and other social news aggregators and discussion sites.

Early in its history, Slashdot (or Chips and Dips as it was initially called) was a fairly simple web page that was updated by Malda to share news stories from other sites. As the number of viewers of the site grew, Malda was unable to keep up with e-mail responses from people reading his site, so he created a commenting system for stories. This was before sites like Livejournal had popularized blogs, and commenting systems were not common in web pages at the time. The number of people making comments grew to a larger scale, and in doing so the discussions started to exhibit some of the same management problems that had been rampant in Usenet. In response, Malda assigned a rating system to comments to help other readers to filter comments based on their preferences. Initially, only Malda and a close group of friends assigned ratings, but as the number of comments grew to thousands per day, he expanded the system to four hundred moderators, and then to all users on the site who had a positive reputation (Chromatic, Aker, and Krieger 2002). Slashdot does not remove comments, so this rating system largely acts as a filter, which readers can use to change the number or type of comments they see. Slashdot moderators get chosen to act on a temporary basis, with a limited number of votes. They vote on a comment by tagging it from a list of descriptors, each of which carries a +1 or −1 weight to the current score of the comment. For example, a comment labeled "Insightful" by a moderator would go up a point, and one labeled as "Troll" would go down by a point. Each comment can have a score ranging from −1 to +5, and the default settings are that only comments +1 and above are shown in their full text. Previous research (Lampe and Resnick 2004) has found that this moderation system largely separates out especially good and bad comments (figure 19.1).

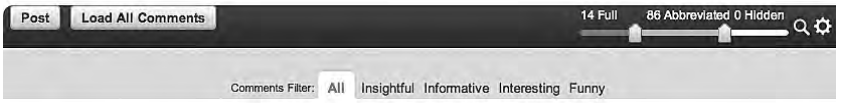


Figure 19.1
Slashdot allows readers to filter comments based on the ratings and tags assigned by other users of the system.

In order to track user behavior, Slashdot created a system for logging in under persistent pseudonymous user accounts, although it still allowed purely anonymous posting with all such comments posted under the name "Anonymous Coward." Slashdot also instantiated a reputation system to both reward prosocial behavior on the site and to help other users have more context about the people with whom they were having conversations. Malda (2007) has said that he initially described this feature as "reputation," but that people became much more interested in it after he had relabeled it as the "karma" system. Users could have negative or positive karma, and higher scores were based on activities like posting comments that had been rated up, reading articles, moderating comments, and other activities that were considered positive by the site owners. Users with good karma have their comments posted with a higher score than those of others and may receive moderation privileges more frequently. Early in the history of the karma system, Slashdot developers removed the exact karma number visible to each user (though it was still visible to administrators) and replaced it with descriptors like "Terrible" and "Excellent." They had found that some users employed the numbers to game the system. One example would be users who worked to gain maximum karma, and then used the privileges associated with high reputation accounts maliciously to attack other users or the moderation system.

Slashdot was one of the first web-based social media systems to have to deal with scale issues in managing a large body of users. The term "slashdotted" was coined to describe server crashes that resulted from being linked by Slashdot, and subsequently receiving massive traffic directed by the site. Like many sites, Slashdot evolved many of its tools over time, borrowing social features from systems that the developers had experienced in other interactive environments. Since Slashdot was started, there has been a much broader adoption not only of social media tools, but the Internet more generally (Hampton et al. 2011). This means that more overall social media use is possible than in the late 1990s, as more people are available. What hasn't changed is that many of these environments struggle with the same issues as Slashdot pioneered. While newer social media sites have added more media and types of interaction, the scale and low barriers to entry and exit in social media sites still require both broad experiences with interactive environments and constant innovation.

Everything2 was created in 1998 primarily by Nate Oostendorp, one of the developers working with Malda on Slashdot. The goal for this site was to "catalog all human knowledge," and it became a type of open contribution encyclopedia. This was three years before Wikipedia had formed, and there were several online communities centered around user-created knowledge repositories, including H2G2,

Interpedia, and Nupedia. Everything2 is an early example of an "open collaboration system," which Forte and Lampe (Forte and Lampe 2013) define as a social media site that "(1) supports the collective production of an artifact (2) through a technologically mediated collaboration platform (3) that presents a low barrier to entry and exit, and (4) supports the emergence of persistent but malleable social structures." Sites like Scratch, DeviantArt, Ravelry, Wikipedia, Quora, and others are all part of this genre of social media. Everything2 still operates, with minor policy and interface changes since the early 2000s, and a relatively large audience of article readers (Lampe et al. 2010; Velasquez et al. 2013).

Everything2 users create articles, or "Write-ups," with topics that can include anything from creative writing to factual, encyclopedia-like entries. Unlike in Wikipedia, Everything2 articles are the sole creation of one author, but each topic entry, or node in the parlance of the site, could have multiple articles. For example, the node for "Homer Simpson" could have multiple Write-ups by different authors.

The developers of Everything2 leaned heavily on common game structures when designing the social features of the site. Write-ups can only be added by registered users on the site, and users with positive reputation can vote on the content of others. As a user's content is voted up, they gain XPs, shorthand for "experience points," and as they gain XPs, they go up in level. Each level has additional user privileges, as shown in figure 19.2. The requirements to reach a level, and the "powers" associated with the levels, have evolved continually over the life of the site.

Besides the rating system for content and users, Everything2 also borrowed heavily from game terminology. For example, the site has regular "quests" to challenge contributors to write about certain topics,

Level	Level Title	XP Req	Writeups Req	Votes per Day	CIs per Day
0	Initiate		0	NONE	NONE
1	Novice	500	10	50	NONE
2	Acolyte	1000	20	50	NONE
3	Scribe	1500	30	50	NONE
4	Wordsmith	2300	40	50	1
5	Crafter	3300	50	50	1
6	**Artificer**	**4500**	**60**	**50**	**1**
7	Chronicler	5900	70	50	1
8	Encyclopedist	7500	80	50	1
9	Archivist	9300	90	50	1
10	Polymath	11300	100	50	1
11	Literatus	13500	110	50	1
12	Laureate	16000	120	50	1

Bugs go to mauler.

Powers:

- Level 0 — Joining Everything2 gives you the ability to contribute writeups, communicate with other members via the Chatterbox and message system, customise your view of the site in User Settings, etc.
- Level 1 — Can vote on E2 writeups by other users. Can give Stars to other users at the E2 Gift Shop. Can display a small (uncopyrighted) image in your home node (Nope, no porn allowed!).
- Level 2 — Can buy additional votes at the E2 Gift Shop; can create categories.
- Level 3 — Can create polls; can post a bounty on Everything's Most Wanted.
- Level 4 — C! power!
- Level 5 — Can display a larger homenode image (up to 400×800), ability to create rooms in the Chatterbox.
- Level 6 — Can reset the chatterbox topic by buying a token at the E2 Gift Shop.
- Level 7 — Can buy easter eggs at the E2 Gift Shop, and give them to other users.
- Level 8 — Can create registries.
- Level 9 — Can give votes to other users at the E2 Gift Shop.
- Level 10 — Can Cloak oneself in the Other Users nodelet.
- Level 11 — Can Sanctify other users with GP.
- Level 12 — Can purchase up to one extra C! per day at the E2 Gift Shop
- Level 15 — Fireball! Can "fireball" other users in the chatterbox using the /fireball command.

Figure 19.2
User levels in Everything2 and associated privileges.

and top-level administrators are called "gods," which is borrowed directly from early MUD terminology. In the case of Everything2, the games metaphor is so strong, it may be confusing or alienating newer users who are not as steeped in role-play–based games.

Everything2 may have more openly borrowed structures and terminology from games than other sites, but the intersection between games and early social media sites made sense for several reasons. First, games provided examples of interaction environments that were bounded in ways similar to online communities, but remained incredibly engaging and compelling. Second, because early participants in social media sites were also very likely to be involved in gaming, game structures provided clear metaphors for social interfaces in the same way terms like *desktop* and *file* had defined interfaces for individual users as a metaphor in the decades before the rise of social computing. In many ways, the current trends in gamification are a rebirth of this earlier design ethos, as developers have long been inspired by other interactive systems they experience, and most software developers have long-term experience as gamers.

Conclusion

Gamification may provide inspiration for how we shape new tools in social media environments that support and extend the work done in those systems. Social media environments are heterogeneous contexts, tied together by the common intersection of social processes like motivation, contribution, and interaction embedded in technical infrastructures that support the remediation of limits imposed by technology, but also extend interactions through the capacities of that same technology.

Which features to add are a matter of intuition and experience on the part of the developers of these systems. As seen with the Slashdot and Everything2 examples, designers of social media sites face challenges that emerge as the sites they create change and grow, and they turn to their experience with other systems for inspiration in how to design tools that support their users. Some of that inspiration comes from other sites they use, but some comes from parallel systems that have the same characteristics as social media sites. Games, especially massively multiplayer games, which share characteristics of social media, have often been a source of inspiration to site designers. As game designers continue to innovate around their own sociotechnical work, there are new opportunities to connect the design insights of game developers and social media creators. More academic work is needed to understand the exact affordances of gamification-like features and how or if they support specific types of interactions.

There are great opportunities in social media, but the inherent characteristics of these sociotechnical systems mean that social features need to be thoughtfully designed in order to realize those opportunities. Social media has changed how people consume news, organize protests, manage relationships, share content, and create art. However, as more and different types of people enter social media spaces, there is a need to address issues like the disparity in participation, managing user conflict, and supporting multicontext activity. Gamification provides an opportunity to think creatively about designed tools that can allow developers to create systems that help us reach the potential offered by these technologies.

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POSITION STATEMENT

COLLABORATION IN THE GAMEFUL WORLD

Peter Williams

The past two centuries of economic development have been primarily a story of *scalable efficiency*: as infrastructures and technology improved, companies grew larger to take advantage of economies of scale. To coordinate ever-larger groups of people, companies created rigid command-and-control hierarchies, silos, and processes. Unfortunately, these institutional architectures have a downside: just as they promote consistency, predictability, and efficiency, they limit an organization's ability to change or try new things. While effective during times of stability, the scalable efficiency rationale faces extreme difficulties during times of rapid change.

Over the past forty years, the emergence of new digital infrastructures and a global liberalization of economic policy have increased the pace of change exponentially. Thus, many companies that were successful under past conditions of stability are now struggling. As the pace of change increases, many executives focus on product and service innovation to stay afloat. However, the reality is that institutions need to rethink not just products and services but also their very architecture. To thrive in a world of exponential change, institutions require a new kind of rationale: *scalable learning*. What is needed are *creation spaces* that facilitate (rather than limit) interactions and relationships, allowing organizations to increase the flow of information within and across their organization's walls to foster learning, adaptability, and downstream product and process innovation (Hagel and Brown 2013).

As part of our research at the Deloitte Center for the Edge, we sought out examples of architectures that enabled such scalable learning across a broad ecosystem. This search led us to mass collaboration and the gameful world. We found that games can foster collaboration skills; that gaming communities are collaborating in ways that produce exponential returns in learning and knowledge; and finally, that organizations can learn from games how to institute scalable learning.

Affinity Spaces and Single-Player Games: Going Deep with Angry Birds Nest

As part of my own research into gaming, I decided to see if I could become a top-end Angry Birds player. I was generally able to get three of three possible stars on every level, but my personal challenge was to

move beyond and get the prized "Score Addict" achievement. The Game Center on my iPad gave me some indication of where I stood in terms of overall scoring. However, to know where I should focus my efforts, I needed more granular information. Some quick online research directed me to Angry Birds Nest. The Nest is what is known as an "affinity space" (Gee 2005): a site of informal learning where newcomers, experts, and everyone in-between interact around a common endeavor. Started and operated solely by Angry Birds fans, the Nest comprises walkthroughs, leaderboards, forums, community tools, blogs, and news. Players can enter their scores for each level and can see whether they are above or below average, where they rank compared to the other Nest members, and what the top score is. This information helps you see which levels give you the most opportunity to improve.[1]

And indeed, by joining the Nest community, accessing hard data in the form of leaderboards and qualitative data in the form of ideas and strategies, I was able rapidly to increase my mastery of the game, at one time ranking eighty-fifth out of 2.5 million players on Apple's Game Center Angry Birds Space.

Collaboration and Multiplayer Games: Halo in the Roof

As a middle-aged gamer cum geek parent, over the years I have heard many fellow parents bemoan how their children never went outside or spoke to anyone because they were all cooped up in their rooms playing computer games. I had a different point of view. Most weekends I observed my son's friends come to my house, armed with Xbox consoles and televisions, to set up eight-versus-eight Halo games. Sixteen kids and four televisions in one bedroom turned out to be pretty crowded, so I built a gaming room in my roof.

Up there, I could observe how my son and his friends were learning leadership and collaboration through gaming: given the collaborative nature of the game and the varying skills of players, they would rapidly coordinate team roles at the beginning of each game. At the end of each game, there was an analysis phase where teams went through each individual's performance. Once the analysis was done, they would mix up the teams, seeking a more competitive balance, and go again. As someone who leads teams in organizations, a few thoughts came to my mind:

- Teams formed and got to a high level of performance in a matter of minutes—in business, this process can take months.
- Weaker team members had active roles, learned quickly from their stronger teammates, and over time got significantly better at the game.
- The collaboration skills of the teenage gamers were significantly more advanced than those I have seen in most business contexts.
- Teams constantly reviewed their results both at a team and individual level and critically analyzed their performance. In business, most individuals have a performance review somewhere between every six to twelve months. Rather than reviewing frequently after every project, providing the opportunity to learn from team members in a transparent manner, the periodic review is done one-on-one behind closed doors.

There are some examples of similar gameful models in business. Particularly in agile software development, teams generally sit together; screens provide real-time information about the success of each build; if there is a problem with a build, teams scrum and fix it together; there are daily review meetings often both at the start and at the end of the day. Yet this is still not the norm for most workers.

The Collaboration Curve: Learning from World of Warcraft

For several years, my colleagues Douglas Thomas and Jagan Nemani have researched accelerated learning across large groups (Thomas and Nemani 2009). One surprising example they found is the massively multiplayer online role-playing game World of Warcraft (2004). In most activities and contexts, people's individual performance improvement describes an S-shaped curve: learning starts slow as one masters the basics, followed by a more rapid increase, only to slow down again as one enters higher levels of expertise. This was not the case with collaboration in World of Warcraft: the more participants and the more interactions between them, the higher the total speed of learning of the community at large. Thomas and Nemani called this effect the *collaboration curve*. So how does it come about?

For World of Warcraft players to be successful, especially at higher levels, they need to participate in groups: longer-lasting guilds, and parties and raids that perform quests or raids together. The size of guilds spans from a few to more than five hundred players, while parties are capped at five, and raids at forty. This means that often, playing knowledge is gained in parties smaller than the guild they belong to. Thus, for the whole guild to benefit from the activity of parties, party members need to share their newly gained knowledge with the rest of the guild.

And unlike the business world where knowledge is often guarded as intellectual property, individuals and guilds freely share their knowledge across the whole World of Warcraft community. Players use existing tools such as wikis, blogs, forums, and online video services such as YouTube to share data, strategies, and "how to" guides. Likewise, they build and share software to aid them in gaming—planning and scheduling tools, but also add-ons that provide quantitative dashboards to optimize game play and carry out post-raid reviews. Thus, when newcomers enter the game today, they can immediately make use of a rich ecology of knowledge and tools, and instead of having to "reinvent the wheel," players can focus their exploration and subsequent contribution to the knowledge commons on those areas of the game not yet well-understood.

Connectivity, Participants, Passion

When we look at the fundamental components enabling this mass collaboration, three key elements stand out.

The first is *connectivity*. Prior to the emergence of Internet infrastructure and cheap or free software to use it, it was unimaginable that knowledge communities with millions of participants could exist and collaborate together in close to real time.

The second element is the scale and composition of *participants*. The more participants there are, the

more interactions and knowledge flows can occur. And the scale of the World of Warcraft community is nothing if not massive: World of Warcraft currently (November 2013) has around 7.6 million active subscribers. Many players are members of guilds that span continents, gender, age, and professions. Thus, guilds can bring together people with diverse skills and knowledge who may in many cases not get to interact regularly. This point was vividly brought home to me when I asked my 13-year-old daughter about her Halo clan member and online friend "Mofo69." She casually replied he was a 22-year-old U.S. Army soldier who was on leave from the war in Iraq. For someone born in the 1960s this was surreal, but to my daughter, the only thing that mattered was whether the individual is "cutting it" when the battle is on.

The third element is *passion*: high-end players of any game tend to be passionate about the game. They want to learn and collaborate with others within and outside their guild to progress and solve challenges. This mirrors the findings of one of the studies we conducted as part of the Deloitte Center for the Edge Shift Index (Hagel, Brown, and Kulasooriya 2011): passionate workers tend to be most motivated to contribute to knowledge flow. Furthermore, they are more likely to seek knowledge and learning from *outside* their organization through social media, forums, conferences, and similar.

Bringing Gameful Collaboration to the Workplace: LiveOps

So what would it look like if organizations want to apply learning from the gameful world to their institutional architecture? One good blueprint is LiveOps.

LiveOps is a provider of multichannel customer services to organizations such as Salesforce, Ebay, and Coca-Cola. Unlike traditional call centers, LiveOps has a workforce of distributed self-employed agents that see themselves as a community of motivated entrepreneurs. Agents operate in the LiveOps agent desktop software that collects all tools and interactions in one place. They access a shared knowledge base and tool set for collaboratively resolving issues or searching answers. The community works as a meritocracy: agents with the best scores get routed more calls and are paid based on completed calls. Agents have levels and scores that are transparent to other agents to enable identification of expertise within the community.

Experts opt to help other operators learn and improve. This increases their personal reputation among the other agents and lends to them being routed more calls. Becoming an expert also provides satisfaction for being a known expert and helping others—indeed, some people we interviewed found that helping others is more fulfilling than actually getting paid for doing the work. Clients choose their key metrics, and agents can see their own performance on these metrics over time, as well as compare their performance to peers. This leads to agents focusing on those areas where they perform best rather than trying to be generalists.

Thus, LiveOps replicates the institutional architecture of gaming communities that drives scalable learning: passionate workers (as our Shift Index research found, self-employed people tend to be significantly more passionate than organization-

employed people), a powerful operational desktop (effectively the game environment), metrics and dashboards, an ever-growing knowledge base maintained by participant contributions and interactions, and a community focused on helping all participants learn and improve.

Conclusion

In these times of rapid change, institutions designed for the industrial age have to reconfigure around new models of collaboration that afford scalable learning. The gameful world shows what fundamentals need to be in place for that:

1. *Clear context*: In games, players can quickly understand the objective of the game and how to improve personally or collectively. Institutions likewise need to have a clear context or purpose so those interacting with them can understand what they are trying to achieve.
2. *Participants*: In gaming communities, as the number of participants scales, so does performance. Institutions need to think beyond their boundaries and consider how to extend their participant ecosystems.
3. *Passion*: When a gamer finds the right game, he or she becomes passionate. The individual wants to learn, share, and improve. Institutions need to think about whether the context they provide engenders such passion in their stakeholders.
4. *A culture of sharing and transparency*: Gamers know that we all get better through sharing strategies and techniques. While there may be a fleeting honor to be at the top of a leaderboard, it is great to learn and teach how to get even better. In my experience, this has been the hardest thing for people outside the gameful world to understand.
5. *Platforms for knowledge flow*: Gamers use whatever the most useful and accessible technology platforms they can find to share and find information, both hard data and soft strategies.

We will see much more research on all these issues in the coming years, and I look forward to that. But what excites me the most is to observe in person what will happen when the masters of collaboration in the gameful world influence the institutions that currently run the mainstream world.

Note

1. See http://www.angrybirdsnest.com/about/.

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20 MASSIVELY MULTIPLAYER RESEARCH: GAMIFICATION AND (CITIZEN) SCIENCE

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Games for Science

These days, massively multiplayer online games are a popular form of entertainment for many gamers. They range from the fantasy World of Warcraft, by far the most popular massively multiplayer online game, with millions of subscribers (IGN 2012), to the space simulation Eve Online with hundreds of thousands of subscribers (PCWorld 2012). Massively multiplayer online game players typically spend their time battling monsters or other players, completing quests, interacting with guild members, and coordinating complex raids. Many of these tasks require significant problem solving and collaboration among players.

This is where science comes in. There are plenty of difficult problems already in the real world, so new games are helping to train, to motivate, and to organize players to solve these real problems and make contributions to science. These are, for the most part, fully fledged games—but can also be seen as game design elements applied to the non-gaming context of science. Traditional game design can be used to make science more fun, more approachable, and more solvable.

Science itself is becoming more open, collaborative, inclusive, and connected in many different ways. As games become a larger part of more areas of science, they will no doubt encounter issues related to ownership, credit, and citation, and well as to rigor and ethics. These are games, but they are not *just* games—the players are doing real work, real research, which could have a major impact not just in their lives but in the lives of others as well. Currently, most scientific games are not based on an immediately pressing need—but they could be.

More people involved means more collaboration—and collaboration is a key aspect of science. When people with different perspectives, backgrounds, and experiences come together, they can work together to generate completely new ideas. Just two people coming together can give rise to new ideas—adding more can magnify this effect. Where else is there the opportunity for such massive collaboration between tens or hundreds of thousands of people than online games? Game communities bring people from all over the world together. Building on massively multiplayer online games, we are at the beginning of an era of massively multiplayer research.

Public Participation in Science, Volunteer Computing, and Human Computation

Games are a growing part of a broader movement to allow public participation in science (PPSR). PPSR engages members of the general public—who may or may not have a scientific background—in scientific research and discovery. Now, games add the possibility of a global network of minds engaged in problem solving.

Astronomy is a good example of a field with a long history of amateur involvement. Amateur astronomers have long been involved in the scientific process of the field, even contributing experiments to be run on the Hubble Space Telescope (HubbleSite 1992). Getting involved in astronomy was relatively easy to an extent—the primary requirements were a telescope and a passion for science. Now, that passion for science can be expressed in a variety of other fields. In particular, the field of phenology, the study of periodic cycles in animal and plant life, has provided a number of opportunities for people to contribute to science through observing their local environments.

With the popularization of the Internet, volunteer computing also arose as a means for the public to contribute to scientific research. Often in the form of screen savers, volunteer computing allowed people to donate the spare computing cycles of their desktop machines to scientific computations. A variety of projects arose, the most prominent of which was SETI@home, which used the extra computational resources to search for extraterrestrial signals in radio telescope data.[1] The possibility of being the person whose computer discovers the first evidence of extraterrestrial life is certainly enticing! Other projects look at protein folding,[2] searching for neutron stars,[3] and breaking encrypted messages.[4]

While contributing to science, volunteer computing is a relatively passive endeavor for those involved. Participants can track their statistics and contributions to various projects. Volunteer computing takes advantage of the vast networks of computers available but misses out on one key factor: directly involving the creative minds of the volunteers in the scientific discovery process.

Other, non-gameful, human computation systems have been developed to harness the power of human minds. Amazon's Mechanical Turk[5] pays participants to carry out small human intelligence tasks, things that computers would have trouble doing. Originally developed to tell humans and computers apart, CAPTCHAs are the squiggly text you have to enter from an image before you sign up for something on the web (von Ahn et al. 2003). This works because computers still have difficulty recognizing text in images, especially if the text is unclear. Following up on CAPTCHA, reCAPTCHA converts this text recognition into useful work by actually pulling those images from a database and having people help transcribe books (von Ahn et al. 2008). People solving CAPTCHAs are motivated by their desire to get at whatever is being protected by the CAPTCHA.

Games provide a new means for the public to get involved in science. They are a means for those who might not otherwise have had an opportunity to get involved and contribute or even to discover new skills they didn't know they had. With purely computational methods, an algorithm may be run in parallel many, many times, but it will still be running that

same strategy or picking from a set of predefined strategies. When people are involved, we're not just searching for solutions to the problems but, in a way, searching for good strategies and good approaches to solving problems. A person may come up with a new approach that no one thought of before.

Why Games? Why Now?

McGonigal defines a game as having four essential elements: rules, a goal, feedback, and voluntary participation (McGonigal 2011). Each of these elements interacts with the scientific aspects of a game.

The rules of the game can be derived from, and must respect, some underlying scientific model. The model can vary from problem to problem and can be more or less abstract. The model could encompass anything from the kinematics of protein chains to the classifications galaxies can take. But, if the results of the game play are going to produce valid scientific outcomes, then the constraints of the model must be respected.

Therein lies one of the key challenges of turning a scientific problem into a game: the game designer doesn't get to make up all the rules. Rather than coming up with rules, the designer has to make the existing rules clear. The designer can, however, augment these with additional rules to add fun.

Different games for science will give the player different goals and use different feedback systems. Typically, the feedback will be structured to direct the player toward the desired goal. However, this can be tricky. Unlike standard games, the goal may be unknown even to the designer! But, the feedback system can also be one of the greatest benefits of turning science into a game. The player can get directly to the core of the problem—the fun part—and interact directly with a fascinating model of the real world. As with all games, participation is voluntary. Players are free to play online games, joining or leaving when they wish.

To tackle the difficult problems of today and the future, we're going to need to apply our resources in the best way possible. Rather than looking at humans versus computers, we should be working together. Humans and computers have a different set of problem-solving abilities. Computers have come a long way in increasing their computational power and working out fast communication, numerical optimization, and visualization. But people still have an advantage in such areas as high-level pattern recognition, spatial reasoning, creativity, understanding emotion and sentiment, image recognition, and drawing on real-world experience. By combining the abilities of humans and computers, we can put that ability toward solving problems that would have been unsolvable by each alone.

Thanks to technology, humanity is more connected now than it ever has been before. People around the world can communicate and collaborate, working on their shared interests. In particular, games as a form of popular mass media have a key feature that previous forms have not: interactivity. Television, radio, movies, and books are all primarily focused on one-way communication: from the creator to the consumer. Games, however, allow the player to "talk back" to the game's creator, giving the player information on how he or she solved the problem contained within the game. The designer creates a

game, and through game play, the player can communicate back to designer through his or her in-game choices.

In addition, science itself is becoming more and more an information technology. Jim Gray argues that we are now entering a "fourth paradigm" of science, based on data exploration, following the previous paradigms of empirical, theoretical, and computational (Hey, Tansley, and Tolle 2009). This new paradigm is defined by the capture, curation, and analysis of scientific data.

Sites like DataONE (DataONE 2012) are working to make more and more scientific data open and available to anyone who wants to access it. DataONE, or Data Observation Network for Earth, has the goal of allowing universal access to data about the earth to enable science. It is a distributed data store where scientific data can be uploaded, shared, reused, and maintained.

It's yet to be seen how far into science games can go, but the concept of data-driven science fits well with games. Players don't have to be in the lab or at a university to participate. They don't have to buy expensive equipment; the tools to participate are in their laptops, tablets, and phones. The data can go to, or come from, wherever the players are: at home, in school, even on the bus. As science becomes more and more of an information technology, more and more aspects of science will be subject to the influence of games.

The games themselves can be seen as a continuous experiment. It is no longer the case that a game has to be shrink-wrapped and shipped. Online games can be continually refined and improved based on the scientific results produced. Updates and "expansion packs" can adjust and add tools, visualizations, and other features in a process of continual improvement. In fact, depending on the difficulty of the problem of focus, it can take time for a game to become useful after it is released: taking the time for the community to learn about the problem and how to apply the tools can be a critical part of the evolution of the game. Some scientific games have taken years after their release to produce discoveries—highlighting the difficulty of the problems these games set out to solve.

Gameful Science

Video games present an exciting new paradigm for mass collaboration in science. They provide an excellent framework for combining what humans are good at with what computers are good at, and putting this combined power toward solving previously unsolvable scientific problems. The way games structure and present problems in increasing levels of difficulty can help to train novice players and bring them to the point where they can contribute to research, and a game can change over time to become a better problem-solving tool. Game mechanics and feedback can make scientific problems engaging and approachable. Game communities can let players get involved who otherwise might not have been able and work on problems that are relevant to them. This approach is not without difficulty, however; the game's designers must face the dual challenge of making a game that is both fun and relevant to science.

Science can benefit from games, but games can also benefit from science. Games for entertainment can

support a number of different motivations. A common way of looking at different types of players is with a classification system develop by Richard Bartle. His classification system, commonly used by game developers, divides players into achievers, killers, socializers, and explorers depending on their motivations for acting on or interacting with the game's world or other players (Bartle 1996). Games for science add a new potential for motivation: involvement with the real world, rather than the game's world. This in turn can lead to a new audience for games. Games for science can provide a new meaning that is absent from purely entertaining games.

Turning a problem into a game presents a number of challenges (Cooper et al. 2010). Which part will people be good at? Which part will computers be good at? At what level of abstraction should the problem be presented? How will players interact with the model? How will the model be visualized? How will the players receive feedback on their progress and solution?

Depending on the difficulty of the problem at hand, there are varying degrees of training required for players. How much is necessary for a player, who may have never encountered the problem before, to get to the point where he or she can make a novel scientific contribution? For a basic image-labeling task, it may require no time at all. But more complicated problems could require weeks or even months of play and expertise development to solve a problem.

For such difficult problems that require the development of expertise, games would do well to recognize, develop, and reward the development of different skills and allow players with different skills to work together. Players can self-organize or games could explicitly try to identify different skills in players and present appropriate problems to group players accordingly.

Applications

We're already seeing a number of exciting applications for games in science. This is only a small sampling of games out of a rapidly growing field. Game researchers have already begun to look at the breadth of games being developed for science and other purposes (Quinn and Bederson 2011).

Biology

Biology has been a particularly fruitful area for new games and science. Biology is a promising early domain because many biological problems are structural in nature. Humans have an edge over computers in some types of reasoning about structural relationships. For computers, which simply see the numbers, the number of spatial possibilities to try is just too great. A good visualization can allow players to see and intuit structural changes and find the right fit or alignment. Further, results of games in biology could be applied to health and medicine and have significant impact.

Foldit

Foldit allows its players to compete and collaborate while manipulating protein structures (Foldit 2012). Proteins are fundamental molecules to life and carry out a number of important functions, from fighting disease to digestion. Understanding their shapes can

help scientists learn more about how life works. Foldit players compete to fold a protein that has a high score—their score is automatically computed. Protein folding is a complicated problem, and the game introduces it through a series of introductory levels of increasing complexity.

Foldit players have produced a number of scientific results. Foldit players were able to determine the structure of a protein experimentalists had been working on for more than a decade—a protein related to AIDS in rhesus monkeys (Khatib et al. 2011). The structure was actually discovered not by an individual, but by a team of players working together and improving on each other's solutions. That team, the Foldit Contenders Group, was listed as an author on the paper that describes the structure—certainly the first time a video game team was listed as authors on a scientific paper describing the solution to a long-standing scientific problem!

In addition to determining the shapes of naturally occurring proteins, Foldit players can also design new, synthetic proteins that can have a variety of purposes, including helping to fight disease. When a player designs a promising protein, it can be synthesized and tested in the wet lab to determine its effectiveness. Foldit players have contributed to the redesign of an existing enzyme, increasing its effectiveness (Eiben et al. 2012). The design process was an iterative collaboration between the players and the scientists, going back and forth to improve on each other's designs.

EteRNA

EteRNA is also concerned with biological molecules, but of a different sort: RNA (EteRNA 2012). The game's goal is to create a large-scale library of synthetic RNA designs in a variety of shapes. This will help to illuminate the principles of RNA design and allow scientists to create even more medically useful biomolecules out of RNA. In the game, players are able to redesign strands of RNA and see what kind of shape they form. The scoring in EteRNA is actually based on the results of experimental validation of players' designs.

Players start with simple shapes to make, and as they solve tutorials and other puzzles, they earn points to unlock more and more areas of the site, eventually being able to work on science problems and vote on which designs should be tested in the lab.

Phylo

Phylo takes another approach to involving players in science (Phylo 2012). Phylo's game play is all about aligning the genome sequences of different organisms: humans, rabbits, horses, and so on. The players slide rows of colorful blocks around, trying to get as many matches as possible in a column while avoiding mismatches. As players progress, more and more sequences to be aligned are added. Phylo introduces the rules of the game to players in a presentation-format online instruction manual.

Correctly aligning genome sequences can help scientist learn about the relationships between the different organisms, how they evolved, and a lot about genetic diseases. Phylo players have been able to improve the sequence alignment of the majority of the sequence blocks they were presented with (Kawrykow et al. 2012).

EyeWire

EyeWire enlists players to map the connectivity of neurons using nanoscale images (EyeWire 2012). This mapping could help scientists better understand the differences in neural connectivity between different

people and provide insights into memories and mental disorders.

In EyeWire, players are given a cube of neurons to look at. They can scan through 2D cube slices, working together with an automated algorithm to correctly color in individual neurons in 3D from the deeply intertwined mass of neurons presented. Players first proceed through a set of tutorial levels to learn how the game and interface work, filling in a progress bar as they correctly locate pieces of the target neuron. The neurons get more complicated and have more branches. After completing the tutorials, players are rewarded with points for each cube they help label.

Biotic Games

Another fascinating intersection of games and science is biotic games (Riedel-Kruse et al. 2011). These are games that operate on biological processes. One such game that has been developed as a prototype is PAC-mecium. Similar to Pac-Man, the player wants to control his or her avatar to collect food while avoiding enemies. However, in PAC-mecium the player uses a game controller actually to control living paramecia. The controller induces an electric field in the environment in which the paramecia are contained, influencing their movement. The environment is captured by a webcam and virtual objects are superimposed on the playfield. The unpredictability of the movement of the paramecia provides an additional challenge.

The project has a number of stated goals. By playing these kinds of games remotely, there is the potential to allow players to carry out real-world experiments and gather data. Were these types of games to become popular, the cost-reducing pressures that have helped to bring down the price of the consumer electronics used to play games might be applied to the equipment needed to play biotic games—which is the same equipment needed to carry out scientific experimentation. And in this case, the experiment involves the control of a living organism, the paramecium—so these games also bring up issues relating to how far this can be taken and initiate a dialog around them.

Astronomy

Building on the long history of amateur involvement, astronomy is another promising scientific domain. New telescopes and sensors are deluging astronomers with more data than they can sort through by hand, and the visual nature of the data prevents automated analysis by machine vision algorithms. Notably, most of these projects are not strictly billed as games, but nevertheless have a number of gameful qualities.

Galaxy Zoo

Galaxy Zoo, one of the most successful astronomy projects built on public participation, was initially built to process the massive streams of data from the Sloan Digital Sky Survey (Galaxy Zoo 2012). The idea is straightforward: classify or otherwise provide information about a galaxy based on its shape, such as a right-hand spiral or an ellipse. Because the player is identifying simple shapes, little initial training is required, and the players can jump right in to classifying galaxies. Many people will classify an individual galaxy, and this redundant information can be used to gather some degree of certainty about the classifications. After classifying a galaxy, a player has the option of discussing the image with other players on the project's forums. Over the lifetime of the project, participants have contributed millions of galaxy classifications.

Galaxy Zoo players have contributed to a number of scientific publications. Among other things, they have helped to carry out a statistical study of the handedness of spiral galaxies (Land et al. 2008). Even more exciting, Galaxy Zoo participants discovered and named a class of galaxies called Green Peas, so called due to their distinctive green color (Cardamone et al. 2009). Participants in the project's online forums, where the participants can raise questions about particularly challenging or interesting classifications, brought the Green Peas to the attention of researchers. This is an interesting and serendipitous discovery, in that it was somewhat tangential to the original goals of the project and something that was discovered by the participants themselves and brought to the attention of researchers. It is possible that the Green Peas could be related to star formation in the early universe.

The success of the project has led to the creation of a new parent site, Zooniverse,[6] which hosts an array of new scientific projects, many of which expand from the initial astronomy domain. These include games such as Old Weather[7] and Snapshot Serengeti.[8]

Stardust@Home

Stardust@Home is another astronomy project designed to search for traces of interstellar dust embedded in an aerogel collector (Stardust@Home 2012). The Stardust spacecraft collected the dust during a mission to rendezvous with a comet. Interstellar dust is believed to play a role in the formation of stars and planets; locating dust particles can help scientists learn about our universe. Project contributors, or "dusters," can use a virtual microscope to examine an aerogel sample—essentially a stack of images that the user can navigate through—and search for tracks left by the dust.

Finding the tracks can be relatively intricate and meticulous, so Stardust@Home has a tutorial sequence that shows users aerogel samples with their tracks labeled as examples. Only after exploring the given examples and taking a short quiz can the player sign up for an account. While looking for tracks, players will sometimes be given calibration samples, for which the track locations are already known. Based on their performance on calibration samples, players are ranked by attributes such as power (all tracks successfully identified) and skill (how often a player can find a track when there is one present). These player attributes are used to give confidence weighting to their identification of unknown tracks.

Players who discover a dust particle are given the opportunity to be authors on the scientific publication and to name the particle! Stardust@Home has begun to produce several candidates of potential dust discoveries (Gainsforth et al. 2012). The first player to discover a particle named his initial discovery Orion.

Computer Science

In computer science, many games have players help to process or label images and other data that computers still have difficulty interpreting. These data could be used to train machine-learning algorithms that can improve the systems that players are using. Further, heuristics and other algorithms could be directly learned or inferred from player strategies.

ESP Game and TagATune

ESP Game was an early game based on generating labels for images (von Ahn and Dabbish 2004). Given

an image, what words would describe the image? This is a difficult problem for computers. A huge data set of images with labels could be used for a variety of purposes, from training computers to be better at generating the labels to improving keyword searches of images.

The ESP Game matches two randomly chosen players online and presents them a sequence of images. The players have no means of communication, so the only information shared between them is the image they are currently looking at. Each player can enter words that describe the image. Once any word entered by a player matches one entered by the other player, both players receive a point and they move on to the next image. Some images have "taboo" words that the players can't use, in order to encourage players to agree on words the system hasn't seen before.

This approach is referred to as output agreement. The players get the same input, and their label is the output. When the players agree on an output, it gives the system some confidence that the word the players agreed on would be a good label for the image.

A similar game, TagATune, uses a complementary approach known as input agreement (Law and von Ahn 2009). In TagATune, the two players are presented with a sequence of audio clips. For each clip, the players can enter words to describe it, which are then seen by the other player. By considering the words the other player has entered, the players must independently decide if they are listening to the same audio or not. After both players decide, they will receive a point if they were both correct, and move on to the next audio clip.

Both the ESP Game and TagATune are collaborative: the two players share a score and enter information meant to help the other player. The players try to get as many points as possible before their time runs out. A whole suite of related games is available as part of a larger effort to gather data labels from game play, called GWAP,[9] or Games With A Purpose.

Pebble It

The suite of games Pebble It tackles a more theoretical computer science problem: the graph pebbling problem (Pebble It 2012). Graph pebbling can be used as a model for other resource allocation problems. A graph is simply a collection of nodes connected by edges, and a node can hold pebbles. A pebbling step consists of removing two pebbles from a node, and adding one pebble to neighboring node (connected to the original by an edge).

In Pebble It, the player is given a visualization of a graph, with nodes appearing as baskets, edges as lines connecting the baskets, and pebbles simply appearing as pebbles. Each problem has a ranked leaderboard where players can see how well their solution compares to those of other players. In one subgame called Reach It, the player can perform pebbling moves to move pebbles around on the graph. Each node starts out with some number of pebbles, and the player can perform pebbling steps to try to determine if it is possible to place a pebble on a given node in the graph. The player's ranking is based on number of moves, with fewer moves representing more efficient solutions, leading to a better ranking. In another subgame, called Confound It, the player is given a graph and tries to place initial pebbles on nodes so that at least one node is unreachable. Here the player's ranking is based on the number of pebbles placed, with more pebbles leading to a higher ranking. The player tries to place as many pebbles as possible, while still stumping the computer, which

tries to solve the pebbling problem given by the player.

One goal of Pebble It is to determine if humans can help solve NP-complete problems, a particular theoretical class of computer science problems that are known to be theoretically solvable by computers, but in practicality they cannot be, because it would simply take too long for computers to find the answer (Cusack et al. 2010). Many practical problems are NP-complete and could theoretically be mapped onto NP-complete graph pebbling problems. Game play could be used to solve particular instances of these problems, or to learn new heuristics to speed up computational algorithms, allowing them to solve larger and larger instances of problems, or to discover shortcomings in existing algorithms. Early on in Pebble It, players were able to find more efficient solutions to graph pebbling problems than those known to the researchers who created the game and to help researchers focus their algorithmic development efforts.

These early results have primarily been applied to relatively small problems; it remains to be seen how players will fare with the very large problems that computers alone are unable to handle. Pebble It has also grown into Algoraph, a suite of games built around other theoretical graph related problems, beyond graph pebbling.

PhotoCity

PhotoCity takes a different approach: its goal is for players to collect data, in the form of images of buildings and architecture (PhotoCity 2012). These images can be used to generate 3D models of the building that could be used for mapping and other applications. Rather than gathering existing images off the web, which might not be useful for reconstruction purposes, PhotoCity encourages players to go out and take the pictures that will be most useful for creating the 3D models of buildings.

The game is essentially a virtual game of capture the flag played in the real world. The game shows players a number of virtual flags attached to buildings on a map, and they can try to capture the flags. As a player takes pictures of where the flags are in the real world, he or she can upload the photos to the game, earning points for good photos, and with enough points can capture flags. As flags are captured, the system generates more flags: here more data are needed, encouraging players to capture the most useful data. Teams of players compete to capture the most territory on the map.

Computer vision algorithms process the uploaded pictures to generate 3D models of the buildings. Players can see the models improve and be refined as they upload photos. In one study, PhotoCity players were able to gather more than 100,000 photos in just six weeks (Tuite et al. 2011). Around 60 percent of the photos were useful in reconstructing building geometry. This is in contracts to the Rome in a Day project (Agarwal et al. 2009), which the PhotoCity team calculated used only about 10 percent of 150,000 photos of Rome found on the web. Clearly, the players' involvement leads to more efficient and useful image capture.

The Future of Games and Science

Games are a growing part of massive collaboration in science. We may now be at the forefront of an explosion of games for science. Scientific games have yet to become quite as popular as World of Warcraft. But it won't be necessary for a single game to dominate: a plethora of smaller games, each focused on their own scientific problem, could contribute a mass of new brainpower to a variety of endeavors; an ecosystem of games that can interoperate, share data, and let players contribute to the scientific problem they are most passionate about; games that are difficult to master, but easy to get into, on Facebook, on phones and tablets, on consoles, all built around a kernel of scientific discovery.

Games can also help to advance scientific awareness and education. More transparency makes science more approachable and accessible, and a better introduction to science can help lead to more people studying science and going into careers in science and technology. Games can help to illuminate the scientific process and make it less mysterious, putting players directly into the shoes of the scientist—forming hypotheses, running experiments, gathering and analyzing data, and drawing their own conclusions. This will be useful, because public sentiment may be even more of a driver of scientific research. Problems that people understand, care about, and are interested in helping to solve will have more involvement from the public, and thus more minds helping to solve them. Science may advance faster in the domains where the crowds go.

Beyond just playing games, we need to work on developing tools for creating games—enabling the public to participate through the creation of games. Just as the creation, editing, and sharing of video, photos, and music have been drastically changed in recent years, new tools will make game creation more and more accessible. Games and other forms of interactive media certainly present their own unique challenges. But as more computational scientific models are developed and made available, it will be possible to use them as the base rules for more and more games. People could make games and communities around problems that are relevant to them. We could see a rise in player-driven science. In addition to contributing to research that scientists are carrying out, players will be using games to conduct their own research, answer their own questions, and solve their own challenges.

As games and science become more intertwined, we'll see a change in the role of the scientist. Part of the role of the scientist has always involved communication with the public. With games, there is an increased opportunity to change this into more of a dialogue. Could video games be part of the global laboratory of the future? There may come a time when many of the interesting problems require this kind of massive collaboration to solve. Why limit a lab to a few or dozens of people when it could be thousands or millions?

Closer integration of games and science will also change the role of the player. Clearly, players should receive credit and ownership of their work and discoveries. Some players are interested in paper authorship, but not all; some players may be satisfied with in-game recognition or the knowledge of contributing to the greater good. In some cases, individual contributions may be difficult to determine or there will simply be too many contributors to list them all. We're already seeing names like "Foldit

Contenders Group," "Phylo Players," and ">30,000 Stardust@Home Dusters" in the author lists of scientific publications. This kind of attribution will only become more common as scientific games progress. Perhaps the loftiest goal for these kinds of games will be to allow such an important discovery that a player could win the highest honor in a field, a Nobel Prize, a Turing Award, a Fields Medal.

Many of the current promising applications in games lie in some form of gathering, curating, and analyzing data. As science moves to be more and more data intensive, these approaches will be applicable to more areas of science. "Zooniverse" projects cover climatology and zoology. Games like PACmecium and PhotoCity show the potential for games to tie in to the physical world, with players potentially controlling experiments or collecting the most needed data. How far can this approach be taken? Could games allow players to run their own experiments in physics? How could a game help paleontologists? Could games organize players to construct formal proofs of long-standing mathematical conjectures? We have yet to see how far games and science will take us if we work to make games that tackle big problems and bring together many creative minds.

Notes

1. See http://setiathome.berkeley.edu/.
2. See http://boinc.bakerlab.org/.
3. See http://www.physicscentral.org/experiment/einsteinathome/index.cfm.
4. See http://www.enigmaathome.net/.
5. See http://www.mturk.com/.
6. See http://www.zooniverse.org/.
7. See http://www.oldweather.org/.
8. See http://www.snapshotserengeti.org/.
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21 GAME STATE? GAMIFICATION AND GOVERNANCE

Greg Lastowka and Constance Steinkuehler

The chapters and position papers in this book provide a variety of perspectives on the extension of games and game play into new frontiers of human life. The editors have asked us to consider the *gamification* of one particular aspect of the world: the government. In short, we are asked to answer a question: can the role of governments be *gamified*, and would the gamification of government be a welcome development?

This is a difficult question. Part of the difficulty can be attributed to the notion of *gamification* itself—what does the term mean, and how might it apply to government? We will break the discussion into two parts. We will first answer our preferred version of the question: can increased state investments in the study and utilization of games improve public welfare, civic participation, and education? We believe the answer to this question is an emphatic "yes." We then turn to the second version of the question: should government be gamified? We are substantially more skeptical about this possibility.

The Civic Potential of Games

Government interest in games is not an entirely new phenomenon. From the Olympic games to the Roman Coliseum, governments have long been involved with games. In recent history, governments and civic leaders studied and used games to enhance educational practice, perfect military strategy, and construct economic policy (von Neumann and Morgenstern, 1947; Piaget 1962; Abt 1970; Perla 1990). Clearly, governments have used games for some purposes. However, despite this history, there is usually some cognitive dissonance encountered when governments are involved with games. Games are associated with relaxation and leisure. Government, in contrast, is viewed as a sober and rational enterprise (ideally) that steers, for example, military force and economic policy. The conflicting cultural expectations surrounding games and governments tend to keep them intellectually separated.

A particular contemporary problem, in our opinion, is the (improper) popular stereotyping of video game play as a wasteful, solitary, and juvenile activity. This negative stereotype of video games has been a significant roadblock to the full utilization of video games by governments. The stereotype fails to conform with reality. Recent research on video games has suggested that they often provide significant

educational benefits (Gee 2003, 13; Jenkins 2006; Steinkuehler 2008). They are often intensely social, textual, and collaborative (Gee 2003, 169; Jenkins 2006; Taylor 2006; Steinkuehler, Compton-Lilly, and King 2010). Multiplayer games often serve to bring together individuals from a variety of backgrounds and ages (Kerr 2006; Steinkuehler and Williams 2006; Taylor 2006; Burri-Nenova 2010). Among other outcomes, video games can build technological and literary fluencies, increase visual acuity and attention control, improve health outcomes, build stronger ties to local and national communities, and increase conceptual understanding in a variety of key domains. In short, despite popular mythologies, video games are creating significant public benefits today for a wide variety of people.

As Dmitri Williams has explained, the history of video game regulation has reflected a "moral panic" on the part of those unfamiliar with the medium. The contemporary fear that video games corrupt youth is very similar to earlier fears of billiards, comic books, and television (Williams 2004). The contemporary reality is that the public (including legislators) increasingly uses networked technologies to facilitate new forms of social organization. We are increasingly communicating, laboring, and playing together by utilizing digital technologies that incorporate game-like features, and game companies strategically employ individuals skilled in creating systems that enable social engagement and productive collaboration. For better or for worse, in effect, successful game developers are functioning today as de facto civic planners.

Of course, today's prevalent new media technologies are often the subject of concern themselves. Platforms such as Facebook and Twitter are sometimes faulted (and sometimes rightfully) for the decline of certain traditional social and civic institutions (Turkle 2011). Others criticize these new platforms as being insufficiently protective of the broader public interest (Lastowka 2010). However, digital networks also raise new possibilities for modes of community and novel forms of positive social capital (Steinkuehler 2006; Steinkuehler and Williams 2006). As national and local governments around the world embrace a more networked world, digital game developers will often possess the very practical knowledge necessary to create the set of tools and rules that can lead to various forms of desired community engagement. Such expertise may prove valuable for governments seeking to promote civic engagement within local and national communities (case study 21.1).

Given the growing literature today on the positive social potential of video games, we believe it is obvious that governments should expand investments in the study and implementation of online games and game design. One of us (Steinkuehler) has been deeply engaged in this sort of work within the White House Office of Science and Technology Policy (OSTP). OSTP created a new senior policy analyst position in 2011 in the first (and now second) term of the Obama Administration to explore how games and game-based technologies might be leveraged to forward work in key areas targeted by the administration. The position was intended to explore certain key policy questions, such as: What games should exist and do not yet? What ecosystem for innovation is necessary for such games to be created? And what new discoveries in the social and behavioral sciences are we ready to make through strategic use of games?

Until the creation of this leadership position within the federal government, investments in game research and development proceeded largely under

Case Study 21.1

Macon Money

Summary

Macon Money is a location-based, hyperlocal, blended (online/offline) game that uses a local, novel currency called "Macon Money" as a vehicle for connecting local citizens across adjacent communities that had largely been disconnected and to encourage residents to frequent local businesses in the College Hill Corridor, an area of town targeted for revitalization. Launched in 2010 in Macon, Georgia, Macon Money bonds, worth a total of $65,000 in $10, $20, $50, or $100 increments at local businesses, were printed and distributed among residents of the city across multiple zip codes and otherwise insular communities. Game play was built on a simple yet powerful concept: Residents received only half a bill in the mail, requiring them to go online (through Facebook, Twitter, or Macon Money online forums) or out into their community at local "Macon Money" events to find the local area resident that held the other half of the bond in order to "make money." This novel approach to civic engagement at the local level successfully reduced socioeconomic segregation by bringing together area residents who might otherwise have never met while at the same time generating an influx of new customers among vital local businesses in the target area.



BoxFigure21.1

Macon, Georgia, residents, after individually receiving only half a bond in the mail or at an area event, coordinate with other local-area residents to find the other half, enabling them both to then "make money" that can be cashed in at targeted local businesses.

Case Study 21.1
(continued)

Facts and Figures

Macon Money launched in October 2010 and officially drew to a close in June 2011. Over its life span, the game engaged nearly three thousand local Macon, Georgia, residents and distributed all of its $65,000 worth in bonds. The game was created by Area/Code as part of the John L. and James S. Knight Foundation's Games for Engagement Initiative. It won the 2011 Future Everything Award for outstanding innovation in art, society, and technology. The game website continues to provide resources for other communities interested in similar endeavors.

Gameful Design Elements

In trying to motivate community residents to interact across disparate communities and to spur economic development along the designated target area, Macon Money leverages these key characteristics:

- Hyperlocal: The game content, mechanics, and time frame were driven by geographic, cultural location, not by imagined global market. The original revitalization project of the College Hill Corridor began in 2002 and was conceptualized and implemented as a "citizen owned, community driven" (Knight Foundation 2012) effort. The Macon Money game was designed to augment those efforts, working directly with partners at the site of implementation.
- Blended: Game play required both online and offline interaction and relied on both digital and analog materials. Such blending enables players where they are most comfortable, be it Facebook or the local VFW.

Our Criteria

1. Voluntary choice: yes
2. Active and enjoyable: presumably
3. Significant public benefits: yes

Outcomes

Macon Money did indeed succeed to impact positively both community and business development in the target area. Evaluation of the game (Knight Foundation 2012) showed a positive correlation between amount of game play and positive attitudes toward the area, with 71 percent of players reporting that they would recognize their matches and greet them on the street. Overwhelmingly, 92 percent of players reported returning to those local businesses targeted in the game, 46 percent of which were entirely new to the player. Efforts to engage African Americans in the game were less successful, with only 16 percent of Macon Money players being African American compared with 66 percent of the local Macon area population.

Further Information

http://www.maconmoney.org/ (official website)
http://www.knightfoundation.org/search/?q=macon+money (Knight foundation website with news and updates on the game)

the radar of civic discussion. In the past, most game-based projects received agency support not because they were game-based but instead, overwhelmingly, *despite* being game-based. Yet over the past decade, millions of dollars have been invested in games across a broad range of agencies including the National Science Foundation, the National Institutes of Health, the U.S. Department of Energy, the U.S. Department of Defense, the National Aeronautics and Space Administration, and the National Endowment for the Arts.

Example investments span a spectrum of applications. There are Web-based browser games designed to facilitate science education. There are mobile games designed to increase activity levels among young people and thereby address the childhood obesity problem. Puzzle games have been created to teach energy-saving practices. The military has created immersive simulation-based video games to train soldiers. There are massively multiplayer online games that are set on distant planets and that use accurate 3D models of space rovers and other gear. There are game-related art installations in national and regional museums.

Coordination of these diverse investments in public games increased recently with the creation of a Federal Games Working Group and a National Science and Technology Committee subcommittee focused on Digital Game Technologies. Through intra-agency groups such as the Federal Games Working Group, nearly two hundred program officers from thirty-three government agencies and four White House offices are developing new ways to collaborate and jointly increase their shared expertise in games. Their shared mission is to identify redundancies and missed opportunities in the portfolio of cross-agency game investments, to share solutions to common challenges in game-based investments across the agencies, and collectively to broaden their national network of experts in both game research and game development as a way to radically improve the quality and scaling of game-based approaches. If indeed successful game developers function today as de facto civic planners, then connecting such expertise to the very agencies whose mission it is to steer a given nation soberly and rationally seems necessary, albeit ironic. From this view, play becomes a vehicle for governance indirectly.

Attributes of Successful Games

In our opinion, merging the knowledge gained by studying game design with the public benefits delivered by governance systems is not a simple task. It entails a rigorous approach and a very practical range of inquiries. Though a full investigation and survey of contemporary state investments in games is beyond the scope of this chapter, we want to highlight three elements that we consider essential to the success of state-sponsored games.

First, participation in state-sponsored games should, ideally, be a matter of *voluntary choice*. Many definitions of games stress that the free and autonomous choice to engage in play is central to the nature of games (Huizinga 1950; Suits 1978; Caillois and Barash, 2001). Even if one can be forced to participate in a game by the state, enforced play is arguably not play at all. Play with ideological, cognitive, and/or behavioral intent made mandatory by the state has

an Orwellian quality to it, eliciting conformity to a regime rather than self-reflective, presumably deliberative choice. Such systems cannot be said to foster civic engagement at all but rather only a kind of sneaky, stolen compliance (case study 21.2).

Second, and following from the first element, state-sponsored games should, ideally, be both *active and enjoyable* for participants who play. An important and valuable part of any game is that the player is not simply a passive recipient, but an agent capable of action. Good games harness this attribute by providing intrinsic benefits that flow to participants from the process of play. They are interactive, designed spaces for choice and action. Games that are "designed by committee" can often stumble on this element, most often by replacing user-center design with committee-centered design. In such contexts, where a priori and external criteria replace play tests as the measure of the merit of a given game, it is easy to design a system that has all the surface features of a game and that provides some measurable outcome of benefit to players, but that is not intrinsically enjoyable to all or many or even some. Simply put, if a game is not active and enjoyable, it will fail as a game by definition.

Third, state-sponsored games should, of course, create *significant public benefits*. The nature of these benefits will vary. Games might teach important skills: collaboration, self-expression, time management, systems knowledge, and so forth (Gee 2003; Steinkuehler 2008; Bogost 2010). They may, as in the case study of Macon Money, help facilitate community solidarity and civic knowledge. They may advance scientific knowledge, foster economic growth, augment existing knowledge, or promote participation in volunteer activities. Games can provide a broad range of benefits. At the very least, state games should contribute to the community spirit of participants, in the long tradition of civic rituals and celebrations.

Given the diversity of ways that games might be used by the state, the selection of particular games for state investments will be (and has been) a political question depending on available funding and state priorities. It may often be the case that private-public partnerships will be the most promising method of state investment in games.

In summary, it seems uncontroversial that governments can, and should, use games, instrumentally, in ways that further the public good. It also seems uncontroversial to us that governments should

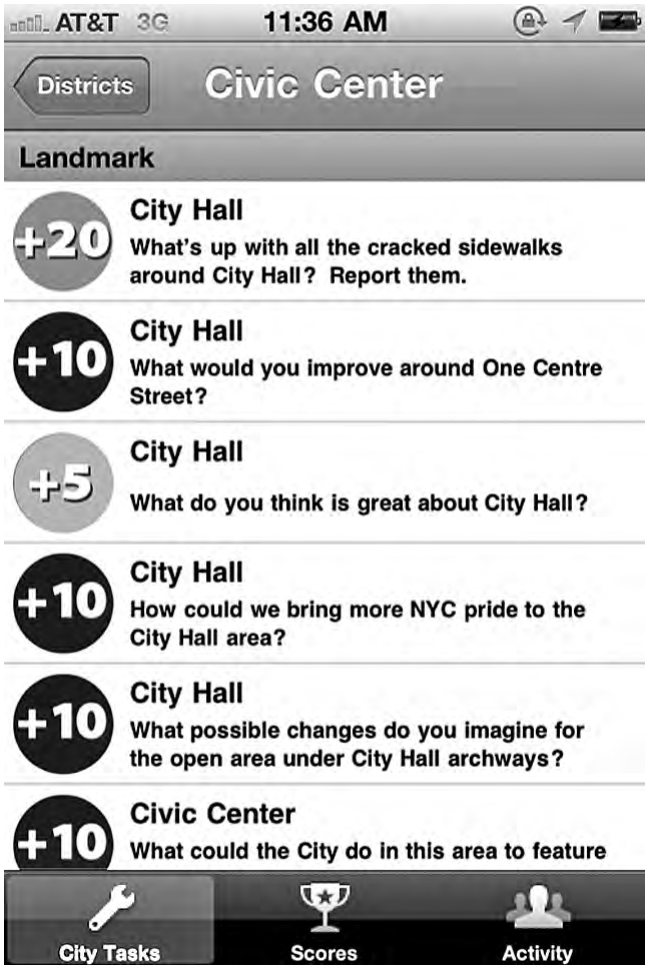
Case Study 21.2

Commons

Summary

Commons is a mobile game for citizen stewardship in urban environments that makes citizen reporting more gameful. During game-play events lasting two hours, players scour a given urban neighborhood and report via the game mobile interface a problem to fix, a suggestion for improvement, or an appreciation. Each City Report includes a brief (100-character) description, the nearest street intersection, a geo-tag, and a photo. Players earn experience points for their City Reports, and the community player base votes on submissions in order to rank players.

Case Study 21.2
(continued)



BoxFigure 21.2
In Commons, players earn their final score based on their experience points earned and community votes.

Case Study 21.2
(continued)

Facts and Figures

Commons, named after "The Tragedy of the Commons" (Hardin 1968), was designed by Suzanne Kirkpatrick, Nien Lam, and Jamie Lin. The game launched on June 19, 2011, in New York City with the opening of the Games for Change conference; the urban neighborhood targeted during its inaugural run was Foley Square in lower Manhattan. It won the first annual Real-World Games For Change Challenge in 2011.

Gameful Design Elements

Commons leverages three common gameful elements as a way to promote citizen stewardship:

- Voting: Player-submitted City Tasks are sent anonymously to a random sample of other players for voting in the "Poll Booth" interface. Voting players are given two City Tasks and asked to choose the one more deserving of city attention and resources. Up votes increase the authoring player's score.
- Leaderboards: Players are ranked on shared leaderboards based on experience points and voting recommendations.
- Titles: Players also earn titles based on their submissions; for example, Mayor (most experience points), City Manager (most popular problem report), Urban Planner (most popular improvement report), and I <3 NYC (most popular appreciation report).

Our Criteria

1. Voluntary choice: yes
2. Active and enjoyable: presumably
3. Significant public benefits: yes

Further Information

http://www.commonsthegame.com/ (official website)

devote more resources to exploring the positive potential of game play, especially emerging forms of digital game play.

In the remainder of this chapter, we will consider a broader question: should government (itself) be gamified? Though we appreciate the utopian ambitions of those who would turn civic life itself into a game, we have serious concerns about overextending the design of games into the political sphere. The use of certain game structures by governments, as tools, is simply sound policy in the twenty-first century. By contrast, the goal of making government itself more *gameful* is, we think, a more radical, ambiguous, and potentially dangerous proposition.

The Limits of Gamification

Given that we both enthusiastically endorse increased government investments in the study and deployment of civic games, we are somewhat reluctant to be critical of those who propose to gamify

government. This is in part because we both find ourselves, fairly often, defending the social value and importance of games to skeptics. It is much easier for us to catalog the positive potentials of games than to explain their inherent limits to audiences primed to view games in a negative light.

Gamification is a very recent neologism. The most popular book on the general subject is surely Jane McGonigal's (2011) *Reality Is Broken*. The success of McGonigal's book speaks to a generational shift, as adults who grew up playing video games and enjoying them are increasingly non-apologetic about their enthusiasm for the medium. McGonigal's stature as a popular speaker testifies to a growing recognition that games can be a significant part of a healthy and positive lifestyle. Her open enthusiasm for video games is an overdue challenge to the negative stereotypes of gamer culture that still persist in the corridors of political power. We generally welcome McGonigal's evangelistic efforts on behalf of video games and gamer culture.

However, we think there are limits to the gamification of non-game activities. While McGonigal and her readers may truly enjoy and benefit from turning household cleaning and medical rehabilitation into games, not everyone will be capable of following suit, and this fact must be acknowledged. If certain citizens don't wish to gamify their lives, the state should certainly not force them to play. It would be nice if games were a panacea for all social ills. They are not.

McGonigal rightly points out that people enjoy playing games. As we noted earlier, one of the reasons that game play is uniquely pleasurable is that it is customarily voluntary. We enjoy games because they are enjoyable when we play them. When we don't enjoy playing games, we don't play games, hence playing games is usually enjoyable. If the state requires you to play a game, that game is far less likely to be enjoyable.

Games have boundaries. Most game theorists have stressed the separateness of both games and play from ordinary experience. For instance, Jean Piaget wrote extensively about play as a natural component of human psychological development, especially the cognitive development of children (Piaget 1962). Johan Huizinga, in his book *Homo Ludens*, suggested that play is the cornerstone element of human civilization, yet maintained that games and play occurred in separate and formally distinct settings from "everyday life" (Huizinga 1950). Roger Caillois agreed with this (Caillois and Barash, 2001, 4). Bernard Suit's book, *The Grasshopper*, examined play from a philosophical perspective, equating it with a special form of inefficient and ultimately utopian activity (Suits 1978). Brian Sutton-Smith's book, *The Ambiguity of Play*, traced various *rhetorics* of play and games through diverse literatures reaching back into history, including rhetorics of childhood development, psychology, imagination, power, fate, and frivolity (Sutton-Smith 2001). What these various definitions share, in their diversity, is the perception that games and play are activities that are *separate* from ordinary experience—if games become completely pervasive and define the world itself, they are not games.

We admit that, given the variety of definitions of *game*, it is possible to define games so that they are indeed pervasive. Almost a century ago, *game theory* was proposed by John von Neumann and Oskar Morgenstern as a mathematical strategy for the analysis of rules designed to promote certain behavioral outcomes in specific situations (von Neumann and Morgenstern, 1947). Since that time, economic and game-theoretical models have been applied to almost

every sort of human behavior: from choosing a marriage partner, to rearing children, to cheating on tests. Yet the definition of *game* in this case is clearly at odds with our common understanding of the term. Most of us would not, normally, consider marriage, parenting, or the educational process "games." Indeed, if the notion of gamification means "exporting game mechanics to non-game settings," then it must follow that these sorts of behavior are *not* games. Otherwise, there would be no non-game settings to which games could be exported.

The various contributions to this volume will no doubt provide the reader with a range of personal interpretations of *gamification*. Because the term is so new, we doubt that it can be authoritatively defined by anyone. However, part of our concern about gamification stems from what we perceive to be a very popular usage of the term. Today, the term is often paired with a consulting practice that purports to increase company profits through game mechanics (Werbach 2011). As evidence of this meaning, in December 2012, an applicant sought to register a U.S. trademark in the term *gamification* for an "incentive awards program ... that can be redeemed for merchandise, discounts, prizes or digital goods and services" (U.S. Patent & Trademark Office App. Serial No. 85795792). This accords with the use of the term in a recent book by Kevin Werbach and Dan Hunter (2012), *For the Win*, which explains gamification as a business strategy focused on awarding points, badges, and virtual prizes to promote consumer loyalty. In short, the most prominent contemporary definition of gamification seems to be centered on a marketing strategy.

Given this, we simply cannot endorse the gamification of government (as opposed to the use of games by governments). Consumers today are well-acquainted with sweepstakes, reward programs, and other manipulative marketing strategies. They are also far too familiar with the insincerity of marketers who offer illusory social status in return for opening their wallets. Gamification promises to increase this familiarity. It is true that many people today choose to participate in online programs that urge them to win points by buying soft drinks or potato chips or to become the virtual "mayor" of a certain donut shop. But these private mechanics are generally tolerated, given that they are easily ignored. Consumers are under no obligation to join in these "games." The benefits conferred by a soap company's point and badge system have little consequence for citizens that decline to participate.

The government, we would hope, would not blithely ally itself with this business consulting practice. As we explained above, we do believe that governments should invest in the creation and use of games to engage citizens and achieve public benefits. Yet the goals of the state are hardly aligned with the goals of gamification's marketers, which consist largely of promoting products and services. What gamification offers to clients is not primarily what good government needs. Worse still, we fear that if particular state actors make the mistake of uncritically partnering with gamification consultants, the failure of such projects might impede more legitimate, scientifically grounded, and publicly beneficial investments in state games. We also fear that companies urging gamification on state actors may be more interested in short-term consulting gains than in promoting the public interest. This is a significant risk.

Of course, for state officials who have done careful research, there are some promising possibilities to be found in the study of game design.

What we have found, in our research, is that there is a fascinating relationship between traditional government and the internal governance of games. A careful study of massively multiplayer games, for instance, reveals that many of them incorporate complex systems of community governance and techniques for building and harnessing the power of online community (Steinkuehler 2006; Lastowka 2010). There are many other researchers also studying the emergence of governance systems from within online games as communities (e.g., Lessig 1999; Castronova 2005; Dibbell 2006; Taylor 2006; Burri-Nenova 2010; Humphreys 2010; Nardi 2010). As commentators have noted, the "architecture" or design of online communities can have important consequences for the scope and manner of political participation (Pool 1983; Lessig 1999; Sunstein 2001; Deterding 2012). Thus, the serious exploration of the "gamification of government," in our opinion, might begin by reviewing the existing literature on the "government of games"—a literature that is already extensive.

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POSITION STATEMENT

HOMO LUDENS (SUBSPECIES ***POLITIKOS***)

William Gaver

As I write this, it seems a bad moment to advocate playfulness.

Disturbing news is all around. Environmental issues loom large, with significant climate change seeming ever more inevitable, large numbers of species becoming endangered or extinct, food shortages threatening, and forests being felled. Honeybee populations crash, antibiotics fail, Western economies falter at the same time that gross inequalities grow, and the only political response seems to be to marketize everything possible. Doctrinal certainty spurs irrational behavior in developed and developing nations alike, and conflict looms in all the old trouble spots with new ones appearing daily. Every age has its problems, to be certain. But still, the world appears a particularly grim place these days.

Surely, in these times, it is irresponsible, even crass, to argue that we have a duty to think of ourselves as playful people?

Things were different a decade ago, when I wrote a manifesto (Gaver 2002, 2009) arguing that we should focus more on designing for play. Computation was poised then to leave the workplace and invade everyday life. I argued that it would be a shame to design domestic technologies just to provide solutions to problems or help people pursue tasks more effectively. Importing values drawn from the workplace (where our understanding of computation had evolved) into our everyday lives would constrain technology to helping with the chores we *have* to do, when it might also help us with the things we *want* to do—the wondering, wandering, pottering, and play that characterize some of our most valued time away from work. Borrowing from Huizinga (1950), I suggested that we consider ourselves as *Homo ludens*, playful creatures, and explore what it would mean to design from that perspective.

Since then, I have been working with my studio to understand what designing for ludic engagement might mean. We have developed a number of computational prototypes that seem to sketch an answer (figure 21a.1). The Drift Table, for instance, has a small porthole showing aerial photography that moves depending on how weights are arranged upon it, creating an experience that a volunteer who used it described as having something like a "digital hot air balloon in the comfort of your own front room." The Local Barometer displays text and images derived from local want ads depending on the speed and direction of wind measured outside the house, giving a sense of the home's sociocultural milieu. The Prayer Companion brings a stream of text drawn from newsfeeds and social networking sites into a cloistered Catholic monastery, providing a resource for the

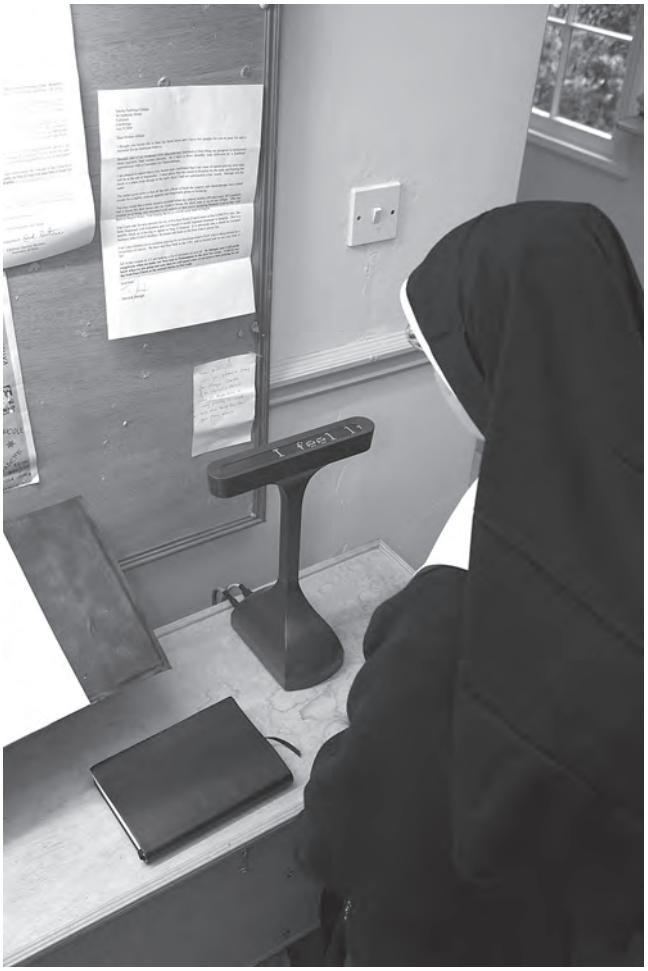


(a)



(b)

Figure 21a.1
Three ludic designs: Drift Table (a), Local Barometer (b), and Prayer Companion (c).



(c)

Figure 21a.1
(continued)

nun's prayers as well as exposing them to a variety of irreverent, humorous, and even offensive material. And so on.

Fundamental to all these designs is that they don't impose a correct way of using them on people, but instead leave ample room for different interpretations and uses. When we've deployed them, often for months at a time, we find that people enjoy that freedom. They play with the devices, trying out different things, trying out different attitudes, and finding different ways to relate them meaningfully to their lives. Often it seems as if our participants are playing with us, as well. We surprise people with our designs, basing them on our understanding of who they are and how they live, but not checking our ideas with them as we go—we are not participatory designers in a traditional sense. Many of the people we've given our designs to seem to like to surprise us in turn, finding new ways to use the designs in a conversation about what ludic engagement might be. After a while, of course, this experimentation tends to decrease, and people settle on preferred ways to engage with the devices. The orientations that stabilize, however, are usually unexpected, both to us and to them. We all learn from their experiences.

Over the course of many projects, we've developed a portfolio of work that seems to map out a space of what ludic design might be. But lately, we've begun to worry. The world is in a serious state. Shouldn't we be more serious too?

Before addressing that concern, perhaps it's best to clarify what I mean by *play*, or *ludic engagement*. To begin with, most games and professionally produced entertainments don't have the open-ended, self-motivated, exploratory feel of what I have in mind. Games have too many rules and too much competition, and most forms of entertainment are too channeled and goal directed—designed to deliver experience as a form of commodity—to lend themselves to ludic engagement. Or as Allan Kaprow (one of my favorite writers about play) put it, "in play one is carefree; in a game one is anxious about winning" (Kaprow 2003, 122). Moreover, games and entertainment exist in a game world, so that the conflicts and activities that take place can only represent real-world ones (insofar as they resemble them at all) and don't "count" in any real way. That's what makes them games. Ludic engagement, in contrast, is much more permeable to the actual world. Half the fun is in skirting between fantasy and accountability: flirting and pipe dreams, mockery and conspiracy theories, all can range from frivolity to deadly serious without crossing any clear boundaries.

Even all the new ways we have to share our thoughts and tell people about our activities, look up random facts and find arcane content, spread video clips and leave digital traces on the real world, sometimes seem to fall prey to a form of social competition that undermines their ability to provide ludic forms of pleasure. After all, it's difficult not to think of hit rates, followers, and trending topics when Kaprow writes:

> The real substance and stimulus of our "fun market," particularly in entertainment and sportive recreation, are superstars, record sales, popularity ratings, prizes, getting somewhere first, catching the biggest fish, beating the house at Las Vegas. Some fun! (Kaprow 2003, 122)[1]

No, the kind of play that interests me is more open-ended, loosely defined, and speculative than that. I have in mind experiences like spinning a fantasy scenario with friends, or wandering unfamiliar city neighborhoods, or trying to imagine how many bathtubs-full of water are flowing by in a rushing

river. Playing with ideas, trying on new identities, finding new perspectives on the world, these are all hallmarks of ludic engagement.

Provisionality is one of play's defining features. If there are goals, they're just for the moment. If there are rules, they can be changed. Such things are just ways to provide temporary structure in engaging the world. The field of play can alter as well. Pretending to be a spy might lead to fanciful speculation about the customs of one's home country, to the deployment of historical facts to support absurd suppositions, then twist to explain, based on a watchband, why the "spy" is really a space alien. Appearances, statistics, imaginary characters, and arcane knowledge: anything can be brought into play and discarded at will, as focus grows and shrinks and shifts depending on the pleasures of the moment. In ludic engagement, everything is in play.

Faced with the enormity of global problems and with an urge to address them meaningfully, the fluidity of ludic engagement, its tendency to shift scale and scope and to refuse a fixed stance might seem frivolous and uncommitted. Isn't it self-evident that, given the serious challenges we confront, our reasoning must be painstaking, and our designs clearly focused on providing tangible benefits? Encouraging play seems self-indulgent.

After all, most of us are aware of self-evident actions we can take in response to large-scale problems. To avoid climate change, we should reduce energy by traveling less, turning down the thermostat, forswearing meat, and buying local vegetables. To live a healthier life, one should give up smoking, drink in moderation, and exercise more. To preserve wildlife, one should give up ecotourism and simply leave nature alone. For many of us, though, solutions such as these haunt us because we cannot or will not live up to them. Often, we take our failures as evidence of individual moral failing, to be addressed by systems to make adherence easier. From this perspective, we need more systems aimed at supporting individual behavior change, such as energy meters and exercise monitors, telecommunications systems, and product trackers.

An alternative perspective, however, suggests that there are sociocultural forces behind our inability to do what we think is right. Turning down the thermostat seems useless when you hear other people talking about their garden heaters, and buying locally or eating healthily is difficult if your supermarket doesn't reveal where produce comes from or offers the best deals on processed ready-meals. Some of the things we might like to do as individuals—reducing motor transport, say, or using more alternative energy—depend on infrastructures, such as bike lanes and wind generators, that almost have to be accomplished collectively. Individual solutions won't work if they go against a cultural grain. From this point of view, most of the big challenges we face, from obesity to economic and ecological breakdowns, are fundamentally political in nature.

Political matters are not necessarily best handled through politics, however. Politics offers no end of solutions to today's problems. Consume less energy. Tax the rich. Become vegetarian. Downsize government. Grow synthetic meat. Limit population. Privatize education. Go nuclear. Nationalize the banks. Make shirkers workers. And so on. The problem is not just that these are simplistic and polarizing, but that the party system tends to gather similarly motivated solutions to form constellations of competing logics. At the extreme, one doesn't favor a given solution so much as become enmeshed in a totalizing worldview. We will only survive in a global market if

we unshackle capital and keep labor costs low. This implies that we must reduce inefficient benefits to the poor (as they are a disincentive to work and inflate wages), understand education as primarily vocational, and either defer environmental problems or look to market-driven technological approaches to solve them. No! To save the environment, we must reduce consumerism and a blind faith in growth, find alternative energy sources, and recognize that happiness resides in an equal, well-educated society that looks after the disadvantaged. Contradictory worldviews such as these, equally viable, arguably, depending on their metrics of success, make it difficult for people to communicate much less agree about solutions to societal problems. Worse, they reduce the political practice of communal action to politics as a form of team sport. Vote Conservative! Vote Liberal!

If politics is like a game, in which ways of seeing and acting about matters of joint concern are bound up in competition, winning and losing, there are other forms of political engagement that more closely resemble play. Many collective decisions—not to litter, for instance, or to avoid swearing in front of children, or to forswear prejudice based on sexual preference—often seem to emerge from a cultural consensus that may never be negotiated explicitly, but instead emerges obliquely through sidelong glances, appreciative remarks, or casual jibes. In Mol's (2009) terms, essentially political decisions like these may emerge through the development of "good taste," in which immediate concerns are informed by wider considerations to the point that activities may become undesirable because they simply "leave a bad taste in one's mouth." Good taste, in Mol's sense, is not decided by cultural arbiters, but rather through a fluid and egalitarian process of communicating about experience, of sensitizing and attuning one's desires to a range of the factors inherent in given situations, and of doing this both individually and together. In short, politics can be pursued through a process of cultural consensus around the aesthetics of living, rather than instrumental political game-playing.

In pursuing the political as a matter of culture, play may serve a fundamental role, as it is in play that we are most open to trying on new interpretations, considering implausible relationships, and even toying with those examples of bad taste that may test the boundaries of the good. From this point of view, if our solutions aren't working, maybe it is our problems that are to blame. Perhaps our logic is too settled, both at collective and individual levels, to allow new ways of looking at things to emerge. Perhaps we need to reengage with complexity, enjoy contradictions, and relax about changing our opinions, rather than seeking and holding to the kinds of simplifying logics that dominate politics. Seen in this light, play is not indulgent after all. Instead, it may be key in helping us move from solutions to "inventive problem making" (Fraser 2010, in Michael 2012). Casting ourselves as *Homo ludens* (subspecies *politikos*) might be essential in freeing us from the attitudes of guilt and triumph that come with institutionalized logics, to find new ways to live with the complicated challenges and opportunities that surround us.

What might it mean to be *Homo ludens politikos*? Clearly, sociopolitical topics are of interest, as well as the more individual and niche diversions that engross all the *Homo ludens* genus. Environmental issues and socioeconomic exploitation, short-term greed and cultural conflict, energy use and consumerism, these are the field of play. They provide fascinating intellectual conundrums, but this does not imply a lack of commitment, or engagement without values or

beliefs. On the contrary, passionately held views provide a standpoint from which to kick off in new directions and motivate speculation. Toying with other perspectives and possibilities, and finding the ridiculous in one's own and others' stances, may change beliefs, but equally, it might enrich and nurture them by trying them against alternatives.

Homo ludens politikos respects facts, but realizes that more and different ones will always come along. So we play with different orientations to matters of importance. We find new connections among things and people, expanding our field to include previously extraneous matters and finding new ways to focus again. We engage empathetically with those who hold different opinions, willing to imagine deeply seeing the world and living the lives of even our worst enemies. Most of all, we play with rules and assumptions. We highlight ambivalence and contradiction, pushing against the conceptual framing of current "solutions" to see where they break and where new perspectives might grow. *Homo ludens politikos* makes fun of authority and ridicules dogma.

Crucially, a ludic approach to politics is inherently social. After all, it is no fun keeping a joke to yourself. Being playful, we are more than happy to share our ideas, seriously or in jest, in ways that might just change our shared culture. The noninstrumental, discursive, profoundly social nature of *Homo ludens politikos*'s political engagement, I suggest, may just produce the new understandings and solutions that can help us escape the reified doctrines of politics.

How can we design to promote *Homo ludens politikos*? Several of our recent design forays hint at what a ludic approach to the political might look like and how we might design to support it.

An early, all but accidental example of a design eliciting what we might call political speculation is the Plane Tracker. This is a freestanding electronic appliance with a screen on the front and a strange, angular aerial on the top (figure 21a.2). The aerial is tuned to pick up transmissions sent repeatedly by all aircraft to identify themselves to air traffic control. The Plane Tracker uses these transmissions to determine the origins and destinations of passing flights and re-creates an approximation of the views that would be seen over the course of their journeys using GoogleEarth. The effect, when the device is located near a busy flight path, is of an endless series of international voyages, reflecting the actual journeys taken by overhead aircraft.

Our inspirations for the Plane Tracker were varied, ranging from the notion that we might compensate people for the disruption caused by passing traffic to urban myths about foreign seeds being released by lowering landing gear to populate gardens below. When we lent the prototype to a family living near Heathrow Airport, they engaged with it in a number of ways. At various times, it elicited discussions of geographic knowledge, memories of personal trips, discussions of foreign cultures, and anecdotes about friends. The piece was open-ended, as we had intended, affording a range of orientations to issues of geography, travel, and flight.

Two reactions, captured in a documentary video we commissioned about the piece, were particularly intriguing. Paul told us:

> It does create an awareness of the amount of travel that takes place and the amount of journeys that are happening very close to us, and sometimes I look up wistfully and wish I was on one of these planes going somewhere or having been somewhere.

For his wife, Gweni, however, this was a matter of some concern:



Figure 21a.2
The Plane Tracker occasioned ambivalence about air travel.

> Because there's been increasingly a debate on how much we travel, particularly this year, because we're all aware of global warming and emissions and so on, it has made me think, you know, here we have a piece of extraordinary technology that in a sense encourages you to fly, and maybe we shouldn't.

It strikes me as suggestive that the Plane Tracker could simultaneously evoke a desire to travel and raise suspicions about that desire. Through its simplicity and unfamiliarity, and the direct yet unfocused way it raises the issue of air travel, it appears to have opened up both sides of the environmental dilemma. It raises the promise that design can create a space for ambivalence to live—in this case, the ambivalence between wanting to protect the environment and wanting to be "going somewhere." Moreover, it suggests that we might design to expose ambivalence about similar issues, ranging from healthy lifestyles to competing desires to control or protect natural phenomena, in ways that might get beyond doctrinal certainty or difficult-to-observe normative stances.

Other clues come from a set of Indoor Weather Stations we designed more recently (figure 21a.3). There are three devices: The Wind Tunnel houses a small wind sensor in its chimney and amplifies the minute breezes it detects to create storms that buffet the stylized vegetation in its transparent Plexiglas chamber. The Temperature Tape is fitted with a needle dial in its casing that shows the difference in temperatures measured by sensors at the end of each of its two 2.5-meter-long ribbons, each of which is

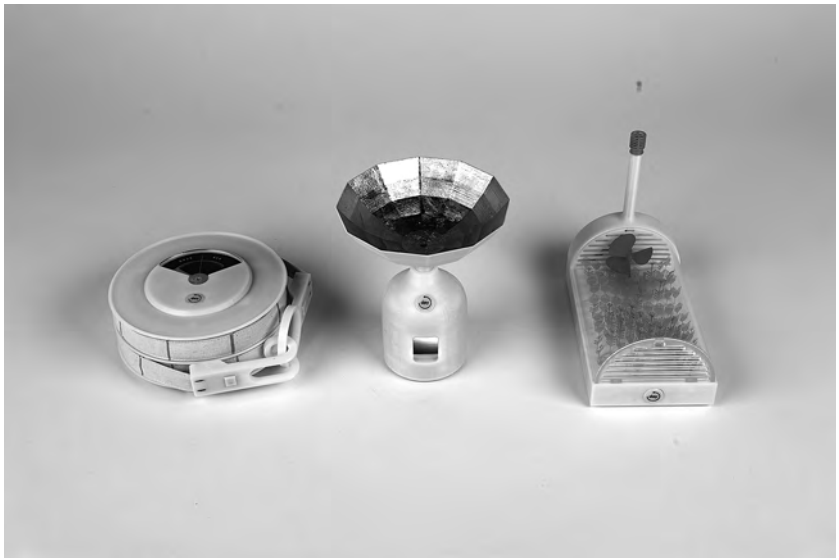


Figure 21a.3
The Indoor Weather Station supports reflection about the domestic microclimate.

also screen-printed with stripes of layered thermochromic ink to show temperature gradients along their length. The Light Collector houses an RGB sensor at the bottom of its funnel, building a striated image of the ambient light around it by adding a strip corresponding to its current readings every five minutes.

We batch-produced more than twenty sets of the Indoor Weather Stations for deployment in our local area. People orient to them in varied ways. Some try to use them to find drafts or assess the effects of insulation, while others play with them to find interesting reactions or simply try to find places in the home where they belong. Even after months of deployment, when interest waned, people told us that the devices made them aware of the home as an enclosed environment, compared them to enigmatic creatures, and speculated about how they would enhance appreciation of the home after years of use. Overall, the Indoor Weather Stations give access to the home's microclimate in a way that balances environmental utilitarianism with aesthetic appreciation—the Light Collector, for instance, could be used to monitor unnecessary lighting or simply to enjoy the changing hues at sundown.

The Indoor Weather Stations were designed as a ludic alternative to the oft-repeated tactic of responding to environmental concerns with resource-demand meters. Such meters are often ineffective or even counterproductive (Abrahamse et al. 2005) and in any case embody a normative persuasive design approach that closes down environmental reasoning. The Indoor Weather Stations, in contrast, were intended to open a space for thinking by presenting their data without imposing interpretation. The home's microclimate might be appreciated in its own

right or by analogy to global weather patterns. Indications of a draft might be interpreted as a problem to be solved or valued as producing fresh air and interesting responses from the Wind Tunnel. A cloud passing over the Sun might be of as much interest as seasonal variations. Like the Plane Tracker, the Indoor Weather Stations create a situation without suggesting a task or meaning: what people make of them is up to them. Only tangentially related to traditional interpretations of "environmental issues," they sought to change perceptions of the domestic environment, emphasizing the continuity of various forms of orientation, in ways that might resonate with how we perceive more global issues.

A final example of what a ludic approach to the political might look like is the Energy Babble, a system that is nearing completion at the time of writing. The Energy Babble has been designed in response to our research with about half a dozen existing communities, each of which is responding in its own way to environmental issues in particular, and questions of energy consumption and production in particular. As we have gotten to know the communities, we have been struck by the way the communities have developed their own accounts of what they believe are important and appropriate ways to address environmental challenges issues—and by the subtle and not-so-subtle differences among the discourses they have evolved.

The Energy Babble is designed to intervene in the ways the communities talk about their energy practices, rather than seeking to affect those practices directly. The device is a kind of mutant radio that plays a stream of spoken comments about energy use and environmental issues in different synthetic voices, punctuated by various jingles and sound effects reminiscent of news broadcasts. Content for the Energy Babble is gathered by scraping the web for statements related to energy use and conservation, focusing mainly on Twitter streams from the communities and from relevant governmental websites, but also on factual information from the United Kingdom's energy grid and related sources. In addition, people can enter their own comments via SMS message or an in-built microphone. These are converted to text, if necessary, and then to synthesized speech, to join them in the common sound world of the system. The result is a continuous stream of talk about energy issues that is sometimes informative, sometimes off-topic, sometimes comical, and sometimes simply annoying.

We are currently building about thirty Energy Babble devices for long-term deployment to the energy communities with whom we have been working (figure 21a.4). The hope is to bring the communities together by sharing their discourses, but also to contrast the different ways of talking about energy issues and ultimately to disrupt them. We do this through combining the communities' own discourses, quotes from other environmental commentators, "facts" that we gather, and what individuals speak or text to the system. Moreover, we add an element of humor and absurdity by including Markov algorithm-generated mash-ups of the content, which, constructed using the probabilities of associated terms, are locally comprehensible but globally nonsensical. The idea behind this is to provide access to a wide variety of ways of thinking and acting in response to environmental issues, but also to disrupt assumptions that any of these ways of thinking and acting are "correct"—or even sensible at all.

What can we learn from these early designs that will be useful for developing longer-term strategies for designing ludic forms of political engagement?



Figure 21a.4
The Energy Babble mixes many forms of talk about energy-related practices.

Michael (2012) has suggested that this form of design can be understood as "proactive idiocy." This is not meant as a pejorative term, but instead is based on Stengers's (2005) discussion of the role that the idiot might play in social enquiry (a discussion that itself draws on Dostoevsky by way of Deleuze). The idiot, according to Stengers, "is the one who always slows the others down, who resists the way the situation is presented and in which emergencies mobilize thought or action" (Stengers 2005, 994). The idiot doesn't present alternative ways of thinking or acting, but insists that "there is something more important" (994), even though what that might be is unknown. Idiotic design, in this account, serves to retard our momentum in thinking and acting about things we think we understand, to insist that there might be "something more important" and to question "what we are busy doing" (Michael 2012) in our customary ways of addressing situations in question. From this point of view, a ludic approach to political issues is valuable in interfering with accepted approaches to issues, encouraging people to move outside typical frameworks for considering them.

Acting as the "idiot" in this sense doesn't necessarily imply playing the role of court jester, however. Standing outside normal conventions to comment critically assumes a privileged role, one which can grate if the results seem glib, condescending, or an in-joke, and particularly if no responsibility is accepted for contributing productively to seeking alternative ways forward. What is needed instead, perhaps, is a design stance in which suggestions for new ways of thinking about issues are made hesitantly and accompanied by sufficiently varied resources that those suggestions are all but hidden in a cloud of other possibilities. At least, I hope that our designs suggest that "idiocy" is not necessarily destructive, but can be achieved in ways that are both engaging and productive of new orientations.

To begin with, our prototypes suggest that political issues can be highlighted without creating partisan designs. This is not the same as creating seemingly neutral designs, which by avoiding overt engagement with political issues tend to support the status quo (Dunne 2005). Designs can explicitly or implicitly raise political questions and provide resources for exploring them, without embodying assumptions about the appropriate ways to address them the way, say, energy-consumption meters do. The Plane Tracker, Indoor Weather Stations, and Energy Babble all occasion significant questions about environmental issues, and while they may tentatively suggest alternative orientations to those issues, they don't insist upon them, but instead make them to open a space for playing with possible attitudes and understandings.

To suggest new orientations without blocking people's independent speculation, our systems create situations that are constrained enough to allow certain issues to come to the foreground, without overly defining any particular problem or insisting on what is important and what is off topic. The Plane Tracker can be understood as problematizing air travel, but equally as allowing exploration of the world. The Indoor Weather Stations can be interpreted as highlighting aspects of the semicontrolled indoor environments we construct, but whether these are to be appreciated aesthetically or in terms of their environmental consequences is left open. The Energy Babble exposes listeners to a barrage of potentially inconsistent environmental statements, but not only does it refrain from privileging any of them, it is ambiguous about whether they should be considered seriously or as a form of parody. Taken as

a rudimentary portfolio (Gaver and Bowers 2012), this collection of designs suggests that if what we are looking for are fresh approaches to current issues, then rather than presenting focused, worked-through perspectives on issues, it may be more fruitful to design situations that offer new orientations while remaining open to irrelevancies and permitting multiple orientations.

Finally, there are a number of more specific tactics that may be valuable in engendering ludic engagement with political matters. Designs can provide new resources to political considerations. These can simply be brought to bear in existing debates, but also might widen their focus, suggest other orientations, or simply confuse prevalent narratives. They can create ambiguity or invite ambivalence. By inviting laughter, they can undermine authority. The list is already long and likely to become longer with further practice. What I hope to have demonstrated here is that far from being inappropriate in these uneasy times, play may have a central role in allowing us to meet them.

Acknowledgments

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Note

1. Perhaps I'm being uncharitable. After all, many of the new media that surround us are truly malleable and open-ended, capable of being used in many ways. If we turn them into arenas for status seeking, maybe it's our own fault. But if developers didn't track these statistics, if the systems didn't keep score, maybe we wouldn't worry so much about how well we're playing; we'd just play.

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22 THE GAMEFUL CITY

Kars Alfrink

Part I: Introducing the Gameful City

Fragment 1: Introduction

This chapter is about gameful design and the city. It is about the physical form of the city but most importantly its function—the way it is used every day by its inhabitants.

I believe a city that allows for play and perhaps even encourages a gaming frame of mind is a humane city—a city that is tolerant and flexible enough to let inhabitants pursue their dreams and desires, in public if they wish. A city that is open to change from below. A city where the unexpected is embraced instead of tamed and shut out. A city, in short, that is livable in the full sense of the word—beyond good public transport, ample green space, and cultural activities.

I long wondered why I responded strongly to parts of the city I call my home and cities that I visited in the Netherlands and abroad. At first, I thought it was about the way certain neighborhoods look. An understandable mistake, as I was introduced to cities through the gorgeous old towns of the Netherlands. But once I had spent time abroad, for instance living for some time in what I guess was a lower-middle-class neighborhood in one of Costa Rica's major cities, my thoughts began to change (figure 22.1). This was not a pretty place—far from it, in fact—but it did feel good, it felt alive. And now I know this is because the neighborhood functioned properly to allow for a complex, varied use by the people in it. This in turn led to a form and experience that I perceived as beautiful.

To a large extent, this chapter is about this experience of livability and even beauty in a city. It is about how that experience is brought about, at least according to contemporary thinking on the topic. But more specifically it is about how gaming and playing affect this experience in various ways. I am mostly interested in how gameful design can contribute to a vibrant livable city. There are pleasant cities for sure, but there are many that can do with improvement. My feeling is that gameful design might help. There are also current developments that might force us to rethink some of the ways in which we have to date given shape to cities. And again, I think gameful design might play a part.

The idea that runs throughout this chapter is the back and forth between institutions and their desire for legible spaces and individuals' local use and knowledge on the ground, which requires a certain



Figure 22.1
The outskirts of Heredia, Costa Rica, 2002.

amount of illegibility. This back and forth and the space in between is where play happens and acts as a generative force for livable cities.

I therefore suggest gamefulness—the extent to which spaces engender or allow for a spirit of gaming and playing—might be considered an additional criterion by which we measure a city's livability. In the spirit of play's innate uncontrollability, illegibility, and bottom-up nature, I believe that the best way to increase gamefulness is by equipping people with gameful tools. These tools can be used to turn ordinary space into gameful space in various ways.

This chapter is divided in three parts. The first part is dedicated to laying a foundation on which I will build my argument. Games and play are closely related to notions of fun. We will therefore start by looking at enjoyment in architecture, how certain places are considered more pleasurable than others. Some of these ideas, taken as ideals, are essentially utopian. So we will briefly survey utopian notions of various kinds. From there, we will move on to an important concept for my argument, which is that of legibility. It is a useful concept for illustrating the shortcomings of planning cities from the top down, as utopian thinking might tempt one to do. In this chapter, I will use it to explain the utility of gamefulness and playfulness in the context of the city. I will conclude the first part with a discussion of two kinds of urbanistic practice—hard and soft urbanism—and also the ideas of layered urban space. These concepts I will use to frame where I see a place for gameful

design in urbanism and how I think gameful design can affect the built environment.

In the second part, we will shifts gears and look at various ways in which a city can be gameful, or perhaps more precisely how gameful design can enable people to shape urban environments in various ways. I have labeled these different approaches appropriation, defamiliarization, socialization, subversion, and formation. Each approach is illustrated with a number of real-world examples of games and playful interventions. Using these examples, I will attempt to show the mechanics of each approach, in the hopes of providing starting points for further work.

The third part, finally, zooms out again and looks to current and near-future developments in architecture and urbanism. The purpose of this final part is to suggest ways in which gameful design can make a meaningful contribution to future cities. I will discuss current issues with megacities, places with different logics from the ones on which much of contemporary urbanistic thinking is based. These megacities are on the rise, and gameful designers with an urbanistic interest will inevitably find themselves working in such places at some point. I will also discuss the smart city idea, which is rather popular with city administrators and large technology firms. My contention is that the gameful city might be a humanistic counterbalance to the techno-utopian notion of the smart city.

A last word on the form I have chosen for this chapter. You will notice I have labeled its basic organizing units "fragments." The name and approach is inspired by James C. Scott's (2012) short, wonderful book *Two Cheers for Anarchism*. These fragments are pieces written from a rather personal perspective—my editors call it an "anecdotal approach." I have chosen this approach as it best fits my personal background as a practicing designer and design entrepreneur who has been active in this space for some time. This has mainly been through making games and playful interventions of various kinds either for clients or for myself. The knowledge I have gained through this practice and the perspective I have on the field is therefore necessarily subjective and first person. I felt it best to have the form of this chapter to reflect this first-person nature. Your reading experience might be a bit collage-like at times; this is one traversal out of many possible ones through a network of ideas. However, I hope it is also enjoyable, in a way perhaps not unlike meandering through a city, familiar or otherwise.

Fragment 2: Enjoyment in Architecture

What makes us find one building ugly and the other beautiful? I have a tendency to appreciate minimalist, even modernist architecture. I still do, even now that I understand modernism might look good, but does not necessarily lead to livable cities.

Alain De Botton makes a compelling argument for the idea that what we find beautiful in architecture is what speaks to those values we hold dearest. The experience of beauty, in other words, is aspirational (De Botton 2006).

This nicely connects to why I appreciate modernist architecture. I am scatter-brained. I like to think that an orderly environment might temper the chaos that characterizes my mind. Similarly, Le Corbusier felt strongly about eliminating much of the apparent fluff of the building styles that came before him (De Botton 2006). He equated the use of modern building materials and a clean and highly legible visual form to this idea of improving the lives of workers.

In other words, what you appreciate about the form of a city relates strongly to your aspirations. The more subtle point is that style, visual form, beyond fads and fashions, serves a purpose. It can serve as a reminder of how we would like to lead our lives and live together (De Botton 2006).

In an attempt to connect enjoyment in architecture to the more specific kind of fun we find in games, David Thomas offers the following perspective: "Fun, as constructed, creates playful ambiguities in the mind of the participant through the creation of an is/is-not contradiction in the perceptions of the world" (Thomas 2012). However, as Thomas points out, not all ambiguity leads to fun. Quite often it just annoys. Human agency seems to be a deciding factor, which is nicely illustrated in the concept of *third space* as described by Edwards Soja, who "uses third space (lived space) to challenge the power structures of what he describes as first space (perceived space) and second space (conceived space), a twist on the concept of emancipation and an open platform for re-examining human geography and meaning" (Thomas 2012).[1]

So fun, play, and games seen in the light of this third space might not only be frivolous but also serve a purpose. It is in this way different from De Botton's notion of aspirational beauty in architecture: "In this sense, fun may very (well) be considered a new aesthetic category. Where beauty relates to truth, fun relates to deliberate uncertainty" (Thomas 2012). The uncertainty here is the space for individuals to produce a lived space as they see fit.

Both De Botton's and Thomas's perspectives on enjoyment in architecture relate to how I think about the connection between gameful design and cities. I propose a more subtle approach to improving on cities than what we have seen so far in the fields of serious games and gamification. As with De Botton, I would like to think gameful designs—the things we make that encourage a spirit of gaming—might be reminders of how we could better live together. And as with Thomas, I think it is openness and an allowance for adaptation by inhabitants that prevents this utopian, prescriptive aspect from getting the overhand. Play in this view is something that balances objective and subjective reality.

This is a different, more generative approach than what we commonly see in serious games and gamification. Both are to me rather prescriptive and, as such, reduce gaming to a utilitarian activity (Alfrink 2012), which simply does not do it justice and, as some argue, even cheapens the cultural form of games as a whole (Lantz 2011).

What I mean by generative is this: the gameful city can be thought of as a tool for players with which they can invent improved ways of living. It is about the creative potential of play. What is most important for this approach is that gameful designs allow for appropriation by players. This is very much in the spirit of Bernard DeKoven's well-played game. Any gameful design should be considered less important than the play community (DeKoven 2002).

Fragment 3: Utopia

Insights into what makes cities enjoyable more or less naturally lead to utopian notions. There is a rich history of schemes aiming to improve the lives of people through a comprehensive reimagining of the city.

Le Corbusier proposed his Ville Radieuse in 1924. It was never realized, but has been at least of some influence on actual buildings and cities, for example with the design of Brasília. Ville Radieuse is an

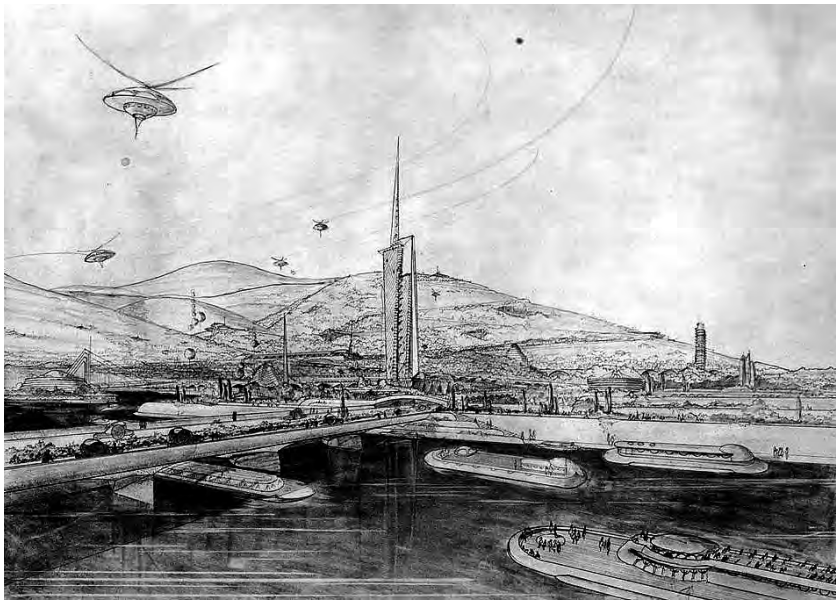


Figure 22.2
Sketch of Broadacre City by Frank Lloyd Wright. (Image courtesy of Kjell Olsen)

example of a kind of utopian urban thinking also represented by Broadacre City (figure 22.2), a concept proposed by Frank Lloyd Wright in his 1932 book *The Disappearing City*, and the Garden City proposed in 1898 by Ebenezer Howard. What all these schemes share is the intent to improve people's lives through a far-reaching restructuring of the built environment along rational, highly legible lines.

Utopian thinking takes on other forms as well. Most notably, a number of ideas for improved cities focus more on responsiveness, processes, and the dynamics of cities as opposed to the spatial structuring typical of the modernist schemes of Le Corbusier and others

New Babylon, first proposed by Constant Nieuwenhuys in 1959, was a proposal for a ludic society that would serve as a counterpoint to the modern city, which Nieuwenhuys thought was dehumanizing in its focus on consumerism, productivity, and utility. In New Babylon "inhabitants play a creation game of movable walls, floors, partitions, ramps, stairs, bridges, and infinitely variable qualities of light, color, ventilation, texture, temperature, and moisture, designing their own environment for adventuring against their designed backdrop" (Walz 2010).

Cedric Price's Fun Palace (1961) shares many of New Babylon's features in that it is also an environment configurable by its inhabitants. Price conceived of it together with theater director Joan Littlewood. Here the main aim seems to have been the encouragement of creativity and adventure and the elimination of competition. Here again, a utopian notion is propagated through a dynamic, in some sense rules-driven architecture (figure 22.3).

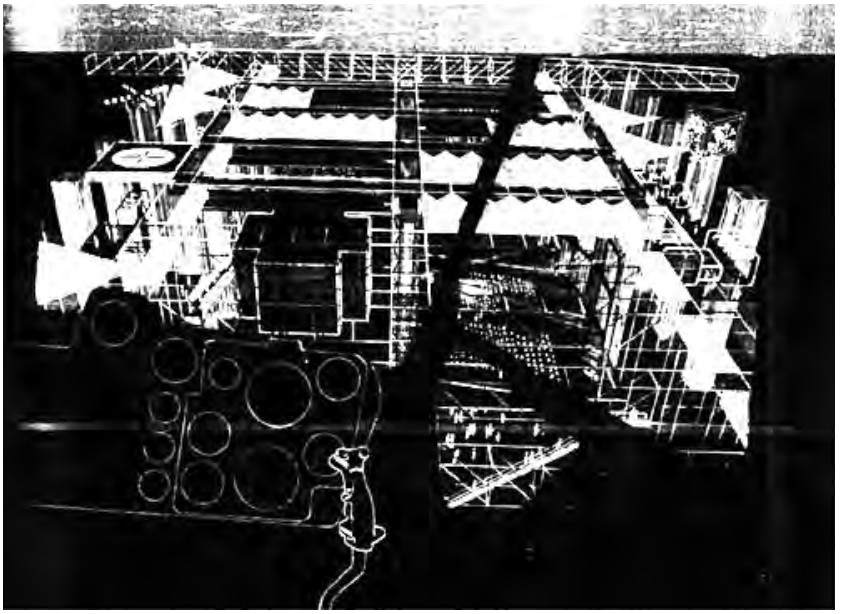


Figure 22.3
Cedric Price's Fun Palace. (Image courtesy of Anil Bawa-Cavia)

Finally, Ron Heron—member of the Archigram collective—proposed the Walking City (1964), which as the name suggests would be able to move across the globe in search of resources and so forth. Walking Cities would be able to connect and combine at the scale of individual buildings. Change and adaptation were also possible. Here rules and procedures seem to be more at the service of utility and less focused on challenging modernist notions. However, the Walking City is certainly not without whimsy and might be considered an example of urban designers at play.

Indeed, any utopia—going as far back as Sir Thomas More's 1516 book of the same title—can be seen as a form of play. By their very nature, utopias are impossible to achieve and have usually served more as guiding lights than blueprints for actual building projects. "Utopia describes not only a physical space meant to entertain those who read about it, but also a perfect living space meant to delight those who inhabit it. A utopia programs perfect behavior and therefore, perfect enjoyment" (Walz 2010).

In any case, it is important to note that—whether modernistic or more ludic in nature—these utopian notions can and have led to policy makers and planners striving for fuller control over urban spaces and their inhabitants. This desire for legibility has its downsides, dangers, and detractors, however.

Fragment 4: Legibility

Nobody has better argued for the failings of modernist urban planning than Jane Jacobs. She compellingly shows how far from being disorderly—although

they look like it—successful city neighborhoods work on the basis of a complex working order (Jacobs 1961). She passionately argues for mixed-use spaces that are underspecified by their designers. This is the direct opposite of the top-down planning of neighborhoods that is still common practice, with its characteristic zoning—separating functions from each other and neatly ordering them. This leads to designs that are easy to execute and manage by governments, but are not conducive to a vibrant urban culture.

James C. Scott has gone on to expand this argument against what he calls a high-modernist way of looking at the world. He shows how the state has a strong tendency for imposing visual order from the top down, which in fact leads to fragile systems that are monofunctional and not future-proof (Scott 2012). The important point is that the more subtle, working order that is found in successful, generative, resilient systems might look messy but is far more sophisticated than any large organization will be able to plan and produce from the top down.

The concept of legibility is also central to Kevin Lynch's (1960) book on the urban form, *The Image of the City*. I would like to emphasize, however, that his legibility is a first-person one, as opposed to the top-down legibility so coveted by city planners. Lynch defines legibility as "the ease with which (the cityscape's) parts can be recognized and can be organized in a coherent pattern." According to Lynch, it is an important property of a beautiful city, albeit not the only one. Legibility not only facilitates way-finding but also offers "emotional satisfaction," a "framework for communication or conceptual organization" and "new depths ... to everyday experience." To achieve legibility, Lynch goes on to say "we must consider not just the city as a thing in itself, but the city being perceived by its inhabitants." Here the difference with legibility in the high-modernist sense becomes clear. Indeed, further on Lynch explicitly states the need for adaptability in the cityscape: "An environment which is ordered in precise and final detail may inhibit new patterns of activity. ... What we seek is not a final but an open-ended order, capable of continuous further development."

So, Lynch's notion of a legible cityscape is in fact at odds with planning in urban spaces from the top down in detail. His starting point is the individual experience of the city. It is entirely conceivable that a city is legible to people in the streets while appearing messy from above. This idea is supported by Lynch's case studies, which show that old areas of Boston, which came about relatively organically, are considered far more legible than parts of Jersey City, which are newer and in fact were planned in at least some detail.

It is my contention that gameful design can contribute to ways of creating urban spaces that are more in line with the ideas of Jacobs, Scott, and Lynch: a generative approach, which involves a diverse range of people at all stages. However, to understand better how gameful design can contribute to urbanism, it is important we gain a more granular understanding of the practice itself first.

Fragment 5: Urbanism, Hard and Soft

Urbanism can be thought of as having two sides, which I will here simply call hard and soft urbanism.[2]

An example of hard urbanism is the effort to plan and build a new city neighborhood. This urbanist practice is concerned with the physical form and its construction, informed of course by an idea of how these new places will be used.



Figure 22.4
Inside Vechtclub XL, a workspace converted from a factory in Utrecht, the Netherlands, 2012. (Photo courtesy of Hein Lagerweij)

Soft urbanism starts from existing spaces and attempts to reimagine the uses of them. This is sometimes called *reprogramming* a space (Burnham 2011). And it usually involves some adaptation of the physical form, although this can be an emergent consequence of people simply starting to use a former factory as a creative workspace (figure 22.4) or an abandoned printing shop as a club (figure 22.5).

Soft urbanism should see an increase in activity in the West, as it is already highly urbanized. We are surrounded by cities, and so often we will want to adapt the existing buildings at our disposal instead of building more.

As Stewart Brand has pointed out, this process is in fact universal and a natural aspect of the life cycle of buildings (Brand 1994). He argues planners and architects rarely take it into account. This is akin to Jacobs's objections, but on the scale of single buildings instead of whole neighborhoods and cities. Brand shows that the buildings that are most loved and that tend to live the longest are the ones that allow the most for adaptation. This requires a loose coupling of different building layers, so that for instance services might be adjusted without the need for a complete overhaul of a building's frame.

What is also interesting in soft urbanism and adaptation is that governance and law comes into sharp focus. Certain spaces might not legally be used for certain purposes, and so designers and users attempting to reprogram a space might first need to engage with law and attempt to have it changed or convince government to apply a more suitable kind of license to their space. An example would be Renew Newcastle, an initiative that has revitalized the inner city of Newcastle, Australia, through clever hacking of existing building law (Hill 2012).



Figure 22.5
The band Caribou playing at TrouwAmsterdam, Amsterdam, the Netherlands, November 26, 2010. (Photo courtesy of Ronny Theeuwes)

Gameful design has a role to play in both sides of urbanism. Hard urbanism might benefit from more participation by future users of the buildings being planned. In this case, gaming might enable nonexperts to articulate their desires or provide feedback on plans.[3]

But I am most interested in the application of gameful design in the second kind of urbanism. Soft urbanism is a less formal process, with more opportunities for direct engagement by people with a space. This makes it a good match for gameful design's participatory nature. Software and digital networks are increasingly a big factor in the experience of buildings, neighborhoods, and cities (McCullough 2005; Greenfield 2006). Because gameful designs can be and often are deployed as code, they are interventions native to such spaces.

The thoughtful design of such systems can lead to new behavior in physical space, which in turn can lead to changes in the built environment. It might seem like a roundabout way of doing things, but in a future in which software has eaten the world (Andreessen 2011) and we find ourselves living in code/space (Kitchin and Dodge 2011), I think it will become impossible to reprogram spaces without also reprogramming the software systems conditioning them.

Fragment 6: Layers

So to summarize, if one is concerned with improving the built environment using gameful design, the first question is: what makes a space enjoyable? On the one hand, the experience of beauty in architecture is aspirational; we prefer to be in environments that speak to our values. On the other hand, the

enjoyment we find in play and games arises from ambiguity; in the case of the built environment, this can be typified as an is/is-not quality. Utopian schemes to improve the lives of urban dwellers through architecture can be roughly divided into modernist attempts to create visual order from the top down and attempts to liberate the individual through playfully reconfigurable spaces. Both are all encompassing and ultimately unachievable but do influence real-world architecture in various ways. Such utopian schemes forgo local knowledge in favor of global legibility. This typically leads to dysfunctional, brittle spaces. My position is that individuals need to be full participants in the spaces they inhabit. This necessarily leads to illegibility but supports local modes of knowing and acting. A gameful approach to the design of spaces might enable this. When doing so, we can distinguish between hard and soft urbanism, the former dealing with the conception of new spaces, the latter with the adaptation of existing ones. With adaptation in particular, the domains of law and software come into sharp focus. Gameful designers wanting to make a real difference will have no choice to engage with those domains.

We are almost ready to turn our attention to the various ways in which a city can be gameful. The stage has been set. What remains to be discussed is this: if a gameful city is a city that encourages a gaming frame of mind, then what are the materials available to designers to facilitate this? I see roughly three categories of material available to designers: (1) the physical form of a city; (2) the digital networks pervading a city—including the ubiquitous media landscape overlapping it; and (3) the social practices of city dwellers.

Usually, gameful designs will use materials from not one but two or even all categories. And even if they don't directly engage with all of the categories, they will influence aspects of all of them, as ultimately they are all interconnected.

Beyond materials, there are many purposes gameful design can be put to. In the second part of this chapter, I will survey the major ones I see as they relate to the built environment. This may not be a comprehensive survey, but I trust it will paint a picture of the many ways in which a city can be gameful, in the hopes of enabling you to dream up many more.

Part II: The Gameful City As ...

In this second part, we will survey various ways in which a city can be gameful, or, to be more precise, ways in which people can approach urban spaces with a playing or gaming mindset, and through these acts of play encourage others to do the same. I have labeled the various approaches appropriation, defamiliarization, socialization, subversion, and formation. Each serves a distinct purpose—although there are connections and overlaps—and each works in a different way. I will discuss them one by one using work by others that I consider exemplary, as well as some work from my own practice that allows me to delve into the more personal experiences I have had working in this domain. My hope is that the fragments that follow serve as starting points for further work by yourself and others.



Figure 22.6
Still from *Dogtown and Z-Boys*.

Fragment 7: The Gameful City as Appropriation

The documentary film *Dogtown and Z-Boys* (Peralta 2001) tells the story of skateboarding's development into its current form. It includes an account of how at some point, the skateboarders profiled in the film—a group calling themselves Z-Boys hailing from the Los Angeles area known as Dogtown—started using empty, bowl-shaped swimming pools (figure 22.6).

Up until that point, the street game could be best described as pavement surfing. Skateboarders rolled down hills, swerved a bit, but that was it. It all changed when, as the result of a drought, the pools so typical of the Los Angeles suburban landscape were emptied. The Z-Boys, looking for new challenges, started using these pools. As soon as they did, the form of skateboarding changed. It became much more about aerial acrobatics and less about traversing distances. It was this style that would soon thereafter be the new standard for skateboarding competitions and also lead to the half pipes and vertical ramps that make up contemporary skate parks.

What the film vividly shows, and what is true of all street sports to some extent, is that the skateboarders derived a huge amount of pleasure from knowing that they were using these pools for an entirely alien purpose. And in fact—as pool owners would most often not approve of the practice—simply locating a pool that could be used is depicted as a major source of fun in and of itself. In the film, Z-Boys tell stories about driving around



Figure 22.7
Traceur Daniel Ilabaca playing parkour in London, England, 2007. (Photo courtesy of Jon Lucas)

neighborhoods, with lookouts on top of the cars peering over fences to find that one empty, unsupervised pool.

This type of play in cities is about the appropriation of space for other ends than what it is planned for. The same applies to more recent practices such as parkour and free running—which are about running through cities and overcoming obstacles by acrobatic means.[4] The latter form adds a bit of showmanship to the mix, but both are very much about using the city in novel ways and challenging the body in the process (figure 22.7). The fact that this happens in public, with nonplayers usually present to witness the antics of players, means performance plays a part as well. But most often, street athletes of various kinds seem to be more preoccupied with the opinions of their peers than passersby.

It is clear that the physical form of the city plays a massive role in these practices. It is ironic that the sprawling, neglected, and often soulless spaces produced by such things as shopping malls, housing developments, and large office parks are the domains

of choice for street athletes (Borden 2003). It appears that the ambiguity these spaces are characterized by is a fertile ground for inventing new uses.[5] Despite all the skate parks and parkour courses available, players still seek out spaces where their play is unauthorized and sometimes actively opposed. There is a pleasure in inventing new uses for physical spaces, and the more ambiguous a space, the more attractive it appears to be.

By appropriating physical space, a kind of resistance is enacted. In *Dogtown and Z-Boys*, skateboarder Wentzl Ruml uses the—admittedly loaded—term "concrete warfare" to describe the social tension inherent in the act of using spaces for things they were not designed for, without permission and with the risk of repercussion. This again explains to a large extent why skateboarding, free running, and its brethren have not disappeared from our streets with the introduction of facilities specialized for the hobby. The pleasure is in pushing the limits imposed by society on what can and cannot be done in public space. With the commercialization of public space, these players have even become somewhat heroic figures, opposing corporate hegemony through daring, nonsensical, and creative acts. As with the ambiguity of physical space, these forms of play operate in the ambiguous terrain between the freedom to put oneself at risk and the collective norm to not annoy and inconvenience others.

In all of these street sports, media and networks play a role primarily as a means of documenting acts of play and sharing it within peer networks for fame—and in some cases fortune. Many of these games start out as small phenomena practiced by isolated groups and then evolve into global phenomena with the help of media and networks. For example, parkour started out as the practice of a small group of young men in Paris in the 1980s and was later dispersed through documentaries and film.

What is interesting about all these forms is that behind an apparently simple set of rules—which makes it easy for them to travel—lies a rich world of actions of increasing complexity. There is also typically a lot of room for individual creativity, and it is not uncommon for new styles to emerge, just as we see in popular music. It is these characteristics—a performative quality that lends itself to documenting play acts, and a simple basic set of rules with generative potential—that enable many of these street sports to become real cultural phenomena.[6]

Fragment 8: The Gameful City as Defamiliarization

There is a rich history of urban play aimed at shifting people's perception of the city. A good starting point would be the set of strategies used by Situationist International, commonly grouped under the header of psycho-geography, which Walz describes as "the playful becoming aware of, reimagining, and exploration of the city; in other words, the affective realization of the city" (Walz 2010).

The practice of the *dérive* is probably the most well known example of psycho-geographic practice. A critique of the disengaged, consumerist use of the city as personified by the *flâneur*—as described by Charles Baudelaire and later expanded upon by Walter Benjamin—the Situationists encouraged active engagement with the city and increased awareness of its effect on the mind (Flanagan 2009). To engage in a *dérive* meant to take the path less traveled, to go against the grain. Trips were documented in the shape of whimsical maps that challenged the accepted way we make sense of urban space.

The *dérive* has spawned a huge tradition of playful strategies for defamiliarization. Nowadays we find most of the examples in the urban and pervasive gaming subculture, which is mostly active in North America and Northwestern Europe and organized around not-for-profit game festivals spanning multiple days. Many games devised in this scene—mostly made up by artists and hobbyists—either explicitly or implicitly see it as their aim to get players to look at their surroundings in a new light. Indeed, it is possibly a tacit agreement that these games derive a large amount of their artistic value from this effect.

Take for example Pieces of Berlin—a game by Erik Burke and Lynn Maharas—which ran at Invisible Playground's You Are Go! in 2011 (figure 22.8). It has players looking for specific locations in the city that match tracings they are given. By standing in the right spot, the tracings align with the view of the cityscape. Pieces of Berlin takes the traditional scavenger hunt as a starting point and by making some adjustments, primarily by changing the way clues are provided, makes it a game that is very much about closely observing the cityscape.

Defamiliarization does not have to be about looking, though. Chromaroma is a continually running game that taps into data produced by the Oyster Card—London's RFID-enabled public transport pass—after receiving permission from the player (figure 22.9). Simply traveling propels the game forward. Each time a player checks into a tube station, she earns points. Various teams battle for control of stations. Players can earn bonuses and unlock content by traveling certain routes. Chromaroma is always on in the background. As such, it is a temporally expanded game (Montola, Stenros and Waern 2009). This means players are shifting their mindsets from the everyday to playing and back again. The mundane, utilitarian task of traveling from A to B is transformed.

Looking at these two examples, it becomes clear the physical form of a city is quite often the subject



Figure 22.8
Pieces of Berlin, 2011 (Burke and Maharas 2011). (Photo courtesy of Invisible Playground)

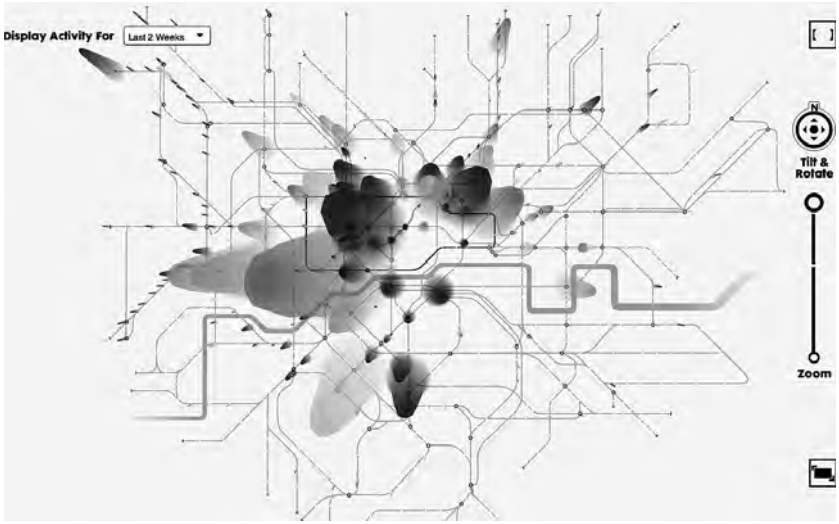


Figure 22.9
Chromaroma, 2010. (Image courtesy of Mudlark 2011)

of play. The idea is that we tend to forget about our surroundings, take it for granted, and stop seeing the wonderful things it has to offer—or stop realizing the wonder of it being there in the first place. Designers take it upon themselves to address this by offering systems that incentivize heightened awareness in various ways. Pieces of Berlin is a straightforward example: looking closely at the city is the way to get to the next checkpoint.

Chromaroma takes a slightly more subtle approach. The public transport system becomes a platform for a game, and to use the system is to play the game. But to play the game *well* means delving deeper into the real-world system, becoming more than just a user but a connoisseur of sorts—knowing about not just the stops on your daily commute but also the others you might normally not visit. Finally, the game tells stories about the public transport system as a reward, sometimes based in fact, but often also including fantasy. All of this combined imbues the London tube with a sense of magic and mystery and makes it more than just a travel system.

The spirit of Situationism is alive in many of these games, perhaps only implicitly and not even to the awareness of the designers. The assumption is that a good urbanite is an urbanite who does not simply let herself be guided by corporate and governmental agendas. Playing these games is an act of resistance. It is also closely related to the earlier notion of appropriation—using the city in a different way makes it less familiar and in turn makes you more aware of how you are typically guided to use the city in a certain way.

With the increasing pervasiveness of information technology and media in the cityscape, these networks themselves are becoming the subject of defamiliarization. Chromaroma is as much about the physical infrastructure of the tube as it is about the

weird and opaque payment mechanism that is the Oyster Card. By using the data generated by this system, players are made more aware of what is gathered on their travel behavior and what one can learn from it. The shape of the data becomes visible.

Looking to the future, one question that arises is how digital city infrastructure might prevent designers from using this strategy of defamiliarization. A lot depends on the openness of the technology itself and the policies surrounding the use of the data generated by it. We will revisit this question in the fragment on subversion.

Fragment 9: The Gameful City as Socialization

A defining aspect of city life is the constant dealings with strangers. There will always be more people in the city you call your home that you do not know than there will be that you do know. As Jacobs (1961) has argued, planners often overlook this fact and tend to perceive cities as nothing more than large towns. They fail to see that the seemingly insignificant interactions between strangers in city streets lead to informal networks of trust. For this to happen, neighborhoods need to allow for a diverse range of uses—shops next to houses, offices next to parks, and so on. Neighborhoods with mixed functions allow for and even encourage these casual interactions between strangers and in turn make a city the kind of vibrant place only a city can be.

However, mixed usage does not guarantee a neighborhood will develop networks of trust between its inhabitants. Mixed backgrounds of inhabitants can lead to a kind of self-imposed social segregation. This is most commonly seen in neighborhoods that go through a process of gentrification. An example of the surreal situations this can lead to is exemplified in the following passage, in which the author, James Meek, describes a visit to Broadway Market in Hackney, London—an area that has undergone significant gentrification:

> As Ghaith [an Iraqi friend of the author] and I walked down the street a disturbance began. A group of about thirty young black kids were moving together, looking anxious and excited. Some had makeshift weapons in their hands, poles and lengths of broken-off wood. After a moment, between a gap in the shops that looked through to the base of a tower block, we saw the reason for their anxiety—two tiny figures on bikes, dressed in black, hooded and masked. As we watched, one of the figures reached into the pocket of his hoodie and lifted—just enough to show—a hand gun, spreading panic among the larger group.
>
> The trouble subsided as quickly as it began and the participants dispersed before the police arrived. Throughout the episode, a young, casually dressed, thoughtful-looking white couple sat at a table outside a wine bar, watching and sipping white wine. The neck of the bottle leaned, misted with condensation, from the rim of an ice bucket on the table. The couple didn't look concerned that the gang confrontation or turf battle, whatever it was, would affect them; the feuding kids didn't seem to see them, either. (Meek 2011)

To be sure, there are many ways to deal with the issue of social fragmentation, and no one approach will guarantee a network of trust to emerge. But in addition to well-constructed neighborhoods, there is a place for play and games in the mix of strategies one might use. Through play, alternative contexts can be constructed within which people interact on different terms—allowing them to leave a lot of baggage at the proverbial door. These gameful interactions—just like Jacobs's usually more utilitarian ones—can



Figure 22.10
Players discussing Koppelkiek photos on display in 2009 (Hubbub 2009).

contribute to the emergence of informal networks of trust.

In 2007, we attempted to put these ideas to the test by creating a game called Koppelkiek (figure 22.10). It was designed for the neighborhood of Hoograven in Utrecht, which was struggling with many of the issues described in Meek's account of Hackney, although perhaps not on such a dramatic scale. The area was characterized by its fragmented nature—various groups of people more or less living in their own housing blocks, rarely meeting or interacting. Koppelkiek was a photo safari game—a common genre of pervasive game—and ran over the course of three weeks. The aim of the game was to take as many photos as possible of oneself with someone else—the titular *koppelkiek,* which translates into English as *couple snapshot*.

These photos were uploaded to the game's website for points. Extra points could be earned by including various bits of scenery in the photos, as described on the website. For example, taking a photo at your partner's front door. Although not a requirement, the game encouraged people to seek out strangers to pair up with on photos. The scenery assignments encouraged players to explore parts of the neighborhood they would otherwise not visit. This combined to create temporarily alternative uses of the streets of Hoograven, which we think contributed—in however small a way—to its social cohesion. The responses from players were encouraging, particularly those



Figure 22.11
Cruel 2 B Kind, 2006. (Photo courtesy of Tracy Fullerton)

who expressed relief at finally having something going on in the neighborhood that was not explicitly about its troubles, but instead was something one might talk about to neighborhood outsiders with a small sense of pride.

Pervasive games that use social expansion (Montola, Stenros and Waern 2009) in this way are a rich source of inspiration for gameful designers. A well-known example is Cruel 2 B Kind, created by Jane McGonigal and Ian Bogost in 2006. This variation on the game of Assassin has players "kill" each other through acts of kindness, such as complimenting someone's shoes or mistaking her for a celebrity (figure 22.11). The catch is that players are not aware of who the opponents are. The game is played in urban areas where participants are mixed in with a large number of nonplayers. The result is that quite often, a player's act of kindness directed at a suspected opponent instead ends up being received by a nonplaying stranger.

Cruel 2 B Kind (McGonigal and Bogost 2006) highlights an important aspect of involving strangers in gaming and playing in public—they do not know a game is being played and so any interactions need to be respectful of them. Annoying or frightening people might seem fun to those behind the curtain and in the know, but if the aim is to increase networks of trust, it simply will not do. Cruel 2 B Kind deals with this in a clever way. Nonplayers are involved in a way that might lead to small moments of surprise and wonder, perhaps puzzlement, but certainly not inconvenience or fear.

Similarly, Koppelkiek wears its heart on its sleeve. The fact that a stranger asks to photograph you might be a bit odd, but it is often also considered flattering. Many people oblige just to help out a neighbor, particularly when they explain the photo's purpose in the game.

This kind of gameful design concerns itself mainly with social practices as a material. The examples discussed here usually use a rule set to shape behavior. This rule set can be conveyed in an analog manner, just as with any kind of folk game or sport.

What has become more common, certainly with the rise of *gamification*, is to use networks and media to communicate the rule set, or even incentivize certain behavior by offering digital rewards. Care needs to be taken with this approach, as it threatens to instrumentalize social interactions—using your fellow urbanites just to get ahead in a game (Bogost 2010). I think any gameful design should be a space in which interacting with strangers becomes more meaningful, not less so.

The service foursquare (2009) serves as a good example. This service encourages its users to *check in* to locations they visit—such as bars and restaurants, museums and workplaces—by awarding points for each check-in. Leaderboards drive competition among users, and badges awarded for visiting a multiple of specific types of locations further encourage continued use. The cumulative use of foursquare offers users a guide to the city generated from the bottom up. Users discover new places to visit, thus driving new encounters between people. As such, it could be argued the service contributes to networks of trust. Here, gameful design is used to incentivize behavior that for each single act by a single individual might not be very interesting or valuable, but that does over time lead to value for the user base as a whole.

Beyond networks, the physical form of the city itself has an important part to play in encouraging sociality between strangers. Gameful designers have so far rarely had the opportunity to intervene on this level, though. Throughout the duration of Koppelkiek, all submitted photos were exhibited in the display windows of various abandoned shops in the neighborhood, as a way to reward players and to expose the game back to the neighborhood. These displays made the small acts of reciprocity tangible to players and nonplayers alike.

This modest attempt to use vacant buildings in Hoograven for alternative purposes for the duration of Koppelkiek does show gameful intervention at the level of bricks and mortar is possible. To be sure, many more approaches are open to further exploration. As noted in the fragment on appropriation, this means designers will have to start engaging with the legal and governance aspects of building use.

Fragment 10: The Gameful City as Subversion

Imagine you have traveled to Berlin. As many tourists do, you visit Checkpoint Charlie—the famous border station controlling travel between West and East Berlin. You are there in the evening and are taking photos. As you inspect one photo of an iconic sign that says "YOU ARE ENTERING THE AMERICAN SECTOR" you notice a text is overlaid that reads "HUNDREDS OF PEOPLE DIED LAST YEAR BY TRYING THIS AT THE US MEXICAN BORDER." The text is not there when you look at the sign, but on your photo it is. How is that possible?

You have just been the target of the Image Fulgurator, a device designed and built by Julius von



Figure 22.12
Image Fulgurator, 2007. (Image courtesy of Richard Wilhelmer)

Bismarck in 2007 (figure 22.12). It is shaped like a gun but mostly consists of parts you would find on any analog single lens reflex camera. It detects photographic flashes and, at that instant, projects an image onto whichever surface you point it at. But because this happens in the blink of an eye, the projection is not visible except in photographs.

Von Bismarck notes on his website that he is mainly interested in using the Fulgurator for subverting large public events in which the photographic memory is the main mechanism through which the masses are reached. The pope and Barack Obama have both been targets of the Fulgurator, as have Tiananmen Square in Beijing and the Berlin Reichstag. According to von Bismarck, "with the Fulgurator it is possible to have a lasting effect on those kinds of individual moments and events that become accessible to the masses only because they are preserved photographically" (von Bismarck 2007).

The Image Fulgurator is an incredibly ingenious instance of subverting the image of the city. More direct and less technological approaches are found in the street art movement—Banksy's elaborate and huge murals that criticize consumerist society being perhaps the most well known example (figure 22.13).

The increasing availability of affordable projection technology has expanded the repertoire of the street artist. For example, Graffiti Research Lab's L.A.S.E.R. Tag, first deployed in 2007, allows anyone to write on the façade of a building using a laser pointer that is tracked by a camera (figure 22.14). A beamer projects an image of simulated spray paint that follows the laser, dripping paint and all. On its website, Graffiti Research Lab calls it a "Weapon of Mass Defacement."



Figure 22.13
Olympic Pole Vaulter by Banksy (2012).

These are all projects that have some link to the culture-jamming techniques of many anti-consumerist social movements. They tend to be about enabling ordinary citizens to intervene in the media landscape of the city in a way that is normally preserved for governments and large corporations. The establishment has embraced some efforts from these communities, in particular projection-mapping techniques. Large ad campaigns are now commonly supported with impressive live interactive projections on building façades (NuFormer 2010). Many contemporary art festivals cannot do without similar efforts, albeit usually with a more artistic slant (figure 22.15; Valbuena 2008).

These examples deal explicitly and almost exclusively with the overlapping media landscape. This layer is probably one of the most recent additions to the city. Western consumerist society has led to a tremendous rise in outdoor advertising of all sorts. Any world capital worth its salt is literally plastered with billboards and increasingly with screens.[7]

It is hard for individuals to make significant changes to the physical form of the city, but in many cases it is equally hard to do this with the layer of media that covers it. It is not unlikely that the gradual replacement of analog signage with digital counterparts will only exacerbate this situation. A billboard or sign can be easily adjusted with some paint. To intervene in what is displayed on a screen, one must have access to the assemblage of technologies driving it.

The project Stickers on the Central Line (2011) is a case in point. These are whimsical stickers placed



Figure 22.14
Graffiti Research Lab's L.A.S.E.R. Tag, Rotterdam, the Netherlands, February 9, 2007. (Photo courtesy of Bennett Williamson)

on top of the way-finding systems in the London tube, adding fictional stops or changing station names (figure 22.16). Following Jacobs (1961), one could argue it is these kinds of nonsensical but mostly harmless acts of human ingenuity that give a city its lifeblood. Will we still be able to do so when the analog displays on the tube are replaced with screens? We might, but if we do, we will need new tools.

Image Fulgurator and L.A.S.E.R. Tag are therefore so interesting because they are not just interventions; they are tools to intervene *with*. Open source software and hardware will enable individuals to build these things themselves, in essence creating the twenty-first-century equivalent of spray paint, stencils, and poster glue. They are examples for gameful designers to aspire to: tools instead of media, things that enable instead of instruct.



Figure 22.15
Video projection on The Hague City Hall by Pablo Valbuena (2008), the Netherlands.



Figure 22.16
Stickers on the Central Line, London, England, April 29, 2011 (Anonymous 2011).

In a historical overview of pre-technological playful spaces, Adriana de Souza e Silva and Larissa Hjorth refer to Robert Luke's concept of the *phoneur*. A kind of dystopian updating of the *flâneur*, the *phoneur* is "unable to break free of a capitalist interpellation. The physical and geographic mobility of the *phoneur* is a misnomer—underscoring this mobility is the inability to escape surveillance and tracking" (de Souza e Silva and Hjorth 2009). However, as de Souza e Silva and Hjorth point out, the same technologies that can be used to pin down the individual can also be used to subvert those efforts. When one performs such acts of playful resistance, "the phoneur shifts from his lineage as a 21st century version of the flâneur's vision and distanced participation in the spectacle, and instead, he partakes in the gestures of locality" (de Souza e Silva and Hjorth 2009).

While entities with a high-modernist agenda work to ever increase the legibility of individual practice, gameful tools of resistance enable those same individuals to introduce noise into the system, to produce illegibility.

Fragment 11: The Gameful City as Formation

So far we have mostly looked at games and play as part of the first-person experience of a city. Now we revisit the issue described in the introduction, of the high-modernist view of the city as something that requires orderly planning from the top down, and the idea that a livable city is something that grows organically and thrives on the edge of legibility (Rao 2010).

What if we intervene in the planning process itself? It is after all here that the decisions get made that have far-reaching consequences for residents on the ground. It is also here that often certain players assert control over others—particularly government and corporate interests.

One interesting example of a gameful intervention in urban planning is the series of workshops devised by urban planner James Rojas (figure 22.17). These work roughly as follows: participants are typically nonexperts, just ordinary citizens with no particular role to play in urban planning. These gather in a space and are provided with large amounts of colorful found materials collected by Rojas. They are then asked a question, for instance: what would make them walk more in their own neighborhood? They provide their answers in the form of constructions built individually. Synthesis of these individual solutions can happen in various ways, from people voting on the nicest contributions to a facilitator summarizing the most notable ones (Haas 2010).

What stands out in Rojas's workshops is the radical simplification of the process of urban planning. Rojas does not seem to be interested in approximating the actual complexities of how neighborhoods are planned, built, and used. Instead, these workshops seem to be staged as a form of brainstorming, in which the suggestions from citizens are a source of inspiration for the urban planner.

A project that resembles Rojas's workshops in some ways but is different in many other aspects is Play the City (Tan 2009). Play the City is a series of gameful participatory planning exercises that departs from the observations that governmental or corporate interests usually dominate contemporary urban planning. Many other actors—activists, entrepreneurs, residents, and so on—have insufficient access to the process. A Play the City session resembles a game in that participants are asked to play the role of one of the typical actors in the process—including government and corporations, but also the more



Figure 22.17
Creations by James Rojas workshop participants, August 25, 2011. (Photo courtesy of the dA Center for the Arts)

marginalized players such as residents themselves. Play the City models the rules and procedures of urban planning and provides participants with a physical model of the city at hand and a large library of model buildings that can be recombined in numerous ways (figure 22.18). In this way, a Play the City session applies the principles of collective intelligence to urban planning issues. The creators cite numerous examples of plans emerging from these sessions finding their way into actual reality.

Compared to Rojas's workshops, Play the City is a more sophisticated approach to applying game principles to the process of urban planning. Of course, because it includes more rules, the question also arises who determines these rules. They can powerfully shape the outcomes of sessions, after all. It might be an interesting exercise to also allow for changes to the rules of urban planning by participants, similar to a game of Nomic (Suber 2003), which models the reflexivity of law. In this way, a gameful planning tool could not only lead to new ideas for neighborhoods and whole cities, but also to new ideas for the way to organize the planning of these neighborhoods and cities.

A big challenge remains in bridging the gap between efforts like James Rojas's workshops and Play the City and the realities of how plans come into being. It is likely that old power structures will assert themselves once a participatory session has ended. Again, it becomes clear that gameful designers will have to address the legal and governmental aspects of urban planning itself—what Dan Hill has aptly



Figure 22.18
A Play the City session in progress, Amsterdam, the Netherlands, August 31, 2012. (Photo courtesy of Ekim Tan)

named dark matter—to effect widespread and sustained change (Hill 2012).

Another point of criticism might be the incidental nature of these activities. It can be contrasted with the far more ambitious plans of Buckminster Fuller, who envisioned the World Game—a continually running computer system that took as its input as much data about the state of the world as it could and allowed players to make projections toward desired improvements. As Walz (2010) points out, the World Game may have served as inspiration for the system designed by cyberneticists Stafford Beer and Fernando Flores to assist Chilean president Salvador Allende with steering the country's socialist economy.

The permanent nature of Fuller's World Game may be admirable, and the direct coupling of Beer and Flores's playful tool to everyday reality is perhaps enviable, but they both do suffer from what Nassim Nicholas Taleb calls the ludic fallacy, "the misuse of games to model real-life situations" (Taleb 2007). Not only is it a fallacy to think one might reliably predict the future with simplified computer models—no matter how much historic data underpins them, as the credit crisis has aptly demonstrated—but also, here once again, as with the schemes of Le Corbusier and others, messy reality must yield to orderly, legible models, at the expense of local knowledge and human dignity.

What these gameful-designs-as-form-givers might all require is an application of Soja's notion of third space, of lived space. So far, it seems they have mostly been concerned with allowing participants to tinker with models of conceived space (idealistic plans for space, for instance with Play the City) and sometimes they have been connected to perceived space (materialistic observations of space, as is the case with the

World Game), but never do they seem to succeed in reconciling both by operationalizing the outcomes into lived space (embodied experience of space) and allowing for feedback from this domain of less formal and more local forms of knowledge.

In any case, gameful designs like the ones discussed in this fragment do engender a culture of play and participation. Citizens exposed to such a culture will expect and demand more from their leaders. Which is as good a starting point as any.

Fragment 12: Looking to the Future

And with that we come to the end of our survey of ways in which the built environment of the city can be gameful. Certainly, not all aspects have been addressed, nor have we cataloged all relevant examples. Still, some of the major themes that run through this domain have been laid bare. Next, we will turn our attention to the future and share some thoughts on what issues we might find ourselves dealing with and how gameful design might be of service.

Part III: Megacities, Smart Cities, Gameful Cities

In this final part, I will discuss the rising phenomenon of megacities and the unique issues they confront us with. This new type of city challenges some of the notions gameful urban designers working in Western cities might take for granted, but it also offers opportunities for valuable new work. I then look at the idea of the smart city and criticize its high-modernist underpinnings. Finally, I propose the gameful city as a counterpoint to the smart city and suggest some possible future direction for the practice of gameful design in the urban context, which is focused on equipping people with gameful tools for playful resistance.

Fragment 13: Megacities

Any visitor to Southeast Asia, South America, or Africa—any place where new megacities have arisen—will know that when it comes to the planning and building of cities, these places are nothing like what we are used to. The massive scale, the speed of development, the incredible contrast between never-ending shantytowns and gleaming shopping malls, the traffic . . . Jakarta is nothing like Amsterdam. Bogotá is nothing like New York.[8]

The experience on the ground often makes me wonder, why are things like this? The mechanisms behind the creation of these cities are clearly off kilter.

Even more so than in the Western cities I have mostly discussed so far, certain interests have gotten the upper hand and left large groups of people effectively unable to intervene in the ways in which their living spaces come into being and are managed—unless it is the complete self-reliance required of one making her home in a slum.

What the place of gameful design in this problem space might be is an open question to me. We see examples of gameful interventions on the street level happen in these new cities. The work if Irwan Ahmett and Tita Salina in Jakarta is one example (Ahmett and Salina 2010). They have for instance staged group-jogging sessions along an abandoned monorail project, called Monorail Slalom (figure 22.19). This and other games staged by Ahmett and Salina are an attempt to reengage citizens with public space in a



Figure 22.19
Monorail Slalom in progress, Jakarta, Indonesia, April 18, 2010. Photo courtesy of Irwan Ahmett.

city that is increasingly at the mercy of self-centered government and corporate interests. Monorail Slalom pokes fun at the sad reality of a city government unable to provide a public transportation service while total gridlock looms imminent.

However, to me such work feels transplanted, an appropriation of Western techniques. On the one hand this is to be expected of the globalized culture Jakarta has very much become part of. On the other hand, such whimsical interventions might make sense in New York, but do they also in Jakarta? The scale and seriousness of the issues being addressed feel just too large for such frivolous temporary events.[9]

These new cities are brittle and fragile and are becoming more so, due mostly to the fact that large parts are planned and built by organizations with a desire for highly legible structures that appear orderly, in the hopes of controlling the public—despite sometimes well-meaning intentions. Such cities will not be able to withstand cataclysms, and indeed the broken social contract, or a total lack thereof, makes it incredibly hard if not impossible to deal with public issues; for example, traffic in Jakarta (Nazeer 2011) or malaria in Mumbai (Roy 2011).

These cities are in dire need of more diversity in the decision-making culture and a reknitting of social fabric on the ground. For this, citizens must be provided with tools that they can use to influence decisions and take action. It is here I see a place for gameful designers: to produce tools or blueprints for tools that can be adopted at scale by urbanites across the globe.

These tools can facilitate the use of gameful strategies as I described in the previous part—appropriation, defamiliarization, socialization, subversion, and formation. It is important to move beyond one-offs and installations because as has been pointed out by Dan Hill, these do not lead to systemic change (Hill 2012). But where Hill argues for placing designers

within organizations and using projects to drive change from within, I argue for gameful designers to consider urbanites across the globe as their clients and to provide them with tools to set up their own projects outside of these organizations. This can act as a counterbalance to the tendency of such large groups of people to drive toward ever-increasing legibility. Citizens with gameful tools can be a generative force, making their cities more resilient, similar to the unseen but very real influence of civil disobedience of various kinds described by James C. Scott (2012).[10]

A recent example of—highly visible—civil disobedience with generative effects would be the global Occupy movement. The most chronicled aspect of the movement was its desire for economic justice, leading to the well-known "we are the 99 percent" meme. An equally important aspect, however, was the desire to gather and care for each other, outside of society's institutions. Even though we might say Occupy ultimately failed in its aim to bring about a new economic order—not only because of the governmental opposition it was faced with, but also because of its inability to reflect critically on what was not functioning within the movement's own emerging institutions, most notably the general assembly—it has demonstrated it can outperform government in caring for citizens, as is the case with Occupy Sandy (Norton 2012). The resilience of New York City, faced with a natural disaster, was thus effectively improved as the result of civil disobedience.[11]

This approach I suggest, of gamefulness being (part of) the solution to megacities' problems, is not the most common perspective you will hear, however. It has become a common view to hold that cities—if they are to remain viable living places—should become smart.

Fragment 14: Smart Cities

Much of the smart cities rhetoric from the likes of IBM, Siemens, and Oracle has been framed in terms of increased efficiency and predictability. This, according to them, is the chief benefit of integrating new technologies in the urban environment. For sure, these are appealing aspects for governments tasked with managing them and for the corporations seeking to turn a profit from them. But let's be honest, is that which you find most lovely about your favorite city or neighborhood really its degree of efficiency and predictability? This to a large extent is just a new variant of what Jane Jacobs called social taxidermy (Jacobs 1961), which aims to "specify and freeze functions by planning fiat" (Scott 2012).

For sure, being able to get from A to B in a timely fashion is an important quality of any city. It also will not do if every time you step out the door, you need to worry about an unknown amount of unpleasant occurrences throughout your day. But all things being equal, once we live in a place that has provided for these things to a reasonable degree, it is other aspects that make a city worth living in.[12]

Granted, I am writing this from a privileged standpoint, living in a charming livable small city—Utrecht—in an equally livable and equally charming small country—the Netherlands. But even if the same cannot be said for your situation, do you really think the smart city technologies on offer from these global companies or the governments desperately and naively seeking for a quick fix to their woes from the same technology will get your city to your desired optimal state? I think one look at the track record of these players will show you what is needed for any kind of livable city to emerge—smart or otherwise—is as much a strong citizenry as a strong state and

market. This is the place where gameful design can really offer value—consider it a balancing force.

Fragment 15: Gameful Cities

Think of a gameful city as a counterpoint to a smart city.[13] Where a smart city promises increased control and legibility to large organizations, a gameful city promises increased autonomy and influence to individuals. Ultimately and in all likelihood, it will be a mixture of both approaches that is best suited to tackle the serious problems we are facing in megacities. Gameful design can enable individuals to act as a trim tab, as described by Buckminster Fuller:

> Something hit me very hard once, thinking about what one little man could do. Think of the Queen Mary—the whole ship goes by and then comes the rudder. And there's a tiny thing at the edge of the rudder called a trim tab.
>
> It's a miniature rudder. Just moving the little trim tab builds a low pressure that pulls the rudder around. Takes almost no effort at all. So I said that the little individual can be a trim tab. Society thinks it's going right by you, that it's left you altogether. But if you're doing dynamic things mentally, the fact is that you can just put your foot out like that and the whole big ship of state is going to go.
>
> So I said, call me Trim Tab. (Farrell 1972)

In this way, the myopia of large organizations is counteracted. Local knowledge—vital in solving megacity problems—balances out the sheer force of smart-city-tech-wielding governments and corporations.

Ultimately then, the question is in what kind of city you would like to live. And hopefully from reading this chapter, you have gotten an appetite to engage with the city in new ways, to inject some illegibility into the urban fabric. And you might want to invent gameful ways of shaping the city. If you do, I hope you will share them with those around you. A gamefully motivated group of citizens with effective and scalable gameful tools is a force to be reckoned with. Cities around the world could use more of them.

Acknowledgments

I would like to thank Alper Çuğun for his contributions to this chapter.

Notes

1. Readers might be reminded of Situationist International's playful resistance, which we will discuss later on.

2. *Soft urbanism* is sometimes used specifically to label that part of urbanist practice that deals with software as it relates to the built environment (Sikiaridi and Vogelaar 2006). Although my use of the term here might include this aspect, it is more defined by the intent of the designer, as opposed to the specific materials she works with.

3. Indeed, we will discuss the approach of involving nonexperts in urban planning through games in more detail in the second part of this chapter.

4. A good introduction to the form is the film *Jump London* (Christie 2003).

5. Here, we are reminded of the ambiguous is/is-not quality of fun spaces as described by Thomas (2012).

6. In this way, games and forms of play can resemble memes (Dawkins 1976). Playful social practices such as BookCrossing and BarCamps as well as social games such as Mafia/Werewolf are all examples of self-spreading rule sets (Alfrink 2011).

7. The exception to this rule is São Paulo, which has removed most of its outdoor billboards, outdoor video screens, and ads in busses following the adoption of the Clean City Law (Harris 2007).

8. Although this fragment deals with new megacities, a strategy similar to the one described in the following pages could be useful in the West's old cities. As opposed to unchecked growth, these are actually experiencing shrinkage. Here as much as anywhere, it is clear government alone won't be able to provide a solution. It is up to citizens to get involved in these issues, and gameful tools might once again enable them to better do so.

9. Although one could argue just as well it is precisely the severity of the situation that calls for a light-hearted approach.

10. This, again, is an approach very different from gamification. For one, because I am arguing for tools that are generative, tools with unspecified outcomes, whereas gamification seeks to incentivize behavior to achieve predetermined outcomes. This once again is a drive toward higher legibility and explains why it has mostly been corporations and other large organizations that have expressed interest in the phenomenon.

11. For an illuminating first-person account of Occupy Sandy, read Adam Greenfield's (2013) "A Diagram of Occupy Sandy." Not only does Greenfield describe the relief effort's workings in some detail, he also provides a starting point of a toolkit that might enable other similar efforts elsewhere. Finally, similar to Norton, he suggests Occupy Sandy exemplifies a direction the larger Occupy movements should have taken: "an ongoing effort to create a fabric of community institutions that live their beliefs matter-of-factly, so that people can experience for themselves the difference between what it feels like to be a consumer and what it feels like to actually participate." I suggest gameful designs can provide similar experiences of participation as opposed to consumption.

12. One could argue that in megacities, such basic amenities are often not even provided, at least not to the masses. Wouldn't a smart city be a desirable alternative? As Scott (2012) rightly points out, top-down planned city centers catering to the happy few can often exist only thanks to the labor from those living in the massive unplanned sprawl at their peripheries. The ordered well-planned nature of such centers is therefore deceptive. I therefore believe it is unlikely smart city technologies—deployed in a similar top-down manner—will make much of a difference in these situations.

13. Watershed, a cross-artform venue and producer based in Bristol, suggests a similar counterpoint to the smart city as part of its Playable City Award: "A Playable City is a city where people, hospitality and openness are key, enabling its residents and visitors to reconfigure and rewrite its services, places and stories" (Reddington 2012). Slightly further afield, not explicitly linked to games and play, Michiel de

Lange and Martijn de Waal also critique smart city rhetoric and propose "social cities" as an alternative, which is also very much about enabling citizens and starting from street level upward: "We propose the 'social city' as an urgent, alternative take on urban design with digital technologies. Other than 'smart city' visions and practices, we urge designers to focus on the active role of citizens and use the city itself as the testbed for experiments" (de Lange and de Waal 2012).

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POSITION STATEMENT

MOBILIZING GAMIFICATION

Paul Coulton

In his 1964 book *Understanding Media*, Marshall McLuhan observed that "we become what we behold. We shape our tools, and thereafter our tools shape us." In a relatively short period, video games have become a major feature of our cultural landscape, even beyond the games themselves. We can nowadays see game aesthetics and iconography represented in the other main forms of media such as films, books, and television—such that we are all becoming more "games literate." In many respects, the emergence of *gamification* is simply a reflection of this expanding aesthetic and literacy.

My own perspective on gamification is that we can use games as a lens when designing interactive systems to produce more gameful designs. This avoids presenting games as some kind of "magic bullet" for user engagement. Rather, gameful design becomes just one more design frame to be considered alongside alternatives through which designers can concretize a solution to meet the needs of the user. I will illustrate this perspective of gameful design in relation to a consumer technology that has arguably had a great impact on society since its appearance in the early 1990s: the mobile phone.

Mobile phones have been part of the gaming landscape since Taneli Armanto programmed a version of Snake for Nokia phones in 1997. With more than six billion phone subscriptions worldwide today, they have not only become ubiquitous, but also constitute a significant proportion of the games market. For example, of the 43 million hours a day currently spent gaming in the United Kingdom, 15 percent is on mobile phones. There is still some discussion what exactly we mean by "mobile game": is it simply required to run on a mobile device? Or is there something more fundamental that needs to be considered? The ITU defines *mobile* as follows: "the term mobile can be distinguished as applying to those systems designed to support terminals in motion while being used."

In other words, it could be argued that unless a game running on a mobile phone uses the mobile network as part of the game play, it is simply a handheld, portable or nomadic game. However, this description would ignore the significant sociotechnological aspects of the device that has become so entrenched in our daily lives and that alongside our keys and wallet is what we check our pockets for before we leave the house. Therefore, I would argue that the actual gaming platform in this case of mobile is significant as it affords opportunities to game in contexts where access to other gaming platforms would not be available or even desirable. Personally, I find it is a lot less obvious that you are not paying

attention in a meeting if you are playing a game on your phone rather than a bespoke mobile gaming device like a Nintendo DS.

In the mobile context, I see a design continuum between gamification on the one end and gameful design on the other that can be represented by two applications. The first, the to-do list application Epic Win is exemplary for the gamification pole. I use this example somewhat reluctantly as its overall design is beautifully crafted but fails to create a meaningful game-like experience. The aspiration of Epic Win is to turn your life into a role-playing game. Players redefine daily chores and to-dos as "quests," allocating experience points (epicness or task difficulty) and a feat category (strength, stamina, intellect, social, spirit). Points for finished to-dos accumulate in the given category and help users evolve their chosen avatar, with total experience points gathered being represented as distance along a map that unlocks loot on the way. The application thus has no actual game mechanics. It simply provides feedback through points and badges (loot) that rely completely on the user engaging sufficiently with the premise to give it meaning. Like many other users, I found the novelty of the aesthetics diminish after a few tries. The application itself became a chore. Whereas in games I might simply "grok" a mechanic to get to a more interesting level, Epic Win has no progression of difficulty, no mastery to be achieved, as there is a disconnect between the actual tasks and the application. Despite trying to improve things by allocating myself maximum points for highly arduous tasks of pondering and breathing, in the hope that the novelty of the unlocked loot would provide further enjoyment, I quickly became bored and removed the app from my phone. Epic Win demonstrates that points and badges are *not* inherently fun. Its beautiful graphics would have perhaps brought me more satisfaction as simply a skin for my normal calendar application.

In stark contrast to this stands the Heineken StarPlayer. While some may describe it as the gamification of watching football on television, in particular the Union of European Football Associations (UEFA) Champions League, I consider it an exemplar of gameful design. The application only becomes available to play ten minutes before the start of a match and automatically synchronizes with the time on the television. When free kicks or corners occur in the match on screen, the player is prompted to make a quick decision whether the situation will result in either a goal, a miss, a save, or a clearance. Points are awarded for a correct guess and vary based on the likelihood of the guessed outcome. Through the course of one match, the player is given eight attempts to anticipate accurately whether either team will score a goal within the next thirty seconds, with more points awarded the longer the time from prediction. The application also caters for the inevitable lulls in a game by including "pop quiz" moments where a quick, correct response can garner more points. The result is a really fun and engaging application. Rather than relying on predicting the total outcome of a game, as in sports betting, it involves anticipating the moment-to-moment action of the football match. Thus, unlike Epic Win, the StarPlayer closely couples task and application to yield ultimately a more satisfying experience. Instead of relying on the iconography of games, it utilizes gameful design as it draws upon an empathy and understanding of the armchair football fan who, like myself, probably normally engages with such anticipation in a match by shouting and occasionally swearing at the television.

23 GAMIFYING GREEN: GAMIFICATION AND ENVIRONMENTAL SUSTAINABILITY

Jon E. Froehlich

In its State of Green Business report, the GreenBiz Group listed gamification as one of the top sustainable business trends of 2012, noting that game mechanics are increasingly used by companies to provide "rewards for making good, green choices" (Makower 2012). In the past few years, we have seen a surge of interest in green gamification touching upon nearly all aspects of our everyday life from cars that rank and reward fuel-efficient driving performance (e.g., the Nissan Leaf) to sanitation services that monitor and reward home recycling behavior (e.g., Recyclebank). As Ashok Kamal, CEO of the green social media marketing company Bennu notes, this movement represents a "tidal wave of green gamification that is capturing the attention of the green community and the business community as a whole" (Cousteau et al. 2012).

Given such vibrant enthusiasm surrounding "green gamification," it is hard not to react with a degree of skepticism. Climate change, pollution, and other human-driven environmental ills are complex, multifaceted problems: Can gamification actually play a serious role in their solution? In this chapter, I attempt to provide a partial answer. I survey the rise of gamification as a mechanism to encourage proenvironmental behavior[1] and connect it back to similar behavior-based movements in environmental psychology and persuasive technology. Though many definitions of *gamification* have previously been offered, here I adopt the definition by Deterding et al. (2011, 1): "the use of game design elements in non-game contexts." And while games—from board to video—have long been used to teach sustainability concepts (e.g., Chris Crawford's 1990 Balance of the Planet), here I specifically emphasize gamification and its link to other behavior-based approaches rather than traditional games themselves.

The chapter begins with a simple argument: that green gamification is a type of *designed persuasion* and that, as such, one must approach it with a *rhetorical* lens to understand how designers think about and apply gamification to motivate behavior. With this argument as a backdrop, I show how gamification elements such as feedback, progress tracking, goals, and rewards were used in the environmental domain long before the rise of gamification to encourage pro-environmental behavior. I determine that one key difference is in how gamification emphasizes *playfulness* in design, which changes the way these elements are operationalized and visualized to the user (i.e., a gamification aesthetic can create a unique user experience). After this contextualization, I offer examples of persuasive technology and green gamification in three domains: home resource consumption,

transportation, and waste disposal. I then close by discussing the ethical implications of gamification, its potential to contribute positively to environmental sustainability, and how the intrinsic versus extrinsic motivation debate within gamification fits in an environmental context.

My goal with this chapter is not to advocate or champion the use of gamification in environmental contexts but rather to provide historical perspective, to provoke thought, and to outline successes and failures. In my own research, I have designed, experimented with, and evaluated technology-mediated game mechanics in the context of "green" behaviors (Froehlich, Dillahunt et al. 2009; Froehlich 2011; Froehlich, Findlater, et al. 2012)—though I have not always identified this work with gamification. Unlike Eric Zimmerman, Ian Bogost, Jane McGonigal, and many others with a high profile in gamification discourse, I am not a game designer nor a game/media scholar but rather a computer scientist who works at the intersection of human behavior and technology, often with respect to environmental sustainability and personal health and wellness (e.g., see Consolvo et al. 2008; Froehlich, Findlater, and Landay 2010; Hekler et al. 2013). My perspectives, then, are not grounded or shaped by game and media scholarship but rather by my disciplinary background and experiences working in human–computer interaction (HCI), computer science, and design.

Contextualizing Green Gamification

Gamification is the application of game design elements to help achieve a particular *designed* agenda or goal; for example, to increase user interest, sustain attention, or provoke specific behaviors. As such, it is a form of designed *persuasion* (or persuasive design). And, because gamification is both enabled and mediated by computation, it is most often manifested as a form of *persuasive technology*. In this section, I first situate gamification within persuasive technology—a field concerned with how computing can be designed to influence behavior—and then contextualize *green* gamification more specifically within eco-feedback and environmental psychology. To start, please refer to figure 23.1, which provides two graphical illustrations for the interconnections between disciplines and concepts discussed in this chapter. These are worth returning to as this section progresses.

Persuasive Technology and Eco-feedback

Although gamification has long existed in various forms (e.g., loyalty marketing programs), it is only through the massive adoption and availability of computing—the Internet, fast computation, cheap and pervasive sensing and output—that gamification has become a viable strategy for influencing *everyday* behaviors. In the late 1990s, B. J. Fogg and his colleagues at Stanford University pioneered research in persuasive technology, which they defined as "any interactive computing system designed to change people's attitudes or behaviors" (Fogg 2003, 1). Since then, persuasive technology has quickly become a common part of the modern technological landscape—from smartphone applications that motivate us to exercise (e.g., Runkeeper) to algorithmically tailored advertisements that turn up in Google search

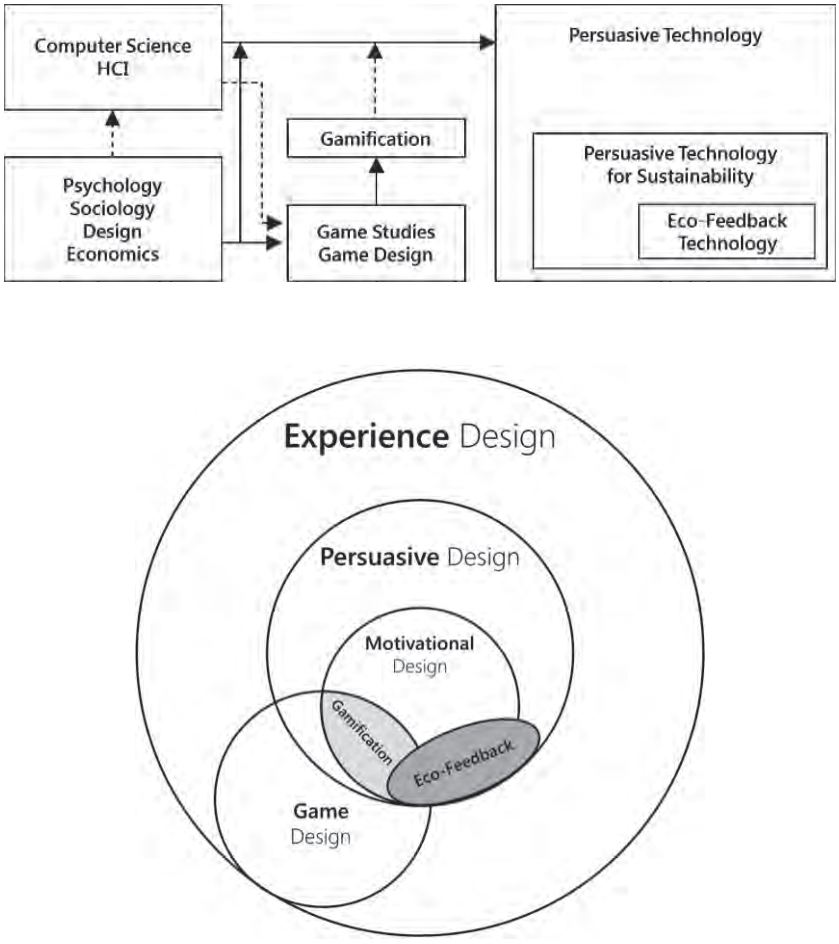


Figure 23.1

Two conceptual illustrations depicting the position and role of gamification in the context of persuasive technology, game design, eco-feedback, and related disciplines.

results and influence our decision to "click"—and serves as the substrate for gamification.

In the context of proenvironmental behaviors, persuasive technology that monitors and provides feedback on individual or group behaviors with a goal of reducing environmental impact is called *eco-feedback* (Froehlich et al. 2010; see also figure 23.1). Many popular commercial examples of green gamification—from the Nest smart thermostat to the Ford SmartGauge (both described in this chapter)—are types of eco-feedback. Eco-feedback is not just about solving an *informational* problem (i.e., a presumption that people would necessarily act more proenvironmentally if they were more aware or more informed) but also a *motivational* one. Thus, a key eco-feedback design question,which is central to green gamification as well,is how to visualize sensed information in a way that not only informs but also motivates action?

To help with this, Fogg provides seven design principles for motivating action through persuasive technology informed largely by work in psychology and sociology (Fogg 2003, 33–49). Most relevant to green gamification are the principles of *tailoring*, *suggestion, conditioning*, and *self-monitoring*. Persuasive technology can automatically (a) *tailor/personalize* information to fit an individual's needs, personality, and usage context to increase its persuasiveness; (b) *suggest* new action or provide new information at opportune "decisional" moments; (c) *apply principles of operant conditioning* such as offering incentives/rewards for certain actions; (d) *automatically sense and feed back* performance information about a target behavior (self-monitoring).

As most green gamification examples in this chapter are self-monitoring systems, it's worth noting that not all such systems are persuasive. Consider a traditional analog speedometer in a car. The display provides instantaneous feedback about speed, but it is meant to be *informative* rather than *persuasive*. In comparison, more recent dashboard designs are specifically built to motivate fuel-efficient driving behaviors—sometimes even employing gamification (e.g., the Ford SmartGauge). This brings about two critical and interwoven issues related to persuasive technology and gamification: (1) there is a spectrum between neutrally conveyed information and information that is conveyed with *persuasive intent*; (2) this intent is ascribed by the designer himself or herself with a specific goal or set of goals in mind. Gamification, as a rhetorical form, can transform an otherwise neutrally portrayed information display to a persuasive one.

Environmental Psychology

Green gamification is related not only to persuasive technology but also to environmental psychology, which is the twofold study of how the environment affects human behavior and how human behavior affects the environment (Pol 2006, 2007).[2] Two particularly applicable threads of work are the identification and study of *determinants* of proenvironmental behavior (i.e., *why* people hold proenvironmental beliefs and behave in a proenvironmental fashion) and the study of financial and nonfinancial motivational techniques for encouraging proenvironmental behavior (i.e., *how* we can influence and effectively promote proenvironmental decision-making and action).

Although numerous theoretical models of proenvironmental behaviors have been developed and studied, no definitive explanation has yet been found (Kollmuss and Agyeman 2002). Still, these models

offer insights into why people act environmentally and thus have direct implications for the design of eco-feedback and green gamification systems. Stern (2011) highlights two dominant theories of proenvironmental behavior: the first emphasizes individualistic motives and presumes that individuals seek to maximize their material welfare, subjective well-being, or utility. The rational-economic model, for example, assumes that people act to maximize rewards and minimize costs. The second set of theories involves factors beyond individualism such as environmental consciousness (e.g., eco-centrism) and altruism. For example, Schwartz's moral norm-activation model (Schwartz 1977) suggests that pro-environmental behavior can be stimulated if a person is aware of the negative consequences for others and ascribes some amount of responsibility for taking ameliorative action. Stern notes that these two perspectives are not mutually exclusive and that both theories have explanatory value across a range of proenvironmental behaviors (Stern 2011). In the context of green gamification, it is worth explicitly thinking about and discussing which behavioral model is more apparent in a design (i.e., how are designers encoding these behavioral models into their designs—whether conscious of it or not?).

Environmental psychologists have also studied the effectiveness of techniques for motivating proenvironmental action, many of which overlap with gamification strategies including feedback, goal setting, social comparisons, modeling, and rewards (e.g., see Froehlich et al. 2010; McKenzie-Mohr 2011). The most popular behavioral interventions are summarized in table 23.1. In many cases, these techniques can be combined with other principles from behavioral psychology and behavioral economics (e.g., Tversky and Kahneman 1981) including loss aversion, social norms, reciprocity, scarcity, and anchoring. For example, a study by Gonzales, Aronson, and Costanzo (1988) found that framing energy choices in terms of avoiding losses rather than achieving gains persuaded homeowners to take greater advantage of financial subsidies for home weatherization. As a second example, Goldstein, Cialdini, and Griskevicius (2008) found that providing a *descriptive social norm*[3] message emphasizing that a *majority* of hotel guests re-use towels increased towel reuse by 34 percent (reducing water and detergent waste). This established towel re-use as a social norm, which influenced behavior. See Wilson and Dowlatabadi (2007) for further discussion of behavioral economics applied to proenvironmental behavior.

Where Does Gamification Fit?

As you may have observed, many persuasive technologies and interventions in environmental psychology use motivational techniques similar to those associated with gamification. Historically, however, these techniques were not explicitly recognized as gamification, but instead were tied to more core theoretical underpinnings from psychology, sociology, and economics. The lack of direct attribution to gamification reflects both the recency with which the term came into being as well as a tendency by some designers to reject the term as hype or ill-fitting (e.g., Bogost 2011). In my experience, it also reflects a general confusion about what is and is not gamification and how gamification differs from other motivational strategies used in persuasive technology.

Deterding and colleagues argue that "gamification does indeed demarcate a distinct but previously unspecified group of phenomena, namely the complex of playfulness, playful interaction, and

Table 23.1
A list of common intervention techniques used in environmental psychology to promote proenvironmental behavior

Intervention Technique	*Description*
Information	*Information* is perhaps the most widely used means to promote proenvironmental behavior. Studies have shown, however, that though presenting information can increase knowledge, it generally has minimal effects on proenvironmental behavior (Gardner and Stern 2002). To maximize information's persuasive potential, it must be easy to understand, prescriptive, trusted, presented in a way that attracts attention and is remembered, and delivered as close as possible in time and place to the relevant choice (Brewer and Stern 2005).
Comparison	A *comparison* between individuals or groups can be useful for motivating action. However, the way in which social comparisons are conducted (e.g., which groups are used for the comparison, whether there is an explicitly structured competition) and how these comparisons are visualized can significantly impact their effectiveness (Siero et al. 1996 versus Haakana, Sillanpää, and Talsi 1997 and Egan 1999). Comparisons can also be to self (e.g., historical performance) or to a goal.
Incentives and rewards / disincentives and punishment	Financial *incentives* and *disincentives* have been used successfully across a broad range of proenvironmental behaviors such as increasing recycling behavior, properly disposing of toxic waste, and installing solar photovoltaic systems (McKenzie-Mohr 2011, 112–115). Nonfinancial incentives can also be used, such as reducing time and effort or offering social recognition (e.g., curbside recycling vs. recycling centers; express lanes for buses and carpools).
Commitments / goal setting	Commitments and goal setting affects behavior primarily by increasing attention, effort, and persistence toward goal-relevant activities (Locke and Latham 2002). Effects vary based on *who* set the goal (e.g., self-set or externally set) and the goal's difficulty (e.g., typically, more challenging goals have greater effectiveness as long as the individual feels that she or he can complete the goal).
Feedback	Many of the above techniques either require or can be enhanced by behavioral *feedback* (e.g., goal setting requires feedback about progress toward a goal). Feedback appears to work because it has both informational and motivational properties: it provides a basis for assessment and action and enables progress toward a goal (Aitken et al. 1994; Fischer 2008). In a meta-review of fifty-seven residential energy usage feedback studies, households reduced electricity consumption by 4 to 12 percent on average (Ehrhardt-Martinez, Donnelly, and Laitner 2010). In another survey, Fischer found that the most effective eco-feedback interfaces provided multiple feedback options (e.g., over various time periods, comparisons), were updated frequently, were interactive, and/or provided an appliance-specific breakdown of usage (Fischer 2008).

Note: For more detail, see Gardner and Stern (2002); Allcott and Mullainathan (2010); McKenzie-Mohr (2011).

gameful design" (Deterding et al. 2011, 2). To me, the clearest distinction between a traditional persuasive technology and a gamified one is this notion of *designed playfulness*. A designer can turn toward games and game-thinking to learn about and apply notions of fun, competition, and narrative in design. Though the term *gamification* itself remains controversial, its emergence has established a common vocabulary and context with which we can describe, study, and discuss related *gamified* systems. This design language, however, is also a source of confusion in the context of persuasive technology because similar phenomena may be referred to with different names: for example, "progress tracking" and "badges" in gamification may be referred to as "feedback" and "virtual rewards" in persuasive technology. Of course, a gamified badge system and a non-gamified progress tracker may be operationalized, implemented, and visualized differently, with the gamified version emphasizing fun and playfulness. In addition, some aspects of game design have been underexplored in persuasive technology, and gamification helps uncover such insights, including the use of uncertainty and unpredictability to create and sustain engagement, the use of storytelling or narrative, and the use of deliberate gaps in a design or message that the user will want to fill (e.g., see Koster 2004; Chatfield 2010; Lockton 2010).

So, while I agree with Deterding and colleagues that gamification demarcates a unique set of design phenomena related to games, this demarcation is more nuanced than the aforementioned quote implies. For example, the Toyota Prius eco-feedback display, which will be discussed in detail later, provides drivers with real-time feedback on "eco-driving performance" but lacks specifically designed gamified elements. In contrast, for newer cars, eco-feedback devices such as the Ford SmartGauge and Honda Eco-Assist were explicitly designed with gamification elements (e.g., competitions, trophies) to motivate fuel-efficient driving behavior. Notably, as you will see in this chapter's survey, in both cases drivers perceived the displays as providing a game-like experience. Thus, simply providing basic mechanisms—like behavioral feedback—can be enough to inspire game-like responses and action.

Gamifying Green Examples

In this section, I survey the use of green gamification. I include historical systems designed prior to the emergence of gamification as well as more recent systems, which may or may not be self-described as gamification but seem to include aspects of game mechanics in their design. I feel it best not to get lost in a philosophical debate between what is and what is not gamification. Instead, I will lean toward inclusion if for no other reason than to provoke discussion and provide historical context.

Though gamification could likely be applied across a broad range of environmental domains (e.g., deforestation and ocean pollution), here I focus on three common ones relatable to the everyday experience: home resource consumption, personal transportation, and waste disposal behaviors. These areas account for a significant amount of energy use and carbon dioxide emissions in the United States (Bin and Dowlatabadi 2005). This survey includes examples from academia and industry. While work in

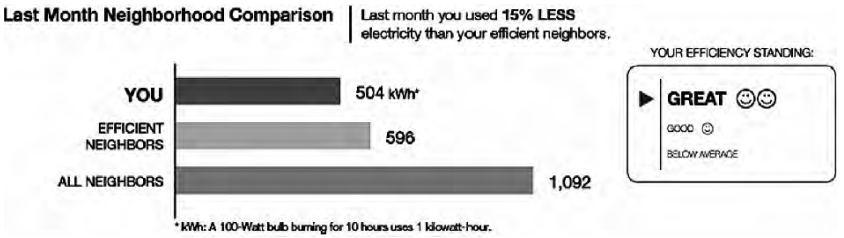


Figure 23.2

The top section of an Opower Home Energy Report, which illustrates the use of normative comparisons (the bar graph comparing the current household's usage with that of its neighbors). Injunctive messaging (the smiley faces) is used to mitigate the boomerang effect where both above-average and below-average users converge toward the norm; see Allcott (2011).

industry is often deployed and evaluated more broadly (e.g., larger studies, longer study length), research in academia is typically more forward-looking and experimental. Thus, there is value in reviewing each.

Home Resource Consumption

Behavior has a significant impact on home resource consumption. Past research has shown that energy use can differ by two to three times in identical homes occupied by people with similar demographics (e.g., Socolow 1978; Schipper et al. 1989; Parker et al. 1996). Dietz et al. (2009) estimate that behavioral interventions could reduce carbon emissions from direct energy use in households by 20 percent, which is equivalent to 120 million metric tons of carbon or more than the entire emissions of France in 2005. With the rise of ubiquitous Internet connectivity, mobile and in-home displays, and smart meters,[4] eco-feedback (with or without gamification) will undoubtedly play an increased role in promoting and supporting proenvironmental behavior in the home.

Opower Home Energy Reports

Opower is often touted as one of the success stories in applying behavioral science techniques to energy efficiency programs. The company works with utilities to provide Home Energy Reports (figure 23.2), which use behavioral science strategies such as loss aversion and tailoring to encourage energy-efficient behaviors (Laskey and Kavazovic 2011). I will focus on two aspects most relevant to green gamification. First, using *descriptive social norms*, the reports compare the receiving household's recent energy usage versus both demographically similar households and energy-efficient households. Second, to combat the *boomerang effect* (Schultz et al. 2007), where low energy users actually increase their energy use to become closer to average, Opower uses *injunctive* messaging—conveying social approval or disapproval. To do so, households are awarded a "Great" or "Good" label on the report plus one or two smiley faces for being efficient or a "Below Average" label otherwise (Allcott 2011). Notably, Opower initially included frownie faces for poor performing households but stopped after customer complaints—people

did not like feeling negatively judged (Kaufman 2009; Allcott 2011). Though it is debatable whether such techniques constitute gamification per se, the findings have implications for the design of badges or other commonly used visual iconic rewards in gamification.

Are Opower's Home Energy Reports successful? The short answer is yes. According to its website, Opower currently has eighty utility clients and has saved more than two terawatt-hours of energy worldwide since its first utility customer in 2007 (DeWitt 2013). These savings are equivalent to removing a city with a population of 500,000 people off the grid for a year. In addition, independent studies of nearly a million treatment and control households across the United States have shown that utilities using Opower's program reduce energy consumption by an average of 2 percent—the highest saving decile decreases usage by about 6 percent (Ayres, Raseman, and Shih 2009; Allcott 2011; Cooney and Provencher 2011; Wu et al. 2012).

Are Opower reports a form of gamification? This is debatable. The reports include progress tracking, iconic rewards (smiley faces), implicit competition support via normative comparisons, and tailoring—all techniques found within gamification. The smiley faces could be perceived as fun and playful, but for the most part the reports favor a more formal, professional aesthetic over a more playful design. With Opower's vast reporting infrastructure in place, however, it has the unique potential to run randomized control trials (RCTs) to investigate how more explicit playful designs may impact energy efficiency behaviors.

Nest Smart Thermostat

The Nest smart thermostat is marketed to save homeowners 20 to 30 percent on utility bills by learning about occupants' behavior and optimizing heating and air conditioning settings accordingly (Nest 2013). Nest also includes persuasive design and gamification elements (figure 23.3). Unlike most programmable thermostats (Peffer et al. 2011), Nest is aesthetically pleasing, easy to use, and engaging. It lights up when approached and illuminates in red or



Figure 23.3

The Nest smart thermostat (left) and its iPhone application (right). Nest uses virtual rewards in the form of an iconic Leaf to incentivize energy-efficient home heating and cooling behaviors.

blue colors depending on whether the home is heating or cooling. For gamification in particular, homeowners are awarded virtual "Leafs" for setting temperatures to energy-efficient levels. This mechanism was added to encourage user control over temperature adjustment after pilot testing found that some users responded negatively to aggressive automatic adjustments (Levy 2011). According to Yoky Matsuoka, Nest's vice president of technology, a majority of "Nesters" use the Leaf to guide temperature selection, and 92 percent check for a Leaf at least once a week (Matsuoka 2012). More impressively, 98.6 percent of households have an automatic schedule set, and of these households, the average energy savings is 19.5 percent with a maximum of 36.1 percent (Matsuoka 2012); however, unlike Opower, these claims have not been independently verified by third parties.

Eco-feedback Beyond Single-Family Residences

While the previous two examples focused on single-family homes, others have explored competitions as a means of raising awareness about environmental issues in shared residences and businesses. As Pierce, Odom, and Blevis (2008) articulate, such contexts are unique because the occupants themselves have little control over energy usage while third parties (e.g., building managers or owners) have a high level of control. In addition, these occupants typically receive little feedback about resource consumption and do not have the same incentives to conserve.

Starbucks, for example, has a pilot program to evaluate how employees respond to real-time energy feedback coupled with "friendly" competition. In a study across ten cafés, the top store saved 9 percent while the bottom increased usage by 2 percent over baseline (Russell 2012). University residence halls (dorms) have also engaged students in energy issues through competitions (Hodge 2010), with typical competitions lasting two to six weeks and eco-feedback provided over the web and/or on shared public displays. For example, a six-week study at Dartmouth combined narrative with a virtual avatar (a polar bear) to represent energy efficiency. As electricity use increased, the bear's happiness decreased and its livelihood became endangered (figure 23.4): a 10 percent reduction in energy use was achieved. These residence hall competitions generally resulted in lower resource consumption (Petersen et al. 2007; Loeb et al. 2010; Sintov, Desario, and Prescott 2010), although it is unclear if the reductions persisted after the relatively short competition period.

Explicitly Designed Home Resource Games

While the previous examples use some aspects of games (e.g., competition, progress tracking, rewards) to help motivate proenvironmental behavior, a few academics and businesses are exploring the use of sensed behavior as *actual input* to video games. Although a few of these games have targeted water use (Ravandi, Mok, and Chignell 2009), most have focused on energy (e.g., Mahmud et al. 2007; Gustafsson, Bång, and Svahn 2009; Kimura and Nakajima 2011; Geelan et al. 2012). As an example, Byron Reeves at Stanford along with his company Serioisity, Inc., created Power House (figure 23.5), a social game that



Figure 23.4

Loeb et al. (2010) used virtual animal characters that respond to real-world energy usage behavior in school buildings. In this case, when electricity use was low, the polar bear was happy, but as use increased, the bear's health and happiness became endangered.

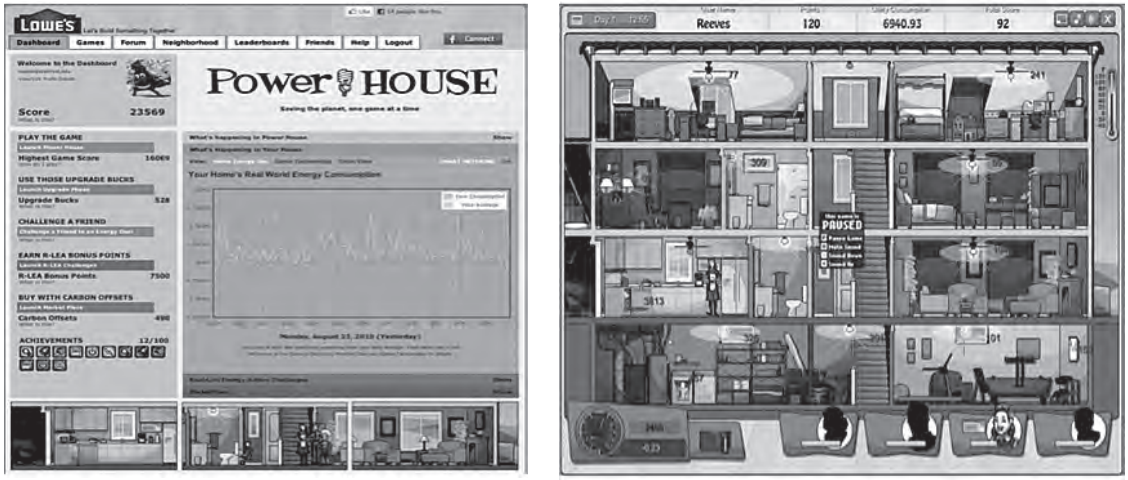


Figure 23.5

The Power House video game prototype uses smart meter data to influence automatically a player's in-game abilities and provide virtual rewards and achievements (Reeves et al. 2012).

uses smart meter data to influence a player's in-game abilities and achievements (Reeves et al. 2012). Periodically, play is interrupted and players are offered an opportunity to learn more about resource consumption and challenge other players to competitions.

The link between real-world sustainability behaviors and virtual worlds is being investigated in industry as well. The green gamification startup, Zema Good Inc., partners with energy utilities to pay consumers *virtual* currency in online games (e.g., Farmville) for verifiable energy savings in the real world. However, none of these gaming models—in academia or industry—have yet been shown to be effective. Both Zema Good Inc. and Power House are currently conducting private trials.

Transportation

For our second green gamification domain, we survey transportation. In 2003, the United States transportation sector accounted for approximately 27 percent of total U.S. greenhouse gas (GHG) emissions, an increase from 24.8 percent in 1990 (EPA 2006) Although airplanes are often considered significant carbon dioxide contributors, roughly 81 percent of all transportation-related GHG emissions in the United States comes from *on-road vehicles* including passenger cars, sport-utility vehicles, motorcycles, and heavy trucks and buses.[5]

Light-duty vehicles, which are used primarily for personal transport, account for 62 percent of overall transportation GHG emissions. While technological improvements continue to increase average fuel efficiency in vehicles, these gains are offset by increases in the total number of miles driven. Since 1980, that number has grown three times faster than the population in the United States (Ewing et al. 2007). In this section, I explore how eco-feedback technology and gamification have been used to promote and encourage environmentally sustainable transit choices (e.g., riding your bike to work versus driving your car) and to motivate economical/ecological driving behavior—so called eco-driving.

UbiGreen: Motivating and Gamifying Green Transportation Decisions

Given the growing prevalence of mobile phones with sensing capabilities, one compelling opportunity to impact human behavior is to offer immediate feedback about how currently sensed or predicted behaviors affect the environment. In 2008, my colleagues and I designed and evaluated a mobile phone–based prototype called UbiGreen, which sensed and fed back information about the user's "green" transportation behaviors through abstract, ambient visualizations (figure 23.6). Transit activities were sensed through a combination of self-report and automatic inference based on cell tower signals and a multimodal sensor platform worn on the user's waist.

The background (wallpaper) of the phone's screen provided feedback with small graphical rewards for taking "green" transportation such as riding the bus or train, walking, biking, or carpooling. Along with a representation of eco-friendly transportation, UbiGreen visualized other goals (e.g., saving money, getting exercise) depending on the sensed transit mode (row of icons in the screenshots of figure 23.6). We created two visual designs. In the first, a tree grows based on green transit activity. At the start of

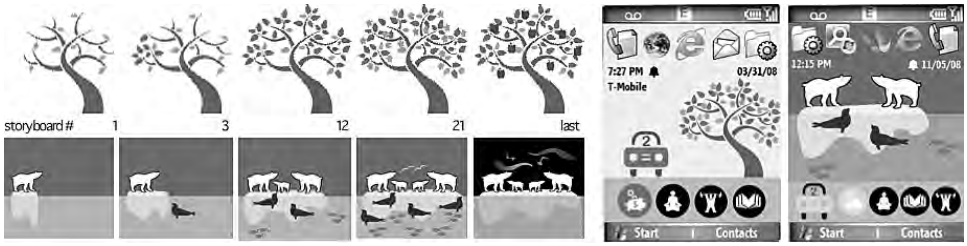


Figure 23.6

(Left) The tree and polar bear UbiGreen designs. (Right) Example screenshots of UbiGreen on the mobile phone—note how the visualization fits unobtrusively in the background of the phone's screen (as "live" wallpaper). For both screenshots, the current activity is carpooling indicated by a car icon with the "2" in the windshield.

each week, the tree is bare, then leaves, blossoms, and eventually apples appear as green transit events are sensed. In the other design, an Arctic ecosystem evolves with green transit to portray a more thriving ecosystem. Both designs follow a linear sequence of images, starting with an initial screen at the beginning of the week and ending with a final visual "reward screen," which is reached by taking green transit. Although UbiGreen includes game-like mechanics (e.g., the use of narrative, tracking progress, rewards, and stages), we did not explicitly conceive of the system as a game in our design meetings. Instead, we were inspired by the behavioral science literature related to feedback (Darby 2006), goals (Locke and Latham 2002), and self-monitoring (e.g., McCann and Bovbjerg 2009, 222).

We conducted a three-week qualitative pilot study of UbiGreen with thirteen participants to assess user reactions to the visual designs and the ways in which UbiGreen stimulated reflection and awareness about green transit. Most relevant to the theme of gamification, users enjoyed the *unfolding narrative* of the image sequences, which provoked anticipation and curiosity as green trips were taken; however, this interest waned over time. Some users suggested being able to download new stories or the use of nonlinear narratives to sustain engagement—it is unclear if this would be sufficient to actually change transportation behaviors.

Although we did not describe UbiGreen as a game to our participants, they still used game-like metaphors when talking about the prototype in poststudy interviews. For example, participants mentioned that engaging in green transit behaviors earned "points" and making it to the last screen was the "final level." One participant even complained that when a trip hadn't been automatically recorded, he was "being cheated out of points." Because so many participants conceptualized UbiGreen as a game, they considered opportunities to "cheat" the system to be problematic. For example, "I don't like incentives for getting points artificially by taking unnecessary trips . . . like trying to beat your own score by taking two more trips just to earn points" (participant no. 11). By virtue of UbiGreen's "reward structure"—where participants earned a new icon for each sensed *green* trip—the application could actually encourage people to take more trips simply to earn more points (some green trips could lead to more emissions than no trip at all). Future designs that

incorporate a more overt gaming model could mitigate these effects by rewarding "more points" for zero-carbon trips such as bicycling and walking and/or by tracking and encouraging progress from week to week.

Shared Bicycling

UbiGreen focused on exploring how activity inference and mobile phone visualizations could increase a person's awareness about his or her transportation habits in general. Here, we look at how persuasive techniques and gamification have been applied to a rapidly growing area of sustainable urban transit: bikeshare. A bicycle sharing system, or bikeshare, is a transit service where bicycles are shared among riders who do not own them, usually for a small fee. Advocates promote bikeshare as a convenient, eco-friendly, and low-cost form of public transportation that simultaneously reduces urban traffic congestion, noise, and air pollution.

The first contemporary bikeshare system began in Lyon, France, in 2005 (Velo'v) and has since spread to more than 130 cities worldwide on six continents including Paris (Vélib'), Montreal (Bixi), and London (Barclays Cycle Hire) (Meddin and DeMaio 2013). Though it is difficult to ascertain the net savings of bikeshare programs on carbon emissions, preliminary research suggests the programs offer a benefit despite the need for electronic bikeshare stations and other support infrastructure. In Paris alone, for example, an average of 110,000 Vélib' trips are made daily, and the number of cyclists in the city has nearly doubled in the past five years (while car-to-bike accidents have actually decreased) (Simons 2012). Preliminary findings also suggests that bike-sharing is displacing between 8 and 16 percent of vehicle trips (McKenzie-Mohr 2011, 123).

With modern bikeshare systems, geographically distributed bicycle docking stations are controlled by computer kiosks linked to a back-end server where bikeshare users can "check out" or "check in" bicycles using a membership identifier (e.g., RFID dongle or membership card) or credit card payment. These transactions create digital records of usage behaviors, which have been used by researchers to analyze and predict city mobility patterns (Froehlich, Neumann, and Oliver 2009) and by operators and application developers for eco-feedback and gamification applications. For example, to help promote bikeshare usage, Montreal's Bixi system publishes a leaderboard of the top riders in the city (the top rider in 2011 logged 1,265 trips). Washington, D.C.'s Capital Bikeshare (CaBi), currently the largest bikeshare system in the United States,[6] provides a "quantified-self" style web page for members to track distances, calories burned, and carbon dioxide emissions saved for each ride. It is unclear, however, what effect—if any—such systems have had on promoting adoption and usage.

Occasionally, system operators will run short-term contests to create publicity and incentivize particular usage patterns. CaBi's Winter Weather Warrior Contest, for example, was an attempt to combat expected drops in usage during the system's first winter in Washington, D.C. The contest included real prizes (e.g., free memberships, gift cards) along with titles such as "Most Saddle Time," "Longest Haul," and "Perfect Attendance." A rider could earn double points for riding in particularly bad weather—below freezing temperatures, in snow or cold rain. During the two-month contest, rides increased by 67 percent (Capitalbikeshare.com 2011). However, as is often the

case with incentivization structures, many riders appeared to "game the system." The overall winner, for example, took advantage of a "double points" day in the final week and earned 204 points by riding between all 104 active stations[7] (Jamieson 2011). Thus, while the Winter Weather Warrior Contest may have been useful for generating positive publicity, it was likely not as successful as the increase in rides implies. This story also reflects the difficulty in setting up an appropriate incentive structure in gamification to encourage particular behaviors. By introducing behavior-based rewards, there is a good chance that someone will try to "game" the game.

Another target of gamification by bikeshare operators is load balancing; that is, ensuring that bikes are adequately spread across stations. Uneven distribution of bikes arises due to commuting directions, major events in the city, or human biases (e.g., a preference for downhill rides). To counter this problem, cities use trucks to carry bicycles between stations, which is costly, creates carbon dioxide emissions, and increases traffic congestion (DeMaio 2009). Some bikeshare programs have thus introduced persuasive tactics to encourage more efficient bike movement across stations. As one example, the Paris bikeshare system Vélib' offers a "bonus" fifteen-minute-ride time credit for users who return a bicycle to one of ~100 designated uphill stations. According to DeMaio (2009), the program is a success and has resulted in increased use of these stations.

The future of bikeshare will likely involve mobile phone–based tools that will help users predict station capacities so that they can plan routes accordingly as well as persuasive systems—which may include gamification—to help make bikeshare more self-sustaining (e.g., by incentivizing load-balancing bike movements).

Eco-driving

The tools discussed earlier are useful to inform and support greener transit decisions, but what if a person has already decided to drive? Research has consistently found that fuel economy for a given vehicle can vary by 10 to 24 percent simply due to differences in driver behavior (Greene 1986; Ford 2008a; Gonder, Earleywine, and Sparks 2012). Economical driving behaviors, called eco-driving, include accelerating moderately, anticipating traffic flow and signals, driving at the speed limit, and eliminating excessive idling. According to recent estimates by Barkenbus (2010, 764), if one third of all U.S. drivers adopted eco-driving behaviors, 33 million metric tons of carbon dioxide would be saved per year (based on a per driver fuel reduction of 10 percent). How can gamification play a role here?

Although some vehicles manufactured in the early 1990s provided fuel efficiency information in their dashboards (e.g., average miles per gallon, distance to an empty tank), it was not until the release of the Toyota Prius hybrid that a mass-manufactured automobile provided instantaneous feedback on fuel efficiency through a visualization display. The original Prius, released in Japan in 1997 and worldwide in 2000, allowed drivers to compare their current fuel efficiency to a bar graph that displayed recent performance (six data points at five-minute intervals) and an overall mileage average based on the trip odometer (figure 23.7). This level of feedback was sufficient to not only improve driver fuel efficiency but, for some, to transform the driving experience into a game (e.g., Bedard 2005). For example, in an early review of the Prius, Fuller notes that "constantly watching the mileage measurements on the Prius's little video screen is really a

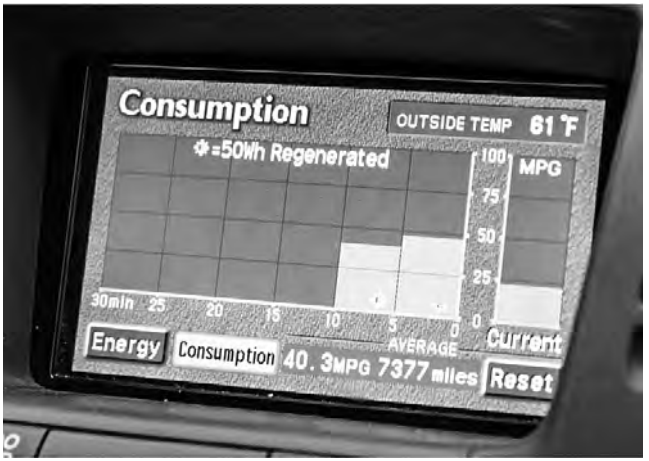


Figure 23.7

The original Toyota Prius eco-driving dashboard display, which includes instantaneous feedback on current fuel efficiency, a moving bar graph of average fuel efficiency in five-minute intervals for the past 30 minutes, the overall fuel efficiency since the trip odometer was last reset, and a "40Wh Regenerated" badge icon for regenerating energy via braking.

mobilized video game. It's NOT simply driving a car" (Fuller 2006, 1).

Qualitative research studies report similar findings (e.g., Kurani et al. 2010; Stillwater and Kurani 2011). In a six-week study of thirty-four households provided with Toyota Priuses,[8] researchers found that drivers attended to the instantaneous fuel economy feedback and reported changing their driving behavior as a result (Kurani et al. 2010). Many drivers also perceived the feedback display as a game and tried to maximize their fuel efficiency (Kurani et al. 2010, 77). More than that though, the researchers stated that it "was not uncommon" for the feedback information to stimulate a competitive game between multiple household drivers. One driver remarked, "Yeah, I looked at that, and so my husband would have a contest, okay, let's see who had the best mileage" (Kurani et al. 2010, 105). So, even though the Prius did not specifically offer features to support gaming or competitions (e.g., by allowing the driver to log in to identify himself or herself, by awarding badges/medals, or by specifically tracking points), these sorts of game-like phenomena arose nonetheless.

In contrast, more recent eco-driving dashboards were specifically designed with games (and gamification) in mind. For example, Steve Bishop of IDEO, the design firm that codesigned the Ford SmartGauge interface, remarked: "When we observed hybrid drivers, we found they were going for high scores, a gaming behavior that has never existed in cars before. We designed to accommodate it" (O'Dell 2009). Both the Nissan Leaf and the Ford interface provide dashboard visualizations of growing leaves and vines that track and reward driver efficiency (figure 23.8). Similar to UbiGreen, this abstract representation serves as both a visual indicator and a reward system for eco-friendly behaviors—the more efficient the driver, the more lush and beautiful the



Figure 23.8

(Left) The Nissan Leaf Carwings system ranks eco-driving performance across drivers in proximal geographic areas (as viewable in a web browser). Efficient drivers earn virtual rewards (trophies and medals). (Copyright © 2010 Nissan. Nissan, Nissan model names, and the Nissan logo are registered trademarks of Nissan.) (Right) The Ford SmartGauge with EcoGuide rewards eco-driving performance with an aesthetically pleasing vine that grows in proportion to fuel efficiency.

leaves and vines (e.g., see Ford 2008b). Others have noted that this "virtual creature" makes eco-driving similar to playing a Tamagotchi-style[9] game where real-world actions affect the vitality of some organism (Makower 2012). In addition, the Nissan Carwings system allows efficient drivers to earn virtual trophies depending on their leaderboard placement compared to other drivers. Notably, for new electric/hybrid cars that do not support leaderboard or social comparison functionality, organic, user-led applications have emerged—so there is a clear demand for this feature. Voltstats.net, for example, is a web application with no official affiliation to Chevrolet but provides leaderboard and achievement tracking to interested Volt drivers.

While the above points illustrate a movement toward more explicitly designed persuasive and gamified eco-driving elements, there is a surprising lack of research quantitatively studying their actual effect on driving behavior. Most studies including Jenness et al. (2009), Manser et al. (2010), and Stillwater and Kurani (2011, 2012) are qualitative relying on user self-report to study changes in eco-driving attitude and behavior or are conducted in short-term simulator environments rather than in the field (Van der Voort et al. 2001). I could find only one large-scale quantitative investigation of a modern eco-driving interface's effect on driving behavior: a corporate report published by Fiat (2010) analyzing real driver data uploaded from Fiat's eco:Drive platform. In analyzing more than 400,000 journeys by 5,697 drivers in 5 countries, Fiat found that drivers reduced their fuel consumption by an average of 6 percent with eco:Drive (the top 10 percent reduced fuel consumption by an average of 16 percent). Fiat also argues in their report that this study provides "evidence that eco-driving behavior is maintained over time, with improvements growing steadily"; however, the study window was thirty days, which seems insufficient to make this claim.



Figure 23.9
(Left) The Infotropism eco-feedback system controls the growth of a live plant to provide feedback about the more active container—trash or recycling (Holstius et al. 2004). (Right) The Bottle Bank Arcade by Volkswagen was part of its "Fun Theory" series of viral videos to show how fun could be used to change people's thoughts and behaviors, though the video shows only anecdotal evidence (Volkswagen 2009b).

Waste Disposal and Recycling

Finally, for our last environmental domain, I survey waste disposal and recycling. In 2007, the United States generated approximately 254 million tons of municipal solid waste—about 4.6 pounds per person per day, and an increase of almost two pounds daily per person since 1960 (EPA 2007). The average American recycles about 1.5 pounds per day (~33 percent); however, the rest ends up in landfills (54 percent) or is combusted for energy (13 percent) (EPA 2007). To increase recycling activity, gamification approaches have ranged from exploring persuasive recycling bins to citywide deployments of gamified recycling programs, most notably Recyclebank.

Persuasive Recycling Bins

HCI researchers and designers have explored a range of persuasive technology for waste. One example is Infotropism, a live, organic eco-feedback system for waste (Holstius et al. 2004). Placed between pairs of trash and recycling containers, directional bursts of light are trigged by trash/recycling deposits, which gradually induces a living plant display to lean toward the more active container (figure 23.9). A preliminary two-week study, however, found negligible effects on recycling activity.

Other projects have used visualizations of trash and recycle bin contents to impact behavior or inform passersby. BinCam captures pictures of discarded refuse and uploads them to Facebook (Thieme et al. 2012). However, studies of BinCam and a household recycling competition based on it have been exploratory, examining usability, notions of surveillance, and user reactions to the system rather than behavior change. Another prototype, Jetsam, projects a time-lapse visualization of a bin's contents outward (e.g., on a sidewalk) to cause passersby to pause and reflect about trash in urban life (Paulos

and Jenkins 2006). Finally, Volkswagen developed a series of viral marketing videos on fun and persuasive technology, two of which were related to waste: *The World's Deepest Bin* (Volkswagen 2009a) and *Bottle Bank Arcade Machine* (Volkswagen 2009b). While neither bin design was studied in a rigorous, scientific manner, the videos were successful as viral marketing videos (figure 23.9). At the time of writing, the two videos have accrued more than 6.5 million views, providing at least anecdotal evidence for interest in the use of fun to motivate routine activities.

Recyclebank

In comparison to the exploratory bin prototypes and studies described earlier, Recyclebank, founded in 2004, is the first and perhaps the most successful wide deployment of waste-related eco-feedback and gamification. Recyclebank tracks household recycling behavior by issuing special curbside recycling bins with a radio frequency identifier (RFID) tag that allows bins to be automatically identified and weighed when they are picked up by waste management trucks. Feedback about recycling volume over time is then provided to households via a website. Households also earn points for recycling, which can be redeemed for products and services from partnering companies (e.g., Walmart, IKEA).

Recyclebank makes money via contracts with municipalities, who save money when waste is diverted from landfills or incineration centers (Lui 2011). The company reports being active in more than three hundred communities across the United States and the United Kingdom, serving more than three million customers who, on average, save more than $130 through their rewards program (Recyclebank 2011). More importantly, from an environmental perspective, the company reports increasing recycling activity by 14 to 155 percent in its partnering cities (Jones 2008; Recyclebank 2011).

Not all municipalities have been happy with Recyclebank, however, often because of a failure to meet performance expectations—leading some cities to drop the program altogether (Davlin 2012; Thomas 2012). In an online survey investigating perceptions and usage of Recyclebank in Ann Arbor, Michigan, the city found that only 17 percent of the 1,012 respondents felt that the Recyclebank Rewards program encouraged them to recycle more, and 63 percent reported not using any of their Recylebank points primarily because none of the rewards were appealing (A2Gov 2011).

Discussion

The previous sections illustrate the use of persuasive technology and green gamification across a range of environmental domains, and in this section we discuss contentious issues and open research questions.

Ethics

Climate change and other environmental ills are ethical dilemmas just as they are technological, economic, political, behavioral, and social dilemmas. As such, any proper discussion of technology designed

to inform or influence environmental behavior must also include a discussion of ethics. Fogg (2003) dedicates a full chapter to the topic in his book. The ethics of persuasive systems—gamification included—becomes even more complicated in areas like public health and environmental sustainability where societal benefits must be weighed against personal costs (e.g., encroachment on individual rights).

Is it ethical for a persuasive technology to deceive individual users for the greater "environmental good?" Imagine a green gamification system that presents falsified data unbeknownst to households but results in 15 percent energy savings and eliminates the need for the neighborhood coal-powered electricity plant. Is this gamification design ethical? Where does one draw the line? Many of the new emerging business models around green gamification require that companies show net reductions using their systems (e.g., Recyclebank and Opower). This creates an attractive but dangerous incentive for companies to adopt gamification strategies in dubious ways to achieve results.

McGonigal provides a rather simple ethical test of a game or gamification system: "If you use the power of games to give people an opportunity to do something they want to do, then you're doing good. If you're using the power of games to get people to do something you want them to do, then you're doing evil" (McGonigal 2011). However, even these high-level qualifications seem insufficient; for example, game players may want to spend a small amount of time playing World of Warcraft but end up unable to break themselves away from the addictive game play (which was largely designed to be that way). Could gamification, in some contexts, be similarly powerful?

Though space prevents us from discussing these and other ethical considerations in greater detail, they are clearly important and should be a center point of any green gamification design process (see also Davis 2012 and the two ethics-related gamification chapters in this volume).

Framing and Behavior Targets

In the same way that research has explored how environmental issues are communicated in public debates (e.g., Macnaghten 1993) and by the mass media (e.g., in television, radio, billboards; see, e.g., Sampei and Aoyagi-Usui 2009) so too must we consider how environmental problems are framed by persuasive technology and gamification systems. As Guttman notes, "an issue becomes a social problem only after it is defined as such by certain stakeholders and the way the problem is framed by them tends to reflect their values, priorities, or ideologies" (Guttman 1997, 98). When gamification designers target a particular environmental problem, they are performing an act of prioritization and judgment. Similarly, their framing of the problem and solution has important implications for actually making a positive impact.

At one end of the spectrum, green gamification is cast as a solution for transforming "ambivalent consumers into eager eco-evangelists" (Aston 2012). The aforementioned green gamification start-up, Zema Good Inc., for example, partners with energy utilities to pay consumers *virtual* currency in online games (e.g., Farmville) for verifiable energy savings in the real world.[10] Cofounder Eric Senunas has touted his company's approach by stating that "You will become [energy] efficient in spite of yourself because you want that farm cash" (Senunas 2012). Senunas's framing disregards proenvironmental attitudinal change completely in favor of the sheer motivational power of rewards on behavior (i.e., you will act in

proenvironmental ways even if your beliefs about the environment do not change). These quotes represent, perhaps, both the vast appeal of gamification and the willingness of many to imbue it with magic-like powers to "save the world."

From a more critical perspective, the quotes from Aston and Senunas also represent rather cynical beliefs and shallow interpretations about what motivates human behavior and the role of gamification therein. The two quotes are concerning not only because of the power they seem to bestow on gamification—a power that has not yet been earned nor shown to be longitudinally viable—but also because of how easy they seem to make attaining environmental sustainability.

This narrow framing of environmental issues is not endemic to persuasive technology. Gardner and Stern (2002) question whether environmental psychologists have targeted the most important environmental problems or simply the most convenient ones. Take recycling, for example, where Gardner and Stern (2002) point out that most behavioral solutions focus on the end of the waste stream—littering and recycling—rather than reducing consumption in the first place, which would lead to the greatest benefits. With this perspective, Recyclebank's approach, though successful in increasing recycling, seems misdirected—indeed, one could argue that its incentive structure actually increases consumption by providing rewards for overall volume or, at least, is set up to support and reinforce a culture of consumption.

From Individual Action to Broad Societal Change

Persuasive technology has largely focused on individual behaviors (e.g., eco-driving, energy use in the home), but there is doubt over whether this focus will lead to broad societal change (DiSalvo, Sengers, and Brynjarsdóttir 2010; Shove 2010; Brynjarsdóttir et al. 2012). John Thøgersen, for example, argues that campaigns for behavior change with small environmental impacts (e.g., turn off the lights) only make sense if they lead to far-reaching impact in the long run (Thøgersen 2011).

Shove goes further contending that we need to consider not just the individual behavior but the context within which that behavior takes place, including institutional, historical, and societal structures, if we are to combat climate change sufficiently (Shove 2010, 1274). Similarly, Stern points out that motivating manufacturers to adopt "greener" production technologies and product designs, bankers and developers to consider environmental criteria in their decisions, and operational policies for plants and commercial buildings would likely have more impact than motivating individual proenvironmental action (Stern 2000, 410).

It is important then to recognize that green gamification is far from a panacea and must be surrounded by social, structural, and regulatory policy to be effective.

Intrinsic versus Extrinsic Motivation

Motivation—the reason or reasons for acting or behaving in a particular way—is one of the most central and perennial topics of inquiry in psychology (Ryan and Deci 2000b). The issue of motivation, where it comes from, how it works, and how it can be used is also an important, often debated topic within persuasive technology, gamification, and environmental psychology communities. Typically, these debates center on the role of *intrinsic motivation,* which refers to doing something because it is inherently

interesting or enjoyable, and *extrinsic motivation*, which refers to doing something for an outcome or reward (Ryan and Deci 2000a). In short, intrinsic motivation comes from within the individual, and extrinsic motivation comes from outside.

Researchers and game scholars argue that the choice and act of playing video games is intrinsically motivating. People play games because they enjoy them—games are fun, gratifying, and pleasurable (Koster 2004, 40; Ryan, Rigby, and Przybylski 2006; Deterding 2010). As Ben Sawyer notes, "play is its own reward" (Sawyer 2010). Gamification, however, largely focuses on *extrinsic* motivation by adding a layer of externally driven incentives and rewards (e.g., leaderboards, points, virtual trophies) on top of typically otherwise mundane tasks (e.g., recycling or running).

This shift from intrinsic to extrinsic is not insignificant and has particular relevancy in the environmental domain. Extrinsically motivated behaviors are often more brittle, they tend to last only as long as the reward is provided (Deci, Koestner, and Ryan 1999), and the rewards themselves must increase in magnitude to trigger the same experience of pleasure and satisfaction (i.e., hedonic adaptation, see Frederick and Loewenstein 1999). Perhaps even more concerning, the very presence of these external rewards can undermine intrinsic motivation (Kohn 1999). This motivational subversion is particularly problematic in the domain of environmental sustainability where, ideally, people would be *intrinsically* motivated to partake in proenvironmental actions long enough until they become routine. In this sense, green gamification does not seem like a promising solution; instead, it feels like a short-term fix.

De Young (1996) argues that it is not necessary to reward or coerce citizens to behave proenvironmentally but instead that approaches should highlight how a conservation-oriented lifestyle is intrinsically satisfying. De Young quotes Seligman and colleagues, who suggest that "individuals can find satisfaction in many of the ways they practice conservation . . . they [become] interested in and challenged by the task of lowering their energy consumption, and [feel] satisfaction not previously present when they did so" (Seligman, Becker, and Darley 1981, 111). A key question moving forward, then, is how can green gamification be used to nurture feelings of intrinsic satisfaction (e.g., by, perhaps, emphasizing the local and global importance of a person's proenvironmental action and the satisfaction one feels for this accomplishment)? Is this possible? If extrinsic motivation is the impetus for initial proenvironmental interest or behavior change, can these extrinsic motivators become internalized?

Surveillance

Dourish argues that technologies designed to monitor and record actions, particularly in the realm of environmental sustainability, are a "natural path for various forms of surveillance and regulation" (Dourish 2010, 5). Even something as benign as a water meter can be used to ensure that householders comply with water-use restrictions (Halich and Stephenson 2009). As behavioral sensing begins pervading nearly all aspects of our lives and finds use in gamification and persuasive technology applications, there is greater opportunity for reappropriation and repurposement—some of which, undoubtedly, will be nefarious.

The difficulty, however, is that it is usually not clear from the onset how a sensing system could be reappropriated and for what purpose. As a rather

simple example, consider the vehicle speedometer, which was first produced to satisfy curiosity about speed, but ultimately became a tool for regulation (Bud and Winter 1997, 567). With the movement toward ubiquitous sensing, similar legislation may emerge for activities that we could not currently imagine (e.g., fines for not driving ecologically, mandates for recycling amounts).

The concern is not simply governmental regulation but also new ways in which corporations may learn about and exploit employee behavior or create new business models that reward a particular way of being. For example, Barkenbus (2010) discusses Progressive Insurance's Snapshot device, which, once installed, allows the company to charge a variable insurance rate based on an individual's safe driving habits. According to Progressive's website, Snapshot monitors driving behaviors not unlike the systems mentioned in the "Eco-driving" subsection—"the better you drive, the more you can save"—up to 30 percent for the most conscientious drivers (Progressive 2013). How far are we from similar insurance programs for our health based on measured activity levels (e.g., from the Fitbit) or exposure to pollutants in the air? Clearly, this topic has serious implications for legislative policies around self-monitoring (be it for gamification purposes or otherwise). It is yet unclear how new business models may emerge around utilizing our pervasively sensed activity data, but I have no doubt that they will emerge.

User Agency

A final controversial aspect of persuasive technologies—gamification included—is whether they reduce a user's agency and sense of self by prescribing particular ways of being—ways that are defined or, at least, constrained by the designers, not by the users (e.g., Purpura et al. 2011). An alternative view is that persuasive technology empowers users by offering support, motivation, and encouragement toward a shared behavioral goal (i.e., a goal shared between the user and the designer). The degree to which either of these views holds depends on the context in which the technology is used (e.g., is it adopted and used by free choice or mandated by a government, doctor, or employer) and how it is designed (e.g., can the user self-set goals, configure motivational strategies, help define "success"). Regardless, designers must be aware that their particular interventions may not fit or benefit all people; for example, an application that rewards water-usage efficiency in the home may not adequately account for consumption based on medical or religious reasons (see Froehlich 2011, 137–141).

Conclusion

We are at the precipice of a new era of computing where small sensors, sophisticated machine learning algorithms, and pervasive computation will allow for monitoring and tracking a breadth of human activity—all of which can be gamified. Some have cast this future with a dystopian twist where no action is performed without some superficial reward (Schell 2010)—from brushing teeth to accomplishing daily chores. My perspective, however, is more balanced. As a computer scientist,

I believe that computational sensing will continue to abound and that these advances will enable new ways of thinking about and understanding human behavior as never before. As digital displays become even more pervasive—in jewelry, glasses, or even under our skin—there will be increased opportunity for sensing and feeding back a variety of signals, from our physiologic well-being to the environmental impact of our actions. However, rather than a source of disillusionment and harm, I see this as a source of empowerment, of previously unrecoverable knowledge that can be used as a basis for good, particularly with regard to personal health and environmental sustainability. To make a gaming analogy, we are at the Atari 2600 stage of persuasive technology and gamification—even the 8-bit NES is but in the distant future—there is much to learn, much to play with, and much to overcome. Though gamification is at the peak of its hype cycle, the ideas and principles it represents will live on in one form or another and, I think, will be a source of positive change.

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Notes

1. It is not always clear what, exactly, constitutes proenvironmental behavior. Here, we borrow from Stern (2000) and Steg and Vlek (2009), who define *environmental behavior* as any behavior that changes the availability of materials or energy from the environment or alters the structure and dynamics of ecosystems or the biosphere and *proenvironmental behavior* as any behavior that harms the environment *as little as possible*.

2. Environment here refers both to the natural and built environments.

3. A descriptive social norm describes what is typical or normal behavior in a particular context: "If everyone is doing it, it must be a sensible thing to do" (Cialdini, Reno, and Kallgren 1990).

4. A smart meter is a digital meter that measures and records usage in near real-time and allows for two-way communication between the utility and customer. In 2012, the Institute for Electric Efficiency found that nearly 33 percent of U.S. households already have an electricity smart meter and that ~65 million will be deployed by 2015 (IEE 2012). Smart meter penetration rates for water and gas are much lower.

5. The difference is scale. There are far more road vehicles and vehicle-based trips than there are airplanes and flights. With that said, a medium-sized car produces roughly 3.5 tons of carbon dioxide emissions a year (based on 9,608 driving miles), which is approximately the same amount produced *per passenger* on a round-trip flight between Chicago and

Frankfurt (Rosenthal 2013). Also, air travel volume is projected to rise significantly in the near future.

6. New York's bikeshare program, Citi Bike, will open in May 2013 with 5,500 bicycles and 300 stations and quickly grow to 10,000 bicycles and 600 stations (see http://citibikenyc.com/).

7. He should have earned 208 but one ride was shorter than five minutes and the other was missed due to an electronic malfunction.

8. Note: These were 2007 and 2008 model Toyota Priuses but modified to be plug-in electric hybrids. However, the researchers did not change the dashboard visualizations (they used "stock" versions). The feedback interface in the 2007/2008 Prius models was largely unchanged from the original Prius.

9. Tamagotchi is a handheld game in the form of a virtual pet simulation. The Tamagotchi creature goes through several stages of growth and develops differently depending on the care that the player provides—better care produces an adult creature that is smarter, happier, and demands less attention. As of 2010, more than 76 million Tamagotchis have been sold worldwide (Bandai 2011).

10. Experienced readers may see similarities between Zema Good Inc. and Kevan Davis' Chorewars, a "chore management system" to help track and encourage housework by earning experience points, power-ups, and other rewards. See http://www.chorewars.com/ or chapter 7 of Jane McGonigal's *Reality Is Broken* (McGonigal 2011).

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24 GAMIFICATION AND HEALTH

Sean A. Munson, Erika Poole, Daniel B. Perry, and Tamara Peyton

In the early twentieth century, infectious diseases were responsible for 30 percent of mortalities in the United States (Centers for Disease Control and Prevention 1999). Thanks to advances in vaccinations, antibiotics, and understanding of the importance of hygiene, people now live longer and healthier lives, with many infectious diseases completely eradicated. Although people are healthier overall, the products of industrialization have caused an entirely different health crisis, correlated to increased dependence on automobiles, reduced physical activity, and a widespread availability of energy-dense foods. With diabetes, cardiovascular disease, and chronic illnesses increasingly common, the prevention and management of chronic conditions presents an enormous health challenge (World Health Organization 2000; Pruitt 2005).

Many consumers and health care providers are turning to software tools to mitigate or solve challenges. As sensors gain greater ubiquity, people can increasingly benefit from tools that help them track, organize, and make sense of their health data. Technologies allowing for health tracking and monitoring may be one of the biggest revolutions in twenty-first-century health care, so long as designers are able to create appealing and effective experiences for users. This is a nontrivial challenge. Although interest is growing in the *quantified-self* movement, which focuses on self-tracking, data analysis, and reflection, personal health systems face barriers to widespread adoption. Grabbing and holding the attention of people who are not health concerned or curious about their own data proves difficult when attempting to encourage use of health tracking technologies. Even people who are already health-interested may struggle with information overload and have trouble collating and making meaning out of data collected about their bodies and health.

To encourage individuals to invest in their personal well-being and to overcome challenges related to interpreting rich and sometimes complex data, some people propose turning tracking and improving one's health into a "game" or adding game-like elements to existing health tracking technologies. Games and gamified systems offer to turn the wash of personal health information into an experience that is meaningful and motivating and to reframe daunting health problems into challenges that are enjoyable to solve.

The potential of health games has not gone unnoticed by researchers, practitioners, and consumers. There are now more than 13,000 consumer health applications available (Dolan 2012). Yet choosing whether to turn a health tracking application into a

game or game-like experience is not always a straightforward decision. Health and wellness information is personal and sensitive data; how such information is collected, stored, analyzed, shared, and displayed can be consequential to individuals who use it for their health decisions or behaviors. The design of consumer health applications presents several design challenges and raises questions about values. How and when should caring for health be something fun? Is making health management into a "game" inappropriate, essentially bribing people to do what they should already do? Because the game designer can craft experiences that reward certain activities, does gamifying health take away an individual's agency with respect to his or her wellness?

In this chapter, we take a critical look at *gamifying* health and well-being. We start with a brief review of behavior change models and discuss important factors for whether and how people can adopt and maintain good health behaviors. Drawing on those factors, we analyze the philosophies behind personal informatics and gamification approaches to encouraging behavior change, and we present some of the advantages and drawbacks of each approach.

Using several case studies, we provide insights into the design space for health games and gamification. We discuss the potential mechanics and attributes of games that shift people's health behaviors and attitudes. Using case studies of exertion/exergames, compliance-oriented games, and educational games, we illustrate some of the choices associated with the broad design space of games and gamified personal informatics tools. We discuss how and why these types of games help achieve health goals and how they differ from a personal informatics approach. We then reflect on critical design and adoption considerations that face designers, medical practitioners, and individuals who are deciding whether a game is an appropriate method for creating, displaying or organizing health information.

Behavior Change

Researchers from many disciplines have studied factors that affect whether someone will adopt and maintain a behavior. In the domain of public health, studies and interventions have focused on models related to cognitive, belief, and intention-based explanations. Fields such as social psychology emphasize habit, emotion, cognition, and neurologic explanations in their models. Both perspectives can help designers understand why existing interventions or gamified strategies work or how they might design new ones. In this chapter, we use these models to discuss which metaphorical levers various interventions might pull, and why and when pulling those levers may or may not be a good idea.

Health Behavior Change Theories

Public health literature offers several models that describe how people make decisions about health behaviors. We highlight two popular health behavior change models, *theory of planned behavior* (Ajzen 1991) and the *health belief model* (Rosenstock 1966; Janz and Becker 1984), and then discuss common elements of these models.

The theory of planned behavior predicts an individual's intended behavior based on several factors, including an individual's attitudes about the behavior, normative beliefs about the behavior, and how the individual weighs his own attitudes against

subjective norms. This intention, moderated by an individual's perceived control over his behavior, predicts his actual behavior. For example, an individual's intent to eat healthier could be influenced by how harmful he believes his current eating behaviors to be relative to the benefits associated with a suggested diet, the food choices modeled by peers and celebrities, and peers' attitudes toward his own food choices. Whether his intent to make healthier food choices translates into a healthier diet would depend on his control over his food choices: are healthy food choices available, and does he believe in his ability to access healthy food?

The health belief model uses individuals' assessments of health threat and target behavior to predict the likelihood of individuals taking a preventative health action, such as screening for a disease. Assessed factors include cues to action, perceived seriousness and susceptibility, cost-benefit analysis associated with the behavior, and the perception of potential barriers. For example, an individual's decision to get a colonoscopy can be influenced by her perceptions of the costs and discomforts associated with the screening, beliefs about the likelihood or severity of colon cancer, belief that a colonoscopy will lead to a better health outcome, and cues to action, such as a celebrity publicly choosing to receive a colonoscopy (the so-called Katie Couric effect; Cram et al. 2003).

Each health behavior change model may be relevant to different health behavior changes. The health belief model is used more commonly to understand one-time behaviors such as screening or immunization, while the theory of planned behavior better models decisions about lifestyle changes. Though the models differ in their elements and applications, they share some common predictors that designers of an intervention might seek to influence:

Perceptions About Susceptibility and Severity

When people believe that their current behavior increases their risk for negative health outcomes, they are more likely to try to change their behavior.

Perceptions About the Costs and Benefits of the New Behavior, Including Individual and Social Perceptions

- *Perceived benefits* Does an individual believe that the target behavior is likely to result in health improvements? For example, do individuals believe that medication is effective at controlling or treating an illness? Or do they believe that increasing physical activity will help them reach their target weight or that reaching this weight will have benefits?
- *Perceived barriers* In terms of time, finances, or negative side effects, do individuals believe that a new behavior would be too expensive? Is the infrastructure to achieve the target behavior available, such as stores that sell fresh food or health facilities with screening equipment? Do people perceive a medication's side effects as worse than the illness it seeks to control, prevent, or treat?
- *Perceived norms* How do people consider whether their current behavior and the target behavior are consistent with what others do and expect? Information that a behavior is prevalent, which is a descriptive norm, can be taken as social proof that it is the right behavior. Also, when an individual particularly likes or feels similar to others, he or she will often try to conform to their behaviors and attitudes (Cialdini 2009).

Perceived Control and Self-Efficacy

Individuals' beliefs in whether they are competent to adopt the target behavior influences both behavior change intent and the likelihood of follow-through (Bandura 1977; Schwarzer and Fuchs 1995). For example, people's belief that they are able to follow an exercise regimen can be predictive of whether they actually will do so, even after controlling for factors affecting their actual ability to follow the regimen.

Systems that help people start, change, or maintain a health-related behavior target one or more of these perceptions. Designing according to these models may not be sufficient, though. Reviews of studies that explain individuals' health decisions using behavioral science models show a gap between individuals' intentions and behavior (e.g., Sutton 1998). While the models can predict both intention and actual behavior, they are far better predictors for what people plan to do than for what people actually do. Some of this gap can be attributed to poor measurement of intention or other variables in the explanatory models, but there are likely additional key factors, such as habit, that affect behaviors but are not included in these models (Norman and Conner 2006; Danner et al. 2008).

Behavior Change and Habit

In contrast to public health research, industry dialogue and persuasive technology research have focused more on habit and its neurologic explanations. Sociocognitive models of behavior best predict infrequently performed behaviors, but past behavior is a better predictor for more frequent behaviors (Verplanken 2010). Behaviors become habitual when people's prior behavior shapes their future behavior without requiring them to appraise the prior behavior cognitively (Verplanken 2009). This habitual behavior consists of three "pillars": frequency, automaticity, and contextual cuing (Orbell and Verplanken 2010; Verplanken 2010).

In persuasive technology, Fogg's behavior model (Fogg 2009) includes motivation and ability, two elements corresponding to the public health models. Fogg's behavior model also emphasizes the importance of triggers to engage in the target behavior. Interventions can be designed to add cues for a desired behavior or disrupt or remove cues that trigger an undesirable, habitual behavior (Orbell and Verplanken 2010). Other interventions might rely less on explicit triggers, drawing instead on knowledge about general human biases in behavior. In a classic example, rearranging a cafeteria so that healthy food is more accessible than unhealthy snacks or desserts can lead people to make better food choices (Thaler and Sunstein 2008; Hanks et al. 2012). This "nudge" relies on individuals' biases toward the more convenient choice.

Personal Informatics, Games, and Gamification

Designers can borrow from models and theories about cognition and about human habit and behavior to influence the systems they design. There are three main approaches to the design of user interfaces for

health tracking applications: personal informatics, gamified interfaces, and games. Designers of health interventions must decide whether their system will take more of a personal informatics approach or more of a game-based approach. If designers choose a personal informatics approach, do they gamify it by borrowing elements of games and grafting them onto the application's design? Each approach can increase the ease of self-monitoring by removing or reducing barriers to data collection and analysis, but each design philosophy differs with respect to interpreting these data and changing behaviors.

Personal Informatics

Personal informatics is a growing area of research and practice focused on tools for collecting and interpreting information about one's self (Li 2011; Li et al. 2010). Some examples of personal informatics tools include step counters, sleep trackers, financial monitoring tools such as Mint, and diabetes management tools like Sweet Spot. Each of these tools promotes the collection and analysis of information about one's own behaviors, and many facilitate sharing or comparing that behavior with a target behavior or peers' behaviors. Through this process, a personal informatics application can support behavior change. For example:

- Data from a device that monitors physical activity may help someone to realize that she is actually far less active than she previously believed.
- A food diary might raise an individual's perceived risk of negative health outcomes associated with his current diet. It might help someone realize, for example, that a daily latte puts him 10 percent over a daily calorie goal. After making this realization, he can work to reduce his latte drinking.
- A historical chart indicating successful or unsuccessful medication adherence can help someone recognize and act on barriers. The barriers may be related to a realization that the individual often misses taking medication at work. This realization may spur the individual to add a daily, automatic reminder to a calendar or mobile reminder application.
- Historical step-count data might highlight previous walking-goal achievements and thus increase someone's self-efficacy with respect to physical activity.

Personal informatics applications can also support making good choices in the moment: Someone who is more aware of his calorie consumption for the day so far can make better choices about what to eat. By providing people with better information about their activities and decisions and their associated consequences, personal informatics can help people to make informed decisions about their behavior and goals and to track progress toward the goals, while still supporting individual autonomy in health decisions.

For a personal informatics application to promote behavior change successfully, an individual must believe that there are negative consequences associated with not altering his behavior. The individual also must believe that he can change his behavior and that the target behavior will have positive health outcomes that outweigh any costs or side effects. Consequently, there is some skepticism around whether a quantified-self approach will lead to successful behavior change across a broad population. For a personal informatics application to support change, individuals must possess or develop self-motivation, beliefs that the behavior change will be

beneficial, and confidence in their ability to change their behavior. Critics argue that quantified-self tools and strategies target a narrow group of people who are highly educated, health-literate, wealthy, skilled, and self-motivated (e.g., Cummings 2012). When individuals do not already have the right beliefs, relevant health education, or motivation, a different approach may be beneficial. This may be one that addresses barriers or that somehow circumvents the need to change beliefs. In some cases, the alternate approach may be a game or gamification.

Games

Games can trigger healthy behaviors. Whether it is an engaging fantasy, a contrived challenge, or a game that provokes curiosity, games for health can help make an activity fun (Malone 1981) by creating cognitively and psychologically appealing experiences around health behaviors or health learning activities. Substituting a game with health benefits, such as Dance Dance Revolution, for another game can also take advantage of existing habits by linking a high probability behavior (playing a game) to a low probability behavior (exercise) (Premack 1965). Games have been used to increase motivation for and knowledge about good health behaviors, perceived health efficacy, and individuals' willingness to engage their social support network on health concerns, such as smoking, diabetes management, and asthma management (Lieberman 1997).

To illustrate the power of reframing a health challenge as a game, consider two important predictors for whether people will adopt a new health behavior: their beliefs about failure and their perception of their chances for success. Framing a challenge as a game can make failure fun and initially expected (McGonigal 2011). A game can also help break a challenge down so that people perceive better odds of eventually succeeding (McGonigal 2011). Framed this way, the challenges and barriers associated with good health behaviors can be motivating. Early failures are okay and are an expected part of the process, rather than a sign that one should quit. Designed to be fun or spectacular, a failure can still reward players for trying. Quick failures in a game environment also offer a much faster feedback cycle than many health behaviors and outcomes. A lost game level can be tried again and perhaps won, while improvements in weight or health outcomes can take much longer to become noticeable. A game that is well designed and well chosen will match challenges to a player's actual and perceived skills. The player may not succeed at first, but progress will be made visible, and the player will eventually triumph at successive challenges. Rapid feedback can reinforce or enhance a players' self-efficacy and can thus increase likelihood of continuing a behavior, even without seeing immediate health benefits (Klimmt and Hartmann 2012).

Players of the mobile application Zombies, Run! participate in missions in which they collect supplies, gather intelligence, and flee from zombies by running. The application presents an audio narrative to the player and uses the phone's sensors to monitor the player's progress. If players do not run fast enough, they can lose their supplies or be caught by zombies. If they do not run long enough, they fail the mission. However, they can try again and they may win. Combined with immediate feedback, this sort of framing can completely change how people see barriers or how they perceive their self-efficacy. After all, it's a game, and so they are supposed to be able to "win"—but not easily.

In addition to reframing challenges, games can add incentives, such as participating in an engaging narrative or earning status, beyond those associated with the target behavior. For some people, living longer or feeling healthier may not actually be sufficient motivation to get them off the couch, whereas being part of an immersive, well-written story in which they help a town fight off the zombie apocalypse may provide the missing cognitive and emotional impetus (Malone 1981).

As Aaron Price (2012) wrote in his review of Zombies, Run!:

> Perhaps the most remarkable thing about this little running app is that I'm emotionally invested in it, almost entirely because of the quality of the acting and writing. I'm the star of this radio play, and every time I hop on the treadmill it's a new episode featuring ME, Runner 5. Running still sucks, but it sucks less with Zombies, Run!

Some games present individuals and groups with the opportunity to display status, which can offer additional motivation for some people. Players might earn badges or other rewards or they may share achievements within the game or on social network sites. They might see where they rank against friends and other players. For competitive individuals, status incentives offer rewarding (or costly!) feedback by attaching their reputation to a health activity. For individuals who prefer cooperative challenges, a collaborative game can offer a social experience or help people feel like a valuable member of a team.

By layering additional incentives over health goals and behaviors, games change the balance of costs and benefits for someone considering a health or wellness activity. Rather than convincing the individual of the health benefits of running or of taking a medication, a game designer can introduce other appealing benefits, though adding these incentives also carries risks. Whether the incentives are trophies, badges, or status, once the system includes and promotes the importance of rewards that are not directly associated with the target behavior, people may adopt behaviors that help them achieve those rewards but not the intended health outcomes. Sunny Consolvo, the researcher who led the design and evaluation of physical activity apps Houston and UbiFit (Consolvo et al. 2006, 2008), describes one such unintended outcome. Once users of a system become focused on beating their friends in the race for the most weekly steps or on earning flowers for the garden on their phones, they may give up on physical activity and instead tie the pedometer to their cat!

Such cheating is an example of what Lepper, Greene, and Nisbett (1973) term the "over justification effect." Adding prizes may increase someone's overall motivation while prizes are available but may ultimately decrease the individual's intrinsic motivation, leading to a reduced overall motivation once the prizes are withdrawn. Once additional rewards and incentives are introduced into an activity, the supporting mechanism may need to be permanently available. These incentives also can cause people to engage in alternative behaviors (e.g., cheating) that will help them achieve the extrinsic goals (the prize) even if their behavior does not support the intrinsically motivated goal (being healthier).

If a game substitutes status or other rewards for positive health outcomes as motivators, is the game actually encouraging people to take responsibility for their health? Or is the game merely enabling individuals to use their health behaviors to achieve some other goal? There are times when the latter may be acceptable. If a game is used to promote a short-term or one-off health behavior, it may be unnecessary to



Figure 24.1
Badges on the Fitbit site for achievement of the following: a 30,000-step day; walking 2,000 miles since starting to use Fitbit; and climbing 25 flights of stairs in one day.

build a connection between that behavior and the individual or societal health benefits, so long as the game gets the target population actually to engage in the desired behavior. For development of sustained health habits, however, such rewards may be problematic, particularly if people use the cat or other cheating as a means of achieving the rewards.

Gamification

Users of Fitbit's line of pedometers can earn badges for achieving personal bests or new milestones (figure 24.1). Users can view leaderboards that compare their performance to the performance of their Facebook friends or similar Fitbit users. These features provide users with social information about their progress and scaffold competition among friends or against the median user. Such gamified systems live in a middle ground between a full-fledged game and a system that simply presents data.

By including gamified elements, designers may add some of the motivators associated with games, such as competition or status opportunities, goals that are perceived as more achievable, and faster feedback. These gamified interfaces may be more motivating than the same system without these elements, but without requiring investment in careful game design. They also come with many of the same questions and disadvantages. What happens when a user who is more motivated by badges or status than by the health outcomes stops earning those badges or becomes bored with them? What happens when the user realizes he can attach the pedometer to the cat or put it in the dryer?

Designers should also avoid the pitfall of believing that it is "easy" to add badges, rewards, or other gamified elements to an existing system. Simply

grafting game elements onto an interface may cause undesirable user reactions. One of the authors added virtual trophies and ribbons to a mobile application that supported physical activity goal setting and self-monitoring, GoalPost. GoalPost rewarded users with trophies and ribbons when they achieved their goals (Munson and Consolvo 2012). Despite the widespread adoption of such features in both health applications (e.g., Fitbit) and other services (e.g., foursquare), the trophies failed to motivate users, and some users felt that trophies trivialized their successes.

The Landscape of Health Behavior Change Applications

Games and gamified systems have been designed to address a large range of health behaviors, including fitness, taking medication as prescribed, and building an understanding of complex disease progressions. Through three exemplar case studies (exergames and games that provide an outlet for physical exertion within the game; compliance games that encourage patients to adhere to prescribed medications; and skill-/ knowledge-building games that provide educational content to inspire behavior change), we highlight the diversity of games and gamified applications and discuss some of the common opportunities and issues that cut across each category.

Gaming Physical Activity

Increased physical activity reduces the risk associated with obesity and many chronic diseases. Despite the well-known benefits, many people still face difficulties in achieving recommended amounts of physical activity. To help with this challenge, designers and researchers have developed systems to help people track and increase their physical activity. For example, exergames (e.g., Müller et al. 2010, 2011) require users to exercise in order to play. There are also several gamified personal informatics systems that help people track their behavior and then celebrate achievements or compare progress with friends by awarding badges or viewing leaderboards. In this subsection, we review examples of games to promote physical activity, focusing on games that do so among children.

Children face many barriers to regular physical activity. Being a member of a sports team or taking lessons often requires some level of commitment from parents. Particularly for children from low socioeconomic status backgrounds, lack of family support can limit the success of health behavior change interventions. Inclement weather or climate, safety concerns, or limited neighborhood walkability can limit unstructured "free" play. The majority of children and adolescents do not meet recommendations that they participate in at least sixty minutes of moderate physical activity most days (Spear et al. 2007; Trost and Loprinzi 2008). At the same time, 83 percent of American children have at least one video game console in their homes, and American children between the ages of eight and ten years spend more than an hour a day playing video games (Biddiss and Irwin 2010). With rising concerns about health risks associated with childhood obesity and sedentary lifestyles, is it possible for games, a medium well liked by children, to support positive health behavior change and increased physical activity?

Many designers and researchers have noted and studied this potential for using exergames, particularly sports simulation and dance games, to promote physical activity in adolescents and youth (e.g., Epstein et al. 2007; Graf et al. 2009; Papastergiou 2009; Staiano and Calvert 2011; Unnithan et al. 2006). Launched in 2006, the Nintendo Wii was the original motion-sensitive gaming console, followed in 2010 by the Kinect peripheral for the Xbox 360. Unlike the Wii, which requires holding a controller, strapping controller devices to the body, or stepping on special floor pads, the Kinect uses cameras to enable naturalistic body-aware gestural interfaces and voice controls. These and other interfaces have enabled body-controlled games for home consoles to rise in popularity. Many dancing games have been released, both for arcades and for home use. The popularity and maturity of dance games makes them a frequent subject of study, with much of the research focusing on Dance Dance Revolution (DDR) (e.g., Tan et al. 2002; Sinclair et al. 2007; Trout and Christie 2007; Chin et al. 2008; Daley 2009; Graf et al. 2009; Tam et al. 2013). This is partially a result of DDR's popularity in K–12 educational settings. Children find the floor pad easy to understand. The pad provides visual cues that help teach the movements required by the games; this helps children link existing motions with more challenging ones as they progress in their game play. The game pad addresses both skill creation and difficulty management, overcoming self-efficacy barriers to exercise. Borja (2006) suggests that the DDR mats and game elements direct children's attention toward game play and away from the perceived drudgery of physical activity.

DDR and other exergames can create social incentives to perform well (Lieberman 2006; Chin et al. 2008; Jin 2009; Papastergiou 2009; Jin 2010; Staiano and Calvert 2011; Tam et al. 2013). Players can gain the encouragement of co-situated others, who may be watching or participating. Players can view their status on leaderboards and share high scores and dance videos on social networks sites. Peer pressure and the preference of young people for social activities can make social exergames particularly compelling. Adolescents who worked out using DDR cite the ability to perform with or against friends and to demonstrate skill as a major reason to play. Institutions can also offer additional support for helping youth start and continue to play exergames. Introducing adolescents and children to dance games in a classroom setting as a fun weekly fitness activity lowers drop-off rates compared to at-home-only players (Chin et al. 2008).

Despite successes, many exergames have been critiqued for not adequately considering everyday life rituals and activities. The majority of games aim simply to maximize performance in a minimal amount of time (Bogost 2007), requiring a defined commitment to carving out the time and space needed to use the console and game regularly. In this way, most console exergames are both prescriptive and restrictive in what sorts of physical activities "count" toward the game and goals.

In contrast, the American Horsepower Challenge was designed to encouraged physical activity without prescribing when and how players "played the game" (case study 24.1). Players could accrue points in a number of ways: walking the dog, shopping with a parent, or running laps on the playground. Individual participants, teams, and the schools that coordinated participation appreciated this flexibility. By awarding points for everyday activities, the American Horsepower Challenge offered players a new lens for viewing their daily life activities and encouraged

Case Study 24.1

The American Horsepower Challenge





Case Study 24.1
(continued)

Summary

The American Horsepower Challenge (AHPC) is a pedometer-based fitness video game for children in secondary schools. The program broadened opportunities for students—even those with little skill in traditional sports—to compete in an athletic competition for their schools. Through a combination of accelerometers, an automatic upload station located within a school classroom, and a web-based game, the AHPC tracked students' daily physical activity over several months and translated it into points in a virtual race between schools enrolled in the program.

Unlike other exercise video games in which points are earned through interacting with the game system itself (e.g., DDR, Wii Fit), players' interaction with the AHPC online game did not earn them any points. Instead, all points came from real-world physical activity separate from the game interface. Over the period of the competition, students wore on-body sensors measuring physical activity within and beyond the school day; each bout of walking or running earned points for the school. To determine each school's rank, step counts from all students on a school's team were aggregated daily to update school rankings. The competition interface was a website that displayed aggregated sensor data. On the website, each school in the program was represented as a bus on a racetrack, and the busses' positions on the track represented the schools' ranks in the competition. Each player was also represented by a horse avatar, which was customizable using a virtual currency earned by taking steps throughout the day. When using the website, players could check bus positions in the race, purchase items for their avatars, update their individual status messages, or view graphs depicting their recent step history. Players could also view their classmates' horse avatars and status messages, but not their classmates' individual step counts.

Facts and Figures

An evaluation of the program in sixty-one schools over a period of three semesters demonstrated modest increases in student physical activity (Poole et al. 2011). However, competitive elements of the game may have encouraged cheating that artificially inflated the measurements of individual physical activity.

Gameful Design Elements

The America Horsepower Challenge uses the following gameful approaches:

- Virtual badges and trophies are awarded for individual and group performance.
- Physical activity is translated into a virtual currency that can be used in the game.
- Players have avatars that can be customized using outfits purchased with the virtual currency.
- A racing metaphor is used.
- A map and leaderboard to show team ranking and position are used.

Issues

Game rivalry can be at odds with health outcomes. Winning the game can become a more important goal than increasing physical activity: In AHPC, players reported "loaning" pedometers to active friends or

Case Study 24.1
(continued)

placing them on active pets or younger siblings to accrue additional points. The AHPC is not alone in having rivalry eclipse health behaviors in importance. Similar results have been seen in other studies involving pedometer- and accelerometer-based interfaces (Morgan et al. 2003; Eyles and Eglin 2007; McCaughtry et al. 2008). In addition to academic reports of unfair play in sensor-based games and multiplayer competitions, there are scores of online resources in which people have inquired about and shared advice about how to cheat when playing commercial accelerometer-based exergames (such as Pokéwalker, a game in which a virtual Pokémon pet is cared for based on real-world physical activity, and Wii Fit). This calls for either finding ways to mitigate cheating or creating compelling game experiences that are more fun than trying to outsmart the system.

Related Cases

There are a number of interventions that use pedometers and competitive elements, both for adults and children.

Further Information

See http://www.slideshare.net/ExergameFitness/the-horsepower-challenge-by-humana-overview.

healthier choices and routines. Players reported thinking about their everyday lives in terms of the game and made efforts to walk more each day. Players also reported trading tips with one another on how to be more active (Xu et al. 2012a, 2012b).

In all of these games and gameful systems, however, it is possible for missteps and unintended outcomes, even in titles and environments that offer high degrees of flexibility or that are well integrated into everyday life. If a player does not set goals for the frequency, duration, or location of physical activity, good intentions about physical activity may be eclipsed by other concerns of daily life. In the American Horsepower Challenge, teachers served as key facilitators that helped keep the children "on track" with the program (Miller et al. 2012). Within-game scaffolding for goal setting, or a game that enables nonplayer coaches to be involved, may increase the effectiveness of games.

Gaming Medication Adherence

Unless Mary Poppins is your nanny, you probably do not consider taking medication to be a particularly fun or playful activity. Yet medication adherence can be encouraged through gamification, and even through video games, resulting in significant health care cost reductions and saving of lives.

Medication adherence is a significant problem worldwide. Roughly half of the 3.5 billion prescriptions given annually are not taken as prescribed (Osterberg and Blaschke 2005). Patients with many common heart-related diseases take their medication only 72 percent of the time (Cramer et al. 2008).

Barriers to medication adherence can be medication-specific and patient-specific, but commonly reported reasons include concern about side effects, forgetfulness, lack of social support (Hudson et al. 2004; Ogedegbe et al. 2004), perceived stigma (Sirey et al. 2001; Hudson et al. 2004), and lack of knowledge of the disease (Ogedegbe et al. 2004). Medication nonadherence increases mortality and a host of health-related maladies. It also contributes to health care costs, amounting to $100 billion annually in the United States alone (Osterberg and Blaschke 2005).

Many of these problems could be avoided simply through stricter medication adherence among patients. We examine several methods of medication adherence in order to provide a broader understanding of the types of game elements that can promote improved outcomes.

Methods to encourage patients to take and refill their medications include education during regular clinical visits, automated reminders and timers, and rewards systems. In a study of high-cholesterol patients, Kalayoglu et al. (2009) found that 73 percent were not taking their prescribed statin medication. Study participants were later provided with the option of calling into an automated phone number when they took their medication as part of the HealthHonors reinforcement platform; upon doing so, they would receive an educational message and the opportunity to earn points redeemable for gift cards. Their adherence rate increased to nearly 98 percent, and the patients described the method as "actually quite fun" and as giving them "an extra boost."

Telefonica Research developed a mobile social game, MoviPill, to increase medication adherence among elderly patients. MoviPill shifted medication adherence from a task of remembering to a social competition. Patients were challenged to earn the highest number of weekly points by taking the correct dosage closest to the doctor's prescribed time. The application played a reminder alarm if participants did not take their medication within fifteen minutes of the scheduled time. At week's end, the mobile application displayed the winner for the week and a new round would begin, wiping away the previous week's points. In a six-week study with elderly patients, MoviPill improved regimen adherence by 43 percent compared to that of patients who simply pressed a button on a mobile device. Additionally, 90 percent of the doses were taken without the alarm reminder (de Oliveira et al. 2010).

A challenge with competitive social games within a health care context is that the competition produces winners and losers, and individuals who feel too far behind could lose motivation. MoviPill's weekly reset feature might frustrate players who prefer persistent characters and histories, such as in massively multiplayer online games where players' characters accrue experience, abilities, and status with time. In the context of the simple social game, however, MoviPill's reset function helped keep the previous week's poor performers motivated, as each week began with a blank slate.

HealthHonors and MoviPill layer other incentive systems on top of tools for tracking medication adherence; they are gamified personal informatics tools. However, this is not the only way to use game elements to improve medication adherence, and Re-Mission is an example of an appealing game that unobtrusively provides health knowledge to the patient-player (case study 24.2). Re-Mission (www.re-mission.net) is a video game intervention for young adults and adolescents undergoing cancer therapy. Re-Mission provides educational content

Case Study 24.2

Re-Mission

Summary

Re-Mission is a fast-paced action, shooter PC game designed to improve behavioral outcomes and medication adherence in young adults and adolescents with cancer. In the video game, players take on the role of "Roxxi," a jet-pack-equipped flying nanobot who fights invading cancer cells and infections by shooting them down. Each mission focuses on the challenges that a young person would face with a commonly diagnosed type of cancer, including acute lymphoblastic leukemia, acute myelogenous leukemia, Hodgkin's lymphoma, non-Hodgkin's lymphoma, osteosarcoma, Ewing's sarcoma, and brain tumors. The in-game actions of Roxxi allow the player to battle harmful cells in twenty different missions that include taking prescription oral antibiotics, ingesting oral chemotherapy, using relaxation techniques to reduce stress, and eating various foods to gain higher amounts of energy.

Facts and Figures

A randomized control study with 375 male and female participants (ages thirteen to twenty-nine years) was conducted in thirty-four medical centers across the United States, Canada, and Australia. A participant's average play for the control and intervention was close to 7.7 hours and 10.7 hours, respectively, for the duration of the three-month study.

Gameful Design Elements

1. Fast-paced action and a third-person shooter perspective and mechanics in a PC game.
2. Clear sensory (visual and sound) feedback and scoring of successful actions.
3. Clear mission objectives.
4. Strong use of character personalities, colloquial language, and humor to build social motivation through in-game character support.
5. A visual map and directional feedback within the game.

Issues

The designers took efforts to ensure that player characters do not "die" within the game, and if players did not successfully complete the mission, the nanobot character simply re-powers and starts over again. Twenty percent of the participants did not play video games before their diagnosis, therefore it is perhaps less surprising that more than 10 percent of the participants did not play the video game at all in the study. While only 28 percent of the participants played for the requested one hour a week, there were no significant differences found between those that played less or more than the one hour requested by researchers (Kato and Beale 2006, Kato et al. 2008), suggesting that amount of playing time did not affect the beneficial outcomes.

Female participants showed a significantly greater positive effect for the intervention group than male participants. While the researchers do not speculate as to this difference in gender, we note that the primary game play characters were both female.

Outcomes

Among two hundred participants taking oral TMP/SMX antibiotics, there was a 16 percent improvement in medication adherence in the intervention video game group compared with that of the control group. There were also significantly higher chemotherapy

Case Study 24.2
(continued)

metabolite levels over time and higher cancer-related knowledge and cancer-specific self-efficacy over time in the intervention group of those playing Re-Mission (Kato et al. 2008).

Related Cases

Additional medication adherence–based incentives and gamified interfaces include the HealthHonors reinforcement platform (Kalayoglu et al. 2009), which provided a call-based incentive system for prescription adherence, and MoviPill (de Oliveira et al. 2010), a mobile game that provided social points–based incentives for elderly patients to take the correct amount of medication closest to their prescribed time.

Further Information

See http://www.Re-Mission.net (official website).

through missions done in a third-person shooter environment. One mission centers around the challenges of a young patient named Ian, who has Ewing's sarcoma and recently received chemotherapy, developing an uncomfortable infection in his mouth. Players are transported inside Ian's mouth as a flying, personified nanobot. They monitor his symptoms and use Ian's prescription of oral antibiotics to blast as many bacteria as possible while in-game characters shout encouragement. Blasted bacteria scores are tallied, and the player is given clear real-time visual feedback regarding successful blasts (figure 24.2).

The game led to significant improvements in adherence among patients taking oral antibiotics when compared to that with the use of a control game (Kato et al. 2008).

Considerations beyond Game Play

Designers face many choices when creating a health intervention. Should it be a game or a personal informatics system? If it will be a personal informatics application, should it include gamified elements? In this section, we highlight several considerations that are more salient for health games than for other games or gamified systems, and we identify areas that merit further study from researchers and practitioners.

Social Context

While lab studies and limited-duration field studies have shown positive outcomes for persuasive health games (Unnithan et al. 2006; Maddison et al. 2007; Graves et al. 2008; Mellecker and McManus 2008; Leatherdale et al. 2010), there has been some evidence that behavior change interventions involving games or gamification do not sustain player interest over time in naturalistic settings (Madsen et al. 2007;

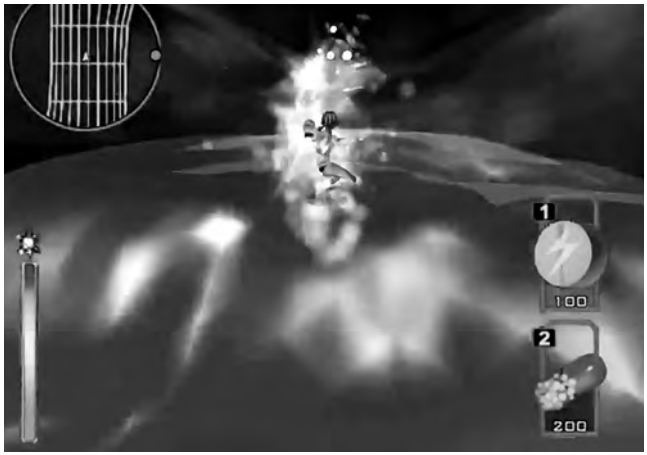


Figure 24.2
A player targeting cancerous cells in the game Re-Mission.

Maloney et al. 2008; Dixon et al. 2010). The gap between controlled and naturalistic settings suggests a need to rethink the metrics for evaluating health gaming interventions in the real world. In lab experiments, players' physical and social contexts are missing, both of which influence people's physical activity level and attitudes (Sallis et al. 2000; Davison and Birch 2001; McGonigal 2006).

Health behaviors are embedded within larger social environments, such as one's community, school, and organizational and family environments (Davison and Birch 2001; Bandura 2004). A game and a target behavior's contexts will have implications for whether an individual or multiplayer game is more appropriate. If it is intended as a multiplayer game, will it foster collaboration or competition? Where and when it should be played (and not played?).

When matched to the social context, a social experience can make a game stickier and more appealing. Among youth, social game play is particularly common: 76 percent of American youth surveyed engage in collaborative or social game play (Lenhart et al. 2008), and 68 percent of European teens report playing games with other people at least sometimes (Nielsen Interactive Entertainment 2008). The American Horsepower Challenge made use of existing social contexts beyond the game interface to support behavior change. Players were accountable to and could support each other both online and offline. The game's hybrid environment provided opportunities to express social support (online and in the hallways, classrooms, and social spaces of the school) from players and from teachers and nonparticipating students. When Dance Dance Revolution was introduced in the classroom, the collaborative social support from the school environment provided additional incentives for game play compared to that when playing at home (Chin et al. 2008). Here, even though the actual activity is bounded to the game, the intervention is situated in a pervasive social context.

Availability and Accessibility

The availability and accessibility of the Internet, gaming devices, and related hardware or technology are important design issues when supporting health-related games. Internet connectivity at home is still limited among certain populations, even within developed regions (Sun et al. 2005), making high-bandwidth requirements or repeated daily logins a challenge. Additionally, while some games address some interface accessibility issues (such as options for players with visual or auditory impairments), many game-play accessibility barriers remain for people with cognitive or physical impairments (Bierre et al. 2005).

Usability and training are other facets of accessibility. Researchers provided the elderly participants in the MoviPill study with a specifically designed pill box and smartphone, and the MoviPill system trained seniors in the mechanics of game play and how to use the pill box (de Oliveira, Cherubini, and Oliver, 2010). Outside of study contexts, where it may not be feasible to provide this level of training, the learnability and accessibility of health interventions will be more important. The HealthHonors system incentive structure used low-tech phone messaging, making it available to a wide variety of users and requiring no training beyond an instructional sheet included with users' medications. Existing community infrastructure can sometimes meet training and support needs: in the American Horsepower Challenge, connectivity and training were accessible in the school environment, and the game was integrated into class activities.

Game Genres: Player Preferences and Design Fit

Not all game genres or narratives will fit all players, and choices around the game's structure, environment, objectives, and narrative affect player engagement (Dondlinger 2007). For example, males are more likely to play action, sports, strategy, or role-playing games, whereas female players are more likely to play puzzle, board, game show, trivia, or card games (Nielsen Interactive Entertainment 2008). Game genre, narrative, and content also correlate to engagement and motivation at different ages. It is questionable whether a rapid-paced action game where the main character yells "you bastards" at she zaps lasers at cancer cells (as is the case with Re-Mission) would be as effective if targeted at elderly patients. While elderly adults are less likely to play digital games than youth, picking an appropriate and appealing genre overcomes barriers to elderly players' adoption of health games (Ijsselsteijn et al. 2007). Whereas a first-person shooter may not resonate with elderly adults, designing for their interest in social activities or simpler puzzle or trivia-like mechanics may lead to games better matched to elderly players' preferences.

The game genre, mechanics of play, and the use of narrative must also match the targeted health behavior or knowledge. Not all game genres will be equally suited to a target health behavior. For example, a mentally challenging turn-based strategy game may be ill-suited for motivating people to engage in aerobic activity, especially when compared to a dancing game. The strategy game, however, may be an excellent match for teaching players how a medication attacks a disease.

Goals: Source and Duration

Goals are most effective when they are self-set rather than assigned; when they are hard but achievable; and when people are able to monitor their progress

toward the goal and receive positive feedback for success (Locke and Latham 2002, 2005). While long-term goals may be more motivating, short-term goals often lead to higher performance. When working toward a series of short-term goals, individuals are able "reset" when they get too far behind, and thus are able to start over or try again, rather than giving up in anticipation of an eventual failure. They are also better able to monitor their progress and sooner able to receive positive feedback from successes. Decisions about the optimal goal period are often highly contextual. Designers may wish to offer users some flexibility to customize goal periods to their lives (Consolvo et al. 2009). In many games, the designers decide what it means to "win" a level, while still giving individual players some choice about how. This choice can come in the form of the difficulty level or through the quest structure, which offers players a range of possible goals and paths to victory. What scaffolding will best balance pushing players toward healthy, challenging, yet achievable goals and behaviors with support for individual choice and buy-in?

Previous research highlights additional challenges for selecting the source of health goals. Individuals are skeptical of goals set by doctors and thus may not find such goals motivating (Consolvo et al. 2009). However, individuals may not have the expertise to set appropriate health goals for themselves. One design strategy is to have people configure a long-term goal, such as the monthly goals in Health Month or long-term weight goals in systems such as LoseIt or Fitbit, and for the application then to decompose these longer-term goals into daily or weekly goals. In some cases, system-generated goals might be presented as suggestions that users can accepted, reject, or modify, a process by which they take ownership of the goal.

Various individuals and groups may also have differing conceptions of what it is to be healthy, further complicating setting appropriate goals. Designers should be sensitive to these different health ideals, though they may still wish to design interventions that either implicitly or explicitly challenge the user's notion of what it is to be healthy. We also caution designers of games and other health interventions against focusing so much on one dimension of health that people are pushed to improve it at the expense of their overall well-being (Purpura et al. 2011).

Design for Single Use or Ongoing Use

What happens after the game and related rewards are removed? Is the goal of the game to promote lasting behavior change or is it sufficient for the behavior to be motivated in the game play itself, and what are the implications of such decisions? Is the intervention designed for ongoing, possibly lifetime use, to promote a one-time behavior, or to be a short-term intervention that leads to long-term behavior changes?

In a video game such as Re-Mission, teaching about healthy behaviors was critical for the short-term health of the patients. The game was designed to achieve its educational objectives and to promote behavior change in a relatively short period of game play, over a short period time, to help patients reach a long-term, stable health state.

Increased physical activity and many other health behavior changes are generally intended to be more enduring. For participants in the American Horsepower Challenge, the game provided a framework for motivating and monitoring healthy activities, but the healthy activities themselves (running, walking, playing sports) occurred outside of the

game environment. Exergames such as Dance Dance Revolution provide similar health benefits to the American Horsepower Challenge, yet the benefits occur in-game, and if the game disappears, so may the healthy activities. In some cases, though, exergames engage participants, and then players discover through game play that various physical activities are enjoyable and may feel increased self-efficacy for participating in those activities outside of the game environment (Schwanda et al. 2011).

These examples highlight that the design and goals of health games need to match the desired long-term or short-term health outcomes. This distinction is made in popular persuasive design frameworks, such as the Fogg behavior grid's contrast between one-time, span, and path behaviors. It is important, though, not to conflate behavior endurance with duration of intervention (Munson 2012). Just because an individual may want to take up running as a new lifelong hobby does not mean that the individual wants to use Zombies, Run! for the rest of his or her life. Games that increase people's self-efficacy for an activity and that help them discover enjoyment of an activity may be particularly promising as short-term interventions that lead to long-term changes (Schwanda et al. 2011).

Privacy, Confidentiality, and Security

Control and access of personal and sensitive health information is a point of concern among the medical informatics community (Yasnoff et al. 2001) but has received notably less attention in game design. Both within the game environment itself and on the back end, we find that transparency and design for appropriate access to personally sensitive health care information is an ethical responsibility that can affect players' willingness to engage in the game. Good designs can balance sharing with privacy. For instance, MoviPill players could view their individual data, and the mobile interface abstracted these data to emoticons for sharing with other players (de Oliveira, Cherubini, and Oliver 2010). Designers need to decide who will have access to data and how to help players understand the terms of this access.

Conclusion

Preventing and managing chronic conditions is one of the great health challenges of the twenty-first century. Successfully coping with these challenges requires that individuals make good health decisions on a daily basis. To help individuals become and sustain their "better selves," many people are turning to applications and devices designed to promote healthy decisions. These systems can sense context and intervene in the moment, can connect people to networks of peers and friends and family to deliver accountability, motivation, and advice, and can engage users in rich narratives offering additional motivation and incentives to participate.

These technologies take a variety of forms, from mobile personal informatics tools to living-room games. Each style of intervention offers different strengths and weaknesses. Considering proposed or existing systems alongside the important behavioral

predictors identified in health behavior change theories can reveal how and for whom an intervention may work and where it may break down. Regardless of a designer's chosen approach, attempting to change or prompt health-related behaviors will make certain design considerations more salient, including those related to accessibility, the source and structure of the goal, and the privacy and sensitivity of personal information.

As discussed in this chapter, when designers are mindful of these considerations, they create games or other applications that improve health behaviors for a large cross-section of a population. Notably, in games such as the Wii Fit, gameful interventions can help people discover an ability to achieve, succeed, and enjoy healthy activities, prompting people to graduate from the "game" and onto other kinds of challenging and fulfilling health activities. When done carelessly, though, a game or gamified elements may cheapen achievements, promote "cheating" rather than actual health behaviors, or cause people to condition the health behavior on the game's rewards, leading to termination of the behavior when those rewards are withdrawn. We believe that games hold great potential for helping some people address certain health behaviors, but we caution that health-oriented games must be designed and studied as carefully as any other health intervention.

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Gameography

Konami. Dance Dance Revolution. Arcade, followed by console (PlayStation, Dreamcast, Nintendo 64, PlayStation 2, PlayStation 3, GameCube, Wii, Xbox, and Xbox 360).

Six to Start. Zombies, Run! Mobile (iOS and Android). See https://www.zombiesrungame.com/.

POSITION STATEMENT

LEARNING TO PIVOT: A PLAY ON POSSIBILITY

Katie Salen Tekinbaş

My dad was once a professional basketball player with a keen eye for extracting big ideas from the rules of a game. In addition to instructing me on the importance of daily practice and kindness toward one's teammates (they are, after all, the ones you'll need to count on when the odds don't fall in your favor), he was the first person to teach me about the beauty of the rule governing my pivot foot—that in its requirement that one foot remain fixed in place while the other roamed freely, it freed a player to carve out a space of possibility on the court. The free foot moving away from and then returning to a single position in space gave rise to new possibilities. The rule, he said, was there to help me understand that to pivot was the most graceful way of increasing my radius of influence, as it allowed me literally to extend my reach.

As I grew up and began to explore design as a space of play, rather than basketball, I took full advantage of my own ability to pivot. And pivot I did from graphic design to animation; from the design of games powered by giant inflatables to a karaoke ice cream truck driven by a squirrel. The point of the pivot, whether I was designing publications or tinklepop music, was always that of *play*. And, more importantly, the participatory and transformative qualities of play: the ability it gave players to push against the rules, to try new things and fail in ways that were useful for future action, and the power of doing so in collaboration with others. To me the design of play was a way to connect people within spaces where they could participate and create together. Play activated a critical space for the possibility of design.

I spent years making all kinds of games (digital, physical, social, conceptual) and worked to understand what it meant to design play experiences that held within them the possibility of transformation. What did an invitation to participate look like, for example? What game design structures enabled people to collaborate in meaningful ways? To break rules? To persist on problems with no simple answers?

And so I pivoted once more, this time into the field of education, where I saw an opportunity for game designers with their exquisite knowledge of the beauty and mechanics of play to rethink what it meant truly to engage young people in the project of learning. What it meant, in other words, to design a possibility space for learning fueled by the transformative qualities of play.

In working at this intersection between game design and learning, I took the concept of the pivot learned all those years ago from my father and

applied it to the creation of two new public secondary schools in New York City and Chicago. They are schools based on principles of game design and play and have been set up to require the daily collaboration between game designers, teachers, and young people. The results have been surprising and somewhat extraordinary for all who have been involved.

The goal of the work from the perspective of the essays contained in this volume was an attempt better to understand and expose both the promises and limitations of games and their design. One strategy taken was to move beyond the idea of games as discrete artifacts in the classroom that students would play to learn content, much like they would use a textbook, video, or online learning resource. Instead, the model argues for a pedagogical approach that frames the design of learning activities over time as play-centric and game-like. This meant developing a model for the creation of learning experiences that drew from the core design principles organizing games and their play.

The design of learning in the schools, as a result of this focus, is centered on the creation of well-structured and engaging challenges that give rise to a "need to know" in students. The game-like challenges that structure the curriculum offer a space of possibility for learners to tinker, explore, hypothesize, and test assumptions. They build in opportunities for authority and expertise to be shared and made reciprocal among learners, peers, and teachers. And they support multiple, overlapping pathways toward mastery. Depending on the class, challenges might be worked on over a series of weeks or last several days; they might focus on a specific content area or skill or require an interdisciplinary integration. Groups of students within a class might be assigned different parts of the challenge to work on, depending on their level and interest.

What I love about this project is that it has done a tremendous job of opening the space of possibility for thinking about the future of learning as interpreted through the lens of game design. The schools offer a new vision for the role of digital media, design, and collaboration in that future. But, more importantly, the very existence of the schools has caused all sorts of productive conflicts to occur, just like in any well-designed game. We have seen traditional teacher practice collide with new learning practices enabled by Web 2.0 tools. We have seen collisions between the languages of learning and design, where terms like iteration, prototyping, agile development, and interaction have taken on new and powerful meanings. Most of all, we have seen a productive conflict between what have been fairly low expectations around what can be achieved by a school (test scores) and the very real possibility of young people transforming their lives and communities through the power of design.

Now the story of these schools has just begun, and it is hard to predict what kind of an impact they might ultimately have. But we know already that the students are changing: they are becoming more agile in their thinking, more creative in their problem solving, and more attuned to their skills as collaborators. They are also all learning to pivot, away from notions that school has to be boring and driven by adult decisions. Away from the idea that learning can't be hard and engaging at the same time. Away from a belief that school is not a place for them.

Designers of all kinds are key players in the game of change that so typifies the opening decades of the

twenty-first century. Called on to imagine, build, guide, demystify, explain, provoke, enable, and inspire, designers deal daily in the currency of transformation—of places, practices, and perspectives. For me, games and the principles that underlie them pivot on a central axis of engagement rooted in the very real power of play. I invite you all to join me in a few designed pivots of your own. There is tremendous possibility in the act of extending ourselves and in the space afforded by collaboration and the subsequent collision such collaboration enables. As my dad would say, "Use your elbows, honey. Own the space you know is yours."

25 GAMIFICATION AND LEARNING

Dennis Ramirez and Kurt Squire

Video games, as a form of intrinsically motivated learning, have been studied by psychologists for decades. As the field of games for learning matured, we've discovered more reasons to use games in the classroom that reach beyond motivation, transforming the ways players interact with content. Just-in-time feedback, well-ordered problems, and the ability to learn through failure, hallmarks of good game design, are also important for effective educational interventions (Gee 2003; Squire 2011). As a result, there is great interest in incorporating games in the classroom and other contexts. For similar reasons, the process of gamification—introducing game-like elements in order to keep students engaged, create a playful environment, and quantify activities—has also become popular in education. The translation of these methods from theory to practice, however, is not a straightforward process—it requires time, resources, careful design, and thoughtful educators.

Gamification means many different things to many people. For the purposes of this chapter, we use the commonly accepted definition that gamification is the use of game design elements in non-game contexts (Walz and Deterding, this volume). We focus on four overlapping design features used in gamification: point systems, achievements, quests and challenges, and narrative structures. Although we do not focus explicitly on designing games for learning or impact here (our main research and development activity), the design challenge of gamification and the design of learning games actually share many features. For example, a common approach to designing games for learning is to create a simulation of a phenomenon and then give it "roles and goals" (Edelson 2002; Norton 2005). We might consider overarching narrative structures and quests as forms of roles and point systems and achievements as examples of goals. These design features can, and often are, used to good effect in games for learning or impact.

Academics have an uneasy relationship with gamification. As Ian Bogost (this volume) points out, gamification is basically a marketing-driven concept designed to commodify the intellectual and social capital of a popular art form. Gamification sells services for consultants and provides business managers easy solutions to the complex issues of confronting a digital, participatory age. As Bogost suggests, it may be the "-ification" that is the most troublesome. Suggesting that a medium could be reduced to a simple formula to be reapplied across contexts is grossly naive. To understand the silliness, imagine someone peddling "televisionication" or "filmification" of business.

At the same time, dismissing gamification out of hand is a mistake. Games exert tremendous social and cultural influence, including how we orient to media. The rapid spread of reputation systems, badges, and virtual currencies suggests that participation structures originating (at least to a degree) from games are having broad impact (see Lampe, this volume). Indeed, if one were to adopt a descriptive (possibly critical) stance toward late capitalism, gamification might be a particularly apt descriptor. Gamification captures how social institutions are restructuring themselves in a "game-like" manner, acknowledging consumer agency (appropriate for an attention economy) and creating participant structures with explicitly detailed mechanisms for increased formal participation (Rey, this volume). This perspective of gamification as "the frontier of social engineering in late capitalism" is one foreseen by game developers such as Raph Koster, who anticipated this development and described his work (and in particular that of the MUD-Dev community) as the experimental petri dish for twenty-first century social thought (Koster, 2013).

However, a second, very important critique of existing gamification efforts is how they co-opt individuals' behavior toward the purposes of organizations that not only disregard, but often exploit their interests (see also Rey, this volume). Chris Franklin in the *Errant Signal* YouTube series offers the following critique (which is notably directed to an audience of gamers):

> The point of gamification is to exploit instinctive human reactions to basic stimuli in order to get you to do something you otherwise wouldn't do. It utilizes collecting and hoarding impulses by providing you a list of badges to complete. It gives you Skinner boxes and experience bars and leveling-up mechanics in order to make you feel more productive. It creates a positive feedback loop where every action you take to the benefit of the systems' owners grants you more rewards. It then compounds all of those features with the need for social recognition or praise by making these badges and levels viewable to everyone establishing a social hierarchy that reinforces the behaviors they are looking to promote. The idea is you take that whole sordid mess of constructs and superglue them onto an established system that you want to encourage interaction with. So users start collecting badges and leveling up in order to make their EP (experience points) feel bigger amongst the other players, while the person who owns the system starts benefiting either through ad revenue, increased productivity, or whatever (Franklin, 2012).

Franklin's critique describes gamification as cultural and economic hegemony, a tool created by the ruling capitalist elite to ensure that participants feed the capitalist system (cf. Apple 1981).

We share Franklin's concern that, when done badly, gamification is little more than a cynical application of behaviorist and other psychological principles toward driving consumption. Gamification as it's typically done provides nothing for the user other than superficial award "points," while providing companies tremendous value. Critiques of gamification—particularly when emanating indigenously from gaming communities—can be a powerful form of resistance to the co-optation of gaming practices by such institutions. Indeed, it is worth remembering that many game design techniques (particularly those from the virtual worlds literature) are rooted in academic computing and grassroots gaming communities such as MUD-Dev (see Bartle 1996, 2004).

As Franklin acknowledges, gamification structures make gamers feel more productive, which, as

Linderoth (2009, 2012) argues, can be an illusion. Linderoth argues that many games create the false impression of growing competence and achievement by increasing the power of player tools, rather than requiring players to be more adept in the game space. However, providing players a feeling of progress is not inherently *bad*, and perhaps even satisfies a core psychological need. People crave feedback and *enjoy* visualized progress. When done correctly, the design techniques that we call gamification (which recall come from games themselves) can be enormously engaging for their players. Using these techniques artfully could (in theory) put players in much greater control of their learning, and even provide opportunities for critical reflection on how their action relates to institutional structures and goals.

We argue here for a *pragmatic* approach to using game design techniques in non-game settings (to build on the definition provided by Deterding et al. 2011). As Lee and Hammer (2011) suggest, one can easily argue for gamification techniques based on cognitive, emotional, and social reasons. Like Deterding et al. (2011), we believe that so-called gamification principles aren't inherently good or bad. Rather, we can examine the consequences of their use and determine their value. As designers and learning scientists who create learning systems with important consequences for learners, we treat issues around gamification techniques as largely empirical questions: How are they working? Whose interests are being served and legitimated? Are they engaging and life enhancing for participants? How might we improve these structures so as to make them more just or equitable?

This chapter examines the application of game design elements in learning systems (which themselves often include games), describing the work we're doing to use game design elements in the social systems that encompass games. It begins by describing our general project of sociocultural learning theory applied to games, in part to contextualize preceding work. We then discuss how so-called gamification techniques are applied "in the wild" with entertainment games to promote learning, and then finally how they are currently being designed into our learning systems. The chapter closes with reflections on using gamification for social change.

Games as Socially and Culturally Situated Systems

Digital games have garnered the attention of educators as sites for studying digital literacies, a medium of expression, and as potential tools for learning (Prensky 2001; Gee 2003, 2005a,b; Bogost 2010; Squire 2011; Steinkuehler, Squire and Barab 2012). Central to one approach, which has been called the socially situated approach, is to view games not as only tools or artifacts, but as sociocultural systems in which tools are a key part (Shaffer, Squire, Halverson, and Gee 2005; Deterding et al. 2011). From this perspective, games are one element of an activity system that includes people as actors, broader social and institutional *constraints*, and other resources (DeVane and Squire 2012). Although games and learning as a field is young and the participatory logic of game-based learning systems contradicts those of modern schooling (particularly in the United States; see Tyack and Cuban 1995; Collins and Halverson 2009), projects

such as Fifth Dimension, Quest Atlantis, Surge, and Citizen Science have demonstrated success working in schools.

Social Learning Theory and Games

Sociocultural (or socially situated) learning theory, for the uninitiated, is a loosely connected family of learning theories (usually rooted in Vygotsky or Dewey) arguing that cognition is inseparable from social context. Although the category *sociocultural learning theory* combines theorists from multiple traditions, each shares some common commitments.

A key idea driving this work is that *most higher learning occurs through social interaction*, and more specifically that understanding occurs first on the external plane before becoming internalized (Vygotsky 1978; Lave and Wenger 1991). Parents, for example, often structure tasks for their children within what Vygotsky called the zone of proximal development (or ZPD). ZPD is the zone of activities that a learner cannot do on her own, but can do with help. A hallmark of learning within the ZPD is that learning is organized around a mutually valued task. Through working together in joint enterprise, the adult (or other more expert person) gradually hands responsibility for that task over to the learner. Subsequent work expands this idea and develops the notion of *cognitive apprenticeships*, which is a common process of *modeling*, *coaching*, *scaffolding*, and *fading* (Brown, Collins, and Newman 1991). Essentially, this is the process by which an adult first *models* competent expert behavior, then *coaches* the learner through just-in-time feedback, next *scaffolds* the learner through supports that enable the learner to work more independently, and then finally *fades* away so that the learner can act independently. Designing these patterns into games is a hallmark of Barab et al's (2009, 2010) work.

Situated Theory for Games

Educators also value a game's capacity to situate learners in complex situations. Situated learning theorists (e.g., Brown, Collins, and Diguid 1989) argue that our thinking is not just *shaped* by, but *constituted* through our experience of situations, including objects, interactions, people, and cultural tools (language, concepts). Classic examples from cognitive science include the way that we use tools (spreadsheets or word processors) to organize thoughts (Pea 1993) and how these tools *transform* thinking. To borrow another simple example from Vygotsky, consider how the axe transforms our understanding of trees or the properties of wood. Not only does it change what we can know about trees; it also changes our conceptions of wood as we create a new conceptual category separate from the trees (the same might be said of man-made objects such as clay and brick, wheat and grain, sheep and wool, or minerals and ore). In educational games such as Citizen Science (Filament Games/GLS), the game positions players as youth seeking to improve a local watershed through using virtual tools (e.g., Secchi disks, chemistry kits) and arguing with virtual characters about the causes of eutrophication and about potential solutions (Gaydos and Squire 2012). The goal of Citizen Science is to use vicarious experience to transform how players think about their natural surroundings, including the use of virtual tools—all through inhabiting the role of a youth trying to change the fate of his local surroundings.

Sociocultural Theory for Games

Games are also situated within game-playing communities, which offers both an intriguing alternative to classroom learning models, as they suggest different trajectories of participation. This work builds on a second branch of sociocultural research pertaining to games, which focuses on learning as participation in social practice. Examples can be found in multiple fields; sociolinguist James Paul Gee (2003), for example, discusses the scripts that we follow in conversations, which both structure interaction and constitute the context in which performance is evaluated. Becoming a competent student, lawyer, game player, or developer is about much more than reciting facts, but being one of these roles, which entails presenting oneself as a member and being accepted as a member of other people in these social groups (Gee 2000/2001). Much work in sociocultural learning theory with games examines this process of *identity formation*, including how students' identities are constructed in their home lifeworlds, and how new contexts might be designed for new kinds of identities. Games, from this perspective, are interesting in that they might provide new models of social organization, ones that perhaps meet our students' needs better than schools. We have argued for online networks such as Apolyton University (Giovanetto and Squire 2008) as one model built around structures of cognitive apprenticeship and problem solving rather than routinized content delivery.

Socially and Situated Critiques of Schooling

Interest in games in schools is often based on long-held sociocultural critiques of schools (cf. Gee 2005a,b). A classic critique (see Lave 1988) is that knowledge in schools is primarily gathered for exchange value (grades, or perhaps economic rewards later in life), rather than for pleasure, joy, fulfillment, or action in the world. Schooling severs knowledge from the contexts in which it was created or in which it is used, treating knowledge as commodities to be mastered (Kirschner and Whitson 1996). Knowledge is conferred by an authority (usually teachers, but increasingly testing companies paid by the state), which robs students of opportunities to develop and test ideas through actions in the world (Lemke 1992). An implication of this view that we always learn new information by linking it with past experiences is that learners with more direct experiences with academic phenomena have a leg up on those without them. For example, learners who are fortunate enough to interact with adults using academic language tend to do better in school where academic language is prevalent. Perhaps most importantly, children that can interact with adults who affiliate with the formal enterprise of schooling also have opportunities to develop identities in relation to the institution of schooling. While these opportunities are not available to all students, video games have the potential to simulate such experiences to a much wider audience (Shaffer 2006).

Games as Models of Situated Learning

Work on sociocultural learning through games is beyond the scope of this chapter and, to some extent, now ingrained in the field of games and learning. Since Gee (2003) laid out an argument for games as a model of situated learning theory in *What Video Games Have to Teach Us About Learning and Literacy*, the basic

notion that game play regularly produces changes in knowledge, skills, and attitudes has gained greater acceptance. Further, Gee's (2003) thirty-six game design principles (such as learning by doing, well-ordered problems, and learning through a projective identity) have been used to design dozens of games for learning, although until recently, relatively few examples of good games for learning existed (see Squire and Patterson 2010 for a good review).

For the purposes of this chapter, two key ideas require explication. First is Gee's (2003) notion of projective identity. Gee argues that anyone who has discussed Pokémon (Game Freak 1996) or more recently Skylanders (Toys for Bob 2011) with a game-playing six-year-old will understand that game play inspires and produces deep knowledge of semiotic systems. Many youth develop this knowledge and even report learning to read complex texts through Pokémon as a pedagogical system (Sefton-Green 2004). Knowledge of Pokémon is mastered in order to do work in the world and as a part of becoming a desirable identity in a game (Pokémon Master), which Gee calls a *projective identity*, and within a player's primary game-playing communities. The robust Pokémon ecology enables players to develop identities as members of Pokémon playing affinity groups.

This notion of a projective identity is *crucial* for gamification in schools. Schools (outside of extracurricular activities) traditionally offer few identities outside of becoming "the good student," which involves mostly reporting back to teachers the information presented to them. There are few opportunities for individuation, becoming expert, or contributing to society in any meaningful way. Through gamification, we might set up trajectories for students in which they become different kinds of people (doctors, lawyers, activists, and so on; Shaffer, Squire, Halverson, and Gee 2005). Game structures might enable students to take on these roles and guide students' transition toward becoming new kinds of people. From this perspective, gamification is rethinking the roles and assessment mechanisms in schools to make them more effective and more democratic.

The preceding section describes a theoretical rather than commercial basis for investigating the potential of games for learning. *Many educators using games level toward schools the same critiques that are leveled toward gamification* and are looking to games as an alternative metaphor for organizing learning. Minimally, rethinking school through the lenses of games raises new questions about whose interests are served, and we believe if games are designed and executed well, they are a tool toward broader educational reform. Most critically, they could provide an alternative credentialing structure that becomes an "end around" for the monopoly in schools enjoyed by the testing regime.

Gamification for Learning through Achievement Systems

As educators, we are constant designers of social systems (more precisely, codesigners along with their participants), and, for that reason, designing participant and reward structures is inescapable. For decades, educators have tackled redesigning social institutions to make them more just, participatory, and responsive to users' needs. Many educators' interest in gamification techniques for learning may stem from this

dissatisfaction with how grades and accreditation systems currently work in schools. The typical course-based learning structure deployed in schools holds time constant (we all start and end at the same time), but then lets achievement vary (Reigeluth 1996). Why not insist on standards of excellence and allow time to completion to vary? Mainstream educators' frustration with contemporary assessment has driven them toward badges as an alternative method for assessing learning, one that could decouple the function of accreditation from courses, tests, and schools (Joseph 2012). In her presidential keynote address "The End of Testing," leading assessment expert Eva Baker (2007) challenged educators to reconceptualize their practice from one of "test making" to one of accomplishment credentialing:

> My image of a qualification is validated accomplishment, obtained inside or outside school. A qualification means simply that, at various levels of challenge, a student has attained a certified, trusted accomplishment. ... Each qualification is not a new test, but an integrated experience with performance requirements. It might look like a course, or a collection, or a musical or sports performance.

Barry Joseph (2012) describes how mainstream assessment experts' interest in qualification credentialing dovetailed with James Paul Gee's (2005b) work on games and assessment. Gee argued that games are the future of assessment because they demonstrate how assessment can be routine, formative, and an inherent component of learning. For example, most games include boss battles, which serve as integrated assessments requiring the player to use the skills he or she has cultivated up until that point in order to progress (see also Collins and Halverson 2009 for an expansion on this idea).

The following sections address the question: how might we make more effective use of those game design elements usually exported by gamification (achievements, badges) within games for learning? These sections builds on earlier classifications of achievements (Blair 2013) by linking design principles culled from the learning sciences with effective uses of achievements *toward the design of learning systems*. Notably, these sections are not a structural analysis of what achievements are (Hamari and Eranti 2011), nor a close examination of participation in gamification structures, such as Jakobsson's (2011) excellent analysis of Xbox 360 players (which is, interestingly, a look at gamers responding to the gamification of game play). We analyze current designs in badges, achievement systems, and narrative structures in games in order to devise the following *design heuristics* for using gamification techniques in learning (Reigeluth 1996; Antin and Churchill 2011; Nicholson 2012; Kapp 2012).

Recording Learning Progress

Achievements serve to indicate whether or not an action has been completed. By recording these achievements, however, they become a persistent history of the user's involvement with the system. Much like merit badges awarded to boy scouts, achievements that reward the completion of a task serve as a record, which is useful for a user to reflect on what he or she has accomplished and might want to do in the future. Such achievement systems suggest easy ways that educators might provide much more detailed learning data than traditional grades or transcripts. Such a model might also encourage deviation from current standardized models in which most students are expected to accomplish all the

same tasks in school or be measured along the same handful of learning trajectories (e.g., science and mathematics course sequences).

Providing Feedback on Progress

Originally used in role-playing games, experience points are a popular way to provide feedback on a user's progress and provide a mechanism for communicating with players. Earning enough points to get to the next level also provides a sense of having accomplished a goal. Unlike grades, point systems are built on a growth model for the user, not a deficit, which is one reason that educators with experience in game design have gravitated toward experience points in lieu of grades (Holman, Aguilar, and Fishman 2013; Sheldon 2012). It should be noted that these experience points are often paired with some type of level system that implies some sort of mastery or prowess. In traditional Role Playing Games (RPGs), a player's attack stat may become higher as he or she levels up simulating mastery through experience, and well-executed point systems do the same. For example, social media websites like Reddit or Digg successfully incorporate this strategy because the point values imply proficiency such as the ability to find unique content or contribute to discussion. The same techniques can be used for discussion forums or contributions via massively open online courses (MOOCs). Further, experience points can be weighted so as to both encourage risk-taking, exploration, or extra practice as an instructor desires. Many implementations of such systems in schools decouple these learning tasks from course timelines, enabling students to learn at their own pace rather than in lockstep. A key for educators is to make the values and assumptions behind these systems transparent and encourage dialogue about them (something rarely done in schools now).

Promoting Transparency in Assessment

Achievements, with the exception of secret achievements to promote exploration, often allow a great deal of transparency when it comes to how a player completes the achievement. (And in fact, a subject of discussion at the 2013 Game Developer's Conference Achievement Roundtable was how to make even these achievements more transparent to users, with one achievement designer declaring, "Secret achievements sucks" to great applause.) We know, from research on rubrics (see Mabry 1999), that making assessment criteria available to students can increase the quality of work and learning. Similarly, designing achievements around such hidden rules can demystify assessment. Indeed, making informal rules explicit and making important academic practices explicit for learners can be a chief democratizing function of achievements.

Motivating Persistence

Not every part of playing a game is fun. For example, grinding in RPGs has often been cited as a dull and repetitive process that most people wish they could fast forward. For this reason, designers use achievement systems to break up the monotony and encourage players to push forward. Pokémon addresses this issue by using the game mechanic of evolution to reward the player for his or her commitment. The game rewards the player with a more powerful monster and updates his or her Pokédex (the player's encyclopedia of Pokémon knowledge) with a new entry, which, like a badge, serves as a persistent

reminder of the player's efforts. Thanks to this reward, players are encouraged to continue the game with the hopes of making their creatures more powerful. Educators have long lamented that many educational systems (and particularly American culture) embrace theories of attribution that attribute achievement to innate talent rather than effort, which ultimately causes many learners falsely to believe that they cannot succeed in domains in which they are not comfortable (Bandura 1982; Ryan and Deci 2000; Dweck and Master 2008). Good achievement systems might explicitly counter these expectations.

Encouraging Mastery

Achievements that highlight prestige or skill are useful when motivating a player not only to continue playing a game but also to master the underlying mechanic. In the original Metroid (Nintendo 1986), players were rewarded with multiple endings depending on how fast the player was able to complete the game as well as how much of the planet was explored. Only the most skilled players were presented with the ending that revealed Samus to be a female. On a similar note, Call of Duty: Modern Warfare 2 (Infinity Ward 2009) awards emblems to its players for completing actions that require a great deal of skill during multiplayer matches. These actions include getting head shots, saving teammates, and maintaining a high kill death ratio. The emblem the player unlocks can then be displayed on the player's profile indicating mastery to the community and encouraging others to attempt mastery. Displays of mastery are core to fandom culture (and an excellent source of achievements). What constitutes mastery of a game or system is one of the most time-honored debates in fan culture, and as educators, we are mostly still waiting for a game that inspires such devotion. Similarly, we are waiting for a game so robust as to produce new practices such as speed runs, which in Metroid for example is one way that players distinguish between expert and competent players.

Encouraging Risk Taking

At the Game Developer Conference (GDC) 2013 roundtable, Stockton discussed how many World of Warcraft (Blizzard Entertainment 2004) players have a deep aversion to Player vs. Player (PvP) and need to be coaxed to even dabble in battlegrounds. Over time, they have tweaked achievements to reward smaller and smaller steps toward experimenting with PvP, so that players can earn achievements for simply queuing for a battleground. Imagine educators targeting analogous areas such as mathematics and designing achievements so as to help students overcome math anxiety, public speaking, or sharing of writing with peers.

Encouraging Exploration of New Systems to Promote Mastery

In Plants Vs. Zombies (PopCap Games 2009), achievements and their respective badges can be earned by playing the game in different ways. For example, one challenge asks a player to complete a level using nothing but man-eating plants. While playing the game normally, man-eating plants are not typically a plant of choice, and may not often be used. In order to unlock this achievement, players must leave their comfort zone and learn how to use "Chompers" effectively. The key to this structure is presenting the

requested action as a challenge to the player. In education, this approach could draw attention to a part of the design that might have been overlooked or could encourage students to leave garden paths (simple solutions to complex problems). Sometimes achievements promoting exploration are presented once the main objective or some similar level of competency is achieved. This way, the exploration of the space in a new way will not be convoluted with the basics.

Reframing the Game Experience to Promote Reflection

Games such as Jetpack Joyride (Halfbrick Studios 2011), a simple side-scrolling action game, contain simple goals (travel as far as possible), but then use achievements to communicate wildly different game goals, which range from destroying scientists to creating "near misses" with lasers. Similarly, strategy games like Civilization routinely include achievements for winning through specific strategies, such as winning through an economic victory or by building only one city. Notably, this challenge (the One City Challenge) emanated from game community forums. More radically, games such as Deus Ex: Human Revolution (Eidos Montreal 2011), which rewards players for beating the game without killing *anyone*, are radical in that they use achievements to reconceptualize the entire game. As educators, we often want our students to reflect on the nature of a game as a representation (McCall 2011). We want our students to see games as designed objects that make conceits and simplifications in representing complex phenomena. Achievements can reframe the game experience and help students reflect on how game rules could be different. Students *authoring* achievements can also invite participation in the system.

Making Knowledge More Flexible (Trying Contrasting Cases)

Achievement systems can be used as a way quickly to describe a set of experiences that a learner has acquired over time. The value of having a flexible achievement system comes from the ability to collect different achievements without having to adhere to collecting all of a predetermined set. By providing a large set of achievements to choose from, players can compare and contrast their profiles, which draws attention to, and often celebrates, the differences. In this way, achievements may be used like a résumé in that it lists your achievements and draws attention to what sets you apart/what makes up your skill set.

Encouraging Collaboration

Collaborative achievements, such as Just Press Play's badge for taking a friend to an off-campus event, can push players toward having new experiences, expanding their social networks, and engaging in discussion and reflection. Some foursquare badges such as the BFF badge (10 check-ins with a friend) are perhaps the most dramatic example of how gamification can lead people to create new experiences for others. Rochester Institute of Technology's Just Press Play (case study 25.1) suggests how educators might leverage these tools, and, hopefully soon, Augmented Reality (AR) systems such as the MITAR (MIT Augmented Reality) engine or the ARIS (Augmented Reality and Interactive Storytelling) engine will enable user-authored badges as well.

Case Study 25.1

Just Press Play Project

Summary

One of the most sophisticated attempts at gamifying education is the Rochester Institute for Technology (RIT) Just Press Play project (https://play.rit.edu/). Just Press Play is a game layer (achievements, quests, profiles, and collectable media) designed to provide a game layer complementing the undergraduate experience in RIT's School of Interactive Games and Media. The goal of Just Press Play is to use game technologies to engage students in a playful way with their educational environment and experiences in a way that can support learning while also improving students' experience of RIT. Notably, Just Press Play currently occurs completely outside of the formal course structure, as its designers felt that it was critical that it be voluntary and outside of the formal curriculum to be truly playful.

Gameful Design Elements

Just Press Play achievements can be attained through course assignments, however, and some leverage course experiences. For example, the Undying achievement is gained by *all participating players* if 90 percent of the students in the Game Software Design class pass the course. Historically, the pass rate had been around 88 percent, but after initiating the achievement, the pass rate bumped to about 92 percent. Notably, Undying worked not because all of a sudden students started studying; rather, as Decker and Lawley (2013) report, this achievement led to the immediate formation of study groups. Upper-class students advised new students to form study groups, and they formed much more quickly than in past years.

Evaluation reports of Just Press Play emphasize that it is working because its designers use it as a tool for interrogating the undergraduate experience, designing quests and achievements that are likely to lead to desirable changes, and empowering students in this process. Other sample quests, such as visit every professor to gain his or her collectable card, attend a social event off-campus, and visit the gym, were design-based factors known to contribute to success as detailed in research, advice from former students, or occasionally data gained from the university.

This last category, data that the institution has about students (such as students who swipe their ID card at breakfast tend to perform better than those who don't), is a fascinating tool for educators to embrace data-driven education. Although most of this information is proprietary and not available to researchers (or even outside evaluators), most school administrators can learn a great deal about the behaviors of the student body, ranging from correlations between GPA and dining habits to exercise and drop-out rates. Our behaviors are nudged in different directions every day by marketers, policy makers, and others (see Thaler and Sunstein 2008). Tools such as achievements and badges, *if done well*, can make these efforts transparent, and possibly even give students a voice in designing achievements.

Reflecting on Just Press Play, questions we wrestle with as educational designers include: How can content production and rollout become a routine, smooth part of educational design when it's outside of course structures? How can we use achievements to facilitate multigenerational communication, so that students from one generation can share their stories, advice, and

Case Study 25.1
(continued)

professional networks with current students? How can we empower students to design, develop, and post achievements for their peers? How can we make this entire process (including the data driving these decisions) transparent to students?

Further Information

See https://play.rit.edu/ (official website).

Modeling for Users What Experts Are and Can Do

Much like a technology tree, achievements can be designed to lay out various competencies required by professionals (or any socially desired role). A primary value of such badges is making tacit forms of expertise explicit. One might imagine, for example, professors designing badges for graduate students to communicate the many "unspoken" parts to succeeding in academics (social networking, professional presentation skills).

Certifying Ongoing Professional Development

While formal educational systems are notoriously difficult to change (Tyack and Cuban 1995), professional development is almost wide open, suggesting opportunities for badges to certify upkeep of skills. Professional training is required in fields from medicine to information technology, but even more mundane domains such as driver's education could benefit from such badges. eBay is one model for doing this. To maintain a *flying star*, eBay vendors need to keep a consistently high rating based on user reviews. Because it needs upkeep (in this case, quality has to remain consistent), it is considered valuable by the community as an example of competence. Some argue against this approach feeling that it's taxing to have to keep up an achievement and that this introduces a lot of pressure. For example, maintaining a diamond rating in Starcraft or keeping your kill death ratio high in Call of Duty changes the way you approach a match. While such "hard core" badges may be problematic in recreation activities, they may have a real role in education.

A primary challenge for game developers is how to use these techniques without undermining motivation (see Hecker 2010). As Hecker wryly observes, the research on intrinsic motivation is complicated and at times contradictory, but we do know that at times, rewards can undermine intrinsic motivation. It may be that intrinsic and extrinsic motivation are largely emergent phenomenon arising at the intersection of person, task, and context, and thus there is significant variance in how people experience such motivation. As design researchers seeking to transform learning experiences, we argue that gamification structures, such as achievements, can function to promote learning if used carefully (for a similar tact on how badges are currently being used in education, see Joseph 2012).

The Potential and Challenge of Gamification as Alternative Accreditation

For many educators, the larger goal of badges is to create an alternative system of assessment and accreditation. Students learn skills in formal and informal spaces unrecognized by institutions of learning. Wikipedia, for example, relies on its community to edit articles and make sure that content is accurate. Because the error rates of Wikipedia and *Encyclopaedia Britannica* are comparable (Giles 2005), we might infer that volunteer editors perform at a level similar to that of professional editors. Within Wikipedia, users are acknowledged and rewarded for their contributions to the site with banners that they can place on their user pages. Unfortunately, there does not currently exist a way to reward users for this expertise outside of Wikipedia. Even if the user is performing at a professional level, his or her work may hold little to no weight when applying to a university program or interviewing for a job (Ochsner and Martin 2013). Ideally, badges should serve as an indicator of expertise to which the user can defer in these situations.

We might expect gamification to thrive in quickly changing domains. Computer science has faced a similar problem of showing expertise without a formal accreditation process, such as a university course. New technologies are released constantly, and proficiency in those technologies quickly becomes a very marketable skill. However, by the time that a university actually offers a class in said technology, there is a good chance that skill is obsolete or that a new technology has replaced it. This results in a time when the skill is in demand but there is no way to hire someone with traditional accreditation. This problem has led to a different way that institutions teach computer science. Rather than teaching a specific computer language, computer science is more geared to programming paradigms, urging its students to develop skills that will allow them to learn new languages rather than just learning one language well. This doesn't mean that experts don't exist—in the meantime, self-taught programmers, who have been working with the technology during the interim, have become experts—just that there is a need for an alternative form of accreditation.

In order to meet this gap, developers, like Sun Microsystems and Microsoft, have created targeted accreditation systems that anyone can take in order to display expertise. Individual programmers have also leveraged services like Github to serve as portfolios. Github is a collaborative coding space where programmers publish their code to the general public to view and modify. As a portfolio, Github serves as a persistent and illustrative example of a programmer's skill as well as an example of how he or she works collaboratively. It's not hard to see how a system of badges can serve a similar purpose.

Already we're seeing online learning forums deploying badges meant to illustrate expertise. Thanks in part to the success of Kahn Academy, Code Academy, and similar communities, it now appears that the question is not "Will gamification be used for accreditation?" but rather "How?" We can learn from these initial adopters by reflecting on the challenges they face implementing such systems in the wild. For example, most "attendance" badges can be gamed easily because they're assigned automatically, triggered by predefined rules. Because of this, a user can

earn a badge by simply playing a video without actually watching it. If we wish to use these badges as alternative forms of assessment, we must keep in mind what we can actually infer about the user given the medium. If we assign a badge for playing a video, we can assume, but cannot be certain, that the user watched the video. Similarly, if our goal is to create meaningful achievements, we must be mindful of not only the ways badges are earned, but also the ways they can be exploited. Badges that require more from the users are not immune to exploits either. An example of this can be found in Kahn Academy's black hole badge.

The black hole badge was meant to be a reward for persistence and proficiency requiring the user to spend a great deal of time on the site completing exercises. Because of its rarity and exclusiveness, a lot of value was placed on obtaining a black hole badge. However, the difficulty associated with getting such a badge resulted in the creation of walk-throughs illustrating ways to game/exploit the system in order to get the badge efficiently. One such exploit is getting nine questions in a row right and then missing the last one on purpose in order to maximize the number of points received (see https://code.google.com/p/khanacademy/issues/detail?id=3083). Originally meant to encourage users who were not quite experts, this point increase instead encouraged students to answer incorrectly questions they might have gotten correct. If the measure of proficiency assumes that users who gets all questions correctly perform better than those who do not, this level of exploit would undermine that assessment. The badge functioned perfectly as an external motivator, but unfortunately, at least in some cases, it led to the reverse engineering of the system in order to earn the badge efficiently without actually having to "learn" or complete the task in the way it was intended.

Just because a system can be exploited doesn't mean it lacks merit, but just that it may need revision. Games are often exploited by players wishing to "min-max" the system. Min-maxing is the process of minimizing undesired experiences or traits, while maximizing desired outcomes. Min-maxing is expected, and it is usually in the interest of the developer to patch, or fix, the problem if it is detrimental to the desired experience. Similarly, a well-executed attempt at gamification will have continual refinement of the approach taking into account how a user interacts with the game.

In addition to circumventing the system, Jenkins (2012) cautions that badges, and similar achievements, can also potentially disrupt organic, existing communities because they assign value to what the designers feel is important, rather than what fan communities value. While this is very important to consider, incorporating player and designer values is not impossible. Taking examples such as Civilization's One City Challenge, the successful incorporation of achievement systems is an asynchronous conversation between players and designers. Designers observe players' activity, see what they value and how they play games (and see also speed runs or head shots), and then acknowledge these practices in future games. Games are, after all, both genres of media and modes of interaction, and we carry over practices from one media experience to the next. Indeed, one of the most rewarding game experiences as a player is when a novel action or strategy is tried (such as jumping off of a cliff in World of Warcraft), and suddenly an achievement pops up recognizing your accomplishment and creating this kind of secret conversation between the player and designer.

Rather than being a static artifact, the incorporation of achievements, especially those that promote mastery or exploration, creates a dynamic system that continues to challenge the user.

Another key question confronting educators is whether to create multiple badge systems (such as a badge system for every museum or informal learning space) or whether to create a education-system-wide, or even global, network of badges. Using a global network has many advantages: issuers/educators don't have to consider whether they have the infrastructure necessary to keep track of and host the achievements, and users have the convenience of having a central repository that catalogs their achievements across a variety of contexts. In contrast, having multiple systems means that a user's achievements are scattered all around the Internet. While those badges may accurately reflect a user's expertise, they run the risk of becoming obscure or esoteric simply because they are harder to share. Unfortunately, an award has little value if no one knows why it was important in the first place. Being unable to share your awards efficiently makes the risk that these achievements will be overlooked more likely. Open systems aggregate these achievements highlighting what users have done without having to pull together resources.

Mozilla's Open Badges project seeks to create an open standard for badges that enables institutions to host their own badges, while also being connected to a larger organization. Mozilla's open Application Programming Interface (API) is a free, "open technical standard any organization can use to create, issue and verify digital badges" (Mozilla Foundation 2011). A museum could, for example, create a badge for completing a game design workshop, which would be displayed alongside achievements designed by NASA, Microsoft, and Disney (see http://community.openbadges.org). Implementing Open Badges is straightforward with many resources and third-party implementations available online. Briefly, the educator's system identifies where the player will be and what the player has to do in order to get the award, and the system (game) informs the hosting accreditation system that the user has met the requirements. The relevant badge is then issued, and a notification is sent via e-mail. This badge can then be displayed on a page, unique to the user, or shared over social networks such as Facebook or Twitter.

Educators hope that through Mozilla's Open Badge initiative, schools, museums, and other formal and informal learning groups might contribute toward this open repository. Particular badges may become respected, so that educators understand that completing a game design workshop or creating a game that achieves a certain rating is valuable. Perhaps, even educational institutions will begin using them as credentials instead of than grades, GPAs, or test scores.

Gamification for Learning through Narratives and Quests

We recognize that objects embody values and can be actors in a system (see Latour 2005). However, we also recognize that such designs are always manifested in actual social contexts and can only be understood in conversation with encompassing social structures (such as culture). Indeed, there exists decades of basic research exploring how game structures contradict those of formal and informal learning spaces (cf.

Brown and Cole 2002; Squire 2004; DeVane and Squire 2012. Most educators from this perspective are using games as a leverage point for systemic change in education and ask, "Can games change how schools operate?" (as opposed to meeting the traditional goals of schooling more effectively). Thus, it is not enough to identify, describe, and reverse engineer design principles from games and game communities or design learning environments based on these principles, but most educators are deeply interested in investigating the impact of these efforts on broader social institutions (Squire and Steinkuehler 2005; Cole et al. 2010; Martin and Steinkuehler 2010). Thus, gamification becomes not simply a tool for organizations to meet old ends more effectively, but rather a tool for transforming institutions like schools.

Needless to say, this is challenging work. Twenty years of learning-sciences research describes how innovations are usually rejected by the system. Ann Brown (1992) developed the term "lethal mutation" to describe how most interventions are changed so that they lack any resemblance to their original design and instantiate the values of encompassing organizations. While the previous sections described classic forms of gamification, it's important to note that learning through narratives and quests, while scarcely covered in the broad reception of gamification in education, has been demonstrated to be highly effective. Projects such as Fifth Dimension and Quest Atlantis demonstrate how such programs work.

Designing Gamified Learning Systems in Informal Spaces: Fifth Dimension

Fifth Dimension is one of the longest running and most well researched gamification-type learning systems (Cole 2006). Fifth Dimension was created in the 1990s as a mechanism for exploring Cole's (1996) theory of cultural psychology. Cole's theory, in short, is that culture is at the center of cognition (particularly the kinds of learning that we care about as educators, such as habits of mind or problem solving). Thus, a primary goal of educators is to create learning contexts, or cultures, with good learning values, such as classrooms based on inquiry or knowledge construction (e.g., Scardamalia and Bereiter 1994). A second (related) goal is to create learning situations in which youth learn through working along with more expert peers or adults on mutually valued tasks (or, in other words, in apprentice-like situations; see Brown, Collins, and Newman 1991).

In Fifth Dimension "clubs," groups of five to fourteen youth work through a maze of quests, which are assigned to them by a fictional wizard. These quests are presented on task cards and frequently are based on a game such as Carmen San Diego, The Incredible Machine, or Oregon Trail. All learning is situated within a narrative experience of helping the wizard, rather than earning grades or points. Adults (usually college students) play Wizard's Assistants, who work as mentors for students and reinforce the values of the clubs (while gaining professional development experience). This Vygotskian-inspired role is crucial for organizing activity and is the mechanism by which youth engage in academically valued expertise with mentors.

More than one hundred research studies and a major evaluation report have been conducted on Fifth Dimension, which has been enacted in sites across the world (Greene Simmons, and Blanton 1999; Blanton, Greene, and Cole 1999; Cole 2006). Evaluation reports show that, through participation in Fifth Dimension, students develop academic skills that can be used across a variety of contexts (Blanton

et al. 1997; Mayer et al. 1997). In a study of the project's implementation in after-school centers, Nicolopoulou and Cole (1993) described how game play as an activity is not purely a function of the game and the player, but is profoundly shaped by local cultural contexts. Local Fifth Dimension cultures were a mixture of the designed Fifth Dimension culture plus the culture of the overlapping institutions. These emergent, hybridized cultural conditions of each setting might be described as a local microculture within the broader culture. It mediates the activities taking place in the environment, crucially affecting the kinds of learning outcomes that result.

These results matter for gamification researchers because they suggest that *game structures may always struggle truly to transform encompassing social institutions*. Meaning, if designers want to use gamification techniques only to change the user experience of a product (e.g., airlines rewards programs) or build a product around a gamification experience (e.g., foursquare), they might anticipate success. If designers want to use gamification techniques to structure learning interactions, they might even expect to produce gains. However, transforming the very culture of an institution, even one as seemingly malleable as a boys & girls club or drop-in library program, has hitherto proved difficult. Recalling that for cultural psychologists, *the transformation of culture* is the core learning goal, we are reminded that cultural transformation is a daunting task that tools themselves can rarely accomplish (Engestrom 1990).

Designing Gamified Learning Systems for Schools: Quest Atlantis

Subsequent projects, such as Quest Atlantis, have created a similar metaverse (that of Atlantis), but built around social commitments such as respect for diversity or creative expression. Quest Atlantis is a transmedia learning experience for late elementary school. Although there are also Quest Atlantis comics, novels, and other media, in Quest Atlantis learners primarily access the world of Atlantis through an online portal (called OTAK) and use it to communicate with citizens of a lost civilization. The world of Quest Atlantis (including its quests) are rendered in real-time 3D (first in Active Worlds, more recently in Unity). Through quests related to these commitments, they improve life for these citizens. The quests range from online to offline behaviors and are usually certified by a teacher through an online dashboard. Students' online profiles include features such as an item inventory, reputation systems, various currencies, and so on. Quest Atlantis is a fascinating example of gamification, in that although it contains a number of "in-game" quests, the majority of its quests occur offline. Most Quest Atlantis quests are traditionally valued academic practices, but are given meaning through a narrative of the Atlantians, communicated via quests. These quests, such as a scrapbooking quest (box 25.1), are academic tasks that many teachers already do, but are given meaning through the Quest Atlantis narrative (Stuckey 2008). The questing structure also serves as a *motivational* meta-game that repositions traditional academic *tasks* as a new *activity* (Barab, Warren, and Ingram-Goble 2009). That is, Quest Atlantis tries to take isolated exercises such as scrapbooking and give them meaning through a connection to a broader narrative of the Quest Atlantis world. Tuzun (2004) describes how students participating in Quest Atlantis were motivated by identity, play, immersion, and social relationships, which for some students transformed academic activity from reward-based activity to

Box 25.1

Exkurs: First Task in Quest Atlantis

The communities of Atlantis are not always appreciated, and we are trying to improve that as much as possible. The Council has asked all of the children of Atlantis to create a scrapbook of their community so that the communities can be once again appreciated. There is one small problem: The children do not know how to make a community scrapbook. We need your help!

Your first task is to decide how your book should look, and then develop it. When you finish, bring your book in your Quest Atlantis Center and show it to a mentor. Also, write a summary about how you created your book and why you chose to include the things you did. You can take a photo of the book and send it to the Council along with your summary.

Your Goal(s):

- Choose four or more categories to feature in your book.
- Create your scrapbook.
- Write a summary about how you created your book and why you included the things you did.
- Bring your book and summary to your Quest Atlantis center.
- Upload and submit your work.

activity driven by a desire to become new kinds of people, to play, or to engage in legitimate social activity. Through Quest Atlantis's many interactions, designers Jackson, Arici, and Barab (2005) find that the Quest Atlantis *narrative* is its most engaging feature.

Quest Atlantis demonstrates how a gamified learning context can lead to learning gains across a variety of contexts in and out of schools. Although Quest Atlantis spans most of the elementary curriculum, much of the research conducted upon it is on scientific reasoning. Quest Atlantis (like many virtual worlds) strives to give players an embodied experience of complex phenomena such as conducting a scientific investigation. Studies of Quest Atlantis use in classrooms show that it can improve students' scientific inquiry, reasoning, and argumentation skills, along both traditional and performance measures (Hickey, Ingram-Goble, and Jameson 2009; Barab et al. 2007, 2010). Barab and colleagues describe this process as "narratizing the curriculum," which suggests one useful way for designers to think about gamifying formal curricular structures. We often think of "narrative wrappers" as little more than motivational contexts, but such narrative contexts, if used well, can attune players to new goals and create a deeper purpose for activity.

The preceding examples illustrate how gamification structures can promote learning by organizing activity around problem-solving or creating narrative contexts for learning. Quest Atlantis includes some traditional "game elements" (i.e., levels such as Taiga, in which the content is encoded within systems or worlds; e.g., Squire 2006), but much of Quest Atlantis focuses on organizing activity *outside* of games. As educators seek to create games for learning or broader impact, opportunities exist to leverage gamification structures in games and in conversation with a broader meta-game system to improve learning.

Conclusion

When designing gamified learning environments, it's important to see it as an ongoing process. Just as good games require that their game mechanics match up with the experience you seek to convey, game mechanics must be used to contribute toward the values and culture that one hopes to foster. We argue for a participatory approach in which all constituents can participate in defining badges, achievements, or structures, but how to design such a system is not straightforward. In short, the execution matters.

Gamification is never inherently "done," but is an ongoing process, the quality of which depends on how it is enacted. We must study empirically the value of such systems in context and for different user groups. Traditional constructs such as "achievement motivation" are problematic because they are rarely contextualized. Motivation is an emergent property defined by person, task, and context, and thus must be constantly studied. Games themselves—which have been used as a reward—are a fascinating problematizer of intrinsic motivation, as they sometimes are motivating and sometimes are not, and even within games, intrinsic and extrinsic properties are frequently used.

Gamification structures will flourish in upcoming years as a response to the open-ended nature of information. As educational systems wrestle with the reality that "knowledge is free on the Internet" and that an increasing amount of learning occurs outside of classrooms, new accreditation mechanisms are clearly needed. We face very real challenges in education. We face very real questions around equity and assessment. Institutions such as testing companies—or even universities—have business models that profit on the status quo, which may or may not be in the best interests of ordinary citizens. We think gamification techniques are a critical set of design tools in an educator's toolbox to address these issues. Whether or not gamification is a good thing is missing the question; good and bad examples of gamification exist. Issues around individual versus institutional values and accreditation versus performance are not new. Our hope is that educators might take advantage of this moment of disruption (and invention of gamification) and work toward more democratic learning organizations.

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POSITION STATEMENT

I'M NOT PLAYFUL, I'M GAMEFUL

Jane McGonigal

I've spent tens of thousands of hours making, researching, and playing games. I love games, and my life is full of game play. However, I am *not* a playful person. In fact, I may be the least playful person you will ever meet. This is not hyperbole. I have actual data to back it up!

Ten years ago, I took a psychology test called the Virtues in Action (or VIA) Inventory of Signature Strengths (viacharacter.org). I answered 240 questions about what I think I'm good at and what I value most in life. Afterward, I received a list of twenty-four character strengths—globally recognized virtues like courage, curiosity, and the ability to love and be loved. These strengths were ranked in the order in which I possessed them.

I had a major "a-ha" moment when I saw my ranked list. This scientifically validated psychology measure had just confirmed all of my research about the quintessential strengths of gamers. The five traits at the top of my list—according to researchers, your top five are your "signature strengths," the character traits and virtues that make you *you*—corresponded perfectly with what I had observed in my ethnographic study of how gamers solve problems and tackle obstacles (McGonigal 2007).

1. *Creativity*: Thinking of novel and productive ways to conceptualize and do things.
2. *Perseverance*: Finishing what one starts; persisting in a course of action in spite of obstacles; taking pleasure in completing tasks.
3. *Love of learning*: Mastering new skills, topics, and bodies of knowledge.
4. *Optimism*: Expecting the best in the future and working to achieve it; believing that a good future is something that can be brought about.
5. *Curiosity*: Taking an interest in ongoing experience for its own sake; exploring and discovering (Petersen and Seligman 2004).

All of these strengths are closely correlated with success in game play and provoked by good game design—so it was no surprise to me that I, a lifelong gamer, should have come to value them so much.

But even more interesting to me than my top five strengths was my *lowest* strength. Ranked all the way at the bottom, dead last, number twenty-four, was the strength of playfulness.

Some friends and colleagues found this result strange and unbelievable. "But you're a game designer!" they said. "You love games! How could you score so low on playfulness?" Indeed, not only had I scored low, I had scored *as low as it is humanly possible to score for playfulness*. I am as unplayful a human being as one can be, at least as measured by the VIA Inventory of Strengths.

This made perfect sense to me, for one reason—a playful state is characterized most quintessentially by being "purposeless," unattached to a particular outcome, and therefore free to explore many different possibilities (Brown 2009). Think of sketching and drawing, sandbox play, improvisational construction with Legos, or jamming with other musicians. In these playful states, we aren't driven by specific goals. Instead, we discover and reinvent our purpose as we go along, constantly evolving our actions with great spontaneity.

Playing a game, in contrast, usually focuses us on a particular outcome—a specific goal that we are trying to achieve, a score we want to beat, a rank we want to obtain, a level we want to complete, a puzzle we want to solve, a part of the map we want to unlock, an opponent we want to defeat, a treasure we want to obtain. Minute by minute within a game, we are full of purpose, motivated by clear goals, and required to develop resilience in the face of obstacles. This goal orientation in the presence of obstacles is what makes a game truly a game, and not just free play (Suits 2005). It is also, according to my research, precisely what builds up the classic gamer strengths. We persevere in the face of obstacles, failing again and again but continuing to play until we improve and succeed. We seek out mastery, or love of learning. Indeed, without the desire to build skills, acquire knowledge, and improve performance, we would all be "level one" players forever. And most crucially, we embrace a form of optimism often referred to as self-efficacy: the belief that we can influence the outcome of an event through our own actions. No gamer will play a game if the gamer believes that he or she is doomed to lose, or fail, or be terrible at it forever. Gamers are fundamentally optimistic about the process of becoming better and achieving their goals.

Of course, games also draw on strengths that have much in common with more playful activities; namely, creativity and curiosity. We often must invent new strategies, or improvise new solutions, or test different approaches. And we must remain curious about what's around a corner, or on the next level, or how a puzzle will be solved. Without creativity and curiosity, game play is essentially just hard work. So, after all, there still must be some *play* in game play. Gaming simply happens to be a form of play that, in addition to creativity and curiosity, *also* requires hard work—thereby building the strengths of perseverance, love of learning, and optimism.

Reviewing my signature strengths results all those years ago, a light bulb went off over my head: I'm not playful, I'm *gameful*.

In other words: I'm highly motivated, goal oriented, and I seek out challenges. I don't give up in the face of obstacles, I love acquiring new skills, and I am optimistic about my own capabilities.

This realization—and how different these traits seem to me, in sum, from what we consider a "playful personality"—is what led me to begin using the term *gameful* in my work.[1]

My original intention in using the word *gameful* was simply to draw attention to the positive qualities of gamers. I wanted to shine a light on the character strengths that games help us develop.

Why shine a light on gamer strengths? According to data from the Pew Internet and American Life research group, 99 percent of boys and 94 percent of girls in the United States under the age of eighteen play games regularly, thirteen hours and eight hours a week, respectively (Pew Internet and American Life Project 2008) And according to my own research compiling demographic reports from dozens of regions worldwide, there are as of early 2014 more

than one billion gamers on the planet spending on average at least an hour a day playing games (McGonigal 2011).

Facing these tremendous demographic shifts, I wanted new language to talk about how game play could become a major asset for the next generation. I wanted to be able to explain to skeptics why a billion gamers could be very good news indeed. Most importantly, I wanted a way to describe to gamers themselves the strengths and virtues that they could bring to tackling real-world challenges, skills and abilities that transcend our game play and become a part of who we are.

In other words, I wanted to take gamers seriously. And the word *playful* just didn't fit the bill. *Playful* doesn't do gamers enough justice. We are more than playful. We are also gameful.

Recently, gameful has just as often, if not more often, been taken up to describe *things* rather than people. Many researchers and designers now use the term *gameful* to talk about designed experiences and systems that work and feel like a game. To be deemed gameful, these experiences and systems must effectively integrate some of the key structural and aesthetic elements of a game—things like clear goals, constant feedback on progress, obstacles that provoke creativity, avatars that get stronger over time, or heroic narratives. (These are, not coincidentally, the very structural and aesthetic elements of a game that most directly help build a gamer's top five signature strengths.)

So instead of just saying "I'm gameful," we're now also saying "*That's* gameful." This complementary usage, it seems to me, performs the same important task of distancing our work and objects of study from the term *playful*. Those of us who are passionate about giving gamers real-world challenges to tackle, and those of us who want to explore serious subjects with games, simply may not in fact be able to do it through *play*.

Why not change the world through play? Our definitions and understanding of *play* are too bound up in the idea of play *not* changing the world; that is, being relatively without real-life consequences. Since the inception of digital game studies as a formal field, circa 2001, the philosophical notion of a "magic circle of play" has been taken up as one of the most crucial defining aspects of games (Salen and Zimmerman 2003).[2] The idea comes from the early twentieth-century philosopher Johan Huizinga, who opined that games are played in special, protected "play-grounds, i.e. forbidden spots, isolated, hedged round, hallowed, within which special rules obtain. All are temporary worlds within the ordinary world, dedicated to the performance of an act apart" (Huizinga 1950).

But with gameful design, we are intentionally stepping outside the magic circle of play—or at the very least, fusing the magic circle with the ordinary world in ways that seek to change it. As a gameful designer, I myself am making games that seek to intervene in players' real lives. My game SuperBetter, for example, is designed to help players overcome anxiety, chronic pain, and depression (superbetter.com). And EVOKE led players to develop business plans for real social enterprises; more than fifty of these business plans became actual businesses when players received funding and mentorship from Global Giving and the World Bank Institute (urgentevoke.com). Like so many new gameful projects, these games are meant to have real impacts, to change players' real lives and to change the world as well.

As gameful designers, we are tackling serious problems. We are harnessing the core strengths of gamers to make a positive impact on reality itself.

The term *gameful* helps us make this intention explicit. It empowers us to leap outside the magic circle of play consciously and deliberately, into a world where our games by design have very real consequences.

This book, *The Gameful World*, and the unique perspectives of its many contributors, will no doubt multiply and challenge these two preliminary definitions of what is gameful. For this reason alone, it does not truly matter—apart from historical interest—what I meant four years ago when I first redefined and began advocating for the term *gameful*. As gameful designers and researchers, together we will make the idea of *gamefulness* explode with new possibilities—new ways of understanding what gamers are good at, and what lies at the heart of a good game. And with any luck, through our collective efforts to unleash the creativity, perseverance, love of learning, curiosity, and optimism of gamers, the world itself will become more gameful—in all the best meanings of the word.

Notes

1. I'm not the first person to use the term *gameful*—you can find it in at least one early sixteenth century Jacobean play, for example, referring to the pleasant time of one's life in which many games are played (Taylor and Lavagnino 2007). But in 2009, I decided to redefine the term to mean "having the positive traits of a gamer" or "having the positive traits of a game," and I adopted it as the primary way I talk about my work as a game designer and game researcher. I added it to the Urban Dictionary in 2009 (http://www.urbandictionary.com/define.php?term=gameful). In 2010, I cofounded Gameful (gameful.org), a social network for gamers and game developers interested in researching and increasing the positive impacts of games on our real lives and on real-world problem solving.

2. For a fascinating and provocative history of use of and debate over "the magic circle of play" in digital game research and game design, see Eric Zimmerman's (2012) "Jerked Around by the Magic Circle," in which he argues that the magic circle is a straw man that has never actually held much weight in either community. Given the pervasiveness of references to the term in both, I'm not sure I agree with Zimmerman, but it's certainly a point worth considering. My own experience is that the critics and researchers who are most opposed to the idea of *gamifying* real work, real problems, and real lives point to the magic circle as a necessary element of games and argue that breaking the magic circle removes the psychological and social safety necessary to play and enjoy a game.

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